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The Theory of Zeta-Functions of Root Systems

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Yasushi Komori · Kohji Matsumoto ·
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The Theory of Zeta-Functions of Root Systems

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Preface

The aim of this book is to give the full account of the present stage of the theory of zeta-functions of root systems, which are certain multiple series associated with root systems.

The theory of multiple zeta-functions has a long history going back to the eighteenth century, but the modern active research began in the 1990s. In this trend, the central role has been played by Euler–Zagier multi-variable multiple zeta-functions. Special values of them, called multiple zeta values (MZVs for short), have close connections with various fields of mathematics such as arithmetic geometry, topology and so on. Some variants of Euler–Zagier multiple zeta-functions, for instance Mordell–Tornheim multiple zeta-functions, have also been introduced and studied extensively.

On the other hand, in the field of quantum gauge theory in mathematical physics, the notion of Witten zeta-functions associated with semisimple Lie algebras was introduced. Witten zeta-functions can be written down as multiple series, and hence those are also multiple zeta-functions, but in one variable.

Zeta-functions of root systems give a unified view in the above landscape of the theory of multiple zeta-functions. The history of the theory of zeta-functions of root systems can be traced back to a paper of L. Tornheim in 1950, but the general definition was given by the authors. The original motivation of the authors was to consider the multi-variable version of Witten zeta-functions; first the definition of the B_2 case was introduced by the second-named author in 2003, and then the definition of the general case was published in 2007. The research of the authors on this theory has been ongoing for decades.

An advantage of introducing the multi-variable version is that we can get more analytic flexibility. Because of this flexibility, we can discuss the recursive structure in the family of Witten zeta-functions, and can achieve the meromorphic continuation of them.

Another important point is that the notion of zeta-functions of root systems is not just the multi-variable generalization of Witten zeta-functions, but also, a generalization of multi-variable multiple zeta-functions of Euler–Zagier type and of Mordell–Tornheim type. Therefore the notion of zeta-functions of root systems is unification of various important classes of multiple zeta-functions.

Moreover, the family of zeta-functions of root systems has various fascinating structures, coming from the structure of the underlying root systems. Using these structures, we can develop the theory of functional relations of zeta-functions of root systems. From the same viewpoint, it is also possible to apply the theory of zeta-functions of root systems to the study of MZVs, and new methods and results on MZVs have been obtained.

Almost all of the contents of the present book is the output from the authors' papers. The main topics treated in this book are special values of zeta-functions of root systems at positive integer points, the meromorphic continuation, the functional relations, applications to MZVs, some variants and generalizations, and so on. We not only give the general theory, but also write down a lot of explicit examples in the case of zeta-functions of root systems of low ranks.

The basic theory of Lie algebras and root systems, necessary to read this book, is gathered in Chapter 2, and further required knowledge is explained here and there in the text.

Many theorems in this book are not merely reproduced from the original papers. Some theorems are proved in a way different from the original one. Some theorems are stated in a generalized form, with a generalized proof. Misprints and inaccuracies included in the original papers are corrected. Furthermore, there are some results in this book which have not been published before.

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December 2022

Yasushi Komori, Kohji Matsumoto and Hirofumi Tsumura

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CHAPTER 1

Introduction

The notion of zeta-functions of root systems can be regarded as a certain subclass of the family of multiple zeta-functions.

The theory of multiple zeta-functions has a long and rich history, going back to the eighteenth century. We begin with a discussion on various kinds of multiple zeta-functions. In this chapter we mainly focus on the historical account, especially on the history of the theory of meromorphic continuation of multiple zeta-functions.

In what follows, $\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the set of all positive integers, non-negative integers, rational integers, rational numbers, real numbers, and complex numbers, respectively. The empty sum is to be understood as 0, and the empty product as 1.

1.1. The Euler double zeta-function

The history of the theory of multiple zeta-functions goes back to private communications between C. Goldbach and L. Euler in 1740s. Double zeta values[1]

$$\zeta_{EZ,2}(k_1,k_2)=\sum_{1\le n_1<n_2}\frac{1}{n_1^{k_1}n_2^{k_2}}=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\frac{1}{m_1^{k_1}(m_1+m_2)^{k_2}}, \tag{1.1.1}$$

where k_1, k_2 are positive integers with $k_2 \ge 2$, were first discussed in their communications. Later, Euler published a paper [42] on several properties of $\zeta_{EZ,2}(k_1,k_2)$, especially the *sum formula*[2]

$$\sum_{k=2}^{m-1}\zeta_{EZ,2}(m-k,k)=\zeta(m)\qquad(m\ge 3), \tag{1.1.2}$$

where $\zeta(\cdot)$ denotes the Riemann zeta-function.

Euler was interested in just the values of double infinite series $\zeta_{EZ,2}(k_1,k_2)$, but if we regard $\zeta_{EZ,2}(k_1,k_2)$ as a special value of the

[1] The meaning of the suffix $EZ,2$ will be clarified in Section 1.3.

[2] A proof of this formula from our viewpoint will be given in Section 12.1.

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double zeta-function

$$\zeta_{EZ,2}(s_1,s_2)=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\frac{1}{m_1^{s_1}(m_1+m_2)^{s_2}}, \tag{1.1.3}$$

where s_1, s_2 are complex variables, we are led to another direction of research, that is, the analytic study of $\zeta_{EZ,2}(s_1,s_2)$. This direction was cultivated much later in Atkinson's paper [11] published in 1949.

Atkinson's motivation was to study the mean square of the Riemann zeta-function. For this purpose, he began with the following simple but important decomposition. When $\Re s_1, \Re s_2 > 1$, we have

$$\begin{aligned}\zeta(s_1)\zeta(s_2)&=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}m_1^{-s_1}m_2^{-s_2}=\sum_{m_1<m_2}+\sum_{m_1=m_2}+\sum_{m_1>m_2}\\&=\zeta_{EZ,2}(s_1,s_2)+\zeta(s_1+s_2)+\zeta_{EZ,2}(s_2,s_1).\end{aligned} \tag{1.1.4}$$

This was also first discovered by Euler, and is called the *harmonic product formula*. At first this formula is proved in the region $\Re s_1, \Re s_2 > 1$, but Atkinson's idea is to use this formula in the case $\Re s_1 = \Re s_2 = 1/2$. Therefore Atkinson was forced to consider the meromorphic continuation of $\zeta_{EZ,2}(s_1,s_2)$ appearing on the right-hand side. By using the Poisson summation formula, Atkinson proved a certain integral expression of $\zeta_{EZ,2}(s_1,s_2)$, which enabled him to obtain the meromorphic continuation.

Later, Y. Matsuoka [152] proved the continuation of $\zeta_{EZ,2}(1,s_2)$, and Apostol and Vu [5] discussed the continuation of $\zeta_{EZ,2}(s_1,s_2)$ with respect to s_1 for each fixed s_2 and also with respect to s_2 for each fixed s_1. Apostol and Vu also considered the series

$$\sum_{m=1}^{\infty}\sum_{m<n}\frac{1}{m^{s_1}n^{s_2}(m+n)}=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\frac{1}{m_1^{s_1}(m_1+m_2)^{s_2}(2m_1+m_2)} \tag{1.1.5}$$

and studied its meromorphic continuation in some cases with one of variables fixed.

Around 1980, several mathematicians, working in the mean value theory of zeta and L-functions, noticed the importance of the method of Atkinson. Generalizations of Atkinson's method to the case of Dirichlet L-functions

$$L(s,\chi)=\sum_{m=1}^{\infty}\frac{\chi(m)}{m^s} \tag{1.1.6}$$

attached to a Dirichlet character χ were discussed by Meurman [155] and Motohashi [158]. In particular, Motohashi developed a new approach to Atkinson's theory, based on a contour integration method.

Katsurada and the second-named author [91] [92] [93] used Motohashi's idea to obtain asymptotic expansion formulas for certain mean values of Dirichlet L-functions and of Hurwitz zeta-functions

$$\zeta(s,\alpha)=\sum_{m=0}^{\infty}\frac{1}{(m+\alpha)^s}\qquad(0<\alpha\le 1). \tag{1.1.7}$$

In the case of Hurwitz zeta-functions, as an analogue of (1.1.4), they obtained the formula

$$\zeta(s_1,\alpha)\zeta(s_2,\alpha)=f(s_1,s_2;\alpha)+\zeta(s_1+s_2,\alpha)+f(s_2,s_1;\alpha), \tag{1.1.8}$$

where

$$f(s_1,s_2;\alpha)=\sum_{m_1=0}^{\infty}\sum_{m_2=1}^{\infty}\frac{1}{(m_1+\alpha)^{s_1}(m_1+m_2+\alpha)^{s_2}}, \tag{1.1.9}$$

which is essential in their argument. This $f(s_1,s_2;\alpha)$ is clearly a generalization of $\zeta_{EZ,2}(s_1,s_2)$.

1.2. Multiple zeta-functions of Barnes and of Mellin

The analytic study of $\zeta_{EZ,2}(s_1,s_2)$ was begun by Atkinson in the middle of the 20th century, but for some other type of multiple zeta-functions, the analytic investigation started much earlier, around 1900, by Barnes and Mellin.

The motivation of Barnes was to develop the theory of multiple gamma functions. For this aim, he [13] [14] introduced the multiple zeta-function of the form

$$\zeta_{B,r}(s;\alpha,w_1,\dots,w_r)=\sum_{m_1=0}^{\infty}\cdots\sum_{m_r=0}^{\infty}\frac{1}{(m_1w_1+\cdots+m_rw_r+\alpha)^s}, \tag{1.2.1}$$

where r is a positive integer and $\alpha,w_1,\dots,w_r$ are complex parameters. This is clearly a multiple generalization of the Hurwitz zeta-function (1.1.7). On the other hand, Mellin [153] [154] considered a more general form

$$\zeta_r(s;P)=\sum_{m_1=0}^{\infty}\cdots\sum_{m_r=0}^{\infty}\frac{1}{P(m_1,\dots,m_r)^s}, \tag{1.2.2}$$

where $P(X_1, \dots, X_r)$ is a polynomial of r indeterminates of complex coefficients with positive real parts. Both Barnes and Mellin considered multiple series, but in one variable s. Both of them proved the meromorphic continuation of their multiple series.

In the 1970s, Shintani [183] [184] [185] noticed that Barnes multiple zeta-functions are useful in the study of construction of class fields over totally real number fields. In his study, Shintani introduced the multiple zeta-function

$$\sum_{m_1=0}^{\infty} \cdots \sum_{m_r=0}^{\infty} \prod_{j=1}^{n} \frac{1}{(w_{j1}(m_1+\alpha_1)+\cdots+w_{jr}(m_r+\alpha_r))^s} \tag{1.2.3}$$

with parameters α_i, w_{ji} $(1 \le i \le r, 1 \le j \le n)$. This is a generalization of (1.2.1). Shintani proved the meromorphic continuation of (1.2.3) under the condition

$$\Re w_{ji} > 0 \qquad \text{for all} \quad i, j. \tag{1.2.4}$$

Cassou-Noguès was inspired by the work of Shintani, and also of Mellin, and wrote a lot of papers (such as [29]) on the meromorphic continuation of $\zeta_r(s; P)$, its values at negative integer points, and applications to algebraic number theory. Her investigations were succeeded by Sargos, Lichtin, Essouabri and others.

So far, multiple zeta-functions defined in this section are in one variable. However multi-variable versions of them may also be considered. Imai [75] and Hida [57] introduced the multi-variable version of Shintani zeta-functions

$$\sum_{m_1=0}^{\infty} \cdots \sum_{m_r=0}^{\infty} \prod_{j=1}^{n} \frac{1}{(w_{j1}(m_1+\alpha_1)+\cdots+w_{jr}(m_r+\alpha_r))^{s_j}}, \tag{1.2.5}$$

where $(s_1, \dots, s_n) \in \mathbb{C}^n$, and proved the meromorphic continuation to the whole space $\mathbb{C}^n$ under the same condition (1.2.4).

The meromorphic continuation of more general multi-variable multiple zeta-functions of the form

$$\zeta_r(\mathbf{s}; P_1, \dots, P_n) = \sum_{m_1=0}^{\infty} \cdots \sum_{m_r=0}^{\infty} \frac{1}{P_1(m_1, \dots, m_r)^{s_1} \cdots P_n(m_1, \dots, m_r)^{s_n}}, \tag{1.2.6}$$

where $\mathbf{s} = (s_1, \dots, s_n)$, was first studied by Lichtin. Assume that $P_j \in \mathbb{R}[X_1, \dots, X_r]$ for $1 \le j \le n$. He [124] proved the meromorphic continuation of $\zeta_r(\mathbf{s}; P_1, \dots, P_n)$ to $\mathbb{C}^n$ if each polynomial P_j $(1 \le j \le n)$ depends explicitly on all variables and is *hypoelliptic*, that

is, $P_j(x_1,\dots,x_r) > 0$ for all $\mathbf{x} = (x_1,\dots,x_r) \in [A,\infty)^r$ (with a certain $A > 0$) and

$$\frac{\partial_{x_1}^{a_1}\cdots\partial_{x_r}^{a_r} P_j}{P_j}(x_1,\dots,x_r) \to 0 \qquad (1 \le j \le n) \tag{1.2.7}$$

as $\mathbf{x} \in [A,\infty)^r$, $|\mathbf{x}| = x_1 + \cdots + x_r \to \infty$, for all non-negative integers $a_1,\dots,a_r$ (not all 0).

Then Essouabri [38] [39] proved[3] the meromorphic continuation of $\zeta_r(\mathbf{s}; P_1,\dots,P_n)$ under a weaker condition,[4] that is, $P_1(\mathbf{x})\cdots P_n(\mathbf{x})$ tends to ∞ as $|\mathbf{x}| \to \infty$ and

$$\frac{\partial_{x_1}^{a_1}\cdots\partial_{x_r}^{a_r} P_j}{P_j}(x_1,\dots,x_r) \text{ is bounded in } [A,\infty)^r \quad (1 \le j \le n). \tag{1.2.8}$$

The method of Essouabri is rather sophisticated, using Hironaka's desingularization theorem on algebraic varieties [58].

In a different context, the second-named author independently noticed the importance of considering the multi-variable situation. The case $r = 2$ of the Barnes zeta-function is

$$\zeta_{B,2}(s;\alpha,w_1,w_2) = \sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} \frac{1}{(m_1w_1 + m_2w_2 + \alpha)^s}. \tag{1.2.9}$$

According to the resemblance of this function and (1.1.9), the double zeta-function in two variables

$$\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} \frac{1}{(m_1w_1+\alpha)^{s_1}(m_1w_1+m_2w_2+\alpha)^{s_2}} \tag{1.2.10}$$

was introduced. This is a generalization of both (1.1.9) and (1.2.9). New asymptotic expansions of the Barnes double zeta-function and related functions were obtained in [130] by analyzing (1.2.10) by the method developed in [91] [92] [93] (mentioned in the preceding section).

An important point here is that, considering the two-variable function (1.2.10), we get more flexibility in the analytic treatment, which enables us to obtain a new result. This is the underlying motivation when the second-named author later introduced various multi-variable multiple zeta-functions, such as the Mordell–Tornheim zeta-function (see (1.4.3)) and the four-variable zeta-function associated with the root system B_2 (see (1.6.2)).

[3] In [39] only the case $n = 1$ is discussed, but in his thesis [38] it is mentioned that his method can be applied to the multi-variable case.

[4] This condition is called H_0S in [38] and [39].

1.3. Euler–Zagier multiple zeta-functions

Another important class of multi-variable multiple zeta-functions is the following generalization of the Euler double zeta-function:

$$\zeta_{EZ,r}(s_1,\dots,s_r) = \sum_{1\le n_1<\cdots<n_r} \frac{1}{n_1^{s_1} n_2^{s_2}\cdots n_r^{s_r}} = \sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty} \frac{1}{m_1^{s_1}(m_1+m_2)^{s_2}\cdots(m_1+\cdots+m_r)^{s_r}}, \tag{1.3.1}$$

which is absolutely convergent in the region

$$\mathcal{A}_r = \{(s_1,\dots,s_r)\in\mathbb{C}^r \mid \Re(s_{r-k+1}+\cdots+s_r) > k \ (1\le k\le r)\} \tag{1.3.2}$$

(see [**131**, Theorem 3]).

The points $(s_1,\dots,s_r) = (k_1,\dots,k_r)$, with $k_1,\dots,k_r \in \mathbb{N}$ and $k_r \ge 2$, are included in this region $\mathcal{A}_r$. Special values of (1.3.1) at these points are called *multiple zeta values* (MZVs). The origin of the study of MZVs is usually attributed to Hoffman [**63**] and Zagier [**215**] (see also [**214**]), independently of each other, and sometimes (1.3.1) is called the *Euler–Zagier multiple zeta-function*.[5] In the work of Drinfel'd, Kontsevich, Zagier and others, it was noticed that MZVs have close connections with various fields of mathematics, such as arithmetic geometry, knot theory, mathematical physics, etc. Therefore, after the papers of Hoffman and Zagier, the study of MZVs has developed quite rapidly.

A closely related object is the *multiple polylogarithm*, defined by

$$\mathrm{Li}_{s_1,\dots,s_r}(x_1,\dots,x_r) = \sum_{1\le n_1<\cdots<n_r} \frac{x_1^{n_1}x_2^{n_2}\cdots x_r^{n_r}}{n_1^{s_1}n_2^{s_2}\cdots n_r^{s_r}} \tag{1.3.3}$$

(see Goncharov [**49**]), which reduces to the classical polylogarithm $\mathrm{Li}_{s_1}(x_1)$ when $r=1$. Usually, special values when $s_1,\dots,s_r$ are positive integers are discussed, but if we regard $s_1,\dots,s_r$ as complex variables, then (1.3.3) is a generalization of $\zeta_{EZ,r}(s_1,\dots,s_r)$.

As a generalization of Euler's classical result $\zeta(2k)\in\mathbb{Q}\pi^{2k}$ $(k\in\mathbb{N})$, it can be shown[6] that

$$\zeta_{EZ,r}(2k,2k,\dots,2k)\in\mathbb{Q}\pi^{2kr}, \tag{1.3.4}$$

whose rational coefficient is determined explicitly.

[5] It should be mentioned that Ecalle [**36**, p. 429], and also Butzer, Markett and Schmidt [**28**] introduced MZVs before them.

[6] A proof of this fact from our viewpoint will be given in Section 12.2.

One of the central unsolved problems is Zagier's conjecture on the dimension of the spaces spanned by MZVs. The values r and $k = k_1 + \cdots + k_r$ are called the depth and the weight, respectively, of $\zeta_{EZ,r}(k_1, \ldots, k_r)$ $(k_1, \ldots, k_r \in \mathbb{N},\ k_r \geq 2)$. Let $\mathcal{Z}_k$ be the vector space over $\mathbb{Q}$ spanned by all MZVs of weight k. If all MZVs of weight k are independent over $\mathbb{Q}$, then $\dim_{\mathbb{Q}} \mathcal{Z}_k = 2^{k-2}$. It seems, however, that the real values of $\dim_{\mathbb{Q}} \mathcal{Z}_k$ are much smaller. Define the sequence $\{d_k\}_{k=0}^{\infty}$ by $d_0 = 1, d_1 = 0, d_2 = 1$, and $d_k = d_{k-2} + d_{k-3}$ for $k \geq 3$. For example,

$$d_3 = d_4 = 1, d_5 = d_6 = 2, d_7 = 3, d_8 = 4, d_9 = 5, d_{10} = 7, \ldots.$$

Then Zagier's conjecture claims that $\dim_{\mathbb{Q}} \mathcal{Z}_k = d_k$.

The upper bound, namely $\dim_{\mathbb{Q}} \mathcal{Z}_k \leq d_k$ has already been proved by Goncharov [51] (see also Deligne-Goncharov [31]) and Terasoma (see [192]) independently. More recently, Brown [25] obtained this inequality by proving the fact that all MZVs are linear combinations of those with arguments only 2 and 3, which are called Hoffman's conjectural basis (see [64]).

On the other hand, the lower bound, namely $\dim_{\mathbb{Q}} \mathcal{Z}_k \geq d_k$ is out of reach at present; we do not even know how to prove that there is a single value of k with $\dim_{\mathbb{Q}} \mathcal{Z}_k \geq 2$, since it belongs instead to the theory of transcendence.

Since d_k is much smaller than 2^{k-2}, the above observation gives that there are lots of non-trivial relation formulas among MZVs. One famous example is the sum formula

$$\sum_{\substack{k_1, \ldots, k_r \in \mathbb{N}, k_r \geq 2 \\ k_1 + \cdots + k_r = k}} \zeta_{EZ,r}(k_1, \ldots, k_r) = \zeta(k), \tag{1.3.5}$$

which is a generalization of (1.1.2). There are many other known relations, such as duality, Ohno's relations, Hoffman's relations, Le-Murakami relations, cyclic sum formulas, derivation relations, and so on. Furthermore, it is conjectured that there exists a family of relations among MZVs from which, with shuffle product procedures, all linear relations of MZVs over $\mathbb{Q}$ can be derived. In fact several families, such as (extended) double shuffle relations, associator relations, confluence relations, Kawashima's relations, integral-series identities, have been found to be candidates for this conjecture.

It is also known that any MZV, whose depth and weight are of different parity, can be written in terms of MZVs of lower depth (Goncharov [49], Ihara, Kaneko and Zagier [72], the third-named author [198]). This type of result is called a *parity result*.

For the details of the theory of MZV, see, for example, Kaneko's survey [82] [84] or Zhao's book [220].

On the other hand, in order to develop the analytic theory of (1.3.1), the first task is to establish its meromorphic continuation. A proof of the meromorphic continuation of (1.3.1) is included in the main general theorem given in 1995 in Essouabri's unpublished thesis [38],[7] mentioned in the preceding section. Another unpublished proof of the meromorphic continuation of (1.3.1) is given in Goncharov's preprint [50].[8]

The paper of Arakawa and Kaneko [7] includes a proof of the meromorphic continuation of (1.3.1) with respect to the last variable s_r, while the other variables are fixed as positive integers.

The first publications on the meromorphic continuation of (1.3.1) as a function in r-variables to the whole space $\mathbb{C}^r$ are Zhao [216] and Akiyama, Egami and Tanigawa [1]. In the latter paper [1] it was also shown that $\zeta_{EZ,r}(s_1,\dots,s_r)$ $(r \geq 2)$ has infinitely many singular hyperplanes

$$\text{(1.3.6)} \qquad \begin{cases} s_r = 1, \quad s_{r-1} + s_r = 2, 1, 0, -2, -4, -6, \dots, \\ s_{r-k+1} + \cdots + s_r = k - n \quad (3 \leq k \leq r, n \in \mathbb{N}_0). \end{cases}$$

The method of Akiyama et al. is elementary, based on the Euler–Maclaurin summation formula. On the other hand, in Zhao's article, the theory of generalized functions of Gel'fand and Shilov is used. But it is possible to remove the Gel'fand–Shilov theory from Zhao's argument; based on Zhao's multiple integral expression of (1.3.1), Onozuka [176] gave a proof of meromorphic continuation, only using elementary calculus.

Now, many alternative proofs of meromorphic continuation of (1.3.1) and its various generalizations are known. Akiyama and Ishikawa [2] applied the method in [1] to *multiple Hurwitz zeta-functions*

$$\text{(1.3.7)} \qquad \sum_{0 \leq n_1 < \cdots < n_r} \frac{1}{(n_1+\alpha_1)^{s_1} \cdots (n_r+\alpha_r)^{s_r}} \qquad (\alpha_1,\dots,\alpha_r \in [0,1))$$

[7]Note that the method of Lichtin [124] cannot be applied to (1.3.1). Shintani's method is also not available in this case, because of the condition (1.2.4) required in his proof.

[8]In [50], it is written that the results described there were obtained by him and by M. Kontsevich in 1995.

and to *multiple Dirichlet L-functions*

(1.3.8)
$$\sum_{1\le n_1<\cdots<n_r} \frac{\chi_1(n_1)\cdots\chi_r(n_r)}{n_1^{s_1}\cdots n_r^{s_r}} \qquad (\chi_1,\dots,\chi_r : \text{Dirichlet characters}).$$

When $r = 1$, these are reduced to the classical Hurwitz zeta-function (1.1.7) and the classical Dirichlet L-function (1.1.6), respectively.[9] Murty and Sinha [**161**] gave an alternative simple proof of the meromorphic continuation of (1.3.7).

Katsurada [**89**] [**90**] discovered an alternative way of deducing asymptotic expansions of mean values of zeta and L-functions (mentioned at the end of Section 1.1), based on the Mellin–Barnes integral formula.[10] Inspired by the papers of Katsurada, the second-named author [**131**] [**132**] [**133**] developed a method of meromorphic continuation of multiple zeta-functions, using the Mellin–Barnes formula. These papers include the treatment of multiple zeta-functions of more generalized form

(1.3.9)
$$\sum_{m_1=0}^{\infty}\cdots\sum_{m_r=0}^{\infty}(m_1w_1+\alpha_1)^{-s_1}(m_1w_1+m_2w_2+\alpha_2)^{-s_2} \times\cdots\times(m_1w_1+\cdots+m_rw_r+\alpha_r)^{-s_r} \quad (w_j,\alpha_j\in\mathbb{C}),$$

which is a generalization of both (1.2.10) and (1.3.7).

This method was further generalized in [**134**] [**137**]. In [**134**], the meromorphic continuation of multiple zeta-functions of the form

(1.3.10)
$$\zeta_r(s_1,\dots,s_n;A_{nr}) = \sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}(a_{11}m_1+\cdots+a_{1r}m_r)^{-s_1} \times\cdots\times(a_{n1}m_1+\cdots+a_{nr}m_r)^{-s_n},$$

was proved, where $A_{nr} = (a_{ij})_{1\le i\le n,1\le j\le r}$ is an (n,r)-matrix of non-negative real elements, all rows and all columns of which include at least one non-zero element. This result will be proved in Chapter 7. In [**137**] a more general case was treated; linear forms on the right-hand side of (1.3.10) are generalized to polynomials of any degree.

Shintani's proof of the meromorphic continuation of his multiple zeta-function, mentioned in Section 1.2, is based on an ingenious contour-integral argument. However, his method cannot be applied to multiple zeta-functions of Euler–Zagier type, because of the condition

[9] The original definition of (1.3.7) in [**2**] is slightly different; they require $n_1 \ge 1$.
[10] See Lemma 6.1.

(1.2.4). The first-named author [97] [98], using a more refined contour-integral expression, removed this restriction and obtained a proof of the meromorphic continuation of multiple series of the form

$$\sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty} e^{c_1m_1+\cdots+c_rm_r}(a_{11}m_1+\cdots+a_{1r}m_r+b_1)^{-s_1} \times\cdots\times(a_{n1}m_1+\cdots+a_{nr}m_r+b_n)^{-s_n} \tag{1.3.11}$$

with complex parameters a_{ij}, b_j, c_j, which includes the case of the Euler–Zagier type.

As mentioned above, there are a lot of relations among MZVs. From the analytic viewpoint, it is natural to ask whether those relations are valid only at integer points, or can be extended continuously to "functional relations" valid also at other values. We may formulate the following:

PROBLEM 1.1. *Is there some functional relation among multiple zeta-functions, whose specializations give relations among MZVs?*

The second-named author raised this problem around 2000, and the first statement of the problem in publication was given in [137].

There is a trivial answer to this problem. The harmonic product formula (1.1.4), and its generalization to $\zeta_{EZ,r}(s_1,\ldots,s_r)$, are clearly functional relations valid for all complex values (except for singular points). Therefore the "right" question is to ask if there exist functional relations which are different from those of harmonic product type. Ikeda and K. Matsuoka [74] proved that, in a certain sense, such relations do not exist, if we only work in the frame of Euler–Zagier multiple zeta-functions. However, it turns out that the situation becomes totally different if we take zeta-functions of root systems into consideration.

1.4. Mordell–Tornheim multiple zeta-functions

In this section we discuss another interesting class of multiple zeta-functions. In 1950, Tornheim [194] considered the infinite series

$$\zeta_{MT,2}(k_1,k_2,k_3)=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\frac{1}{m_1^{k_1}m_2^{k_2}(m_1+m_2)^{k_3}} \tag{1.4.1}$$

$(k_1, k_2, k_3 \in \mathbb{N})$ in the domain of absolute convergence. In 1958, Mordell [157] studied the special case $k_1 = k_2 = k_3$ of (1.4.1), and also considered

$$\sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \frac{1}{m_1 m_2 \cdots m_r (m_1 + m_2 + \cdots + m_r + a)} \qquad (a > -r).$$

Later, Hoffman [63] discussed the properties of

$$\sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \frac{1}{m_1 m_2 \cdots m_r (m_1 + m_2 + \cdots + m_r)^h} \qquad (h \in \mathbb{N}). \tag{1.4.2}$$

The above papers are mainly devoted to the study of special values. The meromorphic continuation of (1.4.1) was first obtained by S. Akiyama and by S. Egami in 1999, but both of their proofs are unpublished. The first publication which gives the meromorphic continuation of (1.4.1) is the second-named author's article [131] (see Theorem 7.1).

As a unified generalization of the above series, the second-named author [134] introduced the multiple series

$$\begin{aligned} &\zeta_{MT,r}(s_1, \ldots, s_r, s_{r+1}) \\ &\quad = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \frac{1}{m_1^{s_1} m_2^{s_2} \cdots m_r^{s_r} (m_1 + m_2 + \cdots + m_r)^{s_{r+1}}}, \end{aligned} \tag{1.4.3}$$

called it the *Mordell–Tornheim multiple zeta-function*, and proved the meromorphic continuation of it. The method in [131] [134] is based on the Mellin–Barnes integral formula.

In [134], the notion of the *Apostol–Vu multiple zeta-function*

$$\begin{aligned} &\zeta_{AV,r}(s_1, \ldots, s_r, s_{r+1}) \\ &\quad = \sum_{1 \le m_1 < \cdots < m_r} \frac{1}{m_1^{s_1} m_2^{s_2} \cdots m_r^{s_r} (m_1 + m_2 + \cdots + m_r)^{s_{r+1}}} \end{aligned} \tag{1.4.4}$$

was also introduced. The series (1.1.5) is clearly a special case $s_3 = 1$ of the Apostol–Vu double zeta-function.

The third-named author [201] proved that the parity result holds in the family of Mordell–Tornheim multiple zeta-functions. Okamoto and Onozuka [169] discovered a certain functional equation for Mordell–Tornheim multiple zeta-functions (see Section 17.1). These results suggest that the class of Mordell–Tornheim multiple zeta-functions also deserves closer investigations.

The third-named author [203] discovered a functional relation among the Mordell–Tornheim double zeta-function and the Riemann zeta-function (see Theorem 8.3). Specializations of this theorem give

relations among MZV (Example 8.18, Proposition 12.4), so this is a solution to Problem 1.1. This was the first affirmative answer to Problem 1.1, and opened the door to the theory of functional relations among zeta-functions of root systems.

1.5. Witten multiple zeta-functions

In the study on two-dimensional quantum gauge theory, Witten [**210**] introduced $\zeta_W(s;\mathfrak{g})$ as the sum of $(\dim\varphi)^{-s}$ over all irreducible representations φ of the Lie algebra[11] $\mathfrak{g}$ (see (3.1.1)), and proved that the volume of a certain moduli space appearing in his theory can be expressed in terms of special values of $\zeta_W(s;\mathfrak{g})$. Zagier [**215**] called it the *Witten zeta-function* attached to $\mathfrak{g}$, and proved

$$\zeta_W(2k;\mathfrak{g}) \in \mathbb{Q}\pi^{2kn} \qquad (k \in \mathbb{N}), \tag{1.5.1}$$

where n denotes the number of positive roots of $\mathfrak{g}$.

The functions $\zeta_W(s;\mathfrak{g})$ will be studied in Chapter 3. Applying Weyl's dimension formula, we can obtain an explicit form of $\zeta_W(s;\mathfrak{g})$, from which we can see that they are a kind of multiple zeta-functions. The explicit form of $\zeta_W(s;\mathfrak{g})$ for some simple Lie algebras of low rank is written in Zagier's paper. When $\mathfrak{g} = \mathfrak{sl}(2)$, the corresponding Witten zeta-function is nothing but the Riemann zeta-function $\zeta(s)$. Other examples:

$$\zeta_W(s;\mathfrak{sl}(3)) = 2^s \sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty} \frac{1}{m_1^s m_2^s (m_1+m_2)^s}, \tag{1.5.2}$$

$$\zeta_W(s;\mathfrak{o}(5)) = 6^s \sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty} \frac{1}{m_1^s m_2^s (m_1+m_2)^s (2m_1+m_2)^s}. \tag{1.5.3}$$

In particular, we see that

$$\zeta_W(s;\mathfrak{sl}(3)) = 2^s \zeta_{MT,2}(s,s,s). \tag{1.5.4}$$

1.6. Zeta-functions of root systems

W. Killing and É. Cartan discovered that semisimple Lie algebras can be classified in terms of root systems. From this viewpoint, we can now arrive at the notion of zeta-functions of root systems. Fundamental definitions and facts in the theory of Lie algebras and root systems are summarized in Chapter 2.

[11] Actually Witten considered, more generally, Dirichlet series attached to compact semisimple Lie groups, which will be discussed in Chapter 15.

Any simple Lie algebra corresponds to a certain root system, which is one of those named as A_r, B_r, C_r $(r \geq 1)$, D_r $(r \geq 3)$, [12] or E_6, E_7, E_8, F_4, G_2. For example, Lie algebras $\mathfrak{sl}(2), \mathfrak{sl}(3)$, and $\mathfrak{o}(5)$ correspond to root systems A_1, A_2, and B_2, respectively.

The notion of zeta-functions of root systems was first introduced as a multi-variable version of Witten zeta-functions. The *zeta-function of the root system* X_r ($X = A, B, C, D, E, F$, or G) is denoted by $\zeta_r(\mathbf{s}; X_r)$, where $\mathbf{s}$ is a complex vector. The general definition of $\zeta_r(\mathbf{s}; X_r)$ will be given in Chapter 3, so here we just mention a few examples.

The zeta-function of A_1 is simply the Riemann zeta-function. Corresponding to A_2 and B_2, we have

(1.6.1)
$$\zeta_2(s_1, s_2, s_3; A_2) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1 + m_2)^{s_3}},$$

(1.6.2)
$$\zeta_2(s_1, s_2, s_3, s_4; B_2) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1 + m_2)^{s_3} (2m_1 + m_2)^{s_4}}.$$

It is obvious that these are essentially multi-variable versions of (1.5.2) and (1.5.3), respectively. The zeta-function of the root system A_2 (1.6.1) is exactly the same as the Mordell–Tornheim double zeta-function (1.4.1).

The zeta-function of the root system B_2 (1.6.2) was introduced in [**134**]. Then the zeta-function of general A_r $(r \in \mathbb{N})$ was introduced in [**146**], and finally, the general notion of zeta-functions of root systems was defined in [**100**].

An advantage of introducing the multi-variable version is that we can get more analytic flexibility. Because of this flexibility, we can discuss the recursive structure in the family of Witten zeta-functions (see Chapter 6).

Another important advantage is that we can get a unified view in the whole theory of multiple zeta-functions. The notion of zeta-functions of root systems is not just a multi-variable generalization of Witten zeta functions, but also includes the multiple zeta-functions of Euler–Zagier type and of Mordell–Tornheim type and the Apostol–Vu double zeta-function among its specializations (see Sections 3.2 and 3.3).

[12] But some of them are isomorphic; see (2.1.1).

In particular, the viewpoint of regarding Euler–Zagier multiple zeta-functions as certain specializations of zeta-functions of root systems of type A gives a remarkable consequence; from this viewpoint, we can show that the double shuffle procedure, which was only expressed as iterated integrals before our theory, can be written using only partial fraction decompositions. This topic will be discussed in Chapter 13. On the other hand, it is also possible to regard Euler–Zagier multiple zeta-functions as specializations of zeta-functions of root systems of type C. Applications of this approach will be given in Chapter 12.

Another important application of the above unified viewpoint is that we can unify (1.3.4) and (1.5.1) to evaluate special values of zeta-functions of root systems at even integer points (see Chapter 4). The values at integer vectors, some of whose components are odd, are more difficult and will be discussed in Chapters 8, 9, 10 and 11.

In the framework of zeta-functions of root systems, such study of relations among special values can naturally be extended to the investigation of functional relations. The functional relation mentioned in Section 1.4 is the first example, and the first positive answer to Problem 1.1. Another affirmative answer to Problem 1.1 will be given in Example 13.12. The theory of functional relations is one of the main topics of the present book, and will be discussed in Chapters 8, 9, 10, 11, 13 and 15.

The reason why the theory of functional relations can be successfully developed in the frame of zeta-functions of root systems is that we can apply the structure theory and the representation theory of Lie algebras and root systems. This is the underlying advantage of the theory of zeta-functions of root systems.[13]

The authors wrote several survey articles on the theory of zeta-functions of root systems (see [**102**] [**106**] [**112**] [**139**]). In those articles, many proofs are briefly sketched, or just omitted. The aim of the present text is to give the full account of the theory, with detailed proofs of all the theorems.

[13]One can compare with the class of multiple zeta-functions of the form (1.3.10). This class may also be regarded as a unification of multiple zeta-functions of Witten, of Euler–Zagier, of Mordell–Tornheim and so on. However, this class is probably too wide, and we cannot expect the theory for (1.3.10) to be as deep as in the case of zeta-functions of root systems.

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CHAPTER 2

Fundamentals of the theory of Lie algebras and root systems

In order to develop the theory of zeta-functions of root systems, it is surely necessary to understand the theory of Lie algebras and root systems. We will explain various known facts on those matters here and there in this text. First of all, in this chapter, we summarize the most basic part of the theory of Lie algebras and root systems, without mentioning the proofs. For the details, see, for example, Bourbaki [20], Humphreys [70] or Samelson [178].

2.1. Lie algebras

We begin with the definition of Lie algebras.

Definition 2.1. A finite-dimensional vector space $\mathfrak{g}$ over $\mathbb{C}$ with a bilinear map $[\ ,\ ] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ is called a *Lie algebra*, if the map satisfies:

(i) (the skew-symmetry) $[X, X] = 0$ for any $X \in \mathfrak{g}$,

(ii) (the Jacobi identity) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for any $X, Y, Z \in \mathfrak{g}$.

The skew-symmetry implies

$$0 = [X + Y, X + Y] = [X, Y] + [Y, X],$$

hence $[X, Y] = -[Y, X]$. In particular, we call $\mathfrak{g}$ *Abelian* if $[X, Y] = 0$ for any $X, Y \in \mathfrak{g}$.

Definition 2.2. Let $\mathfrak{g}$, $\mathfrak{g}'$ be two Lie algebras. A linear mapping $f : \mathfrak{g} \to \mathfrak{g}'$ is called a (Lie algebra) *homomorphism* if $f([X, Y]) = [f(X), f(Y)]$ holds for any $X, Y \in \mathfrak{g}$. If f is further one-to-one and onto, f is called an *isomorphism*, and in this case we call $\mathfrak{g}$ is isomorphic to $\mathfrak{g}'$, and write $\mathfrak{g} \simeq \mathfrak{g}'$.

A subspace $\mathfrak{a}$ of $\mathfrak{g}$ is called a *Lie subalgebra* if it is closed under the above bracket operation. The *normalizer* $N(\mathfrak{a})$ of $\mathfrak{a}$ is the set of all $X \in \mathfrak{g}$ for which

$$[X, \mathfrak{a}] = \{[X, Y] \mid Y \in \mathfrak{a}\} \subset \mathfrak{a}$$

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holds. If $N(\mathfrak{a}) = \mathfrak{g}$, we call $\mathfrak{a}$ an *ideal* of $\mathfrak{g}$.

DEFINITION 2.3. If a non-Abelian Lie algebra $\mathfrak{g}$ has no non-trivial ideals, $\mathfrak{g}$ is called *simple*.

For two Lie subalgebras $\mathfrak{g}_1$ and $\mathfrak{g}_2$ of $\mathfrak{g}$, by $[\mathfrak{g}_1, \mathfrak{g}_2]$ we denote the subspace spanned by all $[X_1, X_2]$ ($X_1 \in \mathfrak{g}_1, X_2 \in \mathfrak{g}_2$). In particular, $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$ is called the *derived Lie subalgebra*. Define the series

$$\mathfrak{g} \supset \mathfrak{g}' \supset \mathfrak{g}'' \supset \cdots \supset \mathfrak{g}^{(n)} \supset \cdots$$

by $\mathfrak{g}^{(n)} = (\mathfrak{g}^{(n-1)})'$. If this series (the *derived series*) goes down to zero for some finite n, we call $\mathfrak{g}$ *solvable*. We also define the *lower central series*, by replacing $\mathfrak{g}^{(n)}$ in the derived series by $\mathfrak{g}^n = [\mathfrak{g}, \mathfrak{g}^{n-1}]$, and call $\mathfrak{g}$ *nilpotent* if the lower central series goes down to zero.

It is known that any $\mathfrak{g}$ contains the unique maximal solvable ideal $\mathfrak{r}$, the *radical* of $\mathfrak{g}$. We call a Lie algebra $\mathfrak{g}$ ($\neq \{0\}$) *semisimple* if its radical is zero. Since an Abelian ideal is solvable, semisimplicity implies that $\mathfrak{g}$ has no non-zero Abelian ideal. A simple Lie algebra is semisimple. It is known that any semisimple Lie algebra can be written as a direct sum of simple Lie algebras.

Here we give several typical examples of semisimple Lie algebras, written in terms of matrices. Denote by $M_n(\mathbb{C})$ the set of all $n \times n$ matrices whose entries are complex numbers. Let E_r be the unit matrix of size r, and define the matrix J_r of size $2r$ by

$$J_r = \begin{pmatrix} 0 & E_r \\ -E_r & 0 \end{pmatrix}.$$

EXAMPLE 2.4. The sets $\mathfrak{g}$ defined by

- (type $A_r, r \geq 1$) $\mathfrak{g} = \mathfrak{sl}(r+1) = \{X \in M_{r+1}(\mathbb{C}) \mid \mathrm{tr}(X) = 0\}$,
- (type $B_r, r \geq 1$) $\mathfrak{g} = \mathfrak{o}(2r+1) = \{X \in M_{2r+1}(\mathbb{C}) \mid {}^tX + X = 0\}$,
- (type $C_r, r \geq 1$) $\mathfrak{g} = \mathfrak{sp}(r) = \{X \in M_{2r}(\mathbb{C}) \mid {}^tXJ_r + J_rX = 0\}$,
- (type $D_r, r \geq 2$) $\mathfrak{g} = \mathfrak{o}(2r) = \{X \in M_{2r}(\mathbb{C}) \mid {}^tX + X = 0\}$

are examples of Lie algebras, with the bracket defined by

$$[X, Y] = X \cdot Y - Y \cdot X,$$

where "dot" means the usual matrix multiplication. Except for the case of D_2, all of these algebras are simple. In the case of D_2, since $\mathfrak{o}(4) \simeq \mathfrak{sl}(2) \oplus \mathfrak{sl}(2)$, the algebra is semisimple.

Later we will introduce the notion of "root systems", and will state that any semisimple Lie algebra is corresponding to a certain root system. The above words "type X_r" (where $X = A, B, C$ or D) are

actually given to the corresponding root system. Two root systems are isomorphic if the corresponding algebras are isomorphic. In this sense, the isomorphism mentioned above can be written as $D_2 \simeq A_1 \oplus A_1$. Moreover, it is known that

$$A_1 \simeq B_1 \simeq C_1, \quad B_2 \simeq C_2, \quad \text{and} \quad A_3 \simeq D_3. \tag{2.1.1}$$

Note that the Lie algebra $\mathfrak{o}(2)$ is Abelian, hence not semisimple, and so there is no root system corresponding to $\mathfrak{o}(2)$.

To classify semisimple Lie algebras, it is sufficient to classify simple algebras. Late in the 19th century, W. Killing and É. Cartan completely classified simple Lie algebras.

THEOREM 2.5 (Killing and Cartan). *Any simple Lie algebra is isomorphic to one of the simple algebras listed in Example 2.4, or the five exceptional simple Lie algebras whose root systems are named as E_6, E_7, E_8, F_4 and G_2.*

Here we do not mention the explicit construction of the above five exceptional algebras.

2.2. Roots

Now we explain some basic parts of representation theory of Lie algebras.

For a finite-dimensional complex vector space U, let $\mathfrak{gl}(U)$ be the set of all linear mappings $U \to U$. This $\mathfrak{gl}(U)$ is a Lie algebra with the product defined by $[A, B] = AB - BA$ for $A, B \in \mathfrak{gl}(U)$.

A *representation* of a Lie algebra $\mathfrak{g}$ on a finite-dimensional complex vector space U is a homomorphism φ of $\mathfrak{g}$ into $\mathfrak{gl}(U)$. Two representations φ_1 and φ_2 (with the underlying vector spaces U_1 and U_2, respectively) are *equivalent* when there is a linear isomorphism $\phi : U_1 \to U_2$ which satisfies $\phi \circ \varphi_1(g) = \varphi_2(g) \circ \phi$ for all $g \in \mathfrak{g}$.

We denote by $\dim \varphi$ the dimension of the representation space U. The most fundamental representation is the *adjoint representation* $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ defined by $(\mathrm{ad}X)Y = [X, Y]$ for any $X, Y \in \mathfrak{g}$. By using the adjoint representation we define the symmetric bilinear form

$$\langle X, Y \rangle = \kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X \circ \mathrm{ad}Y)$$

on $\mathfrak{g}$, which is called the *Killing form*.

FACT 2.6. ([70, Section 5.1, Theorem]) *A Lie algebra $\mathfrak{g}$ is semisimple if and only if its associated Killing form is non-degenerate.*

The following fact is well-known in linear algebra.

FACT 2.7. *Let V be a finite-dimensional vector space, V^* its dual space, and $\langle\ ,\ \rangle$ be a (non-degenerate) inner product defined on V. Then for any $f \in V^*$, there exists an element $v \in V$, for which $f(w) = \langle v, w\rangle$ holds for any $w \in V$.*

Therefore a vector space with non-degenerate inner product can be identified with its dual space. Fact 2.6 implies that this is the case for any semisimple Lie algebra with the associated Killing form.

A *Cartan subalgebra* of a Lie algebra $\mathfrak{g}$ is a nilpotent Lie subalgebra $\mathfrak{h}$ of $\mathfrak{g}$, with the property $N(\mathfrak{h}) = \mathfrak{h}$. (It follows that $\mathfrak{h}$ is maximal nilpotent.) This is not uniquely determined, but its dimension does not depend on the choice of $\mathfrak{h}$. We call this dimension the *rank* of $\mathfrak{g}$.

Now assume that $\mathfrak{g}$ is semisimple. Then any Cartan subalgebra is Abelian, and hence is maximal Abelian. We fix one Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{g}$, and write $r = \operatorname{rank}\mathfrak{g} = \dim\mathfrak{h}$. Let α be a non-zero element of the dual space $\mathfrak{h}^*$ of $\mathfrak{h}$. We call α a *root* of $\mathfrak{g}$ (with respect to $\mathfrak{h}$) if there exists a non-zero $X \in \mathfrak{g}$ such that $\operatorname{ad}(H)X = \alpha(H)X$ for any $H \in \mathfrak{h}$. Denote the set of all such X by $\mathfrak{g}_\alpha$. There are only finitely many roots, and they span $\mathfrak{h}^*$.

DEFINITION 2.8. The set of all roots we denote by $\Delta = \Delta(\mathfrak{g})$, and call it the *root system* of $\mathfrak{g}$.

The root system is a very fundamental notion. One important fact is the decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Delta} \mathfrak{g}_\alpha. \tag{2.2.1}$$

Note that $\dim\mathfrak{g}_\alpha = 1$ for any $\alpha \in \Delta$. The proof of Theorem 2.5 is usually reduced to the classification of root systems.

The Killing form on a semisimple Lie algebra is non-degenerate even if it is restricted on its Cartan subalgebra (which can be seen by, for example, [**70**, Section 8.2, Corollary] and [**70**, Section 15.3, Corollary]). Therefore by Fact 2.7, we can identify $\mathfrak{h}^*$ with $\mathfrak{h}$. This implies that for any $\alpha \in \mathfrak{h}^*$, there exists a unique element $\alpha' \in \mathfrak{h}$ satisfying $\langle\alpha', H\rangle = \alpha(H)$ for any $H \in \mathfrak{h}$. By using this correspondence $\alpha \leftrightarrow \alpha'$, we may transfer the Killing form to $\mathfrak{h}^*$ by putting $\langle\alpha, \beta\rangle := \langle\alpha', \beta'\rangle$.

Let $\mathfrak{h}_0$ be the real subspace of $\mathfrak{h}$ formed by all real linear combinations of α', for $\alpha \in \Delta$. Then $\mathfrak{h}_0$ and its dual space $\mathfrak{h}_0^*$ can be identified by the above way.

Since α' is clearly non-zero, we can define

$$\alpha^\vee = \frac{2}{\langle \alpha', \alpha' \rangle} \alpha', \tag{2.2.2}$$

which we call the *coroot* associated with α. Clearly $\alpha(\alpha^\vee) = \langle \alpha', \alpha^\vee \rangle = 2$. More generally, for any $\alpha, \beta \in \Delta$, it is known that

$$\beta(\alpha^\vee) = \langle \beta', \alpha^\vee \rangle = \frac{2\langle \beta', \alpha' \rangle}{\langle \alpha', \alpha' \rangle} = \frac{2\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}, \tag{2.2.3}$$

which we denote by $a(\beta, \alpha)$ and call the *Cartan integers* of $\mathfrak{g}$. It can be shown that $\beta - a(\beta, \alpha)\alpha$ is again a root. Another important property is that if both α and $c\alpha$ ($c \in \mathbb{C}$) are roots, then $c = \pm 1$.

For any $\alpha, \beta \in \Delta$, there exist $p, q \in \mathbb{N}_0$, such that $\beta + t\alpha \in \Delta$ if and only if $-q \leq t \leq p$. The sequence $\beta - q\alpha, \ldots, \beta + p\alpha$ is called the *α-string through β*. It is known that

$$a(\beta, \alpha) = q - p. \tag{2.2.4}$$

2.3. Abstract root systems

In the preceding section we defined the root system Δ of a Lie algebra $\mathfrak{g}$, and mentioned some basic properties. Choosing those properties as axioms, it is possible to define the notion of root systems axiomatically, independent of Lie algebras.

Let V be an r-dimensional real vector space with an inner product $\langle\ ,\ \rangle$. Write $\|v\| = \langle v, v \rangle^{1/2}$. In view of Fact 2.7, the dual space V^* can be identified with V via the inner product.

For any $\alpha \in V \setminus \{0\}$, let $a(\beta, \alpha) = 2\langle \beta, \alpha \rangle / \langle \alpha, \alpha \rangle$, and define $\sigma_\alpha : V \to V$ by

$$\sigma_\alpha(\beta) = \beta - a(\beta, \alpha)\alpha.$$

From this definition it is obvious that

$$\langle \sigma_\alpha(\beta), \sigma_\alpha(\gamma) \rangle = \langle \beta, \gamma \rangle \tag{2.3.1}$$

for any $\alpha \in V \setminus \{0\}$ and any $\beta, \gamma \in V$.

Definition 2.9. A finite non-empty subset R of V, not containing 0, is called an (abstract) *reduced root system* if it spans V and satisfies:

(i) for $\alpha, \beta \in R$, $a(\beta, \alpha) \in \mathbb{Z}$,

(ii) for $\alpha, \beta \in R$, the vector $\sigma_\alpha(\beta)$ is also in R,

(iii) if both α and $c\alpha$ ($c \in \mathbb{R}$) are in R, then $c = \pm 1$.

Each member $\alpha \in R$ is called a *root*, and $\|\alpha\|$ is called its *length*. We call r the *rank* of the root system R.

Two root systems R_1 and R_2 (with the underlying vector spaces V_1 and V_2, respectively) are *equivalent* when there is a similarity (constant multiple of an isometry) $V_1 \to V_2$ which sends R_1 onto R_2.

Obviously Δ is a reduced root system with the vector space $V = \mathfrak{h}_0^*$. Very important is the following:

FACT 2.10. *There is a bijection between the set of equivalence classes of reduced root systems and the set of isomorphism classes of semisimple Lie algebras.*

Therefore, hereafter, we assume that the root system R corresponds to a semisimple Lie algebra $\mathfrak{g}$, the inner product is given by the Killing form, and write $R = \Delta = \Delta(\mathfrak{g})$ and $V = \mathfrak{h}_0^*$.

In the preceding section $\alpha \in \mathfrak{h}^*$ while $\alpha^\vee \in \mathfrak{h}$, but $\mathfrak{h}^*$ and $\mathfrak{h}$ can be identified, by identifying α' with α. Therefore we may understand that α and $\alpha^\vee$ belong to the same space V. Using this notation, we can write

$$\sigma_\alpha(\beta) = \beta - \frac{2\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle}\alpha = \beta - \langle \beta, \alpha^\vee \rangle \alpha,$$

where $\langle \beta, \alpha^\vee \rangle = \langle \beta', \alpha^\vee \rangle$ via the identification.

The set of all $\alpha^\vee$ again consists of a root system, which is called the *dual* (or *reciprocal*) root system of Δ, and is denoted by $\Delta^\vee$.

Next, we introduce the notion of positive roots and negative roots. For $\alpha \in \Delta$, let

$$H_{\alpha^\vee} = \{\beta \in V \mid \langle \beta, \alpha^\vee \rangle = 0\}.$$

This is the hyperplane orthogonal to $\alpha^\vee$. Since Δ is a finite set, it is obvious that $V \setminus \bigcup_{\alpha \in \Delta} H_{\alpha^\vee}$ is non-empty. The elements of this set are called *regular*. If $\gamma \in V$ is regular, then $\Delta = \Delta_+ \coprod \Delta_-$, where

$$\Delta_+ = \Delta_+(\gamma) = \{\alpha \in R \mid \langle \gamma, \alpha^\vee \rangle > 0\}, \quad \Delta_- = \Delta_-(\gamma) = \{-\alpha \mid \alpha \in \Delta_+\}.$$

We call $\alpha \in \Delta_+$ *decomposable* if $\alpha = \beta_1 + \beta_2$ for some $\beta_1, \beta_2 \in \Delta_+$, and *indecomposable* otherwise. Let

$$\Psi = \Psi(\gamma) = \{\alpha_1, \ldots, \alpha_r\}$$

be the set of all indecomposable elements of Δ_+. Sometimes we also write $\Psi(\Delta)$, or $\Psi(\mathfrak{g})$. It is known that Ψ is a basis of V, and each root β can be written as $\beta = \sum_{\alpha \in \Psi} k_\alpha \alpha$ with integral coefficients k_α, all non-negative or all non-positive. We call β *positive* (resp. *negative*) if all $k_\alpha \geq 0$ (resp. ≤ 0). Then Δ_+ (resp. Δ_-) is the set of all positive (resp. negative) roots. The elements of Ψ are called *simple* or *fundamental*. We call Ψ a *base*, or a *fundamental system*, of Δ.

Define

$$Q = Q(\Delta) = \bigoplus_{j=1}^{r} \mathbb{Z}\alpha_j, \qquad Q^\vee = Q^\vee(\Delta) = \bigoplus_{j=1}^{r} \mathbb{Z}\alpha_j^\vee,$$

and call them the *root lattice* and the *coroot lattice*, respectively.

The mapping σ_α is the reflection with respect to $H_{\alpha^\vee}$. Let $A \subset \Delta$, and let $W(A)$ be the group generated by reflections σ_α for $\alpha \in A$. In particular, we call $W = W(\Delta)$ the *Weyl group* of Δ. This group is generated by $\sigma_j = \sigma_{\alpha_j}$ $(1 \le j \le r)$. Namely, we have $W = W(\Psi)$. If Ψ' is another fundamental system of Δ, then there exists an element $w \in W$ for which $\Psi' = w(\Psi)$ holds. From (2.3.1) it follows that

$$\langle w\beta, w\gamma\rangle = \langle \beta, \gamma\rangle \tag{2.3.2}$$

for any $w \in W$ and any $\beta, \gamma \in V$.

A root system is called *irreducible* if it cannot be written as the union of two proper subsets, each root in one of them being orthogonal to each root in the other. Otherwise a root system is called *reducible*. Irreducible root systems have been completely classified by the Killing–Cartan theory. The result is that any irreducible root system is equivalent to one of the systems named as A_r $(r \ge 1)$, B_r $(r \ge 2)$, C_r $(r \ge 3)$, D_r $(r \ge 4)$, and the five exceptional systems E_6, E_7, E_8, F_4, and G_2. This list exactly corresponds to the list of possible simple Lie algebras. Hereafter, when $\mathfrak{g}$ corresponds to the root system of type X_r ($X = A, B, C, D, E, F$ or G), we sometimes write $\Delta(\mathfrak{g}) = \Delta(X_r)$, $\Psi(\mathfrak{g}) = \Psi(X_r)$, etc. The data of those root systems will be given in the next chapter.

2.4. Weights

Let $\varphi : \mathfrak{g} \to \mathfrak{gl}(U)$ be a representation, where U is a finite-dimensional complex vector space. We call $u \in U$ a *weight vector* if it is a joint eigenvector of all the operators $\varphi(H)$, $H \in \mathfrak{h}$. Hence $\varphi(H)u = \lambda(H)u$, where $\lambda(H) \in \mathbb{C}$. Then $\lambda : H \mapsto \lambda(H)$ is an element of $\mathfrak{h}^*$. We call λ the *weight* of u.

For each $\lambda \in \mathfrak{h}^*$, let U_λ be the subspace of U consisting of 0 and all weight vectors u with weight λ. When U_λ is non-zero, we call λ a *weight* of φ. There is only a finite number of weights. A weight λ is called *dominant* if $\lambda(\alpha^\vee) \ge 0$ for all $\alpha \in \Psi$. If the strict inequality holds for all $\alpha \in \Psi$, then we call it *strongly dominant*.

Define $\lambda_j \in \mathfrak{h}^*$ by

$$\lambda_j(\alpha_i^\vee) = \langle \alpha_i^\vee, \lambda_j\rangle = \delta_{i,j} \quad (\delta : \text{Kronecker's delta})$$

(where the first identitiy comes from the identification $\mathfrak{h}^* = \mathfrak{h}$) for all $\alpha_i \in \Psi$ $(1 \leq i \leq r)$. Then clearly λ_j $(1 \leq j \leq r)$ are dominant. Moreover, $\lambda_j(\alpha') \in \mathbb{R}$ for any $\alpha' \in \mathfrak{h}_0$, and hence $\lambda_j \in \mathfrak{h}_0^* = V$. We call them *fundamental weights*, and write $\Lambda = \{\lambda_1, \ldots, \lambda_r\}$. The weight

$$\rho = \lambda_1 + \cdots + \lambda_r = \frac{1}{2} \sum_{\alpha \in \Delta_+} \alpha \in V \tag{2.4.1}$$

is called the *Weyl vector*, or the *lowest strongly dominant form*. It is known that any dominant weight λ can be written as

$$\lambda = n_1\lambda_1 + \cdots + n_r\lambda_r \qquad (n_1, \ldots, n_r \in \mathbb{N}_0). \tag{2.4.2}$$

Define the *weight lattice*

$$P = P(\Delta) = \bigoplus_{j=1}^{r} \mathbb{Z}\lambda_j,$$

and also we write

$$P_+ = \bigoplus_{j=1}^{r} \mathbb{N}_0\lambda_j, \qquad P_{++} = \bigoplus_{j=1}^{r} \mathbb{N}\lambda_j = P_+ + \rho.$$

Fundamental coweights $\{\lambda_j^\vee\}_{j=1}^r$ are defined by

$$\alpha_i(\lambda_j^\vee) = \langle \lambda_j^\vee, \alpha_i \rangle = \delta_{i,j}$$

for all $\alpha_i \in \Psi$ $(1 \leq i \leq r)$. The *coweight lattice* is

$$P^\vee = P^\vee(\Delta) = \bigoplus_{j=1}^{r} \mathbb{Z}\lambda_j^\vee,$$

whose elements are called *coweights*.

Since $\langle \alpha, \beta^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in \Delta$ (see (2.2.3)), we see that any root $\alpha \in P$, so $Q \subset P$. Moreover, the rank of the lattice Q is the same as the rank of the lattice P $(= r)$, hence Q is a sublattice of P. Similarly, $Q^\vee$ is a sublattice of $P^\vee$.

A representation (φ, U) is called *irreducible* if there is no non-trivial invariant subspace of U. A principal result in representation theory of Lie algebras is as follows:

FACT 2.11. *There is a bijection between the set of (equivalence classes of) finite-dimensional irreducible representations of* $\mathfrak{g}$ *and the set of dominant weights.*

Let φ be a finite-dimensional irreducible representation of $\mathfrak{g}$, and λ the corresponding dominant weight. The following is *Weyl's dimension formula* (see, for example, [**178**, Section 3.8]), which gives the expression of $\dim \varphi$ in terms of the corresponding weight.

THEOREM 2.12. *We have*

$$\dim \varphi = \prod_{\alpha \in \Delta_+} \frac{\langle \alpha^\vee, \lambda + \rho \rangle}{\langle \alpha^\vee, \rho \rangle}. \tag{2.4.3}$$

Using the expression (2.4.2), we obtain

$$\dim \varphi = \prod_{\alpha \in \Delta_+} \frac{\langle \alpha^\vee, (n_1+1)\lambda_1 + \cdots + (n_r+1)\lambda_r \rangle}{\langle \alpha^\vee, \lambda_1 + \cdots + \lambda_r \rangle}. \tag{2.4.4}$$

At first glance it seems that the right-hand sides of (2.4.3) and (2.4.4) depend on the choice of $\mathfrak{h}$ and γ. It is possible, however, to show directly that the right-hand side is independent of those choices, by using the fact that any two Cartan subalgebras are conjugate, and the aforementioned transitivity of the Weyl group on the set of fundamental systems.

CHAPTER 3

Definitions and examples

We give the definition of zeta-functions of root systems, and write down explicit forms of those zeta-functions.

3.1. The definition of zeta-functions of root systems

Let $\mathfrak{g}$ be a complex semisimple Lie algebra, s a complex variable, and define

$$\zeta_W(s;\mathfrak{g}) = \sum_{\varphi} (\dim \varphi)^{-s}, \tag{3.1.1}$$

where the summation runs over all equivalence classes of finite-dimensional irreducible representations φ of $\mathfrak{g}$. Special values of this Dirichlet series were first studied by Witten [**210**] in connection with quantum gauge theory. Zagier [**215**] formulated (3.1.1) and called it the *Witten multiple zeta-function* associated with $\mathfrak{g}$. Some evaluation formulas of $\zeta_W(s;\mathfrak{g})$ at positive even integral arguments were given by Mordell [**157**], Zagier [**215**] and Gunnells and Sczech [**53**].

A more explicit expression of (3.1.1) can be obtained by using Weyl's dimension formula. Let $\Delta = \Delta(\mathfrak{g})$, $\Delta_+ = \Delta_+(\mathfrak{g})$, $\Psi = \Psi(\mathfrak{g}) = \{\alpha_1, \ldots, \alpha_r\}$, $\Lambda = \{\lambda_1, \ldots, \lambda_r\}$ (as subsets of V) and ρ be as in Chapter 2. Recall that any dominant weight can be written as

$$\lambda = n_1\lambda_1 + \cdots + n_r\lambda_r \qquad (n_1, \ldots, n_r \in \mathbb{N}_0). \tag{3.1.2}$$

The dimension of the representation space corresponding to the dominant weight λ is given by (2.4.4). Using (2.4.4) with $m_j = n_j + 1$, we have

$$\begin{aligned} \zeta_W(s;\mathfrak{g}) &= \sum_{\lambda} \prod_{\alpha\in\Delta_+} \left(\frac{\langle \alpha^\vee, m_1\lambda_1 + \cdots + m_r\lambda_r\rangle}{\langle \alpha^\vee, \lambda_1 + \cdots + \lambda_r\rangle} \right)^{-s} \\ &= K(\mathfrak{g})^s \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{\alpha\in\Delta_+} \langle \alpha^\vee, m_1\lambda_1 + \cdots + m_r\lambda_r\rangle^{-s}, \end{aligned} \tag{3.1.3}$$

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Springer Monographs in Mathematics, https://doi.org/10.1007/978-981-99-0910-0_3

where the sum on the second member of the above runs over all dominant weights of the form (3.1.2), and let

$$K(\mathfrak{g}) = K(\Delta) = \prod_{\alpha\in\Delta_+} \langle \alpha^\vee, \lambda_1 + \cdots + \lambda_r\rangle. \tag{3.1.4}$$

The analytic behaviour of $\zeta_W(s;\mathfrak{g})$ is determined essentially by the multiple series part of the right-hand side of (3.1.3). To analyze this multiple series closely, it is more flexible to introduce the following multi-variable version of the series:

$$\zeta_r(\mathbf{s};\mathfrak{g}) = \sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}\prod_{\alpha\in\Delta_+}\langle\alpha^\vee, m_1\lambda_1+\cdots+m_r\lambda_r\rangle^{-s_\alpha}, \tag{3.1.5}$$

where $\mathbf{s} = (s_\alpha)_{\alpha\in\Delta_+} \in \mathbb{C}^{\#(\Delta_+)}$. Series (3.1.5) is obviously convergent absolutely in the region

$$\mathcal{S}_0 = \{\mathbf{s} = (s_\alpha) \in \mathbb{C}^{\#(\Delta_+)} \mid \Re s_\alpha > 1 \quad \text{for all } \alpha\in\Delta_+\}.$$

Since the relation

$$\zeta_W(s;\mathfrak{g}) = K(\mathfrak{g})^s\zeta_r(s,\ldots,s;\mathfrak{g}) \tag{3.1.6}$$

holds, we may understand that $\zeta_r(\mathbf{s};\mathfrak{g})$ is a multi-variable generalization of $\zeta_W(s;\mathfrak{g})$.

Because of the one-to-one correspondence between $\mathfrak{g}$ and Δ, we may write $\zeta_r(\mathbf{s};\mathfrak{g})$ also as $\zeta_r(\mathbf{s};\Delta)$, and call it the *zeta-function of the root system* Δ. The definition (3.1.5) first appeared in [**100**, (1.5)]. Hereafter we frequently use the notation $\zeta_r(\mathbf{s};X_r)$, $K(X_r)$ etc. if Δ is of type X_r (where $X = A$, B, C, D, E, F or G).

For our later purpose, here we introduce the zeta-functions of root systems with twisted factors. Let $\mathbf{y}\in V$, and

$$\zeta_r(\mathbf{s},\mathbf{y};\Delta) = \sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}. \tag{3.1.7}$$

Here, writing $\mathbf{y} = y_1\alpha_1^\vee + \cdots + y_r\alpha_r^\vee$, we understand

$$\langle\mathbf{y},\lambda\rangle = y_1\langle\alpha_1^\vee,\lambda\rangle + \cdots + y_r\langle\alpha_r^\vee,\lambda\rangle = y_1\lambda(\alpha_1^\vee) + \cdots + y_r\lambda(\alpha_r^\vee).$$

This generalized form first appeared in [**101**]. It is clear that, when $\mathbf{y} = \mathbf{0}$, then $\zeta_r(\mathbf{s},\mathbf{y};\Delta)$ coincides with $\zeta_r(\mathbf{s};\Delta)$.

REMARK 3.1. At first glance it may seem that the definitions (3.1.5) and (3.1.7) depend on the choice of Δ_+, hence on the choice of γ in Section 2.3. However, the definitions do not depend on those choices, because any $\Delta_+(\gamma)$ and $\Delta_+(\gamma')$ (γ,γ' are regular) are conjugate under the action of W (see [**71**, Section 1.4]).

REMARK 3.2. Comparing the definition of multiple polylogarithms (1.3.3), we may say that (3.1.7) expresses multiple polylogarithms of a kind associated with root systems. In our theory, it is frequently useful to consider the multiple series with additional factors like $e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle}$ or x^n on the numerators; see Remark 8.19.

To study $\zeta_W(s;\mathfrak{g})$ or $\zeta_r(\mathbf{s},\mathbf{y};\Delta)$, it is sufficient to consider the case of simple Lie algebras, or the corresponding irreducible root systems. In fact, if $\mathfrak{g}$ is the direct sum of two Lie algebras $\mathfrak{g}_1 \oplus \mathfrak{g}_2$, then

$$\zeta_W(s;\mathfrak{g}) = \zeta_W(s;\mathfrak{g}_1)\zeta_W(s;\mathfrak{g}_2). \tag{3.1.8}$$

This can be proved as follows. It is well known that any irreducible representation φ of $\mathfrak{g}$ is equivalent to the tensor product of two irreducible representations φ_1 of $\mathfrak{g}_1$ and φ_2 of $\mathfrak{g}_2$, and conversely if φ_i is an irreducible representation of $\mathfrak{g}_i$ $(i = 1, 2)$ then $\varphi_1 \otimes \varphi_2$ is an irreducible representation of $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ ([**178**, Section 3.4]). Therefore

$$\begin{aligned}\zeta_W(s;\mathfrak{g}) &= \sum_{\varphi_1}\sum_{\varphi_2}(\dim \varphi_1 \otimes \varphi_2)^{-s} \\ &= \sum_{\varphi_1}(\dim \varphi_1)^{-s}\sum_{\varphi_2}(\dim \varphi_2)^{-s} = \zeta_W(s;\mathfrak{g}_1)\zeta_W(s;\mathfrak{g}_2).\end{aligned}$$

From (3.1.8) inductively we can show that $\zeta_W(s;\mathfrak{g})$ for a semisimple $\mathfrak{g}$ is the product of all Witten zeta-functions associated with simple factors of $\mathfrak{g}$.

More generally, if the root system Δ is reducible, and its decomposition into irreducible components is written as $\Delta = \coprod_j \Delta_j$, then $\zeta_r(\mathbf{s},\mathbf{y};\Delta)$ is the product of all zeta-functions corresponding to Δ_j.

This fact can be directly seen from the definition (3.1.7). In fact, let Δ_k be one irreducible component, and $\alpha \in \Delta_k$. Then all roots and weights belonging to other irreducible components are orthogonal to α. Therefore, writing $\lambda = \sum_j \lambda(j)$, where $\lambda(j)$ denotes the linear combination of fundamental weights attached to Δ_j, we find $\langle \alpha^\vee, \lambda\rangle = \langle \alpha^\vee, \lambda(k)\rangle$. The claim immediately follows from this observation.

The function $\zeta_r(\mathbf{s},\mathbf{y};\Delta)$ is also obviously convergent absolutely in $\mathcal{S}_0$, but actually it is convergent absolutely in a wider region. Let $\Delta = \coprod_j \Delta_j$ be the decomposition into irreducible components and r_j be the rank of Δ_j. For $\alpha \in \Delta_j$, let

$$\epsilon_\alpha = \begin{cases} \dfrac{1}{r_j+1} & (r_j > 1), \\ 0 & (r_j = 1) \end{cases} \tag{3.1.9}$$

and

$$\mathcal{S} = \{\mathbf{s} = (s_\alpha) \in \mathbb{C}^{\#(\Delta_+)} \mid \Re s_\alpha > 1 - \epsilon_\alpha \quad \text{for all } \alpha \in \Delta_+\}. \tag{3.1.10}$$

Note that $\mathcal{S}$ is an W-invariant set. The following lemma is a slight refinement of [113, Lemma 9].

LEMMA 3.3. *$\zeta_r(\mathbf{s}, \mathbf{y}; \Delta)$ is absolutely convergent, uniformly on $\mathcal{D} \times V$, where $\mathcal{D}$ is any compact subset of the set $\mathcal{S}$. Hence they are continuous on $\mathcal{S} \times V$, and holomorphic in $\mathbf{s}$ for a fixed $\mathbf{y} \in V$.*

PROOF. If Δ is reducible, then the sum is decomposed into irreducible components and hence it is sufficient to prove the assertion in irreducible cases.

If Δ is of rank 1, then the uniform and absolute convergence on $\mathcal{D} \times V$ is easy. Thus we assume that Δ is irreducible and of rank $r > 1$. Let $\beta^\vee = \alpha_1^\vee + \cdots + \alpha_r^\vee \in \Delta^\vee$. Since for $\mathbf{s} \in \mathcal{D}$, $\alpha \in \Delta_+$ and $\lambda = n_1\lambda_1 + \cdots + n_r\lambda_r \in P_{++}$,

$$\left|\frac{1}{\langle \alpha^\vee, \lambda\rangle^{s_\alpha}}\right| \leq \frac{1}{\langle \alpha^\vee, \lambda\rangle^{\Re s_\alpha}}, \tag{3.1.11}$$

we have

$$\begin{aligned}\left|\prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{s_\alpha}}\right| &\leq \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{\Re s_\alpha}} \\ &\leq \frac{1}{\langle \beta^\vee, \lambda\rangle^{\Re s_\beta}} \prod_{\alpha\in\Psi} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{\Re s_\alpha}} \\ &= \frac{1}{(n_1 + \cdots + n_r)^{\Re s_\beta} n_1^{\Re s_{\alpha_1}} \cdots n_r^{\Re s_{\alpha_r}}} \\ &\leq \frac{r^{-\Re s_\beta}}{n_1^{\Re s_{\alpha_1} + \Re s_\beta/r} \cdots n_r^{\Re s_{\alpha_r} + \Re s_\beta/r}}.\end{aligned} \tag{3.1.12}$$

Note for $\alpha \in \Delta$,

$$\Re s_\alpha + \frac{\Re s_\beta}{r} > \frac{r}{r+1} + \frac{1}{r+1} = 1. \tag{3.1.13}$$

Hence the uniform and absolute convergence on $\mathcal{D} \times V$. □

REMARK 3.4. The region $\mathcal{S}$ is not the largest region in which the statements in Lemma 3.3 and explicit forms given in the rest of this chapter hold, but we work with $\mathcal{S}$ for simplicity. For instance, the region

$$\{\mathbf{s} = (s_\alpha) \in \mathbb{C}^{\#(\Delta_+)} \mid \Re s_\alpha > 1 \quad \text{for } \alpha \in \Psi, \quad \Re s_\alpha \geq 0 \quad \text{otherwise}\}$$

is also a domain of absolute convergence, for which the above proof of Lemma 3.3 also holds. When $\mathbf{y} = \mathbf{0}$, the maximal domain of (absolute)

convergence can be determined by using Zhao and Zhou [221, Proposition 2.1], but we do not use such information in our text. In more general cases, we may need to specify an order of summation if the convergence is conditional. One way to do this is the Q-limit procedure treated in [53].

3.2. Explicit forms for zeta-functions of type A

Here we compute a more explicit form in the case of type A_r, that is the case associated with the algebra $\mathfrak{g} = \mathfrak{sl}(r+1)$ $(r \in \mathbb{N})$. Let ε_j be the j-th coordinate function which assigns to each vector its j-th coordinate. It is known that the fundamental system is $\{\alpha_1, \ldots, \alpha_r\}$ with $\alpha_j = \varepsilon_j - \varepsilon_{j+1}$ $(1 \le j \le r)$, and positive roots are

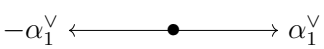
$$\varepsilon_i - \varepsilon_j = \sum_{i \le k < j} \alpha_k \qquad (1 \le i < j \le r+1) \tag{3.2.1}$$

(see, for example, the list at the end of Bourbaki [20]). The corresponding coroots are

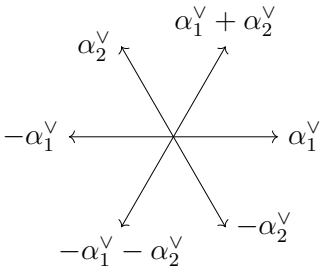
$$(\varepsilon_i - \varepsilon_j)^\vee = e_i - e_j = \sum_{i \le k < j} \alpha_k^\vee \qquad (1 \le i < j \le r+1),$$

where e_j is the standard j-th coordinate vector. All roots are of the same length, so are all coroots.

We illustrate those coroots as vectors in the vector space V, in the cases $r = 1$ and $r = 2$. First, when $r = 1$, since $\Delta(A_1) = \{\alpha_1, -\alpha_1\}$, the figure of $\Delta^\vee(A_1)$ is:



Next, when $r = 2$, the figure of $\Delta^\vee(A_2)$ is:



In the A_r case we have $\langle \alpha^\vee, \lambda \rangle = \langle \alpha, \lambda \rangle$, hence we have

$$\begin{aligned} &\langle (\varepsilon_i - \varepsilon_j)^\vee, m_1\lambda_1 + \cdots + m_r\lambda_r \rangle \\ &= \langle \varepsilon_i - \varepsilon_j, m_1\lambda_1 + \cdots + m_r\lambda_r \rangle \end{aligned}$$

$$= \sum_{i \le k < j} \langle \alpha_k, m_1\lambda_1 + \cdots + m_r\lambda_r \rangle = m_i + \cdots + m_{j-1}.$$

This gives the following explicit form of $\zeta_r(\mathbf{s}; A_r)$, which first appeared in [146].

PROPOSITION 3.5.

$$\zeta_r(\mathbf{s}; A_r) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{1 \le i < j \le r+1} (m_i + \cdots + m_{j-1})^{-s_{ij}}, \tag{3.2.2}$$

for $\mathbf{s} = \mathbf{s}(A_r) = (s_{ij}) \in \mathbb{C}^{r(r+1)/2}$, *where* s_{ij} *is the variable corresponding to the root* $\varepsilon_i - \varepsilon_j$, *and*

$$K(A_r) = \prod_{1 \le i < j \le r+1} (j - i). \tag{3.2.3}$$

In particular, $\zeta_1(\mathbf{s}; A_1)$ is nothing but the Riemann zeta-function $\zeta(s_{12})$, and $\zeta_2(\mathbf{s}; A_2)$ coincides with the Tornheim double series

$$\zeta_{MT,2}(s_1, s_2, s_3) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1} m_2^{-s_2} (m_1 + m_2)^{-s_3}, \tag{3.2.4}$$

with $s_1 = s_{12}$, $s_2 = s_{23}$, and $s_3 = s_{13}$. This series was first introduced by Tornheim [194] in 1950, when s_1, s_2 and s_3 are positive integers.

These two cases for $X = A_1, A_2$ are the only classically known cases of (3.1.5). For more general Δ, the zeta-function of the form (3.1.5) was first introduced by the second-named author [134] in the case of B_2, and then, for any A_r ($r \in \mathbb{N}$) by the second and the third-named authors [146]. The definition (3.1.5) for general Δ was first given in [100].

In the previous section we introduced the notion of zeta-functions of root systems, and noticed that it is a multi-variable generalization of the notion of Witten zeta-functions. Here we remark that the notion of zeta-functions of root systems includes two other important classes of multiple zeta-functions.

First, in (3.2.2), we put $s_{ij} = 0$ if $i \ge 2$. The series is still convergent if $\Re s_{1j} > 1$ for all $2 \le j \le r+1$. Comparing with (1.3.1), we find that this series coincides with the Euler–Zagier multiple zeta-function:

$$\zeta_r(s_{12}, s_{13}, \ldots, s_{1,r+1}, 0, \ldots, 0; A_r) = \zeta_{EZ,r}(s_{12}, s_{13}, \ldots, s_{1,r+1}). \tag{3.2.5}$$

Later we will mention that the Euler–Zagier multiple zeta-function can also be regarded as a special case of the zeta-function of the root system C_r. These facts are quite important when we apply the theory of zeta-functions of root systems to the study of Euler–Zagier multiple zeta-functions (see Chapters 12 and 13).

Next, we discuss the class of Mordell–Tornheim multiple zeta-functions $\zeta_{MT,r}(s_1,\dots,s_r,s_{r+1})$, defined by (1.4.3). The case $r=2$ of this series is clearly (3.2.4).

Let $\mathbf{s}^\sharp=(s_{ij})$ with $s_{ij}=0$ unless $(i,j)=(1,2),(2,3),\dots,(r,r+1)$ and $(i,j)=(1,r+1)$. Substituting this $\mathbf{s}^\sharp$ in (3.2.2) we observe that

$$\zeta_r(\mathbf{s}^\sharp;A_r)=\zeta_{MT,r}(s_{12},s_{23},\dots,s_{r,r+1},s_{1,r+1}). \tag{3.2.6}$$

Therefore $\zeta_{MT,r}(s_1,\dots,s_r,s_{r+1})$ can also be regarded as a specialization of zeta-functions of root systems of type A.

The relations (3.1.6), (3.2.5) and (3.2.6) imply that zeta-functions of root systems are unification of multiple zeta-functions of Witten, of Euler and Zagier, and of Mordell and Tornheim.

In Sections 1.3 and 1.4 it is mentioned that special values of Euler–Zagier multiple zeta-functions, and of Mordell–Tornheim multiple zeta-functions, satisfy parity results. These two subclasses have such interesting properties in the wider family of zeta-functions of root systems (see Section 12.5).

3.3. Explicit forms for zeta-functions of other types

In Section 3.2 we obtained the explicit form of zeta-functions of root systems of type A_r. Now we consider the case of root systems of other types.

Without loss of generality we may restrict our consideration to the case of simple $\mathfrak{g}$.

Therefore now let $\mathfrak{g}$ be simple, and Δ be the associated root system. As in the case of A_r, we can find explicit forms of $\zeta_r(\mathbf{s};\Delta)$ for other Δ.

First, consider the type B_r. The list of Bourbaki [20] shows that the fundamental system is

$$\Psi(B_r)=\{\alpha_j=\alpha_j(B_r)=\varepsilon_j-\varepsilon_{j+1}\ (1\le j\le r-1),\ \alpha_r=\alpha_r(B_r)=\varepsilon_r\}, \tag{3.3.1}$$

and the list of positive roots is

$$\begin{cases}\varepsilon_i=\displaystyle\sum_{i\le k\le r}\alpha_k & (1\le i\le r),\\ \varepsilon_i-\varepsilon_j=\displaystyle\sum_{i\le k<j}\alpha_k & (1\le i<j\le r),\\ \varepsilon_i+\varepsilon_j=\displaystyle\sum_{i\le k<j}\alpha_k+2\sum_{j\le k\le r}\alpha_k & (1\le i<j\le r).\end{cases} \tag{3.3.2}$$

The fundamental coroots are $\alpha_1^\vee = e_1 - e_2, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r$ and $\alpha_r^\vee = 2e_r$. Positive coroots are $2e_i$ $(1 \le i \le r)$, $e_i \pm e_j$ $(1 \le i < j \le r)$. Hence the list (3.3.2) can be modified to the following list of positive coroots:

(3.3.3)
$$\begin{cases} 2e_i = (\varepsilon_i)^\vee = 2\displaystyle\sum_{i\le k<r} \alpha_k^\vee + \alpha_r^\vee & (1 \le i \le r), \\ e_i - e_j = (\varepsilon_i - \varepsilon_j)^\vee = \displaystyle\sum_{i\le k<j} \alpha_k^\vee & (1 \le i < j \le r), \\ e_i + e_j = (\varepsilon_i + \varepsilon_j)^\vee = \displaystyle\sum_{i\le k<j} \alpha_k^\vee + 2\sum_{j\le k<r} \alpha_k^\vee + \alpha_r^\vee & (1 \le i < j \le r). \end{cases}$$

We note that, in the B_r case, there are two different lengths of roots. The length of ε_i $(1 \le i \le r)$ is short, while the length of $\varepsilon_i \pm \varepsilon_j$ $(1 \le i < j \le r)$ is long. Correspondingly, the coroot $2e_i$ $(1 \le i \le r)$ is long, while $e_i \pm e_j$ $(1 \le i < j \le r)$ is short.

Using (3.3.3) we have:

Proposition 3.6 ([**104**, Proposition 2.1]).

(3.3.4)
$$\begin{aligned} &\zeta_r(\mathbf{s}; B_r) \\ &= \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{1\le i\le r} (2(m_i + \cdots + m_{r-1}) + m_r)^{-s_i} \\ &\quad \times \prod_{1\le i<j\le r} (m_i + \cdots + m_{j-1})^{-s_{ij}^-} \\ &\quad \times \prod_{1\le i<j\le r} (m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_{r-1}) + m_r)^{-s_{ij}^+}, \end{aligned}$$

where $\mathbf{s} = \mathbf{s}(B_r) = ((s_i), (s_{ij}^-), (s_{ij}^+)) \in \mathbb{C}^{r^2}$, *and*

$$K(B_r) = \prod_{1\le i\le r} (2r - 2i + 1) \prod_{1\le i<j\le r} (j - i)(2r - i - j + 1). \tag{3.3.5}$$

For example, we have $\zeta_1(s; B_1) = \zeta(s)$, and

$$\zeta_2(\mathbf{s}; B_2) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} (2m_1 + m_2)^{-s_1} m_2^{-s_2} m_1^{-s_{12}^-} (m_1 + m_2)^{-s_{12}^+}, \tag{3.3.6}$$

which was introduced in [**134**, (4.3)]. We remark that the Apostol–Vu double zeta-function

$$\zeta_{AV,2}(s_1, s_2, s_3) = \sum_{m_1<m_2} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1 + m_2)^{s_3}} \tag{3.3.7}$$

$$= \sum_{m_1=1}^{\infty} \sum_{m_2'=1}^{\infty} \frac{1}{m_1^{s_1}(m_1+m_2')^{s_2}(2m_1+m_2')^{s_3}}$$

(see (1.4.4)) is a specialization of the zeta-function of B_2 (obtained by putting $s_2 = 0$ in (3.3.6)). Further,

(3.3.8)

$$\begin{aligned}\zeta_3(\mathbf{s}; B_3) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \sum_{m_3=1}^{\infty} &(2m_1+2m_2+m_3)^{-s_1}(2m_2+m_3)^{-s_2} m_3^{-s_3} \\ &\times m_1^{-s_{12}^-}(m_1+2m_2+m_3)^{-s_{12}^+}(m_1+m_2)^{-s_{13}^-} \\ &\times (m_1+m_2+m_3)^{-s_{13}^+} m_2^{-s_{23}^-}(m_2+m_3)^{-s_{23}^+}.\end{aligned}$$

In the case of type C_r, the fundamental system is

(3.3.9)

$$\Psi(C_r) = \{\alpha_j = \alpha_j(C_r) = \varepsilon_j - \varepsilon_{j+1}\ (1 \le j \le r-1),\ \alpha_r = \alpha_r(C_r) = 2\varepsilon_r\},$$

positive roots are $2\varepsilon_i$ $(1 \le i \le r)$ and $\varepsilon_i \pm \varepsilon_j$ $(1 \le i < j \le r)$, the fundamental coroots are $\alpha_1^\vee = e_1 - e_2, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r$ and $\alpha_r^\vee = e_r$, hence the list of positive coroots is

$$\text{(3.3.10)} \qquad \begin{cases} e_i = \sum_{i \le k \le r} \alpha_k^\vee & (1 \le i \le r), \\ e_i - e_j = \sum_{i \le k < j} \alpha_k^\vee & (1 \le i < j \le r), \\ e_i + e_j = \sum_{i \le k < j} \alpha_k^\vee + 2 \sum_{j \le k \le r} \alpha_k^\vee & (1 \le i < j \le r). \end{cases}$$

Note that there are again short roots and long roots in the C_r case. We have:

Proposition 3.7 ([**104**, Proposition 2.2]).

$$\begin{aligned}\text{(3.3.11)}\quad \zeta_r(\mathbf{s}; C_r) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} &\prod_{1 \le i \le r} (m_i + \cdots + m_r)^{-s_i} \\ &\times \prod_{1 \le i < j \le r} (m_i + \cdots + m_{j-1})^{-s_{ij}^-} \\ &\times \prod_{1 \le i < j \le r} (m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_r))^{-s_{ij}^+},\end{aligned}$$

where $\mathbf{s} = \mathbf{s}(C_r) = ((s_i), (s_{ij}^-), (s_{ij}^+)) \in \mathbb{C}^{r^2}$, *and*

$$\text{(3.3.12)} \qquad K(C_r) = \prod_{1 \le i \le r} (r-i+1) \prod_{1 \le i < j \le r} (j-i)(2r-i-j+2).$$

We see that

(3.3.13)
$$\zeta_2(\mathbf{s};C_2)=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}(m_1+m_2)^{-s_1}m_2^{-s_2}m_1^{-s_{12}^-}(m_1+2m_2)^{-s_{12}^+},$$

which is equal to $\zeta_2(s_{12}^+,s_{12}^-,s_2,s_1;B_2)$, if we change m_1 and m_2 in the above. This is the natural consequence of the isomorphism $B_2\simeq C_2$. The reverse order of variables reflects the fact that the directions of arrows of Dynkin diagrams (see Section 6.4 below) of B_2 and C_2 are opposite. The case $r=3$ gives

(3.3.14)
$$\begin{aligned}\zeta_3(\mathbf{s};C_3)=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\sum_{m_3=1}^{\infty}&(m_1+m_2+m_3)^{-s_1}(m_2+m_3)^{-s_2}m_3^{-s_3}\\&\times m_1^{-s_{12}^-}(m_1+2m_2+2m_3)^{-s_{12}^+}(m_1+m_2)^{-s_{13}^-}\\&\times(m_1+m_2+2m_3)^{-s_{13}^+}m_2^{-s_{23}^-}(m_2+2m_3)^{-s_{23}^+}.\end{aligned}$$

Remark 3.8. From the above data (such as (3.3.2), (3.3.3), (3.3.10)) it is easy to see that $\Delta(B_r)^\vee=\Delta(C_r)$ and $\Delta(C_r)^\vee=\Delta(B_r)$ (if we identify V and V^*). Therefore the root systems $\Delta(B_r)$ and $\Delta(C_r)$ are duals of each other.

Lastly, in the case of type D_r, the fundamental system is

(3.3.15)
$$\Psi(D_r)=$$
$$\{\alpha_j=\alpha_j(D_r)=\varepsilon_j-\varepsilon_{j+1}\ (1\le j\le r-1),\ \alpha_r=\alpha_r(D_r)=\varepsilon_{r-1}+\varepsilon_r\},$$

positive roots are $\varepsilon_i\pm\varepsilon_j$ $(1\le i<j\le r)$, the fundamental coroots are $\alpha_1^\vee=e_1-e_2,\dots,\alpha_{r-1}^\vee=e_{r-1}-e_r$ and $\alpha_r^\vee=e_{r-1}+e_r$, hence the list of positive coroots is

(3.3.16)
$$\begin{cases}e_i+e_r=\displaystyle\sum_{i\le k\le r-2}\alpha_k^\vee+\alpha_r^\vee & (1\le i<r),\\ e_i-e_j=\displaystyle\sum_{i\le k<j}\alpha_k^\vee & (1\le i<j\le r),\\ e_i+e_j=\displaystyle\sum_{i\le k<j}\alpha_k^\vee+2\sum_{j\le k\le r-2}\alpha_k^\vee+\alpha_{r-1}^\vee+\alpha_r^\vee & (1\le i<j<r).\end{cases}$$

All roots are of the same length in the D_r case. From the above list we have:

PROPOSITION 3.9 ([104, Proposition 2.3]).

(3.3.17)

$$\zeta_r(\mathbf{s}; D_r) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{1\le i<r} ((m_i + \cdots + m_{r-2}) + m_r)^{-s_{ir}^+} \times \prod_{1\le i<j\le r} (m_i + \cdots + m_{j-1})^{-s_{ij}^-} \times \prod_{1\le i<j<r} (m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_{r-2}) + m_{r-1} + m_r)^{-s_{ij}^+}$$

for $r \ge 2$, *where* $\mathbf{s} = \mathbf{s}(D_r) = ((s_{ij}^-), (s_{ij}^+)) \in \mathbb{C}^{r(r-1)}$, *and*

$$K(D_r) = \prod_{1\le i\le r} (r-i) \prod_{1\le i<j\le r} (j-i) \prod_{1\le i<j<r} (2r-i-j). \tag{3.3.18}$$

We find that

$$\zeta_2(s_1, s_{12}^-; D_2) = \zeta(s_1)\zeta(s_{12}^-).$$

Since $D_2 \simeq A_1 \oplus A_1$, this agrees with (3.1.8). Also we see that $\zeta_3(\mathbf{s}; D_3)$ is equal to $\zeta_3(\mathbf{s}; A_3)$ (under the suitable renaming of variables), which agrees with the fact $D_3 \simeq A_3$.

In later sections, we sometimes study several zeta-functions of root systems of low rank more explicitly. In such explicit computations, the above naming of variables is not convenient. Therefore we rename the variables of low rank zeta-functions as follows.

First, when $\Delta = A_2$ we use the notation (3.2.4):

$$\zeta_2(s_1, s_2, s_3; A_2) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1} m_2^{-s_2} (m_1 + m_2)^{-s_3}. \tag{3.3.19}$$

Similarly, we rename

$$\begin{aligned} &\zeta_3(s_1, s_2, s_3, s_4, s_5, s_6; A_3) \\ &\quad = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \sum_{m_3=1}^{\infty} m_1^{-s_1} m_2^{-s_2} m_3^{-s_3} (m_1 + m_2)^{-s_4} (m_2 + m_3)^{-s_5} \\ &\qquad \times (m_1 + m_2 + m_3)^{-s_6}, \end{aligned} \tag{3.3.20}$$

$$\begin{aligned} &\zeta_3(s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9, s_{10}; A_4) \\ &\quad = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \sum_{m_3=1}^{\infty} \sum_{m_4=1}^{\infty} m_1^{-s_1} m_2^{-s_2} m_3^{-s_3} m_4^{-s_4} (m_1 + m_2)^{-s_5} \\ &\qquad \times (m_2 + m_3)^{-s_6} (m_3 + m_4)^{-s_7} (m_1 + m_2 + m_3)^{-s_8} \\ &\qquad \times (m_2 + m_3 + m_4)^{-s_9} (m_1 + m_2 + m_3 + m_4)^{-s_{10}}, \end{aligned} \tag{3.3.21}$$

$$\zeta_2(s_1, s_2, s_3, s_4; B_2) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} m^{-s_1} n^{-s_2} (m+n)^{-s_3} (2m+n)^{-s_4}, \tag{3.3.22}$$

$$\zeta_2(s_1, s_2, s_3, s_4; C_2) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} m^{-s_1} n^{-s_2} (m+n)^{-s_3} (m+2n)^{-s_4}, \tag{3.3.23}$$

$$\begin{aligned}\zeta_3(s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9; B_3) &= \sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\sum_{m_3=1}^{\infty} m_1^{-s_1} m_2^{-s_2} m_3^{-s_3} (m_1+m_2)^{-s_4} (m_2+m_3)^{-s_5} \\ &\times (2m_2+m_3)^{-s_6} (m_1+m_2+m_3)^{-s_7} (m_1+2m_2+m_3)^{-s_8} \\ &\times (2m_1+2m_2+m_3)^{-s_9},\end{aligned} \tag{3.3.24}$$

and

$$\begin{aligned}\zeta_3(s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9; C_3) &= \sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\sum_{m_3=1}^{\infty} m_1^{-s_1} m_2^{-s_2} m_3^{-s_3} (m_1+m_2)^{-s_4} (m_2+m_3)^{-s_5} \\ &\times (m_2+2m_3)^{-s_6} (m_1+m_2+m_3)^{-s_7} (m_1+m_2+2m_3)^{-s_8} \\ &\times (m_1+2m_2+2m_3)^{-s_9}.\end{aligned} \tag{3.3.25}$$

When we carry out explicit computations on zeta-functions of low rank, we use these notations.

We additionally mention the case of exceptional algebras. It is also possible to write down the explicit forms of zeta-functions of the root systems of types E_6, E_7, E_8, F_4 and G_2. Here we only treat the simplest case G_2.

Let V be a 2-dimensional real vector space equipped with an inner product $\langle \cdot, \cdot \rangle$. Let $\Delta = \Delta(G_2)$ be a finite reduced root system in V of type G_2. We know that $\Delta_+ = \Delta_+(G_2)$ is given by

$$\Delta_+ = \{\alpha_j\}_{j=1}^{6}, \tag{3.3.26}$$

where $\Psi = \Psi(G_2) = \{\alpha_1, \alpha_2\}$ is the set of fundamental roots and

$$\begin{aligned} \alpha_3 &= 3\alpha_1 + \alpha_2, & \alpha_3^\vee &= \alpha_1^\vee + \alpha_2^\vee \\ \alpha_4 &= 3\alpha_1 + 2\alpha_2, & \alpha_4^\vee &= \alpha_1^\vee + 2\alpha_2^\vee \\ \alpha_5 &= \alpha_1 + \alpha_2, & \alpha_5^\vee &= \alpha_1^\vee + 3\alpha_2^\vee \\ \alpha_6 &= 2\alpha_1 + \alpha_2, & \alpha_6^\vee &= 2\alpha_1^\vee + 3\alpha_2^\vee \end{aligned} \tag{3.3.27}$$

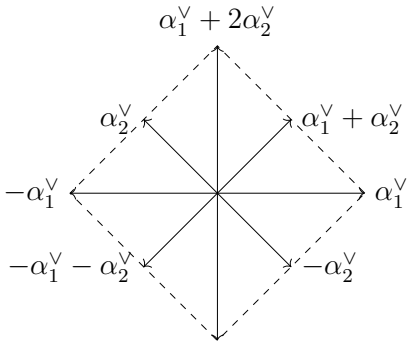
(see, for example, Bourbaki [20]).

From the above data we find that the zeta-function of the root system of type G_2 can be given by
(3.3.28)

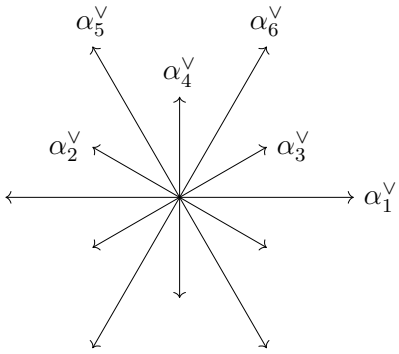
$$\zeta_2(\mathbf{s}; G_2) = \zeta_2(s_1, s_2, s_3, s_4, s_5, s_6; G_2)$$
$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{m^{s_1} n^{s_2} (m+n)^{s_3} (m+2n)^{s_4} (m+3n)^{s_5} (2m+3n)^{s_6}}.$$

REMARK 3.10. We observed that all roots are of the same length in the cases A_r and D_r. Such root systems are called *simply-laced*. Root systems of types B_r and C_r are not simply-laced, and in these cases there are *short roots* and *long roots*. In the exceptional cases, it is known that E_6, E_7, E_8 are simply-laced, while F_4 and G_2 are not. As for the action of the Weyl group $W = W(\Delta)$, the set of all roots of the same length form a single orbit (see [**70**, p.53, Lemma C]). In particular, when Δ is simply-laced, W acts on Δ transitively.

We conclude this section with the illustration of root systems of types C_2 and G_2. (There are four root systems A_2, B_2, C_2, G_2 of rank 2, but we already give the figure for the case A_2 before, and we omit the case B_2 because $B_2 \simeq C_2$.)



$\Delta^\vee(C_2)$



$\Delta^\vee(G_2)$

From these figures we can clearly see that the root systems of types C_2 and G_2 are not simply-laced.

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CHAPTER 4

Values at positive even integer points

The explicit formula for the values of the Riemann zeta-function $\zeta(s)$ at positive even integer points was discovered by Euler. Define *Bernoulli numbers*[1] B_k by

$$\frac{x}{e^x-1}=\sum_{k=0}^{\infty}B_k\frac{x^k}{k!}. \tag{4.0.1}$$

For example, $B_0=1, B_1=-1/2, B_2=1/6, B_4=-1/30, B_6=1/42$, and $B_k=0$ for all odd $k\geq 3$. The celebrated formula of Euler is

$$\zeta(2k)=\frac{(-1)^{k-1}(2\pi)^{2k}}{2(2k)!}B_{2k}\qquad(k\in\mathbb{N}). \tag{4.0.2}$$

Since $B_k\in\mathbb{Q}$, this formula especially implies $\zeta(2k)\in\mathbb{Q}\pi^{2k}$.

This result has been generalized to the case of Euler–Zagier multiple zeta-functions. It is known that

$$\zeta_{EZ,r}(\underbrace{2k,\ldots,2k}_{r})\in\mathbb{Q}\pi^{2kr}, \tag{4.0.3}$$

whose rational coefficients can be explicitly given.[2]

The aim of this chapter is to give an analogue of this fact in the case of zeta-functions of root systems. It especially includes (4.0.2) as a special case (see Section 4.2)[3].

4.1. The sum $S(\mathbf{s},\mathbf{y};\Delta)$

Recall that $H_{\alpha^\vee}=\{\beta\in V\mid\langle\alpha^\vee,\beta\rangle=0\}$ for $\alpha\in\Delta$. This is the hyperplane orthogonal to $\alpha^\vee$ (see the figure in the case of A_2). Recall also that the Weyl group $W=W(\Delta)$ is generated by the reflections σ_α with respect to $H_{\alpha^\vee}$, that is $W=\langle\sigma_\alpha\mid\alpha\in\Delta\rangle$.

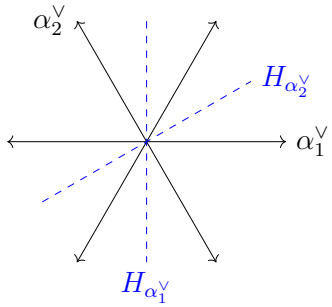
[1] Sometimes called *Seki–Bernoulli numbers*, because T. Seki introduced this notion independently and shortly before J. Bernoulli (see [6]; note that the definition of B_k is slightly different in [6]).

[2] A proof of (4.0.3) from our viewpoint will be given in Theorem 12.5.

[3] Another proof of (4.0.2) will be shown in Chapter 8.

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The open region

$$C = \{v \in V \mid \langle v, \alpha_i^\vee \rangle > 0, 1 \le i \le r\} \tag{4.1.1}$$

is called the *fundamental Weyl chamber*. In the above figure, the fundamental Weyl chamber is the region whose boundary consists of two dotted half-lines in the upper half plane. And generally we call the region of the form $w(C)$, $w \in W$, a *Weyl chamber*. Then

$$V = \bigcup_{w \in W} \overline{w(C)},$$

where $\overline{w(C)}$ means the topological closure of $w(C)$. Two distinct $\overline{w_1(C)}$ and $\overline{w_2(C)}$ $(w_1, w_2 \in W)$ may intersect only on their boundaries. The Weyl group W acts on the set of Weyl chambers simply transitively. Moreover, if $wx = y$ for $w \in W$, $x, y \in \overline{C}$, then $x = y$ (see [71, Section 1.12, Theorem (a)]). The stabilizer W_x of a point $x \in V$ is generated by the reflections which stabilize x. It should be noted that $P_+ = P \cap \overline{C}$, where P and P_+ are defined in Section 2.4.

Denote by $H_{\Delta^\vee}$ the union of all boundaries of Weyl chambers:

$$H_{\Delta^\vee} = \bigcup_{\alpha \in \Delta} H_{\alpha^\vee} = V \setminus \bigcup_{w \in W} w(C). \tag{4.1.2}$$

If $\lambda \in P \setminus H_{\Delta^\vee}$, we have $\langle \alpha^\vee, \lambda \rangle \neq 0$ for any $\alpha \in \Delta_+$, and hence

$$S(\mathbf{s}, \mathbf{y}; \Delta) = \sum_{\lambda \in P \setminus H_{\Delta^\vee}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \langle \alpha^\vee, \lambda \rangle^{-s_\alpha} \tag{4.1.3}$$

for $\mathbf{s} = (s_\alpha)_{\alpha \in \Delta_+}$ and $\mathbf{y} \in V$ can be defined when $\Re s_\alpha$ is sufficiently large. When $\mathbf{y} = \mathbf{0}$, we write $S(\mathbf{s}, \mathbf{0}; \Delta) = S(\mathbf{s}; \Delta)$.

Any $\lambda \in P \setminus H_{\Delta^\vee}$ can be written uniquely as $\lambda = w\lambda'$ with $w \in W$, $\lambda' \in P_{++}$. Therefore, noting $\langle \alpha^\vee, w\lambda' \rangle = \langle w^{-1}\alpha^\vee, \lambda' \rangle$ (see (2.3.2)), we put $w^{-1}\alpha = \beta$ to obtain

$$S(\mathbf{s}, \mathbf{y}; \Delta) = \sum_{w \in W} \sum_{\lambda \in P_{++}} e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y}, \lambda \rangle} \prod_{\beta \in w^{-1}\Delta_+} \langle \beta^\vee, \lambda \rangle^{-s_{w\beta}}.$$

Divide the product as

$$\prod_{\beta\in w^{-1}\Delta_+} = \prod_{\beta\in\Delta_+\cap w^{-1}\Delta_+}\prod_{\beta\in\Delta_-\cap w^{-1}\Delta_+}, \tag{4.1.4}$$

and put $\beta=-\gamma$ ($\gamma\in\Delta_+$) in the second product, then we find

$$\prod_{\beta\in\Delta_-\cap w^{-1}\Delta_+}\langle\beta^\vee,\lambda\rangle^{-s_{w\beta}} = \prod_{\gamma\in\Delta_+\cap w^{-1}\Delta_-}\langle-\gamma^\vee,\lambda\rangle^{-s_{-w\gamma}}.$$

Renaming β,γ to α, and reading $s_\alpha=s_{-\alpha}$ for $\alpha\in\Delta_-$, we have

(4.1.5)

$$\begin{aligned}\prod_{\beta\in w^{-1}\Delta_+}\langle\beta^\vee,\lambda\rangle^{-s_{w\beta}} &= \prod_{\alpha\in\Delta_+\cap w^{-1}\Delta_+}\langle\alpha^\vee,\lambda\rangle^{-s_{w\alpha}}\prod_{\alpha\in\Delta_+\cap w^{-1}\Delta_-}\langle-\alpha^\vee,\lambda\rangle^{-s_{w\alpha}}\\ &= \left(\prod_{\alpha\in\Delta_+\cap w^{-1}\Delta_-}(-1)^{-s_{w\alpha}}\right)\prod_{\alpha\in\Delta_+}\langle\alpha^\vee,\lambda\rangle^{-s_{w\alpha}}.\end{aligned}$$

Therefore, putting

$$\Delta_w=\Delta_+\cap w^{-1}\Delta_-, \tag{4.1.6}$$

we have

$$\begin{aligned}S(\mathbf{s},\mathbf{y};\Delta) &= \sum_{w\in W}\left(\prod_{\alpha\in\Delta_w}(-1)^{-s_{w\alpha}}\right)\sum_{\lambda\in P_{++}}e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\langle\alpha^\vee,\lambda\rangle^{-s_{w\alpha}}\\ &= \sum_{w\in W}\left(\prod_{\alpha\in\Delta_w}(-1)^{-s_{w\alpha}}\right)\zeta_r((s_{w\alpha})_{\alpha\in\Delta_+},w^{-1}\mathbf{y};\Delta).\end{aligned}$$

Since $\zeta_r((s_{w\alpha})_{\alpha\in\Delta_+},w^{-1}\mathbf{y};\Delta)$ is absolutely convergent in $\mathcal{S}$ (see Lemma 3.3), we see that the definition (4.1.3) of $S(\mathbf{s},\mathbf{y};\Delta)$ is also valid in the same region. Define the action of W on $\mathbf{s}$ by $w^{-1}\mathbf{s}=(s_{w\alpha})_{\alpha\in\Delta_+}$. Then the above formula can be written as

(4.1.7)

$$\begin{aligned}S(\mathbf{s},\mathbf{y};\Delta) &= \sum_{w\in W}\left(\prod_{\alpha\in\Delta_w}(-1)^{-s_{w\alpha}}\right)\zeta_r(w^{-1}\mathbf{s},w^{-1}\mathbf{y};\Delta)\\ &= \sum_{w\in W}\left(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\right)\zeta_r(w^{-1}\mathbf{s},w^{-1}\mathbf{y};\Delta)\quad(\mathbf{s}\in\mathcal{S}),\end{aligned}$$

where the second equality is valid because if $\alpha\in\Delta_w$ then $w\alpha\in w\Delta_w=w\Delta_+\cap\Delta_-=-(w\Delta_-\cap\Delta_+)=-\Delta_{w^{-1}}$.

Now assume $\mathbf{s} \in \mathcal{S}_0$. By $\Gamma(s)$ we denote the Euler gamma function. We define a zeta-like function

$$\widetilde{\phi}(s,x) = -\frac{\Gamma(s+1)}{(2\pi\sqrt{-1})^s} \lim_{N\to\infty} \sum_{\substack{n=-N \\ n\neq 0}}^{N} \frac{e^{2\pi\sqrt{-1}nx}}{n^s} \quad (x \in \mathbb{R}), \tag{4.1.8}$$

which is absolutely convergent for $\Re s > 1$. Note that if $x \in \mathbb{R} \setminus \mathbb{Z}$ then $\widetilde{\phi}(1,x)$ can be defined by (4.1.8). In fact, when $s = 1$, the infinite series on the right-hand side of (4.1.8) converges for $x \in \mathbb{R} \setminus \mathbb{Z}$ uniformly in the wider sense (see [**209**, §3.35, Example 1]).

We recall the *Lerch zeta-function* (or the *periodic zeta-function*)

$$\phi(s,x) = \sum_{n\geq 1} e^{2\pi\sqrt{-1}nx} n^{-s} \qquad (x \in \mathbb{R}), \tag{4.1.9}$$

which is absolutely convergent for $\Re s > 1$ and can be meromorphically continued to the whole space $\mathbb{C}$.

We can write

$$\widetilde{\phi}(s,x) = -\frac{\Gamma(s+1)}{(2\pi\sqrt{-1})^s} \left(\phi(s,x) + (-1)^{-s}\phi(s,-x)\right) \quad (x \in \mathbb{R}) \tag{4.1.10}$$

for $\Re s > 1$.

For any non-zero integer a, we have

$$\begin{aligned} a^{-s} &= \sum_{n\neq 0} \int_0^1 \frac{e^{2\pi\sqrt{-1}(n-a)x}}{n^s} dx \\ &= \int_0^1 e^{-2\pi\sqrt{-1}ax} \sum_{n\neq 0} \frac{e^{2\pi\sqrt{-1}nx}}{n^s} dx \\ &= -\frac{(2\pi\sqrt{-1})^s}{\Gamma(s+1)} \int_0^1 \widetilde{\phi}(s,x) e^{-2\pi\sqrt{-1}ax} dx \end{aligned} \tag{4.1.11}$$

for $\Re s > 1$. Therefore, since $\mathbf{s} \in \mathcal{S}_0$, we have

$$\begin{aligned} & S(\mathbf{s},\mathbf{y};\Delta) \\ &= \sum_{\lambda \in P\setminus H_{\Delta^\vee}} e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \langle \alpha^\vee,\lambda\rangle^{-s_\alpha} \\ &= \sum_{\lambda \in P\setminus H_{\Delta^\vee}} e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Psi} \langle \alpha^\vee,\lambda\rangle^{-s_\alpha} \prod_{\alpha\in\Delta_+\setminus\Psi} \langle \alpha^\vee,\lambda\rangle^{-s_\alpha} \\ &= \sum_{\lambda \in P\setminus H_{\Delta^\vee}} e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Psi} \langle \alpha^\vee,\lambda\rangle^{-s_\alpha} \prod_{\alpha\in\Delta_+\setminus\Psi} \frac{-(2\pi\sqrt{-1})^{s_\alpha}}{\Gamma(s_\alpha+1)} \\ &\quad \times \int_0^1 \widetilde{\phi}(s_\alpha,x_\alpha) e^{-2\pi\sqrt{-1}\langle \alpha^\vee,\lambda\rangle x_\alpha} dx_\alpha. \end{aligned}$$

Using the expression $\lambda = \sum_{j=1}^{r} n_j \lambda_j$ we further find that the right-hand side of the above is

$$= \prod_{\alpha\in\Delta_+\backslash\Psi} \frac{-(2\pi\sqrt{-1})^{s_\alpha}}{\Gamma(s_\alpha+1)} \sum_{\substack{n_j\neq 0\\ 1\le j\le r}} \int_0^1 \cdots \int_0^1 \prod_{\alpha\in\Delta_+\backslash\Psi} \widetilde{\phi}(s_\alpha, x_\alpha)$$

$$\times \prod_{j=1}^{r} \exp\left(2\pi\sqrt{-1}\left(\langle \mathbf{y}, \lambda_j\rangle - \sum_{\alpha\in\Delta_+\backslash\Psi} \langle \alpha^\vee, \lambda_j\rangle x_\alpha\right) n_j\right) \prod_{j=1}^{r} n_j^{-s_{\alpha_j}} \prod_{\alpha\in\Delta_+\backslash\Psi} dx_\alpha.$$

Here, we remove the restriction $\lambda \in P\backslash H_{\Delta^\vee}$ to include all λ satisfying $n_j \neq 0$ $(1 \le j \le r)$, which is possible because the right-hand side of (4.1.11) is equal to 0 when $a = 0$. Therefore we now conclude the following theorem.

THEOREM 4.1 ([113, Theorem 6]). *Let $\mathbf{s} \in \mathcal{S}_0$ and $\mathbf{y} \in V$. We have*

$$S(\mathbf{s}, \mathbf{y}; \Delta)$$
$$= (-1)^{\#(\Delta_+)} \left(\prod_{\alpha\in\Delta_+} \frac{(2\pi\sqrt{-1})^{s_\alpha}}{\Gamma(s_\alpha+1)}\right) \int_0^1 \cdots \int_0^1 \left(\prod_{\alpha\in\Delta_+\backslash\Psi} \widetilde{\phi}(s_\alpha, x_\alpha)\right)$$
$$\times \prod_{j=1}^{r} \widetilde{\phi}(s_{\alpha_j}, \langle \mathbf{y}, \lambda_j\rangle - \sum_{\alpha\in\Delta_+\backslash\Psi} x_\alpha \langle \alpha^\vee, \lambda_j\rangle) \prod_{\alpha\in\Delta_+\backslash\Psi} dx_\alpha.$$

The sum $S(\mathbf{s}, \mathbf{y}; \Delta)$ is, as can be seen from (4.1.7), a linear combination of zeta-functions of root systems with signature. Since this sum has a nice symmetric property with respect to the action of Weyl groups, we can prove the above explicit expression in terms of integrals involving the zeta-like function $\widetilde{\phi}(s, x)$.

4.2. The values at even integer points

In this section we consider the case $\mathbf{s} = \mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+}$, with integral values of k_α. Let $\mathcal{K}_0 = \mathcal{S}_0 \cap \mathbb{N}^{\#(\Delta_+)}$ and $\mathcal{K} = \mathcal{S} \cap \mathbb{N}^{\#(\Delta_+)}$. That is, $\mathbf{k} \in \mathcal{K}_0$ if and only if $k_\alpha \ge 2$ for all α, while $\mathbf{k} \in \mathcal{K}$ admits the case $k_\alpha = 1$ for $\alpha \in \Delta_j$ and $r_j > 1$.

Denote by $B_k(x)$ the k-th Bernoulli polynomial, defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{k=0}^{\infty} B_k(x) \frac{t^k}{k!}. \tag{4.2.1}$$

Note that $B_k(0) = B_k$. The function $B_k(\{x\})$, where $\{x\} = x - [x]$ denotes the fractional part of x, is called the (periodic) k-th Bernoulli

function. For $\mathbf{k} \in \mathbb{N}_0^{\#(\Delta_+)}$ and $\mathbf{y} \in V$, we let

(4.2.2)

$$P(\mathbf{k}, \mathbf{y}; \Delta) = \int_0^1 \cdots \int_0^1 \left(\prod_{\alpha \in \Delta_+ \setminus \Psi} B_{k_\alpha}(\{x_\alpha\}) \right) \times \prod_{j=1}^{r} B_{k_{\alpha_j}}(\{\langle \mathbf{y}, \lambda_j \rangle - \sum_{\alpha \in \Delta_+ \setminus \Psi} x_\alpha \langle \alpha^\vee, \lambda_j \rangle\}) \prod_{\alpha \in \Delta_+ \setminus \Psi} dx_\alpha,$$

which we call *Bernoulli functions of root systems*. In particular, when $\mathbf{y} = \mathbf{0}$, we write $P(\mathbf{k}, \mathbf{0}; \Delta) = B(\mathbf{k}; \Delta)$ and call them *Bernoulli numbers of root systems*.

REMARK 4.2. When Δ is of type A_1, since $\Delta_+ = \Psi = \{\alpha_1\}$, clearly $P(\mathbf{k}, \mathbf{y}; A_1) = B_k(\{y\})$ for $\mathbf{k} = (k)$ and $\mathbf{y} = y\alpha_1^\vee$. Therefore $P(\mathbf{k}, \mathbf{y}; \Delta)$ is a generalization of classical Bernoulli functions. We will study $P(\mathbf{k}, \mathbf{y}; \Delta)$ more closely in Chapter 9, where a generalization of Bernoulli polynomials will also be introduced (see Remark 9.28).

It is known that

$$\widetilde{\phi}(k, x) = B_k(\{x\}) \quad (x \in \mathbb{R},\ k \geq 1), \tag{4.2.3}$$

namely

$$\lim_{N \to \infty} \sum_{\substack{m=-N \\ m \neq 0}}^{N} \frac{e^{2\pi\sqrt{-1}m\theta}}{m^k} = -\frac{(2\pi\sqrt{-1})^k}{k!} B_k(\{\theta\}) \quad (k \in \mathbb{N},\ \theta \in \mathbb{R}), \tag{4.2.4}$$

where we let $\theta \in \mathbb{R} \setminus \mathbb{Z}$ when $k = 1$ (see [4, Theorem 12.19]).

Therefore from Theorem 4.1 we obtain the following formula.

THEOREM 4.3 ([113, (123)]). *For $\mathbf{k} \in \mathcal{K}_0$ and $\mathbf{y} \in V$, we have*

$$S(\mathbf{k}, \mathbf{y}; \Delta) = (-1)^{\#(\Delta_+)} \left(\prod_{\alpha \in \Delta_+} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!} \right) P(\mathbf{k}, \mathbf{y}; \Delta). \tag{4.2.5}$$

REMARK 4.4. It is possible to extend (4.2.5) to any $\mathbf{k} \in \mathcal{K}$ (see Theorem 9.11 and (9.4.14)). Therefore we may treat the values at $\mathbf{s} = \mathbf{k} = (k_\alpha)_{\alpha \in \Delta_+}$, where some of $k_\alpha = 1$. However, a much more delicate argument is required to prove this extension. This is because the series (4.1.9) ($x \notin \mathbb{Z}$) at $s = 1$ is convergent but not absolutely, so the interchange of summation and integration performed in (4.1.11) becomes, at $s = 1$, a very delicate matter. A detailed proof of Theorem 9.11 will be described in Chapter 16.

Moreover, we may even treat the case when some of $k_\alpha = 0$. In this case the formula (4.2.5) itself is no longer valid, and hence, it is necessary to show a modified formula (see Section 9.5).

Here we consider the special case when $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+}$ satisfies the condition

$$k_\alpha = k_\beta \quad \text{if} \quad \|\alpha\| = \|\beta\|. \tag{4.2.6}$$

Since the set of all roots of the same length form a single orbit for the action of the Weyl group W (see Remark 3.10), $w^{-1}\mathbf{k} = \mathbf{k}$ for all $w \in W$ under the condition (4.2.6). Therefore (4.1.7) with $\mathbf{y} = \mathbf{0}$ implies

$$\begin{aligned} S(2\mathbf{k};\Delta) &= \sum_{w\in W}\left(\prod_{\alpha\in\Delta_w}(-1)^{-2k_\alpha}\right)\zeta_r(2\mathbf{k};\Delta) \\ &= \#W\cdot\zeta_r(2\mathbf{k};\Delta). \end{aligned}$$

Combining this with (4.2.5), we now arrive at:

THEOREM 4.5. *If* $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathbb{N}^{\#(\Delta_+)}$ *satisfies the condition* (4.2.6)*, we have*

$$\zeta_r(2\mathbf{k};\Delta) = \frac{(-1)^{\#(\Delta_+)}}{\#W}\left(\prod_{\alpha\in\Delta_+}\frac{(2\pi\sqrt{-1})^{2k_\alpha}}{(2k_\alpha)!}\right)B(2\mathbf{k};\Delta). \tag{4.2.7}$$

In particular, for any $k \in \mathbb{N}$*, we have*

$$\zeta_r(\underbrace{2k,\dots,2k}_{\#(\Delta_+)};\Delta) = \frac{(-1)^{\#(\Delta_+)}}{\#W}\left(\frac{(2\pi\sqrt{-1})^{2k}}{(2k)!}\right)^{\#(\Delta_+)}B(\underbrace{(2k,\dots,2k)}_{\#(\Delta_+)};\Delta). \tag{4.2.8}$$

In the next chapter, we will prove:

THEOREM 4.6. $B(\mathbf{k};\Delta) \in \mathbb{Q}$ *for any* $\mathbf{k} \in \mathbb{N}_0^{\#(\Delta_+)}$.

Combining these two theorems, we obtain:

COROLLARY 4.7. *If* $\mathbf{k}$ *satisfies* (4.2.6)*, then we have*

$$\zeta_r(2\mathbf{k};\Delta) \in \mathbb{Q}\pi^{2\sum_{\alpha\in\Delta_+}k_\alpha}, \tag{4.2.9}$$

and especially,

$$\zeta_r(\underbrace{2k,\dots,2k}_{\#(\Delta_+)};\Delta) \in \mathbb{Q}\pi^{2k\#(\Delta_+)} \tag{4.2.10}$$

for any $k \in \mathbb{N}$.

Theorem 4.5, Theorem 4.6 and Corollary 4.7 were proved in [**113**, Theorem 8]. These results generalize (4.0.2) and are analogous to (4.0.3). Moreover, in view of (3.1.6), it follows that the above corollary also implies (1.5.1).

REMARK 4.8. The fact $\zeta_r(2k,\dots,2k;\Delta) \in \mathbb{Q}\pi^{2k\#(\Delta_+)}$ was already mentioned by Witten [**210**] and Zagier [**215**], but the explicit expression (4.2.8) of the rational coefficients was first given in [**113**]. Moreover, our method catches the case when (4.2.6) is satisfied but not all k_α are the same. As for typical examples, see Examples 4.16 and 4.17. It is to be mentioned that Szenes [**190**] [**191**], and also Gunnells and Sczech [**53**], gave certain algorithms for evaluating the coefficients. Their studies are earlier than [**113**] and their methods are different from that in [**113**]. Zhao [**219**] showed a combinatorial approach, based on the ideas of Gunnells and Sczech [**53**], to the evaluation of the coefficients (see Example 4.19).

Define the generating function of $P(\mathbf{k},\mathbf{y};\Delta)$ by

$$F(\mathbf{t},\mathbf{y};\Delta) = \sum_{\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}} P(\mathbf{k},\mathbf{y};\Delta) \prod_{\alpha\in\Delta_+} \frac{t_\alpha^{k_\alpha}}{k_\alpha!} \qquad (t_\alpha \in \mathbb{C}). \tag{4.2.11}$$

This definition first appeared in [**113**]. We first prove the following lemma, which is a special case of [**113**, Lemma 7 (i)].

LEMMA 4.9. *Let $0 < R < 2\pi$. The right-hand side of (4.2.11) is absolutely convergent, uniformly in $\{t \in \mathbb{C} \mid |t| \le R\}^{\#(\Delta_+)}$.*

PROOF. Fix $R' \in \mathbb{R}$ such that $R < R' < 2\pi$. By the Cauchy integral formula

$$\frac{B_k(x)}{k!} = \frac{1}{2\pi\sqrt{-1}} \int_{|z|=R'} \frac{ze^{zx}}{e^z-1} \frac{dz}{z^{k+1}},$$

we have

$$\left|\frac{B_k(x)}{k!}\right| \le \frac{1}{2\pi} \int_{|z|=R'} \left|\frac{ze^{zx}}{e^z-1}\right| \frac{|dz|}{R'^{k+1}} \le \frac{C_{R'}}{R'^k}$$

for $0 \le x \le 1$, where

$$C_{R'} = \max\left\{ \left|\frac{ze^{zx}}{e^z-1}\right| \;\middle|\; |z| = R', 0 \le x \le 1 \right\}.$$

Therefore, when $|t_\alpha| \le R$ for all $\alpha \in \Delta_+$, we have

$$\left| P(\mathbf{k},\mathbf{y};\Delta) \prod_{\alpha\in\Delta_+} \frac{t_\alpha^{k_\alpha}}{k_\alpha!} \right|$$

$$\leq \left(\prod_{\alpha\in\Delta_+}|t_\alpha|^{k_\alpha}\right)\int_0^1\cdots\int_0^1\left(\prod_{\alpha\in\Delta_+\setminus\Psi}\left|\frac{B_{k_\alpha}(\{x_\alpha\})}{k_\alpha!}\right|\right)$$
$$\times\prod_{j=1}^{r}\left|\frac{1}{k_{\alpha_j}!}B_{k_{\alpha_j}}\left(\left\{\langle\mathbf{y},\lambda_j\rangle-\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\right\}\right)\right|\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha$$
$$\leq \left(\prod_{\alpha\in\Delta_+}R^{k_\alpha}\right)\int_0^1\cdots\int_0^1 C_{R'}^{\#(\Delta_+)}\left(\prod_{\alpha\in\Delta_+}\frac{1}{R'^{k_\alpha}}\right)\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha$$
$$= C_{R'}^{\#(\Delta_+)}\prod_{\alpha\in\Delta_+}\left(\frac{R}{R'}\right)^{k_\alpha}.$$

Since $R<R'$, this implies the absolute convergence of the right-hand side of (4.2.11). □

If we can find some closed form of this generating function, then we can compute the values of $P(\mathbf{k},\mathbf{y};\Delta)$ and $B(\mathbf{k};\Delta)$ by using such a form.

First, consider the A_1 case. In this case we have seen in Remark 4.2 that $P(k,y;A_1)=B_k(\{y\})$, the classical Bernoulli function. Therefore

$$F(t,y;A_1)=\frac{te^{t\{y\}}}{e^t-1} \tag{4.2.12}$$

(see (4.2.1)), and Theorem 4.5 implies (4.0.2).

Secondly, consider the A_2 case. In this case $\Delta_+=\{\alpha_1,\alpha_2,\alpha_1+\alpha_2\}$ and $\Psi=\{\alpha_1,\alpha_2\}$. We put $\mathbf{t}=(t_{\alpha_1},t_{\alpha-2},t_{\alpha_1+\alpha_2})=(t_1,t_2,t_3)$ and $\mathbf{y}=y_1\alpha_1^\vee+y_2\alpha_2^\vee$. The generating function is given explicitly as follows:

PROPOSITION 4.10. *We have*

$$F(\mathbf{t},\mathbf{y};A_2)=\frac{t_1t_2t_3}{t_3-t_1-t_2}\left(\frac{e^{\{y_1\}t_1+\{y_2\}t_2}}{(e^{t_1}-1)(e^{t_2}-1)}\right.$$
$$\left.-\frac{e^{\{y_1-y_2\}t_1+\{y_2\}t_3}}{(e^{t_1}-1)(e^{t_3}-1)}-\frac{e^{(1-\{y_1-y_2\})t_2+\{y_1\}t_3}}{(e^{t_2}-1)(e^{t_3}-1)}\right). \tag{4.2.13}$$

This is stated in [**106**, Example 9.2] without the details of the proof. Here we prove this formula by direct computations from the definitions.

PROOF. From (4.2.2) we see that

$$P((k_1,k_2,k_3),\mathbf{y};A_2)=\int_0^1 B_{k_3}(\{x_3\})B_{k_1}(\{y_1-x_3\})B_{k_2}(\{y_2-x_3\})dx_3 \tag{4.2.14}$$

and so, noting (4.2.1), we find

(4.2.15)

$$F(\mathbf{t},\mathbf{y};A_2)=\sum_{k_1,k_2,k_3\geq 0}P((k_1,k_2,k_3),\mathbf{y};A_2)\frac{t_1^{k_1}t_2^{k_2}t_3^{k_3}}{k_1!k_2!k_3!}$$

$$=\int_0^1\sum_{k_1,k_2,k_3\geq 0}B_{k_1}(\{y_1-x\})B_{k_2}(\{y_2-x\})B_{k_3}(\{x\})\frac{t_1^{k_1}t_2^{k_2}t_3^{k_3}}{k_1!k_2!k_3!}dx$$

$$=\frac{t_1t_2t_3}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_3}-1)}\int_0^1 e^{t_1\{y_1-x\}}\cdot e^{t_2\{y_2-x\}}\cdot e^{t_3\{x\}}dx.$$

First, consider the case $\{y_1\}\geq\{y_2\}$. In this case, we split the integral on the last member of the above as

$$\int_0^1=\int_0^{\{y_2\}}+\int_{\{y_2\}}^{\{y_1\}}+\int_{\{y_1\}}^1=I_1+I_2+I_3,$$

say. We have

$$I_1=\int_0^{\{y_2\}}e^{t_1(\{y_1\}-x)+t_2(\{y_2\}-x)+t_3\{x\}}dx$$

$$=e^{\{y_1\}t_1+\{y_2\}t_2}\cdot\frac{e^{(-t_1-t_2+t_3)\{y_2\}}-1}{-t_1-t_2+t_3}.$$

Similarly,

$$I_2=e^{\{y_1\}t_1+(1+\{y_2\})t_2}\cdot\frac{e^{(-t_1-t_2+t_3)\{y_1\}}-e^{(-t_1-t_2+t_3)\{y_2\}}}{-t_1-t_2+t_3}$$

and

$$I_3=e^{(1+\{y_1\})t_1+(1+\{y_2\})t_2}\cdot\frac{e^{-t_1-t_2+t_3}-e^{(-t_1-t_2+t_3)\{y_1\}}}{-t_1-t_2+t_3}.$$

Collecting these results, we obtain

(4.2.16)
$$F(\mathbf{t},\mathbf{y};A_2)=\frac{t_1t_2t_3e^{\{y_1\}t_1+\{y_2\}t_2}}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_3}-1)(t_3-t_1-t_2)}$$
$$\times\Big(e^{(t_3-t_1-t_2)\{y_2\}}-1+e^{t_2}(e^{(t_3-t_1-t_2)\{y_1\}}-e^{(t_3-t_1-t_2)\{y_2\}})$$
$$+e^{t_1+t_2}(e^{t_3-t_1-t_2}-e^{(t_3-t_1-t_2)\{y_1\}})\Big).$$

This formula coincides with [**113**, (249)].

When $\{y_1\}\leq\{y_2\}$, by the same argument we can show an expression quite similar to (4.2.16), but the quantity in the big parentheses must be replaced by

$$e^{(t_3-t_1-t_2)\{y_1\}}-1+e^{t_1}(e^{(t_3-t_1-t_2)\{y_2\}}-e^{(t_3-t_1-t_2)\{y_1\}})$$
$$+e^{t_1+t_2}(e^{t_3-t_1-t_2}-e^{(t_3-t_1-t_2)\{y_2\}}).$$

Therefore the expressions when $\{y_1\} \geq \{y_2\}$ and when $\{y_1\} \leq \{y_2\}$ are slightly different from each other, but we can easily check that these two expressions can be unified in the form (4.2.13). This completes the proof. □

EXAMPLE 4.11. In particular, when $\mathbf{y} = \mathbf{0}$, (4.2.13) is reduced to

$$F((t_1,t_2,t_3);A_2) = \frac{t_1t_2t_3(e^{t_3}-e^{t_1+t_2})}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_3}-1)(t_3-t_1-t_2)}. \tag{4.2.17}$$

The values of $B((k_1,k_2,k_3);A_2)$ can be computed from the Taylor expansion of the right-hand side of (4.2.17), which is

$$\begin{aligned}
&= 1 + \frac{1}{12}(t_1t_2 - t_1t_3 - t_2t_3) + \frac{1}{360}(t_1t_2t_3^2 - t_1^2t_2t_3 - t_1t_2^2t_3) \\
&\quad + \frac{1}{720}(t_1^2t_2^2 + t_1^2t_3^2 + t_2^2t_3^2) + \frac{1}{30240}t_1^2t_2^2t_3^2 + \cdots \\
&\quad - \frac{19}{186810624000}t_1^4t_2^4t_3^4 + \cdots \\
&\quad + \frac{1031}{2554547108585472000}t_1^6t_2^6t_3^6 + \cdots
\end{aligned}$$

From this expression we find, for example,

$$\begin{aligned}
B((2,2,2);A_2) &= \frac{1}{3780}, \\
B((4,4,4);A_2) &= -\frac{19}{13513500}, \\
B((6,6,6);A_2) &= \frac{1031}{6844101264}.
\end{aligned}$$

Combining this value with (4.2.8) we obtain

$$\begin{aligned}
\zeta_2(2,2,2;A_2) &= \frac{1}{2835}\pi^6, \\
\zeta_2(4,4,4;A_2) &= \frac{19}{273648375}\pi^{12}, \\
\zeta_2(6,6,6;A_2) &= \frac{2062}{116937886440375}\pi^{18},
\end{aligned}$$

where the first one was first discovered by Mordell [**157**]. A closed form of $\zeta_2(2m,2m,2m;A_2)$ for general $m \in \mathbb{N}$ was obtained by Subbarao and Sitaramachandrarao [**189**, Section 4] by a more elementary approach (see Corollary 8.20). Several special values were also computed by Gunnells and Sczech [**53**]. By our present method, we can further obtain

$$\begin{aligned}
\zeta_2(8,8,8;A_2) &= \frac{32293}{7060124893837640625}\pi^{24}, \\
\zeta_2(10,10,10;A_2) &= \frac{221912776}{1866223781759038244198906 25}\pi^{30},
\end{aligned}$$

$$\zeta_2(12,12,12;A_2)=\frac{59671681375178}{19298431172012311380638664391992187 5}\pi^{36},$$

$$\zeta_2(14,14,14;A_2)=\frac{574485671887928}{71448581728141180424538527178472675781 25}\pi^{42},$$

$$\zeta_2(16,16,16;A_2)$$
$$=\frac{221137132669842886663}{10576341708012816481744257935466212837209716796875}\pi^{48},$$

$$\zeta_2(18,18,18;A_2)$$
$$=\frac{58016497640927002918411544}{10670494660421747460922820748577252672852020393567626953125}\pi^{54},$$

$$\zeta_2(20,20,20;A_2)$$
$$=\frac{17246016721512586083679663897544}{12197787903592730676654923774020392718892428096047396122039794921875}\pi^{60}.$$

The values of $B((k_1,k_2,k_3);A_2)$ for several other (k_1,k_2,k_3) are listed in [**113**, Table 11.1].

4.3. A closed expression of $F(\mathbf{t},\mathbf{y};\Delta)$ and further examples

The method in the proof of Proposition 4.10 can be applied to the case of other root systems, but the total amount of computation becomes tremendously huge for root systems of higher rank. Therefore in later chapters we will develop two different methods.

The method presented in Chapter 8 is via functional relations. The values at even integer points of $\zeta_2(\mathbf{s};A_2)$ will be again computed by that method in Example 8.18 and Corollary 8.20.

Another method will be fully developed in Chapter 9, but here we announce a part of the result in Chapter 9, which is a closed form of $F(\mathbf{t},\mathbf{y};\Delta)$ for more general cases. To state this result, we need several notations.

Let $\mathscr{V}=\mathscr{V}(\Delta_+)$ be the set of all linearly independent subsets $\mathbf{V}=\{\beta_1,\dots,\beta_r\}\subset\Delta_+$ and let $L(\mathbf{V}^\vee)=\bigoplus_{\beta\in\mathbf{V}}\mathbb{Z}\beta^\vee$. For $\mathbf{V}\in\mathscr{V}$, let $\{\mu_\beta^{\mathbf{V}}\}$ be the dual basis of $\mathbf{V}^\vee=\{\beta^\vee\}$. Let $\mathscr{R}$ be the set of all linearly independent subsets $\mathbf{R}=\{\beta_1,\dots,\beta_{r-1}\}\subset\Delta$, $\mathfrak{H}_{\mathbf{R}^\vee}=\bigoplus_{i=1}^{r-1}\mathbb{R}\,\beta_i^\vee$ the hyperplane passing through $\mathbf{R}^\vee\cup\{0\}$.

We fix a non-zero vector

$$\phi_0\in V\setminus\bigcup_{\mathbf{R}\in\mathscr{R}}\mathfrak{H}_{\mathbf{R}^\vee}, \tag{4.3.1}$$

so that $\langle\phi_0,\mu_\beta^{\mathbf{V}}\rangle\neq 0$ for all $\mathbf{V}\in\mathscr{V}$ and $\beta\in\mathbf{V}$. For $\mathbf{y}\in V$, $\mathbf{V}\in\mathscr{V}$ and $\beta\in\mathbf{V}$, we define[4]

$$\{\mathbf{y}\}_{\mathbf{V},\beta}=\begin{cases}\{\langle\mathbf{y},\mu_\beta^{\mathbf{V}}\rangle\}, & (\langle\phi_0,\mu_\beta^{\mathbf{V}}\rangle>0),\\ 1-\{-\langle\mathbf{y},\mu_\beta^{\mathbf{V}}\rangle\}, & (\langle\phi_0,\mu_\beta^{\mathbf{V}}\rangle<0),\end{cases}\tag{4.3.2}$$

where $\{\cdot\}$ on the right-hand sides denotes the usual fractional part of a real number. This is a generalization of the notion of "fractional parts" of real numbers.

Denote by $\mathrm{Lh}_K[A]$ the linear hull (linear span) of a set A with coefficients in K. Writing $\mathbf{y}=\sum_{i=1}^r y_i\alpha_i^\vee$, we see that

$$\{\mathbf{y}\}_{\mathbf{V},\beta}\in\mathrm{Lh}_{\mathbb{Q}}[1,y_1,\dots,y_r]=\mathrm{Lh}_{\mathbb{Q}}[1,\{y_1\},\dots,\{y_r\}]\tag{4.3.3}$$

holds. In fact, since $\beta\in\mathbf{V}$ is a linear combination of $\{\alpha_j^\vee\}_{j=1}^r$ with coefficients in $\mathbb{Z}$, its dual $\mu_\beta^{\mathbf{V}}$ is a linear combination of $\{\lambda_j\}_{j=1}^r$ with coefficients in $\mathbb{Q}$. Therefore $\langle\mathbf{y},\mu_\beta^{\mathbf{V}}\rangle$ is a linear combination of $\{y_i\}_{i=1}^r$ with coefficients in $\mathbb{Q}$, which implies the claim.

In the definition (4.3.2), the sign of $\langle\phi_0,\mu_\beta^{\mathbf{V}}\rangle$ is important only when $\langle\mathbf{y},\mu_\beta^{\mathbf{V}}\rangle\in\mathbb{Z}$, because

$$\{x\}=1-\{-x\}\qquad(x\in\mathbb{R}\setminus\mathbb{Z}).\tag{4.3.4}$$

Note that $\{x\}$ is right continuous, while $1-\{-x\}$ is left continuous. We see that $\{\mathbf{0}\}_{\mathbf{V},\beta}=0$ or 1 according as $\langle\phi_0,\mu_\beta^{\mathbf{V}}\rangle$ is positive, or negative.

THEOREM 4.12 ([105, Theorem 4.1]). *Assume that Δ is irreducible. Fix a non-zero vector $\phi_0\in V$ satisfying (4.3.1). In particular, if Δ is of type A_1, then choose $\phi_0=c\alpha_1^\vee$ with $c>0$. We have*

(4.3.5)

$$F(\mathbf{t},\mathbf{y};\Delta)=\sum_{\mathbf{V}\in\mathscr{V}}\Bigl(\prod_{\gamma\in\Delta_+\setminus\mathbf{V}}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle}\Bigr)\times\frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))}\sum_{q\in Q^\vee/L(\mathbf{V}^\vee)}\Bigl(\prod_{\beta\in\mathbf{V}}\frac{t_\beta\exp(t_\beta\{\mathbf{y}+q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1}\Bigr).$$

We postpone the proof of this theorem to Chapter 9 and Chapter 16 (see Theorem 9.12).

REMARK 4.13. If Δ is not of type A_1, then the right-hand side of (4.3.5) is independent of the choice of ϕ_0 satisfying (4.3.1). This is a property inherited from its generalizations (see Theorems 9.10, 9.12 and 16.4). On the other hand, in the A_1 case, only the choice $\phi_0=c\alpha_1^\vee$

[4]This notion first appeared in [101].

with $c > 0$ is valid because this choice yields (4.2.12), while the choice with $c < 0$ results in

$$\frac{te^{t(1-\{-y\})}}{e^t - 1},$$

which does not coincide at $y \in \mathbb{Z}$.

REMARK 4.14. The assumption of the irreducibility of Δ is not essential. Since a reducible root system is decomposed into a direct sum of some irreducible root systems, this assumption can be removed by (3.1.8) and the argument below it; The generating function is the product of those for their irreducible components.

REMARK 4.15. When $\mathbf{t}$ satisfies $t_\gamma - \sum_{\beta\in\mathbf{V}} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle = 0$ for some γ, the corresponding denominator on the right-hand side of (4.3.5) is zero. However, in view of Lemma 4.9, we see that this does not give a singularity set for $F(\mathbf{t}, \mathbf{y}; \Delta)$ near the origin.

We conclude this chapter by showing further examples, which are computed by using Theorem 4.12.

EXAMPLE 4.16 ([**105**, Example 1]). The set of positive roots of type $C_2 (\simeq B_2)$ consists of α_1, α_2, $2\alpha_1 + \alpha_2$ and $\alpha_1 + \alpha_2$ and that of the corresponding coroots, $\alpha_1^\vee$, $\alpha_2^\vee$, $\alpha_1^\vee + \alpha_2^\vee$ and $\alpha_1^\vee + 2\alpha_2^\vee$. Let $t_1 = t_{\alpha_1}$, $t_2 = t_{\alpha_2}$, $t_3 = t_{2\alpha_1+\alpha_2}$ and $t_4 = t_{\alpha_1+\alpha_2}$. Let $\phi_0 = \alpha_1^\vee + \varepsilon\alpha_2^\vee$ where $\varepsilon > 0$ is sufficiently small. We have

$$\begin{aligned}\mathscr{V} = \{&\{\alpha_1, \alpha_2\}, \{\alpha_1, 2\alpha_1 + \alpha_2\}, \{\alpha_1, \alpha_1 + \alpha_2\},\\ &\{\alpha_2, 2\alpha_1 + \alpha_2\}, \{\alpha_2, \alpha_1 + \alpha_2\}, \{2\alpha_1 + \alpha_2, \alpha_1 + \alpha_2\}\}.\end{aligned}$$

Since their corresponding coroots are

$$\begin{aligned}&\{\{\alpha_1^\vee, \alpha_2^\vee\}, \{\alpha_1^\vee, \alpha_1^\vee + \alpha_2^\vee\}, \{\alpha_1^\vee, \alpha_1^\vee + 2\alpha_2^\vee\},\\ &\{\alpha_2^\vee, \alpha_1^\vee + \alpha_2^\vee\}, \{\alpha_2^\vee, \alpha_1^\vee + 2\alpha_2^\vee\}, \{\alpha_1^\vee + \alpha_2^\vee, \alpha_1^\vee + 2\alpha_2^\vee\}\},\end{aligned}$$

we have the corresponding dual basis as

$$\begin{aligned}&\{\{\lambda_1, \lambda_2\}, \{\lambda_1 - \lambda_2, \lambda_2\}, \{\lambda_1 - \tfrac{1}{2}\lambda_2, \tfrac{1}{2}\lambda_2\},\\ &\{-\lambda_1 + \lambda_2, \lambda_1\}, \{-2\lambda_1 + \lambda_2, \lambda_1\}, \{2\lambda_1 - \lambda_2, -\lambda_1 + \lambda_2\}\}\end{aligned}$$

and $\#(Q^\vee/L(\mathbf{V}^\vee))$ as

$$1, \quad 1, \quad 2, \quad 1, \quad 1, \quad 1.$$

Thus we have

$$\begin{aligned}F(\mathbf{t}, \mathbf{y}; C_2) &= t_1 t_2 t_3 t_4\\ &\times \Big(\frac{e^{\{y_1\}t_1 + \{y_2\}t_2}}{(e^{t_1} - 1)(e^{t_2} - 1)(t_1 + t_2 - t_3)(t_1 + 2t_2 - t_4)}\end{aligned} \tag{4.3.6}$$

$$
\begin{aligned}
&+\frac{e^{\{y_1-y_2\}t_1+\{y_2\}t_3}}{(e^{t_1}-1)(e^{t_3}-1)(t_1+t_2-t_3)(t_1-2t_3+t_4)}\\
&-\frac{2(e^{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}t_1+\{\frac{y_2}{2}+\frac{1}{2}\}t_4}+e^{\{y_1-\frac{y_2}{2}\}t_1+\{\frac{y_2}{2}\}t_4})}{(e^{t_1}-1)(e^{t_4}-1)(t_1+2t_2-t_4)(t_1-2t_3+t_4)}\\
&-\frac{e^{(1-\{y_1-y_2\})t_2+\{y_1\}t_3}}{(e^{t_2}-1)(e^{t_3}-1)(t_1+t_2-t_3)(t_2+t_3-t_4)}\\
&+\frac{e^{(1-\{2y_1-y_2\})t_2+\{y_1\}t_4}}{(e^{t_2}-1)(e^{t_4}-1)(t_1+2t_2-t_4)(t_2+t_3-t_4)}\\
&+\frac{e^{\{2y_1-y_2\}t_3+(1-\{y_1-y_2\})t_4}}{(e^{t_3}-1)(e^{t_4}-1)(t_2+t_3-t_4)(t_1-2t_3+t_4)}\Big).
\end{aligned}
$$

When $\mathbf{y}=\mathbf{0}$, we expand the above formula to obtain

(4.3.7)

$$
\begin{aligned}
F(\mathbf{t};C_2)=1&+\frac{1}{2880}(2t_1t_2t_3^2-4t_1t_2t_4^2-2t_1t_2^2t_3+4t_1t_2^2t_4-4t_1t_3t_4^2-4t_1t_3^2t_4\\
&\quad+t_1^2t_2t_3-4t_1^2t_2t_4-4t_1^2t_3t_4-t_2t_3t_4^2-2t_2t_3^2t_4-2t_2^2t_3t_4)\\
&+\frac{1}{241920}(3t_1t_2t_3^2t_4^2-3t_1t_2^2t_3t_4^2-3t_1^2t_2t_3^2t_4-3t_1^2t_2^2t_3t_4+2t_1^2t_2^2t_3^2\\
&\quad+8t_1^2t_2^2t_4^2+8t_1^2t_3^2t_4^2+2t_2^2t_3^2t_4^2)\\
&+\frac{1}{9676800}t_1^2t_2^2t_3^2t_4^2+\cdots,
\end{aligned}
$$

from which we find

$$
\zeta_2(2,2,2,2;C_2)=\frac{1}{302400}\pi^8, \tag{4.3.8}
$$

which was first given in [**100**, (3.9)] (see also Example 8.30). Several values of $B((k_1,k_2,k_3,k_4);C_2)$ are given in [**113**, Tables 11.2, 11.3 and 11.4].

Similarly, we can further obtain

$$
\begin{aligned}
\zeta_2(4,4,4,4;C_2)&=\frac{479}{55576160640000}\pi^{16},\\
\zeta_2(6,6,6,6;C_2)&=\frac{830287}{33082502670541868544000}\pi^{24},\\
\zeta_2(8,8,8,8;C_2)&=\frac{15426208283}{2100184151457702642748416000000000}\pi^{32},
\end{aligned}
$$

and also

$$
\begin{aligned}
\zeta_2(2,4,4,2;C_2)&=\frac{53}{6810804000}\pi^{12},\\
\zeta_2(4,2,2,4;C_2)&=\frac{373}{108972864000}\pi^{12},\\
\zeta_2(2,6,6,2;C_2)&=\frac{47}{2404737720000}\pi^{16},
\end{aligned}
$$

$$\zeta_2(6,2,2,6;C_2)=\frac{15343}{4001483566080000}\pi^{16},$$
$$\zeta_2(2,8,8,2;C_2)=\frac{271}{5444726987700000}\pi^{20},$$
$$\zeta_2(4,6,6,4;C_2)=\frac{17279}{784040686228800000}\pi^{20},$$
$$\zeta_2(6,4,4,6;C_2)=\frac{3611}{368960322931200000}\pi^{20},$$
$$\zeta_2(8,2,2,8;C_2)=\frac{291293}{66904805224857600000}\pi^{20}.$$

EXAMPLE 4.17 ([110, Section 2]). The set of positive roots of type G_2 consists of α_1, α_2, $3\alpha_1+\alpha_2$, $3\alpha_1+2\alpha_2$, $\alpha_1+\alpha_2$ and $2\alpha_1+\alpha_2$. Let $t_1=t_{\alpha_1}$, $t_2=t_{\alpha_2}$, $t_3=t_{3\alpha_1+\alpha_2}$, $t_4=t_{3\alpha_1+2\alpha_2}$, $t_5=t_{\alpha_1+\alpha_2}$ and $t_6=t_{2\alpha_1+\alpha_2}$ (see (3.3.26) and (3.3.27)). Let $\phi_0=\alpha_1^\vee+\varepsilon\alpha_2^\vee$ where $\varepsilon>0$ are chosen suitably to verify (4.3.1); that is, take ε much smaller than 1. Then we have

$$\begin{aligned}
&F(\mathbf{t},\mathbf{y};G_2)=t_1t_2t_3t_4t_5t_6\\
&\times\Bigg(\frac{e^{\{y_1\}t_1+\{y_2\}t_2}}{(e^{t_1}-1)(e^{t_2}-1)(t_1+t_2-t_3)(t_1+2t_2-t_4)(t_1+3t_2-t_5)(2t_1+3t_2-t_6)}\\
&+\frac{e^{\{y_1-y_2\}t_1+\{y_2\}t_3}}{(e^{t_1}-1)(e^{t_3}-1)(t_1+t_2-t_3)(t_1-2t_3+t_4)(2t_1-3t_3+t_5)(t_1-3t_3+t_6)}\\
&+\frac{8(e^{\{y_1-\frac{y_2}{2}\}t_1+\{\frac{y_2}{2}\}t_4}+e^{\{y_1-\frac{y_2}{2}+\frac12\}t_1+\{\frac{y_2+1}{2}\}t_4})}{(e^{t_1}-1)(e^{t_4}-1)(t_1+2t_2-t_4)(t_1-2t_3+t_4)(t_1-3t_4+2t_5)(t_1+3t_4-2t_6)}\\
&-\frac{9(e^{\{y_1-\frac{y_2}{3}\}t_1+\{\frac{y_2}{3}\}t_5}+e^{\{y_1-\frac{y_2}{3}+\frac23\}t_1+\{\frac{y_2+1}{3}\}t_5}+e^{\{y_1-\frac{y_2}{3}+\frac43\}t_1+\{\frac{y_2+2}{3}\}t_5})}{(e^{t_1}-1)(e^{t_5}-1)(t_1+3t_2-t_5)(2t_1-3t_3+t_5)(t_1-3t_4+2t_5)(t_1+t_5-t_6)}\\
&-\frac{9(e^{\{y_1-\frac{2y_2}{3}\}t_1+\{\frac{y_2}{3}\}t_6}+e^{\{y_1-\frac{2y_2}{3}+\frac13\}t_1+\{\frac{y_2+1}{3}\}t_6}+e^{\{y_1-\frac{2y_2}{3}+\frac23\}t_1+\{\frac{y_2+2}{3}\}t_6})}{(e^{t_1}-1)(e^{t_6}-1)(t_1+3t_4-2t_6)(2t_1+3t_2-t_6)(t_1+t_5-t_6)(t_1-3t_3+t_6)}\\
&-\frac{e^{-\{y_1-y_2\}t_2+t_2+\{y_1\}t_3}}{(e^{t_2}-1)(e^{t_3}-1)(t_1+t_2-t_3)(t_2+t_3-t_4)(2t_2+t_3-t_5)(t_2+2t_3-t_6)}\\
&-\frac{e^{-\{2y_1-y_2\}t_2+t_2+\{y_1\}t_4}}{(e^{t_2}-1)(e^{t_4}-1)(t_1+2t_2-t_4)(t_2+t_3-t_4)(t_2+t_4-t_5)(t_2-2t_4+t_6)}\\
&+\frac{e^{-\{3y_1-y_2\}t_2+t_2+\{y_1\}t_5}}{(e^{t_2}-1)(e^{t_5}-1)(t_1+3t_2-t_5)(2t_2+t_3-t_5)(t_2+t_4-t_5)(3t_2-2t_5+t_6)}\\
&+\frac{8(e^{-\{\frac{3y_1}{2}-y_2\}t_2+t_2+\{\frac{y_1}{2}\}t_6}+e^{\{-\frac{3y_1}{2}+y_2+\frac12\}t_2+\{\frac{y_1+1}{2}\}t_6})}{(e^{t_2}-1)(e^{t_6}-1)(2t_1+3t_2-t_6)(t_2+2t_3-t_6)(t_2-2t_4+t_6)(3t_2-2t_5+t_6)}\\
&+\frac{e^{\{2y_1-y_2\}t_3-\{y_1-y_2\}t_4+t_4}}{(e^{t_3}-1)(e^{t_4}-1)(t_2+t_3-t_4)(t_1-2t_3+t_4)(t_3-2t_4+t_5)(-t_3-t_4+t_6)}
\end{aligned}$$

$$
\begin{aligned}
&-\frac{8e^{\left\{\frac{3y_1}{2}-\frac{y_2}{2}\right\}t_3-(\left\{\frac{y_1-y_2}{2}\right\}-1)t_5}}{(e^{t_3}-1)(e^{t_5}-1)(2t_2+t_3-t_5)(-2t_1+3t_3-t_5)(t_3-2t_4+t_5)(3t_3+t_5-2t_6)}\\
&+\frac{e^{\{3y_1-2y_2\}t_3-\{y_1-y_2\}t_6+t_6}}{(e^{t_3}-1)(e^{t_6}-1)(3t_3+t_5-2t_6)(t_2+2t_3-t_6)(t_3+t_4-t_6)(t_1-3t_3+t_6)}\\
&+\frac{e^{\{3y_1-y_2\}t_4-\{2y_1-y_2\}t_5+t_5}}{(e^{t_4}-1)(e^{t_5}-1)(t_2+t_4-t_5)(t_3-2t_4+t_5)(t_1-3t_4+2t_5)(-3t_4+t_5+t_6)}\\
&+\frac{e^{-\{3y_1-2y_2\}t_4+t_4+\{2y_1-y_2\}t_6}}{(e^{t_4}-1)(e^{t_6}-1)(t_1+3t_4-2t_6)(t_3+t_4-t_6)(t_2-2t_4+t_6)(-3t_4+t_5+t_6)}\\
&+\frac{9}{(e^{t_5}-1)(e^{t_6}-1)(3t_3+t_5-2t_6)(t_1+t_5-t_6)(-3t_2+2t_5-t_6)(-3t_4+t_5+t_6)}\\
&\times\big(e^{-\left\{y_1-\frac{2y_2}{3}\right\}t_5+t_5+\left\{y_1-\frac{y_2}{3}\right\}t_6}+e^{\left\{-y_1+\frac{2y_2}{3}+\frac{2}{3}\right\}t_5+\left\{y_1-\frac{y_2}{3}+\frac{2}{3}\right\}t_6}\\
&+e^{\left\{-y_1+\frac{2y_2}{3}+\frac{4}{3}\right\}t_5+\left\{y_1-\frac{y_2}{3}+\frac{4}{3}\right\}t_6}\big)\Big),
\end{aligned}
$$

from which we can obtain

$$
\begin{aligned}
\zeta_2(2,2,2,2,2,2;G_2)&=\frac{23}{297904566960}\pi^{12},\\
\zeta_2(4,4,4,4,4,4;G_2)&=\frac{8165653}{144583867612955930599440000 0}\pi^{24},\\
\zeta_2(6,6,6,6,6,6;G_2)&=\frac{55940539974690617}{131888156302530666544150214880458495963616000000}\pi^{36},\\
\zeta_2(8,8,8,8,8,8;G_2)&\\
&=\frac{47346365461279256768015189}{1485697621623958244738368714652675148113575302412190275200000000000}\pi^{48},
\end{aligned}
$$

which were also mentioned in Zhao [**219**]. Further we have

$$
\begin{aligned}
\zeta_2(2,4,4,4,2,2;G_2)&=\frac{467}{213955059990672000}\pi^{18},\\
\zeta_2(4,2,2,2,4,4;G_2)&=\frac{20771}{106061802338575923840}\pi^{18},\\
\zeta_2(2,6,6,6,2,2;G_2)&=\frac{91027}{144934762300631142840000 0}\pi^{24},\\
\zeta_2(6,2,2,2,6,6;G_2)&=\frac{391420483}{770242750118097151820324400000}\pi^{24},\\
\zeta_2(2,8,8,8,2,2;G_2)&=\frac{19152444887}{105645584604256288496564259600000000}\pi^{30},\\
\zeta_2(8,2,2,2,8,8;G_2)&=\frac{1802533972626341}{136430880160239475902213334247183148000000 0}\pi^{30}.
\end{aligned}
$$

Example 4.18 ([**105**, Example 2]). The set of positive roots of type A_3 consists of α_1, α_2, α_3, $\alpha_1+\alpha_2$, $\alpha_2+\alpha_3$ and $\alpha_1+\alpha_2+\alpha_3$. Let $t_1=t_{\alpha_1}$, $t_2=t_{\alpha_2}$, $t_3=t_{\alpha_3}$, $t_4=t_{\alpha_1+\alpha_2}$, $t_5=t_{\alpha_2+\alpha_3}$ and $t_6=t_{\alpha_1+\alpha_2+\alpha_3}$. Let $\phi_0=\alpha_1^\vee+\varepsilon\alpha_2^\vee+\varepsilon'\alpha_3^\vee$ where $\varepsilon,\varepsilon'>0$ are chosen suitably to verify

(4.3.1); that is, take ε much smaller than 1, and take ε' much smaller than ε.

(4.3.9)

$$
\begin{aligned}
F(\mathbf{t},\mathbf{y};A_3) &= t_1t_2t_3t_4t_5t_6 \\
&\times \Bigg(\frac{e^{\{y_1\}t_1+\{y_2\}t_2+\{y_3\}t_3}}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_3}-1)(t_1+t_2-t_4)(t_2+t_3-t_5)(t_1+t_2+t_3-t_6)} \\
&- \frac{e^{\{y_1\}t_1+\{y_2-y_3\}t_2+\{y_3\}t_5}}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_5}-1)(t_1+t_2-t_4)(t_2+t_3-t_5)(t_1+t_5-t_6)} \\
&+ \frac{e^{\{y_1-y_3\}t_1+\{y_2-y_3\}t_2+\{y_3\}t_6}}{(e^{t_1}-1)(e^{t_2}-1)(e^{t_6}-1)(t_1+t_2-t_4)(t_1+t_2+t_3-t_6)(t_1+t_5-t_6)} \\
&+ \frac{e^{\{y_1-y_2\}t_1+\{y_3\}t_3+\{y_2\}t_4}}{(e^{t_1}-1)(e^{t_3}-1)(e^{t_4}-1)(t_1+t_2-t_4)(t_1-t_3-t_4+t_5)(t_3+t_4-t_6)} \\
&- \frac{e^{\{y_1\}t_1+(1-\{y_2-y_3\})t_3+\{y_2\}t_5}}{(e^{t_1}-1)(e^{t_3}-1)(e^{t_5}-1)(t_2+t_3-t_5)(t_1-t_3-t_4+t_5)(t_1+t_5-t_6)} \\
&- \frac{e^{\{y_1-y_2\}t_1+(1-\{y_2-y_3\})t_3+\{y_2\}t_6}}{(e^{t_1}-1)(e^{t_3}-1)(e^{t_6}-1)(t_1+t_2+t_3-t_6)(t_3+t_4-t_6)(t_1+t_5-t_6)} \\
&+ \frac{e^{\{y_1-y_2+y_3\}t_1+\{y_2-y_3\}t_4+\{y_3\}t_5}}{(e^{t_1}-1)(e^{t_4}-1)(e^{t_5}-1)(t_1+t_2-t_4)(t_1-t_3-t_4+t_5)(t_1+t_5-t_6)} \\
&- \frac{e^{\{y_1-y_2\}t_1+\{y_2-y_3\}t_4+\{y_3\}t_6}}{(e^{t_1}-1)(e^{t_4}-1)(e^{t_6}-1)(t_1+t_2-t_4)(t_3+t_4-t_6)(t_1+t_5-t_6)} \\
&+ \frac{e^{(1-\{y_1-y_2\})t_2+\{y_3\}t_3+\{y_1\}t_4}}{(e^{t_2}-1)(e^{t_3}-1)(e^{t_4}-1)(t_1+t_2-t_4)(t_2+t_3-t_5)(t_3+t_4-t_6)} \\
&+ \frac{e^{(1-\{y_1-y_2\})t_2+(1-\{y_1-y_3\})t_3+\{y_1\}t_6}}{(e^{t_2}-1)(e^{t_3}-1)(e^{t_6}-1)(t_2+t_3-t_5)(t_1+t_2+t_3-t_6)(t_3+t_4-t_6)} \\
&- \frac{e^{(1-\{y_1-y_2+y_3\})t_2+\{y_1\}t_4+\{y_3\}t_5}}{(e^{t_2}-1)(e^{t_4}-1)(e^{t_5}-1)(t_1+t_2-t_4)(t_2+t_3-t_5)(t_2-t_4-t_5+t_6)} \\
&+ \frac{e^{(1-\{y_1-y_2\})t_2+\{y_1-y_3\}t_4+\{y_3\}t_6}}{(e^{t_2}-1)(e^{t_4}-1)(e^{t_6}-1)(t_1+t_2-t_4)(t_3+t_4-t_6)(t_2-t_4-t_5+t_6)} \\
&+ \frac{e^{\{y_2-y_3\}t_2+(1-\{y_1-y_3\})t_5+\{y_1\}t_6}}{(e^{t_2}-1)(e^{t_5}-1)(e^{t_6}-1)(t_2+t_3-t_5)(t_1+t_5-t_6)(t_2-t_4-t_5+t_6)} \\
&+ \frac{e^{\{y_1-y_2+y_3\}t_3+\{y_1\}t_4+(1-\{y_1-y_2\})t_5}}{(e^{t_3}-1)(e^{t_4}-1)(e^{t_5}-1)(t_2+t_3-t_5)(t_1-t_3-t_4+t_5)(t_3+t_4-t_6)} \\
&- \frac{e^{(1-\{y_2-y_3\})t_3+(1-\{y_1-y_2\})t_5+\{y_1\}t_6}}{(e^{t_3}-1)(e^{t_5}-1)(e^{t_6}-1)(t_2+t_3-t_5)(t_3+t_4-t_6)(t_1+t_5-t_6)} \\
&- \frac{e^{\{y_2-y_3\}t_4+(1-\{y_1-y_2\})t_5+\{y_1-y_2+y_3\}t_6}}{(e^{t_4}-1)(e^{t_5}-1)(e^{t_6}-1)(t_3+t_4-t_6)(t_1+t_5-t_6)(t_2-t_4-t_5+t_6)} \Bigg),
\end{aligned}
$$

from which we can obtain

$$\zeta_3(2,2,2,2,2,2;A_3)=\frac{23}{2554051500}\pi^{12},$$
$$\zeta_3(4,4,4,4,4,4;A_3)=\frac{14081}{247248455465824312500}\pi^{24},$$
$$\zeta_3(6,6,6,6,6,6;A_3)=\frac{17634884778757}{4155595512373317717297525732408984375 0}\pi^{36},$$
$$\zeta_3(8,8,8,8,8,8;A_3)=\frac{425746382998959473}{13359226211510454534096919557273732895634501953125 00}\pi^{48},$$

which were first given by Gunnells and Sczech [**53**] (see Remark 4.8). See also Subsection 9.8.2.

Example 4.19. As further examples, we gave

$$\zeta_3(2,2,\dots,2;B_3)=\frac{19}{8403115488768000}\pi^{18},$$
$$\zeta_3(2,2,\dots,2;C_3)=\frac{19}{8403115488768000}\pi^{18},$$
$$\zeta_3(2,2,\dots,2;A_4)=\frac{1}{650970015609375}\pi^{20}$$

in [**103**]. We can compute further:

$$\zeta_3(4,4,\dots,4;B_3)$$
$$=\frac{82197390067823}{17429830879929455994931881369545932800000000}\pi^{36},$$
$$\zeta_3(6,6,\dots,6;B_3)$$
$$=\frac{4942776168078403026588037}{48311852818581124252892258054533388775563698764117481881600000000000 0}\pi^{54},$$

and

$$\zeta_3(4,4,\dots,4;C_3)$$
$$=\frac{82197390067823}{17429830879929455994931881369545932800000000}\pi^{36},$$
$$\zeta_3(6,6,\dots,6;C_3)$$
$$=\frac{4942776168078403026588037}{48311852818581124252892258054533388775563698764117481881600000000000 0}\pi^{54},$$

and further

$$\zeta_4(4,4,\dots,4;A_4)$$
$$=\frac{3998447009863}{21084452730904120768138321184810084472656 25}\pi^{40},$$
$$\zeta_4(6,6,\dots,6;A_4)$$
$$=\frac{12779458807351913182141540335 4}{49181663794104443979232802480706833748465053469684541875006283721923828125}\pi^{60}.$$

These values were first published by Zhao [219]. As stated in Remark 4.8, his method is based on the technique in Gunnells and Sczech [53].



CHAPTER 5

Convex polytopes and the rationality

The theory of convex polytopes is another important tool which is necessary to study zeta-functions of root systems. Properties of convex polytopes are used to prove the rationality of Bernoulli polynomials of root systems $B(\mathbf{k};\Delta)$ (Theorem 4.6 stated in Chapter 4), and are also necessary to develop the theory of lattice sums of hyperplane arrangements (Chapter 16). In this chapter we first prove several fundamental properties of convex polytopes, and then show how to prove Theorem 4.6.

5.1. Convex polytopes

In this section we give some definitions and facts about convex polytopes (see [**56**] [**223**]) and their triangulations. For a subset $X \subset \mathbb{R}^N$, we denote by $\mathrm{Conv}(X)$ the convex hull of X. A subset $\mathcal{P} \subset \mathbb{R}^N$ is called a *convex polytope* if $\mathcal{P} = \mathrm{Conv}(X)$ for some finite subset $X \subset \mathbb{R}^N$. Let $\mathcal{P}$ be a d-dimensional polytope. Let $\mathcal{H}$ be a hyperplane in $\mathbb{R}^N$. Then $\mathcal{H}$ divides $\mathbb{R}^N$ into two half-spaces. If $\mathcal{P}$ is entirely contained in one of the two closed half-spaces and $\mathcal{P} \cap \mathcal{H} \neq \emptyset$, then $\mathcal{H}$ is called a *supporting hyperplane*. For a supporting hyperplane $\mathcal{H}$, a subset $\mathcal{F} = \mathcal{P} \cap \mathcal{H} \neq \emptyset$ is called a *face* of the polytope $\mathcal{P}$. If the dimension of a face $\mathcal{F}$ is j, then we call it a j-face $\mathcal{F}$. A 0-face is called a *vertex* and a $(d-1)$-face a *facet*. For convenience, we regard $\mathcal{P}$ itself as its unique d-face. Let $\mathrm{Vert}(\mathcal{P})$ be the set of all vertices of $\mathcal{P}$. Then

$$\mathcal{F} = \mathrm{Conv}(\mathrm{Vert}(\mathcal{P}) \cap \mathcal{F}), \tag{5.1.1}$$

for a face $\mathcal{F}$. A d-dimensional *simple* polytope is a polytope whose vertices are adjacent to exactly d edges.

A *triangulation* of a polytope is a partition of it into simplexes that intersect each other in entire faces. It is known that a convex polytope can be triangulated without adding any vertices (see [**186**]). Here we give an explicit procedure of a triangulation of $\mathcal{P}$. Number all the vertices of $\mathcal{P}$ as $\mathbf{p}_1, \ldots, \mathbf{p}_k$. For a face $\mathcal{F}$, by $\mathcal{N}(\mathcal{F})$ we mean the vertex $\mathbf{p}_j$ whose index j is the smallest in the vertices belonging to $\mathcal{F}$. A *full*

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Y. Komori et al., *The Theory of Zeta-Functions of Root Systems*,
Springer Monographs in Mathematics, https://doi.org/10.1007/978-981-99-0910-0_5

flag Φ is defined by the sequence

$$\Phi : \mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}_{d-1} \subset \mathcal{F}_d = \mathcal{P}, \tag{5.1.2}$$

with j-faces $\mathcal{F}_j$ such that $\mathcal{N}(\mathcal{F}_j) \notin \mathcal{F}_{j-1}$.

THEOREM 5.1 ([186]). *All the collection of the simplexes with vertices* $\mathcal{N}(\mathcal{F}_0)$, $\ldots$, $\mathcal{N}(\mathcal{F}_{d-1})$, $\mathcal{N}(\mathcal{F}_d)$ *associated with full flags gives a triangulation.*

REMARK 5.2. This procedure only depends on the face poset structure of $\mathcal{P}$ (see [56, §5]).

For $\mathbf{a} = {}^t(a_1, \ldots, a_N), \mathbf{b} = {}^t(b_1, \ldots, b_N) \in \mathbb{C}^N$ we write $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + \cdots + a_N b_N$. The definition of polytopes above is that of "V-polytopes". We mainly deal with another representation of polytopes, "H-polytopes", instead. Namely, consider a bounded subset of the form

$$\mathcal{P} = \bigcap_{i \in I} \mathcal{H}_i^+ \subset \mathbb{R}^N, \tag{5.1.3}$$

where $\#I < \infty$ and $\mathcal{H}_i^+ = \{\mathbf{x} \in \mathbb{R}^N \mid \mathbf{a}_i \cdot \mathbf{x} \geq h_i\}$ with $\mathbf{a}_i \in \mathbb{R}^N$ and $h_i \in \mathbb{R}$. The following theorem is intuitively clear but nontrivial (see, for example, [223, Theorem 1.1]).

THEOREM 5.3 (Weyl–Minkowski). *H-polytopes are V-polytopes and vice versa.*

We have a representation of k-faces in terms of hyperplanes $\mathcal{H}_i = \{\mathbf{x} \in \mathbb{R}^N \mid \mathbf{a}_i \cdot \mathbf{x} = h_i\}$.

PROPOSITION 5.4 ([113, Proposition 1]). *Let* $J \subset I$. *Assume that* $\mathcal{F} = \mathcal{P} \cap \bigcap_{j \in J} \mathcal{H}_j \neq \emptyset$. *Then* $\mathcal{F}$ *is a face.*

PROOF. Let $\mathbf{x} \in \mathcal{P}$. Then $\mathbf{x} \in \bigcap_{j \in J} \mathcal{H}_j^+$ and hence $\mathbf{a}_j \cdot \mathbf{x} \geq h_j$ for all $j \in J$. Set $\mathbf{a} = \sum_{j \in J} \mathbf{a}_j$ and $h = \sum_{j \in J} h_j$. Let $\mathcal{H}$ be a hyperplane defined by $\{\mathbf{x} \in \mathbb{R}^N \mid \mathbf{a} \cdot \mathbf{x} = h\}$. Then $\mathbf{a} \cdot \mathbf{x} \geq h$ for $\mathbf{x} \in \mathcal{P}$ and $\mathcal{P} \subset \mathcal{H}^+ = \{\mathbf{x} \in \mathbb{R}^N \mid \mathbf{a} \cdot \mathbf{x} \geq h\}$.

Let $\mathbf{x} \in \mathcal{P} \cap \mathcal{H}$. Then $\mathbf{a} \cdot \mathbf{x} = h$. Since $\mathbf{a}_j \cdot \mathbf{x} = h_j + c_j \geq h_j$ with some $c_j \geq 0$ for all $j \in J$, we have $c_j = 0$ and $\mathbf{x} \in \mathcal{H}_j$ for all $j \in J$. Thus $\mathbf{x} \in \mathcal{F}$. It is easily seen that $\mathbf{x} \in \mathcal{F}$ satisfies $\mathbf{x} \in \mathcal{P} \cap \mathcal{H}$. Therefore $\mathcal{F} = \mathcal{P} \cap \mathcal{H}$ and $\mathcal{H}$ is a supporting hyperplane. □

PROPOSITION 5.5 ([113, Proposition 2]). *Let* $\mathcal{H}$ *be a supporting hyperplane and* $\mathcal{F} = \mathcal{P} \cap \mathcal{H}$ *a* k*-face. Then there exists a set of indices* $J \subset I$ *such that* $\#J = (\dim \mathcal{P}) - k$ *and* $\mathcal{F} = \mathcal{P} \cap \bigcap_{j \in J} \mathcal{H}_j$.

PROOF. We may assume $d = N$ without loss of generality. Let $\mathbf{x} \in \mathcal{F}$. Then $\mathbf{x} \in \mathcal{H}$ and hence $\mathbf{x} \in \partial\mathcal{P}$ since $\mathcal{P} \subset \mathcal{H}^+$. If $\mathbf{x} \in \mathcal{H}_i^+ \setminus \mathcal{H}_i$ for all $i \in I$, then $\mathbf{x}$ is in the interior of $\mathcal{P}$, which contradicts the above. Thus $\mathbf{x} \in \mathcal{H}_j$ for some $j \in I$.

First, we assume that $\mathcal{F} = \mathcal{P} \cap \mathcal{H}$ is a facet. Then there exists a subset $\{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathcal{F}$ such that $\mathbf{x}_2 - \mathbf{x}_1, \dots, \mathbf{x}_N - \mathbf{x}_1$ are linearly independent. Let $\mathcal{C}$ be the convex hull of $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$. We consider that $\mathcal{C} \subset \mathcal{F}$ is equipped with the relative topology. Note that for $\mathbf{x} \in \mathcal{C}$, we have $\mathbf{x} \in \mathcal{H}_{i(\mathbf{x})}$, where $i(\mathbf{x}) \in I$. We show that there exist an open subset $\mathcal{U} \subset \mathcal{C}$ and $i \in I$ such that $\mathcal{H}_i \cap \mathcal{U}$ is dense in $\mathcal{U}$. Fix an order $\{i_1, i_2, \dots\} = I$. If $\mathcal{H}_{i_1} \cap \mathcal{C}$ is dense in $\mathcal{C}$, then we are done. Hence we assume that it is false. Then there exists an open subset $\mathcal{U}_1 \subset \mathcal{C}$ such that $\mathcal{H}_{i_1} \cap \mathcal{U}_1 = \emptyset$. Similarly, we see that $\mathcal{H}_{i_2} \cap \mathcal{U}_1$ is dense in $\mathcal{U}_1$ unless there exists an open subset $\mathcal{U}_2 \subset \mathcal{U}_1$ such that $\mathcal{H}_{i_2} \cap \mathcal{U}_2 = \emptyset$. Since $\#I < \infty$, repeated application of this argument yields the assertion. Thus there exists a subset $\{\mathbf{x}'_1, \dots, \mathbf{x}'_N\} \subset \mathcal{H}_i \cap \mathcal{U}$ such that $\mathbf{x}'_2 - \mathbf{x}'_1, \dots, \mathbf{x}'_N - \mathbf{x}'_1$ are linearly independent. Hence we have $\mathcal{F} \subset \mathcal{H}_i$ and $\mathcal{H} = \mathcal{H}_i$.

For any k-face $\mathcal{F}$, there exists a sequence of faces such that

$$\mathcal{F} = \mathcal{F}_k \subset \mathcal{F}_{k+1} \subset \cdots \subset \mathcal{F}_{N-1} \subset \mathcal{P}, \tag{5.1.4}$$

where $\mathcal{F}_j$ $(k \le j \le N-1)$ is a j-face. Since $\mathcal{F}_j$ is a facet of $\mathcal{F}_{j+1}$, by the induction on dimensions, we have $\mathcal{F} = \mathcal{P} \cap \bigcap_{j \in J} \mathcal{H}_j$ for some $J \subset I$ with $\#J = N - k$. □

LEMMA 5.6 ([113, Lemma 4]). *Let $\mathbf{a} = {}^t(a_1, \dots, a_N) \in \mathbb{C}^N$ and σ be a simplex with vertices $\mathbf{p}_0, \dots, \mathbf{p}_N \in \mathbb{R}^N$ in general position. Then*

$$\begin{aligned} \int_\sigma e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} &= N! \operatorname{Vol}(\sigma) \sum_{m=0}^{N} \frac{e^{\mathbf{a}\cdot\mathbf{p}_m}}{\prod_{j \ne m} \mathbf{a}\cdot(\mathbf{p}_m - \mathbf{p}_j)} \\ &= N! \operatorname{Vol}(\sigma) T(\mathbf{a}\cdot\mathbf{p}_0, \dots, \mathbf{a}\cdot\mathbf{p}_N), \end{aligned} \tag{5.1.5}$$

where

(5.1.6)

$$T(x_0, \dots, x_N) = \det \begin{pmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_N \\ \vdots & \ddots & \vdots \\ x_0^{N-1} & \cdots & x_N^{N-1} \\ e^{x_0} & \cdots & e^{x_N} \end{pmatrix} \Big/ \det \begin{pmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_N \\ \vdots & \ddots & \vdots \\ x_0^{N-1} & \cdots & x_N^{N-1} \\ x_0^{N} & \cdots & x_N^{N} \end{pmatrix},$$

$$\operatorname{Vol}(\sigma) = \int_\sigma 1 d\mathbf{x} = \frac{1}{N!} |\det \tilde{P}|, \tag{5.1.7}$$

and

$$\tilde{P} = \begin{pmatrix} 1 & \cdots & 1 \\ \mathbf{p}_0 & \cdots & \mathbf{p}_N \end{pmatrix}, \tag{5.1.8}$$

with $\mathbf{p}_j$ *regarded as column vectors.*

PROOF. Let τ be the simplex whose vertices consist of the origin and ${}^t(0,\dots,\overset{j}{1},\dots,0)$, $1 \le j \le N$. Then by changing variables from $\mathbf{x}$ to $\mathbf{y}$ as $\mathbf{x} = \mathbf{p}_0 + P\mathbf{y}$ with the $N \times N$ matrix $P = (\mathbf{p}_1 - \mathbf{p}_0, \dots, \mathbf{p}_N - \mathbf{p}_0)$, we have

$$\int_\sigma e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} = e^{\mathbf{a}\cdot\mathbf{p}_0} |\det P| \int_\tau e^{\mathbf{b}\cdot\mathbf{y}} d\mathbf{y}, \tag{5.1.9}$$

where ${}^t\mathbf{b} = (b_1,\dots,b_N) = {}^t\mathbf{a}P$. Since

$$\int_0^{1-y_1-\cdots-y_{m-1}} e^{c+(b_1-c)y_1+\cdots+(b_m-c)y_m} dy_m = \tag{5.1.10}$$
$$\frac{1}{b_m - c}\left(e^{b_m+(b_1-b_m)y_1+\cdots+(b_{m-1}-b_m)y_{m-1}} - e^{c+(b_1-c)y_1+\cdots+(b_{m-1}-c)y_{m-1}}\right),$$

we see that

$$\int_\tau e^{\mathbf{b}\cdot\mathbf{y}} d\mathbf{y} = q_0(\mathbf{b}) + \sum_{m=1}^{N} q_m(\mathbf{b}) e^{b_m}, \tag{5.1.11}$$

where $q_j(\mathbf{b})$ for $0 \le j \le N$ are rational functions in $b_1,\dots,b_N$. In particular, we have

$$q_0(\mathbf{b}) = \frac{(-1)^N}{b_1 \cdots b_N} \tag{5.1.12}$$

and hence

$$\int_\sigma e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} = \frac{|\det(\mathbf{p}_0 - \mathbf{p}_1, \dots, \mathbf{p}_0 - \mathbf{p}_N)|}{\big(\mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{p}_1)\big)\cdots\big(\mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{p}_N)\big)} e^{\mathbf{a}\cdot\mathbf{p}_0} + \sum_{m=1}^{N} \tilde{q}_m(\mathbf{a}) e^{\mathbf{a}\cdot\mathbf{p}_m}, \tag{5.1.13}$$

where $\tilde{q}_m(\mathbf{a})$ are certain rational functions in $a_1,\dots,a_N$. Since $e^{\mathbf{a}\cdot\mathbf{p}_j}$ for $0 \le j \le N$ are linearly independent over the field of rational functions in $a_1,\dots,a_N$, exchanging the roles of the indices 0 and j in the change of variables in (5.1.9) yields

$$\begin{aligned} \int_\sigma e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} &= \sum_{m=0}^{N} \frac{|\det(\mathbf{p}_m - \mathbf{p}_0, \dots, \mathbf{p}_m - \mathbf{p}_{m-1}, \mathbf{p}_m - \mathbf{p}_{m+1}, \dots, \mathbf{p}_m - \mathbf{p}_N)|}{\prod_{j\ne m} \mathbf{a}\cdot(\mathbf{p}_m - \mathbf{p}_j)} e^{\mathbf{a}\cdot\mathbf{p}_m} \\ &= N!\,\mathrm{Vol}(\sigma) \sum_{m=0}^{N} \frac{e^{\mathbf{a}\cdot\mathbf{p}_m}}{\prod_{j\ne m} \mathbf{a}\cdot(\mathbf{p}_m - \mathbf{p}_j)}. \end{aligned} \tag{5.1.14}$$

Generally, we have

$$\sum_{m=0}^{N} \frac{e^{x_m}}{\prod_{j\neq m}(x_m - x_j)} = T(x_0, \dots, x_N) \tag{5.1.15}$$

by the Laplace expansion of the numerator of the right-hand side of (5.1.6) with respect to the last row and hence the result (5.1.5). □

LEMMA 5.7 ([**105**, Lemma 6.5]). *Let $\mathcal{P}$ be an N-dimensional polytope and $\{\mathbf{p}_0, \dots, \mathbf{p}_K\}$ be the vertices of $\mathcal{P}$. Let*

$$E_k = \{j \mid \mathrm{Conv}(\{\mathbf{p}_k, \mathbf{p}_j\}) \text{ is an edge of } \mathcal{P}\}. \tag{5.1.16}$$

Then

$$\int_{\mathcal{P}} e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} = \sum_{k=0}^{K} g_k(\{\mathbf{a}\cdot(\mathbf{p}_k - \mathbf{p}_j)\}_{j\in E_k}) e^{\mathbf{a}\cdot\mathbf{p}_k}, \tag{5.1.17}$$

where g_k is a rational function. In particular, if $\mathcal{P}$ is simple, i.e., $|E_k| = N$ for all k, then we have

$$\int_{\mathcal{P}} e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} = \sum_{k=0}^{K} |\det(\mathbf{p}_k - \mathbf{p}_j)_{j\in E_k}| \frac{e^{\mathbf{a}\cdot\mathbf{p}_k}}{\prod_{j\in E_k} \mathbf{a}\cdot(\mathbf{p}_k - \mathbf{p}_j)}. \tag{5.1.18}$$

REMARK 5.8. The formula (5.1.18) was already shown by Brion and Vergne [**26**] by a different method.

PROOF. Let $\mathcal{H} = \{\mathbf{x} \mid \mathbf{a}\cdot\mathbf{x} = b\}$ be a supporting hyperplane associated with the vertex $\mathbf{p}_0$ and assume that $\mathcal{P} \subset \mathcal{H}^+ = \{\mathbf{x} \mid \mathbf{a}\cdot\mathbf{x} \geq b\}$. There exists a sufficiently small $c > 0$ such that the hyperplane $\mathcal{H}_c = \{\mathbf{x} \mid \mathbf{a}\cdot\mathbf{x} = b + c\}$ divides $\mathcal{P}$ into two polytopes and that one of the two polytopes contains $\mathbf{p}_0$ and no other vertices of $\mathcal{P}$. Let $\mathcal{P}_0$ be the polytope containing $\mathbf{p}_0$ and $\mathcal{P}_1$ the other. Let $\{\mathbf{p}_0, \mathbf{q}_1, \dots, \mathbf{q}_L\}$ be the vertices of $\mathcal{P}_0$. Then $\{\mathbf{p}_1, \dots, \mathbf{p}_K, \mathbf{q}_1, \dots, \mathbf{q}_L\}$ is the vertices of $\mathcal{P}_1$. Regarding $\mathcal{P}$ as a union of simplexes and applying Lemma 5.6 to each of those simplexes, we have

(5.1.19)

$$\begin{aligned}\int_{\mathcal{P}} e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} &= f_0(\mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{p}_1), \dots, \mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{p}_K)) e^{\mathbf{a}\cdot\mathbf{p}_0} + \text{remainder} \\ &= \int_{\mathcal{P}_0} e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} + \int_{\mathcal{P}_1} e^{\mathbf{a}\cdot\mathbf{x}} d\mathbf{x} \\ &= g_0(\mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{q}_1), \dots, \mathbf{a}\cdot(\mathbf{p}_0 - \mathbf{q}_L)) e^{\mathbf{a}\cdot\mathbf{p}_0} + \text{remainder},\end{aligned}$$

where f_0 and g_0 are rational functions (see (5.1.13)). By the linear independence of $e^{\mathbf{a}\cdot\mathbf{q}_j}$'s and $e^{\mathbf{a}\cdot\mathbf{p}_j}$'s over the field of rational functions,

we have

(5.1.20)

$$f_0(\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_1),\dots,\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_K))=g_0(\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{q}_1),\dots,\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{q}_L)).$$

Since f_0 and g_0 do not depend on the lengths of $\mathbf{p}_k-\mathbf{p}_j$ and there exist $c>0$ and j such that $c(\mathbf{p}_0-\mathbf{q}_k)=\mathbf{p}_0-\mathbf{p}_j$, we have

(5.1.21)

$$f_0(\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_1),\dots,\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_K))=g_0(\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_{j_1}),\dots,\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_{j_L})),$$

where j_m runs over indices such that $\mathrm{Conv}(\{\mathbf{p}_0,\mathbf{p}_{j_m}\})$ is an edge of $\mathcal{P}$. In particular, if $|E_0|=N$, then $\mathcal{P}_0$ is a simplex, hence we see that g_0 is calculated as

$$|\det(\mathbf{p}_0-\mathbf{p}_j)_{j\in E_0}|\frac{e^{\mathbf{a}\cdot\mathbf{p}_0}}{\prod_{j\in E_0}\mathbf{a}\cdot(\mathbf{p}_0-\mathbf{p}_j)}. \tag{5.1.22}$$

□

Lemma 5.9 ([**113**, Lemma 5]). *Let $\mathbf{a}\in\mathbb{C}^N$, σ and $\mathbf{p}_0,\dots,\mathbf{p}_N\in\mathbb{R}^N$ are the same as in Lemma 5.6. Then the Taylor expansion with respect to $\mathbf{a}$ is given by*

$$\begin{aligned}\int_\sigma e^{\mathbf{a}\cdot\mathbf{x}}d\mathbf{x}=\mathrm{Vol}(\sigma)\Big(1+\frac{1}{N+1}\sum_{0\le i\le N}\mathbf{a}\cdot\mathbf{p}_i+\cdots\\ +\frac{N!}{(N+k)!}\sum_{\substack{k_0,\dots,k_N\ge 0\\ k_0+\cdots+k_N=k}}(\mathbf{a}\cdot\mathbf{p}_0)^{k_0}\cdots(\mathbf{a}\cdot\mathbf{p}_N)^{k_N}+\cdots\Big).\end{aligned} \tag{5.1.23}$$

Proof. We recall Dirichlet's integral (see [**209**, Chapter XII 12.5]) for nonnegative integers k_j and a continuous function g, that is,

$$\begin{aligned}\int_{\tilde\tau}y_0^{k_0}\cdots y_N^{k_N}g(y_0+\cdots+y_N)dy_0\cdots dy_N\\ =\frac{k_0!\cdots k_N!}{(N+k_0+\cdots+k_N)!}\int_0^1 g(t)t^{k_0+\cdots+k_N+N}dt,\end{aligned} \tag{5.1.24}$$

where $\tilde\tau$ is the $(N+1)$-dimensional simplex with their vertices ${}^t(0,\dots,\overset{j}{1},\dots,0)$, $0\le j\le N$ and the origin. This formula is easily obtained by repeated application of the beta integral.

We calculate

$$f(\mathbf{a})=\frac{1}{k!}\int_\sigma(\mathbf{a}\cdot\mathbf{x})^k d\mathbf{x}. \tag{5.1.25}$$

By multiplying

$$1=(k+1)\int_0^1 s^k ds, \tag{5.1.26}$$

and changing variables as $\mathbf{x}' = s\mathbf{x}$, we obtain

$$f(\mathbf{a}) = \frac{k+1}{k!}\int_{\tilde{\sigma}} (\mathbf{a}\cdot\mathbf{x}')^k s^{-N} ds d\mathbf{x}', \tag{5.1.27}$$

where $\tilde{\sigma} = \bigcup_{0\le s\le 1} \begin{pmatrix} s \\ s\sigma \end{pmatrix}$ is an $(N+1)$-dimensional simplex. Again we change variables as $\tilde{P}\tilde{\mathbf{y}} = \tilde{\mathbf{x}} = \begin{pmatrix} s \\ \mathbf{x}' \end{pmatrix}$. Then

$$\begin{aligned} f(\mathbf{a}) &= \frac{k+1}{k!}|\det \tilde{P}| \int_{\tilde{\tau}} \Big(\sum_{j=0}^{N} (\mathbf{a}\cdot\mathbf{p}_j) y_j\Big)^k \Big(\sum_{j=0}^{N} y_j\Big)^{-N} d\tilde{\mathbf{y}} \\ &= \frac{k+1}{k!}|\det \tilde{P}| \sum_{\substack{k_0,\dots,k_N\ge 0 \\ k_0+\cdots+k_N=k}} (\mathbf{a}\cdot\mathbf{p}_0)^{k_0}\cdots(\mathbf{a}\cdot\mathbf{p}_N)^{k_N} \\ &\qquad \times \frac{k!}{k_0!\cdots k_N!}\int_{\tilde{\tau}} y_0^{k_0}\cdots y_N^{k_N}\Big(\sum_{j=0}^{N} y_j\Big)^{-N} d\tilde{\mathbf{y}}, \end{aligned} \tag{5.1.28}$$

where $\tilde{\mathbf{y}} = {}^t(y_0,\dots,y_N)$. Hence applying (5.1.24), we obtain

$$\begin{aligned} f(\mathbf{a}) &= \frac{k+1}{k!}|\det \tilde{P}| \sum_{\substack{k_0,\dots,k_N\ge 0 \\ k_0+\cdots+k_N=k}} (\mathbf{a}\cdot\mathbf{p}_0)^{k_0}\cdots(\mathbf{a}\cdot\mathbf{p}_N)^{k_N} \frac{k!}{(N+k)!}\frac{1}{k+1} \\ &= \mathrm{Vol}(\sigma)\frac{N!}{(N+k)!} \sum_{\substack{k_0,\dots,k_N\ge 0 \\ k_0+\cdots+k_N=k}} (\mathbf{a}\cdot\mathbf{p}_0)^{k_0}\cdots(\mathbf{a}\cdot\mathbf{p}_N)^{k_N}. \end{aligned} \tag{5.1.29}$$

□

The above proof of Lemma 5.9 is quite computational, but it is also possible to deduce this lemma by the following more conceptual argument.

In fact, applying the Taylor expansion $e^x = \sum_{n=0}^{\infty} x^n/n!$ to the second expression of (5.1.5), we have

$$T(x_0,\dots,x_N) = \sum_{n=0}^{\infty}\frac{1}{n!}\det\begin{pmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_N \\ \vdots & \ddots & \vdots \\ x_0^{N-1} & \cdots & x_N^{N-1} \\ x_0^{n} & \cdots & x_N^{n} \end{pmatrix} \Big/ \det\begin{pmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_N \\ \vdots & \ddots & \vdots \\ x_0^{N-1} & \cdots & x_N^{N-1} \\ x_0^{N} & \cdots & x_N^{N} \end{pmatrix}. \tag{5.1.30}$$

The quotient of two determinants on the right-hand side is 0 if $n \le N-1$, while if $n \ge N$, it is equal to the Schur function $S_{(n-N)}$ corresponding to the partition $(n-N)$ of length 1 (by [126, Chapter I, (3.1)]). It is known that $S_{(n)} = h_n$, where h_n denotes the nth complete symmetric

function ([**126**, Chapter I, (3.9)], which is a special case of the Jacobi–Trudi formula). Therefore

$$T(x_0,\dots,x_N)=\sum_{n=N}^{\infty}\frac{h_{n-N}}{n!}, \tag{5.1.31}$$

and hence the assertion of Lemma 5.9 follows.

5.2. The rationality

The aim of this section is to prove Theorem 4.6. We deduce Theorem 4.6 from the following integral expression of $F(\mathbf{t},\mathbf{y};\Delta)$:

THEOREM 5.10 ([**113**, Theorem 7]). *Assume* $\mathbf{t}=(t_\alpha)_{\alpha\in\Delta_+}\in\mathbb{C}^{\#(\Delta_+)}$ *with* $|t_\alpha|\le R$ *for all* α, *where* $0<R<2\pi$. *We have*

(5.2.1)

$$F(\mathbf{t},\mathbf{y};\Delta)=\Bigl(\prod_{\alpha\in\Delta_+}\frac{t_\alpha}{e^{t_\alpha}-1}\Bigr)\sum_{m_1=0}^{2\langle\rho^\vee,\lambda_1\rangle-1}\cdots\sum_{m_r=0}^{2\langle\rho^\vee,\lambda_r\rangle-1}\exp\Bigl(\sum_{j=1}^{r}t_{\alpha_j}(\{\langle\mathbf{y},\lambda_j\rangle\}+m_j)\Bigr)$$
$$\times\int_{\mathcal{P}_{\mathbf{m},\mathbf{y}}}\exp\Bigl(\sum_{\alpha\in\Delta_+\setminus\Psi}t^*_\alpha x_\alpha\Bigr)\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha,$$

where $\rho^\vee=\frac{1}{2}\sum_{\alpha\in\Delta_+}\alpha^\vee$ *is the positive half sum,* $t^*_\alpha=t_\alpha-\sum_{j=1}^{r}t_{\alpha_j}\langle\alpha^\vee,\lambda_j\rangle$, $\mathbf{m}=(m_1,\dots,m_r)$ *and*

(5.2.2)
$$\mathcal{P}_{\mathbf{m},\mathbf{y}}=\left\{\mathbf{x}=(x_\alpha)_{\alpha\in\Delta_+\setminus\Psi}\;\middle|\;\begin{aligned}&0\le x_\alpha\le 1,\quad(\alpha\in\Delta_+\setminus\Psi)\\&\{\langle\mathbf{y},\lambda_j\rangle\}+m_j-1\le\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\\&\qquad\qquad\le\{\langle\mathbf{y},\lambda_j\rangle\}+m_j,\quad(1\le j\le r)\end{aligned}\right\}$$

is a convex polytope.

PROOF. By (4.2.2) and (4.2.11), we find

(5.2.3)

$$\begin{aligned}F(\mathbf{t},\mathbf{y};\Delta)&=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}}P(\mathbf{k},\mathbf{y};\Delta)\prod_{\alpha\in\Delta_+}\frac{t_\alpha^{k_\alpha}}{k_\alpha!}\\&=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}}\int_0^1\cdots\int_0^1\Bigl(\prod_{\alpha\in\Delta_+\setminus\Psi}B_{k_\alpha}(\{x_\alpha\})\frac{t_\alpha^{k_\alpha}}{k_\alpha!}\Bigr)\\&\quad\times\prod_{j=1}^{r}B_{k_{\alpha_j}}(\{\langle\mathbf{y},\lambda_j\rangle-\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\})\frac{t_{\alpha_j}^{k_{\alpha_j}}}{k_{\alpha_j}!}\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha.\end{aligned}$$

The above interchange of the order of summation and integration can be justified by Lemma 4.9. Applying (4.2.1), we have

(5.2.4)

$$
\begin{aligned}
&F(\mathbf{t},\mathbf{y};\Delta)\\
&=\Big(\prod_{\alpha\in\Delta_+}\frac{t_\alpha}{e^{t_\alpha}-1}\Big)\int_0^1\cdots\int_0^1\Big(\prod_{\alpha\in\Delta_+\setminus\Psi}\exp(t_\alpha x_\alpha)\Big)\\
&\quad\times\Big(\prod_{j=1}^r\exp\Big(t_{\alpha_j}\Big\{\langle\mathbf{y},\lambda_j\rangle-\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\Big\}\Big)\Big)\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha\\
&=\Big(\prod_{\alpha\in\Delta_+}\frac{t_\alpha}{e^{t_\alpha}-1}\Big)\int_0^1\cdots\int_0^1\Big(\prod_{\alpha\in\Delta_+\setminus\Psi}\exp\big((t_\alpha-\sum_{j=1}^r t_{\alpha_j}\langle\alpha^\vee,\lambda_j\rangle)x_\alpha\big)\Big)\\
&\quad\times\Big(\prod_{j=1}^r\exp\Big(t_{\alpha_j}\Big(\langle\mathbf{y},\lambda_j\rangle-\Big[\langle\mathbf{y},\lambda_j\rangle-\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\Big]\Big)\Big)\Big)\prod_{\alpha\in\Delta_+\setminus\Psi}dx_\alpha,
\end{aligned}
$$

where $[x]=x-\{x\}$. For $m_j\in\mathbb{Z}$, the region of $\mathbf{x}$ satisfying

$$\langle\mathbf{y},\lambda_j\rangle-\Big[\langle\mathbf{y},\lambda_j\rangle-\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\Big]=\{\langle\mathbf{y},\lambda_j\rangle\}+m_j \tag{5.2.5}$$

is given by

$$\{\langle\mathbf{y},\lambda_j\rangle\}+m_j-1<\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\le\{\langle\mathbf{y},\lambda_j\rangle\}+m_j. \tag{5.2.6}$$

Since for $0\le x_\alpha\le 1$ ($\alpha\in\Delta_+\setminus\Psi$),

$$0\le\sum_{\alpha\in\Delta_+\setminus\Psi}x_\alpha\langle\alpha^\vee,\lambda_j\rangle\le\sum_{\alpha\in\Delta_+\setminus\Psi}\langle\alpha^\vee,\lambda_j\rangle=2\langle\rho^\vee,\lambda_j\rangle-1 \tag{5.2.7}$$

(where the last equality is by (2.4.1)). Therefore we have $0\le m_j\le 2\langle\rho^\vee,\lambda_j\rangle-1$. □

Now we can immediately prove Theorem 4.6 by specifying $\mathbf{y}=\mathbf{0}$ in the following theorem, which is a part of [**113**, Theorem 13].

THEOREM 5.11. *For any* $\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}$ *and* $\mathbf{y}=\sum_{i=1}^r y_i\alpha_i^\vee\in V$, *we have*

$$P(\mathbf{k},\mathbf{y};\Delta)\in\mathbb{Q}[y_1,\dots,y_r]=\mathbb{Q}[\{y_1\},\dots,\{y_r\}].$$

PROOF. What we have to show is the rationality of the coefficients $P(\mathbf{k},\mathbf{y};\Delta)$ of the Taylor expansion of $F(\mathbf{t},\mathbf{y};\Delta)$ in $\mathbf{t}$. We first decompose the integral on the right-hand side of (5.2.1) into the integrals on simplexes appearing in the triangulation of the polytope $\mathcal{P}_{\mathbf{m},\mathbf{y}}$. Then apply Lemma 5.9 to expand the right-hand side of (5.2.1).

Let $\mathbf{p} = (p_\alpha)_{\alpha \in \Delta_+ \setminus \Psi}$ be a vertex of $\mathcal{P}_{\mathbf{m},\mathbf{y}}$. Then from the expression (5.2.2) it is clear that $p_\alpha \in \mathbb{Q}[\{\langle \mathbf{y}, \lambda_j \rangle\}_{j=1}^r] = \mathbb{Q}[\{y_j\}_{j=1}^r]$ for all α. Since those $\mathbf{p}$ exhaust all vertices of the triangulation of $\mathcal{P}_{\mathbf{m},\mathbf{y}}$, we can conclude that all the coefficients of the above expansion are in $\mathbb{Q}[\{y_j\}_{j=1}^r]$. This proves Theorem 5.11.

□

There is another argument of proving Theorems 4.6 and 5.11. In Chapter 4 we stated Theorem 4.12, which was originally proved in [**105**]. In [**105**], this theorem was deduced from Theorem 5.10. Once we admit Theorem 4.12, then Theorems 4.6 and 5.11 also follow from this theorem by applying the following lemma.

LEMMA 5.12. *Let $f(t_1, \dots, t_n)$ be a holomorphic function in a neighborhood of the origin on $\mathbb{C}^n$ and write its Taylor expansion as*

$$f(t_1, \dots, t_n) = \sum_{k_1=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} a(k_1, \dots, k_n) \frac{t_1^{k_1}}{k_1!} \cdots \frac{t_n^{k_n}}{k_n!}. \tag{5.2.8}$$

If $f(t_1, \dots, t_n)$ vanishes identically on the hyperplane $\mathbf{a} \cdot \mathbf{t} = 0$ for some $\mathbf{a} \in \mathbb{Q}^n \setminus \{\mathbf{0}\}$ through the origin, then

$$\frac{f(t_1, \dots, t_n)}{\mathbf{a} \cdot \mathbf{t}} = \sum_{k_1=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} b(k_1, \dots, k_n) \frac{t_1^{k_1}}{k_1!} \cdots \frac{t_n^{k_n}}{k_n!}, \tag{5.2.9}$$

where each $b(k_1, \dots, k_n)$ is a linear combination of $a(j_1, \dots, j_n)$ with rational coefficients.

PROOF. We may assume $a_1 \neq 0$ without loss of generality. Put $w = \mathbf{a} \cdot \mathbf{t}$ and

$$\begin{aligned} g(w, t_2, \dots, t_n) &= f\Big(\frac{w - (a_2 t_2 + \cdots + a_n t_n)}{a_1}, t_2, \dots, t_n\Big) \\ &\quad (= f(t_1, t_2, \dots, t_n)). \end{aligned}$$

Then

$$\begin{aligned} &g(w, t_2, \dots, t_n) \\ &= \sum_{k_1=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} a(k_1, \dots, k_n) \frac{(w - (a_2 t_2 + \cdots + a_n t_n))^{k_1}}{a_1^{k_1} k_1!} \frac{t_2^{k_2}}{k_2!} \cdots \frac{t_n^{k_n}}{k_n!} \\ &= \sum_{k=1}^{\infty} \Big(\sum_{k_2=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} a'(k, k_2, \dots, k_n) \frac{t_2^{k_2}}{k_2!} \cdots \frac{t_n^{k_n}}{k_n!} \Big) \frac{w^k}{k!}, \end{aligned}$$

where $a'(k, k_2, \dots, k_n)$ is a linear combination of $a(j_1, \dots, j_n)$ with rational coefficients. Here, the term $k = 0$ does not exist because of the

assumption that $g(w, t_2, \dots, t_n) = 0$ if $w = 0$. Therefore we obtain

(5.2.10)
$$\frac{f(t_1, \dots, t_n)}{\mathbf{a} \cdot \mathbf{t}} = \frac{g(w, t_2, \dots, t_n)}{w} = \sum_{k=0}^{\infty} \Big(\sum_{k_2=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} a'(k+1, k_2, \dots, k_n) \frac{t_2^{k_2}}{k_2!} \cdots \frac{t_n^{k_n}}{k_n!} \Big) \frac{(\mathbf{a} \cdot \mathbf{t})^k}{(k+1)!},$$

which implies the assertion. □

Another proof of Theorem 5.11. Let

$$f_0(\mathbf{t}) = \prod_{\mathbf{V} \in \mathscr{V}} \prod_{\gamma \in \Delta_+ \setminus \mathbf{V}} (t_\gamma - \sum_{\beta \in \mathbf{V}} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle),$$

and define the function

$$f(\mathbf{t}) = f_0(\mathbf{t}) F(\mathbf{t}, \mathbf{y}; \Delta). \tag{5.2.11}$$

This function is holomorphic around the origin, which can be seen from the expression (4.3.5) given in Theorem 4.12. Moreover, this function vanishes identically on the hyperplane

$$t_\gamma - \sum_{\beta \in \mathbf{V}} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle = 0,$$

because, as we have already seen in Remark 4.15, $F(\mathbf{t}, \mathbf{y}; \Delta)$ is not singular on this hyperplane. Therefore we can apply Lemma 5.12 (repeatedly) to obtain the desired rationality[1] by using the fact

$$\{\mathbf{y} + q\}_{\mathbf{V}, \beta} \in \mathrm{Lh}_{\mathbb{Q}}[1, y_1, \dots, y_r], \tag{5.2.12}$$

which immediately follows from (4.3.3). □

Remark 5.13. The total degree of $P(\mathbf{k}, \mathbf{y}; \Delta)$, as a polynomial in $\mathbf{y}$, is at most $|\mathbf{k}| = \sum_{\alpha \in \Delta_+} k_\alpha$. In fact, let n_0 be the total degree of $f_0(\mathbf{t})$ as a polynomial in $\mathbf{t}$. Let $P_0(\tau, \mathbf{y})$ be the coefficient of a monomial $\tau = \prod_\alpha t_\alpha^{l_\alpha}$, with $\sum_\alpha l_\alpha = |\mathbf{k}|$, in the Taylor expansion of $f(\mathbf{t})$ around the origin. Then, from (4.3.5) (with (5.2.12)) we may observe that the total degree of $P_0(\tau, \mathbf{y})$ as a polynomial in $\mathbf{y}$ is at most $|\mathbf{k}| - n_0$. Now remove one linear factor from $f_0(\mathbf{t})$ and $f(\mathbf{t})$ to define new functions $\widetilde{f}_0(\mathbf{t})$ and $\widetilde{f}(\mathbf{t})$, and denote by $\widetilde{P}_0(\tau, \mathbf{y})$ the coefficient of τ in the Taylor expansion of $\widetilde{f}(\mathbf{t})$. Then from Lemma 5.12 and its proof we find that the total degree of $\widetilde{P}_0(\tau, \mathbf{y})$ as a polynomial in $\mathbf{y}$ is at most $|\mathbf{k}| - n_0 + 1$. Repeating this procedure n_0 times, we arrive at the assertion.

[1] This proof of Theorem 5.11, as well as Lemma 5.12, was written in the first version of the first-named author's article [**99**], but was not included in the published version. Therefore these materials have not been published before.

Theorem 5.10 can also be used to calculate the Taylor expansion of $F(\mathbf{t};\Delta)$ (and hence the Bernoulli numbers $B(\mathbf{k};\Delta)$). In fact, in [**113**], we computed the expansion of $F(\mathbf{t};\Delta)$ in the cases $\Delta = A_2, A_3$ and C_2 from the expression (5.2.1). However, to carry out such computations, we should first consider the triangulation of the integral. Since the number of simplexes appearing in the triangulation increases very rapidly as the rank of Δ becomes large, this method of computation is not efficient in practice. Therefore the authors searched for some different expression of $F(\mathbf{t};\Delta)$, which would give more efficient tools for computation, and arrived, in [**105**][2], at Theorem 4.12. It was shown in [**105**, Proposition 4.2] that, if Δ is a root system of rank r with the fundamental system $\Psi = \{\alpha_1, \dots, \alpha_r\}$, then

(5.2.13)
$$\#(\mathscr{V}(\Delta_+)) = \frac{\#(W(\Delta))}{2r} \sum_{1\le j\le r} \frac{\#(\mathscr{V}(\Delta_+^j))}{\#(W(\Delta^j))}(\#(\Delta_+) - \#(\Delta_+^j)),$$

where Δ^j denotes the root system generated by $\Psi \setminus \{\alpha_j\}$. This formula gives the number of terms of the right-hand side of (4.3.5), which is much smaller than the number of simplexes.[3] Therefore Theorem 4.12 is much more useful in practical computations.

[2][**105**] was published earlier than [**113**], but written later.

[3]For example, in the cases $\Delta = A_4, B_3$ and C_3, the numbers of simplexes are 5040, 19908 and 20916, respectively (though it may depend on the choice of triangulation), while the numbers of $\mathscr{V}$ given by (5.2.13) are 125, 68 and 68, respectively.

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CHAPTER 6

The recursive structure

The aim of this chapter is to show that a recursive structure exists in the family of zeta-functions of root systems.

6.1. The Mellin–Barnes integral formula

First in this section, we prepare the fundamental tool of producing such a recursive structure, which is the following classical *Mellin–Barnes integral formula* (see [209, Section 14.51]):

LEMMA 6.1. *Let s, λ be complex numbers with $\Re s > 0$, $|\arg \lambda| < \pi$, and $\lambda \neq 0$. Then we have*

$$(1+\lambda)^{-s} = \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s+z)\Gamma(-z)}{\Gamma(s)} \lambda^z dz, \tag{6.1.1}$$

where c is real with $-\Re s < c < 0$, and the path (c) of integration is the vertical line $\Re z = c$.

PROOF. First, let $|\lambda| < 1$. Write $z = x + \sqrt{-1}y$. By Stirling's formula [76, (A.34)][193, (4.12.2)] we have

$$|\Gamma(z)| = \sqrt{2\pi}|y|^{x-\frac{1}{2}} e^{-\frac{\pi}{2}|y|} (1 + O(\frac{1}{|y|})), \tag{6.1.2}$$

hence the integrand on the right-hand side decreases rapidly when $|y| \to \infty$. Therefore we can shift the path of integration to $\Re(z) = n + \frac{1}{2}$ for any $n \in \mathbb{N}_0$. Since the poles of $\Gamma(s+z)$ are $z = -s, -s-1, -s-2, \ldots$ and the poles of $\Gamma(-z)$ are $z = 0, 1, 2, 3, \ldots$, only the latter poles are relevant to the above shifting of the path. The residue at $z = k$ is $(-1)^{k+1}\Gamma(s+k)\frac{\lambda^k}{k!}$, so the right-hand side is equal to

$$-\sum_{k=0}^{n} (-1)^{k+1}\Gamma(s+k)\frac{\lambda^k}{k!} + \frac{1}{2\pi\sqrt{-1}} \int_{(n+\frac{1}{2})} \Gamma(s+z)\Gamma(-z)\lambda^z dz.$$

The first term here is

$$-\sum_{k=0}^{n} (-1)^{k+1}\Gamma(s+k)\frac{\lambda^k}{k!}$$

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$$= \sum_{k=0}^{n} (-1)^k (s+k-1)(s+k-2)\dots(s+1)s\Gamma(s)\frac{\lambda^k}{k!}$$
$$= \Gamma(s)\sum_{k=0}^{n} \frac{(-s)(-s-1)\dots(-s-k+2)(-s-k+1)}{k!}\lambda^k$$
$$= \Gamma(s)\sum_{k=0}^{n} \binom{-s}{k}\lambda^k.$$

Therefore, taking $n \to \infty$, we find that the first term tends to $\Gamma(s)(1+\lambda)^{-s}$, and the second term tends to 0, hence we obtain the desired formula. The proof then completes by the analytic continuation with respect to $\lambda \in \mathbb{C}$. □

The second-named author developed a method of studying analytic properties of various multiple zeta-functions, based on the Mellin–Barnes integral formula. An advantage of using the Mellin–Barnes formula is that via this method, we can naturally find recursive structures in the family of multiple zeta-functions, which can be described by the Mellin–Barnes type of integrals ([**132**] [**133**] [**134**] [**136**]).

In [**146**], it has been shown that such a recursive structure exists in the family of zeta-functions attached to Lie algebras of type A_r. In fact, Theorem 2.2 of [**146**] gives a formula which expresses $\zeta_{r+1}(\mathbf{s}; A_{r+1})$ as a multiple integral involving $\zeta_r(\,\cdot\,; A_r)$. Then in [**104**], the recursive structures for types A_r, B_r, C_r and D_r are more closely discussed. Here we reproduce the results in [**104**]. For this purpose, we introduce the notion of *zeta-functions of root sets*. Let Δ^* be a subset of $\Delta_+ = \Delta_+(\mathfrak{g})$. We call Δ^* a *root set*[1] if the following assumption is satisfied:

ASSUMPTION 6.2. For any λ_j $(1 \le j \le r)$, there exists an element $\alpha \in \Delta^*$ such that $\langle \alpha, \lambda_j \rangle \neq 0$.

We define the multiple zeta-function of Δ^* by

$$\zeta_r(\mathbf{s}; \Delta^*) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{\alpha \in \Delta^*} \langle \alpha^\vee, m_1\lambda_1 + \cdots + m_r\lambda_r \rangle^{-s_\alpha}, \tag{6.1.3}$$

where $\mathbf{s} = \mathbf{s}(\Delta^*) = (s_\alpha)_{\alpha \in \Delta^*} \in \mathbb{C}^{\#\Delta^*}$. This is convergent absolutely for $\mathbf{s} \in \mathcal{S}_0$.

If there is a j for which $\langle \alpha, \lambda_j \rangle = 0$ for all α, then we can write

$$\langle \alpha^\vee, m_1\lambda_1 + \cdots + m_r\lambda_r \rangle = \langle \alpha^\vee, m_1\lambda_1 + \cdots + m_{j-1}\lambda_{j-1} + m_{j+1}\lambda_{j+1} + \cdots + m_r\lambda_r \rangle,$$

[1] This notion was first introduced in [**104**].

hence the sum with respect to m_j on the right-hand side of (6.1.3) obviously diverges. Therefore the above assumption is necessary to define $\zeta_r(\mathbf{s};\Delta^*)$.

REMARK 6.3. In order to establish the recursive structure by using the Mellin–Barnes integrals, it is indispensable to consider multi-variable multiple zeta-functions. Therefore the second-named author introduced various multi-variable multiple zeta-functions such as (1.4.3), (1.6.2) (including zeta-functions of root systems of types A_2 and B_2). This is one of the main motivations why we introduced the multi-variable zeta-functions of root systems. In fact, it is easily seen that many integral expressions which will be proved in this chapter cannot be written without considering multi-variable zeta-functions.

6.2. The recursive structure for A_r

We formulate our recursive structure in terms of these zeta-functions. In this section we consider the case when $\mathfrak{g} = \mathfrak{sl}(r+1)$ $(r \geq 2)$. By (3.2.1) we see that

$$\Delta_+ = \Delta_+(A_r) = \left\{ \varepsilon_i - \varepsilon_j = \sum_{i \leq k < j} \alpha_k \;\middle|\; 1 \leq i < j \leq r+1 \right\}. \tag{6.2.1}$$

Hence

$$\Delta_h^*(A_r) = \{\varepsilon_1 - \varepsilon_j \mid 2 \leq j \leq h\} \cup \{\varepsilon_i - \varepsilon_j \mid 2 \leq i < j \leq r+1\} \tag{6.2.2}$$

$(2 \leq h \leq r+1)$ is a subset of $\Delta_+(A_r)$ satisfying Assumption 6.2. Then there is the relation

$$\Delta^*(A_r) \subset \Delta_2^*(A_r) \subset \cdots \subset \Delta_r^*(A_r) \subset \Delta_{r+1}^*(A_r) = \Delta_+(A_r), \tag{6.2.3}$$

where

$$\Delta^*(A_r) = \{\varepsilon_i - \varepsilon_j \mid 2 \leq i < j \leq r+1\}.$$

The vector $\mathbf{s}(\Delta_h^*(A_r))$ can be written as

$$\mathbf{s}(\Delta_h^*(A_r)) = (s_{12}, \ldots, s_{1h}, \mathbf{s}_2(A_r)), \tag{6.2.4}$$

where $\mathbf{s}_2(A_r) = (s_{ij})_{2 \leq i < j \leq r+1}$.

We now show that this relation gives the Mellin–Barnes recursive structure in the family of zeta-functions of root sets in the A_r case.

Let $3 \leq h \leq r+1$. The set $\Delta_h^*(A_r)$ includes the root

$$\varepsilon_1 - \varepsilon_h = \sum_{1 \leq k < h} \alpha_k,$$

which produces the term $(m_1+\cdots+m_{h-1})^{-s_{1h}}$ on the right-hand side of (6.1.3) for $\Delta^*=\Delta_h^*(A_r)$. We apply formula (6.1.1) to the above term to obtain

$$
\begin{aligned}
(6.2.5)\qquad & (m_1+\cdots+m_{h-1})^{-s_{1h}} \\
&= (m_1+\cdots+m_{h-2})^{-s_{1h}}\left(1+\frac{m_{h-1}}{m_1+\cdots+m_{h-2}}\right)^{-s_{1h}} \\
&= (m_1+\cdots+m_{h-2})^{-s_{1h}}\frac{1}{2\pi\sqrt{-1}}\int_{(c_h)}\frac{\Gamma(s_{1h}+z_h)\Gamma(-z_h)}{\Gamma(s_{1h})} \\
&\qquad\qquad \times\left(\frac{m_{h-1}}{m_1+\cdots+m_{h-2}}\right)^{z_h}dz_h,
\end{aligned}
$$

where $-\Re s_{1h}<c_h<0$. Hence

(6.2.6)

$$
\begin{aligned}
&\zeta_r(\mathbf{s}(\Delta_h^*(A_r));\Delta_h^*(A_r)) \\
&=\sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}\prod_{2\le i<j\le r+1}(m_i+\cdots+m_{j-1})^{-s_{ij}} \\
&\quad\times\prod_{2\le j<h}(m_1+\cdots+m_{j-1})^{-s_{1j}}\frac{1}{2\pi\sqrt{-1}}\int_{(c_h)}\frac{\Gamma(s_{1h}+z_h)\Gamma(-z_h)}{\Gamma(s_{1h})} \\
&\quad\times(m_1+\cdots+m_{h-2})^{-s_{1h}-z_h}m_{h-1}^{z_h}dz_h \\
&=\frac{1}{2\pi\sqrt{-1}}\int_{(c_h)}\frac{\Gamma(s_{1h}+z_h)\Gamma(-z_h)}{\Gamma(s_{1h})} \\
&\quad\times\sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}\prod_{\substack{2\le i<j\le r+1\\(i,j)\ne(h-1,h)}}(m_i+\cdots+m_{j-1})^{-s_{ij}}m_{h-1}^{-s_{h-1,h}+z_h} \\
&\quad\times\prod_{2\le j\le h-2}(m_1+\cdots+m_{j-1})^{-s_{1j}}(m_1+\cdots+m_{h-2})^{-s_{1,h-1}-s_{1h}-z_h}dz_h \\
&=\frac{1}{2\pi\sqrt{-1}}\int_{(c_h)}\frac{\Gamma(s_{1h}+z_h)\Gamma(-z_h)}{\Gamma(s_{1h})}\zeta_r(\mathbf{s}^*(A_r,z_h);\Delta_{h-1}^*(A_r))dz_h,
\end{aligned}
$$

where

$$\mathbf{s}^*(A_r,z_h)=(s_{12},\dots,s_{1,h-2},s_{1,h-1}+s_{1h}+z_h,\mathbf{s}_2^*(A_r,z_h))$$

and $\mathbf{s}_2^*(A_r,z_h)$ is almost the same as $\mathbf{s}_2(A_r)$ but $s_{h-1,h}$ is replaced by $s_{h-1,h}-z_h$. Formula (6.2.6) is an integral expression of $\zeta_r(\mathbf{s};\Delta_h^*(A_r))$ whose integrand includes $\zeta_r(\cdot\,;\Delta_{h-1}^*(A_r))$ $(3\le h\le r+1)$. Repeating this procedure, we find the recursive structure

$$
\begin{aligned}
(6.2.7)\qquad \zeta_r(\cdot\,;A_r)=\zeta_r(\cdot\,;\Delta_{r+1}^*(A_r))&\to\zeta_r(\cdot\,;\Delta_r^*(A_r))\to \\
&\cdots\to\zeta_r(\cdot\,;\Delta_3^*(A_r))\to\zeta_r(\cdot\,;\Delta_2^*(A_r)).
\end{aligned}
$$

Here, the notation $\zeta(\cdot;X) \to \zeta(\cdot;Y)$ means that the zeta-function of a root set X can be expressed as an integral of Mellin–Barnes type whose integrand includes the zeta-function of another root set Y.

In particular, when $h = 3$, the second product on the third member of (6.2.6) is an empty product (hence is equal to 1), hence the sum with respect to m_1 can be separated, which produces the Riemann zeta factor. That is, we have

$$\begin{aligned}(6.2.8)\qquad &\zeta_r(\mathbf{s}^*(A_r,z_3);\Delta_2^*(A_r))\\ &\quad=\zeta_{r-1}(\mathbf{s}_2^*(A_r,z_3);\Delta^*(A_r))\zeta(s_{12}+s_{13}+z_3).\end{aligned}$$

The first factor on the right-hand side is not ζ_r but ζ_{r-1}, because this is defined by an $(r-1)$-fold sum. Hence

$$\begin{aligned}(6.2.9)\quad &\zeta_r(\mathbf{s};\Delta_3^*(A_r))\\ &=\frac{1}{2\pi\sqrt{-1}}\int_{(c_3)}\frac{\Gamma(s_{13}+z_3)\Gamma(-z_3)}{\Gamma(s_{13})}\\ &\qquad\times\zeta_{r-1}(\mathbf{s}_2^*(A_r,z_3);\Delta^*(A_r))\zeta(s_{12}+s_{13}+z_3)dz_3.\end{aligned}$$

This implies that the last step of (6.2.7) can be rewritten as

$$(6.2.10)\qquad \zeta_r(\cdot\,;\Delta_3^*(A_r)) \to \zeta_{r-1}(\cdot\,;\Delta^*(A_r)),$$

by neglecting the Riemann zeta factor.

Renaming ε_i as ε_{i-1} $(2 \le i \le r+1)$, we find that $\Delta^*(A_r)$ is equal to $\Delta_+(A_{r-1})$. Hence we now observe that $\zeta_r(\mathbf{s};A_r)$ can be expressed as an $(r-1)$-fold integral involving $\zeta_{r-1}(\cdot\,;A_{r-1})$, which is exactly the assertion of [**146**, Theorem 2.2].

Summarizing the above argument, we obtain the following refinement of [**146**, Theorem 2.2].

THEOREM 6.4 ([**104**, Theorem 3.1]). *Between* $\zeta_r(\cdot\ ;A_r)$ *and* $\zeta_{r-1}(\cdot\,;A_{r-1})$ *(for* $r \ge 2$*) there is the recursive relation given by* (6.2.7) *and* (6.2.10)*, which can be expressed as the Mellin–Barnes integrals* (6.2.6) *and* (6.2.9)*. This further gives the recursive relation*

$$(6.2.11)\qquad \zeta_r(\cdot\,;A_r) \to \zeta_{r-1}(\cdot\,;A_{r-1}) \to \cdots \to \zeta_2(\cdot\,;A_2) = \zeta_{MT,2} \to \zeta.$$

REMARK 6.5. Among the coroots of A_r listed in the preceding section, the coroot $e_1 - e_{r+1}$ is the highest, and there is the order relation

$$(6.2.12)\qquad e_1 - e_{r+1} > e_1 - e_r > \cdots > e_1 - e_2.$$

The recursive relations (6.2.7) and (6.2.10) correspond to this order relation. In fact, we see that:

(i) in each step (6.2.6) we apply the Mellin–Barnes formula to the sum $m_1 + \cdots + m_{h-1}$ corresponding to the coroot $e_1 - e_h$,
and

(ii) the sum is divided into $m_1 + \cdots + m_{h-2}$ and m_{h-1}, where the former sum corresponds to the next coroot $e_1 - e_{h-1}$.

In the next section we will show that order relations among coroots similar to (6.2.12) also give the recursive structures among zeta-functions in the B_r, C_r and D_r cases.

6.3. The recursive structure for B_r, C_r and D_r

In this section we consider the Mellin–Barnes recursive structure in the B_r, C_r, and D_r cases. First, treat the B_r case ($r \geq 2$). From (3.3.2) we have

$$\Delta_+(B_r) = \{\varepsilon_i \mid 1 \leq i \leq r\} \cup \{\varepsilon_i \pm \varepsilon_j \mid 1 \leq i < j \leq r\}. \tag{6.3.1}$$

We write the vector $\mathbf{s}(B_r)$ as

$$\mathbf{s}(B_r) = (s_1, \ldots, s_r, \mathbf{s}(B_r)^-, \mathbf{s}(B_r)^+),$$

where $\mathbf{s}(B_r)^\pm = (s_{ij}^\pm)_{1 \leq i < j \leq r}$. The highest coroot is $2e_1$, and there is the order relation

$$\begin{aligned} &2e_1 > e_1 + e_2 > e_1 + e_3 > \cdots > e_1 + e_r \\ &> e_1 - e_r > e_1 - e_{r-1} > \cdots > e_1 - e_3 > e_1 - e_2 \end{aligned} \tag{6.3.2}$$

among the coroots involving $\alpha_1^\vee$.

We give the Mellin–Barnes recursive structure according to this relation. Let

$$\begin{aligned} \Delta_h^*(B_r) &= \{\varepsilon_i \mid 2 \leq i \leq r\} \cup \{\varepsilon_1 - \varepsilon_j \mid 2 \leq j \leq h\} \\ &\cup \{\varepsilon_i - \varepsilon_j \mid 2 \leq i < j \leq r\} \cup \{\varepsilon_i + \varepsilon_j \mid 2 \leq i < j \leq r\} \end{aligned} \tag{6.3.3}$$

($2 \leq h \leq r$) and

$$\begin{aligned} \Delta_h^{**}(B_r) &= \{\varepsilon_i \mid 2 \leq i \leq r\} \cup \{\varepsilon_i - \varepsilon_j \mid 1 \leq i < j \leq r\} \\ &\cup \{\varepsilon_1 + \varepsilon_j \mid h \leq j \leq r\} \cup \{\varepsilon_i + \varepsilon_j \mid 2 \leq i < j \leq r\} \end{aligned} \tag{6.3.4}$$

($2 \leq h \leq r$). Then there is the relation

$$\begin{aligned} &\Delta^*(B_r) \subset \Delta_2^*(B_r) \subset \cdots \subset \Delta_r^*(B_r) \\ &\subset \Delta_r^{**}(B_r) \subset \cdots \subset \Delta_2^{**}(B_r) \subset \Delta_+(B_r), \end{aligned} \tag{6.3.5}$$

where

$$\Delta^*(B_r) = \{\varepsilon_i \mid 2 \leq i \leq r\} \cup \{\varepsilon_i \pm \varepsilon_j \mid 2 \leq i < j \leq r\}. \tag{6.3.6}$$

First, the root ε_1 (or the coroot $2e_1$) corresponds to the term

$$(2(m_1+\cdots+m_{r-1})+m_r)^{-s_1}$$

which is, by using (6.1.1), equal to

$$\frac{1}{2\pi\sqrt{-1}}\int_{(c')}\frac{\Gamma(s_1+z')\Gamma(-z')}{\Gamma(s_1)} \times m_1^{z'}(m_1+2(m_2+\cdots+m_{r-1})+m_r)^{-s_1-z'}dz'$$

with $-\Re s_1 < c' < 0$. Hence

(6.3.7)

$$\begin{aligned}
&\zeta_r(\mathbf{s};B_r)\\
&=\frac{1}{2\pi\sqrt{-1}}\int_{(c')}\frac{\Gamma(s_1+z')\Gamma(-z')}{\Gamma(s_1)}\\
&\quad\times\sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}\prod_{2\le i\le r}(2(m_i+\cdots+m_{r-1})+m_r)^{-s_i}\\
&\quad\times m_1^{z'-s_{12}^-}\prod_{\substack{1\le i<j\le r\\(i,j)\ne(1,2)}}(m_i+\cdots+m_{j-1})^{-s_{ij}^-}\\
&\quad\times(m_1+2(m_2+\cdots+m_{r-1})+m_r)^{-s_1-s_{12}^+-z'}\\
&\quad\times\prod_{\substack{1\le i<j\le r\\(i,j)\ne(1,2)}}(m_i+\cdots+m_{j-1}+2(m_j+\cdots+m_{r-1})+m_r)^{-s_{ij}^+}dz'\\
&=\frac{1}{2\pi\sqrt{-1}}\int_{(c')}\frac{\Gamma(s_1+z')\Gamma(-z')}{\Gamma(s_1)}\\
&\qquad\times\zeta_r(\mathbf{s}'(B_r,z');\Delta_2^{**}(B_r))dz',
\end{aligned}$$

where

$$\mathbf{s}'(B_r,z')=(s_2,\dots,s_r,\mathbf{s}'(B_r,z')^-,\mathbf{s}'(B_r,z')^+),$$

$\mathbf{s}'(B_r,z')^-$ is almost the same as $\mathbf{s}(B_r)^-$ but s_{12}^- is replaced by $s_{12}^- - z'$, and $\mathbf{s}'(B_r,z')^+$ is almost the same as $\mathbf{s}(B_r)^+$ but s_{12}^+ is replaced by $s_1+s_{12}^++z'$. This implies the recursive relation

$$\zeta_r(\cdot\,;B_r)\to\zeta_r(\cdot\,;\Delta_2^{**}(B_r)). \tag{6.3.8}$$

Next, consider $\Delta_h^{**}(B_r)$. The corresponding vector can be written as

$$\mathbf{s}(\Delta_h^{**}(B_r))=(s_2,\dots,s_r,\mathbf{s}(B_r)^-,s_{1h}^+,\dots,s_{1r}^+,\mathbf{s}_2(B_r)^+),$$

where $\mathbf{s}_2(B_r)^+=(s_{ij}^+)_{2\le i<j\le r}$. When $2\le h\le r-1$, we apply (6.1.1) to the term

$$(m_1+\cdots+m_{h-1}+2(m_h+\cdots+m_{r-1})+m_r)^{-s_{1h}^+} \tag{6.3.9}$$

$$= (m_1 + \cdots + m_h + 2(m_{h+1} + \cdots + m_{r-1}) + m_r)^{-s^+_{1h}}$$
$$\times \left(1 + \frac{m_h}{m_1 + \cdots + m_h + 2(m_{h+1} + \cdots + m_{r-1}) + m_r}\right)^{-s^+_{1h}},$$

which corresponds to the root $\varepsilon_1 + \varepsilon_h$. Then

(6.3.10)

$$\begin{aligned}
&\zeta_r(\mathbf{s}(\Delta^{**}_h(B_r)); \Delta^{**}_h(B_r)) \\
&= \frac{1}{2\pi\sqrt{-1}} \int_{(c^+_{1h})} \frac{\Gamma(s^+_{1h} + z^+_{1h})\Gamma(-z^+_{1h})}{\Gamma(s^+_{1h})} \\
&\quad \times \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{2\le i\le r} (2(m_i + \cdots + m_{r-1}) + m_r)^{-s_i} \\
&\quad \times m_h^{z^+_{1h}} \prod_{1\le i<j\le r} (m_i + \cdots + m_{j-1})^{-s^-_{ij}} \\
&\quad \times \prod_{h<j\le r} (m_1 + \cdots + m_{j-1} + 2(m_j + \cdots + m_{r-1}) + m_r)^{-s^+_{1j}} \\
&\quad \times (m_1 + \cdots + m_h + 2(m_{h+1} + \cdots + m_{r-1}) + m_r)^{-s^+_{1h} - z^+_{1h}} \\
&\quad \times \prod_{2\le i<j\le r} (m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_{r-1}) + m_r)^{-s^+_{ij}} dz^+_{1h},
\end{aligned}$$

where $-\Re s^+_{1h} < c^+_{1h} < 0$. This is equal to

(6.3.11)

$$\frac{1}{2\pi\sqrt{-1}} \int_{(c^+_{1h})} \frac{\Gamma(s^+_{1h} + z^+_{1h})\Gamma(-z^+_{1h})}{\Gamma(s^+_{1h})} \zeta_r(\mathbf{s}^{**}(B_r, z^+_{1h}); \Delta^{**}_{h+1}(B_r)) dz^+_{1h},$$

where

$$\begin{aligned}
\mathbf{s}^{**}(B_r, z^+_{1h}) = (&s_2, \ldots, s_r, \mathbf{s}^{**}(B_r, z^+_{1h})^-, \\
&s^+_{1h} + s^+_{1,h+1} + z^+_{1h}, s^+_{1,h+2} \ldots, s^+_{1r}, \mathbf{s}_2(B_r)^+),
\end{aligned}$$

and $\mathbf{s}^{**}(B_r, z^+_{1h})^-$ is almost the same as $\mathbf{s}(B_r)^-$ but $s^-_{h,h+1}$ is replaced by $s^-_{h,h+1} - z^+_{1h}$. When $h = r$, we apply (6.1.1) to

$$(m_1 + \cdots + m_r)^{-s^+_{1r}} = (m_1 + \cdots + m_{r-1})^{-s^+_{1r}} \left(1 + \frac{m_r}{m_1 + \cdots + m_{r-1}}\right)^{-s^+_{1r}}.$$

Then we have

(6.3.12)

$$\begin{aligned}
&\zeta_r(\mathbf{s}(\Delta^{**}_r(B_r)); \Delta^{**}_r(B_r)) \\
&\quad = \frac{1}{2\pi\sqrt{-1}} \int_{(c^+_{1r})} \frac{\Gamma(s^+_{1r} + z^+_{1r})\Gamma(-z^+_{1r})}{\Gamma(s^+_{1r})} \zeta_r(\mathbf{s}^{**}(B_r, z^+_{1r}); \Delta^{*}_r(B_r)) dz^+_{1r},
\end{aligned}$$

where

$$\mathbf{s}^{**}(B_r, z_{1r}^+) = (s_2, \dots, s_{r-1}, s_r - z_{1r}^+, \mathbf{s}^{**}(B_r, z_{1r}^+)^-, \mathbf{s}_2(B_r)^+)$$

and $\mathbf{s}^{**}(B_r, z_{1r}^+)^-$ is almost the same as $\mathbf{s}(B_r)^-$ but s_{1r}^- is replaced by $s_{1r}^- + s_{1r}^+ + z_{1r}^+$. Therefore we now find the recursive relation

$$\begin{aligned}(6.3.13)\qquad &\zeta_r(\cdot\,; \Delta_2^{**}(B_r)) \to \zeta_r(\cdot\,; \Delta_3^{**}(B_r)) \to \\ &\cdots \to \zeta_r(\cdot\,; \Delta_r^{**}(B_r)) \to \zeta_r(\cdot\,; \Delta_r^{*}(B_r)).\end{aligned}$$

Lastly, we consider $\Delta_h^*(B_r)$ $(3 \le h \le r)$, with the corresponding vector

$$\mathbf{s}(\Delta_h^*(B_r)) = (s_2, \dots, s_r, s_{12}^-, \dots, s_{1h}^-, \mathbf{s}_2(B_r)^-, \mathbf{s}_2(B_r)^+),$$

where $\mathbf{s}_2(B_r)^- = (s_{ij}^-)_{2 \le i < j \le r}$. Apply (6.1.1) to the term

$$(m_1 + \cdots + m_{h-1})^{-s_{1h}^-} = (m_1 + \cdots + m_{h-2})^{-s_{1h}^-} \left(1 + \frac{m_{h-1}}{m_1 + \cdots + m_{h-2}}\right)^{-s_{1h}^-}$$

to obtain

(6.3.14)

$$\begin{aligned}&\zeta_r(\mathbf{s}(\Delta_h^*(B_r)); \Delta_h^*(B_r)) \\ &= \frac{1}{2\pi\sqrt{-1}} \int_{(c_{1h}^-)} \frac{\Gamma(s_{1h}^- + z_{1h}^-)\Gamma(-z_{1h}^-)}{\Gamma(s_{1h}^-)} \\ &\qquad \times \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{2 \le i \le r} (2(m_i + \cdots + m_{r-1}) + m_r)^{-s_i} \\ &\qquad \times \prod_{2 \le j \le h-1} (m_1 + \cdots + m_{j-1})^{-s_{1j}^-} \\ &\qquad \times (m_1 + \cdots + m_{h-2})^{-s_{1h}^- - z_{1h}^-} m_{h-1}^{z_{1h}^-} \prod_{2 \le i < j \le r} (m_i + \cdots + m_{j-1})^{-s_{ij}^-} \\ &\qquad \times \prod_{2 \le i < j \le r} (m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_{r-1}) + m_r)^{-s_{ij}^+} dz_{1h}^- \\ &= \frac{1}{2\pi\sqrt{-1}} \int_{(c_{1h}^-)} \frac{\Gamma(s_{1h}^- + z_{1h}^-)\Gamma(-z_{1h}^-)}{\Gamma(s_{1h}^-)} \\ &\qquad\qquad \times \zeta_r(\mathbf{s}^*(B_r, z_{1h}^-); \Delta_{h-1}^*(B_r)) dz_{1h}^-,\end{aligned}$$

where $-\Re s_{1h}^- < c_{1h}^- < 0$,

$$\begin{aligned}\mathbf{s}^*(B_r, z_{1h}^-) = (&s_2, \dots, s_r, s_{12}^-, \dots, s_{1,h-2}^-, s_{1,h-1}^- + s_{1h}^- + z_{1h}^-, \\ &\mathbf{s}_2^*(B_r, z_{1h}^-)^-, \mathbf{s}_2(B_r)^+)\end{aligned}$$

and $\mathbf{s}_2^*(B_r, z_{1h}^-)^-$ is almost the same as $\mathbf{s}_2(B_r)^-$ but $s_{h-1,h}^-$ is replaced by $s_{h-1,h}^- - z_{1h}^-$. When $h = 3$, the sum with respect to m_1 on the right-hand

side of (6.3.14) can be separated, which implies

(6.3.15)
$$\zeta_r(\mathbf{s}^*(B_r, z_{13}^-); \Delta_2^*(B_r)) = \zeta_{r-1}(\mathbf{s}_2^*(B_r, z_{13}^-); \Delta^*(B_r))\zeta(s_{12}^- + s_{13}^- + z_{13}^-),$$

where

$$\mathbf{s}_2^*(B_r, z_{13}^-) = (s_2, \dots, s_r, \mathbf{s}_2^*(B_r, z_{13}^-)^-, \mathbf{s}_2(B_r)^+).$$

Hence we have the recursive relation

$$\begin{aligned}(6.3.16)\qquad &\zeta_r(\cdot\,; \Delta_r^*(B_r)) \to \zeta_r(\cdot\,; \Delta_{r-1}^*(B_r)) \to \\ &\cdots \to \zeta_r(\cdot\,; \Delta_3^*(B_r)) \to \zeta_{r-1}(\cdot\,; \Delta^*(B_r))\end{aligned}$$

by neglecting the Riemann zeta factor on the right-hand side of (6.3.15).

REMARK 6.6. The situation similar to (i), (ii) in Remark 6.5 also holds in the B_r case. In fact, the recursive relations (6.3.8), (6.3.13) and (6.3.16) exactly correspond to the order relation (6.3.2), and in each step, the term corresponding to the coroot is divided into two parts, one of which corresponds to the next coroot in the relation (6.3.2).

Renaming ε_i as ε_{i-1} $(2 \le i \le r)$, we see that $\Delta^*(B_r)$ coincides with $\Delta_+(B_{r-1})$. Therefore, collecting the results of the above argument, we now arrive at the following:

THEOREM 6.7 ([**104**, Theorem 4.2]). *Between $\zeta_r(\cdot\ ; B_r)$ and $\zeta_{r-1}(\cdot\ ; B_{r-1})$ (for any $r \ge 2$) there is the recursive relation given by (6.3.8), (6.3.13) and (6.3.16), which can be expressed as the Mellin–Barnes integrals (6.3.7), (6.3.11), (6.3.12) and (6.3.14). This further gives the recursive relation*

$$(6.3.17)\qquad \zeta_r(\cdot\ ; B_r) \to \zeta_{r-1}(\cdot\ ; B_{r-1}) \to \cdots \to \zeta_2(\cdot\ ; B_2) \to \zeta.$$

The treatment in the C_r case $(r \ge 2)$ is similar. In this case

$$(6.3.18)\qquad \Delta_+(C_r) = \{2\varepsilon_i \mid 1 \le i \le r\} \cup \{\varepsilon_i \pm \varepsilon_j \mid 1 \le i < j \le r\},$$

and the order relation is

$$\begin{aligned}(6.3.19)\qquad &e_1 + e_2 > e_1 + e_3 > \cdots > e_1 + e_r > e_1 \\ &> e_1 - e_r > e_1 - e_{r-1} > \cdots > e_1 - e_3 > e_1 - e_2.\end{aligned}$$

Define the set $\Delta_h^{**}(C_r)$ $(2 \le h \le r)$ by replacing $\{\varepsilon_i \mid 2 \le i \le r\}$ in the definition of $\Delta_h^{**}(B_r)$ by $\{2\varepsilon_i \mid 1 \le i \le r\}$, the sets $\Delta_h^*(C_r)$ and

$\Delta^*(C_r)$ $(2 \le h \le r)$ by replacing $\{\varepsilon_i \mid 2 \le i \le r\}$ in the definitions of $\Delta_h^*(B_r)$ and $\Delta^*(B_r)$ respectively by $\{2\varepsilon_i \mid 2 \le i \le r\}$, and

$$\begin{aligned}\Delta^{**}(C_r) =&\{2\varepsilon_i \mid 1 \le i \le r\} \cup \{\varepsilon_i - \varepsilon_j \mid 1 \le i < j \le r\}\\ &\cup \{\varepsilon_i + \varepsilon_j \mid 2 \le i < j \le r\}.\end{aligned} \tag{6.3.20}$$

Then we find

$$\begin{aligned}\Delta^*(C_r) \subset \Delta_2^*(C_r) \subset \cdots \subset \Delta_r^*(C_r) \subset \Delta^{**}(C_r)\\ \subset \Delta_r^{**}(C_r) \subset \cdots \subset \Delta_2^{**}(C_r) \subset \Delta_+(C_r).\end{aligned} \tag{6.3.21}$$

Guided by the same principle as in Remark 6.6, we can easily find the corresponding Mellin–Barnes recursive structure

(6.3.22)

$$\begin{aligned}&\zeta_r(\cdot\,; C_r) \to \zeta_r(\cdot\,; \Delta_2^{**}(C_r)) \to \zeta_r(\cdot\,; \Delta_3^{**}(C_r)) \to\\ &\cdots \to \zeta_r(\cdot\,; \Delta_r^{**}(C_r)) \to \zeta_r(\cdot\,; \Delta^{**}(C_r)) \to \zeta_r(\cdot\,; \Delta_r^*(C_r))\\ &\to \zeta_r(\cdot\,; \Delta_{r-1}^*(C_r)) \to \cdots \to \zeta_r(\cdot\,; \Delta_3^*(C_r)) \to \zeta_{r-1}(\cdot\,; \Delta^*(C_r)).\end{aligned}$$

In the D_r case $(r \ge 3)$, we have

$$\Delta_+(D_r) = \{\varepsilon_i \pm \varepsilon_j \mid 1 \le i < j \le r\}. \tag{6.3.23}$$

Hence the vector $\mathbf{s}(D_r)$ must be written as

$$\mathbf{s}(D_r) = (\mathbf{s}(D_r)^-, \mathbf{s}(D_r)^+),$$

where $\mathbf{s}(D_r)^\pm = (s_{ij}^\pm)_{1 \le i < j \le r}$. Define $\mathbf{s}_2(D_r)^\pm = (s_{ij}^\pm)_{2 \le i < j \le r}$,

$$\Delta_h^*(D_r) = \{\varepsilon_1 - \varepsilon_j \mid 2 \le j \le h\} \cup \{\varepsilon_i \pm \varepsilon_j \mid 2 \le i < j \le r\} \tag{6.3.24}$$

$(2 \le h \le r)$, $\Delta^*(D_r) = \{\varepsilon_i \pm \varepsilon_j \mid 2 \le i < j \le r\}$ and

$$\begin{aligned}\Delta_h^{**}(D_r) =&\{\varepsilon_i - \varepsilon_j \mid 1 \le i < j \le r\} \cup \{\varepsilon_1 + \varepsilon_j \mid h \le j \le r\}\\ &\cup \{\varepsilon_i + \varepsilon_j \mid 2 \le i < j \le r\}\end{aligned} \tag{6.3.25}$$

$(2 \le h \le r)$. Then, corresponding to the order relation

$$\begin{aligned}e_1 + e_2 > e_1 + e_3 > \cdots > e_1 + e_r\\ > e_1 - e_r > \cdots > e_1 - e_3 > e_1 - e_2,\end{aligned} \tag{6.3.26}$$

we have

$$\begin{aligned}\Delta^*(D_r) \subset \Delta_2^*(D_r) \subset \cdots \subset \Delta_r^*(D_r)\\ \subset \Delta_r^{**}(D_r) \subset \cdots \subset \Delta_2^{**}(D_r) \subset \Delta_+(D_r).\end{aligned} \tag{6.3.27}$$

According to this relation, we give the Mellin–Barnes recursive structure. First, by using

$$\begin{aligned}&(m_1 + \cdots + m_{h-1} + 2(m_h + \cdots + m_{r-2}) + m_{r-1} + m_r)^{-s_{1h}^+}\\ &= (m_1 + \cdots + m_h + 2(m_{h+1} + \cdots + m_{r-2}) + m_{r-1} + m_r)^{-s_{1h}^+}\end{aligned}$$

$$\times \left(1+\frac{m_h}{m_1+\cdots+m_h+2(m_{h+1}+\cdots+m_{r-2})+m_{r-1}+m_r}\right)^{-s^+_{1h}}$$

$(2 \le h \le r-2)$ and

$$(m_1+\cdots+m_{r-2}+m_{r-1}+m_r)^{-s^+_{1,r-1}}$$
$$= (m_1+\cdots+m_{r-2}+m_r)^{-s^+_{1,r-1}}\left(1+\frac{m_{r-1}}{m_1+\cdots+m_{r-2}+m_r}\right)^{-s^+_{1,r-1}},$$

we have

(6.3.28)

$$\zeta_r(\mathbf{s}(\Delta^{**}_h(D_r));\Delta^{**}_h(D_r))$$
$$= \frac{1}{2\pi\sqrt{-1}}\int_{(c^+_{1h})}\frac{\Gamma(s^+_{1h}+z^+_{1h})\Gamma(-z^+_{1h})}{\Gamma(s^+_{1h})}\zeta_r(\mathbf{s}^{**}(D_r,z^+_{1h});\Delta^{**}_{h+1}(D_r))dz^+_{1h},$$

for $2 \le h \le r-1$, where

$$\mathbf{s}^{**}(D_r,z^+_{1h}) = (\mathbf{s}^{**}(D_r,z^+_{1h})^-, s^+_{1h}+s^+_{1,h+1}+z^+_{1h}, s^+_{1,h+2},\dots,s^+_{1r},\mathbf{s}_2(D_r)^+)$$

and $\mathbf{s}^{**}_2(D_r,z^+_{1h})^-$ is almost the same as $\mathbf{s}_2(D_r)^-$ but $s^-_{h,h+1}$ is replaced by $s^-_{h,h+1}-z^+_{1h}$.

Next, the term corresponding to the coroot e_1+e_r is

(6.3.29) $$((m_1+\cdots+m_{r-2})+m_r)^{-s^+_{1r}}$$
$$= (m_1+\cdots+m_{r-2})^{-s^+_{1r}}\left(1+\frac{m_r}{m_1+\cdots+m_{r-2}}\right)^{-s^+_{1r}}.$$

Applying (6.1.1) to the above, we obtain

(6.3.30)

$$\zeta_r(\mathbf{s}(\Delta^{**}_r(D_r));\Delta^{**}_r(D_r))$$
$$= \frac{1}{2\pi\sqrt{-1}}\int_{(c^+_{1r})}\frac{\Gamma(s^+_{1r}+z^+_{1r})\Gamma(-z^+_{1r})}{\Gamma(s^+_{1r})}$$
$$\times \prod_{2\le i\le r-2}((m_i+\cdots+m_{r-2})+m_r)^{-s^+_{ir}}m_r^{-s^+_{r-1,r}+z^+_{1r}}$$
$$\times \prod_{1\le i<j\le r}(m_i+\cdots+m_{j-1})^{-s^-_{ij}}(m_1+\cdots+m_{r-2})^{-s^+_{1r}-z^+_{1r}}$$
$$\times \prod_{2\le i<j<r}(m_i+\cdots+m_{j-1}+2(m_j+\cdots+m_{r-2})+m_{r-1}+m_r)^{-s^+_{ij}}dz^+_{1r}$$
$$= \frac{1}{2\pi\sqrt{-1}}\int_{(c^+_{1r})}\frac{\Gamma(s^+_{1r}+z^+_{1r})\Gamma(-z^+_{1r})}{\Gamma(s^+_{1r})}\zeta_r(\mathbf{s}^{**}(D_r,z^+_{1r});\Delta^*_r(D_r))dz^+_{1r},$$

where

$$\mathbf{s}^{**}(D_r,z^+_{1r}) = (\mathbf{s}^{**}(D_r,z^+_{1r})^-, \mathbf{s}^{**}_2(D_r,z^+_{1r})^+),$$

$\mathbf{s}^{**}(D_r, z_{1r}^+)^-$ is almost the same as $\mathbf{s}(D_r)^-$ but $s_{1,r-1}^-$ is replaced by $s_{1,r-1}^- + s_{1r}^+ + z_{1r}^+$, and $\mathbf{s}_2^{**}(D_r, z_{1r}^+)^+$ is almost the same as $\mathbf{s}_2(D_r)^+$ but $s_{r-1,r}^+$ is replaced by $s_{r-1,r}^+ - z_{1r}^+$.

Finally, similar to (6.3.14) we obtain

(6.3.31)

$$\zeta_r(\mathbf{s}(\Delta_h^*(D_r)); \Delta_h^*(D_r)) = \frac{1}{2\pi\sqrt{-1}} \int_{(c_{1h}^-)} \frac{\Gamma(s_{1h}^- + z_{1h}^-)\Gamma(-z_{1h}^-)}{\Gamma(s_{1h}^-)} \zeta_r(\mathbf{s}^*(D_r, z_{1h}^-); \Delta_{h-1}^*(D_r)) dz_{1h}^-$$

$(3 \le h \le r)$, where

$$\mathbf{s}^*(D_r, z_{1h}^-) = (s_{12}^-, \dots, s_{1,h-2}^-, s_{1,h-1}^- + s_{1h}^- + z_{1h}^-, \mathbf{s}_2^*(D_r, z_{1h}^-)^-, \mathbf{s}_2(D_r)^+)$$

and $\mathbf{s}_2^*(D_r, z_{1h}^-)^-$ is almost the same as $\mathbf{s}_2(D_r)^-$ but $s_{h-1,h}^-$ is replaced by $s_{h-1,h}^- - z_{1h}^-$. When $h = 3$, we further find

(6.3.32)

$$\zeta_r(\mathbf{s}^*(D_r, z_{13}^-); \Delta_2^*(D_r)) = \zeta_{r-1}(\mathbf{s}_2^*(D_r, z_{13}^-); \Delta^*(D_r))\zeta(s_{12}^- + s_{13}^- + z_{13}^-)$$

with

$$\mathbf{s}_2^*(D_r, z_{13}^-) = (\mathbf{s}_2^*(D_r, z_{13}^-)^-, \mathbf{s}_2(D_r)^+).$$

Therefore, we now find the recursive structure

(6.3.33)
$$\begin{aligned} &\zeta_r(\cdot\,; D_r) \to \zeta_r(\cdot\,; \Delta_2^{**}(D_r)) \to \zeta_r(\cdot\,; \Delta_3^{**}(D_r)) \to \\ &\quad \cdots \to \zeta_r(\cdot\,; \Delta_r^{**}(D_r)) \to \zeta_r(\cdot\,; \Delta_r^*(D_r)) \to \zeta_r(\cdot\,; \Delta_{r-1}^*(D_r)) \to \\ &\quad \cdots \to \zeta_r(\cdot\,; \Delta_3^*(D_r)) \to \zeta_{r-1}(\cdot\,; \Delta^*(D_r)). \end{aligned}$$

Summarizing the above, we obtain:

THEOREM 6.8 ([**104**, Theorem 4.3]). *In the C_r and D_r cases, there are recursive structures* (6.3.22) *and* (6.3.33)*, which can be expressed as Mellin–Barnes integrals. These further give recursive relations*

(6.3.34) $$\zeta_r(\cdot\,; C_r) \to \zeta_{r-1}(\cdot\,; C_{r-1}) \to \cdots \to \zeta_2(\cdot\,; C_2) \to \zeta$$

and

(6.3.35) $$\zeta_r(\cdot\,; D_r) \to \zeta_{r-1}(\cdot\,; D_{r-1}) \to \cdots \to \zeta_2(\cdot\,; D_2).$$

REMARK 6.9. The situation described in Remarks 6.5 and 6.6 also holds in the C_r case. On the other hand, it should be noted that in the D_r case, the principle of type (ii) is no longer valid at the step $e_1 + e_r \to e_1 - e_r$. In fact, (6.3.29) shows that the sum $(m_1 + \cdots + m_{r-2}) + m_r$, corresponding to $e_1 + e_r$, is divided into $m_1 + \cdots + m_{r-2}$ and m_r, and $m_1 + \cdots + m_{r-2}$ corresponds not to $e_1 - e_r$ but to $e_1 - e_{r-1}$.

6.4. Recursive structures and Dynkin diagrams

Theorems 6.4, 6.7 and 6.8, proved in the previous sections, give certain recursive structures among our zeta-functions of root systems. In this section we discuss that we can find many other recursive relations among those zeta-functions.

In order to describe the situation, it is better to introduce here the viewpoint of Dynkin diagrams. Let $\Psi = \{\alpha_1, \dots, \alpha_r\}$ be the fundamental system of Δ, and define the *Dynkin diagram* $\Gamma(\Delta)$ associated with Δ as follows. First, with each α_i, we associate a vertex, with the weight $\langle \alpha_i, \alpha_i \rangle$. Any two different vertices α_i and α_j are connected by $a(\alpha_i, \alpha_j) \cdot a(\alpha_j, \alpha_i)$ edges (where $a(\cdot, \cdot)$ is the Cartan integer defined by (2.2.3)). The number of edges connecting two vertices is 0, 1, 2, or 3. In the case when the number of edges is 2 or 3, we add an arrow, pointing from the vertex of higher weight to that of lower weight. If $\langle \alpha_i, \alpha_j \rangle = 0$, then there is no edge which connects α_i and α_j. Therefore we see that $\Gamma(\Delta)$ is connected if and only if Δ is irreducible. When Δ is of type X_r, we write $\Gamma(\Delta) = \Gamma(X_r)$.

Let us write the Dynkin diagram $\Gamma(A_r)$. For $i \neq j$,

$$\alpha_i + t\alpha_j = \varepsilon_i - \varepsilon_{i+1} + t(\varepsilon_j - \varepsilon_{j+1})$$

is a root if and only if $t = 0$, or $t = 1$ with $|i - j| = 1$. Therefore if $|i-j| = 1$, the α_j-string through α_i consists of just two elements, hence $a(\alpha_i, \alpha_j) = -1$ by (2.2.4). If $|i - j| \geq 2$, then $a(\alpha_i, \alpha_j) = 0$. Therefore the diagram $\Gamma(A_r)$ is as in Figure 6.4.1.

Theorems 6.4, 6.7 and 6.8 give the recursive relations of the form

$$\zeta_r(\cdot\,; X_r) \to \zeta_{r-1}(\cdot\,; X_{r-1}) \tag{6.4.1}$$

for $X = A, B, C$ and D, by separating m_1 which corresponds to the coroot $\alpha_1^\vee$. In terms of Dynkin diagrams, this is the procedure of cutting off the leftmost edge, that is the edge joining the vertices corresponding to $\alpha_1^\vee$ and $\alpha_2^\vee$, of $\Gamma(X_r)$ (see Figure 6.4.1).

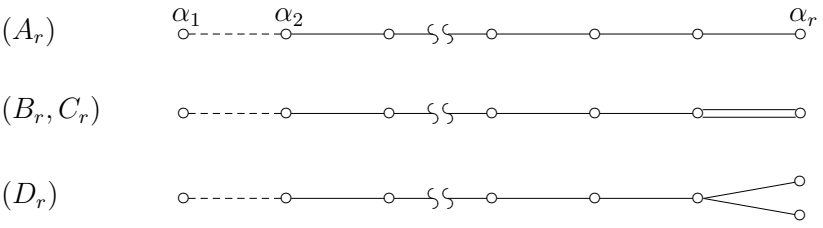


FIGURE 6.4.1.

However, it is also possible to cut off the rightmost edge(s) of $\Gamma(X_r)$. This can be realized by separating m_r corresponding to the coroot $\alpha_r^\vee$. In the case when $X_r = B_r$, the coroots involving $\alpha_r^\vee$ are $2e_i$ $(1 \le i \le r)$ and $e_i + e_j$ $(1 \le i < j \le r)$. For each i $(1 \le i \le r-1)$, we can construct the Mellin–Barnes recursive relation by following

$$2e_i > e_i + e_{i+1} > e_i + e_{i+2} > \cdots > e_i + e_r > e_i - e_r, \tag{6.4.2}$$

similar to the argument in the previous sections. The consequence is that we have a multiple integral expression of $\zeta_r(\mathbf{s}; B_r)$ involving $\zeta_{r-1}(\cdot\,; A_{r-1})$ and ζ. That is, we find the recursive relation

$$\zeta_r(\cdot\,; B_r) \to \zeta_{r-1}(\cdot\,; A_{r-1}). \tag{6.4.3}$$

Similarly, we can obtain the relation

$$\zeta_r(\cdot\,; C_r) \to \zeta_{r-1}(\cdot\,; A_{r-1}), \tag{6.4.4}$$

by using

$$e_i + e_{i+1} > e_i + e_{i+2} > \cdots > e_i + e_r > e_i > e_i - e_r, \tag{6.4.5}$$

and

$$\zeta_r(\cdot\,; D_r) \to \zeta_{r-1}(\cdot\,; A_{r-1}), \tag{6.4.6}$$

by using

$$e_i + e_{i+1} > e_i + e_{i+2} > \cdots > e_i + e_r > e_i - e_{r-1} \tag{6.4.7}$$

(see Figure 6.4.2).

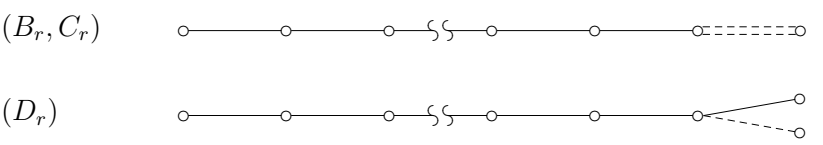


FIGURE 6.4.2.

In the cases of B_r and C_r, we can also cut off only one of the doubled rightmost edges of the diagram, which gives the relation

$$\zeta_r(\cdot\,; B_r) \to \zeta_r(\cdot\,; A_r), \qquad \zeta_r(\cdot\,; C_r) \to \zeta_r(\cdot\,; A_r). \tag{6.4.8}$$

Consider the B_r case. The sum $m_i + \cdots + m_r$, corresponding to the coroot $e_i + e_r$, exists as one of the factors of $\zeta_r(\mathbf{s}; A_r)$. Hence the final step of (6.4.2) is not necessary this time, and we obtain the first relation of (6.4.8) by following

$$2e_i > e_i + e_{i+1} > e_i + e_{i+2} > \cdots > e_i + e_r. \tag{6.4.9}$$

Similarly, by following

$$e_i + e_{i+1} > e_i + e_{i+2} > \cdots > e_i + e_r > e_i, \tag{6.4.10}$$

which is (6.4.5) without the last step, we can obtain the second relation of (6.4.8) (see Figure 6.4.3).

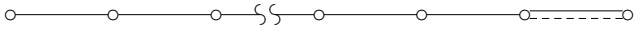
(B_r, C_r) 

FIGURE 6.4.3.

In general, we can cut off any edge of the diagram and reduce a zeta-function for a root system to that for another root system. To state our assertion, we recall the definition concerning root systems and construct embeddings in a non-standard sense (that is, embeddings which do not preserve the inner products).

Let Δ be a reduced root system which may not be irreducible, Γ its Dynkin diagram and $\Psi = \{\alpha_1, \ldots, \alpha_r\}$ its fundamental system. (The j-th vertex from the left on Γ corresponds to α_j.) By $Q^\vee$ we denote the coroot lattice generated by $\Psi^\vee$. Let Γ' be a Dynkin diagram obtained by cutting off some edges from Γ. Let Δ', $\Psi' = \{\alpha'_1, \ldots, \alpha'_n\}$ and $(Q')^\vee$ be the corresponding root system, fundamental system and coroot lattice, respectively. Then we see that the map $f : (\Psi')^\vee \to \Psi^\vee$ defined by $f : (\alpha'_j)^\vee \mapsto \alpha_j^\vee$ is $\mathbb{Z}$-linearly extended to an isomorphism $f : (Q')^\vee \to Q^\vee$ as $\mathbb{Z}$-modules.

For $\beta^\vee = \sum_j c_j \alpha_j^\vee \in Q^\vee$, we denote its height by

$$\operatorname{ht} \beta^\vee = \sum_j c_j. \tag{6.4.11}$$

LEMMA 6.10 ([104, Lemma 5.1]). $f((\Delta')^\vee_+) \subset \Delta^\vee_+$.

PROOF. We show the statement by induction on their heights. We denote by $\beta^\vee \in Q^\vee$ the image $f((\beta')^\vee)$ of $(\beta')^\vee \in (\Delta')^\vee$. We first note that $\langle \alpha'_i, (\alpha'_j)^\vee \rangle \geq \langle \alpha_i, \alpha_j^\vee \rangle$ because the value of $\langle \alpha_i, \alpha_j^\vee \rangle$ is only $0, -1, -2$ or -3 for $i \neq j$ and cutting off some edges of the diagram only produces the effect of increasing the value. Hence in general, we have

$$\langle \alpha'_i, (\beta')^\vee \rangle \geq \langle \alpha_i, \beta^\vee \rangle \tag{6.4.12}$$

for $(\beta')^\vee \in (\Delta')^\vee_+$.

Because $(\beta')^\vee \in (\Delta')^\vee_+$ with $\operatorname{ht}(\beta')^\vee = 1$ implies $(\beta')^\vee = (\alpha'_i)^\vee \in (\Psi')^\vee$, for some i, it follows that $\beta^\vee = \alpha_i^\vee \in \Psi^\vee$ by definition.

Let $m \geq 1$, and assume $\beta^\vee \in \Delta_+^\vee$ for $(\beta')^\vee \in (\Delta')_+^\vee$ with $\mathrm{ht}(\beta')^\vee \leq m$. Let $(\beta')^\vee = \sum_j c'_j(\alpha'_j)^\vee \in (\Delta')_+^\vee$ with $\mathrm{ht}(\beta')^\vee = m+1$. It is known that there exists a decomposition $(\beta')^\vee = (\alpha')^\vee + (\alpha'_i)^\vee$ with $(\alpha')^\vee \in (\Delta')_+^\vee$, $\mathrm{ht}(\alpha')^\vee = m$ and $(\alpha'_i)^\vee \in (\Psi')^\vee$ (see, e.g., [70, Section 10.2, Lemma A] or [178, Section 2.11, Proposition A]).

If $\langle \alpha'_i, (\alpha')^\vee \rangle < 0$, then $\langle \alpha_i, \alpha^\vee \rangle < 0$ by (6.4.12). By the assumption of induction we have $\alpha^\vee \in \Delta_+^\vee$ and the simple reflection

$$r_i(\alpha^\vee) = \alpha^\vee - \langle \alpha_i, \alpha^\vee \rangle \alpha_i^\vee$$

is also $\in \Delta_+^\vee$. Hence $\beta^\vee = \alpha^\vee + \alpha_i^\vee \in \Delta_+^\vee$ since all the roots in the $\alpha_i^\vee$-string through $\alpha^\vee$ belong to $\Delta^\vee$.

If $\langle \alpha'_i, (\alpha')^\vee \rangle \geq 0$, we consider $(\gamma')^\vee = r'_i((\beta')^\vee)$, where r'_i is the simple reflection with respect to α'_i. Then $(\gamma')^\vee \in \Delta^\vee$ because $(\beta')^\vee \in \Delta^\vee$. We see that

$$(\gamma')^\vee = r'_i((\alpha')^\vee + (\alpha'_i)^\vee) = (\alpha')^\vee - (\langle \alpha'_i, (\alpha')^\vee \rangle + 1)(\alpha'_i)^\vee,$$

which implies that $\mathrm{ht}(\gamma')^\vee \leq m-1$. Moreover, we see that $(\gamma')^\vee \in \Delta_+^\vee$; in fact, since $\mathrm{ht}(\beta')^\vee = m+1 \geq 2$, there exists a $j \neq i$ for which c'_j is positive, and so the coefficient of $(\alpha'_j)^\vee$ in $(\gamma')^\vee$ is also positive, which implies $(\gamma')^\vee \in (\Delta')_+^\vee$. Hence by the assumption of induction we have $\alpha^\vee \in \Delta_+^\vee$, $\gamma^\vee \in \Delta_+^\vee$ and

$$r_i(\gamma^\vee) = \alpha^\vee + (\langle \alpha'_i, (\alpha')^\vee \rangle - \langle \alpha_i, \alpha^\vee \rangle + 1)\alpha_i^\vee \in \Delta_+^\vee.$$

Hence $\beta^\vee = \alpha^\vee + \alpha_i^\vee \in \Delta_+^\vee$. □

THEOREM 6.11 ([104, Theorem 5.2]). *Let Δ, Δ' be reduced root systems and Γ, Γ' be their Dynkin diagrams. Then there exists an isomorphism $f : (Q')^\vee \to Q^\vee$ such that $f((\Delta')_+^\vee) \subset \Delta_+^\vee$ and $f : (\alpha'_j)^\vee \mapsto \alpha_j^\vee$ if and only if Γ' is obtained from Γ by cutting off some edges.*

PROOF. By Lemma 6.10, we have only to show $\langle \alpha'_i, (\alpha'_j)^\vee \rangle \geq \langle \alpha_i, \alpha_j^\vee \rangle$ if such f exists. Because $\langle \alpha'_i, (\alpha'_j)^\vee \rangle = \langle \alpha_i, \alpha_j^\vee \rangle = 2$ if $i = j$, we assume $i \neq j$. Since $(\Delta')_+^\vee \ni r'_i(\alpha'_j)^\vee = (\alpha'_j)^\vee - \langle \alpha'_i, (\alpha'_j)^\vee \rangle (\alpha'_i)^\vee$, we have $\Delta_+^\vee \ni f(r'_i(\alpha'_j)^\vee) = \alpha_j^\vee - \langle \alpha'_i, (\alpha'_j)^\vee \rangle \alpha_i^\vee$.

On the other hand, we have $\Delta^\vee \ni r_i \alpha_j^\vee = \alpha_j^\vee - \langle \alpha_i, \alpha_j^\vee \rangle \alpha_i^\vee$. Since the length of the $\alpha_i^\vee$-string through $\alpha_j^\vee$ is $\langle \alpha_i, \alpha_j^\vee \rangle$, it follows that the string consists of $\{\alpha_j^\vee + h\alpha_i^\vee \mid 0 \leq h \leq -\langle \alpha_i, \alpha_j^\vee \rangle\}$. Therefore we have $0 \geq \langle \alpha'_i, (\alpha'_j)^\vee \rangle \geq \langle \alpha_i, \alpha_j^\vee \rangle$. □

Since f is injective, we identify $(\Delta')_+^\vee$ and its image by f, and denote the image by the same symbol $(\Delta')_+^\vee$.

Let $(\Delta^*)^\vee = \Delta_+^\vee \setminus (\Delta')_+^\vee$ and $k = \#\Delta^*$. We fix an order

$$(\Delta^*)^\vee = \{\beta_1^\vee, \beta_2^\vee, \dots, \beta_k^\vee\} \tag{6.4.13}$$

by their heights

$$\operatorname{ht}\beta_1^\vee \le \operatorname{ht}\beta_2^\vee \le \cdots \le \operatorname{ht}\beta_k^\vee. \tag{6.4.14}$$

For $0 \le j \le k$, define

$$(\Delta_j^*)^\vee = (\Delta')_+^\vee \cup \{\beta_1^\vee, \dots, \beta_j^\vee\} \tag{6.4.15}$$

so that

$$(\Delta')_+^\vee = (\Delta_0^*)^\vee \subset (\Delta_1^*)^\vee \subset \cdots \subset (\Delta_{k-1}^*)^\vee \subset (\Delta_k^*)^\vee = \Delta_+^\vee. \tag{6.4.16}$$

Then for $1 \le j \le k$ we have a decomposition $\beta_j^\vee = \alpha_l^\vee + \gamma^\vee$ with some $\alpha_l^\vee \in \Psi^\vee$ and $\gamma^\vee \in (\Delta_{j-1}^*)^\vee$ since the order is determined by their heights. Then, again by the Mellin–Barnes argument, we have

(6.4.17)

$$\begin{aligned}
\zeta_r(\mathbf{s}; \Delta_j^*) &= \sum_\lambda \prod_{\alpha^\vee \in (\Delta_j^*)^\vee} \langle \alpha^\vee, \lambda\rangle^{-s_\alpha} \\
&= \sum_\lambda \left(\prod_{\alpha^\vee \in (\Delta_{j-1}^*)^\vee} \langle \alpha^\vee, \lambda\rangle^{-s_\alpha} \right) \langle \alpha_l^\vee + \gamma^\vee, \lambda\rangle^{-s_{\beta_j}} \\
&= \sum_\lambda \prod_{\alpha^\vee \in (\Delta_{j-1}^*)^\vee \setminus \{\alpha_l^\vee, \gamma^\vee\}} \langle \alpha^\vee, \lambda\rangle^{-s_\alpha} \\
&\quad \times \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_{\beta_j}+z)\Gamma(-z)}{\Gamma(s_{\beta_j})} \langle \alpha_l^\vee, \lambda\rangle^{-s_{\alpha_l}+z} \langle \gamma^\vee, \lambda\rangle^{-s_{\beta_j}-s_\gamma-z} dz,
\end{aligned}$$

where Δ_j^* is the set of positive roots corresponding to $(\Delta_j^*)^\vee$. This implies the following theorem.

THEOREM 6.12 ([104, Theorem 5.3]). *We have*

$$\begin{aligned}
\zeta_r(\mathbf{s}; \Delta_j^*) &= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_{\beta_j}+z)\Gamma(-z)}{\Gamma(s_{\beta_j})} \\
&\quad \times \zeta_r(\dots, s_{\alpha_l} - z, \dots, s_{\beta_j} + s_\gamma + z, \dots; \Delta_{j-1}^*) dz,
\end{aligned} \tag{6.4.18}$$

and, repeating this procedure k times, we obtain the recursive relation

$$\zeta_r(\cdot\,; \Delta_+) \to \zeta_r(\cdot\,; \Delta'_+). \tag{6.4.19}$$

This general result includes all examples discussed above. Let $\Gamma = \Gamma(X_r)$, $X = A, B, C$ or D. When $X = B$ or C and Γ' is obtained

by removing only one of the doubled rightmost edges of Γ, then Γ' is irreducible. These cases are described as (6.4.8).

Except for those cases, any Γ' which is obtained by cutting off edge(s) of Γ is not irreducible, hence the corresponding zeta-function $\zeta_r(\cdot\,;\Delta'_+)$ is the product of two (or more) zeta-functions.

If we cut off the leftmost edge, then $\zeta_r(\cdot\,;\Delta'_+)$ is the product of $\zeta_{r-1}(\cdot\,;X_{r-1})$ and the Riemann zeta-function. These cases are discussed in detail in Sections 6.3 and 6.4. The cases of cutting off the rightmost edge(s) are presented as (6.4.3), (6.4.4) and (6.4.6).

More generally, we can cut off the edge which joins two vertices corresponding to $\alpha^\vee_{\ell-1}$ and $\alpha^\vee_\ell$ ($2 \le \ell \le r$ for $X = A$, $2 \le \ell \le r-1$ for $X = B$ or C, and $2 \le \ell \le r-2$ for $X = D$). Then (6.4.18) implies that $\zeta_r(\cdot\,;X_r)$ can be written as a multiple integral involving $\zeta_{\ell-1}(\cdot\,;A_{\ell-1})$ and $\zeta_{r-\ell+1}(\cdot\,;X_{r-\ell+1})$ (see Figure 6.4.4).

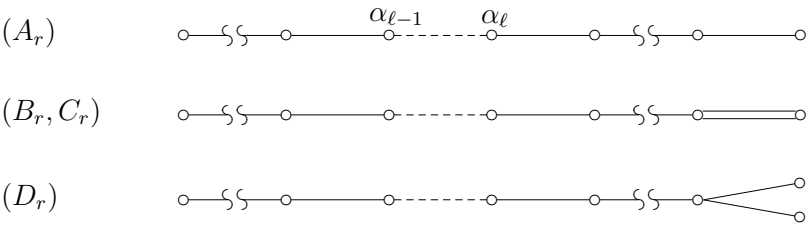


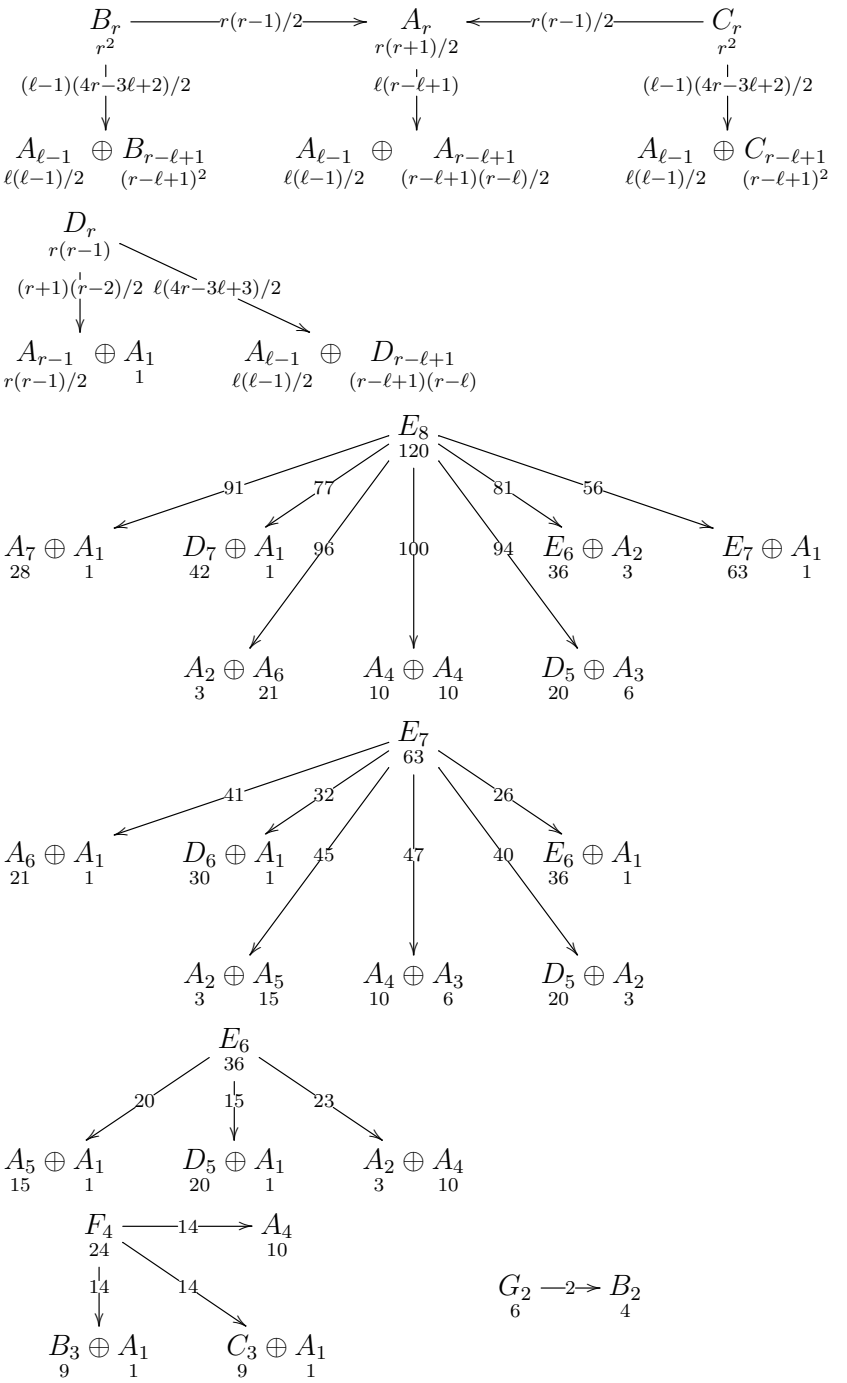
FIGURE 6.4.4.

We can summarize the above argument as follows.

THEOREM 6.13 ([**104**, Theorem 5.4]). *By cutting off any edge of a Dynkin diagram, we find that the zeta-function of the corresponding root system can be written as a (multiple) integral, whose integrand includes zeta-functions of each connected components of the resulting Dynkin diagram.*

The following diagrams show the hierarchy of Lie algebras whose zeta-functions are connected with each other by Mellin–Barnes recursive formulas as above. The number attached to each algebra is the number of positive roots, that is, the number of variables of the associated zeta-function. The number written in the middle of each arrow is the number of iteration of integrals. The horizontal arrow implies that, in the corresponding process, the Dynkin diagram is not divided

into two separate parts. Note that the following diagrams include the cases of exceptional Lie algebras.



CHAPTER 7

The meromorphic continuation

According to the recursive structure proved in the preceding chapter, we may study the zeta-function of a certain root system of rank r by using the information of zeta-functions of root systems of lower rank. In this chapter, as an example of such reduction philosophy, we will show the meromorphic continuation of zeta-functions of root systems by an induction argument.

In this chapter, for two functions f and g, we use Landau's symbol $f = O(g)$ and Vinogradov's symbol $f \ll g$. Both Landau's symbol and Vinogradov's symbol mean that $|f(x)| \leq Cg(x)$ with a certain constant $C > 0$ in some indicated region of x. We call C the implied constant.

7.1. The meromorphic continuation of zeta-functions of root systems

Using the Mellin–Barnes formula, it is possible to prove the meromorphic continuation of zeta-functions of root systems to the whole complex space. In order to explain the basic idea, we first consider the simplest example, that is the case of $\zeta_2(\mathbf{s}, A_2)$, which was already written in the second-named author's article [131].

At first assume $\Re(s_j) > 1$ $(1 \leq j \leq 3)$. Applying Lemma 6.1, we have

(7.1.1)

$$\begin{aligned}
\zeta_2(\mathbf{s}; A_2) &= \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1} m_2^{-s_2} (m_1 + m_2)^{-s_3} \\
&= \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1-s_3} m_2^{-s_2} (1 + \frac{m_2}{m_1})^{-s_3} \\
&= \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1-s_3} m_2^{-s_2} \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_3+z)\Gamma(-z)}{\Gamma(s_3)} \left(\frac{m_2}{m_1}\right)^z dz \\
&= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_3+z)\Gamma(-z)}{\Gamma(s_3)} \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1-s_3-z} m_2^{-s_2+z} dz
\end{aligned}$$

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$$= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_3+z)\Gamma(-z)}{\Gamma(s_3)} \zeta(s_1+s_3+z)\zeta(s_2-z)dz.$$

Here, the two zeta factors in the integrand of the last member of the above are convergent, because $-\Re(s_3) < c < 0$ so

$$\Re(s_1+s_3+z) = \Re(s_1)+\Re(s_3)+c > \Re(s_1) > 1,$$
$$\Re(s_2-z) = \Re(s_2)-c > \Re(s_2) > 1.$$

Stirling's formula (6.1.2) implies that the above integrand is rapidly decreasing when $\Im(z) \to \infty$. Therefore it is possible to shift the path of integration to the right. We shift the path to $\Re(z) = M - \frac{1}{2}$, where $M \in \mathbb{N}$. The locations of the poles of the integrand are

$$\begin{aligned}
&z = -s_3, -s_3-1, -s_3-2, \dots && \text{coming from} \quad \Gamma(s_3+z),\\
&z = 0, 1, 2, \dots && \text{coming from} \quad \Gamma(-z),\\
&z = 1-s_1-s_3 && \text{coming from} \quad \zeta(s_1+s_3+z),\\
&z = s_2-1 && \text{coming from} \quad \zeta(s_2-z).
\end{aligned}$$

Therefore the poles relevant to the above shifting of the path are $z = s_2 - 1$ and $z = \ell$ $(0 \le \ell \le M-1)$. Counting the residues at these poles, we obtain

$$\begin{aligned}
\zeta_2(\mathbf{s}; A_2) = {} & \frac{\Gamma(s_2+s_3-1)\Gamma(1-s_2)}{\Gamma(s_3)} \zeta(s_1+s_2+s_3-1)\\
& + \sum_{\ell=0}^{M-1} \binom{-s_3}{\ell} \zeta(s_1+s_3+\ell)\zeta(s_2-\ell)\\
& + I_M,
\end{aligned} \tag{7.1.2}$$

where

$$I_M = \frac{1}{2\pi\sqrt{-1}} \int_{(M-\frac{1}{2})} \frac{\Gamma(s_3+z)\Gamma(-z)}{\Gamma(s_3)} \zeta(s_1+s_3+z)\zeta(s_2-z)dz.$$

The poles $-s_3-\ell$ are not on the path of integration $\Re z = M - 1/2$ if $\Re(s_3) > -M + \frac{1}{2}$. Similarly, the pole $1-s_1-s_3$ is not on the path if $\Re(s_1)+\Re(s_3) > -M+\frac{3}{2}$, and the pole s_2-1 is not on the path if $\Re(s_2) < M + \frac{1}{2}$. Therefore I_M is holomorphic in the region

$$D_M = \left\{ (s_1,s_2,s_3) \in \mathbb{C}^3 \;\middle|\; \Re(s_3) > -M+\frac{1}{2}, \right.$$
$$\left. \Re(s_1)+\Re(s_3) > -M+\frac{3}{2}, \Re(s_2) < M+\frac{1}{2} \right\}.$$

The terms coming from the residues on the right-hand side of (7.1.2) are clearly meromorphic in the whole space $\mathbb{C}^3$. Since M is arbitrary, we may conclude that $\zeta_2(\mathbf{s}; A_2)$ can be continued meromorphically to the

whole space $\mathbb{C}^3$. Moreover, we can list the candidates of singularities of $\zeta_2(\mathbf{s}; A_2)$ from the terms coming from the residues, which are

$$\begin{aligned}
&s_2 + s_3 = 1 - \ell \ \ (\ell \in \mathbb{N}_0) \qquad \text{coming from} \quad \Gamma(s_2 + s_3 - 1),\\
&s_2 = 1 + \ell \qquad \text{coming from} \quad \Gamma(1 - s_2),\\
&s_1 + s_2 + s_3 = 2 \qquad \text{coming from} \quad \zeta(s_1 + s_2 + s_3 - 1),\\
&s_1 + s_3 = 1 - \ell \qquad \text{coming from} \quad \zeta(s_1 + s_3 + \ell),\\
&s_2 = 1 + \ell \qquad \text{coming from} \quad \zeta(s_2 - \ell).
\end{aligned}$$

However, we can easily check that two possible singularities coming from two factors ($\Gamma(1 - s_2)$ and $\zeta(s_2 - \ell)$) are actually cancelled out. The other candidates in the above list are not cancelled, so they are "true" singularities. Therefore we now arrive at the following:

THEOREM 7.1 ([**131**, Theorem 1]). *The function $\zeta_2(\mathbf{s}; A_2)$ can be meromorphically continued to the whole space $\mathbb{C}^3$, and its singularities are*

$$\text{(7.1.3)} \qquad \begin{cases} s_1 + s_3 = 1 - \ell & (\ell \in \mathbb{N}_0),\\ s_2 + s_3 = 1 - \ell & (\ell \in \mathbb{N}_0),\\ s_1 + s_2 + s_3 = 2. \end{cases}$$

REMARK 7.2. In the case of complex functions of several variables, singularity sets are not isolated points (as in the case of functions of one variable), but hypersurfaces, or more general analytic subsets in the complex space. Typical examples in our present context are (1.3.6), (7.1.3), (7.1.5), and (7.2.1). This produces one of the difficulties in the analytic study of multi-variable multiple zeta-functions.

In the above A_2 case, it is quite easy to distinguish whether the candidates of singularities, appearing in the Mellin–Barnes integral expression, are indeed singularities, or not. In general, however, it is not so easy. For example, in the case of A_3, we have shown in [**146**, Theorem 3.5] that the singularities of

$$\text{(7.1.4)} \qquad \begin{aligned} \zeta_3(\mathbf{s}; A_3) = &\sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \sum_{m_3=1}^{\infty} m_1^{-s_1} m_2^{-s_2} m_3^{-s_3}\\ &\times (m_1 + m_2)^{-s_4} (m_2 + m_3)^{-s_5} (m_1 + m_2 + m_3)^{-s_6}, \end{aligned}$$

where $\mathbf{s} = (s_1, s_2, s_3, s_4, s_5, s_6)$, are given by

$$\text{(7.1.5)} \qquad \begin{cases} s_1 + s_4 + s_6 = 1 - \ell & (\ell \in \mathbb{N}_0), \\ s_3 + s_5 + s_6 = 1 - \ell & (\ell \in \mathbb{N}_0), \\ s_2 + s_4 + s_5 + s_6 = 1 - \ell & (\ell \in \mathbb{N}_0), \\ s_1 + s_2 + s_4 + s_5 + s_6 = 2 - \ell & (\ell \in \mathbb{N}_0), \\ s_1 + s_3 + s_4 + s_5 + s_6 = 2 - \ell & (\ell \in \mathbb{N}_0), \\ s_2 + s_3 + s_4 + s_5 + s_6 = 2 - \ell & (\ell \in \mathbb{N}_0), \\ s_1 + s_2 + s_3 + s_4 + s_5 + s_6 = 3, \end{cases}$$

and all of these are "true" singularities.

In this case, applying Lemma 6.1, we obtain

(7.1.6)

$$\zeta_3(\mathbf{s}; A_3) = \frac{1}{(2\pi\sqrt{-1})^2} \int_{(c)} \int_{(c')} \frac{\Gamma(s_5 + z)\Gamma(-z)\Gamma(s_6 + z')\Gamma(-z')}{\Gamma(s_5)\Gamma(s_6)} \times \zeta(s_1, s_2 + s_5 + z, s_4 + s_6 + z'; A_2)\zeta(s_3 - z - z')dz'dz,$$

where $-\Re s_5 < c < 0$, $-\Re s_6 < c' < 0$ ([**146**, (3.2)]). It is possible to shift the paths of this double integral suitably to show the meromorphic continuation, and then to determine the location of singularities, but the whole argument requires about 20 pages in [**146**].[1]

The singularities of zeta-functions of root systems C_2, B_3, C_3 are discussed in [**104**, Section 6], and those of G_2 are studied in [**110**, Section 3]. In the case of C_2, the "true" singularities are completely determined, but in the other cases, only the lists of "possible" singularities are given. A conjecture which predicts the complete list of singularities of zeta-functions of A_r ($r \geq 4$) is mentioned at the end of [**146**, Section 4].[2]

As we have seen in the proof of Theorem 7.1, to treat the zeta-function of A_2, a vertical shift of the integral is enough. Similarly, vertical shifts work well in the case of C_2, B_3, C_3 and G_2. This is because in these cases, the Mellin–Barnes integral expressions correspond to horizontal lines in the diagram given at the end of Section 6.4 (see [**104**, Remark 6.4]). For example, in [**104**] we use the integral expression of $\zeta_2(\mathbf{s}; C_2)$ involving the zeta-function of A_2, and then in [**110**] we use the integral expression of $\zeta_2(\mathbf{s}; G_2)$ involving the zeta-function of C_2, while the lines between C_2 and A_2, and between G_2 and C_2, are horizontal in the diagram.

[1] There is a misprint in [**146**, p. 1480]. On l. 11, $z_5 = 0, 1, \ldots, N$ should be read as $z_5 = -s_5 - l$ $(0 \leq l \leq N)$.

[2] In the case of the r-fold Mordell–Tornheim zeta-functions, the true singularities are completely determined for any $r \geq 2$ in [**140**].

However, if we have to follow the lines in the diagram which are not horizontal, vertical shifts are not enough. This situation already happened in the study of $\zeta_3(\mathbf{s}; A_3)$ in [146]. In this case we have to use more delicate deformation of the path of integration. Actually, such deformation was already introduced in [134], to prove the meromorphic continuation of more general multiple zeta-functions defined by linear forms.

Let $\mathscr{A} = (a_{ij})_{1\le i\le n, 1\le j\le r}$ be an (n, r)-matrix, where each $a_{ij} \in \mathbb{C}$ satisfies $\Re a_{ij} > 0$ or is equal to 0. Put

$$\theta_i = \max_{\substack{1\le j\le r\\ a_{ij}\neq 0}} \{|\arg a_{ij}|\} < \frac{\pi}{2}.$$

We assume the following:

Assumption 7.3. Each row and each column have at least one nonzero entry.

Let $\mathbf{y} = (y_1, \dots, y_r) \in \mathbb{R}^r$, and define the multiple series associated with $\mathbf{y}$ and $\mathscr{A}$ by

(7.1.7)

$$\begin{aligned}\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A}) &= \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} e^{2\pi\sqrt{-1}(m_1y_1+\cdots+m_ry_r)}(a_{11}m_1 + \cdots + a_{1r}m_r)^{-s_1} \\ &\quad \times \cdots \times (a_{n1}m_1 + \cdots + a_{nr}m_r)^{-s_n}.\end{aligned}$$

Under the above assumption, it is clear that this multiple series is absolutely convergent when $\Re s_j > 1$ for all $j = 1, \dots, n$.

Theorem 7.4. *The function $\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A})$ can be continued meromorphically to the whole space $\mathbb{C}^n$.*

When $y_1 = \cdots = y_r = 0$, this result was included in Essouabri's general theory [38] [39][3] (see Section 1.2), and an alternative proof, based on the Mellin–Barnes formula, was given by the second-named author [134]. The general form of the above theorem was first proved by the first-named author [98]. The proof which we will present in the next section is a generalization of the argument in [134]. An immediate corollary of this theorem is:

[3]The twisted case can also be treated by Essouabri's method; see de Crisenoy [30].

COROLLARY 7.5. *Zeta-functions of root systems* (3.1.5) *and* (3.1.7) *can be continued meromorphically to the whole space* $\mathbb{C}^{\#(\Delta_+)}$.

In the same paper [**134**], the meromorphic continuation of multiple zeta-functions of Mordell–Tornheim type and of Apostol–Vu type (see Section 1.4), and also the continuation of $\zeta_2(\mathbf{s}; B_2)$ are proved. These cases are obviously special cases of Theorem 7.4, but the proofs in [**134**] are simpler. The proof in the case of Mordell–Tornheim multiple zeta-functions is based on the recursive structure

$$\zeta_{MT,r} \to \zeta_{MT,r-1} \to \cdots \to \zeta_{MT,2} = \zeta_2(\cdot; A_2).$$

In the Apostol–Vu case, certain auxiliary multiple series were introduced in [**134**] to construct a recursive structure, but later Okamoto [**167**] discovered a modified argument which does not use such auxiliary series.

More general multiple series of Mordell–Tornheim type and of Apostol–Vu type are studied in Okamoto [**167**], Miyagawa [**156**] and Umezawa [**205**].

As a preparation to the next section, we conclude this section with the following simple estimations of integrals. Let p, A, B, α, β be real numbers with $A + B < 0$, and

$$I = \int_{-\infty}^{\infty} (|y| + 1)^p \exp(A|y + \alpha| + B|y + \beta|)dy.$$

LEMMA 7.6 ([**133**, Lemma 3]). *We have*

$$I = O\left((|\alpha| + 1)^{p+\delta} e^{B|\alpha-\beta|} + (|\beta| + 1)^{p+\delta} e^{A|\alpha-\beta|}\right), \tag{7.1.8}$$

where $\delta = 1$ *or* 0 *according as* $A = B$ *or* $A \neq B$, *and the implied constant depends only on* p, A *and* B.

PROOF. Without loss of generality we may assume $\alpha \geq \beta$. We split the integral I at $y = -\alpha$ and $-\beta$ to obtain

$$\begin{aligned} I &= e^{-A\alpha-B\beta} \int_{-\infty}^{-\alpha} (|y| + 1)^p e^{-(A+B)y} dy \\ &\quad + e^{A\alpha-B\beta} \int_{-\alpha}^{-\beta} (|y| + 1)^p e^{(A-B)y} dy \\ &\quad + e^{A\alpha+B\beta} \int_{-\beta}^{\infty} (|y| + 1)^p e^{(A+B)y} dy \\ &= e^{-A\alpha-B\beta} I_1 + e^{A\alpha-B\beta} I_2 + e^{A\alpha+B\beta} I_3, \end{aligned} \tag{7.1.9}$$

say. Integrating by parts (repeatedly if necessary) we get

$$I_1 \ll (|\alpha| + 1)^p e^{(A+B)\alpha}, \qquad I_3 \ll (|\beta| + 1)^p e^{-(A+B)\beta},$$

and

$$I_2 \ll \max\left\{(|\alpha|+1)^{p+\delta}e^{-(A-B)\alpha}, (|\beta|+1)^{p+\delta}e^{-(A-B)\beta}\right\}.$$

Substituting these estimates into the right-hand side of (7.1.9), we obtain the desired result. □

Next, define

$$J = \int_{-\infty}^{\infty} (|t+y|+1)^p (|y|+1)^q \exp(A|t+y|+B|y|)dy,$$

where t, q are real and p, A, B are as above. Then:

LEMMA 7.7 (([133, Lemma 4]). *We have*

$$J = O\left((1+\tau^p)\tau^{q+\delta}e^{B|t|} + (1+\tau^p)e^{A|t|}\right),$$

where $\tau = |t|+1$, δ *is as in Lemma 7.6 and the implied constant depends only on* p, q, A *and* B.

PROOF. Divide

$$J = \int_{|y|\geq 2\tau} + \int_{|y|<2\tau} = J^* + J^{**},$$

say. Since $|y| \ll |t+y|+1 \ll |y|$ if $|y| \geq 2\tau$ and $1 \leq |t+y|+1 \ll \tau$ if $|y| < 2\tau$, we have

$$J^* \ll \int_{-\infty}^{\infty} (|y|+1)^{p+q} \exp(A|t+y|+B|y|)dy \tag{7.1.10}$$

and

$$J^{**} \ll \max\{1, \tau^p\} \int_{-\infty}^{\infty} (|y|+1)^q \exp(A|t+y|+B|y|)dy. \tag{7.1.11}$$

Applying Lemma 7.6 to (7.1.10) and (7.1.11) with $\alpha = t$ and $\beta = 0$, we obtain the assertion. □

7.2. Multiple zeta-functions defined by linear forms

The first aim of this section is to prove Theorem 7.4. In fact, we prove Theorem 7.4 and the following Theorem 7.8 simultaneously. By $\mathcal{F}(\cdot)$ we denote a quantity, not necessarily the same at each occurrence, which is of polynomial order with respect to the indicated variables.

THEOREM 7.8. (i) *The possible singularities of* $\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A})$ *are located only on hyperplanes of the form*

$$c_1 s_1 + \cdots c_n s_n = u(c_1, \dots, c_n) - \ell \qquad (\ell \in \mathbb{N}_0), \tag{7.2.1}$$

where $c_1, \dots, c_n \in \mathbb{N}_0$, *not all* 0, *and* $u(c_1, \dots, c_n)$ *is an integer determined by* $c_1, \dots, c_n$.

(ii) *In* (7.2.1)*, we may choose* $c_1, \dots, c_n$ *whose greatest common divisor is as small as possible. We call such tuples* $(c_1, \dots, c_n)$ *primitive. Then there are only finitely many primitive tuples which give the (possible) singularity sets* (7.2.1)*.*

(iii) *For each primitive tuples* $(c_1, \dots, c_n)$*, there exists a positive integer* $v(c_1, \dots, c_n)$*, by which the order of the singularity* (7.2.1) *is bounded uniformly in* ℓ*.*

(iv) *Except in neighborhoods of the above singularity sets, we have the estimate*

$$\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A}) = O\left(\mathcal{F}(\Im s_1, \dots, \Im s_n) \prod_{i=1}^{n} e^{\theta_i |\Im s_i|}\right) \tag{7.2.2}$$

uniformly in any "multiple strip" of the form

$$\{(s_1, \dots, s_n) \in \mathbb{C}^n \mid \sigma_{i1} \le \Re s_i \le \sigma_{i2} \ (1 \le i \le n)\},$$

where σ_{i1}, σ_{i2} *are any fixed real numbers with* $\sigma_{i1} < \sigma_{i2}$ $(1 \le i \le n)$*.*

Proof. For each row $\mathbf{a}_i = (a_{i1}, \dots, a_{ir})$ of $\mathscr{A}$, let $\rho(\mathbf{a}_i)$ be the number of non-zero elements of $\mathbf{a}_i$ (which is positive by Assumption 7.3), and define

$$\rho(\mathscr{A}) = \prod_{i=1}^{n} \rho(\mathbf{a}_i).$$

We prove Theorem 7.4 and Theorem 7.8 by induction on $\rho(\mathscr{A})$. We supply some details which are omitted in the original presentation of the proof in [**134**].

First, consider the case $\rho(\mathscr{A}) = 1$. Each row includes only one non-zero element, which we denote by $a_{i,h(i)}$. Then

(7.2.3)

$$\begin{aligned}
&\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A}) \\
&= \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} e^{2\pi\sqrt{-1}(m_1 y_1 + \cdots + m_r y_r)} (a_{1,h(1)} m_{h(1)})^{-s_1} \cdots (a_{n,h(n)} m_{h(n)})^{-s_n} \\
&= (a_{1,h(1)})^{-s_1} \cdots (a_{n,h(n)})^{-s_n} \\
&\qquad \times \left(\sum_{m_1=1}^{\infty} e^{2\pi\sqrt{-1} m_1 y_1} m_1^{-s(1)}\right) \cdots \left(\sum_{m_r=1}^{\infty} e^{2\pi\sqrt{-1} m_r y_r} m_r^{-s(r)}\right) \\
&= (a_{1,h(1)})^{-s_1} \cdots (a_{n,h(n)})^{-s_n} \prod_{j=1}^{r} \phi(s(j), y_j),
\end{aligned}$$

where

$$s(j) = \sum_{h(i)=j} s_i$$

and $\phi(s(j), y_j)$ is the Lerch zeta-function defined by (4.1.9).

When $y \in \mathbb{Z}$, we see that $\phi(s, y) = \zeta(s)$. It is well known that $\phi(s, y)$ can be continued holomorphically to $\mathbb{C}$ ([**121**, Section 2.2]), and is of polynomial order with respect to $\Im s$ ([**121**, Section 3.1, Theorem 1.4]). Therefore $\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A})$ is meromorphic in the whole $\mathbb{C}^n$ space, and the assertions on the order and on the location of singularities are satisfied.

Now consider the case $\rho(\mathscr{A}) \geq 2$. Then $r \geq 2$, and at least one row of $\mathscr{A}$ has at least two non-zero elements. Changing the parameters if necessary, we may assume that $a_{n,r-1} \neq 0$, $a_{nr} \neq 0$.

Let $r^* > 3$ and at first we assume that $(s_1, \dots, s_n)$ is in the region

$$\mathcal{B}^* = \{(s_1, \dots, s_n) \mid \Re s_i > r^* \ (1 \leq i \leq n)\}.$$

Then the series (7.1.7) is absolutely convergent. Rewrite the right-hand side of (7.1.7) as

$$\begin{aligned}
&\sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} e^{2\pi\sqrt{-1}(m_i y_1 + \cdots + m_r y_r)} (a_{11}m_1 + \cdots + a_{1r}m_r)^{-s_1} \cdots \\
&\times (a_{n-1,1}m_1 + \cdots + a_{n-1,r}m_r)^{-s_{n-1}} \times (a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1})^{-s_n} \\
&\times \left(1 + \frac{a_{nr}m_r}{a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1}}\right)^{-s_n},
\end{aligned}$$

and apply Lemma 6.1 with $s = s_n$ and $\lambda = a_{nr}m_r/(a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1})$. Then we have

(7.2.4)

$$\begin{aligned}
&\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A}) \\
&= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_n + z)\Gamma(-z)}{\Gamma(s_n)} \\
&\qquad \times \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} e^{2\pi\sqrt{-1}(m_i y_1 + \cdots + m_r y_r)} (a_{11}m_1 + \cdots + a_{1r}m_r)^{-s_1} \\
&\qquad \times \cdots \times (a_{n-1,1}m_1 + \cdots + a_{n-1,r}m_r)^{-s_{n-1}} \\
&\qquad \times (a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1})^{-s_n - z} (a_{nr}m_r)^z dz \\
&= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_n + z)\Gamma(-z)}{\Gamma(s_n)} \zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}') dz,
\end{aligned}$$

where

$$\mathscr{A}' = \begin{pmatrix} a_{11} & \cdots & a_{1,r-1} & a_{1r} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n-1,1} & \cdots & a_{n-1,r-1} & a_{n-1,r} \\ a_{n1} & \cdots & a_{n,r-1} & 0 \\ 0 & \cdots & 0 & a_{nr} \end{pmatrix}.$$

Here, the choice of c is as follows. To assure the absolute convergence of the multiple series in the integrand, we need

$$\Re s_j > 1 \ (1 \le j \le n-1),\ \Re(s_n + z) > 1,\ \Re z < -1.$$

These inequalities are satisfied if $(s_1, \dots, s_n) \in \mathcal{B}^*$ and $-\Re s_n + 1 < c < -1$. We choose c under the stronger restriction

$$-\Re s_n + 2 < c < -1. \tag{7.2.5}$$

This is possible because $r^* > 3$.

Since $\rho(\mathscr{A}') < \rho(\mathscr{A})$, we can use the induction assumption to find that $\zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}')$ is continued meromorphically, and all the claims of Theorem 7.8 are valid. In particular, possible singularities are of the form

$$c_1 s_1 + \cdots + c_n(s_n + z) - c_{n+1} z = u(c_1, \dots, c_{n+1}) - \ell, \tag{7.2.6}$$

where $c_1, \dots, c_{n+1}, \ell \in \mathbf{N}_0$ and $u(c_1, \dots, c_{n+1}) \in \mathbf{Z}$. If $c_n = c_{n+1}$, then this is

$$c_1 s_1 + \cdots + c_n s_n = u(c_1, \dots, c_n) - \ell \quad (\ell \in \mathbf{N}_0), \tag{7.2.7}$$

which is irrelevant to z. If $c_n - c_{n+1} = d_0 > 0$, then

$$z = d_0^{-1}\{-c_1 s_1 - \cdots - c_n s_n + u(c_1, \dots, c_n, c_n - d_0)\} - d_0^{-1}\ell \quad (\ell \in \mathbf{N}_0), \tag{7.2.8}$$

and if $c_n - c_{n+1} = -e_0 < 0$, then

$$z = e_0^{-1}\{c_1 s_1 + \cdots + c_n s_n - u(c_1, \dots, c_n, c_n + e_0)\} + e_0^{-1}\ell \quad (\ell \in \mathbf{N}_0). \tag{7.2.9}$$

We write the first term on the right-hand side of (7.2.8) (resp. (7.2.9)) as $D(s_1, \dots, s_n; \mathbf{c})$ (resp. $E(s_1, \dots, s_n; \mathbf{c})$) for brevity, where $\mathbf{c} = (c_1, \dots, c_{n+1})$. Denote the set of all primitive tuples $\mathbf{c} = (c_1, \dots, c_{n+1})$ appearing in (7.2.7) (resp. (7.2.8), (7.2.9)) by T_0 (resp. T_D, T_E). These sets are finite because of Theorem 7.8 (ii). The above (7.2.8) and (7.2.9) can be poles, with respect to z, of the integrand on the right-hand side of (7.2.4). The other poles of the integrand are

$$z = -s_n - \ell \quad (\ell \in \mathbf{N}_0) \tag{7.2.10}$$

and

$$z = \ell \quad (\ell \in \mathbf{N}_0), \tag{7.2.11}$$

coming from the gamma factors.

We claim that, if r^* is sufficiently large, then all the poles (7.2.8) and (7.2.10) are on the left of the line $\Re z = c$, while all the poles (7.2.9) and (7.2.11) are on the right of $\Re z = c$. This is immediate when $(c_1, \dots, c_n) \neq (0, \dots, 0)$, because if r^* is large then $\Re s_i$ $(1 \leq i \leq n)$ are also large. When $c_1 = \cdots = c_n = 0$, then $c_{n+1} \neq 0$, and from (7.2.6) we see that the singularities are

$$z = -c_{n+1}^{-1} u(c_1, \dots, c_{n+1}) + c_{n+1}^{-1} \ell. \tag{7.2.12}$$

These singularities form a sequence with the common difference $c_{n+1}^{-1} \leq 1$. Therefore, if one of them is located on the left of $\Re z = c$, there should be one of them in the strip $c - 1 \leq \Re z < c$. But $-\Re s_n + 1 < c - 1$ by (7.2.5), so this strip is in the domain of absolute convergence of $\zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}')$, hence there should not be a singularity in this strip. Therefore all singularities of the form (7.2.12) are on the right of $\Re z = c$, and hence the claim follows.

Now, let $(s_1^0, \dots, s_n^0)$ be any point in the space $\mathbf{C}^n$, and we show that the right-hand side of (7.2.4) can be continued meromorphically to $(s_1^0, \dots, s_n^0)$.

First, remove the singularities of the form (7.2.7) from the integrand. These singularities are cancelled by the factor

$$(c_1 s_1 + \cdots + c_n s_n - u(c_1, \dots, c_n) + \ell)^{v(c_1, \dots, c_n)}$$

(by Theorem 7.8 (iii)). Let L be a sufficiently large positive integer such that, if $\Re s_i \geq \Re s_i^0$ $(1 \leq i \leq n)$,

$$c_1 s_1 + \cdots + c_n s_n = u(c_1, \dots, c_n) - L$$

does not hold for any $\mathbf{c} = (c_1, \dots, c_n) \in T_0$. Define

$$\Phi(s_1, \dots, s_n) = \prod_{\mathbf{c} \in T_0} \prod_{\ell=0}^{L-1} (c_1 s_1 + \cdots + c_n s_n - u(c_1, \dots, c_n) + \ell)^{v(c_1, \dots, c_n)},$$

and rewrite (7.2.4) as

$$\zeta_r(s_1, \dots, s_n; \mathbf{y}; \mathscr{A}) = \Phi(s_1, \dots, s_n)^{-1} J(s_1, \dots, s_n), \tag{7.2.13}$$

where

$$J(s_1, \dots, s_n) = \frac{1}{2\pi\sqrt{-1}} \int_c \frac{\Gamma(s_n + z)\Gamma(-z)}{\Gamma(s_n)} \Phi(s_1, \dots, s_n) \times \zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}') dz. \tag{7.2.14}$$

Then the integrand on the right-hand side of (7.2.14) does not have singularities of the form (7.2.7) in the region $\Re s_i \geq \Re s_i^0$ $(1 \leq i \leq n)$.

Since $\Phi(s_1, \dots, s_n)^{-1}$ is meromorphic in the whole space, in order to complete the proof of the continuation, our remaining task is to show the continuation of $J(s_1, \dots, s_n)$. Let M be a positive integer, and $s_i^* = s_i^0 + M$ $(1 \leq i \leq n)$. We may choose M so large that $(s_1^*, \dots, s_n^*) \in \mathcal{B}^*$. Let $\mathcal{I}_1$ be the set of all imaginary parts of the poles (7.2.8) and (7.2.10), and $\mathcal{I}_2$ be the set of all imaginary parts of the poles (7.2.9) and (7.2.11), for $(s_1, \dots, s_n) = (s_1^*, \dots, s_n^*)$. We divide the discussion into three cases.

Case 1. In the case $\mathcal{I}_1 \cap \mathcal{I}_2 = \emptyset$, we join $D(s_1^*, \dots, s_n^*; \mathbf{c})$ and $D(s_1^0, \dots, s_n^0; \mathbf{c})$ by the segment $S(D; \mathbf{c})$ which is parallel to the real axis. Similarly, join $E(s_1^*, \dots, s_n^*; \mathbf{c})$ and $E(s_1^0, \dots, s_n^0; \mathbf{c})$ by the segment $S(E; \mathbf{c})$, and join $-s_n^*$ and $-s_n^0$ by the segment $S(n)$. Since $\mathcal{I}_1 \cap \mathcal{I}_2 = \emptyset$, we can deform the path $\Re z = c$ to obtain a new path $\mathcal{C}$ from $c - i\infty$ to $c + i\infty$, such that all the segments $S(D; \mathbf{c})$ and $S(n)$ are on the left of $\mathcal{C}$, while all the segments $S(E; \mathbf{c})$ and the poles (7.2.11) are on the right of $\mathcal{C}$ (see Figure 7.2.1). Then we have

$$J(s_1, \dots, s_n) = \frac{1}{2\pi\sqrt{-1}} \int_{\mathcal{C}} \frac{\Gamma(s_n + z)\Gamma(-z)}{\Gamma(s_n)} \Phi(s_1, \dots, s_n) \times \zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}')dz \tag{7.2.15}$$

in a sufficiently small neighborhood of $(s_1^*, \dots, s_n^*)$. Next, on the right-hand side of (7.2.15), we move $(s_1, \dots, s_n)$ from $(s_1^*, \dots, s_n^*)$ to $(s_1^0, \dots, s_n^0)$ with keeping the values of imaginary parts of each s_i. This procedure is possible, because the integrand rapidly decays as $|\Im z|$ tends to $+\infty$ (which is easily seen from the estimate (7.2.2) of $\zeta_r(\cdots; \mathbf{y}; \mathscr{A}')$ in the integrand and Stirling's formula (6.1.2)). During this procedure, the path $\mathcal{C}$ does not cross any poles of the integrand. Hence the expression (7.2.15) gives the holomorphic continuation of $J(s_1, \dots, s_n)$ to a neighborhood of $(s_1^0, \dots, s_N^0)$.

Case 2. Next, consider the case $\mathcal{I}_1 \cap \mathcal{I}_2 \neq \emptyset$. Then the imaginary part of some member of $\{D(s_1^*, \dots, s_n^*; \mathbf{c}),\ -s_n^* \mid \mathbf{c} \in T_D\}$ coincides with the imaginary part of some member of $\{E(s_1^*, \dots, s_n^*; \mathbf{c}),\ 0 \mid \mathbf{c} \in T_E\}$. We consider the case

$$\Im D(s_1^*, \dots, s_n^*; \mathbf{c}_1) = \Im E(s_1^*, \dots, s_n^*; \mathbf{c}_2) \tag{7.2.16}$$

for some $\mathbf{c}_1$ and $\mathbf{c}_2$, because other cases can be treated similarly. The associated poles are $D(s_1^*, \dots, s_n^*; \mathbf{c}_1) - d_0^{-1}\ell_1$ and $E(s_1^*, \dots, s_n^*; \mathbf{c}_1) + e_0^{-1}\ell_2$ $(\ell_1, \ell_2 \in \mathbf{N}_0)$. When $(s_1^*, \dots, s_n^*)$ is moved to $(s_1^0, \dots, s_n^0)$, these

poles are moved to $D(s_1^0,\dots,s_n^0;\mathbf{c}_1)-d_0^{-1}\ell_1$ and $E(s_1^0,\dots,s_n^0;\mathbf{c}_1)+e_0^{-1}\ell_2$, respectively. In the case

$$\Re D(s_1^0,\dots,s_n^0;\mathbf{c}_1)-d_0^{-1}\ell_1\neq\Re E(s_1^0,\dots,s_n^0;\mathbf{c}_1)+e_0^{-1}\ell_2 \tag{7.2.17}$$

for any ℓ_1 and ℓ_2, we modify the argument in Case 1 as follows. Let η be a small positive number, and consider the oriented polygonal path $S'(D;\mathbf{c}_1)$ joining the points $D(s_1^*,\dots,s_n^*;\mathbf{c}_1)$, $D(s_1^*+i\eta,\dots,s_n^*+i\eta;\mathbf{c}_1)$, $D(s_1^0+i\eta,\dots,s_n^0+i\eta;\mathbf{c}_1)$, and then $D(s_1^0,\dots,s_n^0;\mathbf{c}_1)$ in that order. Similarly, define the path $S'(E;\mathbf{c}_2)$ which joins $E(s_1^*,\dots,s_n^*;\mathbf{c}_2)$, $E(s_1^*+i\eta,\dots,s_n^*+i\eta;\mathbf{c}_2)$, $E(s_1^0+i\eta,\dots,s_n^0+i\eta;\mathbf{c}_2)$, and then $E(s_1^0,\dots,s_n^0;\mathbf{c}_2)$. Then $S'(D;\mathbf{c}_1)$ lies on the lower side of the line

$$\mathcal{L}=\{z\mid \Im z=\Im D(s_1^*,\dots,s_n^*;\mathbf{c}_1)=\Im E(s_1^*,\dots,s_n^*;\mathbf{c}_2)\},$$

while $S'(E;\mathbf{c}_2)$ lies on the upper side of $\mathcal{L}$. Because of (7.2.17), we can define the path $\mathcal{C}'$, which is almost the same as $\mathcal{C}$, but near the line $\mathcal{L}$ we draw $\mathcal{C}'$ such that it separates

$$\bigcup_{\ell_1\in\mathbf{N}_0}(S'(D;\mathbf{c}_1)-d_0^{-1}\ell_1)\qquad\text{and}\qquad\bigcup_{\ell_2\in\mathbf{N}_0}(S'(E;\mathbf{c}_2)+e_0^{-1}\ell_2)$$

(see Figure 7.2.2). Then the expression (7.2.15) with $\mathcal{C}$ replaced by $\mathcal{C}'$ is valid in a sufficiently small neighborhood of $(s_1^*,\dots,s_n^*)$.

When $(s_1,\dots,s_n)$ moves along the polygonal path joining $(s_1^*,\dots,s_n^*)$, $(s_1^*+i\eta,\dots,s_n^*+i\eta)$, $(s_1^0+i\eta,\dots,s_n^0+i\eta)$, and then $(s_1^0,\dots,s_n^0)$ in that order, the path $\mathcal{C}'$ encounters no pole, hence we obtain the holomorphic continuation.

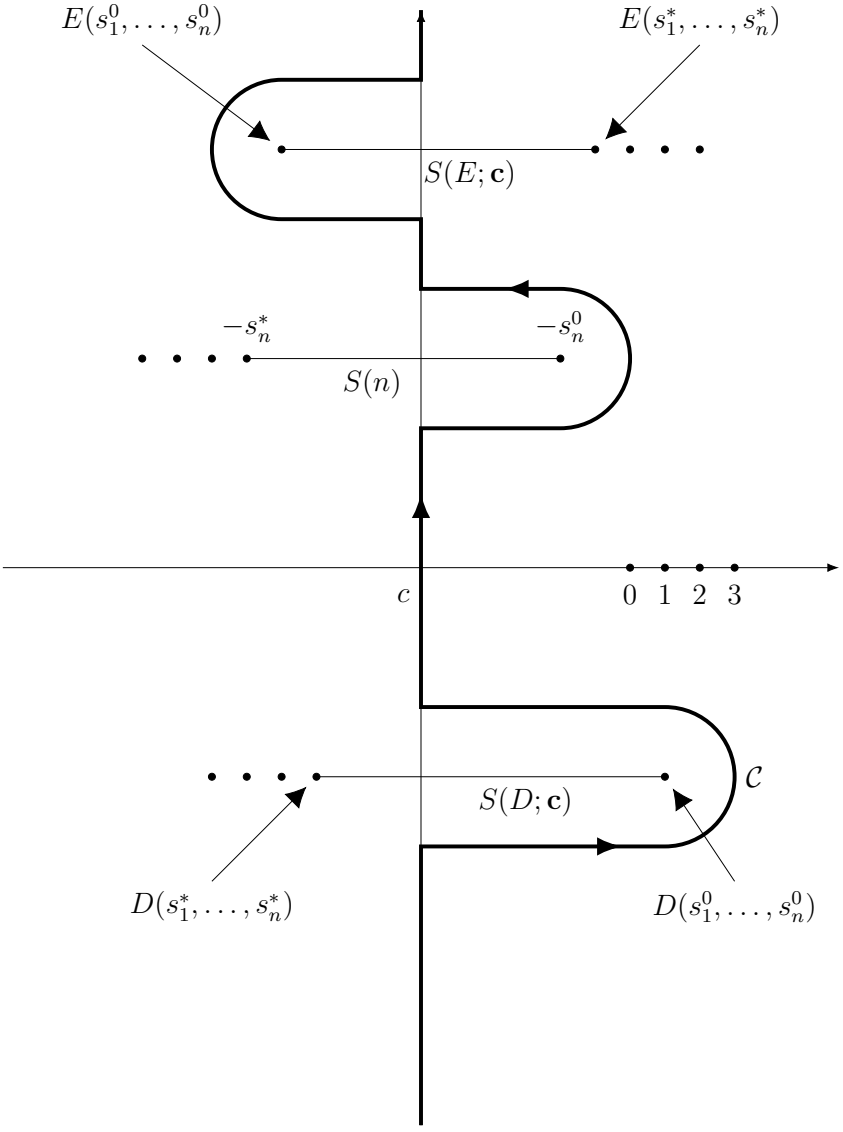
Case 3. The remaining case is that

$$D(s_1^0,\dots,s_n^0;\mathbf{c}_1)-d_0^{-1}\ell_1=E(s_1^0,\dots,s_n^0;\mathbf{c}_2)+e_0^{-1}\ell_2 \tag{7.2.18}$$

holds for some ℓ_1 and ℓ_2. Then this might also hold for some other pairs of (ℓ_1,ℓ_2). In this case we consider the path $\mathcal{C}''$ which is almost the same as $\mathcal{C}$, but near the line $\mathcal{L}$ we only require that $S(D;\mathbf{c}_1)$ is on the left of $\mathcal{C}''$, and that the points $E(s_1^*,\dots,s_n^*;\mathbf{c}_2)+e_0^{-1}\ell_2$, $E(s_1^0,\dots,s_n^0;\mathbf{c}_2)+e_0^{-1}\ell_2$ are not on $\mathcal{C}''$ for any ℓ_2. When we deform the path $\Re z=c$ on the right-hand side of (7.2.14) to $\mathcal{C}''$, we might encounter several poles of the form (7.2.9). Hence, in a small neighborhood of $(s_1^*,\dots,s_n^*)$, the integral $J(s_1,\dots,s_n)$ has the expression

$$\begin{aligned}-R_1(s_1,\dots,s_n)+\frac{1}{2\pi\sqrt{-1}}\int_{\mathcal{C}''}\frac{\Gamma(s_n+z)\Gamma(-z)}{\Gamma(s_n)}\Phi(s_1,\dots,s_n)\\ \times\zeta_r(s_1,\dots,s_{n-1},s_n+z,-z;\mathbf{y};\mathscr{A}')dz,\end{aligned} \tag{7.2.19}$$

FIGURE 7.2.1. The path for Case 1



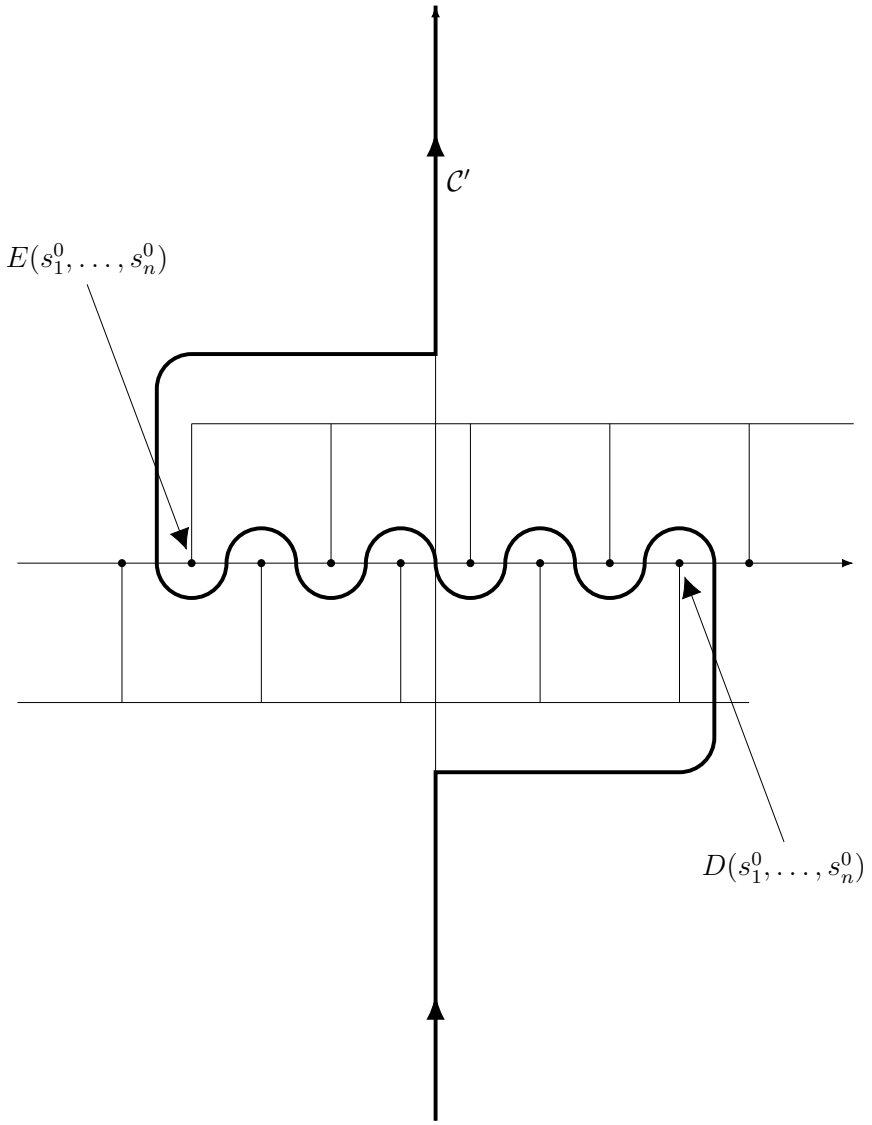
where $R_1(s_1, \dots, s_n)$ is the sum of residues of the above poles. Hence $R_1(s_1, \dots, s_n)$ is a (finite) sum of residues which are of the form

$$\Gamma(s_n)^{-1}\Phi(s_1, \dots, s_n)R(s_1, \dots, s_n; \ell_2), \tag{7.2.20}$$

where

$$R(s_1, \dots, s_n; \ell_2) = \frac{1}{(h-1)!}\frac{d^{h-1}}{dz^{h-1}}\Big\{(z - z(\ell_2))^h\Gamma(s_n + z)\Gamma(-z) \tag{7.2.21}$$

FIGURE 7.2.2. The path for Case 2



$$\times \zeta_r(s_1, \ldots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}') \Big\}\Big|_{z=z(\ell_2)}$$

with $z(\ell_2) = E(s_1, \ldots, s_n; \mathbf{c}_2) + e_0^{-1}\ell_2$, if the order of the pole is h. Note that, as a function in $s_1, \ldots, s_n$, $R(s_1, \ldots, s_n; \ell_2)$ can be continued meromorphically to the whole space $\mathbb{C}^n$.

Next, we move $(s_1, \ldots, s_n)$ from $(s_1^*, \ldots, s_n^*)$ to $(s_1^0, \ldots, s_n^0)$; then the path $\mathcal{C}''$ might encounter several poles of the same type. Suppose

that, during this process, a pole will meet with $\mathcal{C}''$ when $(s_1,\dots,s_n) = (s_1^\bullet,\dots,s_n^\bullet)$, say. In this case, we stop $(s_1,\dots,s_n)$ at $(s_1^\bullet+\varepsilon,\dots,s_n^\bullet+\varepsilon)$, where $\varepsilon>0$ is very small, and deform $\mathcal{C}''$ slightly to make a new path, of which the pole is on the left. The residue produced by this pole is exactly the same form as (7.2.20), with $(s_1,\dots,s_n) = (s_1^\bullet+\varepsilon,\dots,s_n^\bullet+\varepsilon)$. But again, this can be continued meromorphically to the whole space.

We carry out the same deformation each time when we meet poles, and by $R_2(s_1,\dots,s_n)$ we denote the sum of all residues produced by those poles. When finally we arrive at $(s_1^0,\dots,s_n^0)$, we obtain the following expression:

$$\begin{aligned} J(s_1,\dots,s_n) &= -R_1(s_1,\dots,s_n) - R_2(s_1,\dots,s_n) \\ &+ \frac{1}{2\pi\sqrt{-1}}\int_{\mathcal{C}'''} \frac{\Gamma(s_n+z)\Gamma(-z)}{\Gamma(s_n)}\Phi(s_1,\dots,s_n) \\ &\quad\times \zeta_r(s_1,\dots,s_{n-1},s_n+z,-z;\mathbf{y};\mathscr{A}')dz, \end{aligned} \tag{7.2.22}$$

where $\mathcal{C}'''$ is the path after the above indicated deformation. Since this integral on the right-hand side is holomorphic in a sufficiently small neighborhood of $(s_1^0,\dots,s_n^0)$, this expression gives the meromorphic continuation of $J(s_1,\dots,s_n)$ to $(s_1^0,\dots,s_n^0)$. This completes the proof of the meromorphic continuation[4] of $\zeta_r(s_1,\dots,s_n;\mathbf{y};\mathscr{A})$.

Next, we show that all the possible singularity sets of $\zeta_r(s_1,\dots,s_n;\mathbf{y};\mathscr{A})$ are of the form (7.2.1). This is clear for the singularity sets of $\Phi(s_1,\dots,s_n)^{-1}$. The singularity sets of $J(s_1,\dots,s_n)$ only appear in Case 3. Therefore the point $(s_1^0,\dots,s_n^0)$ on the singularity set satisfies (7.2.18), which can be rewritten as

$$\begin{aligned} &(e_0c_{11}+d_0c_{21})s_1^0+\cdots+(e_0c_{1n}+d_0c_{2n})s_n^0 \\ &-e_0u(c_{11},\dots,c_{1n},c_{1n}-d_0)+d_0u(c_{21},\dots,c_{2n},c_{2n}+e_0) = -e_0\ell_1-d_0\ell_2, \end{aligned}$$

where $\mathbf{c}_1 = (c_{11},\dots,c_{1,n+1})$ and $\mathbf{c}_2 = (c_{21},\dots,c_{2,n+1})$. This is of the form (7.2.1). Therefore we obtain Theorem 7.8 (i), (ii) (note that there are only finitely many tuples $\mathbf{c}$, hence also finitely many d_0, e_0). The assertion of Theorem 7.8 (iii) can be verified by observing the form (7.2.21) and using the induction assumption.

Lastly, we show the assertion on the order of $\zeta_r(s_1,\dots,s_n;\mathbf{y};\mathscr{A})$. First, consider Case 1. Since $\zeta_r(s_1,\dots,s_{n-1},s_n+z,-z;\mathbf{y};\mathscr{A}')$ satisfies

[4] If we want to prove only the meromorphic continuation, it is enough to apply the argument in Case 3 to all the cases. However, to prove Theorem 7.8, we have to study more closely when the singularities appear. In fact, Case 1 and Case 2 give the holomorphic continuation. This is the reason why we distinguish the above three cases.

an estimate of the form (7.2.2) by the induction assumption, the integral on the right-hand side of (7.2.15) clearly satisfies the same type of estimate with respect to $\Im s_1, \dots, \Im s_{n-1}$. As for $\Im s_n$, using Stirling's formula we find that the integral is

(7.2.23)
$$\ll \int_{-\infty}^{\infty} \exp\Big(\frac{\pi}{2}(|\Im s_n| - |\Im s_n + y| - |y|) + \theta_n(|\Im s_n + y| + |y|)\Big) \times \mathcal{F}(\Im s_n, y)dy,$$

which is $O(e^{\theta_n|\Im s_n|}\mathcal{F}(\Im s_n))$ by Lemma 7.7. Hence we obtain the desired assertion in Case 1, and the treatment of Case 2 is similar.

In Case 3, we have to estimate $R_1(s_1, \dots, s_n)+R_2(s_1, \dots, s_n)$. Since

$$R(s_1, \dots, s_n; \ell_2) = \frac{1}{2\pi\sqrt{-1}} \int_{\mathcal{K}} \Gamma(s_n + z)\Gamma(-z)\zeta_r(s_1, \dots, s_{n-1}, s_n + z, -z; \mathbf{y}; \mathscr{A}')dz,$$

where $\mathcal{K}$ is a small circle round the point $z(\ell_2)$, it is clear that $R_1(s_1, \dots, s_n) + R_2(s_1, \dots, s_n)$ satisfies an estimate of the form (7.2.2) with respect to $\Im s_1, \dots, \Im s_{n-1}$. As for $\Im s_n$, there appears the exponential factor which is the same as the exponential factor in the integrand of (7.2.23), so again applying Lemma 7.7 we obtain the desired estimate. Thus we now complete the proof of Theorem 7.4 and Theorem 7.8. □

Remark 7.9. Theorem 7.4 can be generalized to the case of multiple series

$$\sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} c_1(m_1) \cdots c_r(m_r)(a_{11}m_1 + \cdots + a_{1r}m_r)^{-s_1} \times \cdots \times (a_{n1}m_1 + \cdots + a_{nr}m_r)^{-s_n} \tag{7.2.24}$$

with arithmetical coefficients $c_1(m_1), \dots, c_r(m_r)$. Besides Assumption 7.3, we further assume that for each $c_j(m_j)$ $(1 \le j \le r)$, the associated Dirichlet series $\sum_{m=1}^{\infty} c_j(m_j)m_j^{-s}$ is absolutely convergent in a certain half-plane $\Re s > d_j$, can be continued meromorphically to $\mathbb{C}$ with the only possible pole at $s = d_j$, and is of polynomial order with respect to $|\Im s|$. Then, by the same argument as above, we can show that the series (7.2.24) can be continued meromorphically to $\mathbb{C}^n$. This is a generalization of the result in [**145**].

For our later purpose, here we give a variant of Theorem 7.4, where each denominator has a shifting parameter. Let $\mathscr{A}$ be as in Theorem 7.4, and let $\mathbf{b} = (b_k)_{1\le k\le n}$ is an n-component complex vector with

$\Re b_k > 0$ $(1 \le k \le n)$. Define the zeta-function associated with $\mathscr{A}$ and $\mathbf{b}$ by

(7.2.25)
$$\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b}) = \sum_{m_1=0}^{\infty}\cdots\sum_{m_r=0}^{\infty}(a_{11}m_1+\cdots+a_{1r}m_r+b_1)^{-s_1} \times\cdots\times(a_{n1}m_1+\cdots+a_{nr}m_r+b_n)^{-s_n}.$$

THEOREM 7.10 ([105, Theorem 8.1]). *The function* $\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b})$ *can be continued meromorphically to the whole space* $\mathbb{C}^n$.

REMARK 7.11. The statement of Theorem 8.1 in [105] is a little more general. Choose arbitrary lines $\ell_i : z = \eta e^{\sqrt{-1}\theta_i}$ $(\eta \in \mathbb{R})$ that pass the origin point $(1 \le i \le n)$. We further choose one of the open half planes partitioned by ℓ_i, and we denote this half plane by $H(\ell_i)$ for each i. The assumption in [105, Theorem 8.1] is that $a_{ij} = 0$ or $a_{ij} \in H(\ell_i)$ satisfying Assumption 7.3, while b_k are arbitrary.

We first prepare the following lemma.

LEMMA 7.12 ([105, Lemma 8.2]). *Let*

(7.2.26)
$$\varphi_k(s_1,\dots,s_k) = \sum_{m=0}^{\infty}(a_1m+b_1)^{-s_1}\cdots(a_km+b_k)^{-s_k}$$

where $\Re a_i > 0$, $\Re b_i > 0$ $(1 \le i \le k)$. *Let* $\theta_i = \max\{|\arg a_i|, |\arg b_i|\} < \pi/2$. *Then* $\varphi_k(s_1,\dots,s_k)$ *can be continued meromorphically to the whole space* $\mathbb{C}^k$, *and its possible singularities are located only on* $s_1+\cdots+s_k = 1-l$ $(l \in \mathbb{N}_0)$. *Moreover, the estimate*

(7.2.27)
$$\varphi_k(s_1,\dots,s_k) = O(\mathcal{F}(\Im s_1,\dots,\Im s_k)\exp(\theta_1|\Im s_1|+\cdots+\theta_k|\Im s_k|))$$

holds uniformly in any fixed vertical multiple strip.

PROOF. First of all, we divide (7.2.26) as

(7.2.28)
$$\varphi_k(s_1,\dots,s_k) = b_1^{-s_1}\cdots b_k^{-s_k} + \sum_{m=1}^{\infty}(a_1m+b_1)^{-s_1}\cdots(a_km+b_k)^{-s_k},$$

and denote the sum on the right-hand side by $\psi_k(s_1,\dots,s_k)$. Moreover, we define

$$\psi_k^*(s_1,\dots,s_k) = \sum_{m=1}^{\infty}(a_1m+b_1)^{-s_1}\cdots(a_{k-1}m+b_{k-1})^{-s_{k-1}}m^{-s_k}.$$

It is enough to prove the assertion of the lemma for $\psi_k(s_1,\dots,s_k)$ instead of $\varphi_k(s_1,\dots,s_k)$. For this purpose, we again use Lemma 6.1.

In fact, the following argument is essentially included in [151], in which the case $a_i = 1$ $(1 \le i \le k)$ is treated.

Write $\sigma_i = \Re s_i$, $t_i = \Im s_i$ $(1 \le i \le k)$. We begin with the situation when $\sigma_1, \dots, \sigma_k > 1$, so $\psi_k(s_1, \dots, s_k)$ is absolutely convergent. Since

$$(a_k m + b_k)^{-s_k} = (a_k m)^{-s_k} \left(1 + \frac{b_k}{a_k m}\right)^{-s_k} \tag{7.2.29}$$

and $|\arg(b_k/a_k m)| < \pi$, we obtain

$$(a_k m + b_k)^{-s_k} = (a_k m)^{-s_k} \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_k + z)\Gamma(-z)}{\Gamma(s_k)} \left(\frac{b_k}{a_k m}\right)^z dz,$$

where $-\sigma_k < c < 0$. Therefore we have

(7.2.30)

$$\begin{aligned}\psi_k(s_1, \dots, s_k) &= \sum_{m=1}^{\infty} (a_1 m + b_1)^{-s_1} \cdots (a_{k-1} m + b_{k-1})^{-s_{k-1}} (a_k m)^{-s_k} \\ &\qquad \times \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_k + z)\Gamma(-z)}{\Gamma(s_k)} \left(\frac{b_k}{a_k m}\right)^z dz \\ &= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_k + z)\Gamma(-z)}{\Gamma(s_k)} a_k^{-s_k - z} b_k^z \\ &\qquad \times \psi_k^*(s_1, \dots, s_{k-1}, s_k + z) dz.\end{aligned}$$

Here, to ensure the convergence of the ψ_k^* factor on the last member of the above, we have to assume that $1 - (\sigma_1 + \cdots + \sigma_k) < c < 0$.

We prove, by induction on k, that $\psi_k^*(s_1, \dots, s_k)$ can be continued meromorphically to the whole space $\mathbb{C}^k$, its possible singularities are located only on $s_1 + \cdots + s_k = 1 - l$ $(l \in \mathbb{N}_0)$, and the estimate

(7.2.31)

$$\psi_k^*(s_1, \dots, s_k) = O(\mathcal{F}(t_1, \dots, t_k) \exp(\theta_1 |t_1| + \cdots + \theta_{k-1} |t_{k-1}|))$$

holds. When $k = 1$, we see that $\psi_1^*(s_1)$ is nothing but the Riemann zeta-function $\zeta(s_1)$, hence the assertion is clear.

Let $k \ge 2$. Using

(7.2.32)

$$\begin{aligned}(a_{k-1} m + b_{k-1})^{-s_{k-1}} &= (a_{k-1} m)^{-s_{k-1}} \left(1 + \frac{b_{k-1}}{a_{k-1} m}\right)^{-s_{k-1}} \\ &= (a_{k-1} m)^{-s_{k-1}} \frac{1}{2\pi\sqrt{-1}} \int_{(c')} \frac{\Gamma(s_{k-1} + z')\Gamma(-z')}{\Gamma(s_{k-1})} \left(\frac{b_{k-1}}{a_{k-1} m}\right)^{z'} dz'\end{aligned}$$

with $-\sigma_{k-1} < c' < 0$, we obtain

(7.2.33)
$$\psi_k^*(s_1,\dots,s_k) = \frac{1}{2\pi\sqrt{-1}}\int_{(c')} \frac{\Gamma(s_{k-1}+z')\Gamma(-z')}{\Gamma(s_{k-1})} a_{k-1}^{-s_{k-1}-z'} b_{k-1}^{z'} \times \psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+z')dz'.$$

To ensure the convergence of the ψ_{k-1}^* factor on the right-hand side, we have to assume that $1-(\sigma_1+\cdots+\sigma_k) < c' < 0$.

Poles of the integrand on the right-hand side of (7.2.33) are

(7.2.34) $$z' = -s_{k-1} - l \quad (l \in \mathbb{N}_0),$$

(7.2.35) $$z' = l \quad (l \in \mathbb{N}_0),$$

(7.2.36) $$z' = 1-(s_1+\cdots+s_k) - l \quad (l \in \mathbb{N}_0),$$

which are coming from $\Gamma(s_{k-1}+z')$, $\Gamma(-z')$ and $\psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+z')$ under the assumption of induction, respectively. Choose an arbitrarily large $M \in \mathbb{N}$ and a sufficiently small $\varepsilon > 0$, and shift the path from $\Re z' = c'$ to $\Re z' = M - \frac{1}{2}$. This shifting is possible because, since $a_{k-1}^{-z'} b_{k-1}^{z'} = O(\exp(2\theta_{k-1}|y'|))$ (where $y' = \Im z'$) and the ψ_{k-1}^* factor is in the domain of absolute convergence when $c' \le \Re z' \le M - \frac{1}{2}$, the integrand is of rapid decay with respect to y'. The only relevant poles are at $z' = l$ $(0 \le l \le M-1)$, with the residue

$$\frac{\Gamma(s_{k-1}+l)}{\Gamma(s_{k-1})} \cdot \mathrm{Res}_{z'=l}\Gamma(-z') \cdot a_{k-1}^{-s_{k-1}-l} b_{k-1}^{l} \psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+l)$$
$$= -\binom{-s_{k-1}}{l} a_{k-1}^{-s_{k-1}-l} b_{k-1}^{l} \psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+l).$$

Therefore, from (7.2.33), we have

(7.2.37)
$$\psi_k^*(s_1,\dots,s_k) = \sum_{l=0}^{M-1} \binom{-s_{k-1}}{l} a_{k-1}^{-s_{k-1}-l} b_{k-1}^{l} \psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+l) + \frac{1}{2\pi\sqrt{-1}}\int_{(M-\frac{1}{2})} \frac{\Gamma(s_{k-1}+z')\Gamma(-z')}{\Gamma(s_{k-1})} a_{k-1}^{-s_{k-1}-z'} b_{k-1}^{z'} \times \psi_{k-1}^*(s_1,\dots,s_{k-2},s_{k-1}+s_k+z')dz'.$$

The integrand on the right-hand side is regular in the region

$$D_k(M) = \left\{(s_1,\dots,s_k) \mid \sigma_{k-1} > -M+\frac{1}{2}, \sigma_1+\cdots+\sigma_k > 1-M+\frac{1}{2}\right\}.$$

Therefore (7.2.37) gives the meromorphic continuation of $\psi_k^*(s_1,\dots,s_k)$ to $D_k(M)$. Since we can choose an arbitrarily large M, this implies that $\psi_k^*(s_1,\dots,s_k)$ can be continued meromorphically to the whole

space $\mathbb{C}^k$. Its possible singularities only come from singularities of $\psi^*_{k-1}(s_1,\dots,s_{k-2},s_{k-1}+s_k+l)$, namely

$$s_1+\cdots+s_{k-2}+(s_{k-1}+s_k+l)=1-l' \quad (l'\in\mathbb{N}_0)$$

from the assumption in the case of $k-1$. Moreover, in the region $D_k(M)$, the integral on the right-hand side of (7.2.37) can be estimated as

(7.2.38)
$$\ll\int_{-\infty}^{\infty}\exp\left(\frac{\pi}{2}(|t_{k-1}|-|t_{k-1}+y'|-|y'|)+\theta_{k-1}(|t_{k-1}+y'|+|y'|)\right)\times\mathcal{F}(t_1,\dots,t_k,y')dy',$$

which is $O(\mathcal{F}(t_1,\dots,t_k)\exp(\theta_{k-1}|t_{k-1}|))$ by Lemma 7.7. Applying this estimate and the $k-1$ case of (7.2.31) to the right-hand side of (7.2.37), we find that (7.2.31) holds for k. Therefore we have now proved all assertions for $\psi^*_k(s_1,\dots,s_k)$.

Lastly, we shift the path of integration on the last member of (7.2.30) from $\Re z=c$ to $\Re z=M-\frac{1}{2}$. Then we obtain

$$\psi_k(s_1,\dots,s_k)=\sum_{l=0}^{M-1}\binom{-s_k}{l}a_k^{-s_k-l}b_k^l\psi^*_k(s_1,\dots,s_{k-1},s_k+l)+\frac{1}{2\pi\sqrt{-1}}\int_{(M-\varepsilon)}\frac{\Gamma(s_k+z)\Gamma(-z)}{\Gamma(s_k)}a_k^{-s_k-z}b_k^z\times\psi^*_k(s_1,\dots,s_{k-1},s_k+z)dz. \tag{7.2.39}$$

Applying the argument quite similar to the above to (7.2.39), we obtain the assertion of the lemma. □

Proof of Theorem 7.10. We make use of the same method as in the proof of Theorem 7.4: that is, by induction on $\rho(\mathscr{A})$, we prove the theorem with all assertions of Theorem 7.8 (with $\zeta_r(s_1,\dots,s_n;\mathscr{A})$ replaced by $\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b})$).

In the case $\rho(\mathscr{A})=1$, we have $\rho_i=1$ for each i. Hence there exists only one nonzero entry $a_{i,h(i)}$ for each i, and

$$\begin{aligned}\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b})&=\sum_{m_1=0}^{\infty}\cdots\sum_{m_r=0}^{\infty}(a_{1,h(1)}m_{h(1)}+b_1)^{-s_1}\\&\quad\times\cdots\times(a_{n,h(n)}m_{h(n)}+b_n)^{-s_n}\\&=\prod_{1\le j\le r}\left(\sum_{m_j=0}^{\infty}\prod_{\substack{1\le i\le n\\ h(i)=j}}(a_{ij}m_j+b_i)^{-s_i}\right).\end{aligned}$$

Therefore in this case the assertion of Theorem 7.10 follows from Lemma 7.12.

Next, we prove the case $\rho(\mathscr{A}) \geq 2$. The condition $\rho(\mathscr{A}) \geq 2$ implies that some row of $\mathscr{A}$ has at least two nonzero entries. Hence, by renaming the indices, we can assume that $a_{n,r-1} \neq 0$ and $a_{nr} \neq 0$. Then, putting

$$\lambda = \frac{a_{nr}m_r + \frac{1}{2}b_n}{a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1} + \frac{1}{2}b_n},$$

we have

$$\begin{aligned}
&(a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1} + a_{nr}m_r + b_n)^{-s_n} \\
&= (a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1} + \frac{1}{2}b_n)^{-s_n}(1+\lambda)^{-s_n} \\
&= (a_{n1}m_1 + \cdots + a_{n,r-1}m_{r-1} + \frac{1}{2}b_n)^{-s_n} \\
&\quad \times \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_n+z)\Gamma(-z)}{\Gamma(s_n)} \lambda^z dz,
\end{aligned}$$

where $|\arg\lambda| < \pi$ because $\Re a_{ij} > 0$ and $\Re b_n > 0$. Therefore we have

$$\begin{aligned}
&\zeta_r(s_1, \ldots, s_n; \mathscr{A}, \mathbf{b}) \\
&= \frac{1}{2\pi\sqrt{-1}} \int_{(c)} \frac{\Gamma(s_n+z)\Gamma(-z)}{\Gamma(s_n)} \zeta_r(s_1, \ldots, s_{n-1}, s_n+z, -z; \mathscr{A}', \mathbf{b}')dz,
\end{aligned}$$

where $\mathscr{A}'$ is as before and

$$\mathbf{b}' = \begin{pmatrix} b_1 \\ \vdots \\ b_{n-1} \\ \frac{1}{2}b_n \\ \frac{1}{2}b_n \end{pmatrix} \in \mathbb{C}^{n+1}.$$

We see that $\Re(\frac{1}{2}b_n) > 0$ and

$$\rho(\mathscr{A}') = \rho(1)\rho(2)\cdots\rho(n-1) \times (\rho(n)-1) \times 1 < \rho(\mathscr{A}).$$

Hence we can apply the assumption of induction to $\zeta_r(s_1, \ldots, s_n; \mathscr{A}', \mathbf{b}')$. By just the same method as in the proof of Theorem 7.4, we arrive at the assertion for $\zeta_r(s_1, \ldots, s_n; \mathscr{A}, \mathbf{b})$. This completes the proof of Theorem 7.10. □



CHAPTER 8

Functional relations (I)

Now we start to describe our theory on functional relations for zeta-functions of root systems.

The original motivation for this theory is Problem 1.1, raised around 2000. When one wants to attack this problem, an obstacle is the fact that the main stream of the research on MZV is based on the Drinfel'd iterated integral expression of MZV. The definition of the Drinfel'd integral will be mentioned in Chapter 13, but the important point here is that the number of iterations of that integral is given by the weight of MZV. It is difficult to extend the definition of such integrals to non-integer values.

The third-named author developed an alternative method of studying MZV, which does not depend on iterated integral expressions. Therefore we may expect that his method can be used to consider Problem 1.1. And in fact, the third-named author [**203**] discovered a new functional relation (Theorem 8.3) by applying his method, which gives the first affirmative answer to Problem 1.1.

This method is sometimes called the *u-method*, because there appears an important parameter u. The third-named author applied this method to evaluate various (multiple) Dirichlet series at positive integers (see [**195**]–[**200**]). An interesting feature of the u-method is that the information of trivial zeros of the Riemann zeta-function plays a vital role. The main aim of the present chapter is to explain the u-method and prove Theorem 8.3 and some other functional relations for zeta-functions of root systems.

Nakamura [**162**], inspired by [**203**], found an alternative proof of Theorem 8.3 (see Remark 8.4). In his proof, Nakamura divided the double sum on the right-hand side of (1.4.1) into several subsums. The first-named author noticed that Nakamura's division can be explained in terms of Weyl chambers attached to the root system of type A_2. This observation inspired our further theory on functional relations, using the action of Weyl groups. This theory, which will be presented in

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Y. Komori et al., *The Theory of Zeta-Functions of Root Systems*,
Springer Monographs in Mathematics, https://doi.org/10.1007/978-981-99-0910-0_8

Chapter 9, gives a more general form of functional relations (Theorem 9.11).

In Chapter 4, we described a method of evaluating the values of zeta-functions of root systems at even integer points. It is a more difficult problem to evaluate the values at integer points $\mathbf{s} = (k_\alpha)_{\alpha\in\Delta_+}$, where some k_α can be odd. It is a remarkable advantage of our theory of functional relations that it catches some methods of evaluating the values of zeta-functions of root systems at such general integer points. Various examples can be found in Example 8.18, Example 8.30, (8.4.2), and Section 9.8.

8.1. A method to evaluate the Riemann zeta-function

First, we go back to the well-known Euler formula $\zeta(2) = \pi^2/6$. Euler sometimes preferred to consider

$$\phi(s) = \phi(s, 1/2) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^s} = \left(2^{1-s} - 1\right)\zeta(s) \tag{8.1.1}$$

instead of $\zeta(s)$, where $\phi(s, x)$ is the Lerch zeta-function defined by (4.1.9). Note that $\phi(2) = -\pi^2/12$. The third member of (8.1.1) is convergent only when $\Re s > 0$, but Euler essentially found $\zeta(0) = \phi(0) = -1/2$ and the trivial zeros $\zeta(-2m) = 0$ and $\phi(-2m) = 0$ $(m \in \mathbb{N})$ by computing divergent series. By the Maclaurin expansions of $\sin x$ and $\cos x$, we formally obtain

$$\sum_{n=1}^{\infty} \frac{(-1)^n \sin(n\theta)}{n^3} = \sum_{j=0}^{\infty} \phi(2-2j)\frac{(-1)^j\theta^{2j+1}}{(2j+1)!} = \phi(2)\theta + \frac{\theta^3}{12},$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cos(n\theta)}{n^2} = \sum_{j=0}^{\infty} \phi(2-2j)\frac{(-1)^j\theta^{2j}}{(2j)!} = \phi(2) + \frac{\theta^2}{4}.$$

In particular, putting $\theta = \pi$, we immediately obtain $\phi(2) = -\pi^2/12$ and $\zeta(2) = \pi^2/6$.

We begin this chapter by justifying these computations and giving an alternative proof of Euler's formula for $\zeta(s)$ at even positive integers, by considering

$$\eta(s; u) = \sum_{n\geq 1} \frac{(-u)^{-n}}{n^s} \quad (u \geq 1) \tag{8.1.2}$$

instead of $\phi(s)$. Note that the above series is convergent for $\Re s > 0$ if $u = 1$ and for any $s \in \mathbb{C}$ if $u > 1$. Obviously $\eta(s; 1) = \phi(s)$. Since

$$\Gamma(s) = \int_0^\infty e^{-x}x^{s-1}dx = n^s \int_0^\infty e^{-nx}x^{s-1}dx \qquad (\Re s > 0, n \in \mathbb{N}),$$

we have

$$\Gamma(s)\eta(s;u) = \int_0^\infty \sum_{n\geq 1} \left(-\frac{1}{u}\right)^n e^{-nx} x^{s-1} dx \tag{8.1.3}$$

$$= \int_0^\infty \sum_{n\geq 1} \left(-\frac{1}{ue^x}\right)^n x^{s-1} dx = -\int_0^\infty \frac{x^{s-1}}{ue^x+1} dx$$

(for $\Re s > 0$, $u \geq 1$), which further implies

$$\eta(s;u) = \frac{-1}{\Gamma(s)(e^{2\pi\sqrt{-1}s}-1)} \int_{\mathcal{C}} \frac{x^{s-1}}{ue^x+1} dx, \tag{8.1.4}$$

where $\mathcal{C}$ denotes the contour consisting of the half-line from $+\infty$ to a small positive number η, a counter-clockwise circle of radius η round the origin, and another half-line from η to $+\infty$. The integral on the right-hand side of (8.1.4) is convergent for all $s \in \mathbb{C}$. Therefore (8.1.4) gives the meromorphic continuation of $\eta(s;u)$ (for $u \geq 1$) to the whole complex plane. Moreover, this is holomorphic because (8.1.2) is convergent for $\Re s > 0$, while in the region $\Re s \leq 0$, the only possible poles $s = -k$ ($k \in \mathbb{N}_0$), coming from the factor $e^{2\pi\sqrt{-1}s} - 1$, are cancelled by the gamma factor.

Now let us state Euler's formula:

PROPOSITION 8.1. *For $m \in \mathbb{N}$,*

$$\zeta(2m) = \frac{(-1)^{m-1} 2^{2m-1} \pi^{2m}}{(2m)!} B_{2m}, \tag{8.1.5}$$

where B_n is the nth Bernoulli number defined by (4.0.1).

The following proof, due to the third-named author [**199**], is not the simplest proof of (8.1.5), but it includes the essence of the u-method.

We prepare some notation and supplemental results as follows.

For a small $\delta > 0$ and $u \in [1, 1+\delta]$, we set

$$F(t;u) = \frac{2e^t}{e^t+u} = \sum_{n=0}^\infty \mathcal{E}_n(u) \frac{t^n}{n!} \qquad (t \in \mathbb{C}, |t| < \pi). \tag{8.1.6}$$

(This Taylor expansion is valid for $|t| < \pi$, because $e^t + u \neq 0$ in this region.) The coefficients $\{\mathcal{E}_n(u)\}$ are rational functions in u. In particular,

$$F(t;1) = \frac{2e^t}{e^t+1} = 2\left(1 - \frac{1}{e^t-1} + \frac{2}{e^{2t}-1}\right)$$

$$= 2 - \frac{2}{t}\left(\frac{t}{e^t-1} - \frac{2t}{e^{2t}-1}\right) = 2 - \frac{2}{t}\sum_{n=0}^\infty \frac{(1-2^n)B_n}{n!} t^n,$$

which implies

$$\mathcal{E}_n(1) = \frac{2(2^{n+1}-1)B_{n+1}}{n+1} \quad (n \in \mathbb{N}), \quad \mathcal{E}_0(1) = 1. \tag{8.1.7}$$

It is noted that $\mathcal{E}_{2m}(1) = 0$ for $m \in \mathbb{N}$, which comes from the fact that $B_{2m+1} = 0$ for $m \in \mathbb{N}$.

For $\gamma \in \mathbb{R}$ with $0 < \gamma < \pi$, and $\mathcal{C}_\gamma : z = \gamma e^{\sqrt{-1}v}$ for $0 \le v \le 2\pi$, it follows from (8.1.6) that

$$\int_{\mathcal{C}_\gamma} F(z;u) z^{-n-1} dz = \frac{(2\pi\sqrt{-1})\mathcal{E}_n(u)}{n!} \quad (n \in \mathbb{N}_0).$$

Since $F(t;u)$ is continuous for $(t,u) \in \mathcal{C}_\gamma \times [1, 1+\delta]$, we can determine

$$M(\gamma) = \max_{(z,u)\in\mathcal{C}_\gamma\times[1,1+\delta]} |F(z,u)|,$$

which is independent of $u \in [1, 1+\delta]$. Hence we obtain

$$\begin{aligned} \frac{|\mathcal{E}_n(u)|}{n!} &\le \frac{1}{2\pi}\int_{\mathcal{C}_\gamma} |F(z;u)|\ |z|^{-n-1}|dz| \\ &\le \frac{M(\gamma)}{\gamma^n} \quad (n \in \mathbb{N}_0, u \in [1, 1+\delta]). \end{aligned} \tag{8.1.8}$$

Now we claim

$$\mathcal{E}_m(u) = -2\eta(-m;u) \qquad (m \in \mathbb{N}_0, u \in [1, 1+\delta]). \tag{8.1.9}$$

To prove this claim, we first assume $u > 1$ and $|t| < \log u$. Then we have

$$\begin{aligned} F(t;u) &= -2\sum_{n\ge1}(-u)^{-n}e^{nt} \\ &= -2\sum_{n\ge1}(-u)^{-n}\sum_{m=0}^{\infty}\frac{(nt)^m}{m!} = -2\sum_{m=0}^{\infty}\eta(-m;u)\frac{t^m}{m!}, \end{aligned}$$

where the change of summations is justified by the absolute convergence (note $|-u^{-1}e^t| < u^{-1}e^{\log u} = 1$). Therefore the claim for $u > 1$ follows. Since both $\mathcal{E}_m(u)$ and $\eta(-m;u)$ are continuous in u when $u \to 1+0$ (the latter follows from (8.1.4)), the claim is also valid for $u = 1$.

In particular, since $\eta(-m;1) = (2^{1+m}-1)\zeta(-m)$, the fact $\mathcal{E}_{2m}(1) = 0$ $(m \in \mathbb{N})$ is equivalent to $\zeta(-2m) = 0$, the trivial zeros of the Riemann zeta-function.

Let us define

$$I_p(\theta;k;u) = \sum_{m=1}^{\infty}\frac{(-u)^{-m}\sin^{(p)}(m\theta)}{m^k}, \tag{8.1.10}$$

for $p \in \mathbb{N}_0$, $\theta \in \mathbb{R}$, $k \in \mathbb{N}$, and $u \in [1, 1+\delta]$, where we let

$$\sin^{(p)}(x) = \frac{d^p}{dx^p} \sin x.$$

The right-hand side of (8.1.10) is convergent for $\theta \in (-\pi, \pi)$ (see [**209**, §3.35, Example 1]). It is convergent trivially for $\theta = \pm\pi$, except for the case where p is odd, $k = 1$ and $u = 1$.

It can be easily confirmed that

$$\begin{aligned} \sin^{(p)}(\theta) &= \frac{(\sqrt{-1})^{p-1}}{2} \left(e^{\sqrt{-1}\theta} + (-1)^{p-1} e^{-\sqrt{-1}\theta} \right) \\ &= (\sqrt{-1})^{p-1} \sum_{n=0}^{\infty} \epsilon_{p+1+n} \frac{(\sqrt{-1}\theta)^n}{n!}, \end{aligned} \tag{8.1.11}$$

where

$$\epsilon_m = \{1 + (-1)^m\}/2 \tag{8.1.12}$$

for $m \in \mathbb{Z}$. Then we can prove the following lemma ([**195**, Lemma 2]).

LEMMA 8.2. (i) *Suppose $u \in (1, 1+\delta]$. Then $I_p(\theta; k; u)$ ($p \in \mathbb{N}_0, k \in \mathbb{N}$) can be analytically continued to $\theta \in \mathbb{C}$ with $|\theta| < \pi$, and satisfies*

$$I_p(\theta; k; u) = (\sqrt{-1})^{p-1} \sum_{n=0}^{\infty} \eta(k-n; u) \epsilon_{p+1+n} \frac{(\sqrt{-1}\theta)^n}{n!}.$$

(ii) *When $\theta \in (-\pi, \pi)$,*

$$\lim_{u \to 1+0} I_p(\theta; k; u) = I_p(\theta; k; 1). \tag{8.1.13}$$

PROOF. (i) Since $u > 1$, we can choose a small $\rho > 0$ such that $u > 1 + \rho$. We first assume $\theta \in \mathbb{R}$ which satisfies $|\theta| < \log(1+\rho)$. Then we obtain from (8.1.10) and (8.1.11) that

(8.1.14)

$$\begin{aligned} I_p(\theta; k; u) &= \frac{(\sqrt{-1})^{p-1}}{2} \sum_{m=1}^{\infty} \frac{(-u)^{-m}}{m^k} \left(e^{m\sqrt{-1}\theta} + (-1)^{p-1} e^{-m\sqrt{-1}\theta} \right) \\ &= (\sqrt{-1})^{p-1} \sum_{m=1}^{\infty} \frac{(-u)^{-m}}{m^k} \sum_{n=0}^{\infty} \epsilon_{p+1+n} \frac{(m\sqrt{-1}\theta)^n}{n!}, \end{aligned}$$

which is absolutely convergent, because

$$\sum_{m=1}^{\infty} \frac{(e^{|\theta|}/u)^m}{m^k} < \infty.$$

Changing the order of summation and using (8.1.9), we have

$$I_p(\theta; k; u) = (\sqrt{-1})^{p-1} \left\{ \sum_{n=0}^{\infty} \eta(k-n; u) \epsilon_{p+1+n} \frac{(\sqrt{-1}\theta)^n}{n!} \right\}$$

$$= (\sqrt{-1})^{p-1}\Big\{\sum_{n=0}^{k-1}\eta(k-n;u)\epsilon_{p+1+n}\frac{(\sqrt{-1}\theta)^n}{n!}$$
$$-\frac{1}{2}\sum_{n=k}^{\infty}\mathcal{E}_{n-k}(u)\epsilon_{p+1+n}\frac{(\sqrt{-1}\theta)^n}{n!}\Big\}.$$

It follows from (8.1.8) that the right-hand side is absolutely convergent for $|\theta|<\pi$, which implies that $I_p(\theta;k;u)$ can be analytically continued to $\theta\in\mathbb{C}$ with $|\theta|<\pi$.

(ii) We can see that if $k\geq 2$ then (8.1.10) is convergent absolutely and uniformly in $u\in[1,1+\delta]$. Hence we obtain (8.1.13) for $k\geq 2$. On the other hand, we should treat the case of $k=1$ more carefully. To explain this, we recall the following classical theorem of Abel ([**209**, §3.71]):

Let $\sum_{n=0}^{\infty}a_nz^n$ $(a_n\in\mathbb{C},\ n\geq 0)$ *be a power series whose radius of convergence is equal to* 1, *and assume that* $\sum_{n=0}^{\infty}a_n$ *converges to* α. *Then* $\lim_{x\to 1-0}\sum_{n=0}^{\infty}a_nx^n=\alpha$.

We can easily see that the right-hand side of (8.1.10) with $k=1$ is convergent absolutely not only for $u\in(1,1+\delta]$ but also for $u\in\mathbb{C}$ with $|u^{-1}|<1$. Also (8.1.10) with $k=u=1$ is convergent for $\theta\in(-\pi,\pi)$. Therefore it follows from Abel's theorem that (8.1.13) holds for $\theta\in(-\pi,\pi)$. Thus we complete the proof. □

Define

(8.1.15)
$$J_p(\theta;k;u)=I_p(\theta;k;u)-(\sqrt{-1})^{p-1}\sum_{n=0}^{k}\eta(k-n;u)\epsilon_{p+1+n}\frac{(\sqrt{-1}\theta)^n}{n!}$$

for $\theta\in\mathbb{C}$ with $|\theta|<\pi$, $k\in\mathbb{N}$ and $u\in(1,1+\delta]$. Lemma 8.2 and (8.1.9) imply

$$J_p(\theta;k;u)=-\frac{(\sqrt{-1})^{p-1}}{2}\sum_{n=1}^{\infty}\mathcal{E}_n(u)\epsilon_{p+1+n+k}\frac{(\sqrt{-1}\theta)^{n+k}}{(n+k)!}. \tag{8.1.16}$$

Because of (8.1.8), this expression is convergent for $u\in[1,1+\delta]$.

Now let $\theta\in(-\pi,\pi)$, and compute $\lim_{u\to 1+0}J_p(\theta;k;u)$ in two ways. Because of (8.1.4), $\eta(k-n;u)$ $(0\leq n\leq k)$ is continuous in $u\in[1,1+\delta]$. Therefore from (8.1.13) and (8.1.15) we see that

(8.1.17)
$$\lim_{u\to 1+0}J_p(\theta;k;u)=I_p(\theta;k;1)-(\sqrt{-1})^{p-1}\sum_{n=0}^{k}\phi(k-n)\epsilon_{p+1+n}\frac{(\sqrt{-1}\theta)^n}{n!},$$

because $\eta(s;1)=\phi(s)$. On the other hand, the estimate (8.1.8) allows us to change the order of the summation and the limit process $u\to 1+0$ on (8.1.16) to obtain

(8.1.18)
$$\lim_{u\to 1+0} J_p(\theta;k;u) = -\frac{(\sqrt{-1})^{p-1}}{2}\sum_{n=1}^{\infty}\mathcal{E}_n(1)\epsilon_{p+1+n+k}\frac{(\sqrt{-1}\theta)^{n+k}}{(n+k)!}.$$

Since $\mathcal{E}_{2m}(1)=0$ $(m\in\mathbb{N})$ and $\epsilon_m=0$ for odd m, we obtain

$$\lim_{u\to 1+0} J_p(\theta;k;u)=0 \quad (\theta\in(-\pi,\pi)) \tag{8.1.19}$$

for $k\in\mathbb{N}$ and $p\in\mathbb{N}_0$ with $k\not\equiv p \pmod 2$. Therefore in this case the right-hand side of (8.1.17) is equal to 0.

It should be noted that (8.1.19) is the key fact in the proof of Proposition 8.1. We stress that the fact $\mathcal{E}_{2m}(1)=0$, or equivalently the existence of trivial zeros of the Riemann zeta-function, is essentially used in the proof of (8.1.19).

Now we can complete the proof of Proposition 8.1. Note that the delicate case $k=1$ in Lemma 8.2 is not necessary in the following proof, but it is prepared for our later purpose.

Proof of Proposition 8.1. Let $h\in\mathbb{N}_0$ and $\theta\in(-\pi,\pi)$. From the above discussion the right-hand side of (8.1.17), with $p=0$ and k replaced by $2h+1$, is equal to 0. We then have

$$I_0(\theta;2h+1;1)-\sum_{j=0}^{h}\phi(2h-2j)\frac{(-1)^j\theta^{2j+1}}{(2j+1)!}=0 \quad (h\in\mathbb{N}_0). \tag{8.1.20}$$

Now assume $h\in\mathbb{N}$. Then $I_0(\theta;2h+1;1)$ is continuous[1] for $\theta\in[-\pi,\pi]$ (because of the uniform convergence of (8.1.10)) and it tends to 0 as $\theta\to\pi$, so

$$\sum_{j=0}^{h}\phi(2h-2j)\frac{(-1)^j\pi^{2j+1}}{(2j+1)!}=0 \quad (h\in\mathbb{N}). \tag{8.1.21}$$

Here we set

(8.1.22)
$$\mathcal{D}_{2m}=\phi(2m)\frac{(2m)!}{(\sqrt{-1}\pi)^{2m}}=(2^{1-2m}-1)\zeta(2m)\frac{(2m)!}{(\sqrt{-1}\pi)^{2m}} \quad (m\in\mathbb{N}_0).$$

[1]For $h=0$, the sum on (8.1.20) tends to $\phi(0)\pi=-\pi/2$ as $\theta\to\pi$, while obviously $I_0(\pi;1;1)=0$. Therefore $I_0(\theta;1;1)$ is not continuous at $\theta=\pi$.

Then (8.1.21) implies that

$$\sum_{j=0}^{h} \binom{2h+1}{2j+1} \mathcal{D}_{2h-2j} = 0 \qquad (h \in \mathbb{N}).$$

Since $\mathcal{D}_0 = \zeta(0) = -1/2$, we obtain

$$\begin{aligned} -\frac{t}{2} &= \sum_{h=0}^{\infty} \left(\sum_{j=0}^{h} \binom{2h+1}{2j+1} \mathcal{D}_{2h-2j} \right) \frac{t^{2h+1}}{(2h+1)!} \\ &= \sum_{m=0}^{\infty} \mathcal{D}_{2m} \frac{t^{2m}}{(2m)!} \sum_{j=0}^{\infty} \frac{t^{2j+1}}{(2j+1)!} = \left(\sum_{m=0}^{\infty} \mathcal{D}_{2m} \frac{t^{2m}}{(2m)!} \right) \frac{e^t - e^{-t}}{2} \end{aligned}$$

for $t \in \mathbb{C}$ with $|t| < \pi$. In fact, we can change the order of summation in the above equation under $|t| < \pi$ because it follows from (8.1.22) that the last member is convergent absolutely for $|t| < \pi$.

We see that

$$\frac{t}{e^t - e^{-t}} = \frac{te^t}{e^{2t} - 1} = \frac{t}{e^t - 1} - \frac{t}{e^{2t} - 1} = \sum_{m=0}^{\infty} \left(1 - 2^{2m-1}\right) B_{2m} \frac{t^{2m}}{(2m)!},$$

so we have $\mathcal{D}_{2m} = (2^{2m-1} - 1)B_{2m}$ for $m \in \mathbb{N}_0$. In view of (8.1.22), we obtain Euler's formula (8.1.5). □

8.2. Functional relations for the zeta-function of A_2

In this section, applying the method similar to that introduced in the previous section, we prove the following functional relation for

$$\zeta_2(s_1, s_2, s_3; A_2) = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} m_1^{-s_1} m_2^{-s_2} (m_1 + m_2)^{-s_3}$$

defined by (3.3.19).

THEOREM 8.3. *For $k, l \in \mathbb{N}$ and $s \in \mathbb{C}$,*

$$\begin{aligned} (8.2.1) \qquad & \zeta_2(k, l, s; A_2) + (-1)^k \zeta_2(k, s, l; A_2) + (-1)^l \zeta_2(l, s, k; A_2) \\ &= 2 \sum_{\rho=0}^{[k/2]} \binom{k+l-2\rho-1}{l-1} \zeta(2\rho)\zeta(s+k+l-2\rho) \\ &\quad + 2 \sum_{\rho=0}^{[l/2]} \binom{k+l-2\rho-1}{k-1} \zeta(2\rho)\zeta(s+k+l-2\rho), \end{aligned}$$

where, and in what follows, $\binom{n}{0} = 1$ for any $n \in \mathbb{Z}$.

REMARK 8.4. This theorem was first proved in a somewhat different form in [**203**, Theorem 4.5]. In [**162**],[2] Nakamura gave its simpler form (8.2.1) using the method based on Zagier's technique (see [**6**, Proposition 4.14]). It can be verified that these two forms are equivalent by an elementary computation (see [**103**, Theorem 3.1]). Theorem 8.3 gives one of the main motivations to construct the theory of zeta-functions of root systems.

From Theorem 8.3, a functional relation involving the Apostol–Vu double zeta-function can be immediately obtained (but has not been published before):

COROLLARY 8.5. *For $k, l \in \mathbb{N}$ and $s \in \mathbb{C}$,*

$$\begin{aligned}(8.2.2)\qquad &\zeta_{AV,2}(k,l,s) + \zeta_{AV,2}(l,k,s)\\ &\quad + (-1)^k \{\zeta_{AV,2}(k,s,l) + \zeta_{AV,2}(s,k,l)\}\\ &\quad + (-1)^l \{\zeta_{AV,2}(l,s,k) + \zeta_{AV,2}(s,l,k)\}\\ &= 2\sum_{\rho=0}^{[k/2]} \binom{k+l-2\rho-1}{l-1} \zeta(2\rho)\zeta(s+k+l-2\rho)\\ &\quad + 2\sum_{\rho=0}^{[l/2]} \binom{k+l-2\rho-1}{k-1} \zeta(2\rho)\zeta(s+k+l-2\rho)\\ &\quad - \left\{2^{-s} + (-1)^k 2^{-l} + (-1)^l 2^{-k}\right\} \zeta(k+l+s),\end{aligned}$$

where $\zeta_{AV,2}(s_1,s_2,s_3)$ is the Apostol–Vu double zeta-function (see (1.4.4)).

PROOF. It is clear from the definitions that

$$\zeta_2(s,t,u;A_2) = \zeta_{AV,2}(s,t,u) + \zeta_{AV,2}(t,s,u) + 2^{-u}\zeta(s+t+u)$$

(see [**131**, (5.6)]). Substituting this formula into (8.2.1), we obtain (8.2.2). □

REMARK 8.6. Apostol and Vu showed

$$\begin{aligned}(8.2.3)\qquad \zeta_{AV,2}(a,a,1) &= \left(\frac{(-1)^{a+1}a}{2} - \frac{1}{4}\right)\zeta(2a+1)\\ &\quad + (-1)^a \sum_{\rho=0}^{[a/2]} \zeta(2\rho)\zeta(2a+1-2\rho)\end{aligned}$$

for $a \in \mathbb{N}$ (see [**5**, (19)]), which was deduced from the functional relation

$$(8.2.4)\qquad \zeta_{AV,2}(a,s,1) + \zeta_{AV,2}(s,a,1)$$

[2] [**162**] was published earlier, but written later than [**203**].

$$= (-1)^{a+1}\zeta_{EZ,2}(1, s+a) + \left((-1)^{a+1} + \frac{1}{2}\right)\zeta(s+a+1)$$
$$+ (-1)^{a+1}\sum_{m=1}^{a-1}(-1)^m\zeta(m+1)\zeta(s+a-m)$$

for $s \in \mathbb{C}$ (see [5, (18)]). It is noted that (8.2.3) can be deduced from (8.2.2) with $(k, l, s) = (1, a, a)$, though (8.2.2) and (8.2.4) look different from each other.

In the rest of this section, we give the proof of Theorem 8.3. We first prepare several lemmas.

LEMMA 8.7 ([140, Lemma 2.1]). *Let $a \in \mathbb{N}$.* (i) *For an arbitrary function $f : \mathbb{N}_0 \to \mathbb{C}$,*

$$\sum_{j=0}^{a}\phi(a-j)\epsilon_{a+j}\sum_{\mu=0}^{[j/2]}f(j-2\mu)\frac{(-1)^\mu\pi^{2\mu}}{(2\mu)!} = \sum_{\ell=0}^{[a/2]}\zeta(2\ell)f(a-2\ell). \tag{8.2.5}$$

(ii) *For an arbitrary function $g : \mathbb{N}_0 \to \mathbb{C}$,*

$$\sum_{j=0}^{a}\phi(a-j)\epsilon_{a+j}\sum_{\mu=0}^{[(j-1)/2]}g(j-2\mu)\frac{(-1)^\mu\pi^{2\mu}}{(2\mu+1)!} = -\frac{1}{2}g(a). \tag{8.2.6}$$

PROOF. First, notice

$$\zeta(2h) = \sum_{\mu=0}^{h}\phi(2h-2\mu)\,\frac{(-1)^\mu\pi^{2\mu}}{(2\mu)!} \quad (h \in \mathbb{N}_0). \tag{8.2.7}$$

In fact, when $h = 0$ this is trivial, because $\phi(0) = \zeta(0) = -1/2$. For $h \geq 1$, we can differentiate (8.1.20) with respect to θ and let $\theta \to \pi$ to get the result.

On the left-hand side of (8.2.5), we change the running index j to ℓ by $j = a + 2\mu - 2\ell$. By $0 \leq j \leq a$ and $0 \leq \mu \leq [j/2]$, we have $0 \leq 2\mu \leq 2\ell \leq a$, and hence $0 \leq \ell \leq [a/2]$. Therefore, since $a - j = 2\ell - 2\mu$ and $j - 2\mu = a - 2\ell$, we see that the left-hand side of (8.2.5) is

$$\sum_{\ell=0}^{[a/2]}f(a-2\ell)\sum_{\nu=0}^{\ell}\phi(2\ell-2\nu)\frac{(-1)^\nu\pi^{2\nu}}{(2\nu)!},$$

which is equal to the right-hand side of (8.2.5) by (8.2.7). Thus we obtain the proof of (8.2.5). Similarly, changing the running indices j to ℓ by $j = a + 2\mu - 2\ell$ on the left-hand side of (8.2.6), and using (8.1.21)

and $\phi(0) = -1/2$, we can see that (8.2.6) holds. This completes the proof. □

COROLLARY 8.8 ([**116**, Lemma 10.1]). *For an arbitrary function* $f : \mathbb{N}_0 \to \mathbb{C}$ *and* $d \in \mathbb{N}$, *we have*

$$\sum_{k=0}^{d} \phi(d-k)\epsilon_{d-k} \sum_{\nu=0}^{k} f(k-\nu)\frac{(\sqrt{-1}\pi)^{\nu}}{\nu!} = -\frac{\sqrt{-1}\pi}{2} f(d-1) + \sum_{\xi=0}^{[d/2]} \zeta(2\xi)\zeta(d-2\xi). \tag{8.2.8}$$

PROOF. Just add two formulas (8.2.5) and (8.2.6) with $g(x) = \sqrt{-1}\pi f(x-1)$. □

The next lemma was first published in [**117**, Lemma 5.1], which is a slight modification of [**203**, Lemma 4.4].[3] The origin of this type of result goes back to [**195**, Lemma 8].

LEMMA 8.9. *Let* $\{\alpha_d\}_{d\in\mathbb{N}_0}$, $\{\beta_d\}_{d\in\mathbb{N}_0}$, $\{\gamma_d\}_{d\in\mathbb{N}_0}$ *be sequences of complex numbers such that*

$$\alpha_d = \sum_{j=0}^{[d/2]} \gamma_{d-2j}\frac{(-1)^j\pi^{2j}}{(2j)!}, \quad \beta_d = \sum_{j=0}^{[d/2]} \gamma_{d-2j}\frac{(-1)^j\pi^{2j}}{(2j+1)!} \quad (d \in \mathbb{N}_0).$$

Then

$$\alpha_d = -2\sum_{\nu=0}^{[d/2]} \zeta(2\nu)\beta_{d-2\nu} \quad (d \in \mathbb{N}_0), \tag{8.2.9}$$

$$\beta_d = \frac{2}{\pi^2}\sum_{\nu=0}^{[d/2]} (2^{2\nu+2}-1)\zeta(2\nu+2)\alpha_{d-2\nu} \quad (d \in \mathbb{N}_0), \tag{8.2.10}$$

PROOF. First, we prove the following fact: *Suppose* $\{a_d\}_{d\in\mathbb{N}_0}$, $\{b_d\}_{d\in\mathbb{N}_0}$, *and* $\{c_d\}_{d\in\mathbb{N}_0}$ *are sequences of complex numbers such that*

$$a_d = \sum_{j=0}^{[d/2]} \binom{d}{2j} c_{d-2j}, \quad b_d = \sum_{j=0}^{[d/2]} \binom{d+1}{2j+1} c_{d-2j} \quad (d \in \mathbb{N}_0). \tag{8.2.11}$$

Then

$$a_d = \frac{1}{d+1}\sum_{\nu=0}^{[d/2]} \binom{d+1}{2\nu} b_{d-2\nu}2^{2\nu}B_{2\nu} \quad (d \in \mathbb{N}_0), \tag{8.2.12}$$

[3] Strictly speaking, in [**203**, Lemma 4.4], only the proof of a formula similar to (8.2.9) was given, while a formula similar to (8.2.10) first appeared in [**110**, Lemma 4.2].

where $\{B_n\}$ are the Bernoulli numbers.

In order to prove this fact, we consider the generating function. By (8.2.11), we have

$$\sum_{d=0}^{\infty} b_d \frac{t^{d+1}}{(d+1)!} = \sum_{d=0}^{\infty} \sum_{\nu=0}^{d+1} \binom{d+1}{\nu} \epsilon_{\nu+1} c_{d+1-\nu} \frac{t^{d+1}}{(d+1)!}$$
$$= \sum_{\nu=0}^{\infty} \epsilon_{\nu+1} \frac{t^\nu}{\nu!} \sum_{d \geq \nu-1} c_{d+1-\nu} \frac{t^{d+1-\nu}}{(d+1-\nu)!} = \frac{e^t - e^{-t}}{2} \sum_{\mu=0}^{\infty} c_\mu \frac{t^\mu}{\mu!}.$$

Write $2t/(e^t - e^{-t}) = \sum_{n \geq 0} \xi_n t^n/(n!)$, where $\xi_n = 0$ for odd n. Then we have

$$\frac{1}{\mu+1} \sum_{n=0}^{\mu} \binom{\mu+1}{n} b_{\mu-n} \xi_n = c_\mu$$

for $\mu \in \mathbb{N}_0$. Therefore from (8.2.11), we have

$$a_d = \sum_{j=0}^{[d/2]} \binom{d}{2j} \frac{1}{d-2j+1} \sum_{n=0}^{d-2j} \binom{d-2j+1}{n} b_{d-2j-n} \xi_n.$$

By letting $r = d - 2j$ and $\nu = 2j + n = d - r + n$ and exchanging the order of the double sum, (noting $r \equiv d \pmod 2$) we have

$$a_d = \sum_{\nu=0}^{d} \sum_{r=d-\nu}^{d} \binom{d}{r} \frac{1}{r+1} \binom{r+1}{r+\nu-d} \epsilon_{d+r} b_{d-\nu} \xi_{\nu+r-d} \tag{8.2.13}$$
$$= \frac{1}{d+1} \sum_{\nu=0}^{d} \binom{d+1}{\nu} b_{d-\nu} \sum_{\rho=0}^{\nu} \binom{\nu}{\rho} \epsilon_{\nu+\rho} \xi_\rho.$$

We can easily check that

$$\frac{2t}{e^t - e^{-t}} \cdot \frac{e^t + e^{-t}}{2} = \frac{2t}{e^{2t} - 1} + t,$$

hence

$$\sum_{\rho+\eta=\nu} \frac{\xi_\rho}{\rho!} \cdot \frac{1 + (-1)^\eta}{2} \cdot \frac{1}{\eta!} = \frac{2^\nu B_\nu}{\nu!} + \delta_{0,1},$$

where $\delta_{0,1}$ is Kronecker's delta. Since $\xi_\rho = 0$ for odd ρ, we obtain

$$\sum_{\rho=0}^{\nu} \binom{\nu}{\rho} \epsilon_{\nu+\rho} \xi_\rho = \begin{cases} 2^\nu B_\nu & (\nu : \text{even}), \\ 0 & (\nu : \text{odd}). \end{cases}$$

Substituting this into (8.2.13), we obtain (8.2.12).

Now define a_d, b_d and c_d by

$$\alpha_d = a_d \frac{(\sqrt{-1}\pi)^d}{d!}, \quad \beta_d = b_d \frac{(\sqrt{-1}\pi)^d}{(d+1)!}, \quad \gamma_d = c_d \frac{(\sqrt{-1}\pi)^d}{d!}$$

for $d \in \mathbb{N}_0$. Then (8.2.11) holds by the assumption. Hence, by (8.2.12) and Euler's formula $\zeta(2m) = -2^{2m-1}(\sqrt{-1}\pi)^{2m} B_{2m}/(2m)!$ for $m \in \mathbb{N}_0$ (see (8.1.5)), we complete the proof of (8.2.9).

Next, we will prove (8.2.10). Combining the well-known Taylor expansion of $\cot x$:

$$x \cot x = 1 - \sum_{m=1}^{\infty} \frac{(-1)^{m-1} B_{2m}}{(2m)!} (2x)^{2k}$$

(see [**209**, § 7.2]) and (8.1.5), we have

$$\pi x \cot(\pi x) = -2 \sum_{m=0}^{\infty} \zeta(2m) x^{2m}, \tag{8.2.14}$$

and hence

$$\begin{aligned} \pi x \tan(\pi x) &= -2\pi x \cot(2\pi x) + \pi x \cot(\pi x) \\ &= 2 \sum_{m=1}^{\infty} (2^{2m} - 1) \zeta(2m) x^{2m}. \end{aligned} \tag{8.2.15}$$

By (8.2.9) and (8.2.14), we have

$$\begin{aligned} \sum_{d=0}^{\infty} \alpha_d x^d &= -2 \sum_{d=0}^{\infty} \left(\sum_{\mu=0}^{d} \zeta(\mu) \epsilon_\mu \beta_{d-\mu} \right) x^d \\ &= -2 \sum_{\mu=0}^{\infty} \zeta(\mu) \epsilon_\mu x^\mu \left(\sum_{d \geq \mu} \beta_{d-\mu} x^{d-\mu} \right) \\ &= \pi x \cot(\pi x) \left(\sum_{m=0}^{\infty} \beta_m \, x^m \right). \end{aligned}$$

Hence, by (8.2.15),

$$\begin{aligned} \sum_{d=0}^{\infty} \beta_d \, x^d &= \frac{2}{(\pi x)^2} \left(\sum_{m=1}^{\infty} (2^{2m} - 1) \zeta(2m) x^{2m} \right) \left(\sum_{n=0}^{\infty} \alpha_n x^n \right) \\ &= \frac{2}{\pi^2} \left(\sum_{\ell=0}^{\infty} (2^{\ell+2} - 1) \zeta(\ell + 2) \epsilon_\ell x^\ell \right) \left(\sum_{n=0}^{\infty} \alpha_n x^n \right) \\ &= \frac{2}{\pi^2} \sum_{d=0}^{\infty} \left(\sum_{\ell=0}^{d} (2^{\ell+2} - 1) \zeta(\ell + 2) \epsilon_\ell \alpha_{d-\ell} \right) x^d. \end{aligned}$$

Thus we complete the proof of (8.2.10). □

The following proof of Theorem 8.3 is essentially under the same line as in [**203**], but here we introduce an additional parameter x.

Let $r, x, u \in \mathbb{R}$, $x \in (0, e^{-\pi})$ and $u \in [1, 1+\delta]$ for a sufficiently small $\delta > 0$. We consider several double series associated with u and x. Let

$$\mathcal{F}(t; r, x; u) = \sum_{n=1}^{\infty} \frac{(-u)^{-n} x^n e^{nt}}{n^r}. \tag{8.2.16}$$

When $|t| < \pi$, since $|xe^t| < 1$, (8.2.16) is absolutely convergent. Hence $\mathcal{F}(t; r, x; u)$ is holomorphic for $|t| < \pi$. Using the Taylor expansion of e^{nt}, we see that

(8.2.17)

$$\mathcal{F}(t; r, x; u) = \sum_{n=1}^{\infty} \frac{(-u)^{-n} x^n}{n^r} \sum_{m=0}^{\infty} \frac{(nt)^m}{m!} = \sum_{m=0}^{\infty} \eta(r-m; u/x) \frac{t^m}{m!}.$$

REMARK 8.10. In [**203**], there was no factor x^n, so the interchange of the summation as in (8.2.17) was verified in a different way. The argument in [**203**] is similar to the proof of Lemma 8.2 (i), and the necessary estimate of $\eta(r-m; u)$ (an analogue of (8.1.8)) was established by using the functional equation for $\eta(s; u)$ (see [**203**, Lemma 2.1]).

For $k \in \mathbb{N}$, set

$$F_2(t; k, r; x; u) = J_{k+1}(t/\sqrt{-1}; k; u) \mathcal{F}(t; r, x; u),$$

which is holomorphic for $|t| < \pi$, because $J_{k+1}(t; k; u)$ is holomorphic in t for $|t| < \pi$ by (8.1.16) when $u \in [1, 1+\delta]$. Also (8.1.19) implies

$$\lim_{u \to 1+0} F_2(t; k, r; x; u) = 0 \quad (|t| < \pi). \tag{8.2.18}$$

REMARK 8.11. This is the key fact in the proof of Theorem 8.3. In Section 8.1, the formula (8.1.19) gives non-trivial relations (8.1.20) and (8.1.21), from which Proposition 8.1 follows. Here, we again use the fundamental fact (8.1.19), but this time we multiply $J_{k+1}(t/\sqrt{-1}; k; u)$ by another series $\mathcal{F}(t; r, x; u)$ to get a new quantity $F_2(t; k, r; x; u)$. The fact (8.2.18), which is the double analogue of (8.1.19), gives non-trivial relations (Propositions 8.15, 8.16 and 8.17). This is the vital step in our proof of Theorem 8.3.

The following double series are necessary to consider the Maclaurin expansion of $F_2(t; k, r; x; u)$. For $u \in [1, 1+\delta]$, $x \in (0, e^{-\pi})$ and $s_1, s_2, s_3 \in \mathbb{C}$ with $\Re s_1, \Re s_3 \geq 1$, let

$$S(s_1, s_2; s_3; x; u) = \sum_{m,n=1}^{\infty} \frac{(-u)^{-m-n} x^n}{m^{s_1} n^{s_2} (m+n)^{s_3}},$$

$$R_1(s_1, s_2, s_3; x; u) = \sum_{m,n=1}^{\infty} \frac{(-u)^{-m-2n} x^n}{m^{s_1} n^{s_2} (m+n)^{s_3}},$$

$$R_2(s_1,s_2,s_3;x;u)=\sum_{m,n=1}^{\infty}\frac{(-u)^{-2m-n}x^{m+n}}{m^{s_1}n^{s_2}(m+n)^{s_3}}.$$

(It can be easily seen that the above series are convergent absolutely[4] even if $u=1$.)

Now assume $u\in(1,1+\delta]$. Then the above series are convergent absolutely for any $s_1,s_2,s_3\in\mathbb{C}$. Using (8.1.15), we find

$$\begin{aligned}(8.2.19)\quad F_2(t;k,r;x;u)&=J_{k+1}(t/\sqrt{-1};k;u)\mathcal{F}(t;r,x;u)\\&\quad-(\sqrt{-1})^k\sum_{j=0}^{k}\eta(k-j;u)\epsilon_{k+j}\frac{t^j}{j!}\mathcal{F}(t;r,x;u).\end{aligned}$$

Apply the second member of (8.1.14) and the definition (8.2.16) to the first term on the right-hand side of the above, while to the second term apply (8.2.17). We obtain

$$\begin{aligned}&F_2(t;k,r;x;u)\\&=\frac{(\sqrt{-1})^k}{2}\sum_{m,n=1}^{\infty}\frac{(-u)^{-m-n}x^n(e^{(m+n)t}+(-1)^ke^{(-m+n)t})}{m^kn^r}\\&\quad-(\sqrt{-1})^k\sum_{j=0}^{k}\eta(k-j;u)\epsilon_{k+j}\frac{t^j}{j!}\sum_{m=0}^{\infty}\eta(r-m;u/x)\frac{t^m}{m!}\\&=F_{21}-F_{22},\end{aligned}$$

say. Put $j+m=N$ in F_{22} to obtain

$$F_{22}=(\sqrt{-1})^k\sum_{N=0}^{\infty}\sum_{j=0}^{k}\binom{N}{j}\eta(k-j;u)\eta(r+j-N;u/x)\epsilon_{k+j}\frac{t^N}{N!},$$

where $\binom{N}{j}=0$ when $N<j$. As for F_{21}, the Taylor expansion of $e^{(m+n)t}$ and $e^{(-m+n)t}$ yields

$$\begin{aligned}F_{21}=\frac{(\sqrt{-1})^k}{2}\Bigg\{&\sum_{m,n=1}^{\infty}\frac{(-u)^{-m-n}x^n}{m^kn^r}\sum_{N=0}^{\infty}\frac{(m+n)^Nt^N}{N!}\\&+\left(\sum_{m>n}+\sum_{m<n}+\sum_{m=n}\right)\frac{(-u)^{-m-n}x^n(-1)^k}{m^kn^r}\sum_{N=0}^{\infty}\frac{(-m+n)^Nt^N}{N!}\Bigg\}.\end{aligned}$$

The first double sum is equal to $\sum_{N=0}^{\infty}S(k,r,-N;x;u)t^N/N!$. On the sum $\sum_{m>n}$, put $m-n=l$ to find that

$$\sum_{m>n}=\sum_{l,n=1}^{\infty}\frac{(-u)^{-l-2n}x^n(-1)^k}{(n+l)^kn^r}\sum_{N=0}^{\infty}\frac{(-l)^Nt^N}{N!}$$

[4]Actually a weaker condition $\Re(s_1+s_3)>1$ is enough to ensure the absolute convergence.

$$= \sum_{N=0}^{\infty} (-1)^{k+N} R_1(-N, r, k; x; u) \frac{t^N}{N!}.$$

Similarly, the sum $\sum_{m<n}$ is equal to

$$\sum_{N=0}^{\infty} (-1)^k R_2(k, -N, r; x; u) t^N / N!.$$

Consequently, putting

$$\begin{aligned} (8.2.20) \qquad & \mathcal{E}_{N,2}(k, r; x; u) \\ &= \frac{1}{2}\Big\{ S(k, r, N; x; u) + (-1)^{k+N} R_1(N, r, k; x; u) \\ &\quad + (-1)^k R_2(k, N, r; x; u) \Big\} \\ &\quad - \sum_{j=0}^{k} \binom{-N}{j} \eta(k-j; u) \eta(r+j+N; u/x) \epsilon_{k+j} \end{aligned}$$

for $N \in \mathbb{Z}$, $k \in \mathbb{N}$, $r \in \mathbb{R}$, $x \in (0, e^{-\pi})$ and $u \in (1, 1+\delta]$, we obtain

$$\begin{aligned} (8.2.21) \qquad F_2(t; k, r; x; u) &= (\sqrt{-1})^k \sum_{N=0}^{\infty} \mathcal{E}_{-N,2}(k, r; x; u) \frac{t^N}{N!} \\ &\quad + \frac{(-\sqrt{-1})^k}{2} \sum_{m=1}^{\infty} \frac{(u^2/x)^{-m}}{m^{k+r}} \end{aligned}$$

for $|t| < \pi$ (where the last sum corresponds to the part $\sum_{m=n}$). This is the Maclaurin expansion of $F_2(t; k, r; x; u)$ in t.

REMARK 8.12. When $N \geq 1$, the above definition of $\mathcal{E}_{N,2}(k, r, x; u)$ is valid for $u \in [1, 1+\delta]$, because the right-hand side of (8.2.20) is absolutely convergent for $(k, N) \in \mathbb{N}$ even if $u = 1$ as mentioned above.

LEMMA 8.13 ([**195**, Lemma 6]). *For $x > 0$, $\theta \in \mathbb{R}$ and $b, c \in \mathbb{N}_0$, we have*

(8.2.22)

$$\sum_{\nu=0}^{c} \frac{(-\theta)^\nu}{\nu!} \binom{b+c-\nu}{c-\nu} \frac{\cos^{(\nu)}(\theta x)}{x^{b+1+c-\nu}} = \sum_{N=0}^{[(b+c)/2]} \binom{b+c-2N}{c} \frac{(\sqrt{-1}\theta)^{2N}}{x^{b+1+c-2N}(2N)!}.$$

PROOF. Let

$$f(x) = \frac{(-1)^c}{c!} \cdot \frac{\cos(\theta x)}{x^{b+1}}.$$

Insert the Taylor expansion $\cos(\theta x) = \sum_{N=0}^{\infty} (\sqrt{-1}\theta x)^{2N}/(2N)!$ to the right-hand side and differentiate c times with respect to x to obtain

$$f^{(c)}(x) = \frac{(-1)^c}{c!} \sum_{N=0}^{\infty} \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!} (2N-b-1) \cdots (2N-b-c) x^{2N-b-c-1}$$

$$= \sum_{N=0}^{[(b+c)/2]} \binom{b+c-2N}{c} \frac{(\sqrt{-1}\theta)^{2N}}{x^{b+1+c-2N}(2N)!}.$$

On the other hand, we have

$$f^{(c)}(x) = \frac{(-1)^c}{c!} \sum_{\nu=0}^{c} \binom{c}{\nu} (\cos(\theta x))^{(\nu)} (x^{-b-1})^{(c-\nu)}$$
$$= (-1)^c \sum_{\nu=0}^{c} \frac{\theta^\nu}{\nu!(c-\nu)!} \cos^{(\nu)}(\theta x)(-b-1)\cdots(-b-c+\nu)x^{-b-c+\nu-1}$$
$$= \sum_{\nu=0}^{c} \frac{(-\theta)^\nu}{\nu!} \binom{b+c-\nu}{c-\nu} \frac{\cos^{(\nu)}(\theta x)}{x^{b+c+1-\nu}},$$

hence the lemma. □

Let $b, c \in \mathbb{N}_0$ and $\theta, r \in \mathbb{R}$. Put $x = m$ in (8.2.22), multiply by $(-u)^{-m} m^{-r}$ and sum up over all $m \in \mathbb{N}$ to get

$$\sum_{\nu=0}^{c} \frac{(-\theta)^\nu}{\nu!} \binom{b+c-\nu}{c-\nu} \sum_{m=1}^{\infty} \frac{(-u)^{-m} \cos^{(\nu)}(m\theta)}{m^{b+1+c+r-\nu}}$$
$$= \sum_{N=0}^{[(b+c)/2]} \binom{b+c-2N}{c} \sum_{m=1}^{\infty} \frac{(-u)^{-m}}{m^{b+1+c+r-2N}} \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!}. \tag{8.2.23}$$

This leads to the following lemma.

LEMMA 8.14. *Let $k, d \in \mathbb{N}$, $r \in \mathbb{R}$, $\theta \in (-\pi, \pi)$, $x \in (0, e^{-\pi})$ and $u \in (1, 1+\delta]$. Then*

$$\frac{1}{2} \sum_{m,n=1}^{\infty} \left\{ \frac{(-u)^{-m-n} x^n \cos((m+n)\theta)}{m^k n^r (m+n)^d} \right.$$
$$+ (-1)^{k+d} \frac{(-u)^{-m-2n} x^n \cos(m\theta)}{m^d n^r (m+n)^k}$$
$$\left. + (-1)^k \frac{(-u)^{-2m-n} x^{m+n} \cos(n\theta)}{m^k n^d (m+n)^r} \right\}$$
$$- \sum_{j=0}^{k} \eta(k-j;u)(-1)^j \epsilon_{k+j} \sum_{\nu=0}^{j} \frac{(-\theta)^\nu}{\nu!} \binom{d-1+j-\nu}{j-\nu}$$
$$\times \sum_{m=1}^{\infty} \frac{(-u)^{-m} x^m \cos^{(\nu)}(m\theta)}{m^{d+j+r-\nu}}$$
$$= \sum_{N=0}^{\infty} \mathcal{E}_{d-2N,2}(k, r; x; u) \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!}. \tag{8.2.24}$$

PROOF. Since $u \in (1, 1+\delta]$, using the Taylor expansion of $\cos(\cdot)$ we can see that

$$\sum_{m,n=1}^{\infty} \frac{(-u)^{-m-n}x^n \cos((m+n)\theta)}{m^{s_1}n^{s_2}(m+n)^{s_3}} = \sum_{N=0}^{\infty} S(s_1, s_2, s_3 - 2N; x; u)\ \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!},$$

$$\sum_{m,n=1}^{\infty} \frac{(-u)^{-m-2n}x^n \cos(m\theta)}{m^{s_1}n^{s_2}(m+n)^{s_3}} = \sum_{N=0}^{\infty} R_1(s_1 - 2N, s_2, s_3; x; u)\ \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!},$$

$$\sum_{m,n=1}^{\infty} \frac{(-u)^{-2m-n}x^{m+n} \cos(n\theta)}{m^{s_1}n^{s_2}(m+n)^{s_3}} = \sum_{N=0}^{\infty} R_2(s_1, s_2 - 2N, s_3; x; u)\ \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!}.$$

Furthermore, by applying (8.2.23) with $(b, c) = (d-1, j)$, we have

$$(-1)^j \sum_{\nu=0}^{j} \frac{(-\theta)^\nu}{\nu!} \binom{d-1+j-\nu}{j-\nu} \sum_{m=1}^{\infty} \frac{(-u)^{-m}x^m \cos^{(\nu)}(m\theta)}{m^{d+j+r-\nu}}$$
$$= \sum_{N=0}^{\infty} \binom{2N-d}{j} \eta(d+j+r-2N; u/x) \frac{(\sqrt{-1}\theta)^{2N}}{(2N)!},$$

because

$$\binom{d-1+j-2N}{j} = (-1)^j \binom{2N-d}{j}.$$

Therefore, by (8.2.20), we obtain the assertion. □

Similar to the case of $\mathcal{E}_n(u)$ in the preceding section, from (8.2.21) we can confirm that for any $\gamma \in (0, \pi)$ there exists $M_2(\gamma) > 0$ depending on (k, r, x) but independent of $u \in [1, 1+\delta]$ such that

$$\frac{|\mathcal{E}_{-N,2}(k, r; x; u)|}{N!} \le \frac{1}{2\pi} \int_{\mathcal{C}_\gamma} |F_2(z; k, r, x; u)|\ |z|^{-N-1}|dz| \le \frac{M_2(\gamma)}{\gamma^N} \quad (N \in \mathbb{N}). \tag{8.2.25}$$

This means that the right-hand side of (8.2.24) is uniformly convergent in $u \in [1, 1+\delta]$. Hence we can let $u \to 1$ on both sides of (8.2.24), namely (8.2.24) holds for $u = 1$ and $\theta \in (-\pi, \pi)$.

It follows from (8.2.18) and (8.2.21) that

$$\lim_{u \to 1+0} \mathcal{E}_{-N,2}(k, r; x; u) = \begin{cases} (-1)^{k+1}2^{-1} \sum_{m=1}^{\infty} x^{-m}m^{-k-r} & (N = 0) \\ 0 & (N \ge 1). \end{cases} \tag{8.2.26}$$

Therefore we have the following result.

PROPOSITION 8.15. *Let $k, d \in \mathbb{N}$, $r, \theta, x \in \mathbb{R}$, $\theta \in (-\pi, \pi)$ and $x \in (0, e^{-\pi})$. Then*

$$
\begin{aligned}
(8.2.27)\qquad & \frac{1}{2}\sum_{m,n=1}^{\infty}\left\{\frac{(-1)^{m+n}x^n\cos((m+n)\theta)}{m^k n^r (m+n)^d}\right. \\
&\qquad + (-1)^{k+d}\frac{(-1)^m x^n\cos(m\theta)}{m^d n^r (m+n)^k} \\
&\qquad \left. + (-1)^k \frac{(-1)^n x^{m+n}\cos(n\theta)}{m^k n^d (m+n)^r}\right\} \\
&- \sum_{j=0}^{k}\phi(k-j)(-1)^j\epsilon_{k+j}\sum_{\nu=0}^{j}\frac{(-\theta)^\nu}{\nu!}\binom{d-1+j-\nu}{j-\nu} \\
&\qquad \times \sum_{m=1}^{\infty}\frac{(-1)^m x^m \cos^{(\nu)}(m\theta)}{m^{d+j+r-\nu}} \\
&= \sum_{N=0}^{[d/2]}\mathcal{E}_{d-2N,2}(k,r;x;1)\frac{(\sqrt{-1}\theta)^{2N}}{(2N)!}.
\end{aligned}
$$

When $d \geq 2$, we can differentiate (8.2.27) with respect to θ because of its uniform convergency. Using the known relation

$$
-\binom{x-1}{y-1} + \binom{x}{y} = \binom{x-1}{y},
$$

we have the following.

PROPOSITION 8.16. *Let $k, d \in \mathbb{N}$ with $d \geq 2$, $r, \theta, x \in \mathbb{R}$, $\theta \in (-\pi, \pi)$ and $x \in (0, e^{-\pi})$. Then*

$$
\begin{aligned}
(8.2.28)\qquad & -\frac{1}{2}\sum_{m,n=1}^{\infty}\left\{\frac{(-1)^{m+n}x^n\sin((m+n)\theta)}{m^k n^r (m+n)^{d-1}}\right. \\
&\qquad + (-1)^{k+d}\frac{(-1)^m x^n\sin(m\theta)}{m^{d-1} n^r (m+n)^k} \\
&\qquad \left. + (-1)^k \frac{(-1)^n x^{m+n}\sin(n\theta)}{m^k n^{d-1} (m+n)^r}\right\} \\
&- \sum_{j=0}^{k}\phi(k-j)(-1)^j\epsilon_{k+j}\sum_{\nu=0}^{j}\frac{(-\theta)^\nu}{\nu!}\binom{d-2+j-\nu}{j-\nu} \\
&\qquad \times \sum_{m=1}^{\infty}\frac{(-1)^m x^m \cos^{(\nu+1)}(m\theta)}{m^{d-1+j+r-\nu}} \\
&= \sum_{N=1}^{[d/2]}\mathcal{E}_{d-2N,2}(k,r;x;1)\frac{(-1)^N\theta^{2N-1}}{(2N-1)!}.
\end{aligned}
$$

From these results, we consequently give the proof of Theorem 8.3. We give a more general result.

PROPOSITION 8.17. *For $k, l \in \mathbb{N}$, $s, x \in \mathbb{C}$ with $|x| < 1$, we have*

$$\begin{aligned}(8.2.29)\qquad &\sum_{m,n=1}^{\infty} \left\{ \frac{x^n}{m^k n^s (m+n)^l} + (-1)^{k+l} \frac{x^n}{m^l n^s (m+n)^k} \right. \\ &\qquad \left. + (-1)^k \frac{x^{m+n}}{m^k n^l (m+n)^s} \right\} \\ &= 2(-1)^k \sum_{\rho=0}^{[k/2]} \zeta(2\rho) \binom{k+l-1-2\rho}{l-1} \sum_{m=1}^{\infty} \frac{x^m}{m^{k+l+s-2\rho}} \\ &\quad + 2(-1)^k \sum_{\rho=0}^{[l/2]} \zeta(2\rho) \binom{k+l-1-2\rho}{k-1} \sum_{m=1}^{\infty} \frac{x^m}{m^{k+l+s-2\rho}}.\end{aligned}$$

PROOF OF PROPOSITION 8.17. Let $k, d \in \mathbb{N}$, $r, x \in \mathbb{R}$ with $x \in (0, e^{-\pi})$. Both sides of (8.2.27) and (8.2.28) are continuous in θ for $\theta \in [-\pi, \pi]$. Letting $\theta \to \pi$ in (8.2.27) we obtain

$$\begin{aligned}(8.2.30)\qquad &\frac{1}{2} \sum_{m,n=1}^{\infty} \left\{ \frac{x^n}{m^k n^r (m+n)^d} + (-1)^{k+d} \frac{x^n}{m^d n^r (m+n)^k} \right. \\ &\qquad \left. + (-1)^k \frac{x^{m+n}}{m^k n^d (m+n)^r} \right\} \\ &- \sum_{j=0}^{k} \phi(k-j) (-1)^j \epsilon_{k+j} \sum_{\mu=0}^{[j/2]} \frac{(-1)^\mu \pi^{2\mu}}{(2\mu)!} \\ &\qquad \times \binom{d-1+j-2\mu}{j-2\mu} \sum_{m=1}^{\infty} \frac{x^m}{m^{d+j+r-2\mu}} \\ &= \sum_{N=0}^{[d/2]} \mathcal{E}_{d-2N,2}(k, r; x; 1) \frac{(\sqrt{-1}\pi)^{2N}}{(2N)!},\end{aligned}$$

and similarly, replacing d by $d+2$ in (8.2.28), dividing by $-\pi$ and letting $\theta \to \pi$, we obtain

$$\begin{aligned}(8.2.31)\qquad &\sum_{j=0}^{k} \phi(k-j) (-1)^j \epsilon_{k+j} \sum_{\mu=0}^{[(j-1)/2]} \frac{(-1)^\mu \pi^{2\mu}}{(2\mu+1)!} \\ &\qquad \times \binom{d-1+j-2\mu}{j-2\mu-1} \sum_{m=1}^{\infty} \frac{x^m}{m^{d+j+r-2\mu}} \\ &= \sum_{N=1}^{[(d+2)/2]} \mathcal{E}_{d+2-2N,2}(k, r; x; 1) \frac{(-1)^{N-1} \pi^{2N-2}}{(2N-1)!}\end{aligned}$$

$$= \sum_{N=0}^{[d/2]} \mathcal{E}_{d-2N,2}(k,r;x;1)\frac{(-1)^N\pi^{2N}}{(2N+1)!}.$$

Denote by α_d and β_d the left-hand sides of (8.2.30) and (8.2.31), respectively. Then we may apply Lemma 8.9 (with $\gamma_d = \mathcal{E}_{d,2}(k,r;x;1)$) to obtain

$$\alpha_d = -2\sum_{\rho=0}^{[d/2]} \beta_{d-2\rho}\zeta(2\rho). \tag{8.2.32}$$

Moreover, by Lemma 8.7, the triple sums on the left-hand sides of (8.2.30) and of (8.2.31) are

$$(-1)^k \sum_{\rho=0}^{[k/2]} \zeta(2\rho)\binom{d-1+k-2\rho}{k-2\rho}\sum_{m=1}^{\infty}\frac{x^m}{m^{d+k+r-2\rho}},$$

and

$$\frac{(-1)^k}{2}\binom{d+k-1}{k-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{d+k+r}},$$

respectively. From these expressions and (8.2.32), we have

$$\begin{aligned}\sum_{m,n=1}^{\infty}\Bigg\{&\frac{x^n}{m^k n^r(m+n)^d} + (-1)^{k+d}\frac{x^n}{m^d n^r(m+n)^k} \\ &+ (-1)^k\frac{x^{m+n}}{m^k n^d(m+n)^r}\Bigg\} \\ &= 2(-1)^k\sum_{\rho=0}^{[k/2]}\zeta(2\rho)\binom{k+d-1-2\rho}{d-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{d+k+r-2\rho}} \\ &+ 2(-1)^k\sum_{\rho=0}^{[d/2]}\zeta(2\rho)\binom{k+d-1-2\rho}{k-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{d+k+r-2\rho}}.\end{aligned} \tag{8.2.33}$$

Replacing d and r by l and s, respectively, we obtain (8.2.29) for $x \in (0, e^{-\pi})$ and $s \in \mathbb{R}$. Since both sides of (8.2.29) are convergent absolutely for all $x, s \in \mathbb{C}$ with $|x| < 1$, we arrive at the assertion. □

The above Lemma 8.14 and Propositions 8.15, 8.16, 8.17 are slight generalizations (that is, with an additional factor x^n) of the results included in [**203**]. Now we are ready to complete the proof of Theorem 8.3.

Proof of Theorem 8.3. Temporarily assume $\Re s \geq 1$. Then (8.2.29) holds for $|x| = 1$ because both sides are convergent uniformly

for $|x| \le 1$. Therefore, putting $x = e^{2\pi\sqrt{-1}y}$ $(y \in \mathbb{R})$ and dividing both sides by $(-1)^k$, we obtain

$$\begin{aligned}(8.2.34)\ &\zeta_2(k,l,s;(y,y);A_2)\\ &\quad+(-1)^k\zeta_2(k,s,l;(1,y);A_2)+(-1)^l\zeta_2(l,s,k;(1,y);A_2)\\ &\quad=2\sum_{\rho=0}^{[k/2]}\binom{k+l-2\rho-1}{l-1}\zeta(2\rho)\phi(s+k+l-2\rho,y)\\ &\quad+2\sum_{\rho=0}^{[l/2]}\binom{k+l-2\rho-1}{k-1}\zeta(2\rho)\phi(s+k+l-2\rho,y),\end{aligned}$$

where $\phi(s,y)$ is defined by (4.1.9). Since all functions on both sides are continued meromorphically (as functions in s) to $\mathbb{C}$ (see Theorem 7.1 and Corollary 7.5), (8.2.34) is valid for any $s \in \mathbb{C}$. In particular, putting $y = 0$, we obtain the assertion of Theorem 8.3. □

EXAMPLE 8.18. Setting $(k,l) = (2,2),\ (3,2)$ in (8.2.1), we have

$$\begin{aligned}(8.2.35)\quad &\zeta_2(2,2,s;A_2)+2\zeta_2(2,s,2;A_2)\\ &\qquad=4\zeta(2)\zeta(s+2)-6\zeta(s+4),\\ (8.2.36)\quad &\zeta_2(3,s,2;A_2)-\zeta_2(3,2,s;A_2)-\zeta_2(2,s,3;A_2)\\ &\qquad=10\zeta(s+5)-6\zeta(2)\zeta(s+3).\end{aligned}$$

Setting $s = 2$ in (8.2.35) and (8.2.36), we have

$$(8.2.37)\quad \zeta_2(2,2,2;A_2)=\frac{1}{2835}\pi^6,$$

$$(8.2.38)\quad \zeta_2(2,2,3;A_2)=6\zeta(2)\zeta(5)-10\zeta(7),$$

respectively, where (8.2.37) was given by Mordell [**157**] and (8.2.38) was given by Tornheim [**194**]. Note that $\zeta_2(k,0,l;A_2) = \zeta_{EZ,2}(k,l)$. Then, setting $s = 0$ in (8.2.36), we have $\zeta_{EZ,2}(3,2) - \zeta_{EZ,2}(2,3) = 10\zeta(5) - 5\zeta(2)\zeta(3)$. On the other hand, a special case of (1.1.4) gives

$$\zeta(2)\zeta(3)=\zeta_{EZ,2}(3,2)+\zeta_{EZ,2}(2,3)+\zeta(5).$$

Combining these results, we recover the known results

$$\begin{aligned}\zeta_{EZ,2}(3,2)&=\frac{9}{2}\zeta(5)-2\zeta(2)\zeta(3),\\ \zeta_{EZ,2}(2,3)&=-\frac{11}{2}\zeta(5)+3\zeta(2)\zeta(3)\end{aligned}$$

(cf. [**15**]). Therefore this example gives an (chronologically first) affirmative answer to Problem 1.1.

REMARK 8.19. We introduced the additional factor x^n in the above proof. There are several advantages; first, it makes the convergence argument simpler (see Remark 8.10). Secondly, it produces a slightly generalized result with twisted factors such as (8.2.34). However, the original motivation of introducing such a factor was different. When we first applied the idea of u-method to zeta-functions of higher ranks (e.g. [**146**]), it only produced functional relations in which some of the variables should be equal to 0. In order to remove this unsatisfactory point, we introduced the additional factor such as x^n on the numerators and consider the (repeated) integration with respect to x to obtain more general functional relations (see [**149**] [**100**]). Later, this idea was refined to the form of Lemma 8.24, which will be proved in the next section. In fact, it can be seen that the numerators in the statement of Lemma 8.24 include the factor $e^{N\sqrt{-1}\theta}$, and repeated integration with respect to θ is performed in its proof. (See also Remark 15.2.)

It should be noted that Theorem 8.3 contains the following fact which is equivalent to [**189**, Theorem 4.1] given by Subbarao and Sitaramachandrarao. However, since their result includes several misprints, we correct them as follows.

COROLLARY 8.20. *For $m, n, p \in \mathbb{N}$,*

$$\begin{aligned}(8.2.39)\quad &\zeta_2(2m,2n,2p;A_2)+\zeta_2(2m,2p,2n;A_2)+\zeta_2(2n,2p,2m;A_2)\\ &=2\sum_{\rho=0}^{m}\binom{2m+2n-2\rho-1}{2n-1}\zeta(2\rho)\zeta(2m+2n+2p-2\rho)\\ &\quad+2\sum_{\rho=0}^{n}\binom{2m+2n-2\rho-1}{2m-1}\zeta(2\rho)\zeta(2m+2n+2p-2\rho).\end{aligned}$$

In particular,

(8.2.40)

$$\zeta_2(2m,2m,2m;A_2)=\frac{4}{3}\sum_{\rho=0}^{m}\binom{4m-2\rho-1}{2m-1}\zeta(2\rho)\zeta(6m-2\rho).$$

Examples listed at the end of Section 4.2 can now be immediately deduced from (8.2.40).

Also, as a corollary of Theorem 8.3, we have the following fact which implies a kind of "parity result" (see Section 12.5). This result was first given by Tornheim [**194**] (see also [**69**]).

COROLLARY 8.21. *For $a, b, c \in \mathbb{N}$ with $2 \nmid (a+b+c)$,*

$$(8.2.41)\qquad \zeta_2(a,b,c;A_2)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}].$$

PROOF. When $2 \nmid (a+b+c)$, a, b, c are all odd, or one of them is odd and the other two are even. As for the former case, setting $(k,l,s) = (2p+1, 2q+1, 2r+1)$ and $(k,l,s) = (2p+1, 2r+1, 2q+1)$ in (8.2.1), we see that the left-hand sides are equal to

$$\begin{aligned}
&\zeta_2(2p+1, 2q+1, 2r+1; A_2) - \zeta_2(2p+1, 2r+1, 2q+1; A_2) \\
&\qquad - \zeta_2(2q+1, 2r+1, 2p+1; A_2), \\
&\zeta_2(2p+1, 2r+1, 2q+1; A_2) - \zeta_2(2p+1, 2q+1, 2r+1; A_2) \\
&\qquad - \zeta_2(2r+1, 2q+1, 2p+1; A_2),
\end{aligned}$$

respectively. By adding them and noting $\zeta_2(a,b,c;A_2) = \zeta_2(b,a,c;A_2)$, we have

$$\zeta_2(2r+1, 2q+1, 2p+1; A_2) \in \mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}] \quad (p,q,r \in \mathbb{N}_0).$$

As for the latter case, for example, setting $(k,l,s) = (2p+1, 2q, 2r)$ and $(k,l,s) = (2p+1, 2r, 2q)$ in (8.2.1), we see that the left-hand sides are equal to

$$\begin{aligned}
&\zeta_2(2p+1, 2q, 2r; A_2) - \zeta_2(2p+1, 2r, 2q; A_2) + \zeta_2(2q, 2r, 2p+1; A_2), \\
&\zeta_2(2p+1, 2r, 2q; A_2) - \zeta_2(2p+1, 2q, 2r; A_2) + \zeta_2(2r, 2q, 2p+1; A_2),
\end{aligned}$$

respectively. Hence we have

$$\zeta_2(2r, 2q, 2p+1; A_2) \in \mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}] \quad (p \in \mathbb{N}_0,\ q,r \in \mathbb{N}).$$

As for other cases, we can prove them in the same way. □

EXAMPLE 8.22. Here we use the method introduced in the proof of Corollary 8.21, that is, combining functional relations in (8.2.1). Then we can obtain

$$\begin{aligned}
\zeta_2(1,1,1;A_2) &= 2\zeta(3), \\
\zeta_2(1,1,3;A_2) &= -\frac{1}{3}\zeta(3)\pi^2 + 4\zeta(5), \\
\zeta_2(1,2,2;A_2) &= \frac{1}{6}\zeta(3)\pi^2 - \frac{3}{2}\zeta(5), \\
\zeta_2(1,3,1;A_2) &= -\frac{1}{6}\zeta(3)\pi^2 + 3\zeta(5), \\
\zeta_2(2,1,2;A_2) &= \frac{1}{6}\zeta(3)\pi^2 - \frac{3}{2}\zeta(5), \\
\zeta_2(2,2,1;A_2) &= \frac{1}{3}\zeta(3)\pi^2 - 3\zeta(5), \\
\zeta_2(3,1,1;A_2) &= -\frac{1}{6}\zeta(3)\pi^2 + 3\zeta(5), \\
\zeta_2(1,1,5;A_2) &= -\frac{1}{45}\zeta(3)\pi^4 - \frac{1}{3}\zeta(5)\pi^2 + 6\zeta(7),
\end{aligned}$$

$$\zeta_2(1,2,4;A_2) = \frac{1}{2}\zeta(5)\pi^2 - 5\zeta(7),$$
$$\zeta_2(1,3,3;A_2) = -\frac{7}{6}\zeta(5)\pi^2 + 12\zeta(7),$$
$$\zeta_2(1,4,2;A_2) = \frac{1}{90}\zeta(3)\pi^4 + \frac{1}{2}\zeta(5)\pi^2 - 6\zeta(7),$$
$$\zeta_2(1,5,1;A_2) = -\frac{1}{90}\zeta(3)\pi^4 - \frac{1}{6}\zeta(5)\pi^2 + 4\zeta(7),$$
$$\zeta_2(2,1,4;A_2) = \frac{1}{2}\zeta(5)\pi^2 - 5\zeta(7),$$
$$\zeta_2(2,2,3;A_2) = \zeta(5)\pi^2 - 10\zeta(7),$$
$$\zeta_2(2,3,2;A_2) = -\frac{1}{6}\zeta(5)\pi^2 + 2\zeta(7),$$
$$\zeta_2(2,4,1;A_2) = \frac{1}{90}\zeta(3)\pi^4 + \frac{1}{3}\zeta(5)\pi^2 - 4\zeta(7),$$
$$\zeta_2(3,1,3;A_2) = -\frac{7}{6}\zeta(5)\pi^2 + 12\zeta(7),$$
$$\zeta_2(3,2,2;A_2) = -\frac{1}{6}\zeta(5)\pi^2 + 2\zeta(7),$$
$$\zeta_2(3,3,1;A_2) = -\frac{1}{3}\zeta(5)\pi^2 + 4\zeta(7),$$
$$\zeta_2(4,1,2;A_2) = \frac{1}{90}\zeta(3)\pi^4 + \frac{1}{2}\zeta(5)\pi^2 - 6\zeta(7),$$
$$\zeta_2(4,2,1;A_2) = \frac{1}{90}\zeta(3)\pi^4 + \frac{1}{3}\zeta(5)\pi^2 - 4\zeta(7),$$
$$\zeta_2(5,1,1;A_2) = -\frac{1}{90}\zeta(3)\pi^4 - \frac{1}{6}\zeta(5)\pi^2 + 4\zeta(7).$$

The symmetric property $\zeta_2(a,b,c;A_2) = \zeta_2(b,a,c;A_2)$ can be easily read off from the above. In fact, this follows from the definition (3.3.19). For more details, see Section 9.3.

REMARK 8.23. Functional relations which connect $\zeta_3(\mathbf{s};A_3)$ with $\zeta_2(\mathbf{s};A_2)$ were given by the authors [**146**, Theorems 5.9, 5.10] [**100**, Theorem 3.4] and also by Nakamura [**163**, Theorem 6.2]. Then, functional relations which connect $\zeta_3(\mathbf{s};A_3)$ with $\zeta(s)$, $\phi(s)$ were first obtained in [**103**, Theorem 7.1] for the case of even integral arguments, and later in [**114**, Theorem 9] for the more general case, by the same principle as in the present section. See Remark 15.26.

8.3. Functional relations for the zeta-function of C_2

By using the u-method explained in the previous section, we can obtain functional relations for zeta-functions of root systems of other

types, for instance type A_3 (see [146] [100] [103] [114]), types C_2, B_3, C_3 (see [103]) and type G_2 (see [110] [117]).

In the previous section, three lemmas of elementary nature (Lemmas 8.7, 8.9 and 8.13) played important roles. In particular, Lemma 8.9 (and its primitive form [195, Lemma 8]) was indispensable in the early stage of the u-method (such as [203] [146] [140] [149] and [110]). Later, however, we discovered the following Lemma 8.24, which is more complicated but is more convenient for studying zeta-functions of root systems of higher ranks. Lemma 8.24 is a generalization of Lemma 8.13, and is proved inductively by using Lemma 8.7 (ii). We may find that the roles of Lemmas 8.7 (ii), 8.9 and 8.13 can be replaced by this single Lemma 8.24. This lemma has been essentially used in later developments of the u-method (such as [103] [110] [114] [116] and [117]). In this section, using Lemma 8.24 and Lemma 8.7 (i), we prove functional relations in the cases of type C_2.

LEMMA 8.24 ([103, Lemma 6.2]). *For $h \in \mathbb{N}$, let*

$$\mathcal{C} = \{C(l) \in \mathbb{C} \,|\, l \in \mathbb{Z},\ l \neq 0\},$$
$$\mathcal{D} = \{D(N;m;\eta) \in \mathbb{R} \,|\, N, m, \eta \in \mathbb{Z},\ N \neq 0,\ m \geq 0,\ 1 \leq \eta \leq h\},$$
$$\mathcal{A} = \{a_\eta \in \mathbb{N} \,|\, 1 \leq \eta \leq h\}$$

be sets of numbers indexed by integers. Assume that

(8.3.1)

$$\begin{aligned}\mathcal{F}(\theta;\mathcal{C};\mathcal{D};\mathcal{A}) &= \sum_{\substack{N\in\mathbb{Z}\\N\neq 0}} (-1)^N C(N) e^{N\sqrt{-1}\theta} \\ &\quad - 2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta} \phi(a_\eta - k)\epsilon_{a_\eta+k} \\ &\quad \times \sum_{\xi=0}^{k} \left\{ \sum_{\substack{N\in\mathbb{Z}\\N\neq 0}} (-1)^N D(N;k-\xi;\eta) e^{N\sqrt{-1}\theta} \right\} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}\end{aligned}$$

converges absolutely and is a constant function with respect to $\theta \in [-\pi, \pi]$. Then, for $d \in \mathbb{N}_0$,

(8.3.2)

$$\begin{aligned}&\sum_{\substack{N\in\mathbb{Z}\\N\neq 0}} \frac{(-1)^N C(N) e^{N\sqrt{-1}\theta}}{N^d} \\ &\quad - 2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta} \phi(a_\eta - k)\epsilon_{a_\eta+k} \sum_{\xi=0}^{k} \left\{ \sum_{\omega=0}^{k-\xi} \binom{\omega+d-1}{\omega} (-1)^\omega \right.\end{aligned}$$

$$\times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{(-1)^m D(m;k-\xi-\omega;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\omega}} \Bigg\} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$+ 2\sum_{k=0}^{d} \phi(d-k)\epsilon_{d+k} \sum_{\xi=0}^{k} \Bigg\{ \sum_{\eta=1}^{h} \sum_{\omega=0}^{a_\eta-1} \binom{\omega+k-\xi}{\omega} (-1)^\omega$$

$$\times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{D(m;a_\eta-1-\omega;\eta)}{m^{k-\xi+\omega+1}} \Bigg\} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} = 0$$

holds for $\theta \in [-\pi,\pi]$, where the left-hand side converges absolutely and uniformly for $\theta \in [-\pi,\pi]$.

PROOF. For $d \in \mathbb{N}_0$, we set

(8.3.3) $G_d(\theta;\mathcal{C};\mathcal{D};\mathcal{A})$

$$= \frac{1}{(\sqrt{-1})^d} \Bigg[\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}} \frac{(-1)^l C(l) e^{l\sqrt{-1}\theta}}{l^d} - 2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta} \phi(a_\eta-k)\epsilon_{a_\eta+k}$$

$$\times \sum_{\xi=0}^{k} \Bigg\{ \sum_{\nu=0}^{\xi} \binom{d-1+\xi-\nu}{\xi-\nu} (-1)^\xi$$

$$\times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{(-1)^m D(m;k-\xi;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\xi-\nu}} \frac{(-\sqrt{-1}\theta)^\nu}{\nu!} \Bigg\} \Bigg]$$

$$= \frac{1}{(\sqrt{-1})^d} \Bigg[\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}} \frac{(-1)^l C(l) e^{l\sqrt{-1}\theta}}{l^d} - 2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta} \phi(a_\eta-k)\epsilon_{a_\eta+k}$$

$$\times \sum_{\nu=0}^{k} \Bigg\{ \sum_{\omega=0}^{k-\nu} \binom{d-1+\omega}{\omega} (-1)^\omega$$

$$\times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{(-1)^m D(m;k-\nu-\omega;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\omega}} \Bigg\} \frac{(\sqrt{-1}\theta)^\nu}{\nu!} \Bigg].$$

It is noted that the second equality of (8.3.3) holds by setting $\omega = \xi - \nu$. Hence (8.3.1) means

$$G_0(\theta;\mathcal{C};\mathcal{D};\mathcal{A}) = R_0 \quad (\theta \in [-\pi,\pi]) \tag{8.3.4}$$

for a constant $R_0 = R_0(\mathcal{C};\mathcal{D};\mathcal{A}) \in \mathbb{C}$, by the relation $\binom{-1+\xi-\nu}{\xi-\nu} = 0$ if $\xi > \nu$. We can confirm that $G_d(\theta;\mathcal{C};\mathcal{D};\mathcal{A})$ converges absolutely with respect to $\theta \in [-\pi,\pi]$ and

$$\frac{d}{d\theta} G_d(\theta;\mathcal{C};\mathcal{D};\mathcal{A}) = G_{d-1}(\theta;\mathcal{C};\mathcal{D};\mathcal{A}) \quad (d \in \mathbb{N}). \tag{8.3.5}$$

In fact, we differentiate term by term the second member of (8.3.3) in θ to obtain

$$\begin{aligned}
&\frac{d}{d\theta}G_d(\theta;\mathcal{C};\mathcal{D};\mathcal{A})\\
&=\frac{1}{(\sqrt{-1})^{d-1}}\Bigg[\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}}\frac{(-1)^l C(l)e^{l\sqrt{-1}\theta}}{l^{d-1}}-2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta}\phi(a_\eta-k)\epsilon_{a_\eta+k}\\
&\quad\times\sum_{\xi=0}^{k}\Bigg\{\sum_{\nu=0}^{\xi}\binom{d-1+\xi-\nu}{\xi-\nu}(-1)^\xi\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{(-1)^m D(m;k-\xi;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\xi-\nu-1}}\frac{(-\sqrt{-1}\theta)^\nu}{\nu!}\\
&\quad-\sum_{\nu=1}^{\xi}\binom{d-1+\xi-\nu}{\xi-\nu}(-1)^\xi\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{(-1)^m D(m;k-\xi;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\xi-\nu}}\frac{(-\sqrt{-1}\theta)^{\nu-1}}{(\nu-1)!}\Bigg\}\Bigg].
\end{aligned}$$

We replace $\nu-1$ by μ in the last double sum and use the property

$$-\binom{m-1}{l-1}+\binom{m}{l}=\binom{m-1}{l}\quad(l,m\in\mathbb{N}).$$

Then the right-hand side of the above can be written as

$$\begin{aligned}
&\frac{1}{(\sqrt{-1})^{d-1}}\Bigg[\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}}\frac{(-1)^l C(l)e^{l\sqrt{-1}\theta}}{l^{d-1}}-2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta}\phi(a_\eta-k)\epsilon_{a_\eta+k}\\
&\quad\times\sum_{\xi=0}^{k}\Bigg\{\sum_{\nu=0}^{\xi}\binom{d-2+\xi-\nu}{\xi-\nu}(-1)^\xi\\
&\qquad\times\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{(-1)^m D(m;k-\xi;\eta)e^{m\sqrt{-1}\theta}}{m^{d+\xi-\nu-1}}\frac{(-\sqrt{-1}\theta)^\nu}{\nu!}\Bigg\}\Bigg],
\end{aligned}$$

which gives (8.3.5). Integrating both sides of (8.3.4) and multipling by $\sqrt{-1}$ on both sides, we obtain

$$\sqrt{-1}G_1(\theta;\mathcal{C};\mathcal{D};\mathcal{A})=R_0\cdot\sqrt{-1}\theta+R_1$$

for a certain constant $R_1=R_1(\mathcal{C};\mathcal{D};\mathcal{A})$. If we repeat this operation and use (8.3.5), then

$$(\sqrt{-1})^d G_d(\theta;\mathcal{C};\mathcal{D};\mathcal{A})=\sum_{\mu=0}^{d}R_{d-\mu}\frac{(\sqrt{-1}\theta)^\mu}{\mu!}\tag{8.3.6}$$

for certain constants $R_\mu=R_\mu(\mathcal{C};\mathcal{D};\mathcal{A})$ $(0\le\mu\le d)$. Now we aim to express $\{R_\mu\}$ explicitly. If we set $\theta=\pm\pi$ in (8.3.6) with $d+1$ $(d\in\mathbb{N}_0)$,

then

(8.3.7)

$$\frac{(\sqrt{-1})^{d+1}}{2\sqrt{-1}\pi}\{G_{d+1}(\pi;\mathcal{C};\mathcal{D};\mathcal{A})-G_{d+1}(-\pi;\mathcal{C};\mathcal{D};\mathcal{A})\}=\sum_{\rho=0}^{[d/2]}R_{d-2\rho}\frac{(\sqrt{-1}\pi)^{2\rho}}{(2\rho+1)!}.$$

By (8.3.3) we see that the left-hand side can be expressed as

(8.3.8)

$$-2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta}\phi(a_\eta-k)\epsilon_{a_\eta+k}$$
$$\times\sum_{\tau=0}^{[(k-1)/2]}\left\{\sum_{\omega=0}^{k-2\tau-1}\binom{d+\omega}{\omega}(-1)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{D(m;k-2\tau-1-\omega;\eta)}{m^{d+\omega+1}}\right\}\frac{(\sqrt{-1}\pi)^{2\tau}}{(2\tau+1)!},$$

where we put $\nu=2\tau+1$. Applying (8.2.6) with

$$g(x)=\sum_{\omega=0}^{x-1}\binom{d+\omega}{\omega}(-1)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{D(m;x-1-\omega;\eta)}{m^{d+\omega+1}},$$

we can express (8.3.7) as

(8.3.9)

$$\sum_{\eta=1}^{h}\sum_{\omega=0}^{a_\eta-1}\binom{d+\omega}{\omega}(-1)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{D(m;a_\eta-1-\omega;\eta)}{m^{d+\omega+1}}=\sum_{\rho=0}^{[d/2]}R_{d-2\rho}\frac{(\sqrt{-1}\pi)^{2\rho}}{(2\rho+1)!}.$$

From (8.3.9) we can deduce

(8.3.10)

$$R_\mu=R_\mu(\mathcal{C};\mathcal{D};\mathcal{A})$$
$$=-2\sum_{\nu=0}^{\mu}\phi(\mu-\nu)\epsilon_{\mu+\nu}\sum_{\eta=1}^{h}\sum_{\omega=0}^{a_\eta-1}\binom{\nu+\omega}{\omega}(-1)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}}\frac{D(m;a_\eta-1-\omega;\eta)}{m^{\nu+\omega+1}}$$

for $\mu\in\mathbb{N}_0$. In fact, putting $d'=d+1$ and $g(x)=R_{x-1}$, we see that the right-hand side of (8.3.9) is equal to

$$\sum_{\rho=0}^{[(d'-1)/2]}g(d'-2\rho)\frac{(-1)^\rho\pi^{2\rho}}{(2\rho+1)!}.$$

Therefore the right-hand side of (8.3.10) is equal to

$$-2\sum_{\nu=0}^{\mu}\phi(\mu-\nu)\epsilon_{\mu+\nu}\sum_{\rho=0}^{[(\nu'-1)/2]}g(\nu'-2\rho)\frac{(-1)^\rho\pi^{2\rho}}{(2\rho+1)!}$$

$$= -2\sum_{\nu'=1}^{\mu'} \phi(\mu'-\nu')\epsilon_{\mu'+\nu'} \sum_{\rho=0}^{[(\nu'-1)/2]} g(\nu'-2\rho)\frac{(-1)^\rho \pi^{2\rho}}{(2\rho+1)!},$$

where $\mu' = \mu+1$ and $\nu' = \nu+1$. Using (8.2.6), we find that this is $g(\mu') = R_\mu$, hence the claim follows.

Combining (8.3.3),(8.3.6) and (8.3.10), we obtain

(8.3.11)

$$\begin{aligned}
&\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}} \frac{(-1)^l C(l) e^{l\sqrt{-1}\theta}}{l^d} - 2\sum_{\eta=1}^{h}\sum_{k=0}^{a_\eta} \phi(a_\eta - k)\epsilon_{a_\eta+k} \\
&\quad \times \sum_{\nu=0}^{k} \left\{ \sum_{\omega=0}^{k-\nu} \binom{d-1+\omega}{\omega} (-1)^\omega \right. \\
&\qquad \left. \times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{(-1)^m D(m; k-\nu-\omega; \eta) e^{m\sqrt{-1}\theta}}{m^{d+\omega}} \right\} \frac{(\sqrt{-1}\theta)^\nu}{\nu!} \\
&= -2\sum_{\mu=0}^{d}\sum_{\nu=0}^{d-\mu} \phi(d-\mu-\nu)\epsilon_{d+\mu+\nu} \\
&\quad \times \sum_{\eta=1}^{h}\sum_{\omega=0}^{a_\eta-1} \binom{\nu+\omega}{\omega}(-1)^\omega \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{D(m; a_\eta - 1 - \omega; \eta)}{m^{\nu+\omega+1}} \frac{(\sqrt{-1}\theta)^\mu}{\mu!}.
\end{aligned}$$

On the right-hand side of (8.3.11), change the running index ν by $k = \mu+\nu$. Then the right-hand side of (8.3.11) can be written as

$$\begin{aligned}
&-2\sum_{k=0}^{d} \phi(d-k)\epsilon_{d+k} \sum_{\mu=0}^{k}\sum_{\eta=1}^{h}\sum_{\omega=0}^{a_\eta-1} \binom{\omega+k-\mu}{\omega}(-1)^\omega \\
&\quad \times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{D(m; a_\eta - 1 - \omega; \eta)}{m^{k-\mu+\omega+1}} \frac{(\sqrt{-1}\theta)^\mu}{\mu!},
\end{aligned}$$

which implies (8.3.2). □

Now we consider

$$\zeta_2(s_1, s_2, s_3, s_4; C_2) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} m^{-s_1} n^{-s_2} (m+n)^{-s_3} (m+2n)^{-s_4},$$

defined by (3.3.23).

Fix $p \in \mathbb{N}$ with $p \geq 2$. It follows from (8.1.10), (8.1.11), (8.1.17) and (8.1.19) that

$$\lim_{u\to 1} J_{p+1}(\theta; p; u)$$

$$= \frac{(\sqrt{-1})^p}{2}\left(\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}}\frac{(-1)^l e^{l\sqrt{-1}\theta}}{l^p} - 2\sum_{j=0}^{p}\phi(p-j)\epsilon_{p+j}\,\frac{(\sqrt{-1}\theta)^j}{j!}\right) = 0$$

for $\theta \in (-\pi, \pi)$. Hence, for $s \in \mathbb{R}$ with $s > 1$ and $x \in \mathbb{C}$ with $|x| = 1$, we have

$$\left(\sum_{\substack{l\in\mathbb{Z}\\ l\neq 0}}\frac{(-1)^l e^{l\sqrt{-1}\theta}}{l^p} - 2\sum_{j=0}^{p}\phi(p-j)\epsilon_{p+j}\,\frac{(\sqrt{-1}\theta)^j}{j!}\right) \times \sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^s} = 0, \tag{8.3.12}$$

which holds for $\theta \in [-\pi, \pi]$ because the left-hand side is continuous for $\theta \in [-\pi, \pi]$. We stress that the principle of the u-method explained in Remark 8.11 is again working here; consider a vanishing quantity, and multiply it by another quantity to get a new non-trivial identity.

From (8.3.12) we have

$$\sum_{\substack{l\in\mathbb{Z},\ l\neq 0\\ m\geq 1\\ l+m\neq 0}}\frac{(-1)^{l+m}x^m e^{(l+m)\sqrt{-1}\theta}}{l^p m^s} - 2\sum_{j=0}^{p}\phi(p-j)\,\epsilon_{p+j}\left\{\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^s}\right\}\frac{(\sqrt{-1}\theta)^j}{j!} = (-1)^{p+1}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p}} \tag{8.3.13}$$

for $\theta \in [-\pi, \pi]$. We apply Lemma 8.24 with $h = 1$, $a_1 = p$, $d = q$ for $q \in \mathbb{N}$ and

$$C(N) = \sum_{\substack{l\neq 0,\ m\geq 1\\ l+m=N}}\frac{x^m}{l^p m^s} \quad (N \in \mathbb{Z},\ N \neq 0),$$

$$D(N;\mu;1) = \begin{cases} \frac{x^N}{N^s} & (\text{if } \mu = 0 \text{ and } N \geq 1),\\ 0 & (\text{otherwise}).\end{cases}$$

Here we fix s and x, and regard the right-hand side of (8.3.13) as a constant. Therefore it follows from (8.3.2) that

$$\sum_{\substack{l\neq 0,\ m\geq 1\\ l+m\neq 0}}\frac{(-1)^{l+m}x^m e^{(l+m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q} - 2\sum_{j=0}^{p}\phi(p-j)\,\epsilon_{p+j}\sum_{\xi=0}^{j}\binom{q-1+j-\xi}{j-\xi} \tag{8.3.14}$$

$$\times (-1)^{j-\xi} \sum_{m=1}^{\infty} \frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+q+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$
$$+ 2 \sum_{j=0}^{q} \phi(q-j)\, \epsilon_{q+j} \sum_{\xi=0}^{j} \binom{p-1+j-\xi}{j-\xi}$$
$$\times (-1)^{p-1} \sum_{m=1}^{\infty} \frac{x^m}{m^{s+p+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} = 0$$

for $\theta \in [-\pi, \pi]$.

Here we replace x by $-xe^{\sqrt{-1}\theta}$ on the left-hand side of (8.3.14). Then

$$\sum_{\substack{l\in\mathbb{Z},\, l\neq 0 \\ m\geq 1,\, l+m\neq 0}} \frac{(-1)^l x^m e^{(l+2m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q}$$
$$- 2 \sum_{j=0}^{p} \phi(p-j)\, \epsilon_{p+j} \sum_{\xi=0}^{j} \binom{q-1+j-\xi}{q-1}$$
$$\times (-1)^{j-\xi} \sum_{m=1}^{\infty} \frac{x^m e^{2m\sqrt{-1}\theta}}{m^{s+q+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$
$$+ 2 \sum_{j=0}^{q} \phi(q-j)\, \epsilon_{q+j} \sum_{\xi=0}^{j} \binom{p-1+j-\xi}{2p-1}$$
$$\times (-1)^{p-1} \sum_{m=1}^{\infty} \frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+p+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} = 0$$

for $\theta \in [-\pi, \pi]$. As for the first term on the left-hand side, we separate the terms corresponding to the condition $l+2m=0$ and move them to the right-hand side. Then we can again apply Lemma 8.24, this time with $(h, a_1, a_2, d) = (2, p, q, r)$ for $r \in \mathbb{N}$,

$$C(N) = \sum_{\substack{l\neq 0,\, m\geq 1 \\ l+m\neq 0, l+2m=N}} \frac{x^m}{l^p m^s (l+m)^q} \quad (N \in \mathbb{Z},\ N \neq 0),$$

$$D(N;\mu;1) = \begin{cases} (-1)^\mu \binom{q-1+\mu}{q-1} x^N (N/2)^{-s-q-\mu} & (\text{if } N \geq 1 : \text{even}), \\ 0 & (\text{otherwise}), \end{cases}$$

$$D(N;\mu;2) = \begin{cases} (-1)^p \binom{p-1+\mu}{p-1} x^N N^{-s-p-\mu} & (\text{if } N \geq 1), \\ 0 & (\text{otherwise}). \end{cases}$$

We consequently obtain

(8.3.15)

$$\sum_{\substack{l\in\mathbb{Z},\, l\neq 0\\ m\geq 1,\, l+m\neq 0\\ l+2m\neq 0}} \frac{(-1)^l x^m e^{(l+2m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q (l+2m)^r}$$
$$- 2\sum_{j=0}^{p} \phi(p-j)\epsilon_{p+j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi} \binom{\omega+r-1}{\omega}(-1)^\omega \binom{q-1+j-\xi-\omega}{q-1}$$
$$\times (-1)^{j-\xi-\omega}\frac{1}{2^{r+\omega}} \sum_{m=1}^{\infty} \frac{x^m e^{2m\sqrt{-1}\theta}}{m^{s+q+j-\xi+r}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$
$$- 2\sum_{j=0}^{q} \phi(q-j)\epsilon_{q+j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi} \binom{\omega+r-1}{\omega}(-1)^\omega \binom{p-1+j-\xi-\omega}{p-1}$$
$$\times (-1)^{p} \sum_{m=1}^{\infty} \frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+p+j-\xi+r}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$
$$+ 2\sum_{j=0}^{r} \phi(r-j)\epsilon_{r+j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{p-1} \binom{\omega+j-\xi}{\omega}(-1)^\omega \binom{p+q-2-\omega}{q-1}$$
$$\times (-1)^{p-1-\omega}\frac{1}{2^{j-\xi+\omega+1}} \sum_{m=1}^{\infty} \frac{x^m}{m^{s+q+j-\xi+p}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$
$$+ 2\sum_{j=0}^{r} \phi(r-j)\epsilon_{r+j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{q-1} \binom{\omega+j-\xi}{\omega}(-1)^\omega \binom{p+q-2-\omega}{p-1}$$
$$\times (-1)^{p} \sum_{m=1}^{\infty} \frac{x^m}{m^{s+p+j-\xi+q}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} = 0$$

for $\theta \in [-\pi, \pi]$. Set $(\theta, x) = (\pi, 1)$ and consider the real part. First, we show:

Claim 8.25. *The first term on the left-hand side of* (8.3.15) *with* $(\theta, x) = (\pi, 1)$ *is equal to*

$$\zeta_2(p,s,q,r;C_2) + (-1)^p\zeta_2(p,q,s,r;C_2)$$
$$+ (-1)^{p+q}\zeta_2(r,q,s,p;C_2) + (-1)^{p+q+r}\zeta_2(r,s,q,p;C_2).$$

Proof. We divide the first term on the left-hand side into the cases $l \geq 1$ and $l \leq -1$. The part $l \geq 1$ gives $\zeta_2(p,s,q,r;C_2)$. As for the latter case, we replace l by $-j$ and divide it into the cases $j < m$ and $j > m$. The part $j < m$ is, by putting $m - j = n$, equal to $(-1)^p\zeta_2(p,q,s,r;C_2)$. On the part $j > m$, we put $j - m = n$ and we further divide into two parts according to $m < n$ and $m > n$. Each

of these two terms can also be written in terms of the zeta-function of C_2. □

As for the latter four terms on the left-hand side of (8.3.15) with $\theta = \pi$, we apply (8.2.5) to the real part, which corresponds to the case that ξ is even. Since the zeta-function of C_2 can be continued meromorphically to the whole space (see Corollary 7.5), we obtain the following.

THEOREM 8.26 ([**103**, Remark 8.1]). *For $p, q, r \in \mathbb{N}$ with $p \geq 2$,*

$$\begin{aligned}
&\zeta_2(p,s,q,r;C_2) + (-1)^p \zeta_2(p,q,s,r;C_2) + (-1)^{p+q} \zeta_2(r,q,s,p;C_2) \\
&\qquad + (-1)^{p+q+r} \zeta_2(r,s,q,p;C_2) \\
&= 2(-1)^p \times \\
&\Bigg\{ \sum_{\ell=0}^{[p/2]} \zeta(2\ell)\zeta(p+q+r-2\ell+s) \sum_{\mu=0}^{p-2\ell} \frac{1}{2^{r+\mu}} \binom{p+q-2\ell-\mu-1}{q-1} \binom{r-1+\mu}{r-1} \\
&\quad + \sum_{\ell=0}^{[q/2]} \zeta(2\ell)\zeta(p+q+r-2\ell+s) \sum_{\mu=0}^{q-2\ell} (-1)^{\mu} \binom{p+q-2\ell-\mu-1}{p-1} \binom{r-1+\mu}{r-1} \\
&\quad + \sum_{\ell=0}^{[r/2]} \zeta(2\ell)\zeta(p+q+r-2\ell+s) \sum_{\mu=0}^{p-1} \frac{1}{2^{r-2\ell+\mu+1}} \binom{p+q-\mu-2}{q-1} \binom{r-2\ell+\mu}{r-2\ell} \\
&\quad + \sum_{\ell=0}^{[r/2]} \zeta(2\ell)\zeta(p+q+r-2\ell+s) \sum_{\mu=0}^{q-1} (-1)^{\mu+1} \binom{p+q-\mu-2}{p-1} \binom{r-2\ell+\mu}{r-2\ell} \Bigg\}
\end{aligned}$$

holds for all $s \in \mathbb{C}$ except for singularities of functions on both sides.

REMARK 8.27. Using a slightly more delicate discussion, we can show that this theorem holds for the case $p = 1$.

REMARK 8.28. Theorem 8.26 (with Remark 8.27) was essentially announced in [**100**, Theorem 2.1 and Example 2.1], though in [**100**] its proof was omitted and the statements are written in terms of the zeta-function of B_2. In [**103**], we gave the proof of the case when p, q, r are even integers in Theorem 8.26, and the general case was just stated. We give here the full proof of the above general form of Theorem 8.26.

It is to be noted that the first published proof of Theorem 8.26 is due to Nakamura [**163**, Theorem 5.5]. His method is different from ours, a generalization of that in [**162**] (see Remark 8.4). In the same paper, Nakamura also gave a functional relation which connects $\zeta_2(\mathbf{s}; C_2)$ with $\zeta_2(\mathbf{s}; A_2)$.

Similar to Corollary 8.20, we obtain the following fact, from which examples listed at the end of Section 4.2 can now be immediately deduced.

COROLLARY 8.29. *For $k,l,m,n\in\mathbb{N}$,*

$$\begin{aligned}(8.3.16)\quad &\zeta_2(2k,2n,2l,2m;C_2)+\zeta_2(2k,2l,2n,2m;C_2)\\ &\quad+\zeta_2(2m,2l,2n,2k;C_2)+\zeta_2(2m,2n,2l,2k;C_2)\\ &=2\Bigg\{\sum_{\nu=0}^{k}\zeta(2\nu)\zeta(2k+2l+2m-2\nu+2n)\\ &\qquad\times\sum_{\mu=0}^{2k-2\nu}\frac{1}{2^{2m+\mu}}\binom{2k+2l-2\nu-\mu-1}{2l-1}\binom{2m-1+\mu}{2m-1}\\ &\quad+\sum_{\nu=0}^{l}\zeta(2\nu)\zeta(2k+2l+2m-2\nu+2n)\\ &\qquad\times\sum_{\mu=0}^{2l-2\nu}(-1)^{\mu}\binom{2k+2l-2\nu-\mu-1}{2k-1}\binom{2m-1+\mu}{2m-1}\\ &\quad+\sum_{\nu=0}^{m}\zeta(2\nu)\zeta(2k+2l+2m-2\nu+2n)\\ &\qquad\times\sum_{\mu=0}^{2k-1}\frac{1}{2^{2m-2\nu+\mu+1}}\binom{2k+2l-\mu-2}{2l-1}\binom{2m-2\nu+\mu}{2m-2\nu}\\ &\quad+\sum_{\nu=0}^{m}\zeta(2\nu)\zeta(2k+2l+2m-2\nu+2n)\\ &\qquad\times\sum_{\mu=0}^{2l-1}(-1)^{\mu+1}\binom{2k+2l-\mu-2}{p-1}\binom{2m-2\nu+\mu}{2m-2\nu}\Bigg\}.\end{aligned}$$

In particular, for $k,l\in\mathbb{N}$,

(8.3.17)

$$\begin{aligned}&2\zeta_2(2k,2l,2l,2k;C_2)\\ &=\Bigg\{\sum_{\nu=0}^{k}\zeta(2\nu)\zeta(4k+4l-2\nu)\sum_{\mu=0}^{2k-2\nu}\frac{1}{2^{2k+\mu}}\binom{2k+2l-2\nu-\mu-1}{2l-1}\binom{2k-1+\mu}{2k-1}\\ &\quad+\sum_{\nu=0}^{l}\zeta(2\nu)\zeta(4k+4l-2\nu)\sum_{\mu=0}^{2l-2\nu}(-1)^{\mu}\binom{2k+2l-2\nu-\mu-1}{2k-1}\binom{2k-1+\mu}{2k-1}\\ &\quad+\sum_{\nu=0}^{m}\zeta(2\nu)\zeta(4k+4l-2\nu)\sum_{\mu=0}^{2k-1}\frac{1}{2^{2k-2\nu+\mu+1}}\binom{2k+2l-\mu-2}{2l-1}\binom{2k-2\nu+\mu}{2k-2\nu}\end{aligned}$$

$$+\sum_{\nu=0}^{m}\zeta(2\nu)\zeta(4k+4l-2\nu)\sum_{\mu=0}^{2l-1}(-1)^{\mu+1}\binom{2k+2l-\mu-2}{2k-1}\binom{2k-2\nu+\mu}{2k-2\nu}\Bigg\}.$$

From (8.3.17) we can immediately obtain the examples listed in Example 4.16.

EXAMPLE 8.30. Setting $(p,q,r)=(2,2,2)$ and $(2,4,2)$ in Theorem 8.26, we have

$$\zeta_2(2,s,2,2;C_2)+\zeta_2(2,2,s,2;C_2) = -\frac{39}{16}\zeta(s+6)+\frac{3}{2}\zeta(2)\zeta(s+4), \tag{8.3.18}$$

$$\zeta_2(2,s,4,2;C_2)+\zeta_2(2,4,s,2;C_2) = -\frac{83}{16}\zeta(s+8)+\frac{5}{2}\zeta(2)\zeta(s+6)+\zeta(4)\zeta(s+4). \tag{8.3.19}$$

Setting $s=2$ in (8.3.18), we recover the result given in Example 4.16:

$$\zeta_2(2,2,2,2;C_2)=\frac{\pi^8}{302400}.$$

Setting $s=4$ in (8.3.19), we obtain

$$\zeta_2(2,4,4,2;C_2)=\frac{53}{6810804000}\pi^{12}.$$

Setting $(p,q,r)=(2,1,1)$ in Theorem 8.26, we have

$$\zeta_2(2,1,s,1;C_2)+\zeta_2(2,s,1,1;C_2)+\zeta_2(1,s,1,2;C_2)-\zeta_2(1,1,s,2;C_2) = \zeta(2)\zeta(s+2)-\frac{11}{8}\zeta(s+4).$$

In particular, setting $s=1$, we have

$$\zeta_2(2,1,1,1;C_2)=\frac{1}{2}\zeta(2)\zeta(3)-\frac{11}{16}\zeta(5). \tag{8.3.20}$$

Similar to Corollary 8.21, we can prove the following fact which includes (8.3.20).

PROPOSITION 8.31 ([**200**, Theorem]). *For $a,b,c,d\in\mathbb{N}$ with $2\nmid(a+b+c+d)$,*

$$\zeta_2(a,b,c,d;C_2)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}]. \tag{8.3.21}$$

REMARK 8.32. Unlike Corollary 8.21, this proposition cannot be derived from Theorem 8.26. We can easily check, however, that several special cases come from Theorem 8.26, for example,

$$\zeta_2(2p,2q+1,2q+1,2r+1;C_2)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}],$$
$$\zeta_2(2p,2q,2q,2r+1;C_2)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}],$$
$$\zeta_2(2p,2q,2r+1,2p;C_2)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\}_{j\in\mathbb{N}}].$$

EXAMPLE 8.33. As examples of Proposition 8.31 (see [200, Example in Section 3]), we can obtain

$$\begin{aligned}
\zeta_2(1,1,1,2;C_2) &= -\frac{3}{8}\zeta(3)\pi^2 + \frac{35}{8}\zeta(5),\\
\zeta_2(2,1,1,1;C_2) &= \frac{1}{12}\zeta(3)\pi^2 - \frac{11}{16}\zeta(5),\\
\zeta_2(1,2,1,3;C_2) &= \frac{61}{48}\zeta(5)\pi^2 - \frac{103}{8}\zeta(7),\\
\zeta_2(2,1,2,2;C_2) &= -\frac{43}{48}\zeta(5)\pi^2 + \frac{73}{8}\zeta(7),\\
\zeta_2(2,3,1,1;C_2) &= -\frac{5}{12}\zeta(5)\pi^2 + \frac{71}{16}\zeta(7),\\
\zeta_2(3,1,1,2;C_2) &= -\frac{53}{96}\zeta(5)\pi^2 + \frac{363}{64}\zeta(7),\\
\zeta_2(1,2,3,3;C_2) &= -\frac{217}{48}\zeta(7)\pi^2 + \frac{1437}{32}\zeta(9),\\
\zeta_2(1,3,2,3;C_2) &= -\frac{265}{48}\zeta(7)\pi^2 + \frac{1755}{32}\zeta(9),\\
\zeta_2(1,3,3,2;C_2) &= -\frac{241}{24}\zeta(7)\pi^2 + \frac{399}{4}\zeta(9),\\
\zeta_2(2,2,2,3;C_2) &= \frac{5}{16}\zeta(7)\pi^2 - \frac{99}{32}\zeta(9),\\
\zeta_2(2,3,2,2;C_2) &= -\frac{235}{48}\zeta(7)\pi^2 + \frac{1557}{32}\zeta(9),\\
\zeta_2(2,3,3,3;C_2) &= -\frac{5}{16}\zeta(9)\pi^2 + \frac{99}{32}\zeta(11),\\
\zeta_2(3,2,3,3;C_2) &= -\frac{991}{192}\zeta(9)\pi^2 + \frac{6531}{128}\zeta(11),\\
\zeta_2(3,3,3,2;C_2) &= -\frac{1021}{96}\zeta(9)\pi^2 + \frac{6729}{64}\zeta(11).
\end{aligned}$$

8.4. An application of Nakamura's method to the zeta-function of G_2

After the discovery of the u-method, some other methods for the proof of functional relations were introduced by several mathematicians. We already mentioned the method of Nakamura in Remark 8.4.

Zhou, Bradley and Cai [222] proposed an alternative method of rather elementary nature, which was further developed by Ikeda and K. Matsuoka [73], who proved certain functional relations among zeta-functions of A_2, A_3 and A_4 (of the form that some of the variables are equal to 0). A method developed in the authors' paper [120, Section

6] for C_2 is also of elementary nature. In the case of A_2, the papers [148] and [175] must also be mentioned.

In this section we show the idea of Nakamura's method, by describing a proof of a certain functional relation for the zeta-function of G_2. This is due to the authors' paper [117].

EXAMPLE 8.34. As stated in (3.3.28), the zeta-function of type G_2 is defined by

$$\begin{aligned}\zeta_2(\mathbf{s};G_2) &= \zeta_2(s_1,s_2,s_3,s_4,s_5,s_6;G_2)\\ &= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{1}{m^{s_1}n^{s_2}(m+n)^{s_3}(m+2n)^{s_4}(m+3n)^{s_5}(2m+3n)^{s_6}}.\end{aligned}$$

Then the functional relation ([117, Example 4.1])

(8.4.1)

$$\begin{aligned}&\zeta_2(s,2,1,1,1,1;G_2)+\zeta_2(s,1,2,1,1,1;G_2)+\zeta_2(1,2,1,1,s,1;G_2)\\ &\quad-\zeta_2(1,1,2,1,1,s;G_2)+\zeta_2(1,1,1,2,1,s;G_2)-\zeta_2(1,1,1,2,s,1;G_2)\\ &=\zeta(2)\zeta(s+4)-\left(\frac{651}{8}-\frac{1}{2^{s+1}}-\frac{5}{2\cdot 3^{s+2}}\right)\zeta(s+6)\\ &\qquad+\frac{9\pi}{2}\sum_{m\geq 1}\frac{\sin(2\pi m/3)}{m^{s+5}}-135\sum_{m\geq 1}\frac{\cos(2\pi m/3)}{m^{s+6}}\\ &=\zeta(2)\zeta(s+4)-\left(\frac{111}{8}-\frac{1}{2^{s+1}}\right)\zeta(s+6)+\frac{81}{4}L(1,\chi_3)L(s+5,\chi_3)\end{aligned}$$

holds for $s\in\mathbb{C}$ except for singularities on both sides, where $L(s,\chi_3)$ is the Dirichlet L-function attached to the primitive odd Dirichlet character χ_3 of conductor 3. The second equality of the above formula depends on the fact[5] that $L(1,\chi_3)=(\pi\sqrt{3})/9$.

In particular, setting $s=1$ in (8.4.1), we obtain that the left-hand side is equal to $2\zeta_2(1,2,1,1,1,1;G_2)$. Hence we have

$$\begin{aligned}&\zeta_2(1,2,1,1,1,1;G_2)\\ &\quad=\frac{1}{2}\zeta(2)\zeta(5)-\frac{109}{16}\zeta(7)+\frac{81}{8}L(1,\chi_3)L(6,\chi_3).\end{aligned}\tag{8.4.2}$$

[5]This is a special case of the well-known formula $L(1,\chi)=q^{-1}\pi\sqrt{-1}\tau(\chi)B_{1,\bar{\chi}}$ for any odd character χ mod q, where $\tau(\chi)$ is the Gauss sum and $B_{1,\chi}=q^{-1}\sum_{a=1}^{q}\chi(a)a$ (see, e.g., [207, Chapter 4]).

We give the proof[6] of (8.4.1) by calculating

$$S=\sum_{m\geq 1}\sum_{\substack{n\neq 0\\ m+n\neq 0\\ m+2n\neq 0\\ m+3n\neq 0\\ 2m+3n\neq 0}}\frac{1}{m^s n^2(m+n)(m+2n)(m+3n)(2m+3n)} \tag{8.4.3}$$

in two ways.

Claim 8.35. (8.4.3) *is equal to the left-hand side of* (8.4.1).

The proof of this claim is similar to the case of C_2 (see Claim 8.25), so we omit it. Therefore the remaining task is to show that (8.4.3) is equal to the right-hand side of (8.4.1). A direct calculation shows that (8.4.3) can be rewritten as

(8.4.4)

$$\sum_{m\geq 1}\sum_{\substack{n\neq 0\\ l_1\neq 0\\ l_2\neq 0\\ l_3\neq 0\\ l_4\neq 0}}\frac{1}{m^s n^2 l_1 l_2 l_3 l_4}\int_0^1\int_0^1\int_0^1\int_0^1 e^{2\pi\sqrt{-1}x_1(m+n-l_1)}e^{2\pi\sqrt{-1}x_2(m+2n-l_2)}$$
$$\times e^{2\pi\sqrt{-1}x_3(m+3n-l_3)}e^{2\pi\sqrt{-1}x_4(2m+3n-l_4)}dx_1dx_2dx_3dx_4.$$

It is possible to compute this value by using (4.2.4).

This idea was initiated by Zagier (see [**6**, Proposition 4.14]) and systematically used by Nakamura (see [**162**] [**163**] and also [**140**] [**141**]). However, if we apply this method directly to (8.4.4), the necessary computations are enormous. Therefore, in order to reduce the total amount of computation, we modify (8.4.3) using partial fraction decompositions, before applying the idea of Zagier–Nakamura.

By the partial fraction decomposition

$$\frac{1}{(m+3n)(2m+3n)}=\frac{1}{m(m+3n)}-\frac{1}{m(2m+3n)},$$

we divide (8.4.3) into two sums,

$$\sum\sum^{*}\frac{1}{m^{s+1}n^2(m+n)(m+2n)(m+3n)}$$
$$-\sum\sum^{*}\frac{1}{m^{s+1}n^2(m+n)(m+2n)(2m+3n)}, \tag{8.4.5}$$

where $\sum\sum^{*}$ denotes the same double sum as in (8.4.3). Then apply

$$\frac{1}{(m+2n)(m+3n)}=\frac{1}{m(m+3n)}-\frac{1}{m(2m+3n)},$$

[6]There are several misprints in the proof of (8.4.1) written in [**117**]. The following proof is the corrected version.

$$\frac{1}{(m+2n)(2m+3n)} = \frac{2}{n(2m+3n)} - \frac{1}{n(m+2n)}$$

to find that (8.4.5) is

$$2\sum\sum^{*} \frac{1}{m^{s+1}n^3(m+n)(m+2n)} - \sum\sum^{*} \frac{1}{m^{s+1}n^3(m+n)(m+3n)} - 2\sum\sum^{*} \frac{1}{m^{s+1}n^3(m+n)(2m+3n)}.$$

Further, we use

$$\frac{1}{(m+n)(m+3n)} = \frac{1}{2n(m+n)} - \frac{1}{2n(m+3n)},$$
$$\frac{1}{(m+n)(2m+3n)} = \frac{1}{n(m+n)} - \frac{2}{n(2m+3n)}.$$

We obtain

(8.4.6)
$$\begin{aligned} S &= 2\sum\sum^{*} \frac{1}{m^{s+1}n^3(m+n)(m+2n)} - \frac{5}{2}\sum\sum^{*} \frac{1}{m^{s+1}n^4(m+n)} \\ &\quad + \frac{1}{2}\sum\sum^{*} \frac{1}{m^{s+1}n^4(m+3n)} + 4\sum\sum^{*} \frac{1}{m^{s+1}n^4(2m+3n)} \\ &= 2\Sigma_1 - \frac{5}{2}\Sigma_2 + \frac{1}{2}\Sigma_3 + 4\Sigma_4, \end{aligned}$$

say.

We divide Σ_1 as

$$\Sigma_1 = \sum\sum^{**} - \sum_{m+3n=0}\sum^{**} - \sum_{2m+3n=0}\sum^{**},$$

where $\sum\sum^{**}$ denotes the sum over $m \geq 1$, $n \neq 0$, $m+n \neq 0$ and $m+2n \neq 0$. Denote the first term by Σ_{11}. Putting $m = 3l$ and $n = -l$ in the second sum, we see that the second term is

$$-\sum_{l=1}^{\infty} \frac{1}{(3l)^{s+1}(-l)^3(3l-l)(3l-2l)} = \frac{1}{2\cdot 3^{s+1}}\zeta(s+6).$$

Similarly, putting $m = 3l$ and $n = -2l$, we see that the third term is $-3^{-s-1}2^{-3}\zeta(s+6)$. Therefore we obtain

$$\Sigma_1 = \Sigma_{11} + \left(\frac{1}{2\cdot 3^{s+1}} - \frac{1}{2^3 3^{s+1}}\right)\zeta(s+6) = \Sigma_{11} + \frac{1}{2^3 3^s}\zeta(s+6). \tag{8.4.7}$$

As for Σ_2, we divide it as

$$\Sigma_2 = \sum\sum^{***} - \sum_{m+2n=0}\sum^{***} - \sum_{m+3n=0}\sum^{***} - \sum_{2m+3n=0}\sum^{***}, \tag{8.4.8}$$

where $\sum\sum^{***}$ denotes the sum over $m \geq 1$, $n \neq 0$, $m+n \neq 0$. Denote the first term by Σ_{21} and evaluate the remaining three sums as above to obtain

$$\Sigma_2 = \Sigma_{21} - \left(\frac{1}{2^{s+1}} + \frac{1}{2^4 3^{s-1}}\right)\zeta(s+6). \tag{8.4.9}$$

Similarly, we can obtain

$$\Sigma_3 = \Sigma_{31} + \left(\frac{1}{2} + \frac{1}{2^{s+1}} + \frac{1}{2^4 3^{s+2}}\right)\zeta(s+6) \tag{8.4.10}$$

and

$$\Sigma_4 = \Sigma_{41} + \left(1 - \frac{1}{2^{s+1}} - \frac{1}{3^{s+2}}\right)\zeta(s+6), \tag{8.4.11}$$

where

$$\Sigma_{31} = \sum_{m\geq 1} \sum_{\substack{n\neq 0 \\ m+3n\neq 0}} \frac{1}{m^{s+1}n^4(m+3n)},$$

$$\Sigma_{41} = \sum_{m\geq 1} \sum_{\substack{n\neq 0 \\ 2m+3n\neq 0}} \frac{1}{m^{s+1}n^4(2m+3n)}.$$

Consequently, we now arrive at the formula

$$S = 2\Sigma_{11} - \frac{5}{2}\Sigma_{21} + \frac{1}{2}\Sigma_{31} + 4\Sigma_{41} + \left(\frac{17}{4} - \frac{1}{2^{s+1}} + \frac{5}{2\cdot 3^{s+2}}\right)\zeta(s+6). \tag{8.4.12}$$

We evaluate the sums Σ_{j1} $(1 \leq j \leq 4)$. First, applying a partial fraction decomposition once more, we obtain

$$\Sigma_{11} = \sum\sum^{**} \frac{1}{m^{s+1}n^4(m+n)} - \sum\sum^{**} \frac{1}{m^{s+1}n^4(m+2n)}.$$

On the right-hand side, we separate the part $m+2n=0$ from the first double sum and separate the part $m+n=0$ from the second double sum. We obtain

$$\Sigma_{11} = \Sigma_{21} - \Sigma^{\sharp} - \left(1 + \frac{1}{2^{s+1}}\right)\zeta(s+6), \tag{8.4.13}$$

where

$$\Sigma^{\sharp} = \sum_{m\geq 1} \sum_{\substack{n\neq 0 \\ m+2n\neq 0}} \frac{1}{m^{s+1}n^4(m+2n)}.$$

Next, we evaluate Σ_{21}. Recall the definition of the zeta-function of the root system A_2

$$\zeta_2(s_1, s_2, s_3; A_2) = \sum_{m,n\geq 1} \frac{1}{m^{s_1}n^{s_2}(m+n)^{s_3}}.$$

The part corresponding to positive n of the sum Σ_{21} is exactly $\zeta_2(s+1,4,1;A_2)$. The part corresponding to negative n can be written as

$$\sum_{\substack{m,n\geq 1\\ m\neq n}} \frac{1}{m^{s+1}n^4(m-n)}.$$

By putting $m-n=l$ when $m>n$ and $n-m=k$ when $m<n$, this is equal to

$$\begin{aligned}&\sum_{n,l\geq 1}\frac{1}{(n+l)^{s+1}n^4l}-\sum_{m,k\geq 1}\frac{1}{m^{s+1}(m+k)^4k}\\ &\quad=\zeta_2(4,s+1,1;A_2)-\zeta_2(1,s+1,4;A_2).\end{aligned}$$

Therefore

(8.4.14)
$$\begin{aligned}\Sigma_{21}&=\zeta_2(s+1,4,1;A_2)+\zeta_2(4,s+1,1;A_2)-\zeta_2(1,s+1,4;A_2)\\ &=-5\zeta(s+6)+2\zeta(2)\zeta(s+4)+2\zeta(4)\zeta(s+2),\end{aligned}$$

where the second equality can be seen by Theorem 8.3.

As for $\Sigma^\sharp$, we apply the idea of Zagier–Nakamura. Write $\Sigma^\sharp$ as

$$\begin{aligned}\Sigma^\sharp&=\sum_{\substack{m\geq 1\\ n\neq 0\\ l\neq 0}}\frac{1}{m^{s+1}n^4l}\int_0^1 e^{2\pi\sqrt{-1}(m+2n-l)\theta}d\theta\\ &=\sum_{m\geq 1}\frac{1}{m^{s+1}}\int_0^1 e^{2\pi\sqrt{-1}m\theta}\sum_{n\neq 0}\frac{e^{2\pi\sqrt{-1}n\cdot 2\theta}}{n^4}\sum_{l\neq 0}\frac{e^{2\pi\sqrt{-1}l(-\theta)}}{l}d\theta,\end{aligned}\tag{8.4.15}$$

where the interchange of the integration and summation is verified because the sum $\sum_{l\neq 0}$ is convergent uniformly in any closed subinterval of $(0,1)$ (see [148, Lemma 4.1]). Applying (4.2.4), we obtain

$$\begin{aligned}\Sigma^\sharp&=\frac{(2\pi\sqrt{-1})^5}{24}\sum_{m\geq 1}\frac{1}{m^{s+1}}\int_0^1 e^{2\pi\sqrt{-1}m\theta}B_4(\{2\theta\})B_1(\{-\theta\})d\theta\\ &=\frac{(2\pi\sqrt{-1})^5}{24}\sum_{m\geq 1}\frac{1}{m^{s+1}}(J_1+J_2),\end{aligned}\tag{8.4.16}$$

where

$$\begin{aligned}J_1&=\int_0^{1/2} e^{2\pi\sqrt{-1}m\theta}B_4(2\theta)B_1(1-\theta)d\theta,\\ J_2&=\int_{1/2}^1 e^{2\pi\sqrt{-1}m\theta}B_4(2\theta-1)B_1(1-\theta)d\theta.\end{aligned}$$

Since $B_1(x)=x-(1/2)$ and $B_4(x)=x^4-2x^3+x^2-(1/30)$, the factors $B_4(2\theta)B_1(1-\theta)$ and $B_4(2\theta-1)B_1(1-\theta)$ are polynomials in θ of degree

5. It is easy to see recursively that

(8.4.17)
$$\int_0^{1/2} e^{2\pi\sqrt{-1}m\theta}\theta^k d\theta = \sum_{j=1}^{k+1} \frac{(-1)^{j-1+m}k!}{(2\pi im)^j 2^{k+1-j}(k+1-j)!} - \frac{(-1)^k k!}{(2\pi im)^{k+1}}$$

and

$$\int_{1/2}^{1} e^{2\pi\sqrt{-1}m\theta}\theta^k d\theta = \sum_{j=1}^{k+1} \frac{(-1)^{j-1}k!}{(2\pi im)^j (k+1-j)!}\left(1 - \frac{(-1)^m}{2^{k+1-j}}\right). \tag{8.4.18}$$

Write $B_4(2\theta)B_1(1-\theta) = \sum_{k=0}^{5} a_k\theta^k$. (We see that $a_0 = -1/60, a_1 = 1/30, a_2 = 2, a_3 = -12, a_4 = 24$ and $a_5 = -16$.) Using (8.4.17), we find that the contribution of J_1 to (8.4.16) is

$$\begin{aligned}
&\frac{(2\pi\sqrt{-1})^5}{24}\sum_{m\geq 1}\frac{1}{m^{s+1}}\sum_{k=0}^{5} a_k \int_0^{1/2} e^{2\pi\sqrt{-1}m\theta}\theta^k d\theta \\
&= \frac{(2\pi\sqrt{-1})^5}{24}\sum_{k=0}^{5} a_k \left\{ \sum_{j=1}^{k+1} \frac{(-1)^{j-1}k!}{(2\pi\sqrt{-1})^j 2^{k+1-j}(k+1-j)!}\phi(s+j+1) \right. \\
&\qquad \left. - \frac{(-1)^k k!}{(2\pi\sqrt{-1})^{k+1}}\zeta(s+k+2)\right\}.
\end{aligned}$$

The contribution of J_2 can be evaluated similarly. Substituting these expressions into (8.4.16) and using (8.1.1), we can obtain

$$\Sigma^\sharp = \frac{\pi^4}{45}\zeta(s+2) + \frac{4\pi^2}{3}\zeta(s+4) - \left(16 + \frac{1}{2^s}\right)\zeta(s+6), \tag{8.4.19}$$

because the terms $\zeta(s+k+2)$ $(k = 1, 3, 5)$ vanish.

Substituting (8.4.14) and (8.4.19) into (8.4.13) we obtain

$$\Sigma_{11} = -\pi^2\zeta(s+4) + \left(10 + \frac{1}{2^{s+1}}\right)\zeta(s+6), \tag{8.4.20}$$

and so

$$\begin{aligned}
2\Sigma_{11} - \frac{5}{2}\Sigma_{21} = &-\frac{\pi^4}{18}\zeta(s+2) - \frac{17\pi^2}{6}\zeta(s+4) \\
&+ \left(\frac{65}{2} + \frac{1}{2^s}\right)\zeta(s+6).
\end{aligned} \tag{8.4.21}$$

The evaluation of Σ_{31} and Σ_{41} is similar to that of $\Sigma^\sharp$, but in these cases, we have to divide the integrals into three parts. As for Σ_{31}, we have

$$\Sigma_{31} = \sum_{\substack{m\geq 1\\ n\neq 0\\ l\neq 0}} \frac{1}{m^{s+1}n^4 l}\int_0^1 e^{2\pi\sqrt{-1}(m+3n-l)\theta}d\theta$$

$$= \frac{(2\pi\sqrt{-1})^5}{24} \sum_{m\geq 1} \frac{1}{m^{s+1}} \int_0^1 e^{2\pi\sqrt{-1}m\theta} B_4(\{3\theta\}) B_1(\{-\theta\}) d\theta$$

$$= \frac{(2\pi\sqrt{-1})^5}{24} \sum_{m\geq 1} \frac{1}{m^{s+1}} (J_1' + J_2' + J_3'),$$

where

$$J_i' = \int_{(i-1)/3}^{i/3} e^{2\pi\sqrt{-1}m\theta} B_4(3\theta - i + 1) B_1(1-\theta) d\theta \qquad (1 \leq i \leq 3).$$

We need, instead of (8.4.17) and (8.4.18), explicit expressions for

$$I_{i,k} = \int_{(i-1)/3}^{i/3} e^{2\pi\sqrt{-1}m\theta} \theta^k d\theta \qquad (1 \leq i \leq 3, k \in \mathbb{N}_0).$$

Such expressions can be obtained recursively by using

$$I_{i,0} = \frac{1}{2\pi\sqrt{-1}m} \left(e^{2\pi\sqrt{-1}i/3} - e^{2\pi\sqrt{-1}(i-1)/3} \right)$$

and

$$I_{i,k} = \frac{1}{2\pi\sqrt{-1}m} \left(e^{2\pi\sqrt{-1}i/3} \left(\frac{i}{3}\right)^k - e^{2\pi\sqrt{-1}(i-1)/3} \left(\frac{i-1}{3}\right)^k \right) - \frac{k}{2\pi\sqrt{-1}m} I_{i,k-1} \qquad (k \geq 1).$$

Using those expressions we can evaluate J_1', J_2' and J_3', and the consequence is

$$\Sigma_{31} = \frac{\pi^4}{45}\zeta(s+2) + 3\pi^2\zeta(s+4) - 189\zeta(s+6) + 18\pi \sum_{m\geq 1} \frac{\sin(2\pi m/3)}{m^{s+5}} - 216 \sum_{m\geq 1} \frac{\cos(2\pi m/3)}{m^{s+6}}. \tag{8.4.22}$$

The evaluation of Σ_{41} is similar. In this case we need the explicit expressions for

$$\int_{(i-1)/3}^{i/3} e^{4\pi\sqrt{-1}m\theta} \theta^k d\theta \qquad (1 \leq i \leq 3, k \in \mathbb{N}_0).$$

The result is

$$\Sigma_{41} = \frac{\pi^4}{90}\zeta(s+2) + \frac{3\pi^2}{8}\zeta(s+4) - \frac{189}{32}\zeta(s+6) - \frac{9\pi}{8} \sum_{m\geq 1} \frac{\sin(2\pi m/3)}{m^{s+5}} - \frac{27}{4} \sum_{m\geq 1} \frac{\cos(2\pi m/3)}{m^{s+6}}. \tag{8.4.23}$$

Therefore

$$\frac{1}{2}\Sigma_{31} + 4\Sigma_{41} = \frac{\pi^4}{18}\zeta(s+2) + 3\pi^2\zeta(s+4) - \frac{945}{8}\zeta(s+6) \tag{8.4.24}$$

$$+\frac{9\pi}{2}\sum_{m\geq 1}\frac{\sin(2\pi m/3)}{m^{s+5}}-135\sum_{m\geq 1}\frac{\cos(2\pi m/3)}{m^{s+6}}.$$

Moreover, it is easy to see that

$$\sum_{m\geq 1}\frac{\cos(2\pi m/3)}{m^{s+6}}=\frac{3^{-s-5}-1}{2}\zeta(s+6),$$

$$\sum_{m\geq 1}\frac{\sin(2\pi m/3)}{m^{s+5}}=\frac{\sqrt{3}}{2}\sum_{m\geq 1}\frac{\chi_3(m)}{m^{s+5}}=\frac{\sqrt{3}}{2}L(s+5,\chi_3).$$

Therefore, from (8.4.12), (8.4.21) and (8.4.24), we can conclude that (8.4.3) coincides with the right-hand side of (8.4.1). This completes the proof of (8.4.1).

REMARK 8.36. Recall that the zeta-function of the root system C_2 is defined by

$$\zeta_2(s_1,s_2,s_3,s_4;C_2)=\sum_{m,n\geq 1}\frac{1}{m^{s_1}n^{s_2}(m+n)^{s_3}(m+2n)^{s_4}}.$$

Divide Σ_{11} into two subsums according to the conditions $n\geq 1$ and $n\leq -1$. Then the former part is exactly $\zeta_2(s+1,3,1,1;C_2)$. The latter is further divided according to the conditions $m-n>0$ and $m-n<0$. The part corresponding to $m-n<0$ is $-\zeta_2(s+1,1,3,1;C_2)$, while the remaining part is again divided into two subsums. The conclusion is that

$$\begin{aligned}\Sigma_{11}=\zeta_2(s+1,3,1,1;C_2)-\zeta_2(s+1,1,3,1;C_2)\\+\zeta_2(1,1,3,s+1;C_2)-\zeta_2(1,3,1,s+1;C_2).\end{aligned}\tag{8.4.25}$$

On the other hand, we have shown that Σ_{11} can be written in terms of $\zeta(s)$ (see (8.4.20)). Combining these two formulas (8.4.25) and (8.4.20), we obtain a functional relation between the zeta-function of C_2 and the Riemann zeta-function, which is different from the previously known relations ([**103**, Section 8] [**163**, Section 5]).

REMARK 8.37. We observed that the results in this chapter especially imply that some values of zeta-functions of root systems can be written in terms of the Riemann zeta values and/or L-values (see (8.2.38), (8.3.20), and (8.4.2)). The problem of *expressing values of zeta-functions of root systems at integer points as a $\mathbb{Q}$-linear combination of MZVs (of the same or lower depth)* was raised in Zhao and Zhou [**221**], and such a result in the A_3 case was given in the same paper. The case of the values of $\zeta_{MT,r}(\mathbf{s})$ was solved by Bradley and

Zhou [**21**]. The case of zeta-functions of B_2 and of G_2 are treated, respectively, by Zhao [**217**] and [**218**], but in those cases, not only MZVs, but also special values of multiple polylogarithms (1.3.3) are necessary to state the results. Later, in the proof of Theorem 13.3 we will see that certain special values of zeta-functions of type A_r can be written in terms of MZVs.

On the other hand, in the examples proved in this chapter, the values of zeta-functions of root systems are expressed by zeta (or L-) values of strictly lower depth. Those are examples of *parity results*, which will be more closely discussed in Section 12.5.

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CHAPTER 9

Functional relations (II)

In this chapter we develop another approach to functional relations for zeta-functions of root systems. As mentioned at the beginning of Chapter 8, this approach was first inspired by Nakamura's alternative proof [**162**] of Theorem 8.3. In this approach, we can see the role of Weyl groups more explicitly.

9.1. A sketch in the case of A_2

The fundamental idea in the present chapter is to introduce a certain sum, which is well behaved under the action of Weyl groups. In Chapter 4, to investigate special values at even integer points, we considered the sum $S(\mathbf{s}, \mathbf{y}; \Delta)$, which runs over the full lattice (see (4.1.3)). In this chapter, we will generalize this idea to introduce the sum $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ (see (9.4.1)) which runs over a part of lattice, and has a nice Weyl-symmetric property. We will show that this sum yields functional relations among zeta-functions of root systems.

Before going into the general formulation, we first explain briefly the essence of the idea in the simplest case of A_2. In this case, (4.1.3) is explicitly given as

$$S(\mathbf{s}; A_2) = \sum_{\substack{m_1, m_2 \in \mathbb{Z} \\ m_1, m_2, m_1+m_2 \neq 0}} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1+m_2)^{s_3}} \tag{9.1.1}$$

(refer to (3.3.19)). Now we define the following partial sum:

$$S(\mathbf{s}, \mathbf{0}; \{1\}; A_2) = \sum_{\substack{m_1 \in \mathbb{N} \\ m_2 \in \mathbb{Z} \\ m_2, m_1+m_2 \neq 0}} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1+m_2)^{s_3}}. \tag{9.1.2}$$

Then by (4.1.11) and the definition of $\widetilde{\phi}(s_2, x_3)$ we have

(9.1.3)

$$S(\mathbf{s}, \mathbf{0}; \{1\}; A_2) = \sum_{m_1 \in \mathbb{N}} \frac{1}{m_1^{s_1}} \sum_{m_2 \in \mathbb{Z} \setminus \{0\}} \frac{1}{m_2^{s_2}} \frac{-(2\pi\sqrt{-1})^{s_3}}{\Gamma(s_3+1)} \int_0^1 \widetilde{\phi}(s_3, x_3) e^{-2\pi\sqrt{-1}(m_1+m_2)x_3} dx_3$$

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$$= \sum_{m_1\in\mathbb{N}} \frac{1}{m_1^{s_1}} \frac{-(2\pi\sqrt{-1})^{s_2}}{\Gamma(s_2+1)} \frac{-(2\pi\sqrt{-1})^{s_3}}{\Gamma(s_3+1)} \int_0^1 \widetilde{\phi}(s_3,x_3)\widetilde{\phi}(s_2,x_3) e^{-2\pi\sqrt{-1}m_1x_3} dx_3.$$

In particular, if $s_2 = k_2, s_3 = k_3 \in \mathbb{N}$, we have

(9.1.4)

$$S(\mathbf{s},\mathbf{0};\{1\};A_2) = \frac{(2\pi\sqrt{-1})^{k_2+k_3}}{k_2!k_3!} \sum_{m_1\in\mathbb{N}} \frac{1}{m_1^{s_1}} \int_0^1 B_{k_2}(x_3)B_{k_3}(x_3) e^{-2\pi\sqrt{-1}m_1x_3} dx_3.$$

For $k \in \mathbb{N}_0$ and $m \in \mathbb{N}$, repeated application of integration by parts yields

$$\int_0^1 x^k e^{-2\pi\sqrt{-1}mx} dx = \sum_{j=1}^{k+1} \frac{a_j}{(\pi\sqrt{-1}m)^j}, \tag{9.1.5}$$

where $a_j \in \mathbb{Q}$. Therefore we obtain

$$\begin{aligned} S(\mathbf{s},\mathbf{0};\{1\};A_2) &= \frac{(2\pi\sqrt{-1})^{k_2+k_3}}{k_2!k_3!} \sum_{m_1\in\mathbb{N}} \frac{1}{m_1^{s_1}} P\Big(\frac{1}{\pi\sqrt{-1}m}\Big) \\ &= (2\pi\sqrt{-1})^{k_2+k_3} \sum_{j=1}^{k_2+k_3+1} \frac{b_j}{(\pi\sqrt{-1})^j} \zeta(s_1+j), \end{aligned} \tag{9.1.6}$$

where $P(x) \in \mathbb{Q}[x]$ of degree k_2+k_3+1 and $b_j \in \mathbb{Q}$. Note that a_j, b_j can be explicitly determined. On the other hand, by dividing the region and changing the running indices (similar to Claims 8.25 and 8.35), $S(\mathbf{s},\mathbf{0};\{1\};A_2)$ is decomposed as

$$\begin{aligned} S(\mathbf{s},\mathbf{0};\{1\};A_2) &= \zeta_2(s_1,k_2,k_3;A_2) \\ &\quad + (-1)^{k_2}\zeta_2(k_3,k_2,s_1;A_2) + (-1)^{k_2+k_3}\zeta_2(k_3,s_1,k_2;A_2). \end{aligned} \tag{9.1.7}$$

Therefore we conclude that a linear combination of $\zeta_2(\mathbf{s};A_2)$ can be written in terms of those of zeta-functions of type A_1, that is, the Riemann zeta-function. In fact, by supplying the details of the above argument (see Subsection 9.8.2), we find that this result coincides with Theorem 8.3.

In the following, we generalize the above method in arbitrary root systems; $P(x)$ in (9.1.6) is a Bernoulli function (see Definition 9.9) and the linear combination in (9.1.7) is the orbit sum over a certain minimum coset representative (see Theorem 9.7).

This chapter and the next chapter are mainly based on [**113**] [**119**] and [**120**].

9.2. Lemmas on root systems and Weyl groups

In this section we introduce further notations on root systems and Weyl groups, and prepare several root-theoretic lemmas.

Let $\mathrm{Aut}(\Delta)$ be the subgroup of $\mathrm{GL}(V)$ which stabilizes Δ (see [**70**, Section 12.2]). Then the Weyl group W is a normal subgroup of $\mathrm{Aut}(\Delta)$ and there exists a subgroup $\Omega \subset \mathrm{Aut}(\Delta)$ such that

$$\mathrm{Aut}(\Delta) = \Omega \ltimes W. \tag{9.2.1}$$

The subgroup Ω is isomorphic to the group $\mathrm{Aut}(\Gamma)$ of automorphisms of the Dynkin diagram Γ. For $w \in \mathrm{Aut}(\Delta)$, we set

$$\Delta_w = \Delta_+ \cap w^{-1}\Delta_- \tag{9.2.2}$$

(which is the same as (4.1.6) when $w \in W$), and define the length function by $\ell(w) = \#\Delta_w$ (see [**71**, Section 1.6]). The subgroup Ω is characterized as $w \in \Omega$ if and only if $\ell(w) = 0$. Note that $w\Delta_w = \Delta_- \cap w\Delta_+ = -\Delta_{w^{-1}}$ and $\ell(w) = \ell(w^{-1})$. For any subset $A \subset \Delta$, let $H_{A^\vee}$ be the set of all $v \in V$ which satisfies $\langle \alpha^\vee, v\rangle = 0$ for some $\alpha \in A$. In particular, $H_{\Delta^\vee}$ is the set of all walls of Weyl chambers, which coincides with $H_{\Delta^\vee}$ defined by (4.1.2).

Let $I \subset \{1, 2, \dots, r\}$, and $\Psi_I = \{\alpha_i \mid i \in I\} \subset \Psi$. Let V_I be the subspace of V spanned by Ψ_I. Then $\Delta_I = \Delta \cap V_I$ is the root system in V_I whose fundamental system is Ψ_I. For Δ_I, we denote the corresponding coroot lattice, weight lattice, etc. by

$$Q_I^\vee = \bigoplus_{i\in I} \mathbb{Z}\alpha_i^\vee, \qquad P_I = \bigoplus_{i\in I} \mathbb{Z}\lambda_i, \tag{9.2.3}$$

etc. Let $\iota : Q_I^\vee \to Q^\vee$ be the natural embedding, and $\iota^* : P \to P_I$ the projection induced from ι; that is, for $\lambda \in P$, $\iota^*(\lambda)$ is defined as a unique element of P_I satisfying $\langle \iota(q), \lambda\rangle = \langle q, \iota^*(\lambda)\rangle$ for all $q \in Q_I^\vee$. Explicitly, we have

$$\iota^*(n_1\lambda_1 + \cdots + n_r\lambda_r) = \sum_{i\in I} n_i\lambda_i. \tag{9.2.4}$$

We define

$$C_I = \{v \in C \mid \langle \Psi_{I^c}^\vee, v\rangle = 0, \quad \langle \Psi_I^\vee, v\rangle > 0\}, \tag{9.2.5}$$

where I^c is the complement of I. Then the dimension of the linear span of C_I is $\#I$ and we have a disjoint union

$$C = \coprod_{I\subset\{1,\dots,r\}} C_I, \tag{9.2.6}$$

where C is the fundamental Weyl chamber defined by (4.1.1). The collection of all sets wC_I for $w \in W$ and $I \subset \{1, \dots, r\}$ is called the Coxeter complex (see [71, Section 1.15]; it should be noted that we use here a slightly different notation), which partitions V and we have a decomposition

$$P_+ = \coprod_{I \subset \{1,\dots,r\}} P_{I++}, \tag{9.2.7}$$

where

$$P_{I++} = P_+ \cap C_I. \tag{9.2.8}$$

In particular, $P_{\emptyset++} = \{0\}$ and $P_{\{1,\dots,r\}++} = P_{++}$.

Let $W_I = W(\Delta_I)$ be the subgroup of W generated by all the reflections associated with the elements in Ψ_I, and

$$W^I = \{w \in W \mid \Delta^\vee_{I+} \subset w\Delta^\vee_+\}. \tag{9.2.9}$$

Then we have the following key lemmas to functional relations among zeta-functions. Note that the statements hold trivially in the case $I = \emptyset$ and hence we deal with $I \neq \emptyset$ in their proofs. Note also that $W(\Delta) = W(\Delta^\vee)$ and hence $\ell(w) = \#(\Delta_+ \cap w^{-1}\Delta_-) = \#(\Delta^\vee_+ \cap w^{-1}\Delta^\vee_-)$.

LEMMA 9.1. ([113, Lemma 1]) *The subset W^I coincides with the minimal (right) coset representatives $\{w \in W \mid \ell(\sigma_i w) > \ell(w)$ for all $i \in I\}$ of W_I.*

PROOF. Let $w \in W^I$. Then $\Delta^\vee_{I+} \subset w\Delta^\vee_+$, which implies $\Delta^\vee_{I+} \cap w\Delta^\vee_- = \emptyset$. In particular, $\alpha_i^\vee \notin w\Delta^\vee_-$ for $i \in I$, which yields $\Delta^\vee_+ \cap w\Delta^\vee_- = (\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap w\Delta^\vee_-$. Therefore

$$\begin{aligned} \sigma_i(\Delta^\vee_+ \cap w\Delta^\vee_-) &= \sigma_i\big((\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap w\Delta^\vee_-\big) \\ &= (\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap \sigma_i w\Delta^\vee_- \\ &\subset \Delta^\vee_+ \cap \sigma_i w\Delta^\vee_- \end{aligned} \tag{9.2.10}$$

and $\ell(\sigma_i w) \geq \ell(w)$. Since $\ell(\sigma_i w) = \ell(w) \pm 1$, we have $\ell(\sigma_i w) = \ell(w) + 1$ and w is a minimal coset representative.

Assume that $w \in W$ satisfies $\ell(\sigma_i w) > \ell(w)$ for all $i \in I$. Then we have

$$\sigma_i(\Delta^\vee_+ \cap w\Delta^\vee_-) = \big((\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap \sigma_i w\Delta^\vee_-\big) \cup \big(\{-\alpha_i^\vee\} \cap \sigma_i w\Delta^\vee_-\big). \tag{9.2.11}$$

Since $\#\sigma_i(\Delta^\vee_+ \cap w\Delta^\vee_-) = \ell(w)$ and $\#((\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap \sigma_i w\Delta^\vee_-) \geq \ell(\sigma_i w) - 1 = \ell(w)$, we have $\#(\{-\alpha_i^\vee\} \cap \sigma_i w\Delta^\vee_-) = 0$. It implies that

$$\sigma_i(\Delta^\vee_+ \cap w\Delta^\vee_-) \subset (\Delta^\vee_+ \setminus \{\alpha_i^\vee\}) \cap \sigma_i w\Delta^\vee_- \subset \Delta^\vee_+, \tag{9.2.12}$$

that is, no element of $\Delta_+^\vee \cap w\Delta_-^\vee$ is sent to $\Delta_-^\vee$ by σ_i for $i \in I$, and hence $\Psi_I^\vee \cap w\Delta_-^\vee = \emptyset$. Since $\langle \Delta_+^\vee, \rho\rangle > 0$ and $\langle \Delta_-^\vee, \rho\rangle < 0$, we see that $\alpha^\vee \in w\Delta_-^\vee$ if and only if $\langle \alpha^\vee, w\rho\rangle < 0$. Thus $\langle \Psi_I^\vee, w\rho\rangle > 0$ and hence $\langle \Delta_{I+}^\vee, w\rho\rangle > 0$. It follows that $\Delta_{I+}^\vee \cap w\Delta_-^\vee = \emptyset$ and $w \in W^I$. □

From Lemma 9.1, or by the definition of W_I and W^I, it follows that

$$W_I W^I = W. \tag{9.2.13}$$

LEMMA 9.2. ([**113**, Lemma 2])

$$\iota^{*-1}(P_{I+}) = P_{I+} \oplus P_{I^c} = \bigcup_{w \in W^I} wP_+. \tag{9.2.14}$$

PROOF. The first equality is clear. We prove the second equality.

Assume $w \in W^I$. Then for $\lambda \in P_+$, we have

$$\langle \Delta_{I+}^\vee, w\lambda\rangle = \langle w^{-1}\Delta_{I+}^\vee, \lambda\rangle \subset \langle \Delta_+^\vee, \lambda\rangle \geq 0. \tag{9.2.15}$$

Hence $wP_+ \subset \iota^{*-1}(P_{I+})$.

Conversely, assume $\lambda \in \iota^{*-1}(P_{I+})$. Since $\#\Delta^\vee < \infty$, it is possible to fix a sufficiently small constant $c > 0$ such that $0 < |\langle \Delta^\vee, c\rho\rangle| < 1$. Then we see that $\lambda + c\rho$ is regular (see [**70**, Section 10.1]), i.e., $0 \notin \langle \Delta^\vee, \lambda + c\rho\rangle$ and the signs of $\langle \alpha^\vee, \lambda\rangle$ and $\langle \alpha^\vee, \lambda + c\rho\rangle$ coincide if $\langle \alpha^\vee, \lambda\rangle \neq 0$, because $\langle \Delta^\vee, \lambda\rangle \subset \mathbb{Z}$. Let $\widetilde{\Delta}_+^\vee = \{\alpha^\vee \in \Delta^\vee \mid \langle \alpha^\vee, \lambda + c\rho\rangle > 0\}$. Then $\widetilde{\Delta}_+^\vee$ is a positive system and hence there exists an element $w \in W$ such that $\widetilde{\Delta}_+^\vee = w\Delta_+^\vee$. Since $\lambda \in \iota^{*-1}(P_{I+})$, we have $\Delta_{I+}^\vee \subset \widetilde{\Delta}_+^\vee$. Hence $\Delta_{I+}^\vee \subset w\Delta_+^\vee$, that is, $w \in W^I$. Moreover, $\langle \widetilde{\Delta}_+^\vee, \lambda + c\rho\rangle > 0$ implies $\langle \Delta_+^\vee, w^{-1}(\lambda + c\rho)\rangle > 0$ and $\langle \Delta_+^\vee, w^{-1}\lambda\rangle \geq 0$ again due to the integrality. Therefore $\lambda \in wP_+$. □

LEMMA 9.3. ([**113**, Lemma 3]) *For $\lambda \in \iota^{*-1}(P_{I+})$, an element $w \in W^I$ satisfying $\lambda \in wP_+$ (whose existence is assured by Lemma 9.2) is unique if and only if $\lambda \notin H_{\Delta^\vee \setminus \Delta_I^\vee}$.*

PROOF. Assume $\alpha^\vee \in \Delta^\vee \setminus \Delta_I^\vee$ and $\lambda \in \iota^{*-1}(P_{I+}) \cap H_{\alpha^\vee}$. Let $w \in W^I$ satisfy $\lambda \in wP_+$. Then $\sigma_\alpha \lambda = \lambda \in wP_+$ and hence $w^{-1}\lambda = \sigma_{w^{-1}\alpha} w^{-1}\lambda \in P_+$, which further implies $w^{-1}\alpha^\vee \in \Delta'^\vee$, where $\Delta'^\vee$ is a coroot system orthogonal to $w^{-1}\lambda$ whose fundamental system is given by $\Psi'^\vee = \{\alpha_i^\vee \in \Psi^\vee \mid \langle \alpha_i^\vee, w^{-1}\lambda\rangle = 0\}$ (see [**71**, Section 1.12]). If $\Psi'^\vee \subset w^{-1}\Delta_I^\vee$, then $W(\Psi'^\vee)\Psi'^\vee = \Delta'^\vee \subset w^{-1}\Delta_I^\vee$, and hence $w^{-1}\alpha^\vee \in w^{-1}\Delta_I^\vee$, which contradicts the assumption $\alpha^\vee \notin \Delta_I^\vee$. Therefore there exists a fundamental coroot $\alpha_i^\vee \in \Psi'^\vee \setminus w^{-1}\Delta_I^\vee$, which satisfies $\sigma_i w^{-1}\lambda = w^{-1}\lambda \in P_+$ by construction. Since $w \in W^I$, we have $w^{-1}\Delta_{I+}^\vee \subset \Delta_+^\vee \setminus \{\alpha_i^\vee\}$. Hence $\sigma_i w^{-1}\Delta_{I+}^\vee \subset \Delta_+^\vee$, because $\sigma_i(\Delta_+^\vee \setminus \{\alpha_i^\vee\}) \subset \Delta_+^\vee$. Then putting $w' = w\sigma_i$, we have $W^I \ni w' \neq w$ such that $\lambda \in wP_+ \cap w'P_+$.

Conversely, assume that there exist $w, w' \in W^I$ such that $w \neq w'$ and $\lambda \in wP_+ \cap w'P_+$. This implies that $w^{-1}\lambda = w'^{-1}\lambda$ is on a wall of C and hence $\lambda \in H_{\Delta^\vee}$. Let $\Delta''^\vee = \{\alpha^\vee \in \Delta^\vee \mid \lambda \in H_{\alpha^\vee}\}$ be a coroot system orthogonal to λ so that $\lambda \in H_{\Delta''^\vee}$. Assume $\Delta''^\vee \subset \Delta_I^\vee$. Then by $\lambda = ww'^{-1}\lambda$, we have $ww'^{-1} \in W_\lambda$ and hence $ww'^{-1} \in W(\Delta_I^\vee)$, because $W_\lambda = W(\Delta''^\vee) \subset W(\Delta_I^\vee)$ by the assumption. Since $\mathrm{id} \neq ww'^{-1} \in W(\Delta_I^\vee)$, there exists a coroot $\alpha^\vee \in \Delta_{I+}^\vee$ such that $\beta^\vee = ww'^{-1}\alpha^\vee \in \Delta_{I-}^\vee$. Then, since $w^{-1}(ww'^{-1})\Delta_{I+}^\vee \subset \Delta_+^\vee$ and $w^{-1}\Delta_{I+}^\vee \subset \Delta_+^\vee$, we have $w^{-1}\beta^\vee \in \Delta_+^\vee$ from the first inclusion and $w^{-1}(-\beta^\vee) \in \Delta_+^\vee$ from the second one, which leads to a contradiction. Therefore $\lambda \in H_{\alpha^\vee}$ for some $\alpha \in \Delta''^\vee \setminus \Delta_I^\vee$. □

Here we give the structures of the Weyl groups of the classical root systems, that is, of types A_r, B_r, C_r and D_r.

$$W(A_r) \simeq \mathfrak{S}_{r+1}, \tag{9.2.16}$$

$$W(B_r) \simeq \mathfrak{S}_r \ltimes (\mathbb{Z}/2\mathbb{Z})^r, \tag{9.2.17}$$

$$W(C_r) \simeq \mathfrak{S}_r \ltimes (\mathbb{Z}/2\mathbb{Z})^r, \tag{9.2.18}$$

$$W(D_r) \simeq \mathfrak{S}_r \ltimes (\mathbb{Z}/2\mathbb{Z})^{r-1}, \tag{9.2.19}$$

where $\mathfrak{S}_r$ denotes the symmetric group of degree r.

9.3. Automorphisms on Dynkin diagrams

For $\mathbf{s} = (s_\alpha)_{\alpha\in\Delta_+} \in \mathbb{C}^{\#\Delta_+}$, we define an action of $w \in \mathrm{Aut}(\Delta)$ by

$$(g\mathbf{s})_\alpha = s_{g^{-1}\alpha}, \tag{9.3.1}$$

where we set $s_{-\alpha} = s_\alpha$ for $-\alpha \in \Delta_-$. This action and the convention are extensions of those already appeared in the proof of (4.1.7).

First, we give simple functional relations related to the group $\mathrm{Aut}(\Gamma)$. Since $g \in \mathrm{Aut}(\Gamma) \subset \mathrm{Aut}(\Delta)$ is a permutation of $\{\alpha_1, \dots, \alpha_r\}$, we have

$$g\Delta_+ = \Delta_+, \qquad gP_{++} = P_{++}. \tag{9.3.2}$$

The following theorem and examples have not been published before.

THEOREM 9.4. *For $\mathbf{s} \in \mathcal{S}$, $\mathbf{y} \in V$ and $g \in \mathrm{Aut}(\Gamma)$, we have*

$$\zeta_r(g^{-1}\mathbf{s}, g^{-1}\mathbf{y}; \Delta) = \zeta_r(\mathbf{s}, \mathbf{y}; \Delta) \tag{9.3.3}$$

PROOF.

$$\zeta_r(g^{-1}\mathbf{s}, g^{-1}\mathbf{y}; \Delta) = \sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle g^{-1}\mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle\alpha^\vee, \lambda\rangle^{s_{g\alpha}}} \tag{9.3.4}$$

$$= \sum_{g\lambda \in gP_{++}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, g\lambda\rangle} \prod_{g\alpha \in g\Delta_+} \frac{1}{\langle g\alpha^\vee, g\lambda\rangle^{s_{g\alpha}}}$$
$$= \sum_{\lambda \in P_{++}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \lambda\rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{s_\alpha}},$$

where in the last equality, we rewrite $g\alpha$, $g\lambda$ as α, λ respectively and use (9.3.2). □

This relation is easily understood by observing the series expressions of zeta-functions as follows.

EXAMPLE 9.5. In the case of the root system of type A_r, by Proposition 3.5, we see the symmetry

$$i \quad \longleftrightarrow \quad r+1-i \tag{9.3.5}$$

in indices, which implies the symmetry in the variables

$$s_{ij} \quad \longleftrightarrow \quad s_{r+1-j,r+1-i}. \tag{9.3.6}$$

In particular, we studied the case $r = 2$ in Example 8.22.

EXAMPLE 9.6. Another interesting case is type D_4. By Proposition 3.9, we have the explicit form of the zeta-function of type D_4 as

(9.3.7)

$$\begin{aligned}\zeta_r(\mathbf{s}; D_4) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_4=1}^{\infty} & (m_1+m_2+m_4)^{-s^+_{14}} (m_2+m_4)^{-s^+_{24}} m_4^{-s^+_{34}} \\ & \times m_1^{-s^-_{12}} (m_1+m_2)^{-s^-_{13}} (m_1+m_2+m_3)^{-s^-_{14}} m_2^{-s^-_{23}} \\ & \times (m_2+m_3)^{-s^-_{24}} m_3^{-s^-_{34}} (m_1+2m_2+m_3+m_4)^{-s^+_{12}} \\ & \times (m_1+m_2+m_3+m_4)^{-s^+_{13}} (m_2+m_3+m_4)^{-s^+_{23}}.\end{aligned}$$

From this expression, we see the $\mathfrak{S}_3$-symmetry with respect to permutations of m_1, m_3, m_4. In fact, this symmetry comes from that of the Dynkin diagram of type D_4 given by Figure 9.3.1.

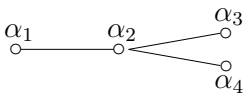


FIGURE 9.3.1.

For example, the permutation (3 4) acts on $\mathbf{s}$ as

$$s^+_{14} \quad \longleftrightarrow \quad s^-_{14}, \qquad s^+_{24} \quad \longleftrightarrow \quad s^-_{24}, \qquad s^+_{34} \quad \longleftrightarrow \quad s^-_{34}. \tag{9.3.8}$$

9.4. General functional relations

For $I \subset \{1, 2, \dots, r\}$, we define

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) = \sum_{\lambda \in \iota^{*-1}(P_{I+}) \setminus H_{\Delta^\vee}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, \lambda \rangle^{s_\alpha}}. \tag{9.4.1}$$

When $I = \emptyset$, we see that $S(\mathbf{s}, \mathbf{y}; \emptyset; \Delta)$ is nothing but $S(\mathbf{s}, \mathbf{y}; \Delta)$ defined by (4.1.3). In the next theorem (which first appeared in [**113**]) we show that $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ converges on $\mathcal{S} \times V$, which is defined in (3.1.10), and is a linear combination of Lerch analogues of the zeta-functions of the root system Δ, which are defined in (3.1.7).

THEOREM 9.7. *For $\mathbf{s} \in \mathcal{S}$ and $\mathbf{y} \in V$, $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ is absolutely convergent and we have*

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) = \sum_{w \in W^I} \Big(\prod_{\alpha \in \Delta_{w-1}} (-1)^{-s_\alpha} \Big) \zeta_r(w^{-1}\mathbf{s}, w^{-1}\mathbf{y}; \Delta). \tag{9.4.2}$$

PROOF. By Lemma 9.2 and the Weyl group-invariance of $H_{\Delta^\vee}$, we have

$$\begin{aligned} \iota^{*-1}(P_{I+}) \setminus H_{\Delta^\vee} &= \Big(\bigcup_{w \in W^I} wP_+ \Big) \setminus H_{\Delta^\vee} \\ &= \bigcup_{w \in W^I} w(P_+ \setminus H_{\Delta^\vee}) \\ &= \bigcup_{w \in W^I} wP_{++}. \end{aligned} \tag{9.4.3}$$

Noting Lemma 9.3, we have

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) = \sum_{w \in W^I} \sum_{\lambda \in P_{++}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, w\lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, w\lambda \rangle^{s_\alpha}}. \tag{9.4.4}$$

Further,

$$\begin{aligned} & e^{2\pi\sqrt{-1}\langle \mathbf{y}, w\lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, w\lambda \rangle^{s_\alpha}} \\ &\qquad = e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle w^{-1}\alpha^\vee, \lambda \rangle^{s_\alpha}} \\ &\qquad = e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y}, \lambda \rangle} \prod_{\alpha \in w^{-1}\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda \rangle^{s_{w\alpha}}}, \end{aligned} \tag{9.4.5}$$

by rewriting α as $w\alpha$. When

$$\alpha \in w^{-1}\Delta_+ \cap \Delta_- = -(\Delta_+ \cap w^{-1}\Delta_-) = -\Delta_w, \tag{9.4.6}$$

we further replace α by $-\alpha$. Then we have

$$\begin{aligned}(9.4.7)\quad &\sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle \mathbf{y},w\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,w\lambda\rangle^{s_\alpha}}\\ &=\Big(\prod_{\alpha\in\Delta_w}(-1)^{-s_{w\alpha}}\Big)\sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_{w\alpha}}}\\ &=\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\Big)\zeta_r(w^{-1}\mathbf{s},w^{-1}\mathbf{y};\Delta),\end{aligned}$$

where we have used the fact that $w\Delta_w=-\Delta_{w^{-1}}$.

The absolute convergence of $S(\mathbf{s},\mathbf{y};I;\Delta)$ is shown by Lemma 3.3. □

Let $\Delta^{(I)}=\Delta_+\setminus\Delta_{I+}$ and $d=\#I^c$. We may find $\mathbf{V}_I=\{\gamma_1,\dots,\gamma_d\}\subset\Delta^{(I)}$ such that $\mathbf{V}=\mathbf{V}_I\cup\Psi_I$ becomes a basis of V. Let $\mathscr{V}_I=\mathscr{V}(\Delta^{(I)})$ be the set of all such bases. In particular, let $\mathscr{V}=\mathscr{V}_\emptyset$ be the set of all linearly independent subsets $\mathbf{V}=\{\beta_1,\dots,\beta_r\}\subset\Delta_+$.

For $\mathbf{V}\in\mathscr{V}_I$, the lattice $L(\mathbf{V}^\vee)=\bigoplus_{\beta\in\mathbf{V}}\mathbb{Z}\beta^\vee$ is a sublattice of $Q^\vee$. Let $\{\mu_\gamma^{\mathbf{V}}\}_{\gamma\in\mathbf{V}}$ be the dual basis of $\mathbf{V}^\vee=\mathbf{V}_I^\vee\cup\Psi_I^\vee$, namely $\langle\gamma_k^\vee,\mu_{\gamma_l}^{\mathbf{V}}\rangle=\delta_{k,l}$, $\langle\alpha_i^\vee,\mu_{\alpha_j}^{\mathbf{V}}\rangle=\delta_{i,j}$, and $\langle\gamma_k^\vee,\mu_{\alpha_i}^{\mathbf{V}}\rangle=\langle\alpha_i^\vee,\mu_{\gamma_k}^{\mathbf{V}}\rangle=0$. Let $p_{\mathbf{V}_I^\perp}$ be the projection defined by

$$(9.4.8)\qquad p_{\mathbf{V}_I^\perp}(v)=v-\sum_{\gamma\in\mathbf{V}_I}\mu_\gamma^{\mathbf{V}}\langle\gamma^\vee,v\rangle=\sum_{\alpha\in\Psi_I}\mu_\alpha^{\mathbf{V}}\langle\alpha^\vee,v\rangle,$$

for $v\in V$. (The second equality can be easily seen by expressing $v=\sum_k a_k\mu_{\gamma_k}^{\mathbf{V}}+\sum_i b_i\mu_{\alpha_i}^{\mathbf{V}}$.) It should be noted that the projection $p_{\mathbf{V}_I^\perp}$ depends only on $\mathbf{V}_I$ in the following sense.

LEMMA 9.8. ([**119**, Lemma 2.1]) *For any linearly independent subset $\Phi_I=\{\beta_1,\dots,\beta_{\#I}\}\subset\Delta_{I+}$ and $\mathbf{U}=\mathbf{V}_I\cup\Phi_I$, we have*

$$(9.4.9)\qquad p_{\mathbf{V}_I^\perp}(v)=v-\sum_{\gamma\in\mathbf{V}_I}\mu_\gamma^{\mathbf{U}}\langle\gamma^\vee,v\rangle=\sum_{\beta\in\Phi_I}\mu_\beta^{\mathbf{U}}\langle\beta^\vee,v\rangle.$$

PROOF. Put $u=\sum_{\beta\in\Phi_I}\mu_\beta^{\mathbf{U}}\langle\beta^\vee,v\rangle$ and we show $p_{\mathbf{V}_I^\perp}(v)=u$. It is enough to check that $\langle\gamma^\vee,p_{\mathbf{V}_I^\perp}(v)\rangle=\langle\gamma^\vee,u\rangle$ for all $\gamma\in\mathbf{U}$. For $\gamma\in\mathbf{V}_I$, we have

$$\langle\gamma^\vee,p_{\mathbf{V}_I^\perp}(v)\rangle=\Big\langle\gamma^\vee,v-\sum_{\gamma'\in\mathbf{V}_I}\mu_{\gamma'}^{\mathbf{V}}\langle\gamma'^\vee,v\rangle\Big\rangle=\langle\gamma^\vee,v\rangle-\langle\gamma^\vee,v\rangle=0,$$

$$\langle\gamma^\vee,u\rangle=\Big\langle\gamma^\vee,v-\sum_{\gamma'\in\mathbf{V}_I}\mu_{\gamma'}^{\mathbf{U}}\langle\gamma'^\vee,v\rangle\Big\rangle=\langle\gamma^\vee,v\rangle-\langle\gamma^\vee,v\rangle=0,$$

because $\mathbf{V}_I\subset\mathbf{V},\mathbf{U}$. For $\beta\in\Phi_I$, we have

$$\langle\beta^\vee,u\rangle=\sum_{\beta'\in\Phi_I}\Big\langle\beta^\vee,\mu_{\beta'}^{\mathbf{U}}\langle\beta'^\vee,v\rangle\Big\rangle=\sum_{\beta'\in\Phi_I}\langle\beta'^\vee,v\rangle\langle\beta^\vee,\mu_{\beta'}^{\mathbf{U}}\rangle=\langle\beta^\vee,v\rangle,$$

while by writing $\beta^\vee = \sum_{\alpha\in\Psi_I} a_\alpha\alpha^\vee$, we have

$$\begin{aligned}\langle\beta^\vee, p_{\mathbf{V}_I^\perp}(v)\rangle &= \sum_{\alpha\in\Psi_I}\langle\beta^\vee,\mu_\alpha^{\mathbf{V}}\rangle\langle\alpha^\vee,v\rangle = \sum_{\alpha\in\Psi_I} a_\alpha\langle\alpha^\vee,v\rangle\\ &= \Big\langle\sum_{\alpha\in\Psi_I} a_\alpha\alpha^\vee, v\Big\rangle = \langle\beta^\vee,v\rangle.\end{aligned}$$

□

Let $\mathscr{V}$, $\mathscr{R}$, ϕ_0 and $\{\cdot\}_{\mathbf{V},\beta}$ be given just before Theorem 4.12. Using these notions, we now define Bernoulli functions of the root system Δ associated with I and their generating functions.

DEFINITION 9.9. Assume that Δ is irreducible. If Δ is of type A_1, then fix $\phi_0 = c\alpha_1^\vee$ with $c > 0$. For $\mathbf{t}_I = (t_\alpha)_{\alpha\in\Delta^{(I)}} \in \mathbb{C}^{\#\Delta^{(I)}}$ and $\lambda \in P_I$, let

(9.4.10)

$$\begin{aligned}&F(\mathbf{t}_I,\mathbf{y},\lambda;I;\Delta)\\ &= \sum_{\mathbf{V}\in\mathscr{V}_I}\left(\prod_{\gamma\in\Delta^{(I)}\setminus\mathbf{V}_I}\frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}_I} t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle - 2\pi\sqrt{-1}\langle\gamma^\vee,p_{\mathbf{V}_I^\perp}(\lambda)\rangle}\right)\\ &\quad\times\frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))}\sum_{q\in Q^\vee/L(\mathbf{V}^\vee)}\exp(2\pi\sqrt{-1}\langle\mathbf{y}+q,p_{\mathbf{V}_I^\perp}(\lambda)\rangle)\\ &\quad\times\prod_{\beta\in\mathbf{V}_I}\frac{t_\beta\exp(t_\beta\{\mathbf{y}+q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1},\end{aligned}$$

and define *Bernoulli functions* $P(\mathbf{k},\mathbf{y},\lambda;I;\Delta)$ *of the root system* Δ *associated with* I by the expansion

$$F(\mathbf{t}_I,\mathbf{y},\lambda;I;\Delta) = \sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Delta^{(I)}}} P(\mathbf{k},\mathbf{y},\lambda;I;\Delta)\prod_{\alpha\in\Delta^{(I)}}\frac{t_\alpha^{k_\alpha}}{k_\alpha!}. \tag{9.4.11}$$

In Chapter 16, we will show:

THEOREM 9.10. (i) $F(\mathbf{t}_I,\mathbf{y},\lambda;I;\Delta)$ *is holomorphic in a neighborhood of the origin with respect to* $\mathbf{t}_I$.

(ii) *If* Δ *is not of type* A_1, *then* $F(\mathbf{t}_I,\mathbf{y},\lambda;I;\Delta)$ *is continuous in* $\mathbf{y}$ *and is independent of the choice of* ϕ_0.

A special case of the last assertion (the independence of the choice of ϕ_0) was already mentioned in Remark 4.13. The irreducibility is not essential either (see Remark 4.14).

The fundamental formula in our theory is the following theorem, which is the first main result of this chapter. The proof of this theorem will also be given in Chapter 16.

THEOREM 9.11. ([119, Theorem 2.3]) *Let $s_\alpha = k_\alpha \in \mathbb{N}$ for $\alpha \in \Delta^{(I)}$ and $s_\alpha \in \mathbb{C}$ for $\alpha \in \Delta_{I+}$. We assume that $\mathbf{s} \in \mathcal{S}$ (see (3.1.10)).*

Then we have

$$
\begin{aligned}
(9.4.12)\qquad & S(\mathbf{s},\mathbf{y};I;\Delta) \\
&= (-1)^{\#\Delta^{(I)}}\Big(\prod_{\alpha\in\Delta^{(I)}}\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big) \\
&\quad\times\sum_{\lambda\in P_{I++}}\Big(\prod_{\alpha\in\Delta_{I+}}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}\Big)P(\mathbf{k},\mathbf{y},\lambda;I;\Delta).
\end{aligned}
$$

In the case $I=\emptyset$, clearly $\Psi_I=\emptyset$, $\mathbf{V}_I=\mathbf{V}$, $\mathscr{V}_I=\mathscr{V}$, $\Delta^{(I)}=\Delta_+$, $P_I=\{0\}$, and hence the only possible λ is $\lambda=0$. Also in this case we write $\mathbf{t}=\mathbf{t}_\emptyset=(t_\alpha)_{\alpha\in\Delta_+}$. Then (9.4.10), (9.4.11) and (9.4.12) are reduced to

$$
\begin{aligned}
(9.4.13)\qquad & F(\mathbf{t},\mathbf{y},0;\emptyset;\Delta)=\sum_{\mathbf{V}\in\mathscr{V}}\Big(\prod_{\gamma\in\Delta_+\setminus\mathbf{V}}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle}\Big) \\
&\quad\times\frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))}\sum_{q\in Q^\vee/L(\mathbf{V}^\vee)}\Big(\prod_{\beta\in\mathbf{V}}\frac{t_\beta\exp(t_\beta\{\mathbf{y}+q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1}\Big) \\
&\quad=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Delta_+}}P(\mathbf{k},\mathbf{y},0;\emptyset;\Delta)\prod_{\alpha\in\Delta_+}\frac{t_\alpha^{k_\alpha}}{k_\alpha!},
\end{aligned}
$$

and

$$
\begin{aligned}
(9.4.14)\qquad & S(\mathbf{k},\mathbf{y};\Delta)=S(\mathbf{k},\mathbf{y};\emptyset;\Delta) \\
&\qquad=(-1)^{\#\Delta_+}\Big(\prod_{\alpha\in\Delta_+}\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)P(\mathbf{k},\mathbf{y},0;\emptyset;\Delta)
\end{aligned}
$$

for $\mathbf{k}=(k_\alpha)_{\alpha\in\Delta_+}\in\mathcal{K}=\mathbb{N}^{\#\Delta_+}\cap\mathcal{S}$ and $\mathbf{y}\in V$. In Chapter 16 we will show:

THEOREM 9.12. *The above $F(\mathbf{t},\mathbf{y},0;\emptyset;\Delta)$ coincides with $F(\mathbf{t},\mathbf{y};\Delta)$ defined by (4.2.11) in Section 4.2.*

This theorem with (9.4.13) clearly implies Theorem 4.12. Comparing the coefficients of expansions (4.2.11) and (9.4.13), we find

$$
(9.4.15)\qquad P(\mathbf{k},\mathbf{y},0;\emptyset;\Delta)=P(\mathbf{k},\mathbf{y};\Delta)\quad(\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}),
$$

where the right-hand side is defined by (4.2.2) in Section 4.2. (We note that (9.4.15) for $\mathbf{k} \in \mathcal{K}_0$ also follows from (4.2.5) and (9.4.14).)

REMARK 9.13. Here we mention some history of our research.

The formula (9.4.14) was already stated in [**113**, (123)] and [**105**, (3.10)]. In those papers, $P(\mathbf{k}, \mathbf{y}; \Delta)$ was defined by the multiple integral (4.2.2) and $F(\mathbf{t}, \mathbf{y}; \Delta)$ was simply defined as the generating function of $P(\mathbf{k}, \mathbf{y}; \Delta)$ (see (4.2.11)). However, in [**105**, Theorem 4.1] an alternative expression of $F(\mathbf{t}, \mathbf{y}; \Delta)$ (Theorem 4.12) was shown, which coincides with the special case $I = \emptyset$ of (9.4.10). The advantage of this form was discussed at the end of Chapter 5.

Since this form has turned out to be fundamental, later in [**119**], we decided to use this form as the definition of $F(\mathbf{t}, \mathbf{y}; \Delta)$. In fact, in this way we defined a more general quantity $F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta)$, which is (9.4.10). Hereafter the whole theory will be constructed on this definition. For example, $P(\mathbf{k}, \mathbf{y}, \lambda; I; \Delta)$ is now defined as the coefficients of the Taylor expansion of $F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta)$.

As mentioned above, the case $I = \emptyset$ of Theorem 9.11 was given in [**105**] [**113**]. The general case, however, becomes much more complicated. To establish Theorem 9.11 for general I, it is necessary to develop a detailed theory on lattice sums of hyperplane arrangements [**115**] (see Remark 16.47).

The fundamental philosophy of the method in the present chapter is to combine two expressions of $S(\mathbf{s}, \mathbf{y}; I; \Delta)$, that is (9.4.2) and (9.4.12). It will produce the relation of the form

(9.4.16)
$$\begin{aligned}&(\text{A signed sum of zeta-functions of root systems})\\ &\quad = (\text{A series involving Bernoulli functions of root systems}).\end{aligned}$$

If the left-hand side of (9.4.16) does not vanish identically, then this relation gives a non-trivial functional relation among zeta-functions of root systems. In the following sections we will show explicit computations on the right-hand side of (9.4.16), and will state several examples. On the other hand, concerning the left-hand side, it is important to know when it does not vanish. A criterion of the non-vanishing in terms of Poincaré polynomials will be given in Chapter 10.

Next, we state the second main result in the present chapter. We mentioned above that the situation for general I is much more complicated than the case $I = \emptyset$. However, the next theorem, in a sense,

asserts the contrary: The generating function $F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta)$ for general I can be deduced, in the following sense, from the case $I = \emptyset$. That is, the generating function for $I = \emptyset$ knows "everything". This "converse" theorem will also be proved in Chapter 16.

THEOREM 9.14. ([119, Theorem 2.4]) *Let $I \subset \{1, \dots, r\}$. Fix an order of the elements of the set Δ_{I+} by numbering as $\beta_1, \dots, \beta_{d(I)}$, where $d(I) = \#\Delta_{I+}$. Then for $\lambda \in P_{I++}$, we have*

$$F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta) = \operatorname*{Res}_{t_{\beta_{d(I)}}=2\pi\sqrt{-1}\langle\beta_{d(I)}^\vee,\lambda\rangle} \cdots \operatorname*{Res}_{t_{\beta_1}=2\pi\sqrt{-1}\langle\beta_1^\vee,\lambda\rangle} \Big(\prod_{\alpha\in\Delta_{I+}} \frac{1}{t_\alpha}\Big) F(\mathbf{t}, \mathbf{y}; \Delta), \tag{9.4.17}$$

which does not depend on the choice of the order of iterated residues.

REMARK 9.15. The point $(2\pi\sqrt{-1}\langle\alpha^\vee, \lambda\rangle)_{\alpha\in\Delta_{I+}}$ is on some singular hyperplane of $(\prod_{\alpha\in\Delta_{I+}} 1/t_\alpha)F(\mathbf{t}, \mathbf{y}; \Delta)$. So when we calculate the residue at $t_{\beta_j} = 2\pi\sqrt{-1}\langle\beta_j^\vee, \lambda\rangle$, the remaining variables $(t_{\beta_{j+1}}, \dots, t_{\beta_{d(I)}})$ are to be located outside any singular hyperplanes.

We postpone the proof of the above four theorems to Chapter 16, because it requires quite a lot of pages. In the remaining part of this chapter we discuss several consequences of those theorems.

9.5. An extension

In the previous section we proved (9.4.14), which is valid for $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathcal{K}$. In this section we present certain extended formulas, which are still valid even if some of k_α's are zero (hence $\mathbf{k}$ is outside of $\mathcal{S}$, because of (3.1.10)). Those formulas are necessary in Chapter 12.

The fundamental idea is to consider the action of a certain differential operator to (9.4.13).

For $\mathbf{v} \in V$, and a differentiable function f on V, let

$$(\partial_{\mathbf{v}} f)(\mathbf{y}) = \lim_{h\to 0} \frac{f(\mathbf{y} + h\mathbf{v}) - f(\mathbf{y})}{h}. \tag{9.5.1}$$

For $\alpha \in \Delta_+$, define

$$\mathfrak{D}_\alpha = \frac{\partial}{\partial t_\alpha}\Big|_{t_\alpha=0} \partial_{\alpha^\vee}.$$

Let $\Delta^* \subset \Delta_+$ be a root set (defined in Section 6.1), and let $A = \Delta_+ \setminus \Delta^* = \{\nu_1, \dots, \nu_N\} \subset \Delta_+$, and define

$$\mathfrak{D}_A = \mathfrak{D}_{\nu_N} \cdots \mathfrak{D}_{\nu_1}.$$

Similarly, we define

$$\mathfrak{D}_{\alpha,2} = \frac{1}{2}\frac{\partial^2}{\partial t_\alpha^2}\Big|_{t_\alpha=0} \partial_{\alpha^\vee}^2,$$
$$\mathfrak{D}_{A,2} = \mathfrak{D}_{\nu_N,2}\cdots\mathfrak{D}_{\nu_1,2}.$$

Further, let $A_j = \{\nu_1,\dots,\nu_j\}$ $(1 \le j \le N-1)$, $A_0 = \emptyset$, and

$$\mathscr{V}_A = \{\mathbf{V} \in \mathscr{V} \mid \nu_{j+1} \notin \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap A_j]\ (0 \le j \le N-1)\},$$

where $\mathrm{Lh}_{\mathbb{R}}[\,\cdot\,]$ denotes the $\mathbb{R}$-linear hull (linear span). Let $\mathscr{R}$ be as before, and let

$$\mathfrak{H}_{\mathscr{R}} = \bigcup_{\substack{\mathbf{R}\in\mathscr{R}\\ q\in Q^\vee}} (\mathfrak{H}_{\mathbf{R}^\vee} + q). \tag{9.5.2}$$

Remark 9.16. It should be noted that $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$ if and only if $\langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}}\rangle \in \mathbb{Z}$ for some $\mathbf{V} \in \mathscr{V}, \beta \in \mathbf{V}, q \in Q^\vee$. In fact, if $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$ then we can write $\mathbf{y} = \sum_{j=1}^{r-1} a_j\beta_j^\vee + q$ $(a_j \in \mathbb{R})$. We can find an element $\beta_r \in \Delta$ such that $\mathbf{V} = \{\beta_1,\dots,\beta_r\} \in \mathscr{V}$. Then $\langle \mathbf{y}-q, \mu_{\beta_r}^{\mathbf{V}}\rangle = 0 \in \mathbb{Z}$. Conversely, assume $\langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}}\rangle = c \in \mathbb{Z}$. Write $\mathbf{V} = \{\beta_1,\dots,\beta_{r-1},\beta\}$. Since this is a basis, we may write $\mathbf{y}+q = \sum_{j=1}^{r-1} a_j\beta_j^\vee + a\beta^\vee$ with $a_j, a \in \mathbb{R}$. Then $c = \langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}}\rangle = a$, especially $a \in \mathbb{Z}$. Therefore $a\beta^\vee - q \in Q^\vee$, which implies $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$.

Definition 9.17. For $\Delta_+ \setminus \Delta^* = A = \{\nu_1,\dots,\nu_N\} \subset \Delta_+$, $\mathbf{t}_{\Delta^*} = \{t_\alpha\}_{\alpha\in\Delta^*}$ and $\mathbf{y} \in V$, we define

(9.5.3)

$$\begin{aligned} F_{\Delta^*}(\mathbf{t}_{\Delta^*},\mathbf{y};\Delta) = \sum_{\mathbf{V}\in\mathscr{V}_A} (-1)^{\#(A\setminus\mathbf{V})} \\ \times \Big(\prod_{\gamma\in\Delta_+\setminus(\mathbf{V}\cup A)} \frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}\setminus A} t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle}\Big) \\ \times \frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))} \sum_{q\in Q^\vee/L(\mathbf{V}^\vee)} \Big(\prod_{\beta\in\mathbf{V}\setminus A} \frac{t_\beta \exp(t_\beta\{\mathbf{y}+q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1}\Big). \end{aligned}$$

We consider the action of $\mathfrak{D}_A$ to F to obtain the following theorem.

Theorem 9.18. ([116, Theorem 3.7]) *Let $\Delta_+ \setminus \Delta^* = A = \{\nu_1,\dots,\nu_N\} \subset \Delta_+$ and $\mathbf{t}_{\Delta^*} = \{t_\alpha\}_{\alpha\in\Delta^*}$.*

(i) *For $\mathbf{y} \in V \setminus \mathfrak{H}_{\mathscr{R}}$, we have*

$$\big(\mathfrak{D}_A F\big)(\mathbf{t}_{\Delta^*},\mathbf{y};\Delta) = \big(\mathfrak{D}_{A,2} F\big)(\mathbf{t}_{\Delta^*},\mathbf{y};\Delta) = F_{\Delta^*}(\mathbf{t}_{\Delta^*},\mathbf{y};\Delta), \tag{9.5.4}$$

and hence is independent of choice of the order of A.

(ii) *For $\mathbf{y} \in V$, the function $F_{\Delta^*}(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta)$ is the continuous extension of $\big(\mathfrak{D}_A F\big)(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta)$ in $\mathbf{y}$ in the sense that $\big(\mathfrak{D}_A F\big)(\mathbf{t}_{\Delta^*}, \mathbf{y}+c\phi_0; \Delta)$ tends continuously to $F_{\Delta^*}(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta)$ when $c \to 0+$, and is holomorphic with respect to $\mathbf{t}_{\Delta^*}$ around the origin.*

To show this theorem, we first prepare the following:

LEMMA 9.19. ([**116**, Lemma 9.1]) *For $B \subset \Delta_+$ and $\mathbf{V} \in \mathscr{V}$, we have*

$$\mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap B] = \{v \in V \mid \langle v, \mu_\beta^{\mathbf{V}} \rangle = 0 \text{ for all } \beta \in \mathbf{V} \setminus B\}. \tag{9.5.5}$$

PROOF. Let v be an element of the right-hand side. We write $v = \sum_{\beta \in \mathbf{V}} c_\beta \beta$ and have $c_\beta = 0$ for all $\beta \in \mathbf{V} \setminus B$ and hence

$$v = \sum_{\beta \in \mathbf{V} \cap B} c_\beta \beta \in \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap B]. \tag{9.5.6}$$

The converse is shown similarly. □

PROOF OF THEOREM 9.18. For $\mathbf{t} = (t_\alpha)_{\alpha \in \Delta_+} \in \mathbf{T}$, $\mathbf{y} \in V$, $\mathbf{V} \in \mathscr{V}$, $B \subset \Delta_+$ and $q \in Q^\vee/L(\mathbf{V}^\vee)$, let

(9.5.7)

$$F(\mathbf{t}, \mathbf{y}; \mathbf{V}, B, q) = (-1)^{\#(B \setminus \mathbf{V})} \Big(\prod_{\gamma \in \Delta_+ \setminus (\mathbf{V} \cup B)} \frac{t_\gamma}{t_\gamma - \sum_{\beta \in \mathbf{V} \setminus B} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle} \Big) \times \Big(\prod_{\beta \in \mathbf{V} \setminus B} \frac{t_\beta \exp(t_\beta \{\mathbf{y}+q\}_{\mathbf{V},\beta})}{e^{t_\beta} - 1} \Big),$$

so that

$$F(\mathbf{t}, \mathbf{y}; \Delta) = \sum_{\mathbf{V} \in \mathscr{V}} \frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))} \sum_{q \in Q^\vee/L(\mathbf{V}^\vee)} F(\mathbf{t}, \mathbf{y}; \mathbf{V}, \emptyset, q). \tag{9.5.8}$$

Assume $\mathbf{y} \in V \setminus \mathfrak{H}_{\mathscr{R}}$, and let

$$F_j = F(\mathbf{t}, \mathbf{y}; \mathbf{V}, A_j, q). \tag{9.5.9}$$

We calculate $\mathfrak{D}_{\nu_{j+1}} F_j$. First, since $\mathbf{y} \notin \mathfrak{H}_{\mathscr{R}}$, noting Remark 9.16 we find that

$$\partial_{\nu_{j+1}^\vee} F_j = \Big(\sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle \Big) F_j. \tag{9.5.10}$$

Consider the case $\nu_{j+1} \in \mathbf{V}$. Then $\langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle = \delta_{\nu_{j+1},\beta}$ and

$$\sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle = t_{j+1}, \tag{9.5.11}$$

where we write $t_{\nu_{j+1}} = t_{j+1}$ for brevity. Hence we have $\partial_{\nu_{j+1}^\vee} F_j = t_{j+1} F_j$. Therefore we obtain

(9.5.12)
$$\mathfrak{D}_{\nu_{j+1}} F_j = (-1)^{\#(A_j \setminus \mathbf{V})} \Big(\prod_{\gamma \in \Delta_+ \setminus (\mathbf{V} \cup A_j)} \frac{t_\gamma}{t_\gamma - \sum_{\beta \in \mathbf{V} \setminus (A_j \cup \{\nu_{j+1}\})} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle} \Big) \times \Big(\prod_{\beta \in \mathbf{V} \setminus (A_j \cup \{\nu_{j+1}\})} \frac{t_\beta \exp(t_\beta \{\mathbf{y} + q\}_{\mathbf{V},\beta})}{e^{t_\beta} - 1} \Big),$$

which is equal to F_{j+1} because $\Delta_+ \setminus (\mathbf{V} \cup (A_j \cup \{\nu_{j+1}\})) = \Delta_+ \setminus (\mathbf{V} \cup A_j)$ and $\#((A_j \cup \{\nu_{j+1}\}) \setminus \mathbf{V}) = \#(A_j \setminus \mathbf{V})$.

Next, consider the case $\nu_{j+1} \notin \mathbf{V}$. If $\langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle = 0$ for all $\beta \in \mathbf{V} \setminus A_j$, then

$$\partial_{\nu_{j+1}^\vee} F_j = \Big(\sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle \Big) F_j = 0 \tag{9.5.13}$$

and hence $\mathfrak{D}_{\nu_{j+1}} F_j = 0$. Otherwise, since

(9.5.14)
$$\frac{\partial}{\partial t_{j+1}}\Big|_{t_{j+1}=0} \Big(\frac{t_{j+1}}{t_{j+1} - \sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle} \Big) = - \frac{1}{\sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \nu_{j+1}^\vee, \mu_\beta^{\mathbf{V}} \rangle},$$

we have

(9.5.15)
$$\mathfrak{D}_\nu F_j = (-1)^{\#(A_j \setminus \mathbf{V}) + 1} \Big(\prod_{\gamma \in \Delta_+ \setminus (\mathbf{V} \cup A_j \cup \{\nu_{j+1}\})} \frac{t_\gamma}{t_\gamma - \sum_{\beta \in \mathbf{V} \setminus A_j} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle} \Big) \times \Big(\prod_{\beta \in \mathbf{V} \setminus A_j} \frac{t_\beta \exp(t_\beta \{\mathbf{y} + q\}_{\mathbf{V},\beta})}{e^{t_\beta} - 1} \Big).$$

By noting $\mathbf{V} \setminus (A_j \cup \{\nu_{j+1}\}) = \mathbf{V} \setminus A_j$ and $\#((A_j \cup \{\nu_{j+1}\}) \setminus \mathbf{V}) = \#(A_j \setminus \mathbf{V}) + 1$, we find that the right-hand side is equal to F_{j+1}.

We see that the condition $\langle \nu_{j+1}, \mu_\beta^{\mathbf{V}} \rangle = 0$ for all $\beta \in \mathbf{V} \setminus A_j$ is equivalent to the condition $\nu_{j+1} \in \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap A_j]$. Therefore the above results can be summarized as

$$\mathfrak{D}_{\nu_{j+1}} F_j = \begin{cases} 0 & (\nu_{j+1} \in \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap A_j]), \\ F_{j+1} & (\nu_{j+1} \notin \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap A_j]). \end{cases} \tag{9.5.16}$$

Hence

$$\mathfrak{D}_A F_0 = \begin{cases} 0 & (\mathbf{V} \notin \mathscr{V}_A), \\ F_N & (\mathbf{V} \in \mathscr{V}_A). \end{cases} \tag{9.5.17}$$

Similar to the above calculations, we see that $\mathfrak{D}_{A,2} F_0$ gives the same result as (9.5.17). Thus, since $F_0 = F(\mathbf{t}, \mathbf{y}; \mathbf{V}, \emptyset, q)$, from (9.5.8) we obtain (9.5.4).

Now let $\mathbf{y} \in V$, and we show

$$\lim_{c\to 0+} \{\mathbf{y}+q+c\phi_0\}_{\mathbf{V},\beta} = \{\mathbf{y}+q\}_{\mathbf{V},\beta}. \tag{9.5.18}$$

In fact, when $\mathbf{y} \in V \setminus \mathfrak{H}_{\mathscr{R}}$ this claim is clear, so we assume $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$. Then $\mathbf{y}+q+c\phi_0 \notin \mathfrak{H}_{\mathscr{R}}$ for all sufficiently small $c > 0$. If $\langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}} \rangle \notin \mathbb{Z}$, then

$$\lim_{c\to 0+} \{\mathbf{y}+q+c\phi_0\}_{\mathbf{V},\beta} = \lim_{c\to 0+} \{\langle \mathbf{y}+q+c\phi_0, \mu_\beta^{\mathbf{V}} \rangle\} = \{\langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}} \rangle\} = \{\mathbf{y}+q\}_{\mathbf{V},\beta}.$$

If $\langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}} \rangle \in \mathbb{Z}$, then

$$\lim_{c\to 0+} \{\mathbf{y}+q+c\phi_0\}_{\mathbf{V},\beta}$$
$$= \begin{cases} \lim\limits_{c\to 0+} \{\langle \mathbf{y}+q+c\phi_0, \mu_\beta^{\mathbf{V}} \rangle\} = \lim\limits_{c\to 0+} \{c\langle \phi_0, \mu_\beta^{\mathbf{V}} \rangle\} = 0 \\ \qquad\qquad (\text{if } \ \langle \phi_0, \mu_\beta^{\mathbf{V}} \rangle > 0), \\ \lim\limits_{c\to 0+} (1 - \{-\langle \mathbf{y}+q+c\phi_0, \mu_\beta^{\mathbf{V}} \rangle\}) = \lim\limits_{c\to 0+} (1 - \{-c\langle \phi_0, \mu_\beta^{\mathbf{V}} \rangle\}) = 1 \\ \qquad\qquad (\text{if } \ \langle \phi_0, \mu_\beta^{\mathbf{V}} \rangle < 0). \end{cases}$$

Therefore we obtain the claim (9.5.18), which yields the assertion on the continuity in $\mathbf{y}$.

Finally, since $F(\mathbf{t}, \mathbf{y}; \Delta)$ is holomorphic with respect to $\mathbf{t}$ around the origin (Theorem 9.10), so is $\big(\mathfrak{D}_A F\big)(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta)$ with respect to $\mathbf{t}_{\Delta^*}$. The proof of Theorem 9.18 is thus complete. □

Define $P_{\Delta^*}(\mathbf{k}_{\Delta^*}, \mathbf{y}; \Delta)$ by

$$F_{\Delta^*}(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta) = \sum_{\mathbf{k}_{\Delta^*} \in \mathbb{N}_0^{\#(\Delta^*)}} P_{\Delta^*}(\mathbf{k}_{\Delta^*}, \mathbf{y}; \Delta) \prod_{\alpha\in\Delta^*} \frac{t_\alpha^{k_\alpha}}{k_\alpha!}.$$

When $\mathbf{y} = \mathbf{0}$, we write $P_{\Delta^*}(\mathbf{k}_{\Delta^*}, \mathbf{0}; \Delta) = B_{\Delta^*}(\mathbf{k}_{\Delta^*}; \Delta)$. We obtain the following extension of (9.4.14).

THEOREM 9.20. ([**116**, Theorem 3.9]) *Let* $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathbb{N}_0^{\#\Delta_+}$ *with* $k_\alpha > 1 - \epsilon_\alpha$ $(\alpha \in \Delta^*$, *where* ϵ_α *is defined by* (3.1.9)) *and* $k_\alpha = 0$ $(\alpha \in \Delta_+ \setminus \Delta^*)$, *and let* $\mathbf{k}_{\Delta^*} = (k_\alpha)_{\alpha\in\Delta^*}$. *We have*

$$\sum_{w\in W} \Big(\prod_{\alpha\in\Delta_+\cap w\Delta_-} (-1)^{k_\alpha} \Big) \zeta_r(w^{-1}\mathbf{k}, w^{-1}\mathbf{y}; \Delta) \tag{9.5.19}$$
$$= (-1)^{\#(\Delta_+)} P_{\Delta^*}(\mathbf{k}_{\Delta^*}, \mathbf{y}; \Delta) \Big(\prod_{\alpha\in\Delta^*} \frac{(2\pi i)^{k_\alpha}}{k_\alpha!} \Big)$$

provided that the series expressions of all the zeta-functions on the left-hand side absolutely converge.

PROOF. First, assume $\mathbf{y} \in V \setminus \mathfrak{H}_{\mathscr{R}}$. Let $\mathbf{k}' = (k'_\alpha)_{\alpha\in\Delta_+}$ with $k'_\alpha = k_\alpha$ ($\alpha \in \Delta^*$), $k'_\alpha = 2$ ($\alpha \in \Delta_+ \setminus \Delta^* = A$). Then clearly $\mathbf{k}' \in \mathcal{K}$, and hence by (4.1.3), (4.1.7) and (9.4.14), we have

$$\begin{aligned} S(\mathbf{k}', \mathbf{y}; \Delta) &= \sum_{\lambda\in P\setminus H_{\Delta^\vee}} e^{2\pi i\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{k'_\alpha}} \\ &= \sum_{w\in W} \Big(\prod_{\alpha\in\Delta_+\cap w\Delta_-} (-1)^{k'_\alpha} \Big) \zeta_r(w^{-1}\mathbf{k}', w^{-1}\mathbf{y}; \Delta) \\ &= (-1)^{\#(\Delta_+)} P(\mathbf{k}', \mathbf{y}; \Delta) \Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi i)^{k'_\alpha}}{k'_\alpha!} \Big). \end{aligned} \tag{9.5.20}$$

Applying $\prod_{\alpha\in A} \partial^2_{\alpha^\vee}$ to the above. From the first line we observe that each $\partial^2_{\alpha^\vee}$ produces the factor $(2\pi i\langle \alpha^\vee, \lambda\rangle)^2$. Hence the factor $\zeta_r(w^{-1}\mathbf{k}', w^{-1}\mathbf{y}; \Delta)$ on the second line is transformed into $(2\pi i)^{2\#A}\zeta_r(w^{-1}\mathbf{k}, w^{-1}\mathbf{y}; \Delta)$. Therefore we have

$$\begin{aligned} &(2\pi i)^{2\#A} \sum_{w\in W} \Big(\prod_{\alpha\in\Delta_+\cap w\Delta_-} (-1)^{k_\alpha} \Big) \zeta_r(w^{-1}\mathbf{k}, w^{-1}\mathbf{y}; \Delta) \\ &\qquad = (-1)^{\#(\Delta_+)} \Big(\prod_{\alpha\in A} \partial^2_{\alpha^\vee} \Big) P(\mathbf{k}', \mathbf{y}; \Delta) \Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi i)^{k'_\alpha}}{k'_\alpha!} \Big). \end{aligned} \tag{9.5.21}$$

Since

$$\Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi i)^{k'_\alpha}}{k'_\alpha!} \Big) = \Big(\prod_{\alpha\in\Delta^*} \frac{(2\pi i)^{k'_\alpha}}{k'_\alpha!} \Big) \Big(\prod_{\alpha\in A} \frac{(2\pi i)^2}{2!} \Big),$$

we have

$$\begin{aligned} &\sum_{w\in W} \Big(\prod_{\alpha\in\Delta_+\cap w\Delta_-} (-1)^{k_\alpha} \Big) \zeta_r(w^{-1}\mathbf{k}, w^{-1}\mathbf{y}; \Delta) \\ &\qquad = (-1)^{\#(\Delta_+)} \Big(\prod_{\alpha\in A} \frac{1}{2}\partial^2_{\alpha^\vee} \Big) P(\mathbf{k}', \mathbf{y}; \Delta) \Big(\prod_{\alpha\in\Delta^*} \frac{(2\pi i)^{k'_\alpha}}{k'_\alpha!} \Big). \end{aligned} \tag{9.5.22}$$

From (4.2.11) it follows that

$$\Big(\prod_{\alpha\in A} \frac{1}{2} \frac{\partial^2}{\partial t_\alpha^2}\Big|_{t_\alpha=0} \partial^2_{\alpha^\vee} \Big) F(\mathbf{t}, \mathbf{y}; \Delta) = \sum_{\substack{\mathbf{m}=(m_\alpha)_{\alpha\in\Delta_+} \\ m_\alpha\in\mathbb{N}_0 (\alpha\in\Delta^*) \\ m_\alpha=2 (\alpha\in A)}} \Big(\prod_{\alpha\in A} \frac{1}{2}\partial^2_{\alpha^\vee} \Big) P(\mathbf{m}, \mathbf{y}; \Delta) \prod_{\alpha\in\Delta^*} \frac{t_\alpha^{m_\alpha}}{m_\alpha!}. \tag{9.5.23}$$

By Theorem 9.18, we see that the left-hand side of (9.5.23) is equal to

$$F_{\Delta^*}(\mathbf{t}_{\Delta^*}, \mathbf{y}; \Delta) = \sum_{\mathbf{m}_{\Delta^*}\in\mathbb{N}_0^{\#(\Delta^*)}} P_{\Delta^*}(\mathbf{m}_{\Delta^*}, \mathbf{y}; \Delta) \prod_{\alpha\in\Delta^*} \frac{t_\alpha^{m_\alpha}}{m_\alpha!}. \tag{9.5.24}$$

Comparing (9.5.23) with (9.5.24) we find that

$$\Big(\prod_{\alpha\in A}\frac{1}{2}\partial_{\alpha^\vee}^2\Big)P(\mathbf{k}',\mathbf{y};\Delta)=P_{\Delta^*}(\mathbf{k}_{\Delta^*},\mathbf{y};\Delta). \tag{9.5.25}$$

Therefore (9.5.22) implies the desired result when $\mathbf{y}\in V\setminus\mathfrak{H}_{\mathscr{R}}$. By the continuity with respect to $\mathbf{y}$, the result is also valid in the case when $\mathbf{y}\in\mathfrak{H}_{\mathscr{R}}$. □

From Theorem 5.11 and (9.5.25), we have the rationality for $P_{\Delta^*}(\mathbf{k}_{\Delta^*},\mathbf{y};\Delta)$ in the following sense.

THEOREM 9.21. *For any* $\mathbf{k}\in\mathbb{N}_0^{\#(\Delta_+)}$ *and* $\mathbf{y}=\sum_{i=1}^r y_i\alpha_i^\vee\in V$, *we have* $P_{\Delta^*}(\mathbf{k}_{\Delta^*},\mathbf{y};\Delta)\in \mathrm{Lh}_{\mathbb{Q}}[1,y_1,\dots,y_r]=\mathrm{Lh}_{\mathbb{Q}}[1,\{y_1\},\dots,\{y_r\}]$.

Next, assume that Δ is not simply-laced. Then we have the disjoint union $\Delta=\Delta_l\cup\Delta_s$, where Δ_l is the set of all long roots and Δ_s is the set of all short roots. The next theorem is for even integer points.[1]

THEOREM 9.22. ([**116**, Theorem 3.10]) *Let* $\Delta_1=\Delta_l$ *(resp.* Δ_s*),* $\Delta_2=\Delta_s$ *(resp.* Δ_l*), and* $\Delta_{j+}=\Delta_j\cap\Delta_+$ $(j=1,2)$. *Then* Δ_{j+} $(j=1,2)$ *is a root subset of* Δ_+. *For* $\mathbf{s}_1=\mathbf{k}_1=(k_\alpha)_{\alpha\in\Delta_{1+}}$ *with* $k_\alpha=k\in 2\mathbb{N}$ *(for all* $\alpha\in\Delta_{1+}$*) and* $\nu\in P^\vee/Q^\vee$, *we have*

$$\zeta_r(\mathbf{k}_1,\nu;\Delta_{1+})=\frac{(-1)^{\#(\Delta_+)}}{\#W}P_{\Delta_{1+}}(\mathbf{k}_1,\nu;\Delta)\Big(\prod_{\alpha\in\Delta_{1+}}\frac{(2\pi i)^{k_\alpha}}{k_\alpha!}\Big). \tag{9.5.26}$$

PROOF. Let $\mathbf{s}=\mathbf{k}=(k_\alpha)_{\alpha\in\Delta_+}$ with $k_\alpha=k\in 2\mathbb{N}$ $(\alpha\in\Delta_{1+})$ and $k_\alpha=0$ $(\alpha\in\Delta_{2+})$. Then obviously $\zeta_r(\mathbf{k}_1,\nu;\Delta_{1+})=\zeta_r(\mathbf{k},\nu;\Delta)$. Noting this fact, and applying Theorem 9.20 to $\Delta^*=\Delta_l$ or Δ_s, we obtain the assertion. □

9.6. The action of Aut(Δ)

In this section we treat the situation $I=\emptyset$, and study the action of $\mathrm{Aut}(\Delta)$ on $S(\mathbf{s},\mathbf{y};\Delta)$, $F(\mathbf{t},\mathbf{y};\Delta)$ and $P(\mathbf{k},\mathbf{y};\Delta)$. Note that $P\setminus H_{\Delta^\vee}$ is an $\mathrm{Aut}(\Delta)$-invariant set, because $H_{\Delta^\vee}$ is $\mathrm{Aut}(\Delta)$-invariant. An action of $\mathrm{Aut}(\Delta)$ is naturally induced on any function f in $\mathbf{s}$ and $\mathbf{y}$ as follows: For $w\in\mathrm{Aut}(\Delta)$,

$$(wf)(\mathbf{s},\mathbf{y})=f(w^{-1}\mathbf{s},w^{-1}\mathbf{y}). \tag{9.6.1}$$

[1] It is written in [**116**, p.49] that if there is an odd k_α, then both sides of (9.5.19) vanish, but this sentence is wrong.

THEOREM 9.23. ([113, Theorem 9]) *For* $\mathbf{s} \in \mathcal{S}$ *and* $\mathbf{y} \in V$, *and for* $w \in \mathrm{Aut}(\Delta)$, *we have*

$$(wS)(\mathbf{s}, \mathbf{y}; \Delta) = \Big(\prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-s_\alpha} \Big) S(\mathbf{s}, \mathbf{y}; \Delta), \tag{9.6.2}$$

if $s_\alpha \in \mathbb{Z}$ *for* $\alpha \in \Delta_{w^{-1}}$.

PROOF. From (9.4.1), we have

$$(wS)(\mathbf{s}, \mathbf{y}; \Delta) = \sum_{\lambda \in P \setminus H_{\Delta^\vee}} e^{2\pi i \langle w^{-1}\mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, \lambda \rangle^{s_{w\alpha}}}. \tag{9.6.3}$$

Rewriting λ as $w^{-1}\lambda$ and noting that $P \setminus H_{\Delta^\vee}$ is $\mathrm{Aut}(\Delta)$-invariant, we have

$$\begin{aligned} (wS)(\mathbf{s}, \mathbf{y}; \Delta) &= \sum_{\lambda \in P \setminus H_{\Delta^\vee}} e^{2\pi i \langle \mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle w\alpha^\vee, \lambda \rangle^{s_{w\alpha}}} \\ &= \sum_{\lambda \in P \setminus H_{\Delta^\vee}} e^{2\pi i \langle \mathbf{y}, \lambda \rangle} \prod_{w^{-1}\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, \lambda \rangle^{s_\alpha}} \\ &= \Big(\prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-s_\alpha} \Big) S(\mathbf{s}, \mathbf{y}; \Delta). \end{aligned} \tag{9.6.4}$$

□

Thus we have:

THEOREM 9.24. ([113, Theorem 10]) *For* $\mathbf{s} \in \mathcal{S}$ *and* $\mathbf{y} \in V$, *we have* $S(\mathbf{s}, \mathbf{y}; \Delta) = 0$ *if there exists an element* $w \in \mathrm{Aut}(\Delta)$ *such that* $w^{-1}\mathbf{s} = \mathbf{s}$, $w^{-1}\mathbf{y} \equiv \mathbf{y} \pmod{Q^\vee}$, $s_\alpha \in \mathbb{Z}$ *for* $\alpha \in \Delta_{w^{-1}}$, *and*

$$\sum_{\alpha \in \Delta_{w^{-1}}} s_\alpha \notin 2\mathbb{Z}. \tag{9.6.5}$$

PROOF. If $w^{-1}\mathbf{s} = \mathbf{s}$ and $w^{-1}\mathbf{y} \equiv \mathbf{y} \pmod{Q^\vee}$, then $(wS)(\mathbf{s}, \mathbf{y}; \Delta) = S(\mathbf{s}, \mathbf{y}; \Delta)$. Therefore by Theorem 9.23

$$\Big(1 - \Big(\prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-s_\alpha} \Big) \Big) S(\mathbf{s}, \mathbf{y}; \Delta) = 0. \tag{9.6.6}$$

Then the assumption (9.6.5) implies $S(\mathbf{s}, \mathbf{y}; \Delta) = 0$. □

We put $t_{-\alpha} = -t_\alpha$ and define an action of $\mathrm{Aut}(\Delta)$ by

$$(w\mathbf{t})_\alpha = t_{w^{-1}\alpha}. \tag{9.6.7}$$

If Δ is of type A_1, then $F(\mathbf{t}, \mathbf{y}; A_1) = te^{t\{y\}}/(e^t - 1)$ in (4.2.12) is an even or, in other words, $\mathrm{Aut}(\Delta)$-invariant function except for $y \in \mathbb{Z}$. In the multiple cases, $F(\mathbf{t}, \mathbf{y}; \Delta)$ is revealed to be really an $\mathrm{Aut}(\Delta)$-invariant function. To show it, we need some lemmas.

LEMMA 9.25 ([113, Lemma 12]). *We have*

$$\mathfrak{H}_{\mathscr{R}} = \bigcup_{w\in W} w\left(\bigcup_{j=1}^{r}(\mathfrak{H}_{\Psi^\vee\setminus\{\alpha_j^\vee\}} + \mathbb{Z}\,\alpha_j^\vee)\right). \tag{9.6.8}$$

The set $\{\mathfrak{H}_{\mathbf{R}^\vee}+q \mid \mathbf{R}\in\mathscr{R}, q\in Q^\vee\}$ *is locally finite, i.e., for any* $\mathbf{y}\in V$, *there exists a neighborhood* $U(\mathbf{y})$ *such that* $U(\mathbf{y})$ *intersects finitely many of these hyperplanes.*

PROOF. Fix $\mathbf{R}\in\mathscr{R}$. Then $\widetilde{\Delta}^\vee = \Delta^\vee\cap\mathfrak{H}_{\mathbf{R}^\vee}$ is a coroot system so that $\mathbf{R}^\vee\subset\widetilde{\Delta}^\vee$. Let μ be a nonzero vector normal to $\mathfrak{H}_{\mathbf{R}^\vee}$. Then there exists an element $w\in W$ such that $w^{-1}\mu\in C$. Put $w^{-1}\mu = \sum_{j=1}^r c_j\lambda_j$ with each $c_j\geq 0$. Then $\alpha^\vee = \sum_{j=1}^r a_j\alpha_j^\vee\in\Delta^\vee$ orthogonal to $w^{-1}\mu$ should satisfy $\sum_{j=1}^r a_jc_j = 0$. Since a_j are all nonpositive or nonnegative, we have $a_j = 0$ for j such that $c_j\neq 0$. Hence $c_j = 0$ except for only one j, because $w^{-1}\widetilde{\Delta}^\vee\subset\Delta^\vee$ is orthogonal to $w^{-1}\mu$ with codimension 1. That is, $w^{-1}\mu = c\lambda_j$ for some $c>0$. Therefore $w(\Psi^\vee\setminus\{\alpha_j^\vee\})$ is a fundamental system of $\widetilde{\Delta}^\vee$ and $\mathfrak{H}_{\mathbf{R}^\vee} = \mathfrak{H}_{w(\Psi^\vee\setminus\{\alpha_j^\vee\})} = w\mathfrak{H}_{\Psi^\vee\setminus\{\alpha_j^\vee\}}$. Moreover, $Q^\vee = wQ^\vee = \bigoplus_{i=1}^r\mathbb{Z}\,w\alpha_i^\vee$, which implies

$$\mathfrak{H}_{\mathbf{R}^\vee} + Q^\vee = w\mathfrak{H}_{\Psi^\vee\setminus\{\alpha_j^\vee\}} + \mathbb{Z}\,w\alpha_j^\vee, \tag{9.6.9}$$

since $\bigoplus_{i=1,i\neq j}^r\mathbb{Z}\,\alpha_i^\vee\subset\mathfrak{H}_{\Psi^\vee\setminus\{\alpha_j^\vee\}}$. This shows that $\mathfrak{H}_{\mathscr{R}}$ is contained in the right-hand side of (9.6.8). The opposite inclusion is clear. The local finiteness follows from the expression (9.6.8) and $\#W<\infty$. □

LEMMA 9.26. *For* $\mathbf{y}\in V\setminus\mathfrak{H}_{\mathscr{R}}$ *and* $q\in Q^\vee$, *we have*

$$\{\mathbf{y}+q\}_{\mathbf{V},\beta} = \{\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\} = 1-\{-\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\}. \tag{9.6.10}$$

PROOF. Put $\mathbf{R} = \mathbf{V}\setminus\{\beta\}$ and write

$$\mathbf{y}+q = a_\beta\beta^\vee + \sum_{\gamma\in\mathbf{R}} a_\gamma\gamma^\vee\in a_\beta\beta^\vee+\mathfrak{H}_{\mathbf{R}^\vee}. \tag{9.6.11}$$

Since $\mathbf{y}\notin\mathfrak{H}_{\mathscr{R}}$ implies $\mathbf{y}+q\notin\mathfrak{H}_{\mathscr{R}}$ by (9.6.8), we have $a_\beta\notin\mathbb{Z}$. Hence $\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle = a_\beta\notin\mathbb{Z}$ and $\{\mathbf{y}+q\}_{\mathbf{V},\beta} = \{\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\} = 1-\{-\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\} = \{a_\beta\}$. □

Let $\mathfrak{D} = \{\mathbf{y}\in V\mid 0\leq\langle\mathbf{y},\lambda_i\rangle\leq 1\quad(1\leq i\leq r)\}$. Due to the local finiteness shown in Lemma 9.25 and the fact $\partial\mathfrak{D}\subset\mathfrak{H}_{\mathscr{R}}$, we may write

$$\mathfrak{D}\setminus\mathfrak{H}_{\mathscr{R}} = \coprod_{\nu\in\mathfrak{J}}\mathfrak{D}^{(\nu)}, \tag{9.6.12}$$

where by $\mathfrak{D}^{(\nu)}$ we denote each open connected component of $\mathfrak{D}\setminus\mathfrak{H}_{\mathscr{R}}$ and $\mathfrak{J}$ is a set of indices. Let $|\mathbf{k}| = \sum_{\alpha\in\Delta_+}k_\alpha$. We denote $\mathbb{Q}[\mathbf{y}] = \mathbb{Q}[y_1,\dots,y_r]$ with $\mathbf{y} = \sum_{i=1}^r y_i\alpha_i^\vee$.

THEOREM 9.27. ([**113**, Theorem 13]) *The function* $P(\mathbf{k},\mathbf{y};\Delta)$ *is analytically continued to a polynomial function* $B_{\mathbf{k}}^{(\nu)}(\mathbf{y};\Delta)\in\mathbb{Q}[\mathbf{y}]$ *from each* $\mathfrak{D}^{(\nu)}$ *to the whole space* $\mathbb{C}\otimes V$ *with its total degree at most* $|\mathbf{k}|$.

In [**113**], the proof of this result was rather long and complicated, because it was based on an integral expression of $F(\mathbf{t},\mathbf{y};\Delta)$ (Theorem 5.10). However, now we can use the explicit expression (4.3.5) (Theorem 4.12), and hence we can give the following simple proof.

PROOF. We see that in (4.3.5), the discontinuity in $\mathbf{y}$ appears only at $\mathbf{y}\in\mathfrak{H}_{\mathscr{R}}$. Thus by Lemma 9.26, for each $\mathfrak{D}^{(\nu)}$, $q\in Q^\vee$ and $\beta\in\mathbf{V}$, there exists an $m=m(\nu,q,\beta)\in\mathbb{Z}$ for which

$$\{\mathbf{y}+q\}_{\mathbf{V},\beta}=m+\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle \tag{9.6.13}$$

holds on $\mathbf{y}\in\mathfrak{D}^{(\nu)}$. Thus from Theorem 5.11 and Remark 5.13 we can conclude that $P(\mathbf{k},\mathbf{y};\Delta)$ is a polynomial function with rational coefficients in $\mathbf{y}\in\mathfrak{D}^{(\nu)}$ whose total degree is at most $|\mathbf{k}|$. □

REMARK 9.28. Polynomials $B_{\mathbf{k}}^{(\nu)}(\mathbf{y};\Delta)$ may be called the *Bernoulli polynomials of root systems*. It is possible to prove several properties of $B_{\mathbf{k}}^{(\nu)}(\mathbf{y};\Delta)$, analogous to well-known properties of classical Bernoulli polynomials (see [**113**, Theorems 15–18]). However, we do not discuss the details of them here, because, to develop our theory, Bernoulli functions of root systems $P(\mathbf{k},\mathbf{y},\lambda;I;\Delta)$ are more convenient than $B_{\mathbf{k}}^{(\nu)}(\mathbf{y};\Delta)$.

An action of $\mathrm{Aut}(\Delta)$ is naturally induced on any function f in $\mathbf{t}$ and $\mathbf{y}$ as follows: For $w\in\mathrm{Aut}(\Delta)$,

$$(wf)(\mathbf{t},\mathbf{y})=f(w^{-1}\mathbf{t},w^{-1}\mathbf{y}). \tag{9.6.14}$$

The following is shown in [**113**, Theorem 11] via the integral expression. Here we show it by using the explicit form (4.3.5).

THEOREM 9.29. *Assume that* Δ *is an irreducible root system. If* Δ *is not of type* A_1, *then*

$$(wF)(\mathbf{t},\mathbf{y};\Delta)=F(\mathbf{t},\mathbf{y};\Delta) \tag{9.6.15}$$

for $\mathbf{t}\in\mathbb{C}^{\#\Delta_+}$ *and* $\mathbf{y}\in V$, *and for* $w\in\mathrm{Aut}(\Delta)$. *Hence for* $\mathbf{k}\in\mathbb{N}_0^{\#\Delta_+}$ *and* $\mathbf{y}\in V$,

$$(wP)(\mathbf{k},\mathbf{y};\Delta)=\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-k_\alpha}\Big)P(\mathbf{k},\mathbf{y};\Delta). \tag{9.6.16}$$

REMARK 9.30. If $\mathbf{k}$ is in the region $\mathcal{S}$ of absolute convergence with respect to $\mathbf{s}$, the relation (9.4.14) and Theorem 9.23 immediately imply (9.6.16), while if $\mathbf{k} \notin \mathcal{S}$, it should be proved independently.

PROOF. First, let $\mathbf{y} \in V \backslash \mathfrak{H}_{\mathscr{R}}$. Let Δ be any irreducible root system. Then $w^{-1}\mathbf{y} + q \in V \setminus \mathfrak{H}_{\mathscr{R}}$ for any $w \in \mathrm{Aut}(\Delta)$ and $q \in Q^\vee/L(\mathbf{V}^\vee)$ (because $\mathfrak{H}_{\mathscr{R}}$ is $\mathrm{Aut}(\Delta)$-invariant). Therefore, for $w \in \mathrm{Aut}(\Delta)$, by Lemma 9.26 we have

(9.6.17)

$$
\begin{aligned}
(wF)(\mathbf{t},\mathbf{y};\Delta) &= \sum_{\mathbf{V}\in\mathscr{V}} \Big(\prod_{\gamma\in\Delta_+\setminus\mathbf{V}} \frac{t_{w\gamma}}{t_{w\gamma} - \sum_{\beta\in\mathbf{V}} t_{w\beta}\langle\gamma^\vee, \mu_\beta^{\mathbf{V}}\rangle} \Big) \\
&\times \frac{1}{|Q^\vee/L(\mathbf{V}^\vee)|} \sum_{q\in Q^\vee/L(\mathbf{V}^\vee)} \Big(\prod_{\beta\in\mathbf{V}} \frac{t_{w\beta}\exp(t_{w\beta}\{\langle w^{-1}\mathbf{y}+q, \mu_\beta^{\mathbf{V}}\rangle\})}{e^{t_{w\beta}}-1} \Big) \\
&= \sum_{\mathbf{V}\in\mathscr{V}} \Big(\prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})} \frac{t_\gamma}{t_\gamma - \sum_{\beta\in w\mathbf{V}} t_\beta\langle\gamma^\vee, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle} \Big) \\
&\times \frac{1}{|Q^\vee/L(\mathbf{V}^\vee)|} \sum_{q\in Q^\vee/L(\mathbf{V}^\vee)} \Big(\prod_{\beta\in w\mathbf{V}} \frac{t_\beta\exp(t_\beta\{\langle \mathbf{y}+wq, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle\})}{e^{t_\beta}-1} \Big).
\end{aligned}
$$

For $\mathbf{V} \in \mathscr{V}$, we put

$$
\mathbf{V}' = (w\mathbf{V} \cap \Delta_+) \cup (-(w\mathbf{V} \cap \Delta_-)) = (w\mathbf{V} \cup (-w\mathbf{V})) \cap \Delta_+. \tag{9.6.18}
$$

Since $\mathbf{V} \in \mathscr{V}$, we see that $\mathbf{V}' \in \mathscr{V}$ and hence this correspondence gives a bijection $\mathscr{V} \to \mathscr{V}$. For $\beta \in w\mathbf{V}$, we write

$$
w\mu_{w^{-1}\beta}^{\mathbf{V}} = \sum_{\gamma\in\mathbf{V}'} a_\gamma \mu_\gamma^{\mathbf{V}'}. \tag{9.6.19}
$$

Then if $\gamma \in w\mathbf{V} \cap \Delta_+$,

$$
a_\gamma = \langle\gamma, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle = \langle w^{-1}\gamma, \mu_{w^{-1}\beta}^{\mathbf{V}}\rangle = \delta_{w^{-1}\beta, w^{-1}\gamma} = \delta_{\beta,\gamma}, \tag{9.6.20}
$$

and if $\gamma \in -(w\mathbf{V} \cap \Delta_-)$,

(9.6.21)

$$
a_\gamma = \langle\gamma, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle = -\langle w^{-1}(-\gamma), \mu_{w^{-1}\beta}^{\mathbf{V}}\rangle = -\delta_{w^{-1}\beta, w^{-1}(-\gamma)} = -\delta_{\beta,(-\gamma)}.
$$

Thus

$$
w\mu_{w^{-1}\beta}^{\mathbf{V}} = \begin{cases} \mu_\beta^{\mathbf{V}'} & (\beta \in w\mathbf{V} \cap \Delta_+) \\ -\mu_{-\beta}^{\mathbf{V}'} & (\beta \in w\mathbf{V} \cap \Delta_-), \end{cases} \tag{9.6.22}
$$

and hence

(9.6.23)

$$
\sum_{\beta\in w\mathbf{V}} t_\beta\langle\gamma^\vee, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle = \sum_{\beta\in w\mathbf{V}\cap\Delta_+} t_\beta\langle\gamma^\vee, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle + \sum_{\beta\in w\mathbf{V}\cap\Delta_-} t_\beta\langle\gamma^\vee, w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle
$$

$$= \sum_{\beta\in w\mathbf{V}\cap\Delta_+} t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle + \sum_{\beta\in w\mathbf{V}\cap\Delta_-}(-t_{-\beta})\langle\gamma^\vee,-\mu_{-\beta}^{\mathbf{V}'}\rangle$$
$$= \sum_{\beta\in\mathbf{V}'} t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle.$$

Therefore

$$\prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in w\mathbf{V}}t_\beta\langle\gamma^\vee,w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle} \tag{9.6.24}$$
$$= \prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle}$$
$$= \prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})\cap\Delta_+}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle}\prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})\cap\Delta_-}\frac{(-t_{-\gamma})}{(-t_{-\gamma})-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle}$$
$$= \prod_{\gamma\in w(\Delta_+\setminus\mathbf{V})\cap\Delta_+}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle}\prod_{\gamma\in w(\Delta_-\setminus(-\mathbf{V}))\cap\Delta_+}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle}$$
$$= \prod_{\gamma\in\Delta_+\setminus\mathbf{V}'}\frac{t_\gamma}{t_\gamma-\sum_{\beta\in\mathbf{V}'}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}'}\rangle},$$

where to show the last equality, we used

$$\begin{aligned}\big(w(\Delta_+\setminus\mathbf{V})\cap\Delta_+\big)\cup\big(w(\Delta_-\setminus(-\mathbf{V}))\cap\Delta_+\big) &= \Delta_+\cap w\big((\Delta_+\setminus\mathbf{V})\cup(\Delta_-\setminus(-\mathbf{V}))\big)\\ &= \Delta_+\cap w\big((\Delta_+\setminus(\mathbf{V}\cup(-\mathbf{V}))\cup(\Delta_-\setminus(\mathbf{V}\cup(-\mathbf{V}))\big)\\ &= \Delta_+\setminus(w\mathbf{V}\cup(-w\mathbf{V}))\\ &= \Delta_+\setminus\mathbf{V}'.\end{aligned} \tag{9.6.25}$$

Also we have

$$\prod_{\beta\in w\mathbf{V}}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle\})}{e^{t_\beta}-1} \tag{9.6.26}$$
$$= \prod_{\beta\in w\mathbf{V}\cap\Delta_+}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle\})}{e^{t_\beta}-1}\prod_{\beta\in w\mathbf{V}\cap\Delta_-}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,w\mu_{w^{-1}\beta}^{\mathbf{V}}\rangle\})}{e^{t_\beta}-1}$$
$$= \prod_{\beta\in w\mathbf{V}\cap\Delta_+}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,\mu_\beta^{\mathbf{V}'}\rangle\})}{e^{t_\beta}-1}\prod_{\beta\in w\mathbf{V}\cap\Delta_-}\frac{-t_{-\beta}\exp(-t_{-\beta}\{\langle\mathbf{y}+wq,-\mu_{-\beta}^{\mathbf{V}'}\rangle\})}{e^{-t_{-\beta}}-1}$$
$$= \prod_{\beta\in w\mathbf{V}\cap\Delta_+}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,\mu_\beta^{\mathbf{V}'}\rangle\})}{e^{t_\beta}-1}\prod_{\beta\in(-w\mathbf{V})\cap\Delta_+}\frac{t_\beta\exp(t_\beta(1-\{-\langle\mathbf{y}+wq,\mu_\beta^{\mathbf{V}'}\rangle\}))}{e^{t_\beta}-1}$$
$$= \prod_{\beta\in\mathbf{V}'}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq\rangle\}_{\mathbf{V}',\beta})}{e^{t_\beta}-1},$$

where on the last equality we again used Lemma 9.26. Since q runs over all $Q^\vee/L(\mathbf{V}^\vee)$, we see that wq runs over all

$$wQ^\vee/wL(\mathbf{V}^\vee) = Q^\vee/L(w\mathbf{V}^\vee) = Q^\vee/L(\mathbf{V}'^\vee). \tag{9.6.27}$$

Therefore we find

(9.6.28)

$$\begin{aligned}&\frac{1}{|Q^\vee/L(\mathbf{V}^\vee)|}\sum_{q\in Q^\vee/L(\mathbf{V}^\vee)}\Big(\prod_{\beta\in w\mathbf{V}}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+wq,w\mu^{\mathbf{V}}_{w^{-1}\beta}\rangle\})}{e^{t_\beta}-1}\Big)\\ &=\frac{1}{|Q^\vee/L(\mathbf{V}'^\vee)|}\sum_{q\in Q^\vee/L(\mathbf{V}'^\vee)}\Big(\prod_{\beta\in\mathbf{V}'}\frac{t_\beta\exp(t_\beta\{\langle\mathbf{y}+q\rangle\}_{\mathbf{V}',\beta}\})}{e^{t_\beta}-1}\Big).\end{aligned}$$

Substituting (9.6.24) and (9.6.28) into (9.6.17), we obtain the assertion.

Lastly, to treat the case $\mathbf{y}\in\mathfrak{H}_{\mathscr{R}}$, we assume that Δ is not of type A_1. Then we can apply the continuity of $F(\mathbf{t},\mathbf{y};\Delta)$ in $\mathbf{y}$ (Theorem 9.10) to obtain the result. The proof is complete. □

REMARK 9.31. The above proof shows that, even in the case when Δ is of type A_1, the assertions of Theorem 9.29 are valid for $\mathbf{y}\in V\setminus\mathfrak{H}_{\mathscr{R}}$ ($= V\setminus Q^\vee$ in this case).

9.7. Explicit generating functions ($\#I = (r-1)$ case)

In the following sections of this chapter we deduce the explicit form of generating functions and functional relations from the fundamental formulas given in Section 9.4. Since some examples in the case $I=\emptyset$ were already presented in Section 4.2, here we treat the case $I\neq\emptyset$. Actually, however, we only discuss the cases $\#I = r-1$ and $\#I = 1$, because the other cases are much more complicated. We use the notation $\delta_\square$, where some condition is inserted in $\square$, defined as $\delta_\square = 1$ if the condition is satisfied, and $=0$ otherwise.

We first introduce the transposes $p^*_{\mathbf{V}_I^\perp}$ of the projections $p_{\mathbf{V}_I^\perp}$ (defined by (9.4.8)) by

$$\begin{aligned}\langle u, p_{\mathbf{V}_I^\perp}(v)\rangle &= \langle u,v\rangle - \sum_{\beta\in\mathbf{V}_I}\langle u,\mu^{\mathbf{V}}_\beta\rangle\langle\beta^\vee,v\rangle\\ &= \langle u-\sum_{\beta\in\mathbf{V}_I}\langle u,\mu^{\mathbf{V}}_\beta\rangle\beta^\vee, v\rangle\\ &= \langle p^*_{\mathbf{V}_I^\perp}(u), v\rangle\end{aligned} \tag{9.7.1}$$

for $v\in V$, that is

$$p^*_{\mathbf{V}_I^\perp}(u) = u - \sum_{\beta\in\mathbf{V}_I}\langle u,\mu^{\mathbf{V}}_\beta\rangle\beta^\vee. \tag{9.7.2}$$

Let $I \subset \{1, \dots, r\}$. In this section we consider the situation $\#I^c = 1$, and put $I^c = \{k\}$. Then $\Psi_I = \{\alpha_i\}_{i \in I}$, and we see that

$$\Delta^{(I)\vee} = \{\alpha^\vee = \sum_{i=1}^r a_i \alpha_i^\vee \in \Delta_+^\vee \mid a_k = \langle \alpha^\vee, \lambda_k \rangle \neq 0\}.$$

Since $\#\mathbf{V}_I = 1$ in the present case, we have

$$\mathscr{V}_I = \{\mathbf{V} = \{\beta\} \cup \Psi_I\}_{\beta \in \Delta^{(I)}}. \tag{9.7.3}$$

For $\mathbf{V} = \{\beta\} \cup \Psi_I \in \mathscr{V}_I$ and $\gamma \in \Delta^{(I)} \setminus \{\beta\}$, from (9.7.2) we have

$$p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) = \gamma^\vee - \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle \beta^\vee. \tag{9.7.4}$$

We put $b_i = b_i(\beta) = \langle \beta^\vee, \lambda_i \rangle$ $(1 \leq i \leq r)$ so that

$$\beta^\vee = \sum_{i=1}^r b_i \alpha_i^\vee. \tag{9.7.5}$$

Then we find

$$\mu_\beta^{\mathbf{V}} = \frac{\lambda_k}{b_k}. \tag{9.7.6}$$

Write $\mathbf{y} = y_1 \alpha_1^\vee + \cdots + y_r \alpha_r^\vee$ and $\lambda = \sum_{\substack{i=1 \\ i \neq k}}^r m_i \lambda_i \in P_I$. Then

$$\begin{aligned} p_{\mathbf{V}_I^\perp}(\lambda) &= \lambda - \frac{\lambda_k}{b_k} \langle \beta^\vee, \lambda \rangle \\ &= \sum_{\substack{i=1 \\ i \neq k}}^r m_i \lambda_i + \Big(- \sum_{\substack{i=1 \\ i \neq k}}^r \frac{b_i}{b_k} m_i \Big) \lambda_k \\ &= \sum_{\substack{i=1 \\ i \neq k}}^r m_i \Big(\lambda_i - \frac{b_i}{b_k} \lambda_k \Big). \end{aligned} \tag{9.7.7}$$

Note that $Q^\vee / L(\mathbf{V}^\vee) = \{a_k \alpha_k^\vee\}_{0 \leq a_k < b_k}$ and $\#(Q^\vee / L(\mathbf{V}^\vee)) = b_k$. For $q = a_k \alpha_k^\vee \in Q^\vee$, from (9.7.7) we obtain

$$\langle \mathbf{y} + q, p_{\mathbf{V}_I^\perp}(\lambda) \rangle = \sum_{\substack{i=1 \\ i \neq k}}^r m_i \Big(y_i - \frac{b_i}{b_k} (y_k + a_k) \Big). \tag{9.7.8}$$

We fix ϕ_0 such that $\langle \phi_0, \lambda_k \rangle > 0$. Then for $q = a_k \alpha_k^\vee \in Q^\vee$,

$$\{\mathbf{y} + q\}_{\mathbf{V}, \beta} = \Big\{ \frac{\langle \mathbf{y} + q, \lambda_k \rangle}{b_k} \Big\} = \Big\{ \frac{y_k + a_k}{b_k} \Big\}. \tag{9.7.9}$$

Substituting the above results into (9.4.10), we obtain the following form of the generating function (under the identification $\mathbf{y} = (y_i)_{1 \leq i \leq r}$ and $\lambda = (m_i)_{1 \leq i (\neq k) \leq r}$):

$$F((t_\beta)_{\beta \in \Delta^{(I)}}, (y_i)_{1 \leq i \leq r}, (m_i)_{1 \leq i (\neq k) \leq r}; I; \Delta) \tag{9.7.10}$$

$$= \sum_{\beta\in\Delta^{(I)}} \prod_{\gamma\in\Delta^{(I)}\setminus\{\beta\}} \frac{t_\gamma}{t_\gamma - \frac{\langle\gamma^\vee,\lambda_k\rangle}{b_k} t_\beta - 2\pi\sqrt{-1}\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee),\lambda\rangle}$$

$$\times \frac{1}{b_k} \sum_{0\le a_k<b_k} \exp\Big(2\pi\sqrt{-1}\sum_{\substack{i=1\\ i\neq k}}^{r} m_i\Big(y_i - \frac{b_i}{b_k}(y_k+a_k)\Big)\Big) \frac{t_\beta \exp\Big(t_\beta\Big\{\frac{y_k+a_k}{b_k}\Big\}\Big)}{e^{t_\beta}-1},$$

where $b_i = b_i(\beta) = \langle\beta^\vee,\lambda_i\rangle$.

For $\gamma^\vee \in \Delta^{(I)\vee}$, we see that

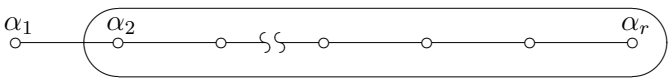
$$p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) = \gamma^\vee - \frac{\langle\gamma^\vee,\lambda_k\rangle}{b_k}\beta^\vee \in \frac{1}{b_k} Q_I^\vee, \tag{9.7.11}$$

which may not be proportional to a coroot in Δ_I. Therefore, observing (9.7.10) with (9.7.11) we see that $S(\mathbf{s},\mathbf{y};I;\Delta)$ may not be necessarily written in terms of zeta-functions of root systems of lower ranks. However, in the classical root systems of types A, B, C, special choices of I give rise to recursive structures among zeta-functions of root systems. To show this, in this section we will give explicit forms of (9.7.11), and will deduce more explicit forms of $F((t_\beta)_{\beta\in\Delta^{(I)}}, (y_i)_{1\le i\le r}, (m_i)_{1\le i(\neq k)\le r}; I;\Delta)$.

9.7.1. A_r **case.** We realize $\Delta^\vee_+(A_r) = \{e_i - e_j \mid 1\le i<j\le r+1\}$, where e_j is the jth unit vector in $\mathbb{R}^{r+1}$. Then the variable $\mathbf{s}$ can be parametrized as $\mathbf{s} = (s_{ij})_{1\le i<j\le r}$, and the zeta-function of type A_r is written as

$$\zeta_r((s_{ij})_{1\le i<j\le r}, (y_i)_{1\le i\le r}; A_r) = \sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty} \frac{\exp(2\pi\sqrt{-1}\sum_{1\le i\le r} m_i y_i)}{\prod_{1\le i<j\le r+1}(m_i+\cdots+m_{j-1})^{s_{ij}}}. \tag{9.7.12}$$

Choose $I = \{2,\dots,r\}$ and $I^c = \{1\}$ as in the following diagram.



(9.7.13)

Then

$$\Psi_I^\vee = \{\alpha_2^\vee = e_2 - e_3, \dots, \alpha_r^\vee = e_r - e_{r+1}\} \tag{9.7.14}$$

and $\Delta^{(I)\vee} = \{e_1 - e_j \mid 2\le j\le r+1\}$. Hence

$$\mathscr{V}_I^\vee = \{\mathbf{V}^\vee = \{e_1 - e_j\}\cup\Psi_I^\vee\}_{2\le j\le r+1}.$$

Temporarily we fix $j\in\{2,3,...,r+1\}$ until the end of (9.7.17), and let

$$\beta^\vee = e_1 - e_j = \alpha_1^\vee + \cdots + \alpha_{j-1}^\vee.$$

Then we see that

$$b_i = \langle \beta^\vee, \lambda_i \rangle = \begin{cases} 1 & (1 \le i < j), \\ 0 & (j \le i \le r), \end{cases} \tag{9.7.15}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 / b_1 = \lambda_1$ by (9.7.6). Since $\mathbf{V}_I^\vee = \{\beta^\vee\}$, for $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$ $(i \ne j)$, from (9.7.4) we have

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 - e_i - \langle e_1 - e_i, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 - e_i - \langle \alpha_1^\vee + \cdots + \alpha_{i-1}^\vee, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 - e_i - (e_1 - e_j) \\ &= e_j - e_i \in \Delta_I^\vee. \end{aligned} \tag{9.7.16}$$

Since

$$e_j - e_i = \begin{cases} -(\alpha_i^\vee + \cdots + \alpha_{j-1}^\vee) & (2 \le i \le j-1), \\ \alpha_j^\vee + \cdots + \alpha_{i-1}^\vee & (j+1 \le i \le r+1), \end{cases}$$

we find that $\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle = -m_{ij}$ for $\lambda = m_2\lambda_2 + \cdots + m_r\lambda_r$, where

$$m_{ij} = \begin{cases} m_i + \cdots + m_{j-1} & (2 \le i \le j-1), \\ -(m_j + \cdots + m_{i-1}) & (j+1 \le i \le r+1). \end{cases} \tag{9.7.17}$$

By putting $t_{e_1 - e_i} = t_i$ for $2 \le i \le r+1$, we can deduce from (9.7.10) (with $k = 1$) the following form of the generating function:

$$\begin{aligned} &F((t_i)_{2\le i\le r+1}, (y_i)_{1\le i\le r}, (m_i)_{2\le i\le r}; \{2, \dots, r\}; A_r) \\ &= \sum_{j=2}^{r+1} \prod_{\substack{2\le i\le r+1 \\ i\ne j}} \frac{t_i}{t_i - t_j + 2\pi\sqrt{-1} m_{ij}} \\ &\quad \times \exp\Bigl(2\pi\sqrt{-1}\Bigl(\sum_{i=2}^{j-1} m_i(y_i - y_1) + \sum_{i=j}^{r} m_i y_i\Bigr)\Bigr) \frac{t_j \exp(t_j\{y_1\})}{e^{t_j} - 1}. \end{aligned} \tag{9.7.18}$$

Next, we calculate the Taylor expansion of the right-hand side of the above formula. First, we note that

$$\begin{aligned} \frac{t_i}{t_i - t_j + 1/x_{ij}} &= \frac{t_i x_{ij}}{1 - (t_j - t_i)x_{ij}} \\ &= \sum_{n=0}^{\infty} x_{ij}^{n+1} t_i (t_j - t_i)^n \\ &= \sum_{n=0}^{\infty} x_{ij}^{n+1} \sum_{\substack{k,l\ge 0 \\ k+l=n}} (-1)^k \binom{n}{l} t_j^l t_i^{k+1} \\ &= \sum_{k\ge 1, l\ge 0} (-1)^{k-1} \binom{k+l-1}{l} x_{ij}^{k+l} t_j^l t_i^k \end{aligned} \tag{9.7.19}$$

(with a certain indeterminate x_{ij}). By use of this result and (4.2.1) we have

$$\Big(\prod_{\substack{2\le i\le r+1\\ i\ne j}}\frac{t_i}{t_i-t_j+1/x_{ij}}\Big)\frac{t_j\exp(t_j\{y_1\})}{e^{t_j}-1}$$

$$=\sum_{\substack{k_i\ge 1\ (i\ne j)\\ l_i\ge 0\ (i\ne j)\\ l_j\ge 0}}\Big(\prod_{\substack{2\le i\le r+1\\ i\ne j}}(-1)^{k_i-1}\binom{k_i+l_i-1}{l_i}x_{ij}^{k_i+l_i}t_j^{l_i}t_i^{k_i}\Big)B_{l_j}(\{y_1\})\frac{t_j^{l_j}}{l_j!}$$

$$=\sum_{\substack{k_i\ge 1\ (i\ne j)\\ k_j\ge 0}}\Big(\sum_{\substack{l_2,\dots,l_{r+1}\ge 0\\ l_2+\cdots+l_{r+1}=k_j}}\frac{B_{l_j}(\{y_1\})}{l_j!}\prod_{\substack{2\le i\le r+1\\ i\ne j}}(-1)^{k_i-1}\binom{k_i+l_i-1}{l_i}x_{ij}^{k_i+l_i}\Big)$$
$$\times\prod_{2\le i\le r+1}t_i^{k_i}.$$

Applying this to the right-hand side of (9.7.18), we obtain the following conclusion.

THEOREM 9.32. ([**119**, Theorem 3.1]) *In the case* $\Delta=\Delta(A_r)$ *and* $I=\{2,\dots,r\}$, *we have*

$$F((t_i)_{2\le i\le r+1},(y_j)_{1\le j\le r},(m_i)_{2\le i\le r};\{2,\dots,r\};A_r)\tag{9.7.20}$$
$$=\sum_{j=2}^{r+1}\prod_{\substack{2\le i\le r+1\\ i\ne j}}\frac{t_i}{t_i-t_j+2\pi\sqrt{-1}m_{ij}}$$
$$\times\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1}m_i(y_i-y_1)+\sum_{i=j}^{r}m_iy_i\Big)\Big)\frac{t_j\exp(t_j\{y_1\})}{e^{t_j}-1}$$
$$=\sum_{k_2,\dots,k_{r+1}\ge 0}P((k_i)_{2\le i\le r+1},(y_i)_{1\le i\le r},(m_i)_{2\le i\le r};\{2,\dots,r\};A_r)\frac{t_2^{k_2}\cdots t_{r+1}^{k_{r+1}}}{k_2!\cdots k_{r+1}!},$$

where

$$P((k_i)_{2\le i\le r+1},(y_i)_{1\le i\le r},(m_i)_{2\le i\le r};\{2,\dots,r\};A_r)\tag{9.7.21}$$
$$=k_2!\cdots k_{r+1}!\sum_{j=2}^{r+1}\Big(\prod_{\substack{i=2\\ i\ne j}}^{r+1}\delta_{k_i\ne 0}\Big)\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1}m_i(y_i-y_1)+\sum_{i=j}^{r}m_iy_i\Big)\Big)$$
$$\times\Big(\sum_{\substack{l_2,\dots,l_{r+1}\ge 0\\ l_2+\cdots+l_{r+1}=k_j}}\frac{B_{l_j}(\{y_1\})}{l_j!}\prod_{\substack{2\le i\le r+1\\ i\ne j}}(-1)^{k_i-1}\binom{k_i+l_i-1}{l_i}\Big(\frac{1}{2\pi\sqrt{-1}m_{ij}}\Big)^{k_i+l_i}\Big).$$

9.7.2. B_r **case.** We realize

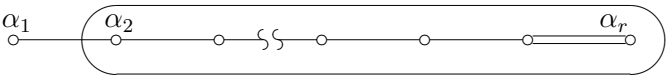
$$\Delta_+^\vee=\{e_i\pm e_j\mid 1\le i<j\le r\}\cup\{2e_j\mid 1\le j\le r\}\tag{9.7.22}$$

and $\Psi^\vee = \{e_1 - e_2, \dots, e_{r-1} - e_r, 2e_r\}$. Then

$$\lambda_i = \begin{cases} e_1 + \cdots + e_i & (1 \le i \le r-1), \\ (e_1 + \cdots + e_r)/2 & (i = r). \end{cases} \tag{9.7.23}$$

Choose $I = \{2, \dots, r\}$ and $I^c = \{1\}$ as in the following diagram.

(9.7.24)



Then

$$\Psi_I^\vee = \{\alpha_2^\vee = e_2 - e_3, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r, \alpha_r^\vee = 2e_r\} \tag{9.7.25}$$

and $\Delta^{(I)\vee} = \{e_1 \pm e_j \mid 2 \le j \le r\} \cup \{2e_1\}$. Hence

(9.7.26)

$$\mathscr{V}_I^\vee = \{\{e_1 - e_j\} \cup \Psi_I^\vee\}_{2 \le j \le r} \cup \{\{e_1 + e_j\} \cup \Psi_I^\vee\}_{2 \le j \le r} \cup \{\{2e_1\} \cup \Psi_I^\vee\}.$$

For $\beta^\vee = e_1 - e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 1 & (1 \le l < j) \\ 0 & (j \le l \le r), \end{cases} \tag{9.7.27}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 \pm e_i - (e_1 - e_j) \\ &= e_j \pm e_i \in \Delta_I^\vee, \end{aligned} \tag{9.7.28}$$

and

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(2e_1) &= 2e_1 - \langle 2e_1, \lambda_1 \rangle (e_1 - e_j) \\ &= 2e_1 - 2(e_1 - e_j) \\ &= 2e_j \in \Delta_I^\vee. \end{aligned} \tag{9.7.29}$$

Next, for $\beta^\vee = e_1 + e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 1 & (1 \le l < j) \\ 2 & (j \le l < r) \\ 1 & (l = r), \end{cases} \tag{9.7.30}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 + e_j) \\ &= e_1 \pm e_i - (e_1 + e_j) \\ &= -e_j \pm e_i \in \Delta_I^\vee, \end{aligned} \tag{9.7.31}$$

and

$$p^*_{\mathbf{V}_I^\perp}(2e_1) = 2e_1 - \langle 2e_1, \lambda_1\rangle(e_1 + e_j) \tag{9.7.32}$$
$$= 2e_1 - 2(e_1 + e_j)$$
$$= -2e_j \in \Delta_I^\vee.$$

Finally, for $\beta^\vee = 2e_1$, we see that

$$b_l = \langle \beta^\vee, \lambda_l\rangle = \begin{cases} 2 & (1 \le l < r) \\ 1 & (l = r), \end{cases} \tag{9.7.33}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1/2 = e_1/2$. Therefore for $\gamma^\vee = e_1 \pm e_i \in \Delta^{(I)\vee}$,

$$p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) = e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1/2\rangle 2e_1 \tag{9.7.34}$$
$$= e_1 \pm e_i - e_1$$
$$= \pm e_i \in \Delta_I^\vee/2.$$

Using the above data of $p^*_{\mathbf{V}_I^\perp}(\gamma^\vee)$ and (9.7.23), we can calculate

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda\rangle = \langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), m_2\lambda_2 + \cdots + m_r\lambda_r\rangle.$$

Let

$$m^\dagger_{j,k} = \begin{cases} m_j + \cdots + m_k & (j \le k), \\ 0 & (\text{otherwise}). \end{cases} \tag{9.7.35}$$

It follows that, when $\beta^\vee = e_1 - e_j$ $(2 \le j \le r)$, then

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda\rangle = \begin{cases} -m^\dagger_{i,j-1} & (\gamma^\vee = e_1 - e_i, i < j) \\ m^\dagger_{j,i-1} & (\gamma^\vee = e_1 - e_i, j < i) \\ m^\dagger_{i,j-1} + 2m^\dagger_{j,r-1} + m_r & (\gamma^\vee = e_1 + e_i, i \le j) \\ m^\dagger_{j,i-1} + 2m^\dagger_{i,r-1} + m_r & (\gamma^\vee = e_1 + e_i, j < i) \\ 2m^\dagger_{j,r-1} + m_r & (\gamma^\vee = 2e_1), \end{cases}$$

when $\beta^\vee = e_1 + e_j$ $(2 \le j \le r)$, then

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda\rangle = \begin{cases} -(m^\dagger_{i,j-1} + 2m^\dagger_{j,r-1} + m_r) & (\gamma^\vee = e_1 - e_i, i \le j) \\ -(m^\dagger_{j,i-1} + 2m^\dagger_{i,r-1} + m_r) & (\gamma^\vee = e_1 - e_i, j < i) \\ m^\dagger_{i,j-1} & (\gamma^\vee = e_1 + e_i, i < j) \\ -m^\dagger_{j,i-1} & (\gamma^\vee = e_1 + e_i, j < i) \\ -(2m^\dagger_{j,r-1} + m_r) & (\gamma^\vee = 2e_1), \end{cases}$$

and when $\beta^\vee = 2e_1$, then

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda\rangle = \pm\left(m^\dagger_{i,r-1} + \frac{1}{2}m_r\right)$$

for $\gamma^\vee = e_1 \pm e_i$.

Therefore by (9.7.10) we now obtain the explicit form of the generating function in the B_r case. We write $t_{e_1 \pm e_i} = t_{\pm i}$ for $2 \le i \le r$ and $t_{e_1} = t_1$. Then the explicit form is as follows.

THEOREM 9.33. ([**120**, (3.17)])

(9.7.36)

$$
\begin{aligned}
&F(t_1, (t_{\pm i})_{2\le i\le r}, (y_j)_{1\le j\le r}, (m_i)_{2\le i\le r}; \{2,\dots,r\}; B_r)\\
&= \sum_{j=2}^{r} \prod_{2\le i<j} \frac{t_{-i}}{t_{-i}-t_{-j}+2\pi\sqrt{-1}m^{\dagger}_{i,j-1}} \prod_{j<i\le r} \frac{t_{-i}}{t_{-i}-t_{-j}-2\pi\sqrt{-1}m^{\dagger}_{j,i-1}}\\
&\quad\times \prod_{2\le i\le j} \frac{t_{+i}}{t_{+i}-t_{-j}-2\pi\sqrt{-1}(m^{\dagger}_{i,j-1}+2m^{\dagger}_{j,r-1}+m_r)}\\
&\quad\times \prod_{j<i\le r} \frac{t_{+i}}{t_{+i}-t_{-j}-2\pi\sqrt{-1}(m^{\dagger}_{j,i-1}+2m^{\dagger}_{i,r-1}+m_r)}\\
&\quad\times \frac{t_1}{t_1-2t_{-j}-2\pi\sqrt{-1}(2m^{\dagger}_{j,r-1}+m_r)}\\
&\quad\times \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1} m_i(y_i-y_1)+\sum_{i=j}^{r} m_i y_i\Big)\Big)\frac{t_{-j}\exp(t_{-j}\{y_1\})}{e^{t_{-j}}-1}\\
&+\sum_{j=2}^{r} \prod_{2\le i\le j} \frac{t_{-i}}{t_{-i}-t_{+j}+2\pi\sqrt{-1}(m^{\dagger}_{i,j-1}+2m^{\dagger}_{j,r-1}+m_r)}\\
&\quad\times \prod_{j<i\le r} \frac{t_{-i}}{t_{-i}-t_{+j}+2\pi\sqrt{-1}(m^{\dagger}_{j,i-1}+2m^{\dagger}_{i,r-1}+m_r)}\\
&\quad\times \prod_{2\le i<j} \frac{t_{+i}}{t_{+i}-t_{+j}-2\pi\sqrt{-1}m^{\dagger}_{i,j-1}} \prod_{j<i\le r} \frac{t_{+i}}{t_{+i}-t_{+j}+2\pi\sqrt{-1}m^{\dagger}_{j,i-1}}\\
&\quad\times \frac{t_1}{t_1-2t_{+j}+2\pi\sqrt{-1}(2m^{\dagger}_{j,r-1}+m_r)}\\
&\quad\times \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1} m_i(y_i-y_1)+\sum_{i=j}^{r-1} m_i(y_i-2y_1)+m_r(y_r-y_1)\Big)\Big)\\
&\quad\times \frac{t_{+j}\exp(t_{+j}\{y_1\})}{e^{t_{+j}}-1}\\
&+\prod_{2\le i\le r} \frac{t_{-i}}{t_{-i}-\frac{1}{2}t_1+\pi\sqrt{-1}(2m^{\dagger}_{i,r-1}+m_r)}\\
&\quad\times \prod_{2\le i\le r} \frac{t_{+i}}{t_{+i}-\frac{1}{2}t_1-\pi\sqrt{-1}(2m^{\dagger}_{i,r-1}+m_r)}
\end{aligned}
$$

$$\times \frac{1}{2}\Bigg(\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{r-1} m_i(y_i - y_1) + m_r(y_r - \frac{1}{2}y_1)\Big)\Big) \times \frac{t_1 \exp(t_1\{\frac{1}{2}y_1\})}{e^{t_1}-1} + \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{r-1} m_i(y_i - y_1) + m_r(y_r - \frac{1}{2}(y_1+1))\Big)\Big) \times \frac{t_1 \exp(t_1\{\frac{1}{2}(y_1+1)\})}{e^{t_1}-1}\Bigg).$$

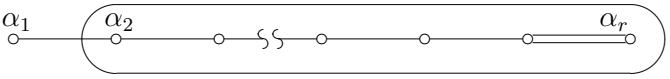
9.7.3. C_r **case.** We realize

$$\Delta_+^\vee = \{e_i \pm e_j \mid 1 \le i < j \le r\} \cup \{e_j \mid 1 \le j \le r\}. \tag{9.7.37}$$

Then $\Psi^\vee = \{\alpha_1^\vee = e_1 - e_2, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r, \alpha_r^\vee = e_r\}$, and the fundamental weights are given by $\lambda_i = e_1 + \cdots + e_i$ for $1 \le i \le r$.

Choose $I = \{2, \dots, r\}$ and $I^c = \{1\}$ as in the following diagram.

(9.7.38)



Then

$$\Psi_I^\vee = \{\alpha_2^\vee = e_2 - e_3, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r, \alpha_r^\vee = e_r\} \tag{9.7.39}$$

and $\Delta^{(I)\vee} = \{e_1 \pm e_j \mid 2 \le j \le r\} \cup \{e_1\}$. Hence

(9.7.40)

$$\mathscr{V}_I^\vee = \{\{e_1 - e_j\} \cup \Psi_I^\vee\}_{2\le j\le r} \cup \{\{e_1 + e_j\} \cup \Psi_I^\vee\}_{2\le j\le r} \cup \{\{e_1\} \cup \Psi_I^\vee\}.$$

For $\beta^\vee = e_1 - e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 1 & (1 \le l < j), \\ 0 & (j \le l \le r), \end{cases} \tag{9.7.41}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 \pm e_i - (e_1 - e_j) \\ &= \begin{cases} e_j \pm e_i \in \Delta_I^\vee & (i \ne j), \\ 2e_j \in 2\Delta_I^\vee & (i = j), \end{cases} \end{aligned} \tag{9.7.42}$$

and

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(e_1) &= e_1 - \langle e_1, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 - (e_1 - e_j) \\ &= e_j \in \Delta_I^\vee. \end{aligned} \tag{9.7.43}$$

Next, for $\beta^\vee = e_1 + e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 1 & (1 \le l < j), \\ 2 & (j \le l \le r), \end{cases} \tag{9.7.44}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 + e_j) \\ &= e_1 \pm e_i - (e_1 + e_j) \\ &= \begin{cases} -e_j \pm e_i \in \Delta_I^\vee & (i \ne j), \\ -2e_j \in 2\Delta_I^\vee & (i = j), \end{cases} \end{aligned} \tag{9.7.45}$$

and

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(e_1) &= e_1 - \langle e_1, \lambda_1 \rangle (e_1 + e_j) \\ &= e_1 - (e_1 + e_j) \\ &= -e_j \in \Delta_I^\vee. \end{aligned} \tag{9.7.46}$$

Lastly, for $\beta^\vee = e_1$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = 1 \qquad (1 \le l \le r), \tag{9.7.47}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 \pm e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle e_1 \\ &= e_1 \pm e_i - e_1 \\ &= \pm e_i \in \Delta_I^\vee. \end{aligned} \tag{9.7.48}$$

Now calculate $\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle$. Consider the case when $\beta^\vee = e_1 - e_j$ $(2 \le j \le r)$. For $\gamma^\vee = e_1 - e_i$ $(i \ne j)$, we have

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle = \langle e_j - e_i, m_2\lambda_2 + \cdots + m_r\lambda_r \rangle = -m_{ij},$$

where m_{ij} is defined in (9.7.17) and the second equality is obtained by using $\lambda_i = e_1 + \cdots + e_i$, or by using (the second member of) the facts

$$\begin{cases} e_i = \sum_{i \le k \le r} \alpha_k^\vee & (1 \le i \le r), \\ e_i - e_j = \sum_{i \le k < j} \alpha_k^\vee & (1 \le i < j \le r), \\ e_i + e_j = \sum_{i \le k < j} \alpha_k^\vee + 2\sum_{j \le k \le r} \alpha_k^\vee & (1 \le i < j \le r). \end{cases} \tag{9.7.49}$$

Using again $\lambda_i = e_1 + \cdots + e_i$ or (9.7.49), for $\gamma^\vee = e_1 + e_i$ we have

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle = \langle e_j + e_i, m_2\lambda_2 + \cdots + m_r\lambda_r \rangle = m_{ijr},$$

where

$$m_{ijr} = \begin{cases} m_i + \cdots + m_{j-1} + 2(m_j + \cdots + m_r) & (2 \le i < j), \\ 2(m_j + \cdots + m_r) & (i = j), \\ m_j + \cdots + m_{i-1} + 2(m_i + \cdots + m_r) & (j < i \le r). \end{cases} \tag{9.7.50}$$

For $\gamma^\vee = e_1$,

$$\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle = \langle e_j, m_2\lambda_2 + \cdots + m_r\lambda_r \rangle = m_j + \cdots + m_r = m_{j,r+1}.$$

Similarly, we evaluate $\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle$ for other $\beta^\vee$.

We now insert those data into (9.7.10). Putting $t_{e_1 \pm e_i} = t_{\pm i}$ for $2 \le i \le r$ and $t_{2e_1} = t_1$, we obtain the following form of the generating function:

THEOREM 9.34. ([**119**, Theorem 3.3]) *In the case* $\Delta = \Delta(C_r)$ *and* $I = \{2, \dots, r\}$ *we have*

(9.7.51)

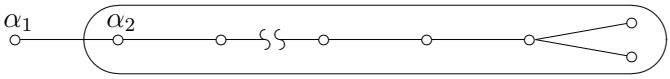
$$\begin{aligned} &F(t_1, (t_{\pm i})_{2\le i\le r}, (y_j)_{1\le j\le r}, (m_i)_{2\le i\le r}; \{2,\dots,r\}; C_r) \\ &= \sum_{j=2}^{r} \prod_{\substack{2\le i\le r \\ i\ne j}} \frac{t_{-i}}{t_{-i} - t_{-j} + 2\pi\sqrt{-1}m_{ij}} \prod_{2\le i\le r} \frac{t_{+i}}{t_{+i} - t_{-j} - 2\pi\sqrt{-1}m_{ijr}} \\ &\qquad \times \frac{t_1}{t_1 - t_{-j} - 2\pi\sqrt{-1}m_{j,r+1}} \\ &\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1} m_i(y_i - y_1) + \sum_{i=j}^{r} m_i y_i\Big)\Big) \frac{t_{-j}\exp(t_{-j}\{y_1\})}{e^{t_{-j}} - 1} \\ &+ \sum_{j=2}^{r} \prod_{2\le i\le j} \frac{t_{-i}}{t_{-i} - t_{+j} + 2\pi\sqrt{-1}m_{ijr}} \prod_{\substack{2\le i\le r \\ i\ne j}} \frac{t_{+i}}{t_{+i} - t_{+j} - 2\pi\sqrt{-1}m_{ij}} \\ &\qquad \times \frac{t_1}{t_1 - t_{+j} + 2\pi\sqrt{-1}m_{j,r+1}} \\ &\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1} m_i(y_i - y_1) + \sum_{i=j}^{r} m_i(y_i - 2y_1)\Big)\Big) \frac{t_{+j}\exp(t_{+j}\{y_1\})}{e^{t_{+j}} - 1} \\ &+ \prod_{2\le i\le r} \frac{t_{-i}}{t_{-i} - t_1 + 2\pi\sqrt{-1}m_{i,r+1}} \prod_{2\le i\le r} \frac{t_{+i}}{t_{+i} - t_1 - 2\pi\sqrt{-1}m_{i,r+1}} \\ &\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{r} m_i(y_i - y_1)\Big)\Big) \frac{t_1\exp(t_1\{y_1\})}{e^{t_1} - 1}. \end{aligned}$$

9.7.4. D_r **case.** We realize $\Delta_+^\vee = \{e_i \pm e_j \mid 1 \le i < j \le r\}$ and $\Psi^\vee = \{e_1 - e_2, \dots, e_{r-1} - e_r, e_{r-1} + e_r\}$. Then

$$\lambda_i = \begin{cases} e_1 + \cdots + e_i & (1 \le i \le r-2), \\ \frac{1}{2}(e_1 + \cdots + e_{r-1} - e_r) & (i = r-1), \\ \frac{1}{2}(e_1 + \cdots + e_{r-1} + e_r) & (i = r). \end{cases} \tag{9.7.52}$$

Choose $I = \{2, \dots, r\}$ and $I^c = \{1\}$ as in the following diagram.

(9.7.53)



Then

$$\Psi_I^\vee = \{\alpha_2^\vee = e_2 - e_3, \dots, \alpha_{r-1}^\vee = e_{r-1} - e_r, \alpha_r^\vee = e_{r-1} + e_r\} \tag{9.7.54}$$

and $\Delta^{(I)\vee} = \{e_1 \pm e_j \mid 2 \le j \le r\}$. Hence

$$\mathscr{V}_I^\vee = \{\{e_1 - e_j\} \cup \Psi_I^\vee\}_{2 \le j \le r} \cup \{\{e_1 + e_j\} \cup \Psi_I^\vee\}_{2 \le j \le r}. \tag{9.7.55}$$

For $\beta^\vee = e_1 - e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 1 & (l < j) \\ 0 & (\text{otherwise}), \end{cases} \tag{9.7.56}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 - e_j) \\ &= e_1 \pm e_i - (e_1 - e_j) \\ &= \begin{cases} e_j \pm e_i \in \Delta_I^\vee & (i \ne j) \\ 2e_i & (i = j). \end{cases} \end{aligned} \tag{9.7.57}$$

Next, for $\beta^\vee = e_1 + e_j$ $(2 \le j \le r)$, we see that

$$b_l = \langle \beta^\vee, \lambda_l \rangle = \begin{cases} 0 & (l = r-1, j = r) \\ 2 & (j \le l \le r-2) \\ 1 & (\text{otherwise}), \end{cases} \tag{9.7.58}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_1 = e_1$. Therefore for $\gamma^\vee = e_1 + e_i \in \Delta^{(I)\vee}$ $(i \ne j)$ and $\gamma^\vee = e_1 - e_i \in \Delta^{(I)\vee}$,

$$\begin{aligned} p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) &= e_1 \pm e_i - \langle e_1 \pm e_i, \lambda_1 \rangle (e_1 + e_j) \\ &= e_1 \pm e_i - (e_1 + e_j) \\ &= \begin{cases} -e_j \pm e_i \in \Delta_I^\vee & (i \ne j) \\ -2e_j & (i = j). \end{cases} \end{aligned} \tag{9.7.59}$$

Using the same notation as in the B_r case, we can now calculate $\langle p^*_{\mathbf{V}_I^\perp}(\gamma^\vee), \lambda \rangle$ and write down the explicit form of the generating function as follows, where $m^\dagger_{j,k}$ is defined in (9.7.35).

THEOREM 9.35. ([120, (3.26)])

(9.7.60)

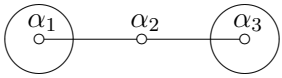
$$
\begin{aligned}
&F((t_{\pm i})_{2\le i\le r},(y_j)_{1\le j\le r},(m_i)_{2\le i\le r};\{2,\dots,r\};D_r)\\
&=\sum_{j=2}^{r-1}\prod_{2\le i<j}\frac{t_{-i}}{t_{-i}-t_{-j}+2\pi\sqrt{-1}m^\dagger_{i,j-1}}\prod_{j<i\le r}\frac{t_{-i}}{t_{-i}-t_{-j}-2\pi\sqrt{-1}m^\dagger_{j,i-1}}\\
&\quad\times\prod_{2\le i\le j}\frac{t_{+i}}{t_{+i}-t_{-j}-2\pi\sqrt{-1}(m^\dagger_{i,j-1}+2m^\dagger_{j,r-2}+m^\dagger_{r-1,r})}\\
&\quad\times\prod_{j<i<r}\frac{t_{+i}}{t_{+i}-t_{-j}-2\pi\sqrt{-1}(m^\dagger_{j,i-1}+2m^\dagger_{i,r-2}+m^\dagger_{r-1,r})}\\
&\quad\times\frac{t_{+r}}{t_{+r}-t_{-j}-2\pi\sqrt{-1}(m^\dagger_{j,r-2}+m_r)}\\
&\quad\times\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1}m_i(y_i-y_1)+\sum_{i=j}^{r}m_iy_i\Big)\Big)\frac{t_{-j}\exp(t_{-j}\{y_1\})}{e^{t_{-j}}-1}\\
&+\prod_{2\le i<r}\frac{t_{-i}}{t_{-i}-t_{-r}+2\pi\sqrt{-1}m^\dagger_{i,r-1}}\\
&\quad\times\prod_{2\le i<r}\frac{t_{+i}}{t_{+i}-t_{-r}-2\pi\sqrt{-1}(m^\dagger_{i,r-2}+m_r)}\\
&\quad\times\frac{t_{+r}}{t_{+r}-t_{-r}-2\pi\sqrt{-1}(m_r-m_{r-1})}\\
&\quad\times\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{r-1}m_i(y_i-y_1)+m_ry_r\Big)\Big)\frac{t_{-r}\exp(t_{-r}\{y_1\})}{e^{t_{-r}}-1}\\
&+\sum_{j=2}^{r-1}\prod_{2\le i\le j}\frac{t_{-i}}{t_{-i}-t_{+j}+2\pi\sqrt{-1}(m^\dagger_{i,j-1}+2m^\dagger_{j,r-2}+m^\dagger_{r-1,r})}\\
&\quad\times\prod_{j<i<r}\frac{t_{-i}}{t_{-i}-t_{+j}+2\pi\sqrt{-1}(m^\dagger_{j,i-1}+2m^\dagger_{i,r-2}+m^\dagger_{r-1,r})}\\
&\quad\times\frac{t_{-r}}{t_{-r}-t_{+j}+2\pi\sqrt{-1}(m^\dagger_{j,r-2}+m_r)}\\
&\quad\times\prod_{2\le i<j}\frac{t_{+i}}{t_{+i}-t_{+j}-2\pi\sqrt{-1}m^\dagger_{i,j-1}}\prod_{j<i\le r}\frac{t_{+i}}{t_{+i}-t_{+j}+2\pi\sqrt{-1}m^\dagger_{j,i-1}}\\
&\quad\times\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1}m_i(y_i-y_1)+\sum_{i=j}^{r-2}m_i(y_i-2y_1)\\
&\qquad+m_{r-1}(y_{r-1}-y_1)+m_r(y_r-y_1)\Big)\Big)
\end{aligned}
$$

$$\times \frac{t_{+j}\exp(t_{+j}\{y_1\})}{e^{t_{+j}}-1}$$
$$+\prod_{2\le i<r}\frac{t_{-i}}{t_{-i}-t_{+r}+2\pi\sqrt{-1}(m^{\dagger}_{i,r-2}+m_r)}$$
$$\times\frac{t_{-r}}{t_{-r}-t_{+r}-2\pi\sqrt{-1}(m_{r-1}-m_r)}$$
$$\times\prod_{2\le i<r}\frac{t_{+i}}{t_{+i}-t_{+r}-2\pi\sqrt{-1}m^{\dagger}_{i,r-1}}$$
$$\times\exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{r-2}m_i(y_i-y_1)+m_{r-1}y_{r-1}+m_r(y_r-y_1)\Big)\Big)$$
$$\times\frac{t_{+r}\exp(t_{+r}\{y_1\})}{e^{t_{+r}}-1}.$$

REMARK 9.36. It should be noted that $\pm 2e_i$ appearing on (9.7.57) and (9.7.59) are not proportional to any coroots. However, they can be regarded as coroots in the root system of type B_r.

9.7.5. $A_1^2 \subset A_3$ **case.** In this subsection, we present another example, due to [**120**, Section 4]. We consider the case of the root system of type A_3 with $I=\{1,3\}$ and $I^c=\{2\}$, as in the following diagram.

(9.7.61)



Then

$$\Psi_I^{\vee}=\{\alpha_1^{\vee},\alpha_3^{\vee}\} \tag{9.7.62}$$

and

$$\Delta^{(I)\vee}=\{\alpha_2^{\vee},\alpha_1^{\vee}+\alpha_2^{\vee},\alpha_2^{\vee}+\alpha_3^{\vee},\alpha_1^{\vee}+\alpha_2^{\vee}+\alpha_3^{\vee}\}=\{e_i-e_j\}_{1\le i\le 2<j\le 4}.$$

Hence

$$\mathscr{V}_I^{\vee}=\{\{\beta^{\vee}\}\cup\Psi_I^{\vee}\}_{\beta^{\vee}\in\Delta^{(I)\vee}} \tag{9.7.63}$$

We see that

$$b_l(\alpha_1+\alpha_2=e_1-e_3)=\langle\alpha_1^{\vee}+\alpha_2^{\vee},\lambda_l\rangle=\begin{cases}1 & (l=1,2)\\ 0 & (l=3),\end{cases} \tag{9.7.64}$$

$$b_l(\alpha_2=e_2-e_3)=\langle\alpha_2^{\vee},\lambda_l\rangle=\begin{cases}1 & (l=2)\\ 0 & (l=1,3),\end{cases} \tag{9.7.65}$$

$$b_l(\alpha_2+\alpha_3=e_2-e_4)=\langle\alpha_2^{\vee}+\alpha_3^{\vee},\lambda_l\rangle=\begin{cases}1 & (l=2,3)\\ 0 & (l=1),\end{cases} \tag{9.7.66}$$

$$b_l(\alpha_1+\alpha_2+\alpha_3=e_1-e_4)=\langle\alpha_1^{\vee}+\alpha_2^{\vee}+\alpha_3^{\vee},\lambda_l\rangle=1, \tag{9.7.67}$$

and in particular $\mu_\beta^{\mathbf{V}} = \lambda_2$. Therefore for $\gamma^\vee \in \Delta^{(I)\vee} \setminus \{\beta^\vee\}$,

$$p^*_{\mathbf{V}_I^\perp}(\gamma^\vee) = \gamma^\vee - \langle \gamma^\vee, \lambda_2 \rangle \beta^\vee = \gamma^\vee - \beta^\vee. \tag{9.7.68}$$

By putting $t_{e_i - e_j} = t_{ij}$ for $1 \le i \le 2 < j \le 4$, we obtain the generating function as follows.

(9.7.69)

$$\begin{aligned}
&F((t_{13}, t_{14}, t_{23}, t_{14}), (y_1, y_2, y_3), (m_1, m_3); \{1,3\}; A_3) \\
&= \frac{t_{23}}{t_{23} - t_{13} + 2\pi\sqrt{-1}m_1} \frac{t_{14}}{t_{14} - t_{13} - 2\pi\sqrt{-1}m_3} \\
&\qquad \times \frac{t_{24}}{t_{24} - t_{13} - 2\pi\sqrt{-1}(m_3 - m_1)} \\
&\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(m_1(y_1 - y_2) + m_3 y_3\Big)\Big) \frac{t_{13}\exp(t_{13}\{y_2\})}{e^{t_{13}} - 1} \\
&+ \frac{t_{13}}{t_{13} - t_{23} - 2\pi\sqrt{-1}m_1} \frac{t_{14}}{t_{14} - t_{23} - 2\pi\sqrt{-1}(m_1 + m_3)} \\
&\qquad \times \frac{t_{24}}{t_{24} - t_{23} - 2\pi\sqrt{-1}m_3} \\
&\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(m_1 y_1 + m_3 y_3\Big)\Big) \frac{t_{23}\exp(t_{23}\{y_2\})}{e^{t_{23}} - 1} \\
&+ \frac{t_{13}}{t_{13} - t_{24} - 2\pi\sqrt{-1}(m_1 - m_3)} \frac{t_{14}}{t_{14} - t_{24} - 2\pi\sqrt{-1}m_1} \\
&\qquad \times \frac{t_{23}}{t_{23} - t_{24} + 2\pi\sqrt{-1}m_3} \\
&\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(m_1 y_1 + m_3(y_3 - y_2)\Big)\Big) \frac{t_{24}\exp(t_{24}\{y_2\})}{e^{t_{24}} - 1} \\
&+ \frac{t_{13}}{t_{13} - t_{14} + 2\pi\sqrt{-1}m_3} \frac{t_{23}}{t_{23} - t_{14} + 2\pi\sqrt{-1}(m_1 + m_3)} \\
&\qquad \times \frac{t_{24}}{t_{24} - t_{14} + 2\pi\sqrt{-1}m_1} \\
&\qquad \times \exp\Big(2\pi\sqrt{-1}\Big(m_1(y_1 - y_2) + m_3(y_3 - y_2)\Big)\Big) \frac{t_{14}\exp(t_{14}\{y_2\})}{e^{t_{14}} - 1}.
\end{aligned}$$

9.8. Explicit functional relations ($\#I = (r-1)$ case)

In this section, using explicit forms of generating functions obtained in the preceding section, we deduce several explicit functional relations.

9.8.1. A_r **case.** First, as an application of Theorem 9.32, we will give the general form of explicit functional relations among zeta-functions of type A_r. We compute (9.4.2) and (9.4.12) in the case of type A_r, with $s_{1j} = k_{1j} \in \mathbb{N}$ for $2 \le j \le r+1$.

Recall that $W(A_r) \simeq \mathfrak{S}_{r+1}$ (see (9.2.16)), as the permutation group on $\{e_1, \dots, e_{r+1}\}$. Via this identification we see that

$$\begin{aligned} W^I =& \{\mathrm{id}, \sigma_1, \sigma_1\sigma_2, \dots, \sigma_1\sigma_2\cdots\sigma_r\} \\ =& \{\mathrm{id}, (1\ 2), (1\ 2\ 3), \dots, (1\ 2\cdots r+1)\}, \end{aligned} \tag{9.8.1}$$

where $\sigma_j = (j\ j+1)$. In fact, we can easily check that

$$w = \sigma_1\sigma_2\cdots\sigma_j = (1\ 2\cdots j+1)$$

for $0 \le j \le r$ satisfies $w^{-1}\Delta^\vee_{I+} \subset \Delta^\vee_+$, and those elements exhaust W^I because $\#W^I = \#W/\#W_I = (r+1)!/r! = r+1$ (see (9.2.13)).

Next, for $2 \le i \le r$ and $1 \le j \le r$, we have

$$\sigma_j\cdots\sigma_1\alpha_1^\vee = e_{j+1} - e_1 = -\alpha_1^\vee - \cdots - \alpha_j^\vee, \tag{9.8.2}$$

$$\sigma_j\cdots\sigma_1\alpha_i^\vee = \begin{cases} \alpha_{i-1}^\vee & (i \le j), \\ \alpha_{i-1}^\vee + \alpha_i^\vee & (i = j+1), \\ \alpha_i^\vee & (i \ge j+2). \end{cases} \tag{9.8.3}$$

Therefore, for $w = \sigma_1\cdots\sigma_j \in W^I$, we have

$$\begin{aligned} w^{-1}\mathbf{y} &= \sigma_j\cdots\sigma_1(y_1\alpha_1^\vee + \cdots + y_r\alpha_r^\vee) \\ &= -y_1(\alpha_1^\vee + \cdots + \alpha_j^\vee) + \sum_{i=2}^{j} y_i\alpha_{i-1}^\vee + y_{j+1}(\alpha_j^\vee + \alpha_{j+1}^\vee) + \sum_{i=j+2}^{r} y_i\alpha_i^\vee \\ &= \sum_{i=1}^{j}(y_{i+1} - y_1)\alpha_i^\vee + \sum_{i=j+1}^{r} y_i\alpha_i^\vee. \end{aligned}$$

Lastly, for $w = \sigma_1\cdots\sigma_j \in W^I$, we can see that

$$\Delta_{w^{-1}} = \{\alpha_1^\vee + \cdots + \alpha_i^\vee = e_1 - e_{i+1} \mid 1 \le i \le j\}. \tag{9.8.4}$$

Therefore from (9.4.2) we obtain

(9.8.5)

$$\begin{aligned} S(\mathbf{s}, \mathbf{y}; \{2, \dots, r\}; A_r) = \sum_{j=0}^{r}\Big(\prod_{i=1}^{j}(-1)^{k_{1,i+1}}\Big) \\ \times \zeta_r((s_{p_jq_j})_{1\le p<q\le r+1}, (y_2 - y_1, \dots, y_{j+1} - y_1, y_{j+1}, \dots, y_r); A_r), \end{aligned}$$

where $p_j = (1\ 2\cdots j+1)p$ and $q_j = (1\ 2\cdots j+1)q$.

On the other hand, (9.4.12) in the present case is written as

$$S(\mathbf{s},\mathbf{y};\{2,\dots,r\};A_r) = (-1)^r \left(\prod_{j=2}^{r+1} \frac{(2\pi\sqrt{-1})^{k_{1j}}}{k_{1j}!}\right) \times \sum_{m_2,\dots,m_r=1}^{\infty} \left(\prod_{2\le p<q\le r+1} \frac{1}{\langle e_p - e_q, m_2\lambda_2 + \cdots + m_r\lambda_r\rangle^{s_{pq}}}\right) \times P(\mathbf{k},\mathbf{y},(m_i)_{2\le i\le r};\{2,\dots,r\};A_r)$$

with $\mathbf{k} = (k_{1j})_{2\le j\le r+1}$. Substituting (9.7.21) (with $k_i = k_{1i}$) into the right-hand side, we obtain

(9.8.6)

$$\begin{aligned} &S(\mathbf{s},\mathbf{y};\{2,\dots,r\};A_r) \\ &= (-1)^r (2\pi\sqrt{-1})^{\sum_{2\le i\le r+1} k_{1i}} \sum_{j=2}^{r+1} \sum_{\substack{l_2,\dots,l_{r+1}\ge 0 \\ l_2+\cdots+l_{r+1}=k_{1j}}} \frac{B_{l_j}(\{y_1\})}{l_j!} \\ &\quad \times \Bigg(\prod_{\substack{2\le i\le r+1 \\ i\ne j}} (-1)^{k_{1i}-1} \binom{k_{1i}+l_i-1}{l_i} \Big(\frac{(-1)^{\delta_{i>j}}}{2\pi\sqrt{-1}}\Big)^{k_{1i}+l_i}\Bigg) \\ &\quad \times \sum_{m_2,\dots,m_r=1}^{\infty} \exp\Big(2\pi\sqrt{-1}\Big(\sum_{i=2}^{j-1} m_i(y_i - y_1) + \sum_{i=j}^{r} m_i y_i\Big)\Big) \\ &\quad \times \prod_{\substack{2\le i\le r+1 \\ i\ne j}} \frac{1}{|m_{ij}|^{k_{1i}+l_i}} \prod_{2\le p<q\le r+1} \frac{1}{(m_p+\cdots+m_{q-1})^{s_{pq}}} \\ &= (-1)^r (2\pi\sqrt{-1})^{\sum_{2\le i\le r+1} k_{1i}} \sum_{j=2}^{r+1} \sum_{\substack{l_2,\dots,l_{r+1}\ge 0 \\ l_2+\cdots+l_{r+1}=k_{1j}}} \frac{(-1)^{k_{1,2}+\cdots+k_{1,j-1}+l_{j+1}+\cdots+l_{r+1}-r+1}}{(2\pi\sqrt{-1})^{l_j+\sum_{2\le i\le r+1} k_{1i}}} \\ &\quad \times \frac{B_{l_j}(\{y_1\})}{l_j!} \Big(\prod_{\substack{2\le i\le r+1 \\ i\ne j}} \binom{k_{1i}+l_i-1}{l_i}\Big) \\ &\quad \times \zeta_{r-1}((s_{pq} + \delta_{p<j}\delta_{q=j}(k_{1p}+l_p) + \delta_{p=j}\delta_{q>j}(k_{1q}+l_q))_{2\le p<q\le r+1}, \\ &\qquad (y_2-y_1,\dots,y_{j-1}-y_1,y_j,\dots,y_r);A_{r-1}), \end{aligned}$$

where in the last equality we have used

$$\sum_{2\le i\le r+1, i\ne j} (k_{1i}+l_i) = -l_j + \sum_{2\le i\le r+1} k_{1i}. \tag{9.8.7}$$

Comparing (9.8.5) and (9.8.6), we now arrive at the following explicit form of functional relation.

THEOREM 9.37. ([**119**, Theorem 3.2]) *For* $(s_{ij})_{1\le i<j\le r+1}$ *with* $s_{1j}=k_{1j}$ $(2\le j\le r+1)$*, we have*

(9.8.8)

$$\begin{aligned}
&\sum_{j=0}^{r}\Big(\prod_{i=1}^{j}(-1)^{k_{1,i+1}}\Big)\\
&\qquad\times\zeta_r((s_{p_jq_j})_{1\le p<q\le r+1},(y_2-y_1,\dots,y_{j+1}-y_1,y_{j+1},\dots,y_r);A_r)\\
&=-\sum_{j=2}^{r+1}\sum_{\substack{l_2,\dots,l_{r+1}\ge 0\\ l_2+\cdots+l_{r+1}=k_{1j}}}(-1)^{k_{1,2}+\cdots+k_{1,j-1}+l_{j+1}+\cdots+l_{r+1}}(2\pi\sqrt{-1})^{l_j}\\
&\qquad\times\frac{B_{l_j}(\{y_1\})}{l_j!}\Big(\prod_{\substack{2\le i\le r+1\\ i\ne j}}\binom{k_{1i}+l_i-1}{l_i}\Big)\\
&\qquad\times\zeta_{r-1}((s_{pq}+\delta_{p<j}\delta_{q=j}(k_{1p}+l_p)+\delta_{p=j}\delta_{q>j}(k_{1q}+l_q))_{2\le p<q\le r+1},\\
&\qquad\qquad(y_2-y_1,\dots,y_{j-1}-y_1,y_j,\dots,y_r);A_{r-1}),
\end{aligned}$$

where $p_j=(1\,2\cdots j+1)p$ *and* $q_j=(1\,2\cdots j+1)q$.

Similarly, from Theorems 9.33, 9.34 and 9.35, we can deduce general form of explicit functional relations for the cases B_r, C_r and D_r. We omit the statement of those general forms, and hereafter in this section we give examples of explicit functional relations of types A_2, A_3 ($\simeq D_3$), B_3 and C_3. These examples are due to [**119**, Sections 3 and 7] [**120**, Section 5].

9.8.2. A_2 **and** A_3 **cases.** ([**119**, Examples 7.1 and 7.2]) Here we treat the types A_2 and A_3. First, we consider the zeta-function of type A_2 (see (3.3.19)). Set $r=2$, $s_{23}\in\mathbb{R}_{>1}$ and $(y_1,y_2)=(0,0)$ in (9.8.8). By considering its real part, we obtain

(9.8.9)

$$\begin{aligned}
&\zeta_2(k_{12},s_{23},k_{13};A_2)+(-1)^{k_{12}}\zeta_2(k_{12},k_{13},s_{23};A_2)\\
&\quad+(-1)^{k_{12}+k_{13}}\zeta_2(k_{13},s_{23},k_{12};A_2)\\
&=2\sum_{j_2=0}^{[k_{12}/2]}(-1)^{k_{12}}\binom{k_{12}+k_{13}-1-2j_2}{k_{13}-1}\zeta(2j_2)\zeta(k_{12}+k_{13}+s_{23}-2j_2)\\
&\quad+2\sum_{j_3=0}^{[k_{13}/2]}(-1)^{k_{13}}\binom{k_{12}+k_{13}-1-2j_3}{k_{12}-1}\zeta(2j_3)\zeta(k_{12}+k_{13}+s_{23}-2j_3),
\end{aligned}$$

where we use Euler's formula (8.1.5). Dividing both sides by $(-1)^{k_{12}}$, we recover Theorem 8.3. This supplies the details to the rough sketch given in Section 9.1.

Next, we consider the zeta-function of type A_3 (see (3.3.20)):

$$\begin{aligned}&\zeta_3(s_1,s_2,s_3,s_4,s_5,s_6;A_3)\\&=\sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty}\sum_{m_3=1}^{\infty} m_1^{-s_1}m_2^{-s_2}m_3^{-s_3}(m_1+m_2)^{-s_4}(m_2+m_3)^{-s_5}\\&\times(m_1+m_2+m_3)^{-s_6}.\end{aligned}$$

Set $r=3$, $(s_{23},s_{24},s_{34})\in(\mathbb{R}_{>1})^3$ and $(y_1,y_2,y_3)=(0,0,0)$ in (9.8.8). By considering its real part, we obtain the following.

THEOREM 9.38. *For $k_{12},k_{13},k_{14}\in\mathbb{N}$,*

(9.8.10)

$$\begin{aligned}&\zeta_3(k_{12},s_{23},s_{34},k_{13},s_{24},k_{14};A_3)\\&+(-1)^{k_{12}}\zeta_3(k_{12},k_{13},s_{34},s_{23},k_{14},s_{24};A_3)\\&+(-1)^{k_{12}+k_{13}}\zeta_3(s_{23},k_{13},k_{14},k_{12},s_{34},s_{24};A_3)\\&+(-1)^{k_{12}+k_{13}+k_{14}}\zeta_3(s_{23},s_{34},k_{14},s_{24},k_{13},k_{12};A_3)\\&=2\sum_{j_2=0}^{[k_{12}/2]}\sum_{\substack{l_3,l_4\geq 0\\ l_3+l_4=k_{12}-2j_2}}(-1)^{k_{12}}\binom{k_{13}+l_3-1}{l_3}\binom{k_{14}+l_4-1}{l_4}\\&\quad\times\zeta(2j_2)\zeta_2(s_{23}+k_{13}+l_3,s_{34},s_{24}+k_{14}+l_4;A_2)\\&+2\sum_{j_3=0}^{[k_{13}/2]}\sum_{\substack{l_2,l_4\geq 0\\ l_2+l_4=k_{13}-2j_3}}(-1)^{k_{12}+l_4}\binom{k_{12}+l_2-1}{l_2}\binom{k_{14}+l_4-1}{l_4}\\&\quad\times\zeta(2j_3)\zeta_2(s_{23}+k_{12}+l_2,s_{34}+k_{14}+l_4,s_{24};A_2)\\&+2\sum_{j_4=0}^{[k_{14}/2]}\sum_{\substack{l_2,l_3\geq 0\\ l_2+l_3=k_{14}-2j_4}}(-1)^{k_{12}+k_{13}}\binom{k_{12}+l_2-1}{l_2}\binom{k_{13}+l_3-1}{l_3}\\&\quad\times\zeta(2j_4)\zeta_2(s_{23},s_{34}+k_{13}+l_3,s_{24}+k_{12}+l_2;A_2).\end{aligned}$$

This formula looks different from the previous result in [**100**, Theorem 4.4]. For example, we obtain from (9.8.10) with $(k_{ij})=(2)$ and $(s_{ij})=(2)$ that

$$\begin{aligned}4\zeta_3(2,2,2,2,2,2;A_3)&=2\zeta(2)\{2\zeta_2(4,2,4;A_2)+\zeta_2(4,4,2;A_2)\}\\&\quad-6\zeta_2(6,2,4;A_2)-6\zeta_2(6,4,2;A_2)-8\zeta_2(5,2,5;A_2)\\&\quad+4\zeta_2(5,5,2;A_2)-6\zeta_2(4,2,6;A_2).\end{aligned}$$

On the other hand, we already obtained from [**100**, (4.28)] that

$$4\zeta_3(2,2,2,2,2,2;A_3) = 8\zeta(2)\{\zeta_2(4,2,4;A_2)+\zeta_2(3,2,5;A_2)\} \\ -12\zeta_2(6,2,4;A_2)-12\zeta_2(5,2,5;A_2)-6\zeta_2(4,2,6;A_2).$$

These two right-hand sides look different. However, we can check that both of the right-hand sides are equal to $887\pi^{12}/3831077250$ by the method of partial fraction decompositions (see [**100**, Example 4.5]).

9.8.3. B_3 **case.** ([**120**, Subsection 5.1]) We write zeta-functions of root systems of type B_2 and type B_3 as

$$\zeta_2(s_1,s_2,s_3,s_4;B_2) = \sum_{m_1,m_2\geq 1} \frac{1}{m_1^{s_1}m_2^{s_2}(m_1+m_2)^{s_3}(2m_1+m_2)^{s_4}},$$

$$\zeta_3(s_1,s_2,s_3,s_4,s_5,s_6,s_7,s_8,s_9;B_3) \\ = \sum_{m_1,m_2,m_3\geq 1} \frac{1}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(2m_2+m_3)^{s_6}} \\ \times \frac{1}{(m_1+m_2+m_3)^{s_7}(m_1+2m_2+m_3)^{s_8}(2m_1+2m_2+m_3)^{s_9}}$$

(see (3.3.22) and (3.3.24)).

Here we give explicit forms of functional relations among them as follows. Let $r=3$, $\Delta=\Delta(B_3)$, $I=\{2,3\}$ and $(y_1,y_2,y_3)=(0,0,0)$ in (9.7.36). Then we have

(9.8.11)

$$F((t_1,t_{\pm 2},t_{\pm 3}),\mathbf{0},(m_2,m_3);\{2,3\};B_3) \\ = \frac{t_{-3}}{t_{-3}-t_{-2}-2\pi\sqrt{-1}m_2}\frac{t_{+2}}{t_{+2}-t_{-2}-2\pi\sqrt{-1}(2m_2+m_3)} \\ \times \frac{t_{+3}}{t_{+3}-t_{-2}-2\pi\sqrt{-1}(m_2+m_3)} \\ \times \frac{t_1}{t_1-2t_{-2}-2\pi\sqrt{-1}(2m_2+m_3)}\frac{t_{-2}}{e^{t_{-2}}-1} \\ + \frac{t_{-2}}{t_{-2}-t_{-3}+2\pi\sqrt{-1}m_2}\frac{t_{+2}}{t_{+2}-t_{-3}-2\pi\sqrt{-1}(m_2+m_3)} \\ \times \frac{t_{+3}}{t_{+3}-t_{-3}-2\pi\sqrt{-1}m_3}\frac{t_1}{t_1-2t_{-3}-2\pi\sqrt{-1}m_3}\frac{t_{-3}}{e^{t_{-3}}-1} \\ + \frac{t_{-2}}{t_{-2}-t_{+2}+2\pi\sqrt{-1}(2m_2+m_3)}\frac{t_{-3}}{t_{-3}-t_{+2}+2\pi\sqrt{-1}(m_2+m_3)} \\ \times \frac{t_{+3}}{t_{+3}-t_{+2}+2\pi\sqrt{-1}m_2}\frac{t_1}{t_1-2t_{+2}+2\pi\sqrt{-1}(2m_2+m_3)}\frac{t_{+2}}{e^{t_{+2}}-1} \\ + \frac{t_{-2}}{t_{-2}-t_{+3}+2\pi\sqrt{-1}(m_2+m_3)}\frac{t_{-3}}{t_{-3}-t_{+3}+2\pi\sqrt{-1}m_3}$$

$$\times \frac{t_{+2}}{t_{+2}-t_{+3}-2\pi\sqrt{-1}m_2}\frac{t_1}{t_1-2t_{+3}+2\pi\sqrt{-1}m_3}\frac{t_{+3}}{e^{t_{+3}}-1}$$
$$+\frac{t_{-2}}{t_{-2}-\frac{1}{2}t_1+\pi\sqrt{-1}(2m_2+m_3)}\frac{t_{-3}}{t_{-3}-\frac{1}{2}t_1+\pi\sqrt{-1}m_3}$$
$$\times \frac{t_{+2}}{t_{+2}-\frac{1}{2}t_1-\pi\sqrt{-1}(2m_2+m_3)}\frac{t_{+3}}{t_{+3}-\frac{1}{2}t_1-\pi\sqrt{-1}m_3}$$
$$\times \frac{1}{2}\Big(\frac{t_1}{e^{t_1}-1}+(-1)^{m_3}\frac{t_1\exp(\frac{1}{2}t_1)}{e^{t_1}-1}\Big).$$

Hence we can compute $P(\mathbf{k},\mathbf{y},\lambda;I;B_3)$. For example, we obtain

(9.8.12)

$$\begin{aligned}
&P((2,1,1,1,1),\mathbf{0},(m_2,m_3);\{2,3\};B_3)\\
&=\frac{1}{16\pi^6m_2^2m_3^3(m_2+m_3)}+\frac{1}{16\pi^6m_2^2(m_2+m_3)(2m_2+m_3)^3}\\
&-\frac{5}{16\pi^6m_2m_3^4(m_2+m_3)}\\
&-\frac{(-1)^{m_3}}{4\pi^6m_3^4(2m_2+m_3)^2}-\frac{1}{4\pi^6m_3^4(2m_2+m_3)^2}-\frac{1}{16\pi^6m_2m_3^3(m_2+m_3)^2}\\
&-\frac{(-1)^{m_3}}{24\pi^4m_3^2(2m_2+m_3)^2}-\frac{(-1)^{m_3}}{4\pi^6m_3^2(2m_2+m_3)^4}-\frac{1}{4\pi^6m_3^2(2m_2+m_3)^4}\\
&+\frac{1}{12\pi^4m_3^2(2m_2+m_3)^2}+\frac{1}{16\pi^6m_2(m_2+m_3)^2(2m_2+m_3)^3}\\
&+\frac{5}{16\pi^6m_2(m_2+m_3)(2m_2+m_3)^4}.
\end{aligned}$$

Therefore, from Theorem 9.11 and by using the relation

$$\begin{aligned}
&\sum_{m,n\geq 1}\frac{(-1)^n}{m^{s_1}n^{s_2}(m+n)^{s_3}(2m+n)^{s_4}}\\
&=-\sum_{m,n\geq 1}\frac{1}{m^{s_1}n^{s_2}(m+n)^{s_3}(2m+n)^{s_4}}\\
&\quad+2\sum_{m,n\geq 1}\frac{1}{m^{s_1}(2n)^{s_2}(m+2n)^{s_3}(2m+2n)^{s_4}}\\
&=-\zeta_2(s_1,s_2,s_3,s_4;B_2)+2^{1-s_2-s_4}\zeta_2(s_2,s_1,s_4,s_3;B_2),
\end{aligned}$$

we obtain the functional relation

(9.8.13)

$$\begin{aligned}
&\zeta_3(1,s_2,s_3,1,s_5,s_6,1,1,2;B_3)-\zeta_3(1,1,s_3,s_2,1,2,s_5,1,s_6;B_3)\\
&+\zeta_3(s_2,1,2,1,1,s_3,1,s_5,s_6;B_3)+\zeta_3(s_2,1,2,1,1,s_3,1,s_5,s_6;B_3)\\
&-\zeta_3(1,1,s_3,s_2,1,2,s_5,1,s_6;B_3)+\zeta_3(1,s_2,s_3,1,s_5,s_6,1,1,2;B_3)
\end{aligned}$$

$$= (-1)^5 \frac{(2\pi\sqrt{-1})^6}{2!1!1!1!1!} \sum_{m_2=1}^{\infty} \sum_{m_3=1}^{\infty} \frac{P((2,1,1,1,1),\mathbf{0},(m_2,m_3);\{2,3\};B_3)}{m_2^{s_2} m_3^{s_3} (m_2+m_3)^{s_5} (2m_2+m_3)^{s_6}}$$

$$= 2\zeta_2(s_2+2,s_3+3,s_5+1,s_6;B_2) + 2\zeta_2(s_2+2,s_3,s_5+1,s_6+3;B_2)$$

$$- 10\zeta_2(s_2+1,s_3+4,s_5+1,s_6;B_2) - \frac{2^{-s_3-s_6}}{4}\zeta_2(s_3+4,s_2,s_6+2,s_5;B_2)$$

$$- 2\zeta_2(s_2+1,s_3+3,s_5+2,s_6;B_2) + 4\pi^2\zeta_2(s_2,s_3+2,s_5,s_6+2;B_2)$$

$$- \frac{2^{-s_3-s_6}\pi^2}{6}\zeta_2(s_3+2,s_2,s_6+2,s_5;B_2) - \frac{2^{-s_3-s_6}}{4}\zeta_2(s_3+2,s_2,s_6+4,s_5;B_2)$$

$$+ 2\zeta_2(s_2+1,s_3,s_5+2,s_6+3;B_2) + 10\zeta_2(s_2+1,s_3,s_5+1,s_6+4;B_2).$$

Setting $(s_2,s_5,s_6) = (1,1,2)$, we obtain

$$\begin{aligned}(9.8.14)\quad & 2\zeta_3(1,1,2,1,1,s_3,1,1,2;B_3)\\ &= 2\zeta_2(3,s_3+3,2,2;B_2) + 2\zeta_2(3,s_3,2,5;B_2)\\ &\quad - 10\zeta_2(2,s_3+4,2,2;B_2) - \frac{2^{-s_3}}{16}\zeta_2(s_3+4,1,4,1;B_2)\\ &\quad - 2\zeta_2(2,s_3+3,3,2;B_2) + 4\pi^2\zeta_2(1,s_3+2,1,4;B_2)\\ &\quad - \frac{2^{-s_3}\pi^2}{24}\zeta_2(s_3+2,1,4,1;B_2) - \frac{2^{-s_3}}{16}\zeta_2(s_3+2,1,6,1;B_2)\\ &\quad + 2\zeta_2(2,s_3,3,5;B_2) + 10\zeta_2(2,s_3,2,6;B_2).\end{aligned}$$

Here we recall the known fact that

$$(9.8.15)\qquad \zeta_2(a,b,c,d;B_2) \in \mathbb{Q}\left[\pi^2, \{\zeta(2j+1)\}_{j\in\mathbb{N}}\right]$$

for $a,b,c,d \in \mathbb{N}$ with $2 \nmid (a+b+c+d)$, which was given by the third-named author (see [**200**]). Similar to [**119**, (100)], setting $s_3 = 2k-1$ ($k \in \mathbb{N}$) in (9.8.14), we obtain from (9.8.15) that

$$(9.8.16)\qquad \zeta_3(1,1,2,1,1,2k-1,1,1,2;B_3) \in \mathbb{Q}\left[\pi^2, \{\zeta(2j+1)\}_{j\in\mathbb{N}}\right]$$

for $k \in \mathbb{N}$. For example, we obtain

$$\zeta_3(1,1,2,1,1,1,1,1,2;B_3) = \frac{9\pi^4}{320}\zeta(7) - \frac{1429\pi^2}{384}\zeta(9) + \frac{4355}{128}\zeta(11),$$

$$\begin{aligned}&\zeta_3(1,1,2,1,1,3,1,1,2;B_3)\\ &\quad = -\frac{7\pi^4}{320}\zeta(9) + \frac{5143\pi^2}{1536}\zeta(11) - \frac{15833}{512}\zeta(13),\end{aligned}$$

$$\begin{aligned}&\zeta_3(1,1,2,1,1,5,1,1,2;B_3)\\ &\quad = \frac{23\pi^8}{2419200}\zeta(7) + \frac{11\pi^6}{20160}\zeta(9) - \frac{941\pi^4}{15360}\zeta(11)\\ &\qquad + \frac{16121\pi^2}{2048}\zeta(13) - \frac{74079}{1024}\zeta(15).\end{aligned}$$

9.8.4. C_3 **case.** ([119, Example 7.3]) We consider zeta-functions of root systems of types C_2 and C_3 (see (3.3.23) and (3.3.25)):

$$\zeta_2(s_1,s_2,s_3,s_4;C_2)=\sum_{m,n\geq 1}\frac{1}{m^{s_1}n^{s_2}(m+n)^{s_3}(m+2n)^{s_4}},$$

$$\begin{aligned}&\zeta_3(s_1,s_2,s_3,s_4,s_5,s_6,s_7,s_8,s_9;C_3)\\&=\sum_{m_1,m_2,m_3\geq 1}\frac{1}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(m_2+2m_3)^{s_6}}\\&\quad\times\frac{1}{(m_1+m_2+m_3)^{s_7}(m_1+m_2+2m_3)^{s_8}(m_1+2m_2+2m_3)^{s_9}}.\end{aligned}$$

Let $r=3$, $\Delta=\Delta(C_3)$, $I=\{2,3\}$ and $(y_1,y_2,y_3)=(0,0,0)$ in (9.7.51). Then we have

$$\begin{aligned}(9.8.17)\quad &F((t_1,t_{\pm 2},t_{\pm 3}),\mathbf{0},(m_2,m_3);\{2,3\};C_3)\\&=\frac{t_{-3}}{t_{-3}-t_{-2}-2\pi\sqrt{-1}m_2}\frac{t_{+2}}{t_{+2}-t_{-2}-2\pi\sqrt{-1}(2(m_2+m_3))}\\&\quad\times\frac{t_{+3}}{t_{+3}-t_{-2}-2\pi\sqrt{-1}(m_2+2m_3)}\frac{t_1}{t_1-t_{-2}-2\pi\sqrt{-1}(m_2+m_3)}\frac{t_{-2}}{e^{t_{-2}}-1}\\&+\frac{t_{-2}}{t_{-2}-t_{-3}+2\pi\sqrt{-1}m_2}\frac{t_{+2}}{t_{+2}-t_{-3}-2\pi\sqrt{-1}(m_2+2m_3)}\\&\quad\times\frac{t_{+3}}{t_{+3}-t_{-3}-2\pi\sqrt{-1}(2m_3)}\frac{t_1}{t_1-t_{-3}-2\pi\sqrt{-1}m_3}\frac{t_{-3}}{e^{t_{-3}}-1}\\&+\frac{t_{-2}}{t_{-2}-t_{+2}+2\pi\sqrt{-1}(2(m_2+m_3))}\frac{t_{-3}}{t_{-3}-t_{+2}+2\pi\sqrt{-1}(m_2+2m_3))}\\&\quad\times\frac{t_{+3}}{t_{+3}-t_{+2}+2\pi\sqrt{-1}m_2}\frac{t_1}{t_1-t_{+2}+2\pi\sqrt{-1}(m_2+m_3)}\frac{t_{+2}}{e^{t_{+2}}-1}\\&+\frac{t_{-2}}{t_{-2}-t_{+3}+2\pi\sqrt{-1}(m_2+2m_3)}\frac{t_{-3}}{t_{-3}-t_{+3}+2\pi\sqrt{-1}(2m_3)}\\&\quad\times\frac{t_{+2}}{t_{+2}-t_{+3}-2\pi\sqrt{-1}m_2}\frac{t_1}{t_1-t_{+3}+2\pi\sqrt{-1}m_3}\frac{t_{+3}}{e^{t_{+3}}-1}\\&+\frac{t_{-2}}{t_{-2}-t_1+2\pi\sqrt{-1}(m_2+m_3)}\frac{t_{-3}}{t_{-3}-t_1+2\pi\sqrt{-1}m_3}\\&\quad\times\frac{t_{+2}}{t_{+2}-t_1-2\pi\sqrt{-1}(m_2+m_3)}\frac{t_{+3}}{t_{+3}-t_1-2\pi\sqrt{-1}m_3}\frac{t_1}{e^{t_1}-1}.\end{aligned}$$

Hence we can compute $P(\mathbf{k},\mathbf{y},\lambda;I;C_3)$ and give some functional relations from Theorem 9.11. For example, we obtain

$$(9.8.18)\quad P((2,1,1,1,1),\mathbf{0},(m_2,m_3);\{2,3\};C_3)=$$

$$\frac{1}{32\pi^6 m_2^2 m_3^3(m_2+2m_3)}+\frac{1}{32\pi^6 m_2^2(m_2+m_3)^3(m_2+2m_3)}$$
$$-\frac{1}{32\pi^6 m_3^4(m_2+m_3)^2}-\frac{5}{64\pi^6 m_2 m_3^4(m_2+2m_3)}$$
$$-\frac{1}{32\pi^6 m_2 m_3^3(m_2+2m_3)^2}-\frac{1}{32\pi^6 m_2^2(m_2+m_3)^4}+\frac{1}{96\pi^4 m_3^2(m_2+m_3)^2}$$
$$+\frac{5}{64\pi^6 m_2(m_2+m_3)^4(m_2+2m_3)}+\frac{1}{32\pi^6 m_2(m_2+m_3)^3(m_2+2m_3)^2}.$$

Hence we have

(9.8.19)

$$\begin{aligned}
&\zeta_3(1,s_2,s_3,1,s_5,s_6,2,1,1;C_3)-\zeta_3(1,1,s_3,s_2,2,1,s_5,s_6,1;C_3)\\
&+\zeta_3(s_2,1,2,1,s_3,1,s_5,1,s_6;C_3)+\zeta_3(s_2,1,2,1,s_3,1,s_5,1,s_6;C_3)\\
&-\zeta_3(1,1,s_3,s_2,2,1,s_5,s_6,1;C_3)+\zeta_3(1,s_2,s_3,1,s_5,s_6,2,1,1;C_3)\\
&=(-1)^5\frac{(2\pi\sqrt{-1})^6}{2!1!1!1!1!}\sum_{m_2=1}^{\infty}\sum_{m_3=1}^{\infty}\frac{P((2,1,1,1,1),\mathbf{0},(m_2,m_3);\{2,3\};C_3)}{m_2^{s_2}m_3^{s_3}(m_2+m_3)^{s_5}(m_2+2m_3)^{s_6}}\\
&=\zeta_2(s_2+2,s_3+3,s_5,s_6+1;C_2)+\zeta_2(s_2+2,s_3,s_5+3,s_6+1;C_2)\\
&\quad-\zeta_2(s_2,s_3+4,s_5+2,s_6;C_2)-\frac{5}{2}\zeta_2(s_2+1,s_3+4,s_5,s_6+1;C_2)\\
&\quad-\zeta_2(s_2+1,s_3+3,s_5,s_6+2;C_2)-\zeta_2(s_2,s_3+2,s_5+4,s_6;C_2)\\
&\quad+\frac{\pi^2}{3}\zeta_2(s_2,s_3+2,s_5+2,s_6;C_2)+\frac{5}{2}\zeta_2(s_2+1,s_3,s_5+4,s_6+1;C_2)\\
&\quad+\zeta_2(s_2+1,s_3,s_5+3,s_6+2;C_2).
\end{aligned}$$

Setting $(s_2,s_5,s_6)=(1,2,1)$, we obtain

(9.8.20)

$$\begin{aligned}
&2\zeta_3(1,1,2,1,s_3,1,2,1,1;C_3)\\
&=\zeta_2(3,s_3+3,2,2;C_2)+\zeta_2(3,s_3,5,2;C_2)-\zeta_2(1,s_3+4,4,1;C_2)\\
&\quad-\frac{5}{2}\zeta_2(2,s_3+4,2,2;C_2)-\zeta_2(2,s_3+3,2,3;C_2)\\
&\quad-\zeta_2(1,s_3+2,6,1;C_2)+\frac{\pi^2}{3}\zeta_2(1,s_3+2,4,1;C_2)\\
&\quad+\frac{5}{2}\zeta_2(2,s_3,6,2;C_2)+\zeta_2(2,s_3,5,3;C_2).
\end{aligned}$$

Since $\zeta_2(s_1,s_2,s_3,s_4;C_2)=\zeta_2(s_2,s_1,s_3,s_4;B_2)$, from (9.8.15) we have

$$\zeta_2(a,b,c,d;C_2)\in\mathbb{Q}\left[\pi^2,\{\zeta(2j+1)\}_{j\in\mathbb{N}}\right] \tag{9.8.21}$$

for $a, b, c, d \in \mathbb{N}$ with $2 \nmid (a+b+c+d)$. If we set $s_3 = 2k-1$ ($k \in \mathbb{N}$) in (9.8.20), we obtain from (9.8.21) that

$$\zeta_3(1,1,2,1,2k-1,1,2,1,1;C_3) \in \mathbb{Q}\left[\pi^2, \{\zeta(2j+1)\}_{j\in\mathbb{N}}\right] \tag{9.8.22}$$

for $k \in \mathbb{N}$. Setting $k = 1, 2$, we have

$$2\zeta_3(1,1,2,1,1,1,2,1,1;C_3) = \frac{3}{20}\zeta(7)\pi^4 - \frac{233}{16}\zeta(9)\pi^2 + \frac{4135}{32}\zeta(11),$$
$$2\zeta_3(1,1,2,1,3,1,2,1,1;C_3) = -\frac{7}{15}\zeta(9)\pi^4 + \frac{681}{16}\zeta(11)\pi^2 - \frac{5995}{16}\zeta(13).$$

More generally, we can prove that

$$\begin{aligned}&\zeta_3(2q-1,2q-1,2r,2p-1,2k-1,2q-1,2r,2p-1,2q-1;C_3)\\ &\qquad \in \mathbb{Q}\left[\pi^2, \{\zeta(2j+1)\}_{j\in\mathbb{N}}\right]\end{aligned} \tag{9.8.23}$$

for $p, q, r, k \in \mathbb{N}$. For example, we can give

$$\begin{aligned}&2\zeta_3(1,1,2,3,1,1,2,3,1;C_3)\\ &\quad = \frac{1}{10080}\zeta(7)\pi^8 + \frac{7}{180}\zeta(9)\pi^6 + \frac{13217}{576}\zeta(11)\pi^4\\ &\quad - \frac{4132493}{1536}\zeta(13)\pi^2 + \frac{24864015}{1024}\zeta(15),\\ &2\zeta_3(1,1,2,3,5,1,2,3,1;C_3)\\ &\quad = \frac{43}{90720}\zeta(11)\pi^8 + \frac{859}{2520}\zeta(13)\pi^6 + \frac{140051}{576}\zeta(15)\pi^4\\ &\quad - \frac{42288073}{1536}\zeta(17)\pi^2 + \frac{253652169}{1024}\zeta(19).\end{aligned}$$

9.8.5. A_3 case and D_3 case. ([120, Subsection 5.2]) Here we deduce an explicit functional relation involving the zeta-function of A_3, from the result proved in Section 9.8. From (9.4.11) and (9.7.69), we can compute $P(\mathbf{k}, \mathbf{y}, \lambda; \{1,3\}; A_3)$. For example, we obtain

$$P((1,1,1,1), \mathbf{0}, (m_1, m_3); \lambda; \{1,3\}; A_3)$$

as follows. When $m_1 \neq m_3$, we have

$$\begin{aligned}&P((1,1,1,1), \mathbf{0}, (m_1,m_3); \{1,3\}; A_3)\\ &\quad = \frac{1}{8\pi^4 m_1^2 m_3(m_1+m_3)} + \frac{1}{8\pi^4 m_1^2 m_3(m_3-m_1)}\\ &\quad - \frac{1}{8\pi^4 m_1 m_3^2(m_3-m_1)} + \frac{1}{8\pi^4 m_1 m_3^2(m_1+m_3)}\\ &\quad - \frac{1}{8\pi^4 m_1 m_3(m_3-m_1)^2} + \frac{1}{8\pi^4 m_1 m_3(m_1+m_3)^2}.\end{aligned} \tag{9.8.24}$$

When $m_1 = m_3 = m$, we have

$$P((1,1,1,1),\mathbf{0},(m,m);\{1,3\};A_3) = \frac{7}{32\pi^4 m^4} - \frac{1}{48\pi^2 m^2}. \tag{9.8.25}$$

We write

$$\begin{aligned}&\zeta_3(s_1,s_2,s_3,s_4,s_5,s_6;A_3)\\ &= \sum_{m_1,m_2,m_3=1}^{\infty} \frac{1}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(m_1+m_2+m_3)^{s_6}}\end{aligned}$$

(see (3.3.20)).

Then, from Theorem 9.11, we obtain the functional relation

$$\begin{aligned}&\zeta_3(s_1,1,s_3,1,1,1;A_3) - \zeta_3(1,1,1,s_1,s_3,1;A_3)\\ &\quad + \zeta_3(1,s_3,1,1,1,s_1;A_3) + \zeta_3(s_3,1,s_1,1,1,1;A_3)\\ &\quad - \zeta_3(1,1,1,s_3,s_1,1;A_3) + \zeta_3(1,s_1,1,1,1,s_3;A_3)\\ &= (2\pi\sqrt{-1})^4\\ &\qquad \times \Bigg(\sum_{\substack{m_1,m_3=1\\ m_1\neq m_3}}^{\infty} \frac{1}{m_1^{s_1}m_3^{s_3}} P((1,1,1,1),\mathbf{0},(m_1,m_3);\{1,3\};A_3)\\ &\qquad\quad + \sum_{m=1}^{\infty} \frac{1}{m^{s_1+s_3}} P((1,1,1,1),\mathbf{0},(m,m);\{1,3\};A_3)\Bigg)\\ &= 2\Big\{ \zeta_2(s_1+2,s_3+1,1;A_2) + \zeta_2(s_1+2,1,s_3+1;A_2)\\ &\quad - \zeta_2(s_3+1,1,s_1+2;A_2) - \zeta_2(s_1+1,1,s_3+2;A_2)\\ &\quad + \zeta_2(s_3+2,1,s_1+1;A_2) + \zeta_2(s_1+1,s_3+2,1;A_2)\\ &\quad - \zeta_2(s_1+1,2,s_3+1;A_2) - \zeta_2(s_3+1,2,s_1+1;A_2)\\ &\quad + \zeta_2(s_1+1,s_3+1,2;A_2)\Big\} + \zeta(s_1+s_3+4)\\ &\quad - \frac{\pi^2}{3}\zeta(s_1+s_3+2).\end{aligned}$$

If we set $(s_1,s_3) = (1,1)$, we see that

$$\{\zeta_2(k,l,m;A_2) \mid k,l,m \in \mathbb{N} \text{ with } k+l+m = 6\}$$

appears on the right-hand side. Using the partial fraction decomposition formula (see Lemma 12.2)

$$\frac{1}{X^p Y^q} = \sum_{i=0}^{p-1} \binom{q-1+i}{i} \frac{1}{X^{p-i}(X+Y)^{q+i}}$$

$$+\sum_{i=0}^{q-1}\binom{p-1+i}{i}\frac{1}{Y^{q-i}(X+Y)^{p+i}}\quad(p,q\in\mathbb{N}),$$

we obtain

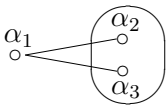
$$\begin{aligned}\zeta_2(k,l,m;A_2)=&\sum_{i=0}^{k-1}\binom{l-1+i}{i}\zeta_{EZ,2}(k-i,l+m+i)\\&+\sum_{i=0}^{l-1}\binom{k-1+i}{i}\zeta_{EZ,2}(l-i,k+m+i)\end{aligned}$$

(see Huard et al [**69**, (1.6)]). It is well-known that $\zeta_{EZ,2}(p,q)$ $(p,q\in\mathbb{N},\ q\geq 2,\ p+q\leq 6)$ can be expressed in terms of $\{\zeta(m)\mid 2\leq m\leq 6\}$, by using EZ-Face, an interface for evaluation of Euler sums (see [**19**]). Using these results, we can consequently obtain

$$2\,\zeta_3(1,1,1,1,1,1;A_3)=4\zeta(3)^2-\frac{31}{5670}\pi^6. \tag{9.8.26}$$

Here we give a comment on the D_3 case. Although generally we consider $r\geq 4$ for D_r cases, the generating function (9.7.60) is valid even when $r=3$. In this case, we see that the Dynkin diagram (9.7.53) reduces to

(9.8.27)



and this coincides with (9.7.61). In fact, if we read $t_{13},t_{14},t_{23},t_{24}$, y_1,y_2,y_3, m_1,m_3 as $t_{-3},t_{+2},t_{-2},t_{+3}$, y_2,y_1,y_3, m_2,m_3 respectively in (9.7.69), then we obtain the generating function for D_3 which comes from (9.7.60).

9.9. Explicit functional relations ($\#I=1$ case)

In this section we consider the case $I=\{i\}$ and $\mathbf{y}=\mathbf{0}$. In this case the result can be stated in terms of the Lerch zeta-function (4.1.9).

THEOREM 9.39. ([**119**, Theorem 4.1]) *Let $s_\alpha=k_\alpha\in\mathbb{Z}_{\geq 2}$ for $\alpha\in\Delta_+\setminus\{\alpha_i\}$ and $s_{\alpha_i}\in\mathbb{C}$. Let $|\mathbf{k}|=\sum_{\alpha\in\Delta_+\setminus\{\alpha_i\}}k_\alpha$. Let $X_i=\{\nu=\{\langle q,\mu_{\alpha_i}^{\mathbf{V}}\rangle\}\mid\mathbf{V}\in\mathscr{V}_I,q\in Q^\vee/L(\mathbf{V}^\vee)\}\subset\mathbb{Q}$. Then we have*

$$\begin{aligned}&\sum_{w\in W^I}\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-k_\alpha}\Big)\zeta_r(w^{-1}\mathbf{s},0;\Delta)\\&=(-1)^{\#\Delta_+-1}\Big(\prod_{\alpha\in\Delta_+\setminus\{\alpha_i\}}\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)\sum_{\nu\in X_i}\sum_{j=0}^{|\mathbf{k}|}\frac{b_{\mathbf{k}\nu j}}{(2\pi\sqrt{-1})^j}\phi(s_{\alpha_i}+j,\nu),\end{aligned} \tag{9.9.1}$$

where $b_{\mathbf{k}\nu j} \in \mathbb{Q}$ *is given by*

$$\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Delta^*}} \sum_{\nu\in X_i} \sum_{j=0}^{|\mathbf{k}|} b_{\mathbf{k}\nu j} x^j y^\nu \prod_{\alpha\in\Delta^*} \frac{t_\alpha^{k_\alpha}}{k_\alpha!} \tag{9.9.2}$$
$$= \sum_{\mathbf{V}\in\mathscr{V}_I} \prod_{\gamma\in\Delta^*\setminus\mathbf{V}_I} \frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}_I} t_\beta \langle\gamma^\vee, \mu_\beta^{\mathbf{V}}\rangle - \langle\gamma^\vee, \mu_{\alpha_i}^{\mathbf{V}}\rangle/x}$$
$$\times \frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))} \sum_{q\in Q^\vee/L(\mathbf{V}^\vee)} y^{\{\langle q,\mu_{\alpha_i}^{\mathbf{V}}\rangle\}} \prod_{\beta\in\mathbf{V}_I} \frac{t_\beta \exp(t_\beta\{q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1},$$

for indeterminates x *and* y.

Proof. Since $I = \{i\}$, we have $\Delta_{I+} = \{\alpha_i\}$ and $\Delta^{(I)} = \Delta_+ \setminus \{\alpha_i\}$. Note that for $\lambda = n\lambda_i \in P_{I++}$, we have $p_{\mathbf{V}_I^\perp}(\lambda) = n\mu_{\alpha_i}^{\mathbf{V}}$ by (9.4.8), and hence for $u \in V$,

$$\langle u, p_{\mathbf{V}_I^\perp}(\lambda)\rangle = n\langle u, \mu_{\alpha_i}^{\mathbf{V}}\rangle. \tag{9.9.3}$$

Therefore

(9.9.4)
$$F(\mathbf{t}, 0, \lambda; I; \Delta) = \sum_{\mathbf{V}\in\mathscr{V}_I} \prod_{\gamma\in\Delta^{(I)}\setminus\mathbf{V}_I} \frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}_I} t_\beta \langle\gamma^\vee, \mu_\beta^{\mathbf{V}}\rangle - 2\pi\sqrt{-1}n\langle\gamma^\vee, \mu_{\alpha_i}^{\mathbf{V}}\rangle}$$
$$\times \frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))} \sum_{q\in Q^\vee/L(\mathbf{V}^\vee)} \exp(2\pi\sqrt{-1}n\{\langle q, \mu_{\alpha_i}^{\mathbf{V}}\rangle\}) \prod_{\beta\in\mathbf{V}_I} \frac{t_\beta \exp(t_\beta\{q\}_{\mathbf{V},\beta})}{e^{t_\beta}-1}.$$

Consider the Taylor expansion of the right-hand side with respect to $\mathbf{t}$. Putting $x = (2\pi\sqrt{-1}n)^{-1}$ and $y = \exp(2\pi\sqrt{-1}n)$, the right-hand side of (9.9.4) is equal to the right-hand side of (9.9.2). We write the Taylor expansion as the left-hand side of (9.9.2) with rational coefficients $b_{\mathbf{k}\nu j}$; here we remark that in (9.9.2), the highest degree of x is at most $|\mathbf{k}|$, because when $\langle\gamma^\vee, \mu_{\alpha_i}^{\mathbf{V}}\rangle \neq 0$,

$$\frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}_I} t_\beta \langle\gamma^\vee, \mu_\beta^{\mathbf{V}}\rangle - \langle\gamma^\vee, \mu_{\alpha_i}^{\mathbf{V}}\rangle/x} \tag{9.9.5}$$
$$= \frac{xt_\gamma}{xt_\gamma - \sum_{\beta\in\mathbf{V}_I} xt_\beta \langle\gamma^\vee, \mu_\beta^{\mathbf{V}}\rangle - \langle\gamma^\vee, \mu_{\alpha_i}^{\mathbf{V}}\rangle}$$

and hence x in the expansion appears necessarily together with t_α.

From (9.9.2) we have

$$P(\mathbf{k}, 0, \lambda; I; \Delta) = \sum_{\nu\in X_i} \sum_{j=0}^{|\mathbf{k}|} \frac{b_{\mathbf{k}\nu j}}{(2\pi\sqrt{-1}n)^j} \exp(2\pi\sqrt{-1}n\nu). \tag{9.9.6}$$

Therefore from Theorem 9.11 we obtain
(9.9.7)

$$S(\mathbf{s}, 0; I; \Delta) = (-1)^{\#\Delta_+ - 1} \left(\prod_{\alpha \in \Delta_+ \setminus \{\alpha_i\}} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!} \right) \sum_{n=1}^{\infty} \frac{1}{n^{s_{\alpha_i}}} P(\mathbf{k}, 0, \lambda; I; \Delta)$$

$$= (-1)^{\#\Delta_+ - 1} \left(\prod_{\alpha \in \Delta_+ \setminus \{\alpha_i\}} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!} \right) \sum_{\nu \in X_i} \sum_{j=0}^{|\mathbf{k}|} \frac{b_{\mathbf{k}\nu j}}{(2\pi\sqrt{-1})^j} \phi(s_{\alpha_i} + j, \nu).$$

Combining this and (9.4.2) we obtain the assertion of the theorem. □



CHAPTER 10

Poincaré polynomials and values at integer points

Now we turn our attention to the left-hand side of (9.4.16). We will give a criterion of the non-vanishing of the left-hand side of (9.4.16), in terms of Poincaré polynomials.

10.1. Poincaré polynomials

We begin with some more preparation on Weyl groups. The Weyl group W is generated by simple reflections $\sigma_j = \sigma_{\alpha_j}$ $(1 \leq j \leq r)$, where $\{\alpha_1, \ldots, \alpha_r\} = \Psi$. Therefore each $w \in W$ can be written as a product of those simple reflections. The number of simple reflections, which are included in such a product, is not uniquely determined. But the smallest among those numbers can be determined, which we denote by $\ell(w)$ and call the *length* of w. The corresponding expression is called the *reduced expression*. It is known that this $\ell(w)$ coincides with $\ell(w) = \#\Delta_w$ defined in Section 9.2 (see [**71**, Section 1.7, Corollary]).

There is a unique element $w_0 \in W$ whose length $\ell(w_0) = \#\Delta_+$. This attains the largest possibility of the length, so we call w_0 the *longest element*.

The product $\sigma_1\sigma_2\cdots\sigma_r$ is called a *Coxeter element* of W. This notion depends on the choice of Ψ, and on the way that the elements of Ψ are numbered. However, it is known that all Coxeter elements are conjugate in W (see [**71**, Section 3.16, Proposition]). Therefore the order h of Coxeter elements is determined uniquely, which is called the *Coxeter number* of W. Then the eigenvalues of Coxeter elements can be written as ζ_h^e, where ζ_h is a primitive h-th root of unity and $0 \leq e < h$. Those e's which indeed appear as the exponents of eigenvalues are called the *exponents* of W. We write them as $e_1, e_2, \ldots, e_r$ $(e_1 \leq e_2 \leq \cdots \leq e_r)$.

Next, consider the ring of W-invariant polynomials in r-variables. Let $f_1, f_2, \ldots, f_r$ be a set of homogeneous, algebraically independent generators of this ring. Such sets are not unique, but their degrees (as polynomials) $d_1, d_2, \ldots, d_r$ $(d_1 \leq d_2 \leq \cdots \leq d_r)$ are uniquely determined (see [**71**, Section 3.7]). These numbers are called the *degrees* of

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W. It is known that $d_1 d_2 \cdots d_r = \#W$ (see [71, Section 3.9, Theorem]), and also, $d_j = e_j + 1$ $(1 \le j \le r$; see [71, Section 3.19, Theorem]).

Let $\mathbf{u} = (u_\alpha)_{\alpha \in \Delta_+}$ be a vector of indeterminates. For $X \subset \mathrm{Aut}(\Delta)$, we define a generalization of Poincaré polynomials due to Macdonald [125] as

$$X(\mathbf{u}) = \sum_{w \in X} \prod_{\alpha \in \Delta_{w^{-1}}} u_\alpha. \tag{10.1.1}$$

When $X = W^I$ and $\mathbf{u} = ((-1)^{-s_\alpha})_{\alpha \in \Delta_+}$, the corresponding Poincaré polynomial is

$$W^I(\mathbf{u}) = \sum_{w \in W^I} \prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-s_\alpha}. \tag{10.1.2}$$

If $\mathbf{s} \in \mathbb{C}$ and $\mathbf{y} \in V$ satisfy $w^{-1}\mathbf{s} = \mathbf{s}$ and $w^{-1}\mathbf{y} \equiv \mathbf{y} \pmod{Q^\vee}$ for all $w \in W^I$, then (9.4.2) reduces to

$$\begin{aligned} S(\mathbf{s}, \mathbf{y}; I; \Delta) &= \Big(\sum_{w \in W^I} \prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-s_\alpha} \Big) \zeta_r(\mathbf{s}, \mathbf{y}; \Delta) \\ &= W^I(\mathbf{u}) \zeta_r(\mathbf{s}, \mathbf{y}; \Delta). \end{aligned} \tag{10.1.3}$$

If $W^I(\mathbf{u})$ is not zero, then by (9.4.12), $\zeta_r(\mathbf{s}, \mathbf{y}; \Delta)$ is expressed by a series of lower rank. This is a non-trivial functional equation among zeta-functions of root systems. Therefore it is important to find some criterion when $W^I(\mathbf{u})$ is not zero, which will be discussed in the next section.

To obtain a simple description of $W^I(\mathbf{u})$, we begin with the relation $W_I W^I = W$ (see (9.2.13)). Therefore $w \in W$ is written as $w = xy$ with $x \in W_I$, $y \in W^I$, and this decomposition is unique [71, Section 1.10, Proposition (c)]. Although the following is well-known, we supply a proof.

LEMMA 10.1. *For $w = xy \in W$ with $x \in W_I$, $y \in W^I$,*

$$\Delta_{w^{-1}} = x \Delta_{y^{-1}} \coprod \Delta_{x^{-1}}. \tag{10.1.4}$$

PROOF. Let $x = \sigma_{\nu_1} \cdots \sigma_{\nu_j}$, $y = \sigma_{\nu_{j+1}} \cdots \sigma_{\nu_k}$ be reduced expressions by simple reflections. Then $w = \sigma_{\nu_1} \cdots \sigma_{\nu_k}$ is also a reduced expression because of the length identity $l(w) = l(x) + l(y)$ ([71, Section 1.10, Proposition (c)]). Let $\beta_{im} = \sigma_{\nu_m} \sigma_{\nu_{m-1}} \cdots \sigma_{\nu_{i+1}}(\alpha_{\nu_i})$ $(1 \le i \le m-1)$ and $\beta_{mm} = \alpha_{\nu_m}$. Then $\Delta_w = \{\beta_{1k}, \ldots, \beta_{kk}\}$ ([71, Section 1.7, p. 14]). Similarly, we have $\Delta_x = \{\beta_{1j}, \ldots, \beta_{jj}\}$ and $\Delta_y = \{\beta_{j+1,k}, \ldots, \beta_{kk}\}$. Noting $y^{-1}\beta_{ij} = \sigma_{\nu_k} \cdots \sigma_{\nu_{j+1}} \beta_{ij} = \beta_{ik}$, we have $\Delta_w = y^{-1}\Delta_x \coprod \Delta_y$. Applying this argument to $w^{-1} = y^{-1}x^{-1}$, we obtain (10.1.4). □

LEMMA 10.2 (cf. [71, Section 1.11, p. 21]). *Assume* $u_\alpha = u_\beta$ *if* $\|\alpha\| = \|\beta\|$. *Then*

$$W_I(\mathbf{u})W^I(\mathbf{u}) = W(\mathbf{u}). \tag{10.1.5}$$

PROOF. The assumption for u_α implies $u_\alpha = u_{x^{-1}\alpha}$. Therefore, using (9.2.13) and (10.1.4) we have

(10.1.6)

$$\begin{aligned} W(\mathbf{u}) &= \sum_{w\in W}\prod_{\alpha\in\Delta_{w^{-1}}} u_\alpha = \sum_{x\in W_I}\sum_{y\in W^I}\Big(\prod_{\alpha\in x\Delta_{y^{-1}}} u_\alpha\Big)\Big(\prod_{\alpha\in\Delta_{x^{-1}}} u_\alpha\Big) \\ &= \sum_{x\in W_I}\sum_{y\in W^I}\Big(\prod_{\alpha\in\Delta_{y^{-1}}} u_\alpha\Big)\Big(\prod_{\alpha\in\Delta_{x^{-1}}} u_\alpha\Big) = W_I(\mathbf{u})W^I(\mathbf{u}). \end{aligned}$$

□

Assume $u_\alpha = u$ for all $\alpha \in \Delta_+$, that is, $\mathbf{u} = (u, \dots, u)$, which we write (u) for brevity. By Lemma 10.2, we have

$$W^I((u)) = \frac{W((u))}{W_I((u))}. \tag{10.1.7}$$

In this case the numerator and the denominator of the right-hand side are classical Poincaré polynomials, which have the following product expression (due to Chevalley):

$$W((u)) = \prod_{i=1}^{r}\frac{u^{d_i}-1}{u-1} = \prod_{i=1}^{r}(1+\cdots+u^{e_i}), \tag{10.1.8}$$

$$W_I((u)) = \prod_{i\in I}\frac{u^{d'_i}-1}{u-1} = \prod_{i\in I}(1+\cdots+u^{e'_i}), \tag{10.1.9}$$

where d_i and d'_i (resp. e_i and e'_i) are degrees (resp. exponents) of the Weyl groups W and W_I. In fact, the first equalities are [71, Section 3.15, Theorem], and then the second equalities are immediate in view of [71, Section 3.19, Theorem]. The following is the table of degrees ([71, Section 3.7]). Here we note that $W(B_r) \simeq W(C_r)$ (see (9.2.17) and (9.2.18)) and hence their degrees agree.

Type	$\{d_1, \dots, d_r\} = \{e_1 + 1, \dots, e_r + 1\}$
A_r	$2, 3, \dots, r+1$
B_r	$2, 4, \dots, 2r$
C_r	$2, 4, \dots, 2r$
D_r	$2, 4, \dots, 2r-2, r$
E_6	$2, 5, 6, 8, 9, 12$
E_7	$2, 6, 8, 10, 12, 14, 18$
E_8	$2, 8, 12, 14, 18, 20, 24, 30$
F_4	$2, 6, 8, 12$
G_2	$2, 6$

Assume that Δ is a root system, which is not simply-laced, and $\Delta = \Delta_1 \cup \Delta_2$, where each Δ_i consists of all roots of the same length. Then each Δ_i is a root system. Let $\Delta_{1+} = \Delta_+ \cap \Delta_1$ and $\Delta_{1-} = \Delta_- \cap \Delta_1$. Let J be the set of indices determined by $\Psi_2 = \Psi \cap \Delta_2 = \{\alpha_i\}_{i \in J}$. Note that $W_J \subsetneq W(\Delta_2)$ in general.

PROPOSITION 10.3 ([119, Proposition 5.2]). *We have* $W(\Delta)((u), (1)) = (\#W_J)W(\Delta_1)((u))$, *where the left-hand side means that* $u_\alpha = u$ *for* $\alpha \in \Delta_1$ *and* $u_\alpha = 1$ *for* $\alpha \in \Delta_2$.

PROOF. We show $W(\Delta_1) \cap W_J = \{\mathrm{id}\}$. Let $w \in W(\Delta_1) \cap W_J$. Then $w = \sigma_{i_1} \cdots \sigma_{i_l}$ with $i_k \in J$. Since $\sigma_{i_k}\Delta_+ = (\Delta_+ \setminus \{\alpha_{i_k}\}) \cup \{-\alpha_{i_k}\}$, we have $\sigma_{i_k}\Delta_- = (\Delta_- \setminus \{-\alpha_{i_k}\}) \cup \{\alpha_{i_k}\}$ and hence $\sigma_{i_k}\Delta_{1-} = \Delta_{1-}$ because $\pm\alpha_{i_k} \notin \Delta_1$. This implies that $\Delta_{1+} \cap w^{-1}\Delta_{1-} = \emptyset$. Therefore the length of w in $W(\Delta_1)$ is 0 and hence $w = \mathrm{id}$.

We define $f : W(\Delta_1) \to W^J$ by $f(w) = w^J$ using the decomposition $w = w_J w^J$ with $w_J \in W_J$ for $w \in W(\Delta_1)$. We show that f is bijective. For $w, \tilde{w} \in W(\Delta_1)$, assume $f(w) = f(\tilde{w})$. Then $w_J^{-1}w = \tilde{w}_J^{-1}\tilde{w}$ and $\tilde{w}w^{-1} = \tilde{w}_J w_J^{-1} \in W(\Delta_1) \cap W_J = \{\mathrm{id}\}$ by the previous paragraph, which implies $w = \tilde{w}$ and the injectivity of f. Fix $y \in W^J$. Let $y = \sigma_{i_1} \cdots \sigma_{i_m}$ be a reduced expression. We decompose $(i_1, \dots, i_m)$ into two subsequences as $(j_1, \dots, j_q)$ and $(k_1, \dots, k_{m-q})$ such that $j_l \in J$ and $k_l \notin J$. Let $x = \sigma_{j_q} \cdots \sigma_{j_1} \in W_J$ and consider $w = xy$. Then by use of $\sigma_\alpha\sigma_\beta = \sigma_{\sigma_\alpha\beta}\sigma_\alpha$, we carry each σ_{j_l} in x to the right until it cancels the same element at the original position in y. Hence we see that w can be written as $\sigma_{\beta_1} \cdots \sigma_{\beta_{m-q}}$ with $\beta_p = \sigma_{j_q} \cdots \sigma_{j_h}\alpha_{k_p} \in \Delta_1$ for some h. Thus we have $w = xy \in W(\Delta_1)$. By the uniqueness of the decomposition, we see that f is surjective and hence bijective.

In the first paragraph of the proof we have seen that $w_J\Delta_{1+} = \Delta_{1+}$ for $w_J \in W_J$. Noting this fact and the bijectivity of f, we obtain

$$
\begin{aligned}
W(\Delta_1)((u)) &= \sum_{w\in W(\Delta_1)} \prod_{\alpha\in(\Delta_1)_{w^{-1}}} u \\
&= \sum_{w\in W(\Delta_1)} \prod_{\alpha\in\Delta_{1+}\cap w_J w^J \Delta_{1-}} u \\
&= \sum_{w\in W(\Delta_1)} \prod_{\alpha\in w_J^{-1}\Delta_{1+}\cap w^J\Delta_{1-}} u \\
&= \sum_{y\in W^J} \prod_{\alpha\in\Delta_{1+}\cap y\Delta_{1-}} u.
\end{aligned} \tag{10.1.10}
$$

Furthermore, we have

$$
\begin{aligned}
W(\Delta)((u),(1)) &= \sum_{w\in W(\Delta)} \Big(\prod_{\alpha\in\Delta_{w^{-1}}\cap\Delta_1} u\Big)\Big(\prod_{\alpha\in\Delta_{w^{-1}}\cap\Delta_2} 1\Big) \\
&= \sum_{w\in W(\Delta)} \prod_{\alpha\in\Delta_{w^{-1}}\cap\Delta_1} u \\
&= \sum_{x\in W_J}\sum_{y\in W^J} \prod_{\alpha\in\Delta_{1+}\cap xy\Delta_{1-}} u \\
&= \sum_{x\in W_J}\sum_{y\in W^J} \prod_{\alpha\in x^{-1}\Delta_{1+}\cap y\Delta_{1-}} u \\
&= \sum_{x\in W_J}\sum_{y\in W^J} \prod_{\alpha\in\Delta_{1+}\cap y\Delta_{1-}} u.
\end{aligned} \tag{10.1.11}
$$

Combining (10.1.10) and (10.1.11), we obtain the result. □

We denote $W(X_r) = W(\Delta)$ for a root system of type X_r.

Corollary 10.4 ([**119**, Corollary 5.3]).

$$
\begin{aligned}
W(B_r)((u),(1)) &= W(C_r)((1),(u)) = (\#W(A_1))W(D_r)((u)) = 2W(D_r)((u)), \\
W(C_r)((u),(1)) &= W(B_r)((1),(u)) = (\#W(A_{r-1}))W(A_1^r)((u)) = r!(W(A_1)((u)))^r, \\
W(F_4)((u),(1)) &= W(F_4)((1),(u)) = (\#W(A_2))W(D_4)((u)) = 6W(D_4)((u)), \\
W(G_2)((u),(1)) &= W(G_2)((1),(u)) = (\#W(A_1))W(A_2)((u)) = 2W(A_2)((u)).
\end{aligned}
$$

Proof. These are simple consequences of Proposition 10.3. In this proof, for each root system of type $X_r = B_r, C_r, F_4, G_2$, let $\Delta_1 = \Delta_l(X_r)$ be the set of all long roots, $\Delta_2 = \Delta_s(X_r)$ that of all short roots. Then we just notice that:

(i) $\Delta_l(B_r) \simeq \Delta_s(C_r)$ and $\Delta_s(B_r) \simeq \Delta_l(C_r)$,

(ii) $\Delta_l(B_r) \simeq \Delta(D_r)$, $\Delta_l(C_r) \simeq \Delta(A_1^r)$, $\Delta_l(F_4) \simeq \Delta(D_4)$, $\Delta_l(G_2) \simeq \Delta(A_2)$, and $\Psi_2(B_r) \simeq \Psi(A_1)$, $\Psi_2(C_r) \simeq \Psi(A_{r-1})$, $\Psi_2(F_4) \simeq \Psi(A_2)$,

$\Psi_2(G_2) \simeq \Psi(A_1)$ (which can be seen from the list of roots of each system; see [**20**, Planche][**178**, Section 2.14]),

(iii) $W(A_r) \simeq \mathfrak{S}_{r+1}$ (see (9.2.16)), and hence $\#W(A_r) = (r+1)!$. □

10.2. The non-vanishing of the sum of coefficients

Now we carry out the case study when (10.1.2) vanishes, or does not vanish. In this section, assume that $s_\alpha = s_{w\alpha}$ for all $w \in W^I$, $\alpha \in \Delta$ and that $s_\alpha = k_\alpha \in \mathbb{N}$ for all $\alpha \in \Delta \setminus \Delta_I$. Let $u_\alpha = (-1)^{-k_\alpha}$ for $\alpha \in \Delta \setminus \Delta_I$.

LEMMA 10.5 ([**119**, Lemma 6.1]). *Let $y \in W^I$. Then $\Delta_{y^{-1}} \cap \Delta_{I+} = \emptyset$.*

PROOF. Assume $\alpha \in \Delta_{y^{-1}} \cap \Delta_{I+}$. Then $s_\alpha \in W_I$. Let $w = s_\alpha y$; this expression gives the unique decomposition of w. Then $\Delta_{w^{-1}} = s_\alpha \Delta_{y^{-1}} \coprod \Delta_{s_\alpha}$ by (10.1.4). We have $\Delta_{w^{-1}} \subset \Delta_+$ by definition, while $-\alpha = s_\alpha \alpha \in s_\alpha \Delta_{y^{-1}}$, which leads to a contradiction. □

From the above lemma, we see that the product on the right-hand side of

$$W^I(\mathbf{u}) = \sum_{w \in W^I} \prod_{\alpha \in \Delta_{w^{-1}}} u_\alpha \tag{10.2.1}$$

consists of only $\alpha \in \Delta_+ \setminus \Delta_{I+}$.

Now consider several cases. Assume $I \subsetneq \{1, \dots, r\}$.

Case 1:

This is the trivial case. If all k_α are even, then $u_\alpha = 1$ for all $\alpha \in \Delta$, and we have $W^I((1)) = \#W^I > 0$.

Case 2:

If all k_α are odd, then we have to evaluate (10.1.7) at $\mathbf{u} = (-1)$. Let K and K_I be the sets of indices of even degrees (i.e., odd exponents) of the Weyl groups W and W_I respectively, given as

$$K = \{i \mid 1 \le i \le r, d_i \in 2\mathbb{Z}\} = \{i \mid 1 \le i \le r, e_i \in 2\mathbb{Z}+1\}, \tag{10.2.2}$$

$$K_I = \{i \mid i \in I, d'_i \in 2\mathbb{Z}\} = \{i \mid i \in I, e'_i \in 2\mathbb{Z}+1\}. \tag{10.2.3}$$

From (10.1.8) and (10.1.9) we see that $W((-1)) = 0$ if $K \neq \emptyset$, and $W_I((-1)) = 0$ if $K_I \neq \emptyset$. Comparing the orders of zeros of both sides of (10.1.5), we find that

$$\#K \ge \#K_I, \tag{10.2.4}$$

hence $W^I((-1))$ does not vanish if and only if

$$\#K = \#K_I. \tag{10.2.5}$$

If it holds, then by (10.1.8), (10.1.9) and l'Hôpital's rule

$$\begin{aligned} W^I((-1)) &= \lim_{u\to -1} W^I((u)) \\ &= \lim_{u\to -1} \frac{\prod_{i\in K^c}(1+\cdots+u^{e_i})}{\prod_{i\in K_I^c}(1+\cdots+u^{e'_i})} \frac{\prod_{i\in K}(u^{d_i}-1)}{\prod_{i\in K_I}(u^{d'_i}-1)} \\ &= \frac{\prod_{i\in K^c} 1}{\prod_{i\in K_I^c} 1} \frac{\prod_{i\in K} d_i}{\prod_{i\in K_I} d'_i} \\ &= \frac{\prod_{i\in K} d_i}{\prod_{i\in K_I} d'_i} \in \mathbb{N}. \end{aligned} \tag{10.2.6}$$

The above argument especially implies the following:

Claim 10.6. *If all degrees of a root system X_r are even, then $W^I((-1))$ vanishes for any choice of $I \subsetneq \{1,\dots,r\}$.*

This is obvious from (10.2.5), because for any $I \subsetneq \{1,\dots,r\}$, we have $\#K > \#K_I$. From this claim and the table in Section 10.1 we find that $W^I((-1))$ always vanishes for $B_r, C_r, D_{2k}, E_7, E_8, F_4$, and G_2. Therefore the only root systems for which (10.2.6) can be applied are A_r, D_{2k+1}, and E_6.

Remark 10.7. The following statements for a root system X_r are equivalent [**71**, Section 3.7, Exercise]: (1) All degrees are even; (2) the longest element of the Weyl group is -1.

Theorem 10.8 ([**119**, Theorem 6.4]). *$W^I((-1))$ does not vanish only in one of the following cases.*

Type of Δ	*Type of* Δ_I	$W^I((-1))$
A_r	$A_{r_1}\times\cdots\times A_{r_n}$	$\dfrac{[(r+1)/2]!}{[(r_1+1)/2]!\cdots[(r_n+1)/2]!}$
D_{2k+1}	D_{2k}	$\dfrac{2\cdot 4\cdots(4k-2)\cdot 4k}{2\cdot 4\cdots(4k-2)\cdot 2k} = 2$
E_6	D_5	$\dfrac{2\cdot 6\cdot 8\cdot 12}{2\cdot 4\cdot 6\cdot 8} = 3$
E_6	D_4	$\dfrac{2\cdot 6\cdot 8\cdot 12}{2\cdot 4\cdot 6\cdot 4} = 6$

where $r_1,\dots,r_n$ satisfy $[(r_1+1)/2]+\cdots+[(r_n+1)/2] = [(r+1)/2]$.

PROOF. Assume that Δ is of type E_6. Then Δ_I must be a subroot system such that there are 4 even degrees in it, which implies that Δ_I is either of type D_4 or D_5.

Assume that Δ is of type D_{2k+1}. Then Δ_I must be a subroot system such that there are $2k$ even degrees in it, which implies that Δ_I is of type D_{2k}.

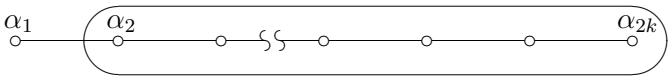
Assume that Δ is of type A_r. By noting that there are $[(r+1)/2]$ even degrees and they are $2, 4, \ldots, 2[(r+1)/2]$, and that their product is

$$2 \cdot 4 \cdot \cdots \cdot (2[(r+1)/2]) = 2^{[(r+1)/2]}[(r+1)/2]!, \tag{10.2.7}$$

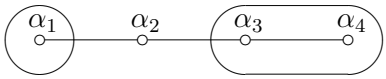
we obtain the result. □

For example, see the following diagrams for the pairs (A_{2k}, A_{2k-1}) and $(A_4, A_1 \times A_2)$, where the set of enclosed nodes is Ψ_I.

(10.2.8)



(10.2.9)



Here are some examples of the case when Δ is of type A.

Type of Δ	Type of Δ_I	$W((-1))$
A_{2k}	A_{2k-1}	$2 \cdot 4 \cdots 2k/2 \cdot 4 \cdots 2k = 1$
A_3	A_1^2	$2 \cdot 4/2 \cdot 2 = 2$
A_4	A_1^2	$2 \cdot 4/2 \cdot 2 = 2$
A_4	$A_1 \times A_2$	$2 \cdot 4/2 \cdot 2 = 2$
A_5	A_1^3	$2 \cdot 4 \cdot 6/2 \cdot 2 \cdot 2 = 6$

Case 3:

Assume k_α are odd for $\alpha \in \Delta_1$ and k_β are even for $\beta \in \Delta_2$. Let $\mathbf{u} = ((u), (1))$. Then we have

$$W^I(\mathbf{u}) = \frac{W((u),(1))}{W_I((u),(1))} = \frac{(\#W_J)W(\Delta_1)((u))}{(\#W_{I\cap J})W(\Delta_1 \cap \Delta_I)((u))} \tag{10.2.10}$$

by Proposition 10.3. By using an argument similar to Case 2, we can calculate $W^I((-1),(1))$ as follows. Let K_1 and K_{1I} be the sets of indices of even degrees of the Weyl groups $W(\Delta_1)$ and $W(\Delta_1 \cap \Delta_I)$ respectively, given as

$$K_1 = \{i \mid \alpha_i \in \Delta_1, d_i \in 2\mathbb{Z}\}, \tag{10.2.11}$$

$$K_{1I} = \{i \mid \alpha_i \in \Delta_1 \cap \Delta_I, d_i' \in 2\mathbb{Z}\}. \tag{10.2.12}$$

Then we see that $W^I((-1),(1))$ does not vanish if and only if

$$\#K_1 = \#K_{1I}. \tag{10.2.13}$$

If it holds, then

(10.2.14)

$$\begin{aligned} W^I((-1),(1)) &= \lim_{u\to -1} W^I((u),(1)) \\ &= \frac{\#W_J}{\#W_{I\cap J}} \lim_{u\to -1} \frac{\prod_{i\in K_1^c}(1+\cdots+u^{e_i})}{\prod_{i\in K_{1I}^c}(1+\cdots+u^{e_i'})} \frac{\prod_{i\in K_1}(u^{d_i}-1)}{\prod_{i\in K_{1I}}(u^{d_i'}-1)} \\ &= \frac{\#W_J}{\#W_{I\cap J}} \frac{\prod_{i\in K_1^c} 1}{\prod_{i\in K_{1I}^c} 1} \frac{\prod_{i\in K_1} d_i}{\prod_{i\in K_{1I}} d_i'} \\ &= \frac{\#W_J}{\#W_{I\cap J}} \frac{\prod_{i\in K_1} d_i}{\prod_{i\in K_{1I}} d_i'} \in \mathbb{N}, \end{aligned}$$

where the complements of sets of indices mean those in $\{i \mid \alpha_i \in \Delta_1\}$.

THEOREM 10.9 ([**119**, Theorem 6.5]). *$W^I((-1),(1))$ does not vanish only in one of the following cases.*

Type of Δ	*Type of* Δ_I	Δ_1	$W^I((-1),(1))$
B_{2k+1}	B_{2k}	Δ_l	$\frac{2}{2}\frac{2\cdot 4\cdots(4k-2)\cdot 4k}{2\cdot 4\cdots(4k-2)\cdot 2k} = 2$
C_{2k+1}	C_{2k}	Δ_s	$\frac{2}{2}\frac{2\cdot 4\cdots(4k-2)\cdot 4k}{2\cdot 4\cdots(4k-2)\cdot 2k} = 2$
G_2	A_1 *(long)*	Δ_l	$\frac{2}{1}\frac{2}{2} = 2$
G_2	A_1 *(short)*	Δ_s	$\frac{2}{1}\frac{2}{2} = 2$

PROOF. The root systems which have roots of two (long and short) lengths are of types B_r, C_r, F_4, G_2.

When Δ is of type F_4, by Corollary 10.4, Δ_1 is of type D_4. Since the degrees of $W(D_4)$ are all even, we see that $W^I((-1),(1))$ vanishes for any choice of I in $\Delta(F_4)$.

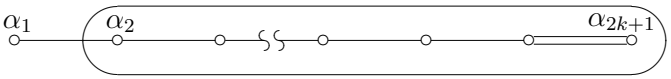
Assume that Δ is of type B_r. Then by Corollary 10.4, Δ_1 must be Δ_l and r must be odd because otherwise all the degrees are even. When $\Delta_1 = \Delta_l$, $(r-1)$ degrees are even. Hence Δ_I must be of type B_{r-1} whose degrees are all even, which is the unique choice such that $\#K_1 = \#K_{1I}$.

In the case of C_r, a similar argument works and we obtain the result.

Assume that Δ is of type G_2. Then Δ_1 is either Δ_l or Δ_s. However, Δ_I must be determined uniquely so that $\#K_1 = \#K_{1I}$, that is, $\Delta_I \subset \Delta_1$. □

For example, see the following diagram for the pair (B_{2k+1}, B_{2k}), where the set of enclosed nodes is Ψ_I.

(10.2.15)



REMARK 10.10. The special case $s_3 = 2$ of (9.8.20) is an example of the assertion in Case 3 of Section 10.2 corresponding to $\Delta = \Delta(C_3)$, $\Delta_I = \Delta(C_2)$ with $I = \{1\}$ and $W^I((-1), (1)) = 2$, because the left-hand side does not vanish and its coefficient is 2.

Check for updates

CHAPTER 11

The case of the exceptional algebra G_2

In the previous chapter, we studied special values at odd integers via Poincaré polynomials. However, we still have chances to calculate special values by other methods. In this chapter, we give a criterion to obtain a certain reduction method of evaluating special values and apply it to the simplest exceptional case, which is the G_2 case [**117**]. The values at even integer points of $\zeta_2(\mathbf{s}; G_2)$ were discussed in Example 4.17, and a certain functional relation was given in Example 8.34, but in this chapter we will develop a more general study on $\zeta_2(\mathbf{s}; G_2)$.

11.1. A criterion of reduction

The following theorem implies that under certain conditions, the number of terms on the right-hand side of (9.4.2) can be reduced.

THEOREM 11.1 ([**117**, Theorem 2.1]). *Assume that there exist* $w_1 \in \mathrm{Aut}(\Delta)$, $\mathbf{s} \in \mathcal{S}$ *and* $\mathbf{y} \in V$ *which satisfy the conditions that* $s_\alpha \in \mathbb{Z}$ *for* $\alpha \in \Delta_{w_1^{-1}}$,

$$\Big(\prod_{\alpha \in \Delta_{w_1^{-1}}} (-1)^{s_\alpha}\Big) = -1 \tag{11.1.1}$$

and

$$w_1^{-1}\mathbf{s} = \mathbf{s}, \qquad w_1^{-1}\mathbf{y} = \mathbf{y}. \tag{11.1.2}$$

Then we have

(11.1.3)

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) = \frac{1}{2}\Big(\sum_{v \in W^I \backslash w_1 W^I} \Big(\prod_{\alpha \in \Delta_{v^{-1}}} (-1)^{-s_\alpha}\Big) \zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y}; \Delta) + \sum_{v \in W^I \backslash w_1^{-1} W^I} \Big(\prod_{\alpha \in \Delta_{v^{-1}}} (-1)^{-s_\alpha}\Big) \zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y}; \Delta)\Big).$$

Furthermore, if $w_1^{-1} W^I = w_1 W^I$, *then*

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) = \sum_{v \in W^I \backslash w_1 W^I} \Big(\prod_{\alpha \in \Delta_{v^{-1}}} (-1)^{-s_\alpha}\Big) \zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y}; \Delta). \tag{11.1.4}$$

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PROOF. We first prove that

(11.1.5)

$$\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)S(w^{-1}\mathbf{s},w^{-1}\mathbf{y};I;\Delta)$$
$$=\sum_{v\in wW^I\cap W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}\Big)\zeta_r(v^{-1}\mathbf{s},v^{-1}\mathbf{y};\Delta)$$
$$+\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\sum_{v\in wW^I\setminus W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}w}}(-1)^{-s_{w\alpha}}\Big)\zeta_r(v^{-1}\mathbf{s},v^{-1}\mathbf{y};\Delta)$$

for any $w\in\mathrm{Aut}(\Delta)$. Since $\iota^{*-1}(P_{I+})=\bigcup_{v\in W^I}vP_+$ by Lemma 9.2, we can write the left-hand side of (11.1.5) as

$$\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\sum_{u\in W^I}\sum_{\lambda\in uP_{++}}e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_{w\alpha}}}. \tag{11.1.6}$$

The inner sum of (11.1.6) is equal to

$$\sum_{\lambda\in uP_{++}}e^{2\pi\sqrt{-1}\langle\mathbf{y},w\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle(w\alpha)^\vee,w\lambda\rangle^{s_{w\alpha}}}=\sum_{\lambda\in wuP_{++}}e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}\prod_{\alpha\in w\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}},$$

and concerning the last product, since $w\Delta_+\cap\Delta_-=w\Delta_w=-\Delta_{w^{-1}}$, we have

$$\prod_{\alpha\in w\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}=\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\Big)\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}. \tag{11.1.7}$$

Using this expression when $wu\in W^I$, we see that (11.1.6) is equal to

(11.1.8)

$$\sum_{\substack{u\in W^I\\ wu\in W^I}}\sum_{\lambda\in wuP_{++}}e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}$$
$$+\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\sum_{\substack{u\in W^I\\ wu\notin W^I}}\sum_{\lambda\in uP_{++}}e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_{w\alpha}}}$$
$$=\Sigma_1+\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\Sigma_2,$$

say. Putting $wu=v$, we have

$$\Sigma_1=\sum_{v\in wW^I\cap W^I}\sum_{\lambda\in vP_{++}}e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}$$
$$=\sum_{v\in wW^I\cap W^I}\sum_{\lambda\in P_{++}}e^{2\pi\sqrt{-1}\langle\mathbf{y},v\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,v\lambda\rangle^{s_\alpha}}$$
$$=\sum_{v\in wW^I\cap W^I}\sum_{\lambda\in P_{++}}e^{2\pi\sqrt{-1}\langle v^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in v^{-1}\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_{v\alpha}}},$$

and, as in (11.1.7), the signature appears from the last product when $\alpha \in v^{-1}\Delta_+ \cap \Delta_- = -\Delta_v$, which is

$$\prod_{\alpha\in\Delta_v}(-1)^{-s_{v\alpha}} = \prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}$$

because $v\Delta_v = -\Delta_{v^{-1}}$. Therefore we obtain

(11.1.9)

$$\begin{aligned}\Sigma_1 &= \sum_{v\in wW^I\cap W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}\Big)\sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle v^{-1}\mathbf{y},\lambda\rangle}\prod_{\alpha\in\Delta_+}\frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_{v\alpha}}}\\ &= \sum_{v\in wW^I\cap W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}\Big)\zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y};\Delta).\end{aligned}$$

Similarly, we can show

$$\Sigma_2 = \sum_{v\in wW^I\setminus W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}w}}(-1)^{-s_{w\alpha}}\Big)\zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y};\Delta). \tag{11.1.10}$$

Substituting (11.1.9) and (11.1.10) into (11.1.8), we obtain (11.1.5).

Now consider the equation

(11.1.11)

$$2S(\mathbf{s},\mathbf{y};I;\Delta) = S(\mathbf{s},\mathbf{y};I;\Delta) - \Big(\prod_{\alpha\in\Delta_{w_1^{-1}}}(-1)^{s_\alpha}\Big)S(w_1^{-1}\mathbf{s}, w_1^{-1}\mathbf{y};I;\Delta),$$

which trivially follows from the assumptions (11.1.1) and (11.1.2). Substitute the expansions (9.4.2) and (11.1.5) (with $w = w_1$) to the right-hand side of (11.1.11). The first sum on the right-hand side of (11.1.5) is cancelled with the part $v \in w_1W^I \cap W^I$ of (9.4.12), and hence

(11.1.12)

$$\begin{aligned}2S(\mathbf{s},\mathbf{y};I;\Delta) &= \sum_{v\in W^I\setminus wW^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}\Big)\zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y};\Delta)\\ &\quad - \Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\sum_{v\in wW^I\setminus W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}w}}(-1)^{-s_{w\alpha}}\Big)\zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y};\Delta).\end{aligned}$$

We see that the second term on the right-hand side of (11.1.12) is, after renaming $w^{-1}v$ by v and using (11.1.1) and (11.1.2), equal to

(11.1.13)

$$\begin{aligned}&\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{s_\alpha}\Big)\sum_{v\in W^I\setminus w^{-1}W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_{w\alpha}}\Big)\zeta_r(v^{-1}w^{-1}\mathbf{s}, v^{-1}w^{-1}\mathbf{y};\Delta)\\ &\quad = -\sum_{v\in W^I\setminus w^{-1}W^I}\Big(\prod_{\alpha\in\Delta_{v^{-1}}}(-1)^{-s_\alpha}\Big)\zeta_r(v^{-1}\mathbf{s}, v^{-1}\mathbf{y};\Delta).\end{aligned}$$

The desired results follow from (11.1.12) and (11.1.13). □

REMARK 11.2. Since the right-hand sides of (11.1.3) and (11.1.4) include signature factors, sometimes the right-hand side might be zero. If so, then Theorem 11.1 gives no useful information. In the next section we will give examples where the right-hand side does not vanish. This is the key to why we can sometimes treat the situation when some of the variables are odd integers.

11.2. Application of Theorem 11.1 to the case G_2

Hereafter in the present chapter we concentrate on the study of the zeta-function of the root system G_2 (see (3.3.28)).

Here we show an application of Theorem 11.1 to the G_2 case. Assume s_i are all positive integers (≥ 2). Let $I = \{2\}$, $\mathbf{y} = \mathbf{0}$ and $w_1 = w_0\sigma_1$ with the longest element

$$w_0 = \sigma_1\sigma_2\sigma_1\sigma_2\sigma_1\sigma_2 = \sigma_2\sigma_1\sigma_2\sigma_1\sigma_2\sigma_1 = -1.$$

Then we have $w_1 = w_1^{-1} = \sigma_2\sigma_1\sigma_2\sigma_1\sigma_2 = -\sigma_1$, so $w_1\alpha_1 = \alpha_1$, $w_1\alpha_2 = -\alpha_3$, $w_1\alpha_3 = -\alpha_2$, $w_1\alpha_4 = -\alpha_4$, $w_1\alpha_5 = -\alpha_6$, $w_1\alpha_6 = -\alpha_5$, and hence $w_1^{-1}\mathbf{s} = (s_1, s_3, s_2, s_4, s_6, s_5)$. Therefore, when $s_2 = s_3$ and $s_5 = s_6$, we have $w_1^{-1}\mathbf{s} = \mathbf{s}$. Since

$$\Delta_{w_1^{-1}} = \Delta_+ \cap w_1\Delta_- = \Delta_+ \cap \sigma_1\Delta_+ = \Delta_+ \setminus \{\alpha_1\},$$

we have

$$\Big(\prod_{\alpha\in\Delta_{w_1^{-1}}} (-1)^{s_\alpha}\Big) = (-1)^{s_2+s_3+s_4+s_5+s_6}.$$

Therefore we can apply Theorem 11.1 when $s_2 = s_3$, $s_5 = s_6$, and $s_2 + s_3 + s_4 + s_5 + s_6$ is odd. It is easy to see that

$$W_I = \{1, \sigma_2\},$$
$$W^I = \{1, \sigma_1, \sigma_1\sigma_2, \sigma_1\sigma_2\sigma_1, \sigma_1\sigma_2\sigma_1\sigma_2, \sigma_1\sigma_2\sigma_1\sigma_2\sigma_1\},$$
$$w_1W^I = \{\sigma_2\sigma_1\sigma_2\sigma_1\sigma_2, \sigma_1\sigma_2\sigma_1\sigma_2\sigma_1\sigma_2, \sigma_1\sigma_2\sigma_1\sigma_2\sigma_1, \sigma_1\sigma_2\sigma_1\sigma_2, \sigma_1\sigma_2\sigma_1, \sigma_1\sigma_2\},$$

and hence $W^I \setminus w_1W^I = \{1, \sigma_1\}$. Thus by (11.1.4), we have

$$S(\mathbf{s}, \mathbf{0}; \{2\}; G_2) = \zeta_2(\mathbf{s}, \mathbf{0}; G_2) + (-1)^{s_1}\zeta_2(\sigma_1^{-1}\mathbf{s}, \mathbf{0}; G_2).$$

Since $s_2 = s_3$ and $s_5 = s_6$, we have $\sigma_1^{-1}\mathbf{s} = \mathbf{s}$. Therefore if s_1 is even then the right-hand side of the above is $2\zeta_2(\mathbf{s}, \mathbf{0}; G_2)$. The conclusion is as follows.

PROPOSITION 11.3 ([117, Proposition 3.1]). *Let $p, q, r, u \in \mathbb{N}_{\geq 2}$ with even p and odd r. Then*

$$S((p,q,q,r,u,u),\mathbf{0};\{2\};G_2) = 2\zeta_2(p,q,q,r,u,u;G_2). \tag{11.2.1}$$

Similarly, we can treat the case $I = \{1\}$, $\mathbf{y} = \mathbf{0}$. Then $W_I = \{1, \sigma_1\}$ and

$$W^I = \{1, \sigma_2, \sigma_2\sigma_1, \sigma_2\sigma_1\sigma_2, \sigma_2\sigma_1\sigma_2\sigma_1, \sigma_2\sigma_1\sigma_2\sigma_1\sigma_2\}. \tag{11.2.2}$$

In this case we choose $w_1 = w_0\sigma_2$. Then $W^I \setminus w_1 W^I = \{1, \sigma_2\}$. We can apply Theorem 11.1 when $s_1 = s_5$, $s_3 = s_4$, and $s_1 + s_3 + s_4 + s_5 + s_6$ is odd. We obtain:

PROPOSITION 11.4 ([117, Proposition 3.2]). *Let $p, q, r, u \in \mathbb{N}_{\geq 2}$ with even q and odd u. Then*

$$S((p,q,r,r,p,u),\mathbf{0};\{1\};G_2) = 2\zeta_2(p,q,r,r,p,u;G_2). \tag{11.2.3}$$

These propositions give non-trivial value relations involving odd integer values.

11.3. Some lemmas

The results in the previous sections are valid only in the case $\mathbf{s} \in \mathcal{S}$. However, it is possible to obtain relations which include the case when some of the variables take the value 1.

Already in Section 8.4 we proved a functional relation of such type (see (8.4.1)) which corresponds, in some sense, to the case $I = \{1\}$ in the preceding section. In the next section, we will deduce a functional relation corresponding to the case $I = \{2\}$. In this section, we prepare several lemmas which are necessary in the next section.

First, we prepare the following lemma, which is of the same flavor as Lemma 8.24, and is a slight generalization of [110, Lemma 4.4].

LEMMA 11.5 ([117, Lemma 5.3]). *Let $h \in \mathbb{N}$,*

$$\mathcal{A} = \{\alpha(l) \in \mathbb{C} \,|\, l \in \mathbb{Z},\ l \neq 0\},$$
$$\mathcal{B} = \{\beta(N;m;\eta) \in \mathbb{R} \,|\, N, m, \eta \in \mathbb{Z},\ N \neq 0,\ m \geq 0,\ 1 \leq \eta \leq h\},$$
$$\mathcal{C} = \{c_\eta \in \mathbb{N} \,|\, 1 \leq \eta \leq h\}$$

be sets of numbers indexed by integers, and

(11.3.1)

$$R^{\pm}(\theta) = \sum_{\substack{m \in \mathbb{Z} \\ m \neq 0}} (\pm\sqrt{-1})^m \alpha(m) e^{m\sqrt{-1}\theta/2} - 2\sum_{\eta=1}^{h}\sum_{k=0}^{c_\eta} \phi(c_\eta - k)\lambda_{c_\eta - k}$$

$$\times \sum_{\xi=0}^{k} \left\{ \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} (\pm\sqrt{-1})^m \beta(m; k-\xi; \eta) e^{m\sqrt{-1}\theta/2} \right\} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}.$$

Assume that both of the right-hand sides of $R^{\pm}(\theta)$ *in* (11.3.1) *are absolutely convergent for* $\theta \in [-\pi, \pi]$, *and that both* $R^{+}(\theta)$ *and* $R^{-}(\theta)$ *are constant functions on* $[-\pi, \pi]$. *Then, for* $d \in \mathbb{N}$,

(11.3.2)

$$\begin{aligned}
&\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\alpha(m)}{m^d} - 2\sum_{\eta=1}^{h} \sum_{k=0}^{[c_\eta/2]} \zeta(2k) \sum_{\omega=0}^{c_\eta-2k} \binom{\omega+d-1}{\omega} (-2)^\omega \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\beta(m; c_\eta-2k-\omega; \eta)}{m^{d+\omega}} \\
&\quad + 2\sum_{\eta=1}^{h} \sum_{k=0}^{[d/2]} \zeta(2k) 2^{-2k} \sum_{\omega=0}^{c_\eta-1} \binom{\omega+d-2k}{\omega} (-2)^\omega \\
&\qquad \times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{((-1)^m+1)\beta(m; c_\eta-1-\omega; \eta)}{m^{d-2k+\omega+1}} \\
&\quad - 2\sum_{\eta=1}^{h} \sum_{k=0}^{[(d+1)/2]} \zeta(2k) \left(1-2^{-2k}\right) \sum_{\omega=0}^{c_\eta-1} \binom{\omega+d-2k}{\omega} (-2)^\omega \\
&\qquad \times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{((-1)^m-1)\beta(m; c_\eta-1-\omega; \eta)}{m^{d-2k+\omega+1}} = 0
\end{aligned}$$

for $\theta \in [-\pi, \pi]$, *where the infinite series appearing on the left-hand side of (11.3.2) are absolutely convergent for* $\theta \in [-\pi, \pi]$.

Proof. Put

(11.3.3)

$$\begin{aligned}
\mathcal{G}_N^{\pm}(\theta) &= \mathcal{G}_N^{\pm}(\theta; \mathcal{A}; \mathcal{B}; \mathcal{C}) \\
&= \left(\frac{2}{\sqrt{-1}}\right)^N \left\{ \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((\sqrt{-1})^m \pm (\sqrt{-1})^{-m}\right) \alpha(m) e^{m\sqrt{-1}\theta/2}}{m^N} \right. \\
&\quad - 2\sum_{\eta=1}^{h} \sum_{k=0}^{c_\eta} \phi(c_\eta - k) \epsilon_{c_\eta-k} \sum_{\rho=0}^{k} \sum_{\omega=0}^{k-\rho} \binom{N-1+\omega}{\omega} (-2)^\omega \\
&\quad \left. \times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((\sqrt{-1})^m \pm (\sqrt{-1})^{-m}\right) \beta(m; k-\rho-\omega; \eta) e^{m\sqrt{-1}\theta/2}}{m^{N+\omega}} \frac{(\sqrt{-1}\theta)^\rho}{\rho!} \right\}
\end{aligned}$$

for $N \in \mathbb{N}_0$, where $\epsilon_{c_\eta-k}$ is defined by (8.1.12). From the assumption, we see that $\mathcal{G}_0^{\pm}(\theta) = R^{+}(\theta) \pm R^{-}(\theta)$ are constant functions for $\theta \in [-\pi, \pi]$.

By using the relation

$$-\binom{m-1}{l-1}+\binom{m}{l}=\binom{m-1}{l} \quad (l,m\in\mathbb{N}),$$

we can confirm that

$$\frac{d}{d\theta}\mathcal{G}_N^{\pm}(\theta)=\mathcal{G}_{N-1}^{\pm}(\theta) \quad (N\in\mathbb{N}). \tag{11.3.4}$$

Therefore, repeating the indefinite integration, we can write

$$\left(\frac{\sqrt{-1}}{2}\right)^N\mathcal{G}_N^{\pm}(\theta)=\sum_{j=0}^{N}\mathcal{C}_{N-j}^{\pm}\frac{(\sqrt{-1}\theta/2)^j}{j!} \quad (N\in\mathbb{N}_0) \tag{11.3.5}$$

for some $\{\mathcal{C}_n^{\pm}\in\mathbb{C}\,|\,n\in\mathbb{N}_0\}$.

Let $d\in\mathbb{N}$. Put $N=d+1$ in the "plus" case of (11.3.5). Taking the difference of two cases $\theta=\pi$ and $\theta=-\pi$, we obtain

$$\frac{(\sqrt{-1})^d}{2\pi}\left\{\mathcal{G}_{d+1}^{+}(\pi)-\mathcal{G}_{d+1}^{+}(-\pi)\right\}=\sum_{\nu=0}^{[d/2]}\mathcal{C}_{d-2\nu}^{+}2^{d-2\nu}\frac{(\sqrt{-1}\pi)^{2\nu}}{(2\nu+1)!}. \tag{11.3.6}$$

Similarly, putting $N=d$ and $\theta=\pm\pi$ in the "plus" case of (11.3.5), we have

$$\frac{(\sqrt{-1})^d}{2}\left\{\mathcal{G}_{d}^{+}(\pi)+\mathcal{G}_{d}^{+}(-\pi)\right\}=\sum_{\nu=0}^{[d/2]}\mathcal{C}_{d-2\nu}^{+}2^{d-2\nu}\frac{(\sqrt{-1}\pi)^{2\nu}}{(2\nu)!}. \tag{11.3.7}$$

Therefore we can apply Lemma 8.9 with $\gamma_d=\mathcal{C}_d^{+}2^d$ to obtain

$$\begin{aligned}&\frac{1}{2}\left\{\mathcal{G}_{d}^{+}(\pi)+\mathcal{G}_{d}^{+}(-\pi)\right\}\\&\quad=-\frac{1}{\pi}\sum_{\tau=0}^{[d/2]}\zeta(2\tau)(-1)^{\tau}\left\{\mathcal{G}_{d-2\tau}^{+}(\pi)-\mathcal{G}_{d-2\tau}^{+}(-\pi)\right\}.\end{aligned} \tag{11.3.8}$$

We will calculate each side of (11.3.8) explicitly as follows. Note that

$$\begin{aligned}&\left((\sqrt{-1})^m\pm(\sqrt{-1})^{-m}\right)^2=2\left\{(-1)^m\pm1\right\},\\&\left((\sqrt{-1})^m+(\sqrt{-1})^{-m}\right)\left((\sqrt{-1})^m-(\sqrt{-1})^{-m}\right)=0,\end{aligned}$$

and hence

$$\begin{aligned}&((\sqrt{-1})^m+(\sqrt{-1})^{-m})((\sqrt{-1})^m+(-1)^{\rho}(\sqrt{-1})^{-m})\\&\qquad=\begin{cases}2\left\{(-1)^m\pm1\right\} & \text{if}\quad \rho:\text{even}\\ 0 & \text{if}\quad \rho:\text{odd.}\end{cases}\end{aligned}$$

Therefore, from the definition (11.3.3) (with putting $\rho=2\mu$) we have

(11.3.9)

$$\mathcal{G}_{d}^{+}(\pi)+\mathcal{G}_{d}^{+}(-\pi)=(\sqrt{-1})^{-d}2^d\left\{2\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\alpha(m)}{m^d}\right.$$

$$- 4\sum_{\eta=1}^{h}\sum_{j=0}^{c_\eta}\phi(c_\eta-j)\epsilon_{c_\eta-j}\sum_{\mu=0}^{[j/2]}\sum_{\omega=0}^{j-2\mu}\binom{d-1+\omega}{\omega}(-2)^\omega \times\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\beta(m;j-2\mu-\omega;\eta)}{m^{d+\omega}}\frac{(\sqrt{-1}\pi)^{2\mu}}{(2\mu)!}\Bigg\}.$$

Applying (8.2.5) with

$$f(x)=\sum_{\omega=0}^{x}\binom{d-1+\omega}{\omega}(-2)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\beta(m;x-\omega;\eta)}{m^{d+\omega}}$$

to the right-hand side, we obtain

(11.3.10)

$$\mathcal{G}_d^+(\pi)+\mathcal{G}_d^+(-\pi)=(\sqrt{-1})^{-d}2^{d+1}\Bigg\{\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\alpha(m)}{m^d} - 2\sum_{\eta=1}^{h}\sum_{\xi=0}^{[c_\eta/2]}\zeta(2\xi)\sum_{\omega=0}^{c_\eta-2\xi}\binom{d-1+\omega}{\omega}(-2)^\omega \times\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\beta(m;c_\eta-2\xi-\omega;\eta)}{m^{d+\omega}}\Bigg\}.$$

Similarly, by using (8.2.6), we have

(11.3.11)

$$\mathcal{G}_{d+1}^+(\pi)-\mathcal{G}_{d+1}^+(-\pi) =(\sqrt{-1})^{-d}2^{d+2}\pi\sum_{\eta=1}^{h}\sum_{\omega=0}^{c_\eta-1}\binom{d+\omega}{\omega}(-2)^\omega\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\beta(m;c_\eta-1-\omega;\eta)}{m^{d+1+\omega}}.$$

Substituting (11.3.10) and (11.3.11) into (11.3.8), we have

(11.3.12)

$$\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\alpha(m)}{m^d} - 2\sum_{\eta=1}^{h}\sum_{\xi=0}^{[c_\eta/2]}\zeta(2\xi)\sum_{\omega=0}^{c_\eta-2\xi}\binom{d-1+\omega}{\omega}(-2)^\omega \times\sum_{\substack{m\in\mathbb{Z}\\ m\neq0}}\frac{((-1)^m+1)\,\beta(m;c_\eta-2\xi-w;\eta)}{m^{d+\omega}} + 4\sum_{\eta=1}^{h}\sum_{\xi=0}^{[d/2]}\sum_{\omega=0}^{c_\eta-1}\zeta(2\xi)2^{-2\xi}\binom{d-2\xi+\omega}{\omega}(-2)^\omega$$

$$\times \sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((-1)^m+1\right)\beta(m;c_\eta-1-w;\eta)}{m^{d-2\xi+1+\omega}} = 0.$$

Next, we consider the "minus" case of (11.3.5), and proceed as above. Similar to (11.3.8), using Lemma 8.9 with $\gamma_d = \mathcal{C}_d^- 2^d$ we obtain

(11.3.13)

$$\frac{1}{2}\left\{\mathcal{G}_d^-(\pi)-\mathcal{G}_d^-(-\pi)\right\} = -\frac{1}{\pi}\sum_{\tau=1}^{[(d+1)/2]}\left(2^{2\tau}-1\right)\zeta(2\tau)(-1)^\tau\left\{\mathcal{G}_{d+1-2\tau}^-(\pi)+\mathcal{G}_{d+1-2\tau}^-(-\pi)\right\}.$$

Similar to (11.3.10) and (11.3.11), we can compute $\mathcal{G}_d^-(\pi)-\mathcal{G}_d^-(-\pi)$ and $\mathcal{G}_{d+1-2\tau}^-(\pi)+\mathcal{G}_{d+1-2\tau}^-(-\pi)$. Hence, as well as (11.3.12), we obtain

(11.3.14)

$$\begin{aligned}
&\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((-1)^m-1\right)\alpha(m)}{m^d} - 2\sum_{\eta=1}^{h}\sum_{\xi=0}^{[c_\eta/2]}\zeta(2\xi)\sum_{\omega=0}^{c_\eta-2\xi}\binom{\omega+d-1}{\omega}(-2)^\omega\\
&\times\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((-1)^m-1\right)\beta(m;c_\eta-2\xi-w;\eta)}{m^{d+\omega}}\\
&+4\sum_{\eta=1}^{h}\sum_{\xi=1}^{[(d+1)/2]}\sum_{\omega=0}^{c_\eta-1}\zeta(2\xi)\left(1-2^{-2\xi}\right)\binom{\omega+d-2\xi}{\omega}(-2)^\omega\\
&\times\sum_{\substack{m\in\mathbb{Z}\\ m\neq 0}} \frac{\left((-1)^m-1\right)\beta(m;c_\eta-1-w;\eta)}{m^{d-2\xi+\omega+1}} = 0.
\end{aligned}$$

Combining (11.3.12) and (11.3.14), we obtain (11.3.2).

□

Next, we prove the following lemma. Note that the case when p and q are even has already been proved in (8.3.14).

LEMMA 11.6 ([118, Lemma 5.3]). *For $p\in\mathbb{N}$, $s\in\mathbb{R}$ with $s>1$ and $x\in\mathbb{C}$ with $|x|\leq 1$,*

(11.3.15)

$$\begin{aligned}
&\sum_{\substack{l\neq 0,\, m\geq 1\\ l+m\neq 0}} \frac{(-1)^{l+m}x^m e^{(l+m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q}\\
&-2\sum_{j=0}^{p}\phi(p-j)\epsilon_{p-j}\sum_{\xi=0}^{j}\binom{q-1+j-\xi}{q-1}(-1)^{j-\xi}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+q+j-\xi}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}
\end{aligned}$$

$$+2\sum_{j=0}^{q}\phi(q-j)\epsilon_{q-j}\sum_{\xi=0}^{j}\binom{p-1+j-\xi}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}=0$$

holds for $\theta\in[-\pi,\pi]$.

PROOF. For the case $p\geq 2$, we can prove this assertion in exactly the same way as in the proof of (8.3.14). Hence we omit it here.

We now prove the case $p=1$. Replacing x by $-xe^{\sqrt{-1}\theta}$ in (11.3.15) for the case $p=2$, we have

(11.3.16)

$$\sum_{\substack{l\neq 0,\ m\geq 1\\ l+m\neq 0}}\frac{(-1)^l x^m e^{l\sqrt{-1}\theta}}{l^2 m^s (l+m)^q}$$
$$-2\sum_{j=0}^{2}\phi(2-j)\epsilon_{2-j}\sum_{\xi=0}^{j}\binom{q-1+j-\xi}{q-1}(-1)^{j-\xi}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+q+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}$$
$$+2\sum_{j=0}^{q}\phi(q-j)\epsilon_{q-j}\sum_{\xi=0}^{j}\binom{1+j-\xi}{1}(-1)^{1}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{-m\sqrt{-1}\theta}}{m^{s+2+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}=0$$

for $\theta\in[-\pi,\pi]$. We denote the first, the second and the third term on the left-hand side of (11.3.16) by $I_1(\theta)$, $I_2(\theta)$ and $I_3(\theta)$, respectively. We differentiate these terms in θ. We can easily compute $I_1'(\theta)$ and $I_2'(\theta)$. As for $I_3'(\theta)$, we have

$$I_3'(\theta)=2\sum_{j=0}^{q}\phi(q-j)\epsilon_{q-j}$$
$$\times\left\{-\sqrt{-1}\sum_{\xi=0}^{j}(1+j-\xi)(-1)\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{-m\sqrt{-1}\theta}}{m^{s+1+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\right.$$
$$\left.+\sqrt{-1}\sum_{\xi=1}^{j}(1+j-\xi)(-1)\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{-m\sqrt{-1}\theta}}{m^{s+2+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi-1}}{(\xi-1)!}\right\}.$$

Note that as to the second member in the curly brackets on the right-hand side, ξ may also run from 1 to $j+1$ because $1+j-(j+1)=0$ in the summand. Hence, replacing $\xi-1$ by ξ, we have

$$I_3'(\theta)=2\sqrt{-1}\sum_{j=0}^{q}\phi(q-j)\epsilon_{q-j}\sum_{\xi=0}^{j}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{-m\sqrt{-1}\theta}}{m^{s+1+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}.$$

Thus, we see that $(I_1'(\theta)+I_2'(\theta)+I_3'(\theta))/\sqrt{-1}$, replacing x by $-xe^{\sqrt{-1}\theta}$, gives (11.3.15) in the case $p=1$. This completes the proof. □

11.4. A functional relation corresponding to $I = \{2\}$

Using the lemmas in the previous section, we now construct a functional relation among $\zeta_2(\boldsymbol{s}; G_2)$, $\zeta(s)$ and $\phi(s) = (2^{1-s} - 1)\zeta(s)$, which corresponds to the case $I = \{2\}$ in Section 11.2.

THEOREM 11.7 ([**117**, Theorem 6.1]). *For $p, q, r, u, v \in \mathbb{N}$,*

(11.4.1)
$$\begin{aligned}
&\zeta_2(p,s,q,r,u,v;G_2) + (-1)^p\zeta_2(p,q,s,r,v,u;G_2) + (-1)^{p+q}\zeta_2(v,q,r,s,p,u;G_2)\\
&\quad + (-1)^{p+q+v}\zeta_2(v,r,q,s,u,p;G_2) + (-1)^{p+q+r+v}\zeta_2(u,r,s,q,v,p;G_2)\\
&\quad + (-1)^{p+q+r+u+v}\zeta_2(u,s,r,q,p,v;G_2)\\
&\quad + I_1 + I_2 + \cdots + I_8 = 0
\end{aligned}$$

holds for all $s \in \mathbb{C}$ except for singularities of functions on the left-hand side, where I_j $(1 \le j \le 8)$, defined below, are linear combinations of $\zeta(s)$ and $\phi(s)$.

The definition of I_j is given by

$$I_j = A_j + B_{1j} + B_{2j} \qquad (1 \le j \le 8),$$

where A_j, B_{1j}, B_{2j} $(1 \le j \le 8)$ are defined as follows:

$$\begin{aligned}
A_j &= 2(-1)^{p+a_1} \sum_{k=0}^{[a_2/2]} \zeta(2k) \sum_{\sigma=0}^{a_2-2k} \binom{\sigma+v-1}{\sigma} \sum_{\rho=0}^{a_3-a_4} \binom{\rho+u-a_5}{\rho}\\
&\quad \times \sum_{\omega=0}^{a_6-a_7} \binom{\omega+r-a_8}{\omega} \binom{p+q-1-\omega-a_7}{a_9-1}\\
&\quad \times (-1)^{a_{10}} 2^{a_{11}} 3^{a_{12}} \zeta(s+p+q+r+u+v-2k),
\end{aligned}$$

$$\begin{aligned}
B_{1j} &= 2(-1)^{p+b_1} \sum_{k=0}^{[v/2]} 2^{-2k}\zeta(2k) \sum_{\sigma=0}^{a_2-1} \binom{\sigma+v-2k}{\sigma} \sum_{\rho=0}^{a_3-b_4} \binom{\rho+u-b_5}{\rho}\\
&\quad \times \sum_{\omega=0}^{a_6-b_7} \binom{\omega+r-b_8}{\omega} \binom{p+q-1-\omega-b_7}{a_9-1} (-1)^{a_{10}} 2^{b_{11}} 3^{b_{12}}\\
&\quad \times \{\zeta(s+p+q+r+u+v-2k) + \phi(s+p+q+r+u+v-2k)\},
\end{aligned}$$

and

$$\begin{aligned}
B_{2j} &= 2(-1)^{p+b_1} \sum_{k=0}^{[(v+1)/2]} (1-2^{-2k})\zeta(2k) \sum_{\sigma=0}^{a_2-1} \binom{\sigma+v-2k}{\sigma} \sum_{\rho=0}^{a_3-b_4} \binom{\rho+u-b_5}{\rho}\\
&\quad \times \sum_{\omega=0}^{a_6-b_7} \binom{\omega+r-b_8}{\omega} \binom{p+q-1-\omega-b_7}{a_9-1} (-1)^{a_{10}} 2^{b_{11}} 3^{b_{12}}
\end{aligned}$$

$$\times\{\zeta(s+p+q+r+u+v-2k)-\phi(s+p+q+r+u+v-2k)\},$$

where $a_l = a_l(j)$, $b_l = b_l(j)$ are as follows: According to $j = 1, \dots, 8$, a_l $(1 \le l \le 12)$ take the values

$$\begin{aligned}
a_1 &= 1, v+1, 1, v, v+1, v, v, v+1,\\
a_2 &= p, u, q, u, r, u, r, u,\\
a_3 &= p, p, q, q, r, r, r, r,\\
a_4 &= 2k+\sigma, 1, 2k+\sigma, 1, 2k+\sigma, 1, 2k+\sigma, 1,\\
a_5 &= 1, 2k+\sigma, 1, 2k+\sigma, 1, 2k+\sigma, 1, 2k+\sigma,\\
a_6 &= p, p, q, q, p, p, q, q,\\
a_7 &= 2k+\sigma+\rho, 1+\rho, 2k+\sigma+\rho, 1+\rho, 1, 1, 1, 1,\\
a_8 &= 1, 1, 1, 1, 2k+\sigma+\rho, 1+\rho, 2k+\sigma+\rho, 1+\rho,\\
a_9 &= q, q, p, p, q, q, p, p,\\
a_{10} &= 0, 0, \sigma+\rho+\omega, \rho+\omega, \rho, \rho, \rho+\omega, \rho+\omega,
\end{aligned}$$

$$\begin{aligned}
a_{11} = \sigma-r-\omega, \sigma-r-\omega, \sigma-u-\rho, -u+2k+2\sigma-\rho-1,\\
-r+2k+2\sigma+\rho-\omega-1, -r+\sigma+\rho-\omega, \sigma, \sigma,
\end{aligned}$$

and

$$a_{12} = -u-v-\sigma-\rho, -u-v+2k-\rho-1, 0, -v-\sigma, 0, -v-\sigma, 0, -v-\sigma.$$

Next, define b_l $(l = 1, 4, 5, 7, 8, 11, 12)$. First,

$$b_1 = 1, v+1, 0, v, v+1, v, v, v+1.$$

The definitions of b_4, b_5, b_7, b_8 are similar to a_4, a_5, a_7, a_8, but all $2k+\sigma$ are replaced by $1+\sigma$. Finally,

$$\begin{aligned}
b_{11} = \sigma-r-\omega, \sigma-r-\omega, \sigma-u-\rho, -u+2\sigma-\rho,\\
-r+2\sigma+\rho-\omega, -r+\sigma+\rho-\omega, \sigma, \sigma,
\end{aligned}$$

and

$$\begin{aligned}
b_{12} = -u-v+2k-\sigma-\rho-1, -u-v+2k-\rho-1, 0, -v+2k-\sigma-1,\\
0, -v+2k-\sigma-1, 0, -v+2k-\sigma-1.
\end{aligned}$$

REMARK 11.8. When p, q, r, u, v are even, formula (11.4.1) coincides with the result given in [**110**, Theorem 5.1].[1] In particular, we can obtain the explicit formula for even integer values of the zeta-function of G_2, similar to (8.2.40) and (8.3.17).

[1]There are misprints in the statement of [**110**, Theorem 5.1]. On line 8 of page 202, we should replace $\binom{2p+2q-2-\rho-\omega}{2q-1}$ by $\binom{2p+2q-2-\rho-\omega}{2p-1}$. On lines 12 and 16 of page 203, we should replace $3^{-2v-\sigma}$ by $3^{-2v-\sigma-1+2k}$.

PROOF OF THEOREM 11.7. The basic principle of the proof of this theorem is the same as in the proof of Theorem 8.26.

From Lemma 11.6, we have

(11.4.2)

$$\sum_{\substack{l\neq 0,\, m\geq 1\\ l+m\neq 0\\ l+2m\neq 0}} \frac{(-1)^{l+m}x^m e^{(l+m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q}$$

$$-2\sum_{j=0}^{p}\phi(p-j)\epsilon_{p-j}\sum_{\xi=0}^{j}\binom{q-1+j-\xi}{q-1}(-1)^{j-\xi}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+q+j-\xi}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$+2\sum_{j=0}^{q}\phi(q-j)\epsilon_{q-j}\sum_{\xi=0}^{j}\binom{p-1+j-\xi}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+j-\xi}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$=-\frac{(-1)^{p+q}}{2^p}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+q}}$$

for $p, q \in \mathbb{N}$, $\theta \in [-\pi, \pi]$, $s \in \mathbb{R}$ with $s > 1$ and $x \in \mathbb{C}$ with $|x| = 1$. By Lemma 8.24 with $d = r \in \mathbb{N}$, we have

$$\sum_{\substack{l\in\mathbb{Z},\, l\neq 0\\ m\geq 1\\ l+m\neq 0\\ l+2m\neq 0}} \frac{(-1)^l x^m e^{(l+2m)\sqrt{-1}\theta}}{l^p m^s (l+m)^q (l+2m)^r}$$

$$-2\sum_{j=0}^{p}\phi(p-j)\,\epsilon_{p-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi}\binom{\omega+r-1}{\omega}(-1)^\omega$$
$$\times\binom{q-1+j-\xi-\omega}{b-1}(-1)^{j-\xi-\omega}\frac{1}{2^{r+\omega}}\sum_{m=1}^{\infty}\frac{x^m e^{2m\sqrt{-1}\theta}}{m^{s+q+j-\xi+r}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$+2\sum_{j=0}^{q}\phi(q-j)\,\epsilon_{q-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi}\binom{\omega+r-1}{\omega}(-1)^\omega$$
$$\times\binom{p-1+j-\xi-\omega}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+p+r+j-\xi}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$+2\sum_{j=0}^{r}\phi(r-j)\,\epsilon_{r-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{p-1}\binom{\omega+j-\xi}{\omega}(-1)^\omega$$
$$\times\binom{p+q-2-\omega}{q-1}(-1)^{p-1-\omega}\frac{1}{2^{j-\xi+\omega+1}}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+q+j-\xi}}\frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$-2\sum_{j=0}^{r}\phi(r-j)\,\epsilon_{r-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{q-1}\binom{\omega+j-\xi}{\omega}(-1)^\omega$$

$$\times \binom{p+q-2-\omega}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+q+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}=0$$

for $\theta\in[-\pi,\pi]$, $p,q,r\in\mathbb{N}$, $s\in\mathbb{R}$ with $s>1$ and $x\in\mathbb{C}$ with $|x|\le 1$. Here, we replace x by $-xe^{\sqrt{-1}\theta}$ and move the terms corresponding to $l+3m=0$ of the first member on the left-hand side of the above equation to the right-hand side. Then we have

$$\begin{aligned}
&\sum_{\substack{l\in\mathbb{Z},\,l\neq 0\\ m\ge 1\\ l+m\neq 0\\ l+2m\neq 0\\ l+3m\neq 0}}\frac{(-1)^{l+m}x^m e^{(l+3m)\sqrt{-1}\theta}}{l^p m^s(l+m)^q(l+2m)^r}\\
&\quad-2\sum_{j=0}^{p}\ \phi(p-j)\ \epsilon_{p-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi}\binom{\omega+r-1}{\omega}(-1)^{\omega}\\
&\qquad\times\binom{q-1+j-\xi-\omega}{q-1}(-1)^{j-\xi-\omega}\frac{1}{2^{r+\omega}}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{3m\sqrt{-1}\theta}}{m^{s+q+r+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\\
&\quad+2\sum_{j=0}^{q}\ \phi(q-j)\ \epsilon_{q-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi}\binom{\omega+r-1}{\omega}(-1)^{\omega}\\
&\qquad\times\binom{p-1+j-\xi-\omega}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{x^m e^{2m\sqrt{-1}\theta}}{m^{s+p+r+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\\
&\quad+2\sum_{j=0}^{r}\ \phi(r-j)\ \epsilon_{r-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{p-1}\binom{\omega+j-\xi}{\omega}(-1)^{\omega}\\
&\qquad\times\binom{p+q-2-\omega}{q-1}(-1)^{p-1-\omega}\frac{1}{2^{j-\xi+\omega+1}}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+p+q+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\\
&\quad+2\sum_{j=0}^{r}\ \phi(r-j)\ \epsilon_{r-j}\sum_{\xi=0}^{j}\sum_{\omega=0}^{q-1}\binom{\omega+j-\xi}{\omega}(-1)^{\omega}\\
&\qquad\times\binom{p+q-2-\omega}{p-1}(-1)^{p-1}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{m\sqrt{-1}\theta}}{m^{s+p+q+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\\
&=-\frac{(-1)^{p+q+r}}{3^{2p}2^{2q}}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+q+r}}.
\end{aligned}$$

We again apply Lemma 8.24 with $d=u\in\mathbb{N}$ to the above equation. Then we have

$$\sum_{\substack{l\in\mathbb{Z},\,l\neq 0\\ m\ge 1\\ l+m\neq 0\\ l+2m\neq 0\\ l+3m\neq 0}}\frac{(-1)^{l+m}x^m e^{(l+3m)\sqrt{-1}\theta}}{l^p m^s(l+m)^q(l+2m)^r(l+3m)^u}\tag{11.4.3}$$

$$+ J_1(\theta;x) + J_2(\theta;x) + J_3(\theta;x) + J_4(\theta;x) = 0,$$

where

$$\begin{aligned}
&J_1(\theta;x)\\
&= -2\sum_{j=0}^{p}\phi(p-j)\ \epsilon_{p-j}\sum_{\xi=0}^{j}\sum_{\rho=0}^{j-\xi}\binom{\rho+u-1}{\rho}(-1)^{\rho}\sum_{\omega=0}^{j-\xi-\rho}\binom{\omega+r-1}{\omega}(-1)^{\omega}\\
&\quad\times 3^{-u-\rho}\binom{q-1+j-\xi-\rho-\omega}{q-1}\frac{(-1)^{j-\xi-\rho-\omega}}{2^{r+\omega}}\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{3m\sqrt{-1}\theta}}{m^{s+q+r+u+j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}\\
&+2\sum_{j=0}^{u}\ \phi(u-j)\ \epsilon_{u-j}\sum_{\xi=0}^{j}\sum_{\rho=0}^{p-1}\binom{\rho+j-\xi}{\rho}(-1)^{\rho}\sum_{\omega=0}^{p-1-\rho}\binom{\omega+r-1}{\omega}(-1)^{\omega}\\
&\quad\times 3^{-j+\xi-\rho-1}\binom{p+q-2-\rho-\omega}{q-1}\frac{(-1)^{p-1-\rho-\omega}}{2^{r+\omega}}\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p+q+r+2j-\xi}}\frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}.
\end{aligned}$$

We can similarly write $J_2(\theta;x)$, $J_3(\theta;x)$ and $J_4(\theta;x)$, but they are omitted for the purpose of saving space.

Next, setting $x = \pm ie^{-3\sqrt{-1}\theta/2}$ in (11.4.3) and moving the terms corresponding to $2l+3m=0$ of the first member on the left-hand side to the right-hand side, we have

$$\begin{aligned}
&\sum_{\substack{l\in\mathbb{Z},\, l\neq 0\\ m\geq 1\\ l+m\neq 0\\ l+2m\neq 0\\ l+3m\neq 0\\ 2l+3m\neq 0}}\frac{(-1)^{l+m}(\pm\sqrt{-1})^m e^{(2l+3m)\sqrt{-1}\theta/2}}{l^p m^s (l+m)^q (l+2m)^r (l+3m)^u}\\
&\quad + J_1(\theta;\pm\sqrt{-1}e^{-3\sqrt{-1}\theta/2}) + J_2(\theta;\pm\sqrt{-1}e^{-3\sqrt{-1}\theta/2})\\
&\quad + J_3(\theta;\pm\sqrt{-1}e^{-3\sqrt{-1}\theta/2}) + J_4(\theta;\pm\sqrt{-1}e^{-3\sqrt{-1}\theta/2})\\
&\quad = -\sum_{\substack{l,m=1\\ 2l=3m}}^{\infty}\frac{1}{(-l)^p m^s (-l+m)^q (-l+2m)^r (-l+3m)^u}.
\end{aligned}\tag{11.4.4}$$

Note that $(-1)^{l+m}(\pm\sqrt{-1})^m = (\pm\sqrt{-1})^{2l+3m}$. Hence we can apply Lemma 11.5 with $d = v \in \mathbb{N}$ to (11.4.4), because we can see that the left-hand side of (11.4.4) is of the same form as the right-hand side of (11.3.1). Consequently, we obtain the equation given from (11.3.2). The first term on the left-hand side of the obtained equation is

$$\sum_{\substack{l\in\mathbb{Z},\ l\neq 0\\ m\geq 1\\ l+m\neq 0\\ l+2m\neq 0\\ l+3m\neq 0\\ 2l+3m\neq 0}}\frac{1}{l^p m^s (l+m)^q (l+2m)^r (l+3m)^u (2l+3m)^v},\tag{11.4.5}$$

while the remaining terms on the left-hand side of the obtained equation can be expressed explicitly in terms of the Riemann zeta-function, which are $I_1 + \cdots + I_8$ in the statement of the theorem. On the other hand, we see that (11.4.5) is equal to

(11.4.6)
$$\begin{aligned}&\zeta_2(p,s,q,r,u,v;G_2) + (-1)^p\zeta_2(p,q,s,c,v,u;G_2)\\ &\quad + (-1)^{p+q}\zeta_2(v,q,r,s,p,u;G_2) + (-1)^{p+q+v}\zeta_2(v,r,q,s,u,p;G_2)\\ &\quad + (-1)^{p+q+r+v}\zeta_2(u,r,s,q,v,a;G_2) + (-1)^{p+q+r+u+v}\zeta_2(u,s,r,q,p,v;G_2).\end{aligned}$$

In fact, since (11.4.5) coincides with $S(p,s,q,r,u,v;\mathbf{0};\{2\};G_2)$ (see (9.4.1)), (11.4.6) simply follows from (9.4.12). Thus we obtain the assertion of the theorem. □

Setting $(p,q,r,u,v) = (2a,b,2c-1,d,d)$ for $a,b,c,d \in \mathbb{N}$ in (11.4.1), we see that

(11.4.7)
$$\begin{aligned}&\zeta_2(2a,s,b,2c-1,d,d;G_2) + \zeta_2(2a,b,s,2c-1,d,d;G_2)\\ &+ (-1)^b\zeta_2(d,b,2c-1,s,2a,d;G_2) + (-1)^{b+d}\zeta_2(d,2c-1,b,s,d,2a;G_2)\\ &- (-1)^{b+d}\zeta_2(d,2c-1,s,b,d,2a;G_2) - (-1)^b\zeta_2(d,s,2c-1,b,2a,d;G_2)\end{aligned}$$

is expressed in terms of $\zeta(s)$ and $\phi(s)$. As we noted above (see (11.4.6)), (11.4.7) coincides with

$$S((2a,s,b,2c-1,d,d),\mathbf{0};\{2\};G_2).$$

In particular when $s=b$, it is equal to $2\zeta_2(2a,b,b,2c-1,d,d;G_2)$ (see (11.2.1)). Therefore we have the following.

Corollary 11.9. *For $a,b,c,d \in \mathbb{N}$,*

$$\zeta_2(2a,b,b,2c-1,d,d;G_2) \in \mathbb{Q}[\{\zeta(j) \mid j \in \mathbb{N}_{\geq 2}\}]. \tag{11.4.8}$$

Example 11.10. Putting $(p,q,r,u,v) = (2,1,1,1,1)$ in (11.4.1), we have

$$\begin{aligned}&\zeta_2(2,s,1,1,1,1;G_2) + \zeta_2(2,1,s,1,1,1;G_2) - \zeta_2(1,1,1,s,2,1;G_2)\\ &+ \zeta_2(1,1,1,s,1,2;G_2) - \zeta_2(1,1,s,1,1,2;G_2) + \zeta_2(1,s,1,1,2,1;G_2)\\ &- \frac{1}{9}\zeta(2)\zeta(s+4) + \frac{109}{648}\zeta(s+6) = 0.\end{aligned}$$

Setting $s=1$, we obtain a special case of (11.4.8), that is,

$$\zeta_2(2,1,1,1,1,1;G_2) = \frac{1}{18}\zeta(2)\zeta(5) - \frac{109}{1296}\zeta(7), \tag{11.4.9}$$

which agrees with Zhao's numerical value

$$\zeta_2(2,1,1,1,1,1;G_2) = 0.0099527234\cdots \tag{11.4.10}$$

given in [218]. Similarly, we can compute

$$\zeta_2(4,1,1,1,1,1;G_2) = \frac{1}{18}\zeta(4)\zeta(5) + \frac{145}{648}\zeta(2)\zeta(7) - \frac{19753}{46656}\zeta(9),$$
$$\zeta_2(4,2,2,1,1,1;G_2) = \frac{1}{18}\zeta(4)\zeta(7) + \frac{595}{648}\zeta(2)\zeta(9) - \frac{73201}{46656}\zeta(11),$$
$$\zeta_2(2,1,1,5,3,3;G_2) = \frac{5}{4}\zeta(4)\zeta(11) + \frac{1043857}{23328}\zeta(2)\zeta(13) - \frac{41971423}{559872}\zeta(15),$$
$$\zeta_2(4,2,2,1,4,4;G_2) = \frac{61441}{209952}\zeta(4)\zeta(13) + \frac{600677}{944784}\zeta(2)\zeta(15) - \frac{23172773}{17006112}\zeta(17),$$
$$\zeta_2(2,4,4,3,3,3;G_2) = \frac{1}{8}\zeta(4)\zeta(15) + \frac{281221}{23328}\zeta(2)\zeta(17) - \frac{11177971}{559872}\zeta(19).$$

EXAMPLE 11.11. More generally, setting $(p,s,q,r,u,v) = (2a, 2b-1, 2b-1, 2b-1, 2a, 2a)$ $(a,b \in \mathbb{N})$ in (11.4.1), we see that the first six terms of (11.4.1) are reduced to

$$2\zeta_2(2a, 2b-1, 2b-1, 2b-1, 2a, 2a; G_2).$$

This is an example of the assertion in Case 3 of Section 10.2 corresponding to $\Delta = \Delta(G_2)$, $\Delta_I = \Delta(A_1)$ with $I = \{2\}$ and $W^I((-1),(1)) = 2$. For example, we can compute that

$$2\zeta_2(2,1,1,1,2,2;G_2) = -\frac{187}{972}\zeta(7)\pi^2 + \frac{11149}{5832}\zeta(9),$$
$$2\zeta_2(4,1,1,1,4,4;G_2) = -\frac{15337}{4723920}\zeta(11)\pi^4 - \frac{157303}{2834352}\zeta(13)\pi^2 + \frac{14696765}{17006112}\zeta(15),$$
$$2\zeta_2(2,3,3,3,2,2;G_2) = -\frac{16171}{3888}\zeta(13)\pi^2 + \frac{957697}{23328}\zeta(15).$$

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CHAPTER 12

Applications to multiple zeta values (I)

In this chapter and the next chapter, we discuss certain applications of zeta-functions of root systems to the study of multiple zeta values. The main aim of the present chapter is to show that multiple zeta values are regarded as special values of the zeta-function of type C_r. An analogy in the case of type B_r is also discussed.

12.1. The case of type A_2

As stated in Section 1.6 and Section 3.2, the multiple zeta value $\zeta_{EZ,r}(k_1, k_2, \ldots, k_r)$ for $k_1, \ldots, k_r \in \mathbb{N}$ with $k_r > 1$ is a special value of $\zeta_r(\mathbf{s}; A_r)$ (see (3.2.5)). For example,

$$\zeta_2(k_1, 0, k_2; A_2) = \zeta_2(0, k_1, k_2; A_2) = \zeta_{EZ,2}(k_1, k_2).$$

As an application coming from this viewpoint, we can give an alternative proof of Euler's sum formula (1.1.2) for double zeta values:

PROPOSITION 12.1. *For $p \in \mathbb{N}$,*

$$\sum_{j=0}^{p-1} \zeta_{EZ,2}(p-j, j+2) = \zeta(p+2). \tag{12.1.1}$$

For the proof, we need the following formula of the partial fraction decomposition which was already known by Euler ([42]; see also Weil [208, Chapter II, (2)], Borwein et al. [15]).

LEMMA 12.2. *For $\alpha, \beta \in \mathbb{N}_0$ with $\alpha + \beta \geq 1$,*

$$\frac{1}{X^\alpha Y^\beta} = \sum_{j=0}^{\alpha-1} \binom{\beta-1+j}{j} \frac{1}{X^{\alpha-j}(X+Y)^{\beta+j}} + \sum_{j=0}^{\beta-1} \binom{\alpha-1+j}{j} \frac{1}{Y^{\beta-j}(X+Y)^{\alpha+j}}. \tag{12.1.2}$$

PROOF. We will prove (12.1.2) by induction on $\alpha + \beta \geq 1$.

The case $\alpha + \beta = 1$ holds trivially. Next, we assume (12.1.2) holds for the case $\alpha + \beta = N\ (\geq 1)$, and consider the case $\alpha + \beta = N + 1$.

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It is obvious that (12.1.2) holds for the case $\alpha = 0$ or $\beta = 0$. Hence we assume $\alpha, \beta \in \mathbb{N}$. By the induction hypothesis, the left-hand side of (12.1.2) is equal to

$$\begin{aligned}
&\frac{1}{X^{\alpha}Y^{\beta-1}(X+Y)} + \frac{1}{X^{\alpha-1}Y^{\beta}(X+Y)} \\
&= \sum_{j=0}^{\alpha-1} \binom{\beta-2+j}{j} \frac{1}{X^{\alpha-j}(X+Y)^{\beta+j}} \\
&\quad + \sum_{j=0}^{\beta-2} \binom{\alpha-1+j}{j} \frac{1}{Y^{\beta-1-j}(X+Y)^{\alpha+1+j}} \\
&\quad + \sum_{j=0}^{\alpha-2} \binom{\beta-1+j}{j} \frac{1}{X^{\alpha-1-j}(X+Y)^{\beta+1+j}} \\
&\quad + \sum_{j=0}^{\beta-1} \binom{\alpha-2+j}{j} \frac{1}{Y^{\beta-j}(X+Y)^{\alpha+j}}.
\end{aligned}$$

Replacing $j+1$ by j in the second and the third member on the right-hand side, and using the property

$$\binom{x}{j-1} + \binom{x}{j} = \binom{x+1}{j} \quad (j \in \mathbb{N}),$$

we obtain that (12.1.2) holds. This completes the proof. □

Proof of Proposition 12.1. The following proof is mentioned in [147, Section 2] and [148, Remark 2.4].

It follows from Theorem 8.3 with $k = l = 1$ and $s = p$ that

$$\sum_{m,n=1}^{\infty} \frac{1}{mn(m+n)^p} - 2\sum_{m,n=1}^{\infty} \frac{1}{mn^p(m+n)} = -2\zeta(p+2). \tag{12.1.3}$$

On the other hand, putting $(X, Y, \alpha, \beta) = (m, n, 1, 1)$ in (12.1.2), dividing by $(m+n)^p$ and summing up, we have

$$\sum_{m,n=1}^{\infty} \frac{1}{mn(m+n)^p} = 2\zeta_{EZ,2}(1, p+1). \tag{12.1.4}$$

Similarly, putting $\alpha = 1, \beta = p$ and dividing by $m+n$,

$$\sum_{m,n=1}^{\infty} \frac{1}{mn^p(m+n)} = \zeta_{EZ,2}(1, p+1) + \sum_{j=0}^{p-1} \zeta_{EZ,2}(p-j, j+2).$$

Substituting these relations into (12.1.3), we obtain (12.1.1). □

REMARK 12.3. Using the above method, we can generalize the sum formula to the relation among certain double series (for the details, see [150]). For example, we can obtain

$$\sum_{j=0}^{p-1}\sum_{\substack{m,n=1\\ m\equiv n\ (\mathrm{mod}\ 3)}}^{\infty}\frac{1}{m^{p-j}(m+n)^{j+2}}=\frac{1}{3^{p+2}}\zeta(p+2)+\frac{\pi}{3\sqrt{3}}\sum_{m=1}^{\infty}\frac{\chi_3(m)}{m^{p+1}}$$

for $p\in\mathbb{N}$, where χ_3 is the Dirichlet character of conductor 3.

Euler showed the following fact, which is called the *parity result* for double zeta values mentioned in Section 1.3:

$$\zeta_{EZ,2}(p,q)\in\mathbb{Q}[\pi^2,\ \{\zeta(2j+1)\mid j\in\mathbb{N}\}] \tag{12.1.5}$$

for $p,q\in\mathbb{N}$ with $q\geq 2$ satisfying that $p+q$ is odd. For example,

$$\zeta_{EZ,2}(1,2)=\zeta(3),\quad \zeta_{EZ,2}(2,3)=\frac{9}{2}\zeta(5)-2\zeta(2)\zeta(3),\ \ldots$$

It should be emphasized that by combining Theorem 8.3 and the harmonic product formula (1.1.4), we can prove (12.1.5) in the following explicit form.

As for the case $\zeta_{EZ,2}(1,q)$, combining the sum formula (12.1.1) and (1.1.4), we can show

$$\zeta_{EZ,2}(1,q)=\frac{q}{2}\zeta(q+1)-\frac{1}{2}\sum_{j=1}^{q-2}\zeta(q-j)\zeta(j+1), \tag{12.1.6}$$

which holds for any $q\in\mathbb{N}$ with $q\geq 2$. In fact, using (1.1.4) we have

$$\begin{aligned}&\frac{1}{2}\sum_{j=1}^{q-2}\zeta(q-j)\zeta(j+1)\\ &\quad=\frac{1}{2}\sum_{j=1}^{q-2}(\zeta_{EZ,2}(q-j,j+1)+\zeta_{EZ,2}(j+1,q-j))+\frac{1}{2}(q-2)\zeta(q+1)\\ &\quad=\sum_{j=1}^{q-2}\zeta_{EZ,2}(q-j,j+1)+\frac{1}{2}(q-2)\zeta(q+1),\end{aligned}$$

and the sum on the right-hand side is $\zeta(q+1)-\zeta_{EZ,2}(1,q)$ by (12.1.1), hence we obtain (12.1.6).

As for $\zeta_{EZ,2}(p,q)$ for $p,q\geq 2$, setting $(k,l,s)=(p,q,0)$ in (8.2.1), we have

$$\begin{aligned}&\zeta_{EZ,2}(p,q)+(-1)^{p+q}\zeta_{EZ,2}(q,p)\\ &\quad=2(-1)^p\sum_{\rho=0}^{[p/2]}\binom{p+q-2\rho-1}{q-1}\zeta(2\rho)\zeta(p+q-2\rho)\end{aligned}$$

$$+ 2(-1)^p \sum_{\rho=0}^{[q/2]} \binom{p+q-2\rho-1}{p-1} \zeta(2\rho)\zeta(p+q-2\rho)$$
$$- (-1)^p \zeta(p)\zeta(q).$$

For the case that $p+q$ is odd, combining this and (1.1.4), we obtain

PROPOSITION 12.4. *Let $p, q \in \mathbb{N}$ with $p, q \geq 2$. If $p+q$ is odd,*

$$\zeta_{EZ,2}(p,q) = (-1)^p \sum_{\rho=0}^{[p/2]} \binom{p+q-2\rho-1}{q-1} \zeta(2\rho)\zeta(p+q-2\rho)$$
$$+ (-1)^p \sum_{\rho=0}^{[q/2]} \binom{p+q-2\rho-1}{p-1} \zeta(2\rho)\zeta(p+q-2\rho)$$
$$+ \frac{(1-(-1)^p)}{2}\zeta(p)\zeta(q) - \frac{1}{2}\zeta(p+q).$$

Euler proved (12.1.6) and Proposition 12.4 in [42], which explicitly give the parity result (12.1.5). It is a natural question whether this kind of parity result holds for multiple zeta values of other types, or not. We will discuss this question in the following sections.

12.2. The case of type C_r

The basic idea of the previous section is to use (3.2.5), that is, to understand $\zeta_{EZ,r}(\mathbf{s})$ as a special case of the zeta-function of A_r. This idea will be further developed in Chapter 13, but here, we discuss an alternative viewpoint; we regard $\zeta_{EZ,r}(\mathbf{s})$ as a special case of the zeta-function of C_r.

Since the root system $\Delta(C_r)$ of type C_r is not simply-laced, we have the disjoint union $\Delta(C_r) = \Delta_l(C_r) \cup \Delta_s(C_r)$, where $\Delta_l(C_r)$ (resp. $\Delta_s(C_r)$) is the set of all long (resp. short) roots. Correspondingly, we have the disjoint union $\Delta_+^\vee = (\Delta_{l+})^\vee \cup (\Delta_{s+})^\vee$, where $\Delta_{l+}(C_r) = \Delta_l(C_r) \cap \Delta_+(C_r)$, $\Delta_{s+}(C_r) = \Delta_s(C_r) \cap \Delta_+(C_r)$, and

$$(\Delta_{l+})^\vee(C_r) = \{\alpha_r^\vee,\ \alpha_{r-1}^\vee + \alpha_r^\vee,\ \alpha_{r-2}^\vee + \alpha_{r-1}^\vee + \alpha_r^\vee,\ \dots, \alpha_1^\vee + \cdots + \alpha_r^\vee\}$$
$$= \{e_r, e_{r-1}, e_{r-2}, \dots, e_1\}.$$

Hence it follows from (3.3.11) with $\mathbf{s} = ((s_{r+1-i})_{1\leq i\leq r}, (0), (0))$ (that is, all $s_{ij}^- = s_{ij}^+ = 0$) that

$$\zeta_r(\mathbf{s}; \Delta(C_r)) = \zeta_r(s_1, \dots, s_r; \Delta_{l+}(C_r)) \tag{12.2.1}$$
$$= \sum_{m_1,\dots,m_r=1}^{\infty} \prod_{i=1}^{r} \frac{1}{(\sum_{j=r-i+1}^{r-1} m_j + m_r)^{s_i}},$$

which is exactly the Euler–Zagier multiple zeta-function $\zeta_{EZ,r}(s_1,\dots,s_r)$ (see (1.3.1)). In the expression (12.2.1), only the variables corresponding to long roots (or equivalently, short coroots) exist. This is an essentially important point here, because $\Delta(C_r)$ is not simply-laced, and the set of all long roots form an orbit of the corresponding Weyl group (see Remark 3.10).

The Weyl group $W(C_r)$ can be realized as $(\mathbb{Z}/2\mathbb{Z})^r \rtimes \mathfrak{S}_r$, where $\mathfrak{S}_r$ is the symmetric group of degree r which permutes bases, and the j-th $\mathbb{Z}/2\mathbb{Z}$ flips the sign of e_j. Since the sign flips act trivially on the variables $(s_\alpha)_{\alpha\in\Delta_{l+}}$, from Theorem 9.20 with $\mathbf{y}=\mathbf{0}$, $\Delta=\Delta(C_r)$ and $\Delta^*=\Delta_{l+}(C_r)$, we obtain the following formulas.

THEOREM 12.5 ([**116**, Corollaries 4.1 and 4.2]). *For* $(2k_1,\dots,2k_r)\in(2\mathbb{N})^r$, *we have*

(12.2.2)

$$\sum_{\sigma\in\mathfrak{S}_r}\zeta_{EZ,r}(2k_{\sigma^{-1}(1)},\dots,2k_{\sigma^{-1}(r)}) = \frac{(-1)^r}{2^r}B_{\Delta_{l+}}((2k_1,\dots,2k_r);C_r)\left(\prod_{j=1}^{r}\frac{(2\pi\sqrt{-1})^{2k_j}}{(2k_j)!}\right)\in\mathbb{Q}\cdot\pi^{2\sum_{j=1}^r k_j}.$$

In particular, for $k\in\mathbb{N}$,

(12.2.3)
$$\zeta_{EZ,r}(2k,2k,\dots,2k) = \frac{(-1)^r}{2^r r!}B_{\Delta_{l+}}((2k,\dots,2k);C_r)\frac{(2\pi\sqrt{-1})^{2kr}}{((2k)!)^r}\in\mathbb{Q}\cdot\pi^{2kr}.$$

REMARK 12.6. The fact

$$\sum_{\sigma\in\mathfrak{S}_r}\zeta_{EZ,r}(2k_{\sigma^{-1}(1)},\dots,2k_{\sigma^{-1}(r)})\in\mathbb{Q}\cdot\pi^{2\sum_{j=1}^r k_j}$$

was first proved by Hoffman [**63**, Theorem 2.2]. This includes

$$\zeta_{EZ,r}(2k,2k,\dots,2k)\in\mathbb{Q}\cdot\pi^{2kr},$$

which was given by Zagier [**215**, Section 9] independently. A certain explicit expression of (12.2.3) was given in [**16**]. It is also known that

$$\zeta_{EZ,r}(2k,\dots,2k)=\mathcal{C}_r^{(k)}\frac{(2\pi\sqrt{-1})^{2kr}}{(2kr)!},$$

where

$$\mathcal{C}_0^{(k)}=1,\quad \mathcal{C}_n^{(k)}=\frac{1}{2n}\sum_{j=1}^{n}(-1)^j\binom{2nk}{2jk}B_{2jk}\mathcal{C}_{n-j}^{(k)}\quad(n\geq 1),$$

which was first published by Arakawa and Kaneko in their lecture notes [**9**] [**10**].

Next, we give an alternative proof of the parity result for the triple zeta value whose indices are greater than 1 (see Corollary 12.8). Actually, we first prove the following more general result.

THEOREM 12.7 ([**116**, Theorem 5.2]). *For $a, b, c \in \mathbb{N}_{\geq 2}$,*

$$
\begin{aligned}
&(1+(-1)^a)\zeta_{EZ,3}(a,b,c) + (1+(-1)^b)\{\zeta_{EZ,3}(b,a,c) + \zeta_{EZ,3}(b,c,a)\} \\
&\qquad + (-1)^b(1+(-1)^c)\zeta_{EZ,3}(c,b,a) \\
&= 2\Bigg\{ \sum_{j=0}^{[a/2]} \zeta(2j) \sum_{\omega=0}^{a-2j} \binom{\omega+b-1}{\omega} \\
&\qquad\qquad \times \binom{a+c-2j-\omega-1}{c-1} \zeta_{EZ,2}(b+\omega, a+c-2j-\omega) \\
&\quad + \sum_{j=0}^{[b/2]} \zeta(2j) \sum_{\ell=0}^{a-1} \binom{\ell+b-2j}{\ell} \\
&\qquad\qquad \times \binom{a+c-\ell-2}{c-1} \zeta_{EZ,2}(b-2j+\ell+1, a+c-1-\ell) \\
&\quad + (-1)^b \sum_{j=0}^{[c/2]} \zeta(2j) \sum_{\ell=0}^{c-2j} \binom{\ell+b-1}{\ell} \\
&\qquad\qquad \times \binom{a+c-2j-\ell-1}{a-1} \zeta_{EZ,2}(b+\ell, a+c-2j-\ell) \\
&\quad + (-1)^b \sum_{j=0}^{[b/2]} \zeta(2j) \sum_{\ell=0}^{c-1} \binom{\ell+b-2j}{\ell} \\
&\qquad\qquad \times \binom{a+c-\ell-2}{a-1} \zeta_{EZ,2}(b-2j+\ell+1, a+c-1-\ell) \Bigg\} \\
&\quad - \zeta_{EZ,2}(a+b,c) - (1+(-1)^b)\zeta_{EZ,2}(b,a+c) - (-1)^b\zeta_{EZ,2}(b+c,a).
\end{aligned}
$$

PROOF. From (8.1.17) and (8.1.19) we find

$$
I_p(\theta; a; 1) - (\sqrt{-1})^{p-1} \sum_{j=0}^{a} \phi(a-j)\epsilon_{p+1+j} \frac{(\sqrt{-1}\theta)^j}{j!} = 0
$$

for $a \in \mathbb{N}, p \in \mathbb{N}_0, a \not\equiv p \pmod 2$ and $\theta \in (-\pi, \pi)$, where $\phi(s)$ is defined by (8.1.1). Choose $p = a-1$, and rewrite $I_p(\theta; a; 1)$ using (8.1.10) and (the second member of) (8.1.11) to obtain

$$
\lim_{N\to\infty} \sum_{\substack{n=-N \\ n\neq 0}}^{N} \frac{(-1)^n e^{\sqrt{-1}n\theta}}{n^a} - 2\sum_{j=0}^{a} \phi(a-j)\epsilon_{a-j} \frac{(\sqrt{-1}\theta)^j}{j!} = 0. \tag{12.2.4}
$$

Now assume $a \geq 2$. Then (12.2.4) is also valid for $\theta = \pm\pi$ because of the continuity. For $x, y \in (-1, 1)$, we multiply (12.2.4) by

$$\sum_{l,m=1}^{\infty} (-1)^{l+m} x^l y^m e^{\sqrt{-1}(l+m)\theta}. \tag{12.2.5}$$

Separating the terms corresponding to $l + m + n = 0$, we obtain

$$\begin{aligned}
&\sum_{l,m=1}^{\infty} \sum_{\substack{n\neq 0\\ l+m+n\neq 0}} \frac{(-1)^{l+m+n} x^l y^m e^{\sqrt{-1}(l+m+n)\theta}}{n^a} \\
&- 2\sum_{j=0}^{a} \phi(a-j)\epsilon_{a-j} \sum_{l,m=1}^{\infty} (-1)^{l+m} x^l y^m e^{\sqrt{-1}(l+m)\theta} \frac{(\sqrt{-1}\theta)^j}{j!} \\
&= -(-1)^a \sum_{l,m=1}^{\infty} \frac{x^l y^m}{(l+m)^a}
\end{aligned}$$

for $\theta \in [-\pi, \pi]$. We regard the right-hand side as a constant function with respect to θ. Applying Lemma 8.24 with $h = 1$, $a_1 = a$, $d = c \geq 2$,

$$C(N) = \sum_{\substack{l,m\geq 1, n\neq 0\\ l+m+n=N}} \frac{x^l y^m}{n^a},$$

$$D(N; r; 1) = \begin{cases} \sum_{\substack{l,m\in\mathbb{N}\\ l+m=N}} x^l y^m & (N \geq 2, r = 0), \\ 0 & (\text{otherwise}), \end{cases}$$

we obtain

$$\begin{aligned}
&\sum_{l,m=1}^{\infty} \sum_{\substack{n\in\mathbb{Z}, n\neq 0\\ l+m+n\neq 0}} \frac{(-1)^{l+m+n} x^l y^m e^{\sqrt{-1}(l+m+n)\theta}}{n^a (l+m+n)^c} \\
&- 2\sum_{j=0}^{a} \phi(a-j)\epsilon_{a-j} \sum_{\xi=0}^{j} \binom{j-\xi+c-1}{j-\xi} (-1)^{j-\xi} \\
&\quad \times \sum_{l,m=1}^{\infty} \frac{(-1)^{l+m} x^l y^m e^{\sqrt{-1}(l+m)\theta}}{(l+m)^{c+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} \\
&+ 2\sum_{j=0}^{c} \phi(c-j)\epsilon_{c-j} \sum_{\xi=0}^{j} \binom{j-\xi+a-1}{a-1} (-1)^{a-1} \\
&\quad \times \sum_{l,m=1}^{\infty} \frac{x^l y^m}{(l+m)^{a+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!} = 0.
\end{aligned}$$

We replace x by $-xe^{-\sqrt{-1}\theta}$ and separate the term corresponding to $m + n = 0$ in the first member on the left-hand side of the above

equation, and again apply Lemma 8.24 with $d = b \geq 2$. Consequently, we have

$$\begin{aligned}
&\sum_{l,m=1}^{\infty} \sum_{\substack{n\in\mathbb{Z},n\neq 0\\ m+n\neq 0\\ l+m+n\neq 0}} \frac{(-1)^{m+n}x^l y^m e^{\sqrt{-1}(m+n)\theta}}{n^a(m+n)^b(l+m+n)^c} \\
&= 2\sum_{j=0}^{a} \phi(a-j)\epsilon_{a-j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi} \binom{\omega+b-1}{\omega}(-1)^{\omega}\binom{j-\xi-\omega+c-1}{c-1} \\
&\qquad \times (-1)^{j-\xi-\omega} \sum_{l,m=1}^{\infty} \frac{(-1)^m x^l y^m e^{\sqrt{-1}m\theta}}{m^{b+\omega}(l+m)^{c+j-\xi-\omega}} \frac{(\sqrt{-1}\theta)^{\xi}}{\xi!} \\
&- 2\sum_{j=0}^{b} \phi(b-j)\epsilon_{b-j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{a-1} \binom{\omega+j-\xi}{\omega}(-1)^{\omega}\binom{a-1-\omega+c-1}{c-1} \\
&\qquad \times (-1)^{a-1-\omega} \sum_{l,m=1}^{\infty} \frac{x^l y^m}{m^{j-\xi+\omega+1}(l+m)^{a+c-1-\omega}} \frac{(\sqrt{-1}\theta)^{\xi}}{\xi!} \\
&- 2\sum_{j=0}^{c} \phi(c-j)\epsilon_{c-j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{j-\xi} \binom{\omega+b-1}{\omega}(-1)^{\omega}\binom{j-\xi-\omega+a-1}{a-1} \\
&\qquad \times (-1)^{a-1} \sum_{l,m=1}^{\infty} \frac{(-1)^l x^l y^m e^{-\sqrt{-1}l\theta}}{(-l)^{b+\omega}(l+m)^{a+j-\xi-\omega}} \frac{(\sqrt{-1}\theta)^{\xi}}{\xi!} \\
&+ 2\sum_{j=0}^{b} \phi(b-j)\epsilon_{b-j} \sum_{\xi=0}^{j}\sum_{\omega=0}^{c-1} \binom{\omega+j-\xi}{\omega}(-1)^{\omega}\binom{a-1-\omega+c-1}{a-1} \\
&\qquad \times (-1)^{a-1} \sum_{l,m=1}^{\infty} \frac{x^l y^m}{(-l)^{j-\xi+\omega+1}(l+m)^{a+c-1-\omega}} \frac{(\sqrt{-1}\theta)^{\xi}}{\xi!}.
\end{aligned}$$

Since $a, b, c \geq 2$, we can let $x, y \to 1$ on both sides. Then set $\theta = \pi$. We will compute the left-hand side, and then the right-hand side, of the resulting formula.

First, we deform the left-hand side, which is a kind of twisted polylogarithm associated with zeta values of the root system of type C_3. We extract $(-1)^a\zeta_{EZ,2}(a+b,c)$ from the terms corresponding to $m+2n=0$, and

$$(-1)^a \sum_{\substack{m,n\in\mathbb{N}\\ m\neq n, m<2n}} \frac{1}{n^{a+c}(m-n)^b},$$

from the terms corresponding to $l+m+2n=0$ (where we rewrite $-n$ by n), which can be written as $(-1)^a(1+(-1)^b)\zeta_{EZ,2}(b,a+c)$ by considering the cases $n<m<2n$ and $0<m<n$ separately. Also the

contribution of the terms corresponding to $l+2m+2n=0$ is

$$(-1)^a \sum_{\substack{m,n\in\mathbb{N}\\ n>m}} \frac{1}{n^a(m-n)^b(n-m)^c} = (-1)^{a+b}\zeta_{EZ,2}(b+c,a).$$

The remaining part of the left-hand side is

$$\sum_{l,m=1}^{\infty} \sum_{\substack{n\in\mathbb{Z},n\neq 0\\ m+n\neq 0\\ m+2n\neq 0\\ l+m+n\neq 0\\ l+m+2n\neq 0\\ l+2m+2n\neq 0}} \frac{1}{n^a(m+n)^b(l+m+n)^c}$$

$$= \zeta_{EZ,3}(a,b,c) + (-1)^a \sum_{l,m=1}^{\infty} \sum_{\substack{n\geq 1\\ m\neq n\\ m\neq 2n\\ l+m\neq n\\ l+m\neq 2n\\ l+2m\neq 2n}} \frac{1}{n^a(m-n)^b(l+m-n)^c}.$$

In the above double sum, replace $j=m-n$ and $k=n-m$ corresponding to $m>n$ and $m<n$, respectively. In the part corresponding to $m>n$, we further divide the sum into three parts according to $l+j<n$, $j<n<l+j$, and $n<j$, and find that the contribution of this part is

$$(-1)^a\{\zeta_{EZ,3}(b,c,a)+\zeta_{EZ,3}(b,a,c)+\zeta_{EZ,3}(a,b,c)\}.$$

Similarly, we treat the part $m<n$. Consequently, we see that the left-hand side is

$$\begin{aligned}(-1)^a\Big\{&(1+(-1)^a)\zeta_{EZ,3}(a,b,c)+(1+(-1)^b)\left(\zeta_{EZ,3}(b,a,c)+\zeta_{EZ,3}(b,c,a)\right)\\ &+(-1)^b(1+(-1)^c)\zeta_{EZ,3}(c,b,a)+\zeta_{EZ,2}(a+b,c)\\ &+(1+(-1)^b)\zeta_{EZ,2}(b,a+c)+(-1)^b\zeta_{EZ,2}(b+c,a)\Big\}.\end{aligned}$$

Secondly, we deform the right-hand side. Applying Corollary 8.8, we can rewrite it as

$$\begin{aligned}2(-1)^a\Bigg\{&\sum_{\xi=0}^{[a/2]}\zeta(2\xi)\sum_{\omega=0}^{a-2\xi}\binom{\omega+b-1}{\omega}\\ &\times\binom{a+c-2\xi-\omega-1}{c-1}\zeta_{EZ,2}(b+\omega,a+c-2\xi-\omega)\\ &+\sum_{\xi=0}^{[b/2]}\zeta(2\xi)\sum_{\omega=0}^{a-1}\binom{\omega+b-2\xi}{\omega}\\ &\times\binom{a+c-\omega-2}{c-1}\zeta_{EZ,2}(b-2\xi+\omega+1,a+c-1-\omega)\end{aligned}$$

$$+(-1)^b\sum_{\xi=0}^{[c/2]}\zeta(2\xi)\sum_{\omega=0}^{c-2\xi}\binom{\omega+b-1}{\omega}$$
$$\times\binom{a+c-2\xi-\omega-1}{a-1}\zeta_{EZ,2}(b+\omega,a+c-2\xi-\omega)$$
$$+(-1)^b\sum_{\xi=0}^{[b/2]}\zeta(2\xi)\sum_{\omega=0}^{c-1}\binom{\omega+b-2\xi}{\omega}$$
$$\times\binom{a+c-\omega-2}{a-1}\zeta_{EZ,2}(b-2\xi+\omega+1,a+c-1-\omega)\Big\}.$$

Thus we complete the proof of Theorem 12.7. □

Now we give the parity result for triple zeta values, which was first proved by Borwein and Girgensohn [**18**] (under the weaker assumption that $a+b+c$ is even or ≤ 10).

COROLLARY 12.8. *For $a,b,c\in\mathbb{N}$ with $a,b,c\ge 2$ satisfying that $a+b+c$ is even,*

$$\zeta_{EZ,3}(a,b,c)\in\mathbb{Q}[\{\zeta_{EZ,2}(j,k+1),\ \zeta(l+1)\mid j,k,l\in\mathbb{N}\}]. \tag{12.2.6}$$

PROOF. We first notice the following triple analogue of the harmonic product formula (1.1.4):

$$\begin{aligned}\zeta_{EZ,3}(a,b,c)+\zeta_{EZ,3}(b,a,c)+\zeta_{EZ,3}(b,c,a)\\ =\zeta(a)\zeta_{EZ,2}(b,c)-\zeta_{EZ,2}(b,c+a)-\zeta_{EZ,2}(a+b,c)\end{aligned} \tag{12.2.7}$$

for $a,b,c\in\mathbb{N}_{\ge 2}$. Similar to (1.1.4), this formula can be shown by considering the decomposition

$$\begin{aligned}\zeta(a)\zeta_{EZ,2}(b,c)&=\sum_{l=1}^{\infty}\frac{1}{l^a}\sum_{m<n}\frac{1}{m^bn^c}\\ &=\sum_{l<m<n}+\sum_{l=m<n}+\sum_{m<l<n}+\sum_{m<l=n}+\sum_{m<n<l}.\end{aligned}$$

Changing variables a and c in (12.2.7), we also obtain

$$\begin{aligned}\zeta_{EZ,3}(c,b,a)+\zeta_{EZ,3}(b,a,c)+\zeta_{EZ,3}(b,c,a)\\ =\zeta(c)\zeta_{EZ,2}(b,a)-\zeta_{EZ,2}(b,c+a)-\zeta_{EZ,2}(c+b,a).\end{aligned} \tag{12.2.8}$$

Let $a,b,c\in\mathbb{N}_{\ge 2}$ satisfying that $a+b+c$ is even. First, we assume that a,b,c are all even. Then, substituting (12.2.8) into the left-hand side of Theorem 12.7, we obtain (12.2.6). Next, we assume that a is even and b,c are odd. Then, directly from Theorem 12.7, we obtain (12.2.6). As for other cases, we can similarly obtain the assertion by

using Theorem 12.7, (12.2.7) and (12.2.8). Thus we complete the proof. □

REMARK 12.9 ([**116**, Proposition 5.1]). As for the type C_2, we can similarly obtain that for $p, q \in \mathbb{N}_{\geq 2}$,

$$\begin{aligned}(12.2.9)\quad &(1+(-1)^p)\,\zeta_{EZ,2}(p,q)+(1+(-1)^q)\,\zeta_{EZ,2}(q,p)\\ &=2\sum_{j=0}^{[p/2]}\binom{p+q-2j-1}{q-1}\zeta(2j)\zeta(p+q-2j)\\ &\quad+2\sum_{j=0}^{[q/2]}\binom{p+q-2j-1}{p-1}\zeta(2j)\zeta(p+q-2j)-\zeta(p+q),\end{aligned}$$

which recovers the parity result (12.1.5) for $\zeta_{EZ,2}(p,q)$ with $p, q \geq 2$.

EXAMPLE 12.10. We give explicit formulas coming from Theorem 12.7. Setting $(a,b,c)=(2,2,4)$, we have

$$\begin{aligned}&2\zeta_{EZ,3}(2,2,4)+2\{\zeta_{EZ,3}(2,2,4)+\zeta_{EZ,3}(2,4,2)\}+2\zeta_{EZ,3}(4,2,2)\\ &=2\zeta(4)\zeta_{EZ,2}(2,2)\\ &\quad+\zeta(2)\{8\zeta_{EZ,2}(4,2)+12\zeta_{EZ,2}(3,3)+16\zeta_{EZ,2}(2,4)+16\zeta_{EZ,2}(1,5)\}\\ &\quad-16\zeta_{EZ,2}(6,2)-20\zeta_{EZ,2}(5,3)-25\zeta_{EZ,2}(4,4)-24\zeta_{EZ,2}(3,5)\\ &\quad-17\zeta_{EZ,2}(2,6).\end{aligned}$$

Therefore, using (12.2.7) we obtain, as an example of Corollary 12.8,

$$\begin{aligned}\zeta_{EZ,3}(4,2,2)&=\zeta(4)\zeta_{EZ,2}(2,2)\\ &\quad+\zeta(2)\{4\zeta_{EZ,2}(4,2)+6\zeta_{EZ,2}(3,3)+7\zeta_{EZ,2}(2,4)+8\zeta_{EZ,2}(1,5)\}\\ &\quad-8\zeta_{EZ,2}(6,2)-10\zeta_{EZ,2}(5,3)-\frac{23}{2}\zeta_{EZ,2}(4,4)-12\zeta_{EZ,2}(3,5)\\ &\quad-\frac{15}{2}\zeta_{EZ,2}(2,6).\end{aligned}$$

12.3. The case of type B_r

In this section, we construct the B_r-analogue of the C_r-case results shown in the preceding section.

We have

$$\begin{aligned}(\Delta_{s+}(B_r))^\vee=\{&\alpha_r^\vee,\ 2\alpha_{r-1}^\vee+\alpha_r^\vee, 2\alpha_{r-2}^\vee+2\alpha_{r-1}^\vee+\alpha_r^\vee,\\ &\ldots, 2\alpha_1^\vee+\cdots+2\alpha_{r-1}^\vee+\alpha_r^\vee\}.\end{aligned}$$

Hence it follows from (3.3.4) with $\mathbf{s}=((s_{r+1-i})_{1\leq i\leq r},(0),(0))$ that

$$(12.3.1)\qquad \zeta_r(\mathbf{s};\Delta(B_r))=\zeta_r(s_1,\ldots,s_r;\Delta_{s+}(B_r))$$

$$= \sum_{m_1,\dots,m_r=1}^{\infty} \prod_{i=1}^{r} \frac{1}{(2\sum_{j=r-i+1}^{r-1} m_j + m_r)^{s_i}},$$

which is a partial sum of $\zeta_{EZ,r}(s_1,\dots,s_r)$.

It follows from the viewpoint of zeta-functions of root systems that values of (12.3.1) at positive integers can be regarded as the objects dual to multiple zeta values in the sense that B_r and C_r are dual of each other (see Remark 3.8). In view of this observation, we denote (12.3.1) by $\zeta_r^{\sharp}(s_1,\dots,s_r)$. For example,

$$\zeta_2^{\sharp}(s_1,s_2) = \sum_{l,m=1}^{\infty} \frac{1}{m^{s_1}(2l+m)^{s_2}},$$

$$\zeta_3^{\sharp}(s_1,s_2,s_3) = \sum_{l,m,n=1}^{\infty} \frac{1}{n^{s_1}(2m+n)^{s_2}(2l+2m+n)^{s_3}}.$$

It follows from Theorem 9.20 with $\mathbf{y} = \mathbf{0}$, $\Delta = \Delta(B_r)$ and $\Delta^* = \Delta_{s+}(B_r)$ that:

THEOREM 12.11 ([**116**, Theorem 6.1]). *Let* $\Delta = \Delta(B_r)$. *For* $k_1,\dots,k_r \in \mathbb{N}$,

(12.3.2)

$$\sum_{\sigma\in\mathfrak{S}_r} \zeta_r^{\sharp}(2k_{\sigma^{-1}(1)},\dots,2k_{\sigma^{-1}(r)})$$
$$= \frac{(-1)^r}{2^r} B_{\Delta_{s+}}((2k_1,\dots,2k_r);B_r)\left(\prod_{j=1}^{r}\frac{(2\pi\sqrt{-1})^{2k_j}}{(2k_j)!}\right) \in \mathbb{Q}\cdot\pi^{2\sum_{j=1}^{r}k_j}.$$

Hence we obtain a B_r-type analogue of (12.2.3).

COROLLARY 12.12 ([**116**, Corollary 6.2]). *For* $k \in \mathbb{N}$,

$$\zeta_r^{\sharp}(2k,\dots,2k) = \frac{(-1)^r}{2^r r!} B_{\Delta_{s+}}((2k,\dots,2k);B_r)\frac{(2\pi\sqrt{-1})^{2kr}}{((2k)!)^r} \in \mathbb{Q}\cdot\pi^{2kr}.$$

EXAMPLE 12.13. We can explicitly obtain

$$\zeta_2^{\sharp}(2,2) = \sum_{m,n=1}^{\infty}\frac{1}{n^2(2m+n)^2} = \frac{1}{320}\pi^4,$$

$$\zeta_2^{\sharp}(4,4) = \sum_{m,n=1}^{\infty}\frac{1}{n^4(2m+n)^4} = \frac{23}{14515200}\pi^8,$$

$$\zeta_2^{\sharp}(6,6) = \sum_{m,n=1}^{\infty}\frac{1}{n^6(2m+n)^6} = \frac{1369}{871782912000}\pi^{12}.$$

These formulas can be obtained by calculating the generating function of type B_2, and also from Theorem 12.14 below with $(p,q)=(2k,2k)$. However, unlike the double zeta values of Euler–Zagier type, these cannot be directly obtained from the harmonic product formula (1.1.4).

Similarly, from the generating function of type B_3, we can explicitly give examples of Corollary 12.12:

$$\zeta_3^\sharp(2,2,2)=\sum_{l,m,n=1}^{\infty}\frac{1}{n^2(2m+n)^2(2l+2m+n)^2}=\frac{1}{40320}\pi^6,$$

$$\zeta_3^\sharp(4,4,4)=\sum_{l,m,n=1}^{\infty}\frac{1}{n^4(2m+n)^4(2l+2m+n)^4}=\frac{23}{697426329600}\pi^{12},$$

$$\zeta_3^\sharp(6,6,6)=\sum_{l,m,n=1}^{\infty}\frac{1}{n^6(2m+n)^6(2l+2m+n)^6}=\frac{1997}{17030314057236480000}\pi^{18}.$$

Also, as an analogue of (12.2.9), we can prove the following.

THEOREM 12.14 ([**116**, Theorem 6.4]). *For $p,q\in\mathbb{N}$ with $p,q\geq 2$,*

(12.3.3)

$$\begin{aligned}&(1+(-1)^p)\zeta_2^\sharp(p,q)+(1+(-1)^q)\zeta_2^\sharp(q,p)\\&=2\sum_{j=0}^{[p/2]}\frac{1}{2^{p+q-2j}}\binom{p+q-1-2j}{q-1}\zeta(2j)\zeta(p+q-2j)\\&+2\sum_{j=0}^{[q/2]}\frac{1}{2^{p+q-2j}}\binom{p+q-1-2j}{p-1}\zeta(2j)\zeta(p+q-2j)-\zeta(p+q).\end{aligned}$$

PROOF. Let $p\in\mathbb{N}_{\geq 2}$ and $s\in\mathbb{R}_{>1}$. It follows from (8.3.13) that

$$\begin{aligned}&\sum_{\substack{l\neq 0,m\geq 1\\ l+m\neq 0}}\frac{(-1)^{l+m}x^m e^{\sqrt{-1}(l+m)\theta}}{l^p m^s}\\&\quad-2\sum_{j=0}^{p}\phi(p-j)\varepsilon_{p-j}\left\{\sum_{m=1}^{\infty}\frac{(-1)^m x^m e^{\sqrt{-1}m\theta}}{m^s}\right\}\frac{(\sqrt{-1}\theta)^j}{j!}\\&\quad+(-1)^p\sum_{m=1}^{\infty}\frac{x^m}{m^{s+p}}=0\end{aligned}$$

for $\theta\in[-\pi,\pi]$ and $x\in\mathbb{C}$ with $|x|\leq 1$. Setting $x=-e^{\sqrt{-1}\theta}$ on both sides and separating the term corresponding to $l+2m=0$ of the first

term on the left-hand side, we have

$$\sum_{\substack{l\neq 0,m\geq 1\\ l+m\neq 0\\ l+2m\neq 0}} \frac{(-1)^l e^{\sqrt{-1}(l+2m)\theta}}{l^p m^s} - 2\sum_{j=0}^{p} \phi(p-j)\varepsilon_{p-j} \left\{ \sum_{m=1}^{\infty} \frac{e^{2\sqrt{-1}m\theta}}{m^s} \right\} \frac{(\sqrt{-1}\theta)^j}{j!}$$

$$+ (-1)^p \sum_{m=1}^{\infty} \frac{(-1)^m e^{\sqrt{-1}m\theta}}{m^{s+p}} = -\sum_{m=1}^{\infty} \frac{1}{(-2m)^p m^s}.$$

By Lemma 8.24 with $d = q \geq 2$, we obtain

(12.3.4)

$$\sum_{\substack{l\neq 0,m\geq 1\\ l+m\neq 0\\ l+2m\neq 0}} \frac{(-1)^l e^{\sqrt{-1}(l+2m)\theta}}{l^p m^s (l+2m)^q}$$

$$= 2\sum_{j=0}^{p} \phi(p-j)\varepsilon_{p-j} \sum_{\xi=0}^{j} \binom{j-\xi+q-1}{j-\xi} \frac{(-1)^{j-\xi}}{2^{q+j-\xi}} \sum_{m=1}^{\infty} \frac{e^{2\sqrt{-1}m\theta}}{m^{s+q+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$- 2\sum_{j=0}^{q} \phi(q-j)\varepsilon_{q-j} \sum_{\xi=0}^{j} \binom{j-\xi+p-1}{j-\xi} \frac{(-1)^{p-1}}{2^{p+j-\xi}} \sum_{m=1}^{\infty} \frac{1}{m^{s+p+j-\xi}} \frac{(\sqrt{-1}\theta)^\xi}{\xi!}$$

$$- (-1)^p \sum_{m=1}^{\infty} \frac{(-1)^m e^{\sqrt{-1}m\theta}}{m^{s+p+q}}.$$

Setting $\theta = \pi$ and using Corollary 8.8, we see that the right-hand side of (12.3.4) is equal to

(12.3.5)

$$2(-1)^p \sum_{\xi=0}^{[p/2]} \frac{1}{2^{p+q-2\xi}} \binom{p+q-1-2\xi}{q-1} \zeta(2\xi)\zeta(s+p+q-2\xi)$$

$$+ 2(-1)^p \sum_{\xi=0}^{[q/2]} \frac{1}{2^{p+q-2\xi}} \binom{p+q-1-2\xi}{p-1} \zeta(2\xi)\zeta(s+p+q-2\xi)$$

$$- (-1)^p \zeta(s+p+q).$$

We confirm here that the left-hand side of (12.3.4) can be written in terms of the zeta-function of B_2. Here we use the notation

$$\zeta_2(s_1,s_2,s_3,s_4;B_2) = \sum_{m_1=1}^{\infty}\sum_{m_2=1}^{\infty} \frac{1}{m_1^{s_1} m_2^{s_2} (m_1+m_2)^{s_3} (2m_1+m_2)^{s_4}}$$

(see (3.3.22)). The contribution of the terms with $l > 0$ to the left-hand side is obviously $\zeta_2(s,p,0,q;B_2)$. As for the terms with $l < 0$, we rewrite $-l$ by l, divide the sum into three parts according to the conditions $l < m$, $m < l < 2m$ and $l > 2m$, and evaluate each part in

terms of the zeta-function of B_2. Then we find that the left-hand side of (12.3.4) is

(12.3.6)
$$\zeta_2(s,p,0,q;B_2)+(-1)^p\zeta_2(0,p,s,q;B_2)+(-1)^p\zeta_2(0,q,s,p;B_2)+(-1)^{p+q}\zeta_2(s,q,0,p;B_2).$$

Combine (12.3.5) and (12.3.6), and multiply by $(-1)^p$. We can set $s=0$ because (12.3.5) and (12.3.6) are absolutely convergent for $s>-1$. Therefore, since $\zeta_2(0,p,0,q;B_2)=\zeta_2^{\sharp}(p,q)$, we complete the proof of Theorem 12.14. □

Remark 12.15. Theorem 12.14 was applied by Kaneko and Tasaka [**85**] to their study on the dimension of the linear space spanned by double zeta values of level 2 (in connection with cusp forms of level 2). Bachmann [**12**] considered the q-analogue of

$$\zeta_{AV,2}(0,s_2,s_3)=\zeta_2(0,0,s_2,s_3;B_2)$$

(see (3.3.7), (3.3.22)) and expressed cusp forms as linear combinations of them.

Theorem 12.14 for the case that $p+q$ is odd gives the following parity result for $\zeta^{\sharp}$ with $p,q\geq 2$ (see [**116**, Corollary 6.5]).

Proposition 12.16. *Let $p,q\in\mathbb{N}$ with $q\geq 2$ and suppose $p+q$ is odd. Then*

$$\zeta_2^{\sharp}(p,q)\in\mathbb{Q}\left[\pi^2,\,\{\zeta(2j+1)\,|\,j\in\mathbb{N}\}\right]. \tag{12.3.7}$$

Proof. When $p,q\geq 2$, we obtain the assertion from Theorem 12.14 immediately. When $p=1$, this property can be given by combining the known facts for double zeta values and for their alternating series

$$\phi_2(s_1,s_2)=\sum_{m,n=1}^{\infty}\frac{(-1)^m}{n^{s_1}(m+n)^{s_2}}.$$

Actually, we see that

$$\zeta_2^{\sharp}(s_1,s_2)=\frac{1}{2}\left\{\zeta_{EZ,2}(s_1,s_2)+\phi_2(s_1,s_2)\right\}.$$

When $p+q$ is odd ($p,q\in\mathbb{N}$ and $q\geq 2$), Euler proved (12.1.5) and Borwein et al. proved that

$$\phi_2(p,q)\in\mathbb{Q}\left[\pi^2,\,\{\zeta(2j+1)\,|\,j\in\mathbb{N}\}\right]$$

(see [**16**]), from which Proposition 12.16 follows. □

EXAMPLE 12.17. We see that (12.3.3) gives more explicit information on the parity result for $\zeta_2^{\sharp}(p,q)$.

For example, setting $(p,q)=(3,2)$ in (12.3.3), we have

$$\zeta_2^{\sharp}(2,3)=\sum_{m,n=1}^{\infty}\frac{1}{n^2(2m+n)^3}=-\frac{21}{32}\zeta(5)+\frac{3}{8}\zeta(2)\zeta(3).$$

Combining the obtained results, we can prove the following parity result for the Apostol–Vu double zeta-functions, which has not been published before.

PROPOSITION 12.18. *Let $p,q,r\in\mathbb{N}$ and suppose $p+q+r$ is odd. Then*

$$\zeta_{AV,2}(p,q,r)\in\mathbb{Q}\left[\pi^2,\ \{\zeta(2j+1)\,|\,j\in\mathbb{N}\}\right]. \tag{12.3.8}$$

PROOF. By (3.3.7) and (3.3.22), we can regard $\zeta_{AV,2}(p,q,r)=\zeta_2(p,0,q,r;B_2)$. Let $\mathcal{V}=\mathbb{Q}\left[\pi^2,\ \{\zeta(2j+1)\,|\,j\in\mathbb{N}\}\right]$ and suppose $2\nmid(p+q+r)$. We use the partial fraction decomposition

$$\frac{1}{X(X+Y)}=\frac{1}{XY}-\frac{1}{Y(X+Y)} \tag{12.3.9}$$

several times in the following proof.

First, assume $p\geq 2$. Using (12.3.9) with $X=2m, Y=n$, we have

$$\zeta_{AV,2}(p,q,r)=\zeta_2(p,1,q,r-1;B_2)-2\zeta(p-1,1,q,r;B_2),$$

which belongs to $\mathcal{V}$ because of (8.2.41) and (8.3.21).

Next, consider the case $p=1$. By (12.3.9) with $X=m+n, Y=m$ we have

$$\zeta_{AV,2}(1,q,r)=\zeta_{AV,2}(2,q,r-1)-\zeta_{AV,2}(2,q-1,r), \tag{12.3.10}$$

which belongs to $\mathcal{V}$ if $q,r\geq 2$ by the fact just proved above. When $q=r=1$, we obtain from (8.2.3) that $\zeta_{AV,2}(1,1,1)\in\mathcal{V}$. If $q=1$ and $r\geq 2$, then we set $(k,l,s)=(1,r,1)$ for odd r in (8.2.2). Then we obtain $\zeta_{AV,2}(1,1,r)\in\mathcal{V}$. Finally, we consider the case $q\geq 2$ and $r=1$. Using (12.3.9) twice, we have for odd q,

$$\begin{aligned}\zeta_{AV,2}(1,q,1)&=\zeta_2(1,1,q;A_2)-2\zeta_2(0,1,q,1;B_2)\\&=\zeta_2(1,1,q;A_2)-2\left\{\zeta_2(1,1,q-1,1;B_2)-\zeta_{AV,2}(1,q,1)\right\}.\end{aligned}$$

Hence $\zeta_{AV,2}(1,q,1)\in\mathcal{V}$ because of (8.2.41) and (8.3.21). Thus we complete the proof. □

EXAMPLE 12.19. By the method stated in the proof of Proposition 12.18, we obtain, for example,

$$\begin{aligned}
\zeta_{AV,2}(1,1,1) &= \frac{3}{4}\zeta(3),\\
\zeta_{AV,2}(1,1,3) &= -\frac{1}{6}\zeta(3)\pi^2 + \frac{31}{16}\zeta(5),\\
\zeta_{AV,2}(1,2,2) &= -\frac{3}{4}\zeta(3)\pi^2 + \frac{69}{8}\zeta(5),\\
\zeta_{AV,2}(2,1,2) &= \frac{11}{12}\zeta(3)\pi^2 - \frac{83}{8}\zeta(5),\\
\zeta_{AV,2}(2,2,1) &= \frac{1}{6}\zeta(3)\pi^2 - \frac{7}{4}\zeta(5),\\
\zeta_{AV,2}(3,1,1) &= -\frac{1}{2}\zeta(3)\pi^2 + \frac{25}{4}\zeta(5).
\end{aligned}$$

REMARK 12.20. The Apostol–Vu double zeta-function can be regarded as a specialization of the zeta-function of B_2, and is also connected with the zeta-function of A_2 (see Corollary 8.5). Both of these two viewpoints are used in the proof of Proposition 12.18. Here we mention another viewpoint. Comparing (1.4.4) and (3.3.28), we can see that

$$\zeta_{AV,2}(s,t,u) = \zeta_2(0,s,t,u,0,0;G_2) \tag{12.3.11}$$

for $s,t,u \in \mathbb{C}$, where non-zero indices on the right-hand side correspond to $\Delta_{l+}(G_2)$. Hence, using the same method as in this and the preceding sections, we can give certain value-relations, or more generally functional relations, among the Apostol–Vu double zeta-function and the Riemann zeta-function. Those relations essentially coincide with those given in Corollary 8.5. Note that it is impossible to put $(p,u,v) = (0,0,0)$ in Theorem 11.7, because its proof is not available for the case $p = u = v = 0$.

REMARK 12.21. Recently, Hoffman [67] defined the odd variant of the multiple zeta value by

(12.3.12)

$$\begin{aligned}
t(a_1,\dots,a_r) &= \sum_{0<m_1<\cdots<m_r,\ m_j:\text{odd}} m_1^{-a_1}\cdots m_r^{-a_r}\\
&= \sum_{n_1\geq 0,\ n_2,\dots,n_r>0} (2n_1+1)^{-a_1}(2n_1+2n_2+1)^{-a_2}\cdots\\
&\qquad \times\cdots\times (2n_1+\cdots+2n_r+1)^{-a_r}
\end{aligned}$$

for $a_1,\dots,a_r \in \mathbb{N}$ with $a_r > 1$. Note that the order of summation of (12.3.12) is opposite to the original one in [67]. Comparing with

(12.3.1), we have

$$t(a_1,\dots,a_r)=\zeta_r^{\sharp}(a_1,\dots,a_r)-2^{-a_1-\cdots-a_r}\zeta_{EZ,r}(a_1,\dots,a_r).$$

12.4. Restricted sum formulas

In this section, we give a certain refinement of restricted sum formulas for multiple zeta values. Gangl, Kaneko and Zagier [47] proved the following restricted sum formulas, which "divide" (12.1.1) for even p into two parts with $N=(p+2)/2$:

$$\sum_{j=1}^{N-1}\zeta_{EZ,2}(2N-2j,2j)=\frac{3}{4}\zeta(2N)\in\mathbb{Q}\cdot\pi^{2N}, \tag{12.4.1}$$

$$\sum_{j=1}^{N-1}\zeta_{EZ,2}(2N-2j-1,2j+1)=\frac{1}{4}\zeta(2N)\in\mathbb{Q}\cdot\pi^{2N}$$

for $N\in\mathbb{Z}_{\geq 2}$. As their analogues, Shen and Cai [182] gave the following restricted sum formulas for triple and quadruple zeta values:

$$\sum_{\substack{a_1,a_2,a_3\in\mathbb{N}\\ a_1+a_2+a_3=N}}\zeta_{EZ,3}(2a_1,2a_2,2a_3)=\frac{5}{8}\zeta(2N)-\frac{1}{4}\zeta(2)\zeta(2N-2)\in\mathbb{Q}\cdot\pi^{2N} \tag{12.4.2}$$

for $N\in\mathbb{Z}_{\geq 3}$, and

$$\sum_{\substack{a_1,a_2,a_3,a_4\in\mathbb{N}\\ a_1+a_2+a_3+a_4=N}}\zeta_{EZ,4}(2a_1,2a_2,2a_3,2a_4) \tag{12.4.3}$$

$$=\frac{35}{64}\zeta(2N)-\frac{5}{16}\zeta(2)\zeta(2N-2)\in\mathbb{Q}\cdot\pi^{2N}$$

for $N\in\mathbb{Z}_{\geq 4}$. Then Hoffman [66] established the formula for the general case. His formula is

$$\sum_{\substack{a_1,\dots,a_k\in\mathbb{N}\\ a_1+\cdots+a_k=N}}\zeta_{EZ,k}(2a_1,\dots,2a_k)=\frac{1}{2^{2(k-1)}}\binom{2k-1}{k}\zeta(2N) \tag{12.4.4}$$

$$-\sum_{j=1}^{[(k-1)/2]}\frac{1}{2^{2k-3}(2j+1)B_{2j}}\binom{2k-2j-1}{k}\zeta(2j)\zeta(2N-2j).$$

Related articles were published by many authors, such as Machide [127] [128], Ding and Liu [33], Yuan and Zhao [213], Genčev [48], and Li and Qin [123]. Yuan and Zhao treated the same sum as the case $d=2$ of our Theorem 12.22 below. As for a different type of restricted sum formulas, see Eie's book [37].

In the restricted sum formulas stated above, the summations are taken over all tuples $(a_1,\dots,a_r)$ satisfying $a_1+\cdots+a_r=N$. On

the other hand, the summations in the formulas of Theorems 12.5 and 12.11 are running over much smaller range, namely, just all the permutations of one fixed $(a_1, \dots, a_r)$ with $a_1 + \cdots + a_r = N$. Therefore these theorems give subdivisions, or refinements, of known restricted sum formulas. Summing these formulas for all tuples $(a_1, \dots, a_r)$ satisfying $a_1 + \cdots + a_r = N$, we can obtain the r-fold version of (12.4.1), (12.4.2) and (12.4.3).

We prove a further generalization, which gives a new type of restricted sum formulas. For $d \in \mathbb{N}$ and $N \in \mathbb{N}$, let

$$I_r(d, N) = \{(2da_1, \dots, 2da_r) \in (2d\mathbb{N})^r \,|\, a_1 + \cdots + a_r = N\}.$$

Denote by Π_r the set of all partitions of r, namely

$$\Pi_r = \bigcup_{\nu=1}^{r} \{(j_1, \cdots, j_\nu) \in \mathbb{N}^\nu \,|\, j_1 + \cdots + j_\nu = r\}.$$

For $J = (j_1, \cdots, j_\nu) \in \Pi_r$, we set

$$\mathcal{A}_r(d, N, J) = \left\{((2dh_1)^{[j_1]}, \dots, (2dh_\nu)^{[j_\nu]}) \in I_r(d, N) \,|\, h_1 < \cdots < h_\nu\right\},$$

where $(2h)^{[j]} = (2h, \dots, 2h) \in (2\mathbb{N})^j$. Then we have the following restricted sum formulas of depth r.

THEOREM 12.22 ([116, Theorem 7.1]). *For $d \in \mathbb{N}$ and $N \in \mathbb{N}$ with $N \geq r$,*

(12.4.5)

$$\sum_{\substack{a_1, \dots, a_r \in \mathbb{N} \\ a_1 + \cdots + a_r = N}} \zeta_{EZ,r}(2da_1, \dots, 2da_r)$$
$$= \frac{(-1)^r}{2^r} \sum_{J=(j_1, \cdots, j_\nu) \in \Pi_r} \frac{1}{j_1! \cdots j_\nu!}$$
$$\times \sum_{(2d\mathbf{k})_l \in \mathcal{A}_r(d,N,J)} B_{\Delta_{l+}}((2d\mathbf{k})_l; \Delta(C_r)) \prod_{\rho=1}^{r} \frac{(2\pi\sqrt{-1})^{2dk_\rho}}{(2k_\rho)!} \in \mathbb{Q} \cdot \pi^{2dN}.$$

REMARK 12.23. The cases $d = 1$ and $r = 2, 3, 4$ essentially coincide with (12.4.1), (12.4.2) and (12.4.3). Also, for the case $N = r$, we obtain (12.2.3). In the case $d = 1$ and $r \geq 2$, Muneta [159] conjectured an explicit expression of the left-hand side of (12.4.5) in terms of $\zeta(2k)$, $k \in \mathbb{N}$. The above theorem gives an answer to Muneta's conjecture.

PROOF OF THEOREM 12.22. Let $(2da_1, \dots, 2da_r) \in I_r(d, N)$. Denote a set of different elements in $\{a_1, \dots, a_r\}$ by $\{h_1, \dots, h_\nu\}$, and put $j_\mu = \#\{a_m \,|\, a_m = h_\mu\}$ $(1 \leq \mu \leq \nu)$. We may assume $h_1 < \cdots < h_\nu$.

We can check that there exist $\sigma \in \mathfrak{S}_r$ and $((2dh_1)^{[j_1]}, \dots, (2dh_\nu)^{[j_\nu]}) \in \mathcal{A}_r(d,N,J)$ with $J = (j_1, \cdots, j_\nu) \in \Pi_r$ such that

$$(2da_1, \dots, 2da_r) = ((2dh_1)^{[j_1]}, \dots, (2dh_\nu)^{[j_\nu]})^\sigma,$$

where we use the notation

$$(k_1, \dots, k_r)^\sigma = (k_{\sigma(1)}, \dots, k_{\sigma(r)}).$$

On the other hand, the set $\{\left((2dh_1)^{[j_1]}, \dots, (2dh_\nu)^{[j_\nu]}\right)^\tau \mid \tau \in \mathfrak{S}_r\}$ contains $j_1! \cdots j_\nu!$-copies of each element. Actually, if we denote by $\mathfrak{S}(1, \dots, j_1)$ the set of all permutations among $\{1, \dots, j_1\}$, then

$$\mathfrak{X}(J) = \mathfrak{S}(1, \dots, j_1) \times \mathfrak{S}(j_1+1, \dots, j_1+j_2) \times \cdots \times \mathfrak{S}(\sum_{\rho=1}^{\nu-1} j_\rho + 1, \dots, \sum_{\rho=1}^{\nu} j_\rho) \subset \mathfrak{S}_r$$

forms the stabilizer subgroup of $((2dh_1)^{[j_1]}, \dots, (2dh_\nu)^{[j_\nu]})$, and hence $\#\mathfrak{X}(J) = j_1! \cdots j_\nu!$. Therefore, using Theorem 12.5, we have

$$\begin{aligned}
&\sum_{\substack{a_1,\dots,a_r \in \mathbb{N} \\ a_1+\cdots+a_r=N}} \zeta_{EZ,r}(2da_1, \dots, 2da_r) = \sum_{(2da_1,\dots,2da_r) \in I_r(d,N)} \zeta_{EZ,r}(2da_1, \dots, 2da_r) \\
&= \sum_{J=(j_1,\cdots,j_\nu)\in P_r} \frac{1}{j_1! \cdots j_\nu!} \sum_{\substack{(2dk_1,\dots,2dk_r) \\ \in \mathcal{A}_r(d,N,J)}} \sum_{\sigma \in \mathfrak{S}_r} \zeta_{EZ,r}(2dk_{\sigma(1)}, \dots, 2dk_{\sigma(r)}) \\
&= \frac{(-1)^r}{2^r} \sum_{J=(j_1,\cdots,j_\nu)\in \Pi_r} \frac{1}{j_1! \cdots j_\nu!} \\
&\quad \times \sum_{(2d\mathbf{k})_l \in \mathcal{A}_r(d,N,J)} B_{\Delta_{l+}(C_r)}((2d\mathbf{k})_l; \Delta) \prod_{\rho=1}^{r} \frac{(2\pi\sqrt{-1})^{2dk_\rho}}{(2dk_\rho)!}.
\end{aligned}$$

Thus we obtain the proof. □

Similarly, using Theorem 12.11, we obtain the following.

THEOREM 12.24 ([116, Theorem 7.3]). *For $d \in \mathbb{N}$ and $N \in \mathbb{N}$ with $N \geq r$,*

$$\begin{aligned}
&\sum_{\substack{a_1,\dots,a_r \in \mathbb{N} \\ a_1+\cdots+a_r=N}} \zeta_r^\sharp(2da_1, \dots, 2da_r) \\
&= \frac{(-1)^r}{2^r} \sum_{J=(j_1,\cdots,j_\nu)\in \Pi_r} \frac{1}{j_1! \cdots j_\nu!} \\
&\quad \times \sum_{(2d\mathbf{k})_s \in \mathcal{A}_r(d,N,J)} B_{\Delta_{s+}}((2d\mathbf{k})_s; \Delta(B_r)) \prod_{\rho=1}^{r} \frac{(2\pi\sqrt{-1})^{2dk_\rho}}{(2k_\rho)!} \in \mathbb{Q} \cdot \pi^{2dN}.
\end{aligned}$$

12.5. Parity results

As introduced in Section 1.3, the parity result for the multiple zeta value can be stated as

"$\zeta_{EZ,r}(p_1, p_2, \dots, p_r)$ can be expressed as a rational linear combination of products of multiple zeta values of depth lower than r, provided its depth r and its weight $\sum_{j=1}^{r} p_j$ are of different parity".

As stated in Section 12.2, Euler's result (12.1.5) is the prototype. The above result was shown by Goncharov [**49**], Ihara, Kaneko and Zagier [**72**], and the third-named author [**198**] independently.

In the 1950s, Tornheim [**194**] proved the parity result for double zeta values of the root system of type A_2, which were originally called Tornheim's double sums or Mordell–Tornheim double zeta values (see Corollary 8.21). Later Huard, Williams and Zhang [**69**] gave explicit expressions of these values in terms of the Riemann zeta values like Proposition 12.4. As its generalization, the third-named author [**201**] proved the parity result for Mordell–Tornheim multiple zeta values.

From these observations, we can formulate the parity result for multiple series as

"the multiple series can be expressed as a rational linear combination of products of (the same type of) multiple series of lower depth, provided its depth and its weight are of different parity".

Note that the parity condition may sometimes be replaced by "of the same parity".

It is unclear whether it is possible to show some parity-type result for more general zeta-functions of root systems as in the case of type A_2. Proposition 12.16 asserts that the dual of A_2, namely $\zeta_2^{\sharp}$, has the parity result. As for the type $C_2(\simeq B_2)$, the third-named author [**200**] proved the parity result (Proposition 8.31) by using partial fraction decompositions similar to [**69**]. Furthermore, the latter result implies the parity result for the Apostol–Vu double zeta-function (Proposition 12.18).

Inspired by this result and the results in Nakamura [**163**] and Onodera [**174**], Okamoto [**168**] considered the double series defined by

$$\begin{aligned}&\zeta_{\mathbf{a}_r,\mathbf{b}_r}(s_1, s_2, \dots, s_{r+2})\\ &= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} m^{-s_1} n^{-s_2} (a_1 m + b_1 n)^{-s_3} \cdots (a_r m + b_r n)^{-s_{r+2}}\end{aligned}$$

for a positive integer r, where $\mathbf{a}_r = (a_1, a_2, \dots, a_r)$, $\mathbf{b}_r = (b_1, b_2, \dots, b_r) \in \mathbb{N}^r$. This includes the zeta-functions of A_2, C_2 and G_2 as special

cases. Okamoto proved that some special values of this double series can be expressed in terms of Clausen-type functions defined by

$$S_r(x) = \sum_{m\geq 1} \frac{\sin(2\pi m x)}{m^r} \quad \text{or} \quad C_r(x) = \sum_{m\geq 1} \frac{\cos(2\pi m x)}{m^r} \qquad (r \in \mathbb{N})$$

with $x = j/l \in \mathbb{Q}$ $(l \in \mathbb{N},\ 0 \leq j < l)$ under a certain parity condition. Moreover, in his formula, in the case of G_2, only the cases $l = 1, 2, 3, 4, 6$ and 12 of Clausen-type functions appear. For $l = 1, 2, 3$ and 6, the values $S_r(j/l)$ and $C_r(j/l)$ can be written in terms of the values of $\zeta(s)$ and $L(s, \chi_3)$. Therefore, Okamoto's result implies that *if* $2 \nmid (p + q + r + u + v + w)$, *then the value* $\zeta_2(p, q, r, u, v, w; G_2)$ *can be written in terms of* $\zeta(s)$, $L(s, \chi_3)$, $S_r(j/l)$ *and* $C_r(j/l)$ *for* $l = 4, 12$ and $0 < j < l$ *with* $(j, l) = 1$.

On the other hand, in view of Example 8.34 (especially (8.4.2)), we may guess that the following statement is plausible:

$$\zeta_2(p, q, r, u, v, w; G_2) \overset{?}{\in} \mathbb{Q}[\{\zeta(j+1),\ L(j, \chi_3) \mid j \in \mathbb{N}\}] \tag{12.5.1}$$

for $p, q, r, u, v, w \in \mathbb{N}$ with $2 \nmid (p + q + r + u + v + w)$.

This conjecture was raised in [**117**, Section 7]. We can resolve this conjecture if we can remove $S_r(j/l)$ and $C_r(j/l)$ from Okamoto's statement. This was done by Kadota, Okamoto and Tasaka [**80**], and hence the above conjecture (12.5.1) is now completely established.

This result (12.5.1) may be regarded as a kind of "generalized parity result" (involving Dirichlet L-values). It seems that for the Witten zeta-function of G_2, the parity-type result is valid only in this generalized form. On the other hand, our Examples 11.10 and 11.11 show that sometimes the value $\zeta_2(p, q, r, u, v, w; G_2)$, with $2 \nmid (p+q+r+u+v+w)$, can be expressed only by the values of $\zeta(s)$. It is an interesting problem to determine when such a restricted form of the parity result holds.

In a more generalized framework, Kadota [**79**] formulated a certain parity result for multiple Dirichlet series, which includes zeta-functions of root systems. His proof is based on the theory developed in Chapter 16. From this result, however, it is not easy to give explicit expressions of parity results.



CHAPTER 13

Applications to multiple zeta values (II)

In this chapter, we briefly introduce an algebraic setup for the theory of MZVs defined by Hoffman, and give a new interpretation of the shuffle product of MZVs obtained in [109]. In fact, we show that the procedure of shuffle products essentially coincides with that of partial fraction decompositions of zeta values of root systems of type A_r.

13.1. Hoffman's algebra

We begin by constructing Hoffman's algebraic setup for the theory of MZVs with no proof. For the details, see [64].

Let $\mathfrak{H} = \mathbb{Q}\langle x, y\rangle$ be the non-commutative polynomial algebra over $\mathbb{Q}$ in indeterminates x, y. The monomial $u_1u_2\cdots u_d \in \mathfrak{H}$, where each $u_j = x$ or y, is often called a "word". Let $\mathfrak{H}^1$ and $\mathfrak{H}^0$ be its subalgebras $\mathbb{Q} + y\mathfrak{H}$ and $\mathbb{Q} + y\mathfrak{H}x$, respectively. Let $Z : \mathfrak{H}^0 \to \mathbb{R}$ be the $\mathbb{Q}$-linear map defined by

$$Z\left(yx^{k_1-1}yx^{k_2-1}\cdots yx^{k_r-1}\right) = \zeta_{EZ,r}\left(k_1, k_2, \cdots, k_r\right). \tag{13.1.1}$$

Then it is known that

$$Z(u_1u_2\cdots u_d) = \int\cdots\int_{0<t_1<\cdots<t_d<1} \omega_{u_1}(t_1)\omega_{u_2}(t_2)\cdots\omega_{u_d}(t_d), \tag{13.1.2}$$

where $\omega_x(t) = dt/t$ and $\omega_y(t) = dt/(1-t)$, and the right-hand side of (13.1.2) is called the *Drinfel'd integral*. By this setup, Hoffman gave several relation formulas for MZVs (see [64, 65, 68]).

Remark 13.1. We note here that the above setup is a slight modification of the original setup of Hoffman. In fact, in Hoffman's paper, the order of x and y is reversed in $\mathfrak{H}$ corresponding to the MZV defined by the reverse order of summation, that is,

$$\sum_{m_1>m_2>\cdots>m_r\geq 1} \frac{1}{m_1^{k_1}m_2^{k_2}\cdots m_r^{k_r}} \quad (k_1,\ldots,k_r \in \mathbb{N};\ k_1 > 1).$$

(This reverse order of MZV is also used in our original paper [109].)

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For simplicity, we let $z_p = yx^{p-1} \in \mathfrak{H}^1$ $(p \in \mathbb{N})$. It follows that $\{z_p \,|\, p \in \mathbb{N}\}$ and 1 are the generators of $\mathfrak{H}^1$. On $\mathfrak{H}^1$, the harmonic product $*$ is defined by

$$1 * w_1 = w_1 * 1 = w_1, \tag{13.1.3}$$
$$w_1 z_p * w_2 z_q = (w_1 * w_2 z_q) z_p + (w_1 z_p * w_2) z_q + (w_1 * w_2) z_{p+q},$$

for any $p, q \in \mathbb{N}$ and $w_1, w_2 \in \mathfrak{H}^1$, and extended by $\mathbb{Q}$-bilinearity. Here, $w_1 z_p * w_2 z_q$ should be read as $(w_1 z_p) * (w_2 z_q)$, but we omit the parentheses. We will use the same convention for $\sqcup\!\sqcup$ products. With respect to the harmonic product, it is known that $\mathfrak{H}^1$ becomes a commutative algebra, $\mathfrak{H}^0$ is its subalgebra, and the map Z defined by (13.1.2) is an algebra-homomorphism, that is,

$$Z(w_1 * w_2) = Z(w_1) Z(w_2) \tag{13.1.4}$$

for $w_1, w_2 \in \mathfrak{H}^0$.

By repeated application of (13.1.3), we find that $w_1 * w_2$ $(w_1, w_2 \in \mathfrak{H}^0)$ can be written as a linear combination of words in $\mathfrak{H}^0$. We call this procedure the *harmonic product procedure* (HPP) for $w_1 * w_2$. For example, we have $z_p * z_q = z_p z_q + z_q z_p + z_{p+q}$ from definition (13.1.3). Hence, by (13.1.1) and (13.1.4), we obtain the harmonic product relation (see (1.1.4)), that is,

$$\zeta(p)\zeta(q) = \zeta_{EZ,2}(p, q) + \zeta_{EZ,2}(q, p) + \zeta(p + q). \tag{13.1.5}$$

On the other hand, the shuffle product $\sqcup\!\sqcup$ is defined by

$$1 \sqcup\!\sqcup w_1 = w_1 \sqcup\!\sqcup 1 = w_1, \tag{13.1.6}$$
$$w_1 u_1 \sqcup\!\sqcup w_2 u_2 = (w_1 \sqcup\!\sqcup w_2 u_2) u_1 + (w_1 u_1 \sqcup\!\sqcup w_2) u_2,$$

for $w_1, w_2 \in \mathfrak{H}^1$ and $u_1, u_2 \in \{x, y\}$, and extended by $\mathbb{Q}$-bilinearity. This product can also be defined on $\mathfrak{H}$. Similar to the case of $*$, with respect to $\sqcup\!\sqcup$, we see that $\mathfrak{H}^1$ becomes a commutative algebra, $\mathfrak{H}^0$ is its subalgebra, and the map Z is an algebra-homomorphism, namely,

$$Z(w_1 \sqcup\!\sqcup w_2) = Z(w_1) Z(w_2) \tag{13.1.7}$$

for $w_1, w_2 \in \mathfrak{H}^0$ (see [68, Theorem 4.1]). Applying (13.1.6) repeatedly, we find that $w_1 \sqcup\!\sqcup w_2$ $(w_1, w_2 \in \mathfrak{H}^0)$ can be written as a linear combination of words in $\mathfrak{H}^0$. We call this procedure the *shuffle product procedure* (SPP) for $w_1 \sqcup\!\sqcup w_2$. Combining (13.1.4) and (13.1.7), we obtain

$$Z(w_1 * w_2) = Z(w_1 \sqcup\!\sqcup w_2). \tag{13.1.8}$$

Combining (13.1.8) with HPP for $w_1 * w_2$ and SPP for $w_1 \sqcup\!\sqcup w_2$, we obtain non-trivial relations for MZVs, which are called (finite) *double shuffle relations*.

The harmonic product $*$, via the mapping Z, corresponds to harmonic product formulas for MZVs such as (1.1.4), (12.2.7). Those harmonic product formulas can be shown by using the series expression of MZVs.

On the other hand, a well-known method of performing SPP is via Drinfel'd integrals. The main aim of this chapter is to give a new interpretation of SPP, which can be realized by series expressions (of zeta-functions of root systems). From definition (13.1.6), we can obtain

(13.1.9)
$$z_p \sqcup\!\sqcup z_q z_r = \sum_{i=0}^{r-1} \binom{p-1+i}{i} z_q z_{r-i} z_{p+i} + \sum_{i=0}^{p-1} \binom{r-1+i}{i} (z_{p-i} \sqcup\!\sqcup z_q) z_{r+i}$$

for $p, q, r \in \mathbb{Z}$ with $p, r \geq 2$. On the other hand, we will prove in Section 13.3 that

(13.1.10)
$$\begin{aligned} \zeta(p)\zeta_{EZ,2}(q,r) = &\sum_{i=0}^{r-1} \binom{p-1+i}{i} \sum_{l,m,n=1}^{\infty} \frac{1}{m^q (m+n)^{r-i} (l+m+n)^{p+i}} \\ &+ \sum_{i=0}^{p-1} \binom{r-1+i}{i} \sum_{l,m,n=1}^{\infty} \frac{1}{l^{p-i} m^q (l+m+n)^{r+i}}, \end{aligned}$$

by using the partial fraction decomposition. The left-hand side of (13.1.9) corresponds to that of (13.1.10) via the Z-map (by (13.1.1), (13.1.7)). Also, the first sum on the right-hand side of (13.1.9) corresponds to that on the right-hand side of (13.1.10) via the Z-map. Hence the second sum on the right-hand side of (13.1.9) corresponds to that on the right-hand side of (13.1.10) via the Z-map. Therefore the following correspondence is plausible:

$$Z((z_{p-i} \sqcup\!\sqcup z_q) z_{r+i}) = \sum_{l,m,n=1}^{\infty} \frac{1}{l^{p-i} m^q (l+m+n)^{r+i}}. \tag{13.1.11}$$

From this observation, we can expect that there exists an explicit correspondence between SPP and the partial fraction decomposition. In this chapter we will confirm this expectation.

13.2. Extension of Hoffman's setup

We recall here the zeta-function of the root system of type A_r defined by (3.2.2), namely

$$\zeta_r(\mathbf{s}; A_r) = \sum_{l_1=1}^{\infty} \cdots \sum_{l_r=1}^{\infty} \prod_{1\le i\le j\le r} (l_i + \cdots + l_j)^{-s_{ij}}, \tag{13.2.1}$$

where $\mathbf{s} = (s_{ij})_{1\le i\le j\le r} \in \mathbb{C}^{(r+1)r/2}$. In particular,

$$\zeta_r(s_{11}, s_{12}, \ldots, s_{1r}, 0, 0, \ldots, 0; A_r) = \zeta_{EZ,r}(s_{11}, s_{12}, \ldots, s_{1r}).$$

We aim to extend Hoffman's setup. For this aim, we first prepare the following proposition, which is a generalization of (13.1.9).

Proposition 13.2 ([**109**, Proposition 1]). *Let $a, b \in \mathbb{N}$, and let $p_1, \ldots, p_a, q_1, \ldots, q_b \in \mathbb{N}$. We have*

$$\begin{aligned} &z_{p_1} \cdots z_{p_a} \sqcup\!\sqcup z_{q_1} \cdots z_{q_b} \\ &\quad = \sum_{\tau=0}^{q_b-1} \binom{p_a - 1 + \tau}{\tau} (z_{p_1} \cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}} z_{q_b-\tau}) z_{p_a+\tau} \\ &\qquad + \sum_{\tau=0}^{p_a-1} \binom{q_b - 1 + \tau}{\tau} (z_{p_1} \cdots z_{p_{a-1}} z_{p_a-\tau} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}}) z_{q_b+\tau}. \end{aligned} \tag{13.2.2}$$

Consequently, SPP is performed by applying this proposition repeatedly. Then we can see that only terms of the form

$$(z_{p_1} \cdots z_{p_a} \sqcup\!\sqcup z_{q_1} \cdots z_{q_b}) z_{r_1} \cdots z_{r_c}$$

appear in SPP. Hence, from (13.2.2), we find that each step of SPP is expressed by

(13.2.3)

$$\begin{aligned} &(z_{p_1} \cdots z_{p_a} \sqcup\!\sqcup z_{q_1} \cdots z_{q_b}) z_{r_1} \cdots z_{r_c} \\ &= \sum_{\tau=0}^{q_b-1} \binom{p_a - 1 + \tau}{\tau} (z_{p_1} \cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}} z_{q_b-\tau}) z_{p_1+\tau} z_{r_1} \cdots z_{r_c} \\ &\quad + \sum_{\tau=0}^{p_a-1} \binom{q_b - 1 + \tau}{\tau} (z_{p_1} \cdots z_{p_{a-1}} z_{p_a-\tau} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}}) z_{q_b+\tau} z_{r_1} \cdots z_{r_c}. \end{aligned}$$

For each term on both sides, we can determine its image via the Z-map as follows.

Theorem 13.3 ([**109**, Theorem 1]). *Let $a, b \in \mathbb{N}$ and $c \in \mathbb{N}_0$. For $(p_\eta) \in \mathbb{N}^a$, $(q_\xi) \in \mathbb{N}^b$, $(r_\sigma) \in \mathbb{N}^c$,*

$$Z\left((z_{p_1} \cdots z_{p_a} \sqcup\!\sqcup z_{q_1} \cdots z_{q_b}) z_{r_1} \cdots z_{r_c}\right) = \zeta_{a+b+c}\left(\{d_{ij}\}; A_{a+b+c}\right) \tag{13.2.4}$$

holds under the assumption that the right-hand side of (13.2.4) *is convergent, where*

$$d_{ij} = \begin{cases} p_j & (i = 1;\ 1 \le j \le a) \\ q_{j-a} & (i = a+1;\ a+1 \le j \le a+b) \\ r_{j-a-b} & (i = 1;\ a+b+1 \le j \le a+b+c) \\ 0 & (\textit{otherwise}). \end{cases}$$

Furthermore, each step of the shuffle product procedure (SPP) given by (13.2.3) *can be realized as the corresponding step of partial fraction decompositions (PFD) of zeta values of root systems.*

Proofs of Proposition 13.2 and Theorem 13.3 will be given in Section 13.4.

Remark 13.4. Theorem 13.3 can be illustrated as the following commutative diagram of procedures:

$$\begin{array}{ccc} \boxed{\mathfrak{H}^0} & \xrightarrow{Z} & \boxed{\text{Zeta values of root systems}} \\ \text{SPP}\downarrow & & \downarrow\text{PFD} \\ \boxed{\mathfrak{H}^0} & \xrightarrow{Z} & \boxed{\text{Zeta values of root systems}} \\ \text{SPP}\downarrow & & \downarrow\text{PFD} \end{array}$$

where SPP implies the step determined by (13.2.3) and PFD implies that of a partial fraction decomposition which corresponds to SPP. The explicit correspondence between SPP and PFD will be given in the proof of Theorem 13.3.

Remark 13.5. In Theorem 13.3, we only consider the case that the right-hand side of (13.2.4) is convergent. We can explicitly state the condition of this case as follows: $p_a \ge 2$, $q_b \ge 2$ if $c = 0$; $r_c \ge 2$ if $c \ge 1$.

Remark 13.6. Each step of SPP is given by (13.2.3), but each step itself is actually obtained by applying repeatedly the fundamental shuffle product relation (13.1.6). It is possible to show that each (13.1.6) can also be realized as a PFD of zeta values of root systems; see [**109**, Section 6]. This is a refinement of Theorem 13.3.

We recall here the algebra-homomorphism property (13.1.7) of Z with respect to $\sqcup\!\sqcup$:

$$Z(w_1 \sqcup\!\sqcup w_2) = Z(w_1)Z(w_2) \quad (w_1, w_2 \in \mathfrak{H}^0).$$

In Section 13.4 we will also prove the following theorem. Note that Goncharov observed this fact in [50, Section 7].

THEOREM 13.7 ([109, Theorem 2]). *The algebra-homomorphism property of the Z-map with respect to $\sqcup\!\sqcup$:*

$$Z(w_1 \sqcup\!\sqcup w_2) = Z(w_1)Z(w_2) \quad (w_1, w_2 \in \mathfrak{H}^0) \tag{13.2.5}$$

can be proved by using only the partial fraction decompositions.

REMARK 13.8. Ihara, Kaneko and Zagier [72, Proposition 1] defined two algebra homomorphisms

$$Z^{\sqcup\!\sqcup} : \mathfrak{H}^1 \to \mathbb{R}[T], \quad Z^* : \mathfrak{H}^1 \to \mathbb{R}[T]$$

with respect to $\sqcup\!\sqcup$ and $*$, which are uniquely characterized by the properties that they both extend the Z-map $\mathfrak{H}^0 \to \mathbb{R}$ and send y to an indeterminate T. It should be noted that the extended double shuffle relations are expressed as

$$Z^{\sqcup\!\sqcup}(w_1 * w_2 - w_1 \sqcup\!\sqcup w_2) = 0 \quad (w_1 \in \mathfrak{H}^1,\ w_2 \in \mathfrak{H}^0), \tag{13.2.6}$$

$$Z^*(w_1 * w_2 - w_1 \sqcup\!\sqcup w_2) = 0 \quad (w_1 \in \mathfrak{H}^1,\ w_2 \in \mathfrak{H}^0) \tag{13.2.7}$$

(see [72, Theorem 2]). We emphasize here that we proved (13.2.5) without using Drinfel'd integral expressions for MZVs, by using only the partial fraction decompositions, and also proved (13.1.8) by using only the series representation. Therefore, combining these results and the results in [72, Section 3], we can consequently obtain the extended double shuffle relations (13.2.6) and (13.2.7) without using Drinfel'd integral expressions for MZVs.

13.3. Double and triple zeta values

In this section, we discuss double and triple zeta values. Though results in this section are included in Theorem 13.3, we insert this section so that the readers can easily catch the essence of our idea from the description of these simple examples.

First, we consider the case of double series. It follows from (13.1.6) that SPP for $z_p \sqcup\!\sqcup z_q$ gives

(13.3.1)

$$z_p \sqcup\!\sqcup z_q = \sum_{\tau=0}^{q-1} \binom{p-1+\tau}{\tau} z_{q-\tau} z_{p+\tau} + \sum_{\tau=0}^{p-1} \binom{q-1+\tau}{\tau} z_{p-\tau} z_{q+\tau},$$

which is mapped to

$$Z(z_p \sqcup\!\sqcup z_q) = \sum_{\tau=0}^{q-1} \binom{p-1+\tau}{\tau} \zeta_{EZ,2}(q-\tau, p+\tau) \tag{13.3.2}$$

$$+\sum_{\tau=0}^{p-1}\binom{q-1+\tau}{\tau}\zeta_{EZ,2}(p-\tau,q+\tau)$$

via the Z-map for $p,q\in\mathbb{Z}$ with $p,q\geq 2$. Here we recall the formula of the partial fraction decomposition given in Lemma 12.2:

$$\frac{1}{X^{\alpha}Y^{\beta}}=\sum_{\tau=0}^{\beta-1}\binom{\alpha-1+\tau}{\tau}\frac{1}{Y^{\beta-\tau}(X+Y)^{\alpha+\tau}}+\sum_{\tau=0}^{\alpha-1}\binom{\beta-1+\tau}{\tau}\frac{1}{X^{\alpha-\tau}(X+Y)^{\beta+\tau}}. \tag{13.3.3}$$

It follows from (13.3.3) that

$$\zeta(p)\zeta(q)=\sum_{\tau=0}^{q-1}\binom{p-1+\tau}{\tau}\zeta_{EZ,2}(q-\tau,p+\tau)+\sum_{\tau=0}^{p-1}\binom{q-1+\tau}{\tau}\zeta_{EZ,2}(p-\tau,q+\tau). \tag{13.3.4}$$

Comparing (13.3.2) and (13.3.4), we obtain

$$Z(z_p \sqcup\!\sqcup z_q)=\zeta(p)\zeta(q)=Z(z_p)Z(z_q).$$

The above argument implies the assertion in the double case of Theorem 13.7 and also implies that the shuffle product (13.3.1) corresponds to the partial fraction decomposition (13.3.3). It is noted that Euler already noticed the correspondence between (13.3.3) and (13.3.4) (see Borwein et al. [**17**]).

Secondly, we consider the case of triple series. Setting $(X,Y)=(l,m+n)$ and $(\alpha,\beta)=(p,r)$ in the partial fraction decomposition (13.3.3), multiplying both sides by m^{-q}, and then summing up with respect to l,m and n, we obtain (13.1.10). Next, we prove (13.1.11). From (13.3.1), we can see that

$$(z_{p-i} \sqcup\!\sqcup z_q)z_{r+i}=\sum_{\tau=0}^{q-1}\binom{p-i-1+\tau}{\tau}z_{q-\tau}z_{p-i+\tau}z_{r+i}+\sum_{\tau=0}^{p-i-1}\binom{q-1+\tau}{\tau}z_{p-i-\tau}z_{q+\tau}z_{r+i}.$$

The right-hand side is mapped to

$$\sum_{\tau=0}^{q-1}\binom{p-i-1+\tau}{\tau}\zeta_{EZ,3}(q-\tau,p-i+\tau,r+i) \tag{13.3.5}$$

$$+\sum_{\tau=0}^{p-i-1}\binom{q-1+\tau}{\tau}\zeta_{EZ,3}(p-i-\tau,q+\tau,r+i)$$

via the Z-map. On the other hand, setting $(X,Y)=(l,m)$ and $(\alpha,\beta)=(p-i,q)$ in the partial fraction decomposition (13.3.3), multiplying both sides by $(l+m+n)^{-r-i}$, and then summing up with respect to l,m and n, we can confirm that (13.3.5) is equal to the right-hand side of (13.1.11). Therefore we have (13.1.11), namely[1]

$$Z((z_{p-i}\sqcup z_q)z_{r+i})=\sum_{l,m,n=1}^{\infty}\frac{1}{l^{p-i}m^q(l+m+n)^{r+i}}=\zeta_3(p-i,q,0,0,0,r+i;A_3). \tag{13.3.6}$$

Hence we obtain the case $(a,b,c)=(1,1,1)$ of (13.2.4) in Theorem 13.3. Moreover, (13.3.6) implies that the right-hand side of (13.1.9) is mapped to the right-hand side of (13.1.10) via the Z-map. Therefore the left-hand side of (13.1.9) should be mapped to the left-hand side of (13.1.10) via the Z-map, namely

$$Z(z_p\sqcup z_rz_q)=\zeta(p)\zeta_{EZ,2}(r,q)=Z(z_p)Z(z_rz_q). \tag{13.3.7}$$

In Section 13.1, we assume (13.3.7) as a special case of (13.1.7), and deduce the correspondence of the left-hand sides of (13.1.9) and (13.1.10). Here we proceed backwards to obtain (13.3.7) as a conclusion. Hence we obtain the triple case in Theorem 13.7.

13.4. Proofs of results in Section 13.2

In this section, we will give the proofs of results stated in Section 13.2.

Proof of Proposition 13.2. We give the proof by induction on $a+b\geq 2$. For the case $a+b=2$, that is, $a=b=1$, the assertion is given from (13.3.1).

Next, we consider the case $a+b>2$. For this case, we further use the induction on $p_a+q_b\geq 2$. For the case $p_a+q_b=2$, that is, $p_a=q_b=1$, we have $z_{p_a}=z_{q_b}=y$. Hence, it follows from (13.1.6) that

$$z_{p_1}\cdots z_{p_a}\sqcup z_{q_1}\cdots z_{q_b}=(z_{p_1}\cdots z_{p_{a-1}}\sqcup z_{q_1}\cdots z_{q_b})z_{p_a}+(z_{p_1}\cdots z_{p_a}\sqcup z_{q_1}\cdots z_{q_{b-1}})z_{q_b},$$

which gives (13.2.2) for the case $p_a=q_a=1$.

[1]Here, the order of variables on the right-hand side is that given in (3.3.20), hence is different from that used in [109].

Now we consider the general case. Here we need to check the following three cases:

(Case 1) $p_a = 1$ and $q_a > 1$,
(Case 2) $p_a > 1$ and $q_a = 1$,
(Case 3) $p_a > 1$ and $q_a > 1$.

First, we consider Case 1. Since $p_a = 1$, namely $z_{p_a} = y$, we have from (13.1.6) that

(13.4.1)

$$z_{p_1}\cdots z_{p_a} \sqcup\!\sqcup z_{q_1}\cdots z_{q_b} = (z_{p_1}\cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1}\cdots z_{q_b})z_1 + (z_{p_1}\cdots z_{p_{a-1}}z_1 \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}}z_{q_b-1})x.$$

By the induction hypothesis for the case of $p_a + q_b - 1$, the second term on the right-hand side is

$$\sum_{\tau=0}^{q_b-2}(z_{p_1}\cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}}z_{q_b-1-\tau})z_{1+\tau}x + (z_{p_1}\cdots z_{p_{a-1}}z_1 \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}})z_{q_b-1}x$$
$$= \sum_{\rho=1}^{q_b-1}(z_{p_1}\cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}}z_{q_b-\rho})z_{1+\rho} + (z_{p_1}\cdots z_{p_{a-1}}z_1 \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}})z_{q_b},$$

by putting $\rho = \tau + 1$ in the first sum on the left-hand side and using $z_a x = z_{a+1}$. Therefore, by (13.4.1), we obtain Case 1 of (13.2.2). Similarly, we can prove Case 2.

Now we consider Case 3. It follows from (13.1.6) and the induction hypothesis for the case of $p_a + q_b - 1$ that

(13.4.2)

$$\begin{aligned}
&z_{p_1}\cdots z_{p_a} \sqcup\!\sqcup z_{q_1}\cdots z_{q_b}\\
&\quad = (z_{p_1}\cdots z_{p_{a-1}}z_{p_a-1} \sqcup\!\sqcup z_{q_1}\cdots z_{q_b})x + (z_{p_1}\cdots z_{p_a} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}}z_{q_b-1})x\\
&\quad = \sum_{\tau=0}^{q_b-1}\binom{p_a-2+\tau}{\tau}(z_{p_1}\cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1}z_{q_{b-1}}z_{q_b-\tau})z_{p_a-1+\tau}x\\
&\qquad + \sum_{\tau=0}^{p_a-2}\binom{q_b-1+\tau}{\tau}(z_{p_1}\cdots z_{p_{a-1}}z_{p_a-1-\tau} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}})z_{q_b+\tau}x\\
&\qquad + \sum_{\tau=0}^{q_b-2}\binom{p_a-1+\tau}{\tau}(z_{p_1}\cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}}z_{q_b-1-\tau})z_{p_a+\tau}x\\
&\qquad + \sum_{\tau=0}^{p_a-1}\binom{q_b-2+\tau}{\tau}(z_{p_1}\cdots z_{p_{a-1}}z_{p_a-\tau} \sqcup\!\sqcup z_{q_1}\cdots z_{q_{b-1}})z_{q_b-1+\tau}x.
\end{aligned}$$

Replace $\tau+1$ by τ in the second and the third sums on the right-hand side of (13.4.2), and use the relation

$$\binom{m+n-1}{n}+\binom{m+n-1}{n-1}=\binom{m+n}{n} \quad (m,n\in\mathbb{N})$$

in order to unite the first and the third sums, and the second and the fourth sums, respectively. Then, noting $z_p x = z_{p+1}$, we obtain (13.2.2) for the case of p_a+q_b. Thus, by induction, we complete the proof of Proposition 13.2. □

Proof of Theorem 13.3. For our assertion of the former part of Theorem 13.3, we first prove that

(13.4.3)

$$\zeta_{a+b+c}(\{d_{ij}\};A_{a+b+c})=\sum_{\substack{l_1,\dots,l_a\in\mathbb{N}\\ m_1,\dots,m_b\in\mathbb{N}\\ n_1,\dots,n_c\in\mathbb{N}}}\prod_{\eta=1}^{a}\left(\sum_{\mu=1}^{\eta}l_\mu\right)^{-p_\eta}\prod_{\xi=1}^{b}\left(\sum_{\rho=1}^{\xi}m_\rho\right)^{-q_\xi}\times\prod_{\sigma=1}^{c}\left(\sum_{\mu=1}^{a}l_\mu+\sum_{\rho=1}^{b}m_\rho+\sum_{\nu=1}^{\sigma}n_\nu\right)^{-r_\sigma},$$

where

$$d_{ij}=\begin{cases}p_j & (i=1;\ 1\le j\le a)\\ q_{j-a} & (i=a+1;\ a+1\le j\le a+b)\\ r_{j-a-b} & (i=1;\ a+b+1\le j\le a+b+c)\\ 0 & (\text{otherwise}).\end{cases}$$

In fact, comparing (13.4.3) and (13.2.1) with $r=a+b+c$, we see that m_ρ and n_ν correspond to $l_{a+\rho}$ and $l_{a+b+\nu}$, respectively. Hence we immediately obtain (13.4.3).

Now we show

(13.4.4)

$$Z((z_{p_1}\cdots z_{p_a}\sqcup z_{q_1}\cdots z_{q_b})z_{r_1}\cdots z_{r_c})=\sum_{\substack{l_1,\dots,l_a\in\mathbb{N}\\ m_1,\dots,m_b\in\mathbb{N}\\ n_1,\dots,n_c\in\mathbb{N}}}\prod_{\eta=1}^{a}\left(\sum_{\mu=1}^{\eta}l_\mu\right)^{-p_\eta}\prod_{\xi=1}^{b}\left(\sum_{\rho=1}^{\xi}m_\rho\right)^{-q_\xi}\prod_{\sigma=1}^{c}\left(\sum_{\mu=1}^{a}l_\mu+\sum_{\rho=1}^{b}m_\rho+\sum_{\nu=1}^{\sigma}n_\nu\right)^{-r_\sigma}$$

by induction on $a+b\ge 2$ ($a,b\in\mathbb{N}$). We first consider the case $a+b=2$, namely, $(a,b)=(1,1)$. It follows from (13.1.1) and (13.3.1) that

(13.4.5)

$$Z((z_{p_1}\sqcup z_{q_1})z_{r_1}\cdots z_{r_c})$$

$$
\begin{aligned}
&= \sum_{\tau=0}^{q_1-1} \binom{p_1-1+\tau}{\tau} Z\left(z_{q_1-\tau} z_{p_1+\tau} z_{r_1} \cdots z_{r_c}\right) \\
&\quad + \sum_{\tau=0}^{p_1-1} \binom{q_1-1+\tau}{\tau} Z\left(z_{p_1-\tau} z_{q_1+\tau} z_{r_1} \cdots z_{r_c}\right) \\
&= \sum_{\tau=0}^{q_1-1} \binom{p_1-1+\tau}{\tau} \sum_{\substack{l,m \\ n_1,\ldots,n_c}} \frac{1}{m^{q_1-\tau}(l+m)^{p_1+\tau} \left\{ \prod_{\xi=1}^{c} \left(l+m+\sum_{\nu=1}^{\xi} n_\nu \right)^{r_\xi} \right\}} \\
&\quad + \sum_{\tau=0}^{p_1-1} \binom{q_1-1+\tau}{\tau} \sum_{\substack{l,m \\ n_1,\ldots,n_c}} \frac{1}{l^{p_1-\tau}(l+m)^{q_1+\tau} \left\{ \prod_{\xi=1}^{c} \left(l+m+\sum_{\nu=1}^{\xi} n_\nu \right)^{r_\xi} \right\}}.
\end{aligned}
$$

Using the partial fraction decomposition (13.3.3) with $(X, Y) = (l, m)$ and $(\alpha, \beta) = (p_1, q_1)$, we see that the right-hand side of (13.4.5) coincides with

$$
\sum_{\substack{l,m \\ n_1,\ldots,n_c}} \frac{1}{l^{p_1} m^{q_1} \left\{ \prod_{\xi=1}^{c} \left(l+m+\sum_{\nu=1}^{\xi} n_\nu \right)^{r_\xi} \right\}}.
$$

This implies (13.4.4) for the case $a + b = 2$.

Next, we consider the general case. Mapping each side of (13.2.3) via the Z-map, we have

(13.4.6)

$$
\begin{aligned}
&Z\left((z_{p_1} \cdots z_{p_a} \sqcup\!\sqcup z_{q_1} \cdots z_{q_b}) z_{r_1} \cdots z_{r_c}\right) \\
&= \sum_{\tau=0}^{q_b-1} \binom{p_a-1+\tau}{\tau} Z\left((z_{p_1} \cdots z_{p_{a-1}} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}} z_{q_b-\tau}) z_{p_a+\tau} z_{r_1} \cdots z_{r_c}\right) \\
&\quad + \sum_{\tau=0}^{p_a-1} \binom{q_b-1+\tau}{\tau} Z\left((z_{p_1} \cdots z_{p_{a-1}} z_{p_a-\tau} \sqcup\!\sqcup z_{q_1} \cdots z_{q_{b-1}}) z_{q_b+\tau} z_{r_1} \cdots z_{r_c}\right).
\end{aligned}
$$

By the induction hypothesis, the right-hand side of (13.4.6) is

(13.4.7)

$$
\begin{aligned}
&\sum_{\tau=0}^{q_b-1} \binom{p_a-1+\tau}{\tau} \sum_{\substack{l_1,\ldots,l_{a-1} \\ m_1,\ldots,m_b \\ n_0,n_1,\ldots,n_c}} \prod_{\eta=1}^{a-1} \left(\sum_{\mu=1}^{\eta} l_\mu \right)^{-p_\eta} \prod_{\xi=1}^{b-1} \left(\sum_{\rho=1}^{\xi} m_\rho \right)^{-q_\xi} \left(\sum_{\rho=1}^{b} m_\rho \right)^{-q_b+\tau} \\
&\qquad \times \left(\sum_{\mu=1}^{a-1} l_\mu + \sum_{\rho=1}^{b} m_\rho + n_0 \right)^{-p_a-\tau} \prod_{\sigma=1}^{c} \left(\sum_{\mu=1}^{a-1} l_\mu + \sum_{\rho=1}^{b} m_\rho + n_0 + \sum_{\nu=1}^{\sigma} n_\nu \right)^{-r_\sigma}
\end{aligned}
$$

$$+\sum_{\tau=0}^{p_a-1}\binom{q_b-1+\tau}{\tau}\sum_{\substack{l_1,\dots,l_a\\ m_1,\dots,m_{b-1}\\ n_0,n_1,\dots,n_c}}\prod_{\eta=1}^{a-1}\left(\sum_{\mu=1}^{\eta}l_\mu\right)^{-p_\eta}\left(\sum_{\mu=1}^{a}l_\mu\right)^{-p_a+\tau}\prod_{\xi=1}^{b-1}\left(\sum_{\rho=1}^{\xi}m_\rho\right)^{-q_\xi}$$
$$\times\left(\sum_{\mu=1}^{a}l_\mu+\sum_{\rho=1}^{b-1}m_\rho+n_0\right)^{-q_b-\tau}\prod_{\sigma=1}^{c}\left(\sum_{\mu=1}^{a}l_\mu+\sum_{\rho=1}^{b-1}m_\rho+n_0+\sum_{\nu=1}^{\sigma}n_\nu\right)^{-r_\sigma}.$$

On the right-hand side, we rewrite n_0 to l_a in the first part, and to m_b in the second part. Then, by using the partial fraction decomposition (13.3.3) of with $(X,Y)=\left(\sum_{\mu=1}^{a}l_\mu,\ \sum_{\rho=1}^{b}m_\rho\right)$ and $(\alpha,\beta)=(p_a,q_b)$, we obtain (13.4.4). Thus we obtain the proof of the former part.

The element $(z_{p_1}\cdots z_{p_a}\sqcup\!\sqcup z_{q_1}\cdots z_{q_b})z_{r_1}\cdots z_{r_c}$ is, if we first apply the step (13.2.3) of SPP and then operate the Z-map, transformed to the right-hand side of (13.4.6). On the other hand, the same element is transformed to the right-hand side of (13.4.4) via the Z-map, and then to (13.4.7) by partial fraction decompositions. This implies that the diagram of procedures in Remark 13.4 is commutative. Thus we obtain the latter part of our assertion. We complete the proof. □

The explicit correspondence between SPP and the procedure of partial fraction decompositions is given in (13.4.7). It should be emphasized that we can successfully formulate the proof of Theorem 13.3, especially (13.4.4) and (13.4.7), because we have the framework of zeta-functions of root systems defined by (13.2.1), which is larger than the framework of MZVs.

Remark 13.9. Yamamoto [**212**] invented a new method of expressing multiple series by integrals of rather combinatorial nature, which are now called *Yamamoto's integral expressions*. In particular, he proved that the zeta-function of the form (13.4.3) can be written in terms of Yamamoto integrals (see [**212**, Section 3.3]).

Lastly, we give a proof of Theorem 13.7.

Proof of Theorem 13.7. We emphasize that in the proofs of Proposition 13.2 and Theorem 13.3, we used only definition (13.1.1) and the linearity of the Z-map. By setting $c=0$ in (13.4.4), we can obtain (13.2.5), namely, (13.1.7). On the other hand, Hoffman proved (13.1.4) without using (13.1.2) (see [**64**, Theorem 4.2]). Therefore the double shuffle relation (13.1.8) can be deduced without using (13.1.2). □

13.5. Functional relations including double shuffle relations

From (13.1.1) and (13.1.2) we find that the MZV $\zeta_{EZ,r}(k_1,\dots,k_r)$ can be expressed as a Drinfel'd integral. It seems that this expression cannot be extended to the case when some of k_j's are not integers. Therefore the proof of (extended) double shuffle relations, based on Drinfel'd integrals, cannot be generalized to the case of non-integral arguments. However, Theorem 13.7 with (13.1.8) supplies a proof of (extended) double shuffle relations without using Drinfel'd integrals, using only partial fraction decompositions. This argument can be generalized to the non-integral case. From this viewpoint, we can give some functional relations for zeta-functions of root systems of type A_r which include double shuffle relations for MZVs. In this section, we will explain the principle of the method by illustrating the triple and the quadruple cases.

In the triple case, we see that (13.1.10) holds for $p,r\in\mathbb{N}$ with $p,r\geq 2$ and any $q=s\in\mathbb{C}$ with $\Re(s)>1$, that is,

(13.5.1)
$$\begin{aligned}\zeta(p)\zeta_{EZ,2}(s,r)&=\sum_{i=0}^{r-1}\binom{p-1+i}{i}\zeta_{EZ,3}(s,r-i,p+i)\\&\quad+\sum_{i=0}^{p-1}\binom{r-1+i}{i}\zeta_3(p-i,s,0,0,0,r+i;A_3).\end{aligned}$$

Similarly, we obtain the following.

LEMMA 13.10 ([**109**, Lemma 1]). *For $p,u\in\mathbb{N}$ with $p,u\geq 2$ and $s_1,s_2\in\mathbb{C}$ with $\Re(s_1),\Re(s_2)>1$, we have*[2]

(13.5.2)
$$\begin{aligned}\zeta(p)\zeta_{EZ,3}(s_1,s_2,u)&=\sum_{i=0}^{u-1}\binom{p-1+i}{i}\zeta_{EZ,4}(s_1,s_2,u-i,p+i)\\&\quad+\sum_{i=0}^{p-1}\binom{u-1+i}{i}\zeta_4(p-i,s_1,0,0,0,s_2,0,0,0,u+i;A_4).\end{aligned}$$

PROOF. It follows from (13.1.6) that

$$\begin{aligned}&z_p\sqcup z_qz_rz_u\\&=\sum_{i=0}^{u-1}\binom{p-1+i}{i}z_qz_rz_{u-i}z_{p+i}+\sum_{i=0}^{p-1}\binom{u-1+i}{i}(z_{p-i}\sqcup z_qz_r)z_{u+i}.\end{aligned}$$

[2]The order of variables of the zeta-function of A_4 is that given in (3.3.21).

By (13.2.4) in Theorem 13.3 and (13.2.5) in Theorem 13.7, we have

$$\zeta(p)\zeta_{EZ,3}(q,r,u)=\sum_{i=0}^{u-1}\binom{p-1+i}{i}\zeta_{EZ,4}(q,r,u-i,p+i)+\sum_{i=0}^{p-1}\binom{u-1+i}{i}\zeta_4(p-i,q,0,0,0,r,0,0,0,u+i;A_4),$$

where we have used only the partial fraction decompositions. Hence this holds for $(q,r)=(s_1,s_2)\in\mathbb{C}^2$ with $\Re(s_1),\Re(s_2)>1$. □

As stated in Section 7.1, all functions on both sides of (13.5.1) and (13.5.2) can be meromorphically continued to the whole complex space. Hence (13.5.1) and (13.5.2) hold for all $s,s_1,s_2\in\mathbb{C}$ except for the singularities.

On the other hand, the harmonic product procedure (13.1.3) gives

$$\zeta(p)\zeta_{EZ,2}(s,r)=\zeta_{EZ,3}(s,r,p)+\zeta_{EZ,3}(s,p,r)+\zeta_{EZ,3}(p,s,r)+\zeta_{EZ,2}(s,p+r)+\zeta_{EZ,2}(p+s,r) \tag{13.5.3}$$

(see (12.2.7)), and similarly

$$\begin{aligned}\zeta(p)\zeta_{EZ,3}(s_1,s_2,u)&=\zeta_{EZ,4}(s_1,s_2,u,p)+\zeta_{EZ,4}(s_1,s_2,p,u)+\zeta_{EZ,4}(s_1,p,s_2,u)\\&\quad+\zeta_{EZ,4}(p,s_1,s_2,u)+\zeta_{EZ,3}(s_1,s_2,p+u)\\&\quad+\zeta_{EZ,3}(s_1,p+s_2,u)+\zeta_{EZ,3}(p+s_1,s_2,u).\end{aligned} \tag{13.5.4}$$

Note that (13.5.3) and (13.5.4) hold not only for integer points but also for any complex points.

Combining (13.5.1)–(13.5.4), we obtain the following functional relations which really include double shuffle relations.

THEOREM 13.11 ([**109**, Theorem 3]). *For $p_1,p_2\in\mathbb{N}$ with $p_1,p_2\geq 2$,*

$$\begin{aligned}&\sum_{i=0}^{p_2-1}\binom{p_1-1+i}{i}\zeta_{EZ,3}(s_1,p_2-i,p_1+i)\\&\quad+\sum_{i=0}^{p_1-1}\binom{p_2-1+i}{i}\zeta_3(p_1-i,s_1,0,0,0,p_2+i;A_3)\\&=\zeta_{EZ,3}(s_1,p_2,p_1)+\zeta_{EZ,3}(s_1,p_1,p_2)+\zeta_{EZ,3}(p_1,s_1,p_2)\\&\quad+\zeta_{EZ,2}(s_1,p_1+p_2)+\zeta_{EZ,2}(p_1+s_1,p_2),\end{aligned} \tag{13.5.5}$$

and

$$\sum_{i=0}^{p_2-1}\binom{p_1-1+i}{i}\zeta_{EZ,4}(s_1,s_2,p_2-i,p_1+i)$$

$$+\sum_{i=0}^{p_1-1}\binom{p_2-1+i}{i}\zeta_4(p_1-i,s_1,0,0,0,s_2,0,0,0,p_2+i;A_4)$$
$$=\zeta_{EZ,4}(s_1,s_2,p_2,p_1)+\zeta_{EZ,4}(s_1,s_2,p_1,p_2)+\zeta_{EZ,4}(s_1,p_1,s_2,p_2)$$
$$+\zeta_{EZ,4}(p_1,s_1,s_2,p_2)+\zeta_{EZ,3}(s_1,s_2,p_1+p_2)$$
$$+\zeta_{EZ,3}(s_1,s_2+p_1,p_2)+\zeta_{EZ,3}(s_1+p_1,s_2,p_2)$$

hold for all $s_1,s_2\in\mathbb{C}$ *except for the singularities.*

EXAMPLE 13.12. We set $(p_1,p_2)=(2,2)$ in (13.5.5). Then

$$\zeta_3(2,s_1,0,0,0,2;A_3)+2\zeta_3(1,s_1,0,0,0,3;A_3)$$
$$=\zeta_{EZ,3}(s_1,2,2)-2\zeta_{EZ,3}(s_1,1,3)+\zeta_{EZ,3}(2,s_1,2)$$
$$+\zeta_{EZ,2}(s_1,4)+\zeta_{EZ,2}(2+s_1,2). \tag{13.5.6}$$

From (13.3.3), we obtain (if $b,c\in\mathbb{N}$)

$$\zeta_3(b,c,0,0,0,d;A_3)=\sum_{i=0}^{c-1}\binom{b-1+i}{i}\zeta_{EZ,3}(c-i,b+i,d)$$
$$+\sum_{i=0}^{b-1}\binom{c-1+i}{i}\zeta_{EZ,3}(b-i,c+i,d).$$

Therefore, when $s_1=k\in\mathbb{N}$, we can see that (13.5.6) gives the following double shuffle relation for MZVs:

$$\sum_{i=0}^{k-1}(1+i)\zeta_{EZ,3}(k-i,2+i,2)+2\sum_{i=0}^{k-1}\zeta_{EZ,3}(k-i,1+i,3)$$
$$+\zeta_{EZ,3}(2,k,2)+k\zeta_{EZ,3}(1,k+1,2)+2\zeta_{EZ,3}(1,k,3)$$
$$=\zeta_{EZ,3}(k,2,2)-2\zeta_{EZ,3}(k,1,3)+\zeta_{EZ,3}(2,k,2)$$
$$+\zeta_{EZ,2}(k,4)+\zeta_{EZ,2}(2+k,2)\quad(k\in\mathbb{N}).$$

This is an answer to Problem 1.1. In particular, when $k=1$, we have

$$6\zeta_{EZ,3}(1,1,3)+\zeta_{EZ,3}(1,2,2)=\zeta_{EZ,2}(1,4)+\zeta_{EZ,2}(3,2).$$

Similarly, combining (13.2.2), (13.2.4) and the harmonic product formulas, we can produce functional relations for zeta-functions of the root system of A_r type for any r, including double shuffle relations.

REMARK 13.13. The idea of proving shuffle relations by using only partial fraction decompositions, developed in the present chapter, can be applied to the situation of "finite/symmetric analogues". The notions of *finite multiple zeta values* and *symmetric multiple zeta values* were first introduced by Kaneko and Zagier ([**87**]; see also [**83**][**84**]),

and have been studied very extensively in recent years. Ono and Yamamoto [**173**] considered *finite multiple polylogarithms*, and proved a shuffle relation formula for them, by an argument analogous to that in the present chapter (hence their main tool was (13.3.3), that is Lemma 12.2). Their result was further used by Ono [**171**] to show some functional equations for finite multiple polylogarithms.

The full power of Lemma 12.2 is used in our argument, and also in the proof of Ono and Yamamoto [**173**]. The case $\alpha = \beta = 1$ of the lemma (and its multi-variable generalization), however, is much simpler and had been used by many people (such as Goncharov [**50**]). Applying the case $\alpha = \beta = 1$ of Lemma 12.2, Ono [**170**] and Ono, Seki and Yamamoto [**172**] gave proofs of shuffle relations of finite and symmetric multiple zeta values, using only partial fraction decompositions.



CHAPTER 14

L-functions

As is seen from (1.3.8), Akiyama and Ishikawa [2] considered the multiple Dirichlet L-function (of Euler–Zagier type) defined by

$$L(\{s_j\},\{\chi_j\}) = \sum_{1\le n_1<\cdots<n_r} \frac{\chi_1(n_1)\cdots\chi_r(n_r)}{n_1^{s_1}\cdots n_r^{s_r}} \tag{14.0.1}$$

for Dirichlet characters $\chi_1,\dots,\chi_r$, and observed its analytic continuation.[1] A little later, Arakawa and Kaneko [8] considered two types of multiple L-values defined by

$$\begin{aligned} &L_{\text{ш}}(k_1,\dots,k_r;f_1,\dots,f_r) \\ &\quad = \sum_{1\le n_1<\cdots<n_r} \frac{f_1(n_1)f_2(n_2-n_1)\cdots f_r(n_r-n_{r-1})}{n_1^{k_1}\cdots n_r^{k_r}}, \\ &L_*(k_1,\dots,k_r;f_1,\dots,f_r) \\ &\quad = \sum_{1\le n_1<\cdots<n_r} \frac{f_1(n_1)f_2(n_2)\cdots f_r(n_r)}{n_1^{k_1}\cdots n_r^{k_r}} \end{aligned}$$

for periodic functions $f_1,\dots,f_r$ and $k_1,\dots,k_r\in\mathbb{N}$ with $k_r>1$ if necessary for convergence. They established Hoffman's setup for these series.

In this chapter, we study the multiple L-functions of root systems

$$L_r(\mathbf{s},\boldsymbol{\chi};X_r) = \sum_{m_1=1}^{\infty}\cdots\sum_{m_r=1}^{\infty}\prod_{\alpha\in\Delta_+}\frac{\chi_\alpha(\langle\alpha^\vee,m_1\lambda_1+\cdots+m_r\lambda_r\rangle)}{\langle\alpha^\vee,m_1\lambda_1+\cdots+m_r\lambda_r\rangle^{s_\alpha}} \tag{14.0.2}$$

for a set of Dirichlet characters $\boldsymbol{\chi}=(\chi_\alpha)_{\alpha\in\Delta_+}$, which can be regarded as character analogues of $\zeta_r(\mathbf{s};X_r)$. In the case of A_2, (14.0.2) is studied by the third-named author [202] and Nakamura [163], and then, the general case was introduced and studied in [105].

Note that as well as (12.2.1), we can see that

$$L_r(\{s_j\},\{\chi_j\};\Delta_{l+}(C_r)) = L(\{s_j\},\{\chi_j\})$$

[1] Analogous results for the Mordell–Tornheim and Apostol–Vu types were studied by Wu [211]. This work is unpublished, but the statement of theorems and the brief sketch of the proofs can be found in [137].

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if $s_{ij}^- = s_{ij}^+ = 0$, where the right-hand side is (14.0.1).

In Theorems 14.2 and 14.6 below, we will find that special values of $L_r(\mathbf{s}, \boldsymbol{\chi}; X_r)$ at positive integers can be expressed in terms of Bernoulli functions $P(\mathbf{k}, \mathbf{y}; X_r)$ of root systems defined by (4.2.11) (see also (9.4.13), (9.4.15)). Those formulas are multiple generalizations of well-known formulas for the values of ordinary Dirichlet L-functions $L(2k, \chi)$ and $L(2k+1, \chi)$, and may also be regarded as character analogues of Witten's volume formula (1.5.1). It should be emphasized that in our formulas, the values of rational coefficients can be determined completely.

14.1. L-functions of root systems

Let Δ be the root system of rank r. Let χ_α be a Dirichlet character modulo $f_\alpha \in \mathbb{N}$ for $\alpha \in \Delta_+$, and set $\boldsymbol{\chi} = (\chi_\alpha)_{\alpha\in\Delta_+}$, $\mathbf{f} = (f_\alpha)_{\alpha\in\Delta_+}$. For $\alpha \in \Delta_-$, we understand that $f_\alpha = f_{-\alpha}$ and $\chi_\alpha = \chi_{-\alpha}$. We define actions of $w \in \mathrm{Aut}(\Delta)$ on characters and on moduli by

$$(w\boldsymbol{\chi})_\alpha = \chi_{w^{-1}\alpha}, \qquad (w\mathbf{f})_\alpha = f_{w^{-1}\alpha}. \tag{14.1.1}$$

The purpose of this chapter is to define the L-function twisted by $\boldsymbol{\chi}$ associated with Δ, and to study its properties. We define the L-function by

$$L_r(\mathbf{s}, \boldsymbol{\chi}; \Delta) = \sum_{\lambda\in P_{++}} \prod_{\alpha\in\Delta_+} \frac{\chi_\alpha(\langle\alpha^\vee, \lambda\rangle)}{\langle\alpha^\vee, \lambda\rangle^{s_\alpha}}, \tag{14.1.2}$$

which is the same as (14.0.2). This series is absolutely convergent in $\mathcal{S}$, as in Lemma 3.3. More generally, we define the L-function of any root set $\Delta^* \subset \Delta_+$ by

$$L_r(\mathbf{s}, \boldsymbol{\chi}; \Delta^*) = \sum_{\lambda\in P_{++}} \prod_{\alpha\in\Delta^*} \frac{\chi_\alpha(\langle\alpha^\vee, \lambda\rangle)}{\langle\alpha^\vee, \lambda\rangle^{s_\alpha}}, \tag{14.1.3}$$

which is absolutely convergent in $\mathcal{S}_0$.

THEOREM 14.1 ([**105**, Theorem 3.1]). *The L-function $L_r(\mathbf{s}, \boldsymbol{\chi}; \Delta^*)$ can be continued meromorphically to the whole $\mathbb{C}^{n^*}$ space, where $n^* = \#\Delta^*$.*

PROOF. Let $\Delta^* = \{\beta_1, \dots, \beta_{n^*}\}$ and let χ_{β_j} be defined modulo f_j for each $\beta_j \in \Delta^*$. Then we can write

$$L_r(\mathbf{s}, \boldsymbol{\chi}; \Delta^*) = \sum_{a_1=1}^{f_1} \cdots \sum_{a_{n^*}=1}^{f_{n^*}} \left\{ \prod_{j=1}^{n^*} \chi_{\beta_j}(a_j) \right\} \sum_{\lambda}^{\sharp} \prod_{1\le j\le n^*} \frac{1}{\langle\beta_j^\vee, \lambda\rangle^{s_{\beta_j}}}, \tag{14.1.4}$$

where $\sum^{\sharp}$ means the summation over all $\lambda \in P_{++}$ satisfying $\langle \beta_j^{\vee}, \lambda \rangle \equiv a_j \pmod{f_j}$ for all $j = 1, \dots, n^*$.

For $A = (a_1, \dots, a_{n^*})$, let

$$\Lambda_A = \{\lambda \in P \mid \langle \beta_j^{\vee}, \lambda \rangle \equiv a_j \pmod{f_j}\}. \tag{14.1.5}$$

Let c be the least common multiple of $\{f_j\}_{1 \le j \le n^*}$, and

$$P_{++}(c) = \left\{ \sum_{i=1}^{r} b_i \lambda_i \in P_{++} \;\middle|\; 1 \le b_i \le c \ (1 \le i \le r) \right\}.$$

We show that

$$\Lambda_A = \bigcup_{1 \le l \le k} (\mu_l + cP) \tag{14.1.6}$$

for some $k \in \mathbb{N}_0$ and elements $\mu_l \in P_{++}(c)$ $(1 \le l \le k)$ which are distinct from each other. If Λ_A is empty, then $k = 0$. Assume that $\Lambda_A \neq \emptyset$ and let $\nu \in \Lambda_A$. Then there exists an element $\mu_1 \in \Lambda_A \cap P_{++}(c)$ because $\nu \in \Lambda_A$ implies

$$\langle \beta_j^{\vee}, \nu + cP \rangle \equiv a_j + c\langle \beta_j^{\vee}, P \rangle \equiv a_j \pmod{f_j} \tag{14.1.7}$$

and hence $\nu + cP \subset \Lambda_A$. If $\Lambda_A \setminus (\mu_1 + cP)$ is not empty, then we apply the same argument to $\Lambda_A \setminus (\mu_1 + cP)$. Repeating this procedure, we arrive at (14.1.6), because $P_{++}(c)$ is a finite set.

Now we see that

$$\begin{aligned} \sum_{\lambda}^{\sharp} \prod_{1 \le j \le n^*} \frac{1}{\langle \beta_j^{\vee}, \lambda \rangle^{s_{\beta_j}}} &= \sum_{\lambda \in \Lambda_A \cap P_{++}} \prod_{1 \le j \le n^*} \frac{1}{\langle \beta_j^{\vee}, \lambda \rangle^{s_{\beta_j}}} \\ &= \sum_{1 \le l \le k} \sum_{\lambda \in \mu_l + cP_+} \prod_{1 \le j \le n^*} \frac{1}{\langle \beta_j^{\vee}, \lambda \rangle^{s_{\beta_j}}} \\ &= \sum_{1 \le l \le k} \sum_{\lambda \in P_+} \prod_{1 \le j \le n^*} \frac{1}{\langle \beta_j^{\vee}, \mu_l + c\lambda \rangle^{s_{\beta_j}}}, \end{aligned} \tag{14.1.8}$$

which is a finite sum of the series of type (7.2.25). Thus we have the assertion from Theorem 7.10. □

Now we state the formula of expressing the special value $L_r(\mathbf{k}, \boldsymbol{\chi}; \Delta)$ in terms of the Bernoulli functions $P(\mathbf{k}, \mathbf{y}; \Delta)$ (see (4.2.2), (9.4.15)). We write

$$\chi_\alpha(m) = \sum_{a_\alpha = 1}^{f_\alpha} c_\alpha(a_\alpha) e^{2\pi\sqrt{-1} a_\alpha m / f_\alpha}, \tag{14.1.9}$$

with

$$c_\alpha(m) = \frac{1}{f_\alpha} \sum_{a_\alpha = 1}^{f_\alpha} \chi_\alpha(a_\alpha) e^{-2\pi\sqrt{-1} a_\alpha m / f_\alpha}. \tag{14.1.10}$$

Note that if χ_α is a primitive character of conductor f_α, we have

$$c_\alpha(m) = \frac{\overline{\chi_\alpha}(m)}{g(\overline{\chi_\alpha})} \tag{14.1.11}$$

(see [4, Chpater 8]), where for a primitive character χ of conductor f, $g(\chi)$ denotes the Gauss sum defined by

$$g(\chi) = \sum_{m=0}^{f-1} \chi(m) e^{2\pi\sqrt{-1}m/f}. \tag{14.1.12}$$

For $\mathbf{a} = (a_\alpha)_{\alpha\in\Delta_+} \in \mathbb{Z}^{\#\Delta_+}$ and $\mathbf{f} = (f_\alpha)_{\alpha\in\Delta_+} \in \mathbb{N}^{\#\Delta_+}$, let

$$\mathbf{y}(\mathbf{a};\mathbf{f}) = \sum_{\alpha\in\Delta_+} a_\alpha \alpha^\vee / f_\alpha. \tag{14.1.13}$$

Then, in Section 14.2, we will prove the following result.

THEOREM 14.2. *Let $k_\alpha \in \mathbb{N}$ for $\alpha \in \Delta_+$ and assume $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathcal{S}$ and*

$$k_\alpha = k_\beta, \chi_\alpha = \chi_\beta \quad \text{if } \|\alpha\| = \|\beta\|, \qquad (-1)^{-k_\alpha}\chi_\alpha(-1) = 1. \tag{14.1.14}$$

Then we have

$$L_r(\mathbf{k},\boldsymbol{\chi};\Delta) = \frac{(-1)^{\#\Delta_+}}{\#W}\Big(\prod_{\alpha\in\Delta_+}\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+} c_\alpha(a_\alpha)\Big)P(\mathbf{k},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta), \tag{14.1.15}$$

and in particular, if all χ_α's are primitive,

$$L_r(\mathbf{k},\boldsymbol{\chi};\Delta)\prod_{\alpha\in\Delta_+} g(\overline{\chi_\alpha}) = \frac{(-1)^{\#\Delta_+}}{\#W}\Big(\prod_{\alpha\in\Delta_+}\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+} \overline{\chi_\alpha}(a_\alpha)\Big)P(\mathbf{k},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta). \tag{14.1.16}$$

This is [**105**, Theorem 3.5], but in [**105**] it is stated by using Bernoulli polynomials (see Remark 9.28) instead of Bernoulli functions as above.

REMARK 14.3. When Δ is of type A_1, noting Remark 4.2 we find that (14.1.16) is reduced to the classical formula

$$L(k,\chi)g(\overline{\chi}) = \frac{-1}{2}\frac{(2\pi\sqrt{-1})^k}{k!}\sum_{a=1}^{f}\overline{\chi}(a)B_k(a/f). \tag{14.1.17}$$

Note that this formula follows immediately from (14.1.22) and (14.1.25) below.

Let

$$G(\mathbf{t},\boldsymbol{\chi};\Delta)=\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}\chi_\alpha(a_\alpha)/f_\alpha\Big)F(\mathbf{f}\,\mathbf{t},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta), \tag{14.1.18}$$

where $\mathbf{f}\,\mathbf{t}=(f_\alpha t_\alpha)_{\alpha\in\Delta_+}$ and $F(\cdot,\cdot;\Delta)$ is given by (4.2.11) and Theorem 9.12. We define *generalized Bernoulli numbers* $B(\mathbf{k},\boldsymbol{\chi};\Delta)$ *of the root system* Δ by their generating function $G(\mathbf{t},\boldsymbol{\chi};\Delta)$ as

$$G(\mathbf{t},\boldsymbol{\chi};\Delta)=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Delta_+}}B(\mathbf{k},\boldsymbol{\chi};\Delta)\prod_{\alpha\in\Delta_+}\frac{t_\alpha^{k_\alpha}}{k_\alpha!}. \tag{14.1.19}$$

THEOREM 14.4. *For* $\mathbf{k}\in\mathbb{N}_0^{\#\Delta_+}$ *we have*

(14.1.20)

$$B(\mathbf{k},\boldsymbol{\chi};\Delta)=\Big(\prod_{\alpha\in\Delta_+}f_\alpha^{k_\alpha-1}\Big)\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}\chi_\alpha(a_\alpha)\Big)P(\mathbf{k},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta).$$

This is [**105**, Theorem 3.6], again stated there in terms of Bernoulli polynomials.

PROOF. From (4.2.11) we have

$$F(\mathbf{f}\,\mathbf{t},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta)=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Delta_+}}P(\mathbf{k},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta)\prod_{\alpha\in\Delta_+}f_\alpha^{k_\alpha}\frac{t_\alpha^{k_\alpha}}{k_\alpha!}. \tag{14.1.21}$$

Substituting this into the right-hand side of (14.1.18) and comparing the coefficients, we obtain the result. □

REMARK 14.5. This can be regarded as a multiple analogue of

$$B_{k,\chi}=f^{k-1}\sum_{a=1}^{f}\chi(a)B_k(a/f), \tag{14.1.22}$$

where $B_{k,\chi}$ is the k-th classical generalized Bernoulli number (see [**207**, Proposition 4.1][**77**, p.10]).

By combining Theorems 14.2, 14.4 and the formula

$$\chi(-1)g(\chi)g(\overline{\chi})=f \tag{14.1.23}$$

for a primitive character χ of conductor f, we immediately obtain the following theorem.

THEOREM 14.6 ([105, Theorem 3.7]). *Assume* (14.1.14) *and that all χ_α's are primitive. Then, for any $\mathbf{k} \in \mathbb{N}^{\#\Delta_+} \cap \mathcal{S}$,*

(14.1.24)
$$L_r(\mathbf{k}, \boldsymbol{\chi}; \Delta) = \frac{(-1)^{|\mathbf{k}|+\#\Delta_+}}{\#W} \Bigg(\prod_{\alpha \in \Delta_+} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha! f_\alpha^{k_\alpha}} g(\chi_\alpha) \Bigg) B(\mathbf{k}, \overline{\boldsymbol{\chi}}; \Delta).$$

REMARK 14.7. This can be regarded as a multiple analogue of

(14.1.25)
$$L(k, \chi) = \frac{(-1)^{k+1}}{2} \frac{(2\pi\sqrt{-1})^k}{k! f^k} g(\chi) B_{k,\overline{\chi}}$$

(see [77, p. 12]).

Next, we give a sufficient condition for $B_{\mathbf{k},\overline{\boldsymbol{\chi}}}(\Delta) = 0$. For this purpose, we first consider the action of the Weyl group. We defined actions of $\mathrm{Aut}(\Delta)$ on the variables $\mathbf{s}$ and $\mathbf{t}$ in (9.3.1) and (9.6.7) respectively, and on the variables $\boldsymbol{\chi}$ and $\mathbf{f}$ in (14.1.1). Here we define the action on $\mathbf{a}$ by

(14.1.26)
$$(w\mathbf{a})_\alpha = a_{w^{-1}\alpha}$$

for $w \in \mathrm{Aut}(\Delta)$, under the convention $a_\alpha = a_{-\alpha}$ for $\alpha \in \Delta_-$. Further for $w \in \mathrm{Aut}(\Delta)$, we define

(14.1.27)
$$(wG)(\mathbf{t}, \boldsymbol{\chi}; \Delta) = G(w^{-1}\mathbf{t}, w^{-1}\boldsymbol{\chi}; \Delta).$$

THEOREM 14.8 ([105, Theorem 3.8]). *Assume that Δ is an irreducible root system. Moreover, assume that $f_\alpha > 1$ if Δ is of type A_1. Then for $w \in \mathrm{Aut}(\Delta)$,*

(14.1.28)
$$(wG)(\mathbf{t}, \boldsymbol{\chi}; \Delta) = \Big(\prod_{\alpha \in \Delta_{w^{-1}}} \chi_\alpha(-1) \Big) G(\mathbf{t}, \boldsymbol{\chi}; \Delta),$$

(14.1.29)
$$B(w^{-1}\mathbf{k}, w^{-1}\boldsymbol{\chi}; \Delta) = \Big(\prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-k_\alpha} \chi_\alpha(-1) \Big) B(\mathbf{k}, \boldsymbol{\chi}; \Delta).$$

THEOREM 14.9 ([105, Theorem 3.9]). *Under the same assumption as in Theorem 14.8, we have $B(\mathbf{k}, \boldsymbol{\chi}; \Delta) = 0$ if there exists an element $w \in \mathrm{Aut}(\Delta)$ such that $w^{-1}\mathbf{k} = \mathbf{k}$, $w^{-1}\boldsymbol{\chi} = \boldsymbol{\chi}$ and*

(14.1.30)
$$\prod_{\alpha \in \Delta_{w^{-1}}} (-1)^{-k_\alpha} \chi_\alpha(-1) \neq 1.$$

REMARK 14.10. This can be regarded as a multiple analogue of the fact that

(14.1.31)
$$B_{k,\chi} = 0$$

if $(-1)^{-k}\chi(-1) \neq 1$.

Proofs of Theorems 14.8 and 14.9 will be given in the next section.

14.2. Proofs of theorems

For $w \in \mathrm{Aut}(\Delta)$ and $\alpha \in \Delta_+$, let

$$\mathrm{sgn}(w,\alpha) = \begin{cases} -1, & \alpha \in \Delta_w, \\ 1, & \alpha \notin \Delta_w. \end{cases} \tag{14.2.1}$$

For $w \in \mathrm{Aut}(\Delta)$, we define

$$(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta) = L(w^{-1}\mathbf{s}, w^{-1}\boldsymbol{\chi};\Delta). \tag{14.2.2}$$

Lemma 14.11 ([105, Lemma 5.1]). *For $\mathbf{s} \in \mathcal{S}$, we have*

$$\sum_{\lambda \in P \setminus H_{\Delta^\vee}} \prod_{\alpha\in\Delta_+} \frac{\chi_\alpha(\langle\alpha^\vee,\lambda\rangle)}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}} = \sum_{w\in W} \Big(\prod_{\alpha\in\Delta_{w^{-1}}} (-1)^{-s_\alpha}\chi_\alpha(-1)\Big)(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta). \tag{14.2.3}$$

Proof. Note that $P \setminus H_{\Delta^\vee} = \bigcup_{w\in W} wP_{++}$. We have

$$\begin{aligned} \sum_{\lambda\in wP_{++}} \prod_{\alpha\in\Delta_+} \frac{\chi_\alpha(\langle\alpha^\vee,\lambda\rangle)}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}} &= \sum_{\lambda\in P_{++}} \prod_{\alpha\in\Delta_+} \frac{\chi_\alpha(\langle w^{-1}\alpha^\vee,\lambda\rangle)}{\langle w^{-1}\alpha^\vee,\lambda\rangle^{s_\alpha}} \\ &= \sum_{\lambda\in P_{++}} \prod_{\alpha\in w^{-1}\Delta_+} \frac{\chi_{w\alpha}(\langle\alpha^\vee,\lambda\rangle)}{\langle\alpha^\vee,\lambda\rangle^{s_{w\alpha}}}. \end{aligned}$$

We divide the product in the last member into two subproducts as (4.1.4), and replace α by $-\alpha$ in the second subproduct. Then, by the computations similar to (4.1.5) — (4.1.7), we find that the above is

$$\begin{aligned} &= \sum_{\lambda\in P_{++}} \prod_{\alpha\in\Delta_+} \frac{\chi_{w\alpha}(\mathrm{sgn}(w,\alpha)\langle\alpha^\vee,\lambda\rangle)}{\big(\mathrm{sgn}(w,\alpha)\langle\alpha^\vee,\lambda\rangle\big)^{s_{w\alpha}}} \\ &= \Big(\prod_{\alpha\in\Delta_w} (-1)^{-s_{w\alpha}}\chi_{w\alpha}(-1)\Big) \sum_{\lambda\in P_{++}} \prod_{\alpha\in\Delta_+} \frac{\chi_{w\alpha}(\langle\alpha^\vee,\lambda\rangle)}{\langle\alpha^\vee,\lambda\rangle^{s_{w\alpha}}} \\ &= \Big(\prod_{\alpha\in\Delta_{w^{-1}}} (-1)^{-s_\alpha}\chi_\alpha(-1)\Big)(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta). \end{aligned}$$

□

Lemma 14.12 ([105, Lemma 5.2]). *For $\mathbf{s} \in \mathcal{S}$, we have*

$$\begin{aligned} &\sum_{w\in W} \Big(\prod_{\alpha\in\Delta_{w^{-1}}} (-1)^{-s_\alpha}\chi_\alpha(-1)\Big)(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta) \\ &\qquad = \sum_{\substack{a_\alpha=1 \\ \alpha\in\Delta_+}}^{f_\alpha} \Big(\prod_{\alpha\in\Delta_+} c_\alpha(a_\alpha)\Big) S(\mathbf{s},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta). \end{aligned} \tag{14.2.4}$$

In particular, if all χ_α's are primitive,

$$\sum_{w\in W}\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\chi_\alpha(-1)\Big)(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta)\prod_{\alpha\in\Delta_+}g(\overline{\chi_\alpha})$$
$$=\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}\overline{\chi_\alpha}(a_\alpha)\Big)S(\mathbf{s},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta). \tag{14.2.5}$$

Proof. By Lemma 14.11 we have

$$\begin{aligned}
&\sum_{w\in W}\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\chi_\alpha(-1)\Big)(wL_r)(\mathbf{s},\boldsymbol{\chi};\Delta)\\
&\quad=\sum_{\lambda\in P\setminus H_{\Delta^\vee}}\prod_{\alpha\in\Delta_+}\frac{\chi_\alpha(\langle\alpha^\vee,\lambda\rangle)}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}\\
&\quad=\sum_{\lambda\in P\setminus H_{\Delta^\vee}}\prod_{\alpha\in\Delta_+}\frac{\sum_{a_\alpha=1}^{f_\alpha}c_\alpha(a_\alpha)e^{2\pi\sqrt{-1}a_\alpha\langle\alpha^\vee,\lambda\rangle/f_\alpha}}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}\\
&\quad=\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}c_\alpha(a_\alpha)\Big)\sum_{\lambda\in P\setminus H_{\Delta^\vee}}\prod_{\alpha\in\Delta_+}\frac{e^{2\pi\sqrt{-1}a_\alpha\langle\alpha^\vee,\lambda\rangle/f_\alpha}}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}\\
&\quad=\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}c_\alpha(a_\alpha)\Big)S(\mathbf{s},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta).
\end{aligned} \tag{14.2.6}$$

□

We see that if

$$\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-s_\alpha}\chi_\alpha(-1)=1,\qquad\text{for all } w\in W, \tag{14.2.7}$$

and $s_\alpha=s_\beta$, $\chi_\alpha=\chi_\beta$ for $\alpha,\beta\in\Delta_+$ such that $\|\alpha\|=\|\beta\|$, then from Lemma 14.12 we have

$$L_r(\mathbf{s},\boldsymbol{\chi};\Delta)=\frac{1}{\#W}\sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha}\Big(\prod_{\alpha\in\Delta_+}c_\alpha(a_\alpha)\Big)S(\mathbf{s},\mathbf{y}(\mathbf{a};\mathbf{f});\Delta). \tag{14.2.8}$$

We give a simple condition equivalent to (14.2.7).

Lemma 14.13 ([**105**, Lemma 5.4]). *Condition* (14.2.7) *is equivalent to the condition*

$$\begin{aligned}
&s_\alpha=k_\alpha\in\mathbb{Z},\\
&(-1)^{-k_\alpha}\chi_\alpha(-1)=1,
\end{aligned} \tag{14.2.9}$$

for all $\alpha\in\Delta_+$.

PROOF. It is clear that (14.2.9) implies (14.2.7). Conversely, assume that (14.2.7) holds. Let $w_0 = \sigma_{i_n} \cdots \sigma_{i_1}$ be a reduced expression of the longest element with $n = \#\Delta_+$ (see Section 10.1). Let $v_k = \sigma_{i_k} \cdots \sigma_{i_1}$, $1 \le k \le n$ (hence $v_n = w_0$). We have $\ell(v_k) = k$ and

$$\prod_{\beta \in \Delta_{v_k}} (-1)^{-s_\beta} \chi_\beta(-1) = 1 \qquad (1 \le k \le n) \tag{14.2.10}$$

by the assumption. Also we have

$$\Delta_{v_k} = \{\beta_1, \dots, \beta_k\} = \Delta_{v_{k-1}} \cup \{\beta_k\} \qquad (2 \le k \le n) \tag{14.2.11}$$

and $\Delta_{v_1} = \{\beta_1\}$, where $\beta_m = \sigma_{i_1} \cdots \sigma_{i_{m-1}} \alpha_{i_m}$ for $2 \le m \le n$ and $\beta_1 = \alpha_{i_1}$ (see [**71**, Section 1.7]).

The case $k = 1$ of (14.2.10) implies $(-1)^{-s_{\beta_1}} \chi_{\beta_1}(-1) = 1$, and hence $s_{\beta_1} \in \mathbb{Z}$. Combining this result and the case $k = 2$ of (14.2.10), we then obtain $(-1)^{-s_{\beta_2}} \chi_{\beta_2}(-1) = 1$ with $s_{\beta_2} \in \mathbb{Z}$. Repeating this procedure, we arrive at (14.2.9) for all $\alpha \in \Delta_+$. □

Under assumption (14.2.9), substituting (9.4.14) into the right-hand side of (14.2.8), we obtain (14.1.15). When all χ_α's are primitive, then from (14.1.15) and (14.1.11) we obtain (14.1.16). This completes the proof of Theorem 14.2.

PROOF OF THEOREM 14.8. Let $w \in \mathrm{Aut}(\Delta)$. It is shown in Theorem 9.29 that

$$(wF)(\mathbf{t}, \mathbf{y}; \Delta) = F(\mathbf{t}, \mathbf{y}; \Delta) \tag{14.2.12}$$

for all $\mathbf{y} \in V$ if Δ is not of type A_1, while (14.2.12) holds for $\mathbf{y} \in V \setminus Q^\vee$ if Δ is of type A_1 (see Remark 9.31). By use of this relation, we have

(14.2.13)

$$\begin{aligned}
&(wG)(\mathbf{t}, \boldsymbol{\chi}; \Delta) \\
&= \sum_{\substack{a_{w\alpha}=1 \\ \alpha \in \Delta_+}}^{f_{w\alpha}} \Big(\prod_{\alpha \in \Delta_+} \chi_{w\alpha}(a_{w\alpha})/f_{w\alpha} \Big) F((w^{-1}\mathbf{f})(w^{-1}\mathbf{t}), \mathbf{y}(w^{-1}\mathbf{a}; w^{-1}\mathbf{f}); \Delta) \\
&= \sum_{\substack{a_{w\alpha}=1 \\ \alpha \in \Delta_+}}^{f_{w\alpha}} \Big(\prod_{\alpha \in \Delta_+} \chi_{w\alpha}(a_{w\alpha})/f_{w\alpha} \Big) F(\mathbf{f}\,\mathbf{t}, w\mathbf{y}(w^{-1}\mathbf{a}; w^{-1}\mathbf{f}); \Delta).
\end{aligned}$$

Here, (14.2.12) is used to show the second equality. When Δ is of type A_1, by the assumption we know $f_{w\alpha} > 1$, so if $a_{w\alpha} = f_{w\alpha}$ then $\chi_{w\alpha}(a_{w\alpha}) = 0$. Therefore in the above summation we may understand that $a_{w\alpha} < f_{w\alpha}$. This implies that $\mathbf{y}(w^{-1}\mathbf{a}; w^{-1}\mathbf{f}) \in V \setminus Q^\vee$, and hence (14.2.12) can be applied to the A_1 case.

We see that

(14.2.14)
$$\begin{aligned} w\mathbf{y}(w^{-1}\mathbf{a}; w^{-1}\mathbf{f}) &= w\Big(\sum_{\alpha\in\Delta_+} a_{w\alpha}\alpha^\vee/f_{w\alpha}\Big) \\ &= \sum_{\alpha\in\Delta_+} a_{w\alpha} w\alpha^\vee/f_{w\alpha} \\ &= \sum_{\alpha\in\Delta_+} \operatorname{sgn}(w^{-1},\alpha) a_\alpha\alpha^\vee/f_\alpha \\ &\equiv \sum_{\alpha\in\Delta_{w^{-1}}} \alpha^\vee + \sum_{\alpha\in\Delta_+} \operatorname{sgn}(w^{-1},\alpha) a_\alpha\alpha^\vee/f_\alpha \pmod{Q^\vee} \\ &= \sum_{\alpha\in\Delta_{w^{-1}}} (f_\alpha - a_\alpha)\alpha^\vee/f_\alpha + \sum_{\alpha\in\Delta_+\setminus\Delta_{w^{-1}}} a_\alpha\alpha^\vee/f_\alpha. \end{aligned}$$

(To show the third equality, we write $\beta = w\alpha \in w\Delta_+$ and again use the decomposition similar to (4.1.4)). Since F is periodic mod $Q^\vee$ with respect to $\mathbf{y}$, by replacing a_α with $f_\alpha - a_\alpha$ for $\alpha \in \Delta_{w^{-1}}$, we have

(14.2.15)
$$\begin{aligned} (wG)(\mathbf{t},\boldsymbol{\chi};\Delta) &= \sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha} \Big(\prod_{\alpha\in\Delta_+} \chi_\alpha(\operatorname{sgn}(w^{-1},\alpha)a_\alpha)/f_\alpha\Big) F(\mathbf{f}\,\mathbf{t}, \mathbf{y}(\mathbf{a};\mathbf{f});\Delta) \\ &= \Big(\prod_{\alpha\in\Delta_{w^{-1}}} \chi_\alpha(-1)\Big) \sum_{\substack{a_\alpha=1\\ \alpha\in\Delta_+}}^{f_\alpha} \Big(\prod_{\alpha\in\Delta_+} \chi_\alpha(a_\alpha)/f_\alpha\Big) F(\mathbf{f}\,\mathbf{t}, \mathbf{y}(\mathbf{a};\mathbf{f});\Delta) \\ &= \Big(\prod_{\alpha\in\Delta_{w^{-1}}} \chi_\alpha(-1)\Big) G(\mathbf{t},\boldsymbol{\chi};\Delta), \end{aligned}$$

and hence (14.1.28). Since

(14.2.16)
$$\begin{aligned} G(w^{-1}\mathbf{t}, w^{-1}\boldsymbol{\chi};\Delta) &= \sum_{\mathbf{k}\in\mathbb{N}_0^n} B(\mathbf{k}, w^{-1}\boldsymbol{\chi};\Delta) \prod_{\alpha\in\Delta_+} \frac{t_{w\alpha}^{k_\alpha}}{k_\alpha!} \\ &= \sum_{\mathbf{k}\in\mathbb{N}_0^n} B(w^{-1}\mathbf{k}, w^{-1}\boldsymbol{\chi};\Delta) \prod_{\alpha\in\Delta_+} \frac{t_{w\alpha}^{k_{w\alpha}}}{k_{w\alpha}!} \\ &= \sum_{\mathbf{k}\in\mathbb{N}_0^n} \Big(\prod_{\alpha\in\Delta_{w^{-1}}} (-1)^{k_\alpha}\Big) B(w^{-1}\mathbf{k}, w^{-1}\boldsymbol{\chi};\Delta) \prod_{\alpha\in\Delta_+} \frac{t_\alpha^{k_\alpha}}{k_\alpha!}, \end{aligned}$$

we have (14.1.29). □

PROOF OF THEOREM 14.9. Assume (14.1.30). Then by (14.1.29)

$$\left(1-\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{-k_\alpha}\chi_\alpha(-1)\Big)\right)B(\mathbf{k},\boldsymbol{\chi};\Delta)=0 \tag{14.2.17}$$

implies $B(\mathbf{k},\boldsymbol{\chi};\Delta)=0$. □

14.3. Special values of L-functions

EXAMPLE 14.14. In the case of C_2, from (4.3.6), we can explicitly calculate that

(14.3.1)

$$\begin{aligned}
&P((2,2,2,2),(y_1,y_2);C_2)\\
&=\frac{23\{y_1\}^8}{40320}-\frac{\{2y_1-y_2\}\{y_1\}^7}{840}-\frac{17\{y_1\}^7}{10080}+\frac{\{2y_1-y_2\}^2\{y_1\}^6}{1440}\\
&+\frac{\{2y_1-y_2\}\{y_1\}^6}{288}+\frac{\{y_1\}^6}{1440}-\frac{\{2y_1-y_2\}^2\{y_1\}^5}{480}-\frac{\{2y_1-y_2\}\{y_1\}^5}{480}\\
&+\frac{\{y_1\}^5}{576}+\frac{\{2y_1-y_2\}^2\{y_1\}^4}{576}-\frac{\{2y_1-y_2\}\{y_1\}^4}{576}-\frac{\{y_1\}^4}{960}\\
&+\frac{\{2y_1-y_2\}\{y_1\}^3}{720}-\frac{\{y_1\}^3}{1440}-\frac{\{2y_1-y_2\}^2\{y_1\}^2}{2880}+\frac{\{2y_1-y_2\}\{y_1\}^2}{2880}\\
&+\frac{13\{y_1\}^2}{40320}-\frac{\{2y_1-y_2\}\{y_1\}}{5040}+\frac{\{y_1\}}{10080}+\frac{23\{y_1-y_2\}^8}{40320}\\
&+\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^8}{2520}+\frac{\{\frac{y_2}{2}\}^8}{2520}-\frac{17\{y_1-y_2\}^7}{10080}-\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^7}{630}-\frac{\{\frac{y_2}{2}\}^7}{630}\\
&+\frac{\{y_1-y_2\}^6}{1440}+\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}^2\{\frac{y_2}{2}+\frac{1}{2}\}^6}{180}-\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}\{\frac{y_2}{2}+\frac{1}{2}\}^6}{180}\\
&+\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^6}{360}+\frac{\{y_1-\frac{y_2}{2}\}^2\{\frac{y_2}{2}\}^6}{180}-\frac{\{y_1-\frac{y_2}{2}\}\{\frac{y_2}{2}\}^6}{180}+\frac{\{\frac{y_2}{2}\}^6}{360}\\
&+\frac{\{y_1-y_2\}^5}{576}-\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}^2\{\frac{y_2}{2}+\frac{1}{2}\}^5}{60}+\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}\{\frac{y_2}{2}+\frac{1}{2}\}^5}{60}\\
&-\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^5}{360}-\frac{\{y_1-\frac{y_2}{2}\}^2\{\frac{y_2}{2}\}^5}{60}+\frac{\{y_1-\frac{y_2}{2}\}\{\frac{y_2}{2}\}^5}{60}-\frac{\{\frac{y_2}{2}\}^5}{360}-\frac{\{y_1-y_2\}^4}{960}\\
&+\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}^2\{\frac{y_2}{2}+\frac{1}{2}\}^4}{72}-\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}\{\frac{y_2}{2}+\frac{1}{2}\}^4}{72}+\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^4}{720}\\
&+\frac{\{y_1-\frac{y_2}{2}\}^2\{\frac{y_2}{2}\}^4}{72}-\frac{\{y_1-\frac{y_2}{2}\}\{\frac{y_2}{2}\}^4}{72}+\frac{\{\frac{y_2}{2}\}^4}{720}-\frac{\{y_1-y_2\}^3}{1440}+\frac{13\{y_1-y_2\}^2}{40320}\\
&+\frac{\{y_1-y_2\}^6\{2y_1-y_2\}^2}{1440}-\frac{\{y_1-y_2\}^5\{2y_1-y_2\}^2}{480}+\frac{\{y_1-y_2\}^4\{2y_1-y_2\}^2}{576}\\
&-\frac{\{y_1-y_2\}^2\{2y_1-y_2\}^2}{2880}+\frac{\{2y_1-y_2\}^2}{30240}+\frac{\{y_1-\frac{y_2}{2}\}^2}{7560}+\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}^2}{7560}
\end{aligned}$$

$$-\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}^2\{\frac{y_2}{2}+\frac{1}{2}\}^2}{360}+\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}\{\frac{y_2}{2}+\frac{1}{2}\}^2}{360}-\frac{\{\frac{y_2}{2}+\frac{1}{2}\}^2}{5040}$$
$$-\frac{\{y_1-\frac{y_2}{2}\}^2\{\frac{y_2}{2}\}^2}{360}+\frac{\{y_1-\frac{y_2}{2}\}\{\frac{y_2}{2}\}^2}{360}-\frac{\{\frac{y_2}{2}\}^2}{5040}+\frac{\{y_1-y_2\}}{10080}$$
$$-\frac{\{y_1-y_2\}^7\{2y_1-y_2\}}{840}+\frac{\{y_1-y_2\}^6\{2y_1-y_2\}}{288}-\frac{\{y_1-y_2\}^5\{2y_1-y_2\}}{480}$$
$$-\frac{\{y_1-y_2\}^4\{2y_1-y_2\}}{576}+\frac{\{y_1-y_2\}^3\{2y_1-y_2\}}{720}+\frac{\{y_1-y_2\}^2\{2y_1-y_2\}}{2880}$$
$$-\frac{\{y_1-y_2\}\{2y_1-y_2\}}{5040}-\frac{\{2y_1-y_2\}}{30240}-\frac{\{y_1-\frac{y_2}{2}\}}{7560}-\frac{\{y_1-\frac{y_2}{2}+\frac{1}{2}\}}{7560}-\frac{1}{67200}.$$

By substituting (14.3.1) into (14.1.20) with

$$\mathbf{y}(\mathbf{a};\mathbf{f})=(a_1/f_1+a_3/f_3+a_4/f_4,\,a_2/f_2+a_3/f_3+2a_4/f_4), \tag{14.3.2}$$

where the indices are the same as in Example 4.16, and applying Theorem 14.6, we can compute $L_2(2,2,2,2;\chi,\psi,\psi,\chi;C_2)$ with primitive characters χ and ψ, though it is not so easy to complete this task.

For example, let χ_5 be the quadratic character of conductor 5, namely $\chi_5(1)=\chi_5(4)=1$ and $\chi_5(2)=\chi_5(3)=-1$. Since $g(\overline{\chi_5})=\sqrt{5}$, it follows from Theorem 14.6 with (14.3.1) that

$$L_2(2,2,2,2;\chi_5,\chi_5,\chi_5,\chi_5;C_2)=\frac{92}{29296875}\pi^8. \tag{14.3.3}$$

Note that we gave a certain functional relation for $L_2(\boldsymbol{s};\boldsymbol{\chi};C_2)$ which includes (14.3.3) as a value relation (see [**105**, Example 5]).

Similarly, calculating $P(\mathbf{k},\mathbf{y};C_2)$ from the Taylor expansion of (4.3.6), we can compute

$$L_2(2,4,4,2;\chi_5,\chi_5,\chi_5,\chi_5;C_2)=\frac{133676}{17303466796875}\pi^{12},$$
$$L_2(4,4,4,4;\chi_5,\chi_5,\chi_5,\chi_5;C_2)=\frac{66531503}{7732486724853515625}\pi^{16},$$
$$L_2(6,6,6,6;\chi_5,\chi_5,\chi_5,\chi_5;C_2)=\frac{52885564435709}{21073442727327346801757812500000}\pi^{24}.$$

In the case of conductor 7, we denote by ρ_7 the cubic character of conductor 7 defined by

$$\rho_7(1)=\rho_7(6)=1,\ \rho_7(2)=\rho_7(5)=e^{2\pi i/3},\ \rho_7(3)=\rho_7(4)=e^{4\pi i/3}.$$

Then we can obtain

$$L_2(2,2,2,2;\rho_7,\rho_7,\rho_7,\rho_7;C_2)=\frac{\pi^8}{g(\overline{\rho_7})^4}\left(-\frac{3406}{86472015}-\frac{1294\sqrt{-3}}{17294403}\right),$$
$$L_2(4,4,4,4;\rho_7,\rho_7,\rho_7,\rho_7;C_2)$$
$$=\frac{\pi^{16}}{g(\overline{\rho_7})^4}\left(-\frac{2166724453}{307949627830150706400}-\frac{4482132121\sqrt{-3}}{307949627830150706400}\right),$$

$$L_2(6,6,6,6;\rho_7,\rho_7,\rho_7,\rho_7;C_2)$$
$$=\frac{\pi^{24}}{g(\overline{\rho_7})^4}\Bigg(-\frac{889132946513450957}{27093701320378391838996016360200000}$$
$$-\frac{123533320486750319\sqrt{-3}}{18062467546918927892664010906800 0}\Bigg),$$

where

$$g(\overline{\rho_7})=2\left(\cos(2\pi/7)+e^{4\pi i/3}\cos(4\pi/7)+e^{2\pi i/3}\cos(6\pi/7)\right).$$

By the same method as above, we can evaluate $L_2(2p,2q,2q,2p;\mathbf{1},\chi,\chi,\mathbf{1};C_2)$, where $\mathbf{1}$ is the trivial character and $\chi(-1)=1$. For example, we can obtain

$$L_2(2,2,2,2;\mathbf{1},\chi_5,\chi_5,\mathbf{1};C_2)=-\frac{3679}{1230468750}\pi^8,$$
$$L_2(4,4,4,4;\mathbf{1},\chi_5,\chi_5,\mathbf{1};C_2)=-\frac{4463343161}{5196231070910156250000}\pi^{16}.$$

Furthermore, let χ_8 be the quadratic character of conductor 8 defined by $\chi_8(1)=\chi_8(7)=1$ and $\chi_8(3)=\chi_8(5)=-1$. Then we can easily see that $L_2(p,q,r,s;\chi_8,\chi_8,\chi_8,\chi_8;C_2)=0$. On the other hand, we can obtain

$$L_2(2,2,2,2;\mathbf{1},\chi_8,\chi_8,\mathbf{1};C_2)=-\frac{31}{165150720}\pi^8,$$
$$L_2(2,4,4,2;\mathbf{1},\chi_8,\chi_8,\mathbf{1};C_2)=-\frac{106189}{502269581721600}\pi^{12},$$
$$L_2(4,4,4,4;\mathbf{1},\chi_8,\chi_8,\mathbf{1};C_2)=-\frac{15691883}{4680348870314557440 00}\pi^{16},$$
$$L_2(6,6,6,6;\mathbf{1},\chi_8,\chi_8,\mathbf{1};C_2)=-\frac{857222938145797}{1399364578750218821941077422899200 00}\pi^{24}.$$

EXAMPLE 14.15. By the same method, we can give explicit formulas for the L-functions of A_3-type:

(14.3.4)

$$L_3(s_1,s_2,s_3,s_4,s_5,s_6;\psi_1,\psi_2,\psi_3,\psi_4,\psi_5,\psi_6;A_3)$$
$$=\sum_{l,m,n=1}^{\infty}\frac{\psi_1(l)\psi_2(m)\psi_3(n)\psi_4(l+m)\psi_5(m+n)\psi_6(l+m+n)}{l^{s_1}m^{s_2}n^{s_3}(l+m)^{s_4}(m+n)^{s_5}(l+m+n)^{s_6}}.$$

In fact, from (4.3.9), we can calculate $P(\mathbf{k},\mathbf{y};A_3)$ as for (14.3.1), though it is too complicated to write here. Hence we list only a few examples as follows. For the quadratic character χ_5 of conductor 5, we have

$$L_3(2,2,2,2,2,2;\chi_5,\chi_5,\chi_5,\chi_5,\chi_5,\chi_5;A_3)=-\frac{1856}{213623046875}\pi^{12},$$
$$L_3(4,4,4,4,4,4;\chi_5,\chi_5,\chi_5,\chi_5,\chi_5,\chi_5;A_3)$$

$$= -\frac{71581861479424}{125722243543714284896850585937 5}\pi^{24}.$$

Let χ_{13} be the unique quadratic character of conductor 13 defined by

$$\chi_{13}(n) = \begin{cases} 1 & (n \equiv 1, 3, 4, 9, 10, 12 \pmod{13}) \\ -1 & (n \equiv 2, 5, 6, 7, 8, 11 \pmod{13}) \\ 0 & (n \equiv 0 \pmod{13}). \end{cases}$$

Since $g(\overline{\chi_{13}}) = \sqrt{13}$, we have

$$L_3(2, 2, 2, 2, 2, 2; \chi_{13}, \chi_{13}, \chi_{13}, \chi_{13}, \chi_{13}, \chi_{13}; A_3) = \frac{12155648}{1467779362716303}\pi^{12}.$$

We can regard these formulas as χ-analogues of the results of Gunnells and Sczech [**53**].



CHAPTER 15

Zeta-functions of Lie groups

Witten zeta-functions (3.1.1) defined by Zagier are the series associated with semisimple Lie algebras. However, as remarked in Section 1.5 (footnote), Witten originally considered more general series associated with semisimple Lie groups. Inspired by this point, in this chapter we introduce the notion of zeta-functions of weight lattices of Lie groups, which gives a generalization of our zeta-functions of root systems. In this chapter we assume some basic knowledge of the theory of Lie groups.

15.1. The background and the motivation

Let M be a compact real 2-dimensional manifold, G a compact connected semisimple Lie group[1] acting as a gauge group, and E a G-bundle over M. Motivated by 2-dimensional quantum gauge theories, Witten [**210**] evaluated the volume of the moduli space $\mathcal{M}$ of flat connections on E up to gauge transformations. Such a result can be regarded as a limit of Verlinde's formula [**206**] when M is orientable, but Witten developed a more elementary method, based on the decomposition of M into three-holed spheres. The main result in [**210**] is now called Witten's volume formula, which expresses the volume of $\mathcal{M}$ in terms of special values at even integer points of the Dirichlet series

$$\zeta_W(s;G)=\sum_{\psi}(\dim\psi)^{-s}, \tag{15.1.1}$$

where ψ runs over all isomorphism classes of finite-dimensional irreducible representations of G.

Let $\mathfrak{g}=\mathbb{C}\otimes\mathrm{Lie}(G)$ be the complexified Lie algebra of G. Recall the definition (3.1.1):

$$\zeta_W(s;\mathfrak{g})=\sum_{\varphi}(\dim\varphi)^{-s},$$

where the summation runs over all isomorphism classes of finite-dimensional irreducible representations φ of $\mathfrak{g}$. When G is simply-connected,

[1] This G is a real Lie group, because any compact complex Lie group is abelian (see, for example, [**43**, Exercise 8.44] or [**122**, Theorem 1.19]).

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then there is a one-to-one correspondence between φ and ψ. In fact, each φ is obtained from the differential of a certain ψ, and so

$$\zeta_W(s;\mathfrak{g}) = \zeta_W(s;G). \tag{15.1.2}$$

As explained in Section 1.5 and Corollary 4.7, Witten's volume formula especially implies

$$\zeta_W(2k;\mathfrak{g}) = C_W(2k,\mathfrak{g})\pi^{2k\#(\Delta_+)} \tag{15.1.3}$$

for $k \in \mathbb{N}$, where $C_W(2k,\mathfrak{g})$ is a rational number.

In Chapters 4 and 9, we introduced a root-system theoretic generalization of Bernoulli numbers and periodic Bernoulli functions, and express $C_W(2k,\mathfrak{g})$ explicitly in terms of Bernoulli numbers of root systems (see Theorem 4.5). Therefore we now have sufficiently explicit information on formula (15.1.3). Moreover, in Chapters 8, 9, 10 and 11 we proved various functional relations among zeta-functions (3.1.1), which include evaluation formulas like (15.1.3) as special cases.[2]

However, the group G is not necessarily simply-connected in Witten's paper [**210**]. (In fact, this point is emphasized by Witten himself on [**210**, p. 182].) When G is not simply-connected, relation (15.1.2) does not hold. It is the aim of the present chapter to consider such a situation; that is, to study the zeta-functions and volume formulas in the sense of the original formulation of Witten [**210**]. The contents of the present chapter were originally developed in [**114**] [**118**].[3]

For this purpose, we introduce the multi-variable version of $\zeta_W(s;G)$ [**118**]. From (3.1.5) we see that the multi-variable version of $\zeta_W(s;\mathfrak{g})$ can be regarded as the zeta-function of the weight lattice of $\mathfrak{g}$. Similarly, in the present chapter we will define a multi-variable zeta-function $\zeta_r(\mathbf{s},\mathbf{y};G)$ associated with the weight lattice of G. The definition is given in Section 15.3, and the volume formula for this zeta-function is stated as Theorem 15.7, which gives an explicit formula for the values

[2]By the theory in those chapters, special values of $\zeta_W(k;\mathfrak{g})$ for an odd integer k can sometimes be computed, but it does not fit the original motivation of Witten. In Witten's theory values at odd integers appear in the unorientable case, and the associated series is different from $\zeta_W(k;\mathfrak{g})$.

[3]The article [**118**] was published later than [**114**], but actually written earlier. It was already posted to the arXiv in 2010 (arXiv:1011.0323). We presented the contents of [**118**] at a conference on analytic number theory and spatial tessellations held in Kyiv (September 16–20, 2013), and submitted the paper to the Proceedings of that conference. However, because of the political chaotic situation in Ukraine, the publication of the Proceedings was delayed, and finally published in 2015 as a special issue of Šiauliai Mathematical Seminar.

of this zeta-function at $\mathbf{s} = 2\mathbf{k}$, where $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathbb{N}^{\#(\Delta_+)}$ satisfying $k_\alpha = k_\beta$ if α and β are of the same length. As explicit examples, in Section 15.4, we consider the cases of types A_r, B_r and C_r $(r \le 3)$. The explicit forms in those cases show that $\zeta_r(\mathbf{s}, \mathbf{y}; G)$ is a partial sum of $\zeta_r(\mathbf{s}, \mathbf{y}; \Delta)$, restricted by a certain congruence condition.

Since Theorem 15.7 is a formula for $\mathbf{s} = 2\mathbf{k}$, it is not useful when we consider the values at odd integer points. In order to study such cases, in Section 15.6, we give some functional relations among zeta-functions of types A_2 and $C_2(\simeq B_2)$ in a way similar to that in Chapter 8. Those relations produce explicit formulas for special values of zeta-functions at some points of the form $\mathbf{s} = \mathbf{l} = (l_\alpha)_{\alpha\in\Delta_+}$, where $l_\alpha \in \mathbb{N}$ and some of them are odd. Those results include not only evaluation formulas given in Section 15.4 but also another type of evaluation formulas which can be regarded as certain extensions of the results in previous chapters such as Examples 4.11, 4.16, 4.17, 4.18, 4.19, 8.22, 8.30, 8.33, 11.10, and 11.11.

15.2. A general form of zeta-functions

For a set X, denote by $\mathfrak{F}(X)$ the set of all complex-valued functions on X. Note that an action of W is induced on $\mathfrak{F}(P)$ as $(wf)(\lambda) = f(w^{-1}\lambda)$, where P is the weight lattice. We will also use the notation $P^\vee$, Q and $Q^\vee$ defined in Chapter 2.

Let $f \in \mathfrak{F}(P/Q)$. Since $P^\vee/Q^\vee$ is regarded as the dual of P/Q over $\mathbb{Q}/\mathbb{Z}$, f can be expanded as follows:

$$f(\lambda) = \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu) e^{2\pi\sqrt{-1}\langle\mu,\lambda\rangle}, \tag{15.2.1}$$

where $\langle\ ,\ \rangle$ is regarded as the natural pairing, and $\widehat{f} : P^\vee/Q^\vee \to \mathbb{C}$ is given by

$$\widehat{f}(\mu) = \frac{1}{\#(P/Q)} \sum_{\lambda\in P/Q} f(\lambda) e^{-2\pi\sqrt{-1}\langle\mu,\lambda\rangle}, \tag{15.2.2}$$

because for $\nu \in P^\vee/Q^\vee$ we have

$$\sum_{\lambda\in P/Q} e^{2\pi\sqrt{-1}\langle\nu,\lambda\rangle} = \#(P/Q)\delta_{\nu,0}. \tag{15.2.3}$$

Note that f is automatically W-invariant because for $\lambda \in P$ we have

$$\sigma_\alpha(\lambda) = \lambda - \langle\alpha^\vee, \lambda\rangle\alpha \equiv \lambda \pmod{Q}. \tag{15.2.4}$$

Remark 15.1. A coweight $\mu \in P^\vee$ is said to be *minuscule* if $0 \le \langle\mu, \alpha\rangle \le 1$ for all $\alpha \in \Delta$. It is known that a minuscule coweight is

one of fundamental coweights and as a system of representatives for $P^\vee/Q^\vee$, we can take $\{0\} \cup \{\lambda_j^\vee\}_{j\in J}$, where J is the set of all indices of minuscule coweights (see Bourbaki [**20**, Chapitre 6, §2.2], Kirillov [**94**, Lemma 2.2]). Therefore we may use these representatives in (15.2.1).

For $\mathbf{s} = (s_\alpha) \in \mathbb{C}^n$, $\mathbf{y} \in V$ and $f \in \mathfrak{F}(P/Q)$, we define

$$\zeta_r(\mathbf{s}, \mathbf{y}, f; \Delta) = \sum_{\lambda\in P_{++}} f(\lambda) e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{s_\alpha}}. \tag{15.2.5}$$

Note that $\zeta_r(\mathbf{s}, \mathbf{y}, 1; \Delta)$ coincides with $\zeta_r(\mathbf{s}, \mathbf{y}; \Delta)$ defined in (3.1.7) and is studied in previous chapters. For $w \in \mathrm{Aut}(\Delta)$, define the action of w on ζ_r by

$$(w\zeta_r)(\mathbf{s}, \mathbf{y}, f; \Delta) = \zeta_r(w^{-1}\mathbf{s}, w^{-1}\mathbf{y}, w^{-1}f; \Delta), \tag{15.2.6}$$

where $w^{-1}\mathbf{s} = (s_{w\alpha})_{\alpha\in\Delta_+}$. Then it is easy to see that for $w \in \mathrm{Aut}(\Gamma)$,

$$(w\zeta_r)(\mathbf{s}, \mathbf{y}, f; \Delta) = \zeta_r(\mathbf{s}, \mathbf{y}, f; \Delta). \tag{15.2.7}$$

Remark 15.2. Here we note the role of the exponential factor $e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle}$ in the definition (15.2.5). This is analogous to the Lerch zeta-function $\phi(s, \alpha)$ defined by (4.1.9), and more generally, to the twisted zeta-functions of root systems (3.1.7). As in the case of multiple L-functions with Dirichlet characters studied in Chapter 14, it is clear that the form (15.2.5) with an exponential factor is useful in the study of multiple series with twisting factors, such as the series discussed in Example 15.9 and Remark 15.11 below. Moreover, the existence of this exponential factor simplifies our argument in various places. In the argument below, (15.2.8), (15.2.12) etc. are impossible to show without using this factor. (See also Remark 8.19.)

Remark 15.3. It should be noted that we may regard $\mathbf{y} \in V/Q^\vee$ in (15.2.5). This is because when $a \in Q^\vee$ we have $\langle a, \lambda\rangle \in \mathbb{Z}$, hence $e^{2\pi\sqrt{-1}\langle \mathbf{y}+a,\lambda\rangle} = e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle}$, for any $\lambda \in P_{++}$.

Proposition 15.4 ([**118**, Proposition 2.1]). *The function* $\zeta_r(\mathbf{s}, \mathbf{y}, f; \Delta)$*, as a function in* $\mathbf{s}$*, can be continued meromorphically to the whole space* $\mathbb{C}^n$*.*

Proof. By use of the expression (15.2.1), noting Remark 15.3, we obtain

$$\zeta_r(\mathbf{s}, \mathbf{y}, f; \Delta) = \sum_{\lambda\in P_{++}} \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu) e^{2\pi\sqrt{-1}\langle \mu,\lambda\rangle} e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee, \lambda\rangle^{s_\alpha}} \tag{15.2.8}$$

$$= \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu) \sum_{\lambda\in P_{++}} e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mu,\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee,\lambda\rangle^{s_\alpha}}$$
$$= \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu)\zeta_r(\mathbf{s},\mathbf{y}+\mu,1;\Delta).$$

In Chapter 7 we showed that $\zeta_r(\mathbf{s},\mathbf{y},1;\Delta)$ can be continued meromorphically to the whole space (Corollary 7.5), hence, so can be $\zeta_r(\mathbf{s},\mathbf{y},f;\Delta)$ from (15.2.8). □

Let

$$S(\mathbf{s},\mathbf{y},f;\Delta) = \sum_{\lambda\in P\setminus H_{\Delta^\vee}} f(\lambda)e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee,\lambda\rangle^{s_\alpha}}. \tag{15.2.9}$$

Then, in the same way as above, we obtain

$$S(\mathbf{s},\mathbf{y},f;\Delta) = \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu)S(\mathbf{s},\mathbf{y}+\mu,1;\Delta). \tag{15.2.10}$$

We have already obtained

$$S(\mathbf{k},\mathbf{y},1;\Delta) = (-1)^{\#\Delta_+}\Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)P(\mathbf{k},\mathbf{y};\Delta) \tag{15.2.11}$$

for $\mathbf{k}\in\mathbb{N}^{\#\Delta_+}\cap\mathcal{S}$ (see (9.4.14)).

Suggested by (15.2.10), we define generalized Bernoulli functions associated with f and Δ by

$$P(\mathbf{k},\mathbf{y},f;\Delta) = \sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu)P(\mathbf{k},\mathbf{y}+\mu;\Delta). \tag{15.2.12}$$

Then by (15.2.10), (15.2.11) and (15.2.12), we have

$$S(\mathbf{k},\mathbf{y},f;\Delta) = (-1)^{\#\Delta_+}\Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)P(\mathbf{k},\mathbf{y},f;\Delta) \tag{15.2.13}$$

for $\mathbf{k}\in\mathbb{N}^{\#\Delta_+}\cap\mathcal{S}$.

By (9.4.13) we can evaluate $P(\mathbf{k},\mathbf{y}+\mu;\Delta)$ from the expansion of $F(\mathbf{t},\mathbf{y};\Delta)$. In particular, we find that $P(\mathbf{k},\mu;\Delta)\in\mathbb{Q}$ for any $\mu\in P^\vee/Q^\vee$.

THEOREM 15.5 ([118, Theorem 2.2]). *For* $\mathbf{s}=\mathbf{k}=(k_\alpha)_{\alpha\in\Delta_+}\in\mathbb{N}^{\#\Delta_+}\cap\mathcal{S}$, $\mathbf{y}\in V$ *and* $f\in\mathfrak{F}(P/Q)$,

$$\sum_{w\in W}\Big(\prod_{\alpha\in\Delta_{w^{-1}}}(-1)^{k_\alpha}\Big)\zeta_r(w^{-1}\mathbf{k},w^{-1}\mathbf{y},f;\Delta) \tag{15.2.14}$$
$$= (-1)^{\#\Delta_+}\Big(\prod_{\alpha\in\Delta_+} \frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!}\Big)P(\mathbf{k},\mathbf{y},f;\Delta).$$

PROOF. Since $P \setminus H_{\Delta^\vee} = \bigcup_{w\in W} w(P_{++})$, we have

(15.2.15)
$$\begin{aligned} S(\mathbf{s},\mathbf{y},f;\Delta) &= \sum_{\lambda\in P\setminus H_{\Delta^\vee}} f(\lambda)e^{2\pi\sqrt{-1}\langle \mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee,\lambda\rangle^{s_\alpha}} \\ &= \sum_{w\in W}\sum_{\lambda\in P_{++}} f(w\lambda)e^{2\pi\sqrt{-1}\langle \mathbf{y},w\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle \alpha^\vee,w\lambda\rangle^{s_\alpha}} \\ &= \sum_{w\in W}\sum_{\lambda\in P_{++}} (w^{-1}f)(\lambda)e^{2\pi\sqrt{-1}\langle w^{-1}\mathbf{y},\lambda\rangle} \prod_{\alpha\in\Delta_+} \frac{1}{\langle w^{-1}\alpha^\vee,\lambda\rangle^{s_\alpha}} \\ &= \sum_{w\in W}\Big(\prod_{\alpha\in\Delta_{w^{-1}}} (-1)^{-s_\alpha}\Big)\zeta_r(w^{-1}\mathbf{s},w^{-1}\mathbf{y},w^{-1}f;\Delta), \end{aligned}$$

where the last equality follows by rewriting α to $w\alpha$, and then following the argument similar to (4.1.4)–(4.1.7).

Combining (15.2.13) and the W-invariance of f, we have the result. □

In the following sections, we treat some special cases. Let $\mathcal{A} \subset P$ with $\mathcal{A}+Q=\mathcal{A}$. Let $\iota_{\mathcal{A}} : P \to \{0,1\}$ be the characteristic function of $\mathcal{A}$ defined by

$$\iota_{\mathcal{A}}(\lambda) = \begin{cases} 1 & (\lambda\in\mathcal{A}), \\ 0 & (\lambda\notin\mathcal{A}). \end{cases} \tag{15.2.16}$$

Then $\iota_{\mathcal{A}}$ can be regarded as a function on P/Q. Hence (15.2.1) and (15.2.2) implies that

$$\iota_{\mathcal{A}}(\lambda) = \sum_{\mu\in P^\vee/Q^\vee} \widehat{\iota_{\mathcal{A}}}(\mu)e^{2\pi\sqrt{-1}\langle\mu,\lambda\rangle}, \tag{15.2.17}$$

where $\widehat{\iota_{\mathcal{A}}} : P^\vee/Q^\vee \to \mathbb{C}$ is given by

(15.2.18)
$$\widehat{\iota_{\mathcal{A}}}(\mu) = \frac{1}{\#(P/Q)}\sum_{\lambda\in P/Q} \iota_{\mathcal{A}}(\lambda)e^{-2\pi\sqrt{-1}\langle\mu,\lambda\rangle} = \frac{1}{\#(P/Q)}\sum_{\lambda\in\mathcal{A}/Q} e^{-2\pi\sqrt{-1}\langle\mu,\lambda\rangle}.$$

15.3. Zeta-functions of weight lattices of Lie groups

Now we define zeta-functions of weight lattices of Lie groups. Let $\widetilde{G}$ be a simply-connected compact semisimple Lie group, and $\mathfrak{g} = \mathrm{Lie}(\widetilde{G})$. There is a one-to-one correspondence between a connected compact semisimple Lie group G whose universal covering group is $\widetilde{G}$, and a lattice L with $Q(\Delta(\mathfrak{g})) \subset L \subset P(\Delta(\mathfrak{g}))$ up to automorphisms by taking $L = L(G)$ as the *weight lattice of* G (see Remark 15.8 below). Let $L_+ = P_+ \cap L$.

We define the *zeta-function of the weight lattice* $L = L(G)$ *of* G by (15.3.1)

$$\zeta_r(\mathbf{s}, \mathbf{y}; G) = \zeta_r(\mathbf{s}, \mathbf{y}; L; \Delta) = \sum_{\lambda \in L_+ + \rho} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle} \prod_{\alpha \in \Delta_+} \frac{1}{\langle \alpha^\vee, \lambda \rangle^{s_\alpha}}.$$

This is the case $f = \iota_{\mathcal{A}}$, $\mathcal{A} = L + \rho$ and Δ of (15.2.5), and so, by Proposition 15.4, we see that this zeta-function can be continued meromorphically to $\mathbb{C}^n$. When $\mathbf{y} = \mathbf{0}$, we sometimes write this zeta-function as $\zeta_r(\mathbf{s}; G)$ or $\zeta_r(\mathbf{s}; L; \Delta)$ for brevity. It should be noted that as a generalization of (3.1.6), we have

$$K(\Delta)^s \zeta_r(s, \dots, s; G) = \zeta_W(s; G) \tag{15.3.2}$$

and if $G = \widetilde{G}$, then $L = P$ and $\zeta_r(\mathbf{s}; \widetilde{G})$ coincides with $\zeta_r(\mathbf{s}; \mathfrak{g})$ defined in (3.1.5).

For any lattice M, we define $M^* = \mathrm{Hom}(M, \mathbb{Z})$. Since $Q^* = P^\vee$ and $P^* = Q^\vee$, from $Q \subset L \subset P$ we obtain

$$P^\vee = Q^* \supset L^* \supset P^* = Q^\vee. \tag{15.3.3}$$

We define

$$\delta_{L^*/Q^\vee}(\mu) = \begin{cases} 1 & (\mu \in L^*/Q^\vee), \\ 0 & (\mu \notin L^*/Q^\vee). \end{cases} \tag{15.3.4}$$

Proposition 15.6 ([118, Proposition 3.1]). *Let L be a lattice satisfying $Q \subset L \subset P$. For $\mu \in P^\vee/Q^\vee$, we have*

$$\widehat{\iota_{L+\rho}}(\mu) = \frac{(-1)^{\langle \mu, 2\rho \rangle}}{\#(P/L)} \delta_{L^*/Q^\vee}(\mu) \in \mathbb{Q}. \tag{15.3.5}$$

Proof. We have

$$\sum_{\lambda \in (L+\rho)/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda \rangle} = (-1)^{\langle \mu, 2\rho \rangle} \sum_{\lambda \in L/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda \rangle}.$$

Note that $(-1)^{\langle \mu, 2\rho \rangle} \in \{1, -1\}$ due to $\rho \in Q/2$. We can show that

$$\sum_{\lambda \in L/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda \rangle} = \begin{cases} \#(L/Q) & (\mu \in L^*/Q^\vee), \\ 0 & (\mu \notin L^*/Q^\vee). \end{cases} \tag{15.3.6}$$

This is shown as follows: if $\mu \in L^*/Q^\vee$, then $\langle \mu, \lambda \rangle \in \mathbb{Z}$ for all $\lambda \in L/Q$, and if $\mu \notin L^*/Q^\vee$, then there exists $\lambda_0 \in L$ such that $\langle \mu, \lambda_0 \rangle \notin \mathbb{Z}$ and

$$\sum_{\lambda \in L/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda \rangle} = \sum_{\lambda \in L/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda + \lambda_0 \rangle} \tag{15.3.7}$$

implies

$$(1 - e^{-2\pi\sqrt{-1}\langle \mu, \lambda_0 \rangle}) \sum_{\lambda \in L/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda \rangle} = 0, \tag{15.3.8}$$

hence (15.3.6).

Therefore (15.2.18) gives

$$\widehat{\iota_{L+\rho}}(\mu) = (-1)^{\langle \mu, 2\rho\rangle} \frac{\#(L/Q)}{\#(P/Q)} \delta_{L^*/Q^\vee}(\mu).$$

□

In particular,

$$\widehat{\iota_{P+\rho}}(\mu) = \frac{1}{\#(P/Q)} \sum_{\lambda \in (P+\rho)/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda\rangle} = \delta_{\mu,0} \tag{15.3.9}$$

(because $\delta_{P^*/Q^\vee}(\mu) = \delta_{Q^\vee/Q^\vee}(\mu) = 1$ only if $\mu = 0$), and

$$\widehat{\iota_{Q+\rho}}(\mu) = \frac{1}{\#(P/Q)} \sum_{\lambda \in (Q+\rho)/Q} e^{-2\pi\sqrt{-1}\langle \mu, \lambda\rangle} = \frac{(-1)^{\langle \mu, 2\rho\rangle}}{\#(P/Q)}. \tag{15.3.10}$$

We define

$$P(\mathbf{k}, \mathbf{y}; L; \Delta) = P(\mathbf{k}, \mathbf{y}, \iota_{L+\rho}; \Delta). \tag{15.3.11}$$

When $\mathbf{y} = \mathbf{0}$, we write $P(\mathbf{k}, \mathbf{0}; L; \Delta) = B(\mathbf{k}; L; \Delta)$. Note that since $P/L \simeq L^*/Q^\vee \simeq \pi_1(G)$ (see, e.g., Sepanski [**181**, Section 6.3.3]), this can also be written as

$$P(\mathbf{k}, \mathbf{y}; L; \Delta) = \frac{1}{\#\pi_1(G)} \sum_{\mu \in \pi_1(G)} (-1)^{\langle \mu, 2\rho\rangle} P(\mathbf{k}, \mathbf{y} + \mu; \Delta) \tag{15.3.12}$$

by (15.2.12) and Proposition 15.6.

We can compute $P(\mathbf{k}, \mathbf{y}; L; \Delta)$ explicitly by (15.3.12). In particular, we have for $\nu \in P^\vee/Q^\vee$,

$$P(\mathbf{k}, \nu; L; \Delta) \in \mathbb{Q}. \tag{15.3.13}$$

From this fact we can deduce the following.

THEOREM 15.7 ([**118**, Theorem 3.2]). *Let G be a compact connected semisimple Lie group, and $L = L(G)$ be its weight lattice. Let $\mathbf{k} = (k_\alpha)_{\alpha \in \Delta_+} \in \mathbb{N}^{\#\Delta_+}$ satisfying $k_\alpha = k_\beta$ whenever $\|\alpha\| = \|\beta\|$. Let $\kappa = \sum_{\alpha \in \Delta_+} 2k_\alpha$. Then we have for $\nu \in P^\vee/Q^\vee$,*

$$\begin{aligned} \zeta_r(2\mathbf{k}, \nu; G) &= \zeta_r(2\mathbf{k}, \nu; L; \Delta) \\ &= \frac{(-1)^{\#\Delta_+}}{\#W} \left(\prod_{\alpha \in \Delta_+} \frac{(2\pi\sqrt{-1})^{2k_\alpha}}{(2k_\alpha)!} \right) P(2\mathbf{k}, \nu; L; \Delta) \in \mathbb{Q} \cdot \pi^\kappa. \end{aligned} \tag{15.3.14}$$

PROOF. By Theorem 15.5 with $\mathbf{s} = 2\mathbf{k}$ and $\mathbf{y} = \nu$, we obtain

$$\sum_{w \in W} \zeta_r(w^{-1}(2\mathbf{k}), w^{-1}\nu; L; \Delta) = (-1)^{\#\Delta_+} \left(\prod_{\alpha \in \Delta_+} \frac{(2\pi\sqrt{-1})^{2k_\alpha}}{(2k_\alpha)!} \right) P(2\mathbf{k}, \nu; L; \Delta).$$

Since roots of the same length form a single Weyl-orbit and $w^{-1}\nu \equiv \nu \pmod{Q^\vee}$ (similar to (15.2.4)), the left-hand side of the above is

$$\sum_{w\in W} \zeta_r(2\mathbf{k}, \nu; L; \Delta) = (\#W)\zeta_r(2\mathbf{k}, \nu; L; \Delta).$$

The assertion follows from this and (15.3.13). □

This theorem is the explicit form of the volume formula for the zeta-function of the lattice $L = L(G)$. In the case when $L = P$ and $\nu = 0$, (15.3.14) coincides with (4.2.7) in Theorem 4.5.

REMARK 15.8. The correspondence between Lie groups and lattices is a well-known fact, but here we sketch the demonstration for the convenience of readers. Let G be a compact connected semisimple Lie group, whose universal covering group is $\widetilde{G}$. There is a one-to-one correspondence between isomorphism classes of finite-dimensional irreducible representations of G and dominant analytically integral forms (for example, [**95**, Theorem 5.110]). The set of dominant analytically integral forms produces a sublattice $L = L(G)$ of the weight lattice (or the lattice of algebraically integral forms) $P = P(\Delta(\mathfrak{g}))$, and L includes the root lattice $Q = Q(\Delta(\mathfrak{g}))$ ([**95**, (4.63)]). In particular, $L(\widetilde{G}) = P$. Conversely, let $\widetilde{G}$ be a simply-connected Lie group, $\mathfrak{g} = \mathrm{Lie}(\widetilde{G})$, and let L be a lattice satisfying $Q = Q(\Delta(\mathfrak{g})) \subset L \subset P = P(\Delta(\mathfrak{g}))$. Then (15.3.3) holds. Since $P^\vee/Q^\vee$ is isomorphic to the center $\widetilde{Z}$ of $\widetilde{G}$ by the mapping

$$\Phi \ : \ P^\vee \ni \mu \mapsto \exp_{\widetilde{G}}(2\pi\sqrt{-1}\mu) \in \widetilde{Z}$$

(where $\exp_{\widetilde{G}}$ means the exponential mapping associated with $\widetilde{G}$), we may regard $L^*/Q^\vee$ as a subgroup of $\widetilde{G}$. Define $G = \widetilde{G}/(L^*/Q^\vee)$. We show that the lattice corresponding to G is L. Write $L_1 = L(G)$. Take a maximal torus K of G, and $\mathfrak{k} = \mathrm{Lie}(G)$. Let $\lambda \in L$ and $H \in \mathfrak{k}$ with $\exp_G(H) = 1$. The last condition implies $\exp_{\widetilde{G}}(H) \in \Phi(L^*)$, so $H \in 2\pi\sqrt{-1}L^*$, $\lambda(H) \in 2\pi\sqrt{-1}\mathbb{Z}$. Therefore $\lambda \in L_1$ by [**95**, Proposition 4.58], hence $L \subset L_1$. On the other hand, $(L_1^* : Q^\vee) = (L_1^* : P^*) = (P : L_1)$, but the right-hand side is equal to $(L^* : Q^\vee)$ by [**95**, Proposition 4.67]. Therefore $(L_1^* : Q^\vee) = (L^* : Q^\vee)$, and we can conclude that $L_1 = L$.

15.4. Explicit forms of zeta-functions

In this section, we will give several examples of explicit forms of zeta-functions defined by (15.3.1). When $L = P$, the zeta-function is

nothing but $\zeta_r(\mathbf{s};\mathfrak{g})$, so our main concern is the case $P \supsetneq L \supset Q$. We use the following standard notation:

$$SU(r) = \left\{ A \in M_r(\mathbb{C}) \mid \det A = 1, \overline{{}^tA}A = E_r \right\}$$

(where E_r is the unit matrix of degree r), and

$$SO(r) = \left\{ A \in M_r(\mathbb{R}) \mid \det A = 1, {}^tAA = E_r \right\}.$$

EXAMPLE 15.9 ([118, Example 4.1]). We first study the case of A_2 type. Let $\Delta = \Delta(A_2)$ with $\Psi = \{\alpha_1, \alpha_2\}$, $\Delta_+ = \{\alpha_1, \alpha_2, \alpha_1+\alpha_2\}$, $P = \mathbb{Z}\lambda_1 + \mathbb{Z}\lambda_2$, $Q = \mathbb{Z}\alpha_1 + \mathbb{Z}\alpha_2$, and $\rho = \lambda_1 + \lambda_2$. It is known that $(P : Q) = 3$ (see [20, Planche I]). Therefore the only lattice L with $P \supsetneq L \supset Q$ is Q. Then $Q_+ = P_+ \cap Q$. We show

$$Q_+ + \rho = \{ m_1\lambda_1 + m_2\lambda_2 \mid m_1, m_2 \in \mathbb{N},\ m_1 \equiv m_2 \pmod 3 \}. \tag{15.4.1}$$

To show this, first note that

$$\lambda_1 = \frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_2, \quad \lambda_2 = \frac{1}{3}\alpha_1 + \frac{2}{3}\alpha_2. \tag{15.4.2}$$

In fact, write $\lambda_1 = u\alpha_1 + v\alpha_2$. Since $\alpha_i^\vee = 2\alpha_i/\langle\alpha_i,\alpha_i\rangle$ and $\langle\alpha_i^\vee, \lambda_j\rangle = \delta_{i,j}$, we have

$$1 = \langle\alpha_1^\vee, \lambda_1\rangle = \frac{2}{\langle\alpha_1,\alpha_1\rangle}(u\langle\alpha_1,\alpha_1\rangle + v\langle\alpha_1,\alpha_2\rangle) = 2u - v,$$

and $0 = \langle\alpha_2^\vee, \lambda_1\rangle = -u + 2v$, from which we obtain $u = 2/3$, $v = 1/3$, so $\lambda_1 = (2/3)\alpha_1 + (1/3)\alpha_2$. The case of λ_2 is similar.

Let $\lambda = m_1\lambda_1 + m_2\lambda_2 \in Q_+ + \rho$ $(m_1, m_2 \in \mathbb{N})$. Then $\lambda - \rho = n_1\lambda_1 + n_2\lambda_2 \in Q_+$, where $n_j = m_j - 1$ $(j = 1, 2)$. From (15.4.2) we have

$$n_1\lambda_1 + n_2\lambda_2 = \frac{2n_1 + n_2}{3}\alpha_1 + \frac{n_1 + 2n_2}{3}\alpha_2.$$

Since this belongs to Q, we have $(2n_1+n_2)/3 \in \mathbb{Z}$ and $(n_1+2n_2)/3 \in \mathbb{Z}$, which are equivalent to $n_1 \equiv n_2 \pmod 3$. This implies (15.4.1).

The simply-connected group $\widetilde{G}$ in the A_2 case is the special unitary group $SU(3)$. This is compact, connected and simply-connected ([95, Proposition 1.136]). The corresponding Lie algebra $\mathrm{Lie}(SU(3))$ is $\mathfrak{su}(3) = \{X \in \mathfrak{sl}(3) \mid \overline{{}^tX} + X = 0\}$, and hence the complexified Lie algebra is $\mathbb{C} \otimes \mathfrak{su}(3) = \mathfrak{sl}(3)$ ([181, Section 6.1.2]).

Let $\widetilde{Z}$ be the center of $\widetilde{G}$. The group corresponding to Q is $\widetilde{G}/\widetilde{Z}$, which is the projective unitary group $PU(3)$. The zeta-function corresponding to P is

$$\zeta_2((s_1,s_2,s_3),\mathbf{y};SU(3)) = \zeta_2((s_1,s_2,s_3),\mathbf{y};P;A_2) = \sum_{m,n=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle\mathbf{y},m\lambda_1+n\lambda_2\rangle}}{m^{s_1}n^{s_2}(m+n)^{s_3}}, \tag{15.4.3}$$

which, when $\mathbf{y} = \mathbf{0}$, is the classical Mordell–Tornheim double sum. In the case $\mathbf{y} = \lambda_1^\vee = \frac{2}{3}\alpha_1^\vee + \frac{1}{3}\alpha_2^\vee$, we have

$$\zeta_2((s_1,s_2,s_3),\lambda_1^\vee;SU(3)) = \sum_{m,n=1}^{\infty} \frac{\varrho^{2m+n}}{m^{s_1}n^{s_2}(m+n)^{s_3}}, \tag{15.4.4}$$

where $\varrho = e^{2\pi\sqrt{-1}/3}$ is the cube root of unity. The zeta-function corresponding to Q is, by (15.3.1) and (15.4.1),

(15.4.5)

$$\zeta_2((s_1,s_2,s_3),\mathbf{y};PU(3)) = \zeta_2((s_1,s_2,s_3),\mathbf{y};Q;A_2)$$
$$= \sum_{\lambda\in Q_+ +\rho} \frac{e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}}{\langle\alpha_1^\vee,\lambda\rangle^{s_1}\langle\alpha_2^\vee,\lambda\rangle^{s_2}\langle\alpha_1^\vee+\alpha_2^\vee,\lambda\rangle^{s_3}} = \sum_{\substack{m,n=1\\ m\equiv n \pmod 3}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle\mathbf{y},m\lambda_1+n\lambda_2\rangle}}{m^{s_1}n^{s_2}(m+n)^{s_3}}.$$

This can be regarded as a kind of "partial zeta-function" of type A_2, the double analogue of the partial (Riemann) zeta-function.

For $\mathbf{y} = y_1\alpha_1^\vee + y_2\alpha_2^\vee$, we can compute $P(\mathbf{k},\mathbf{y};A_2)$ from their generating function $F(\mathbf{t},\mathbf{y};A_2)$, whose explicit form is given in Proposition 4.10. For example, the result for $P((2,2,2),\mathbf{y};A_2)$ is explicitly written as [**106**, (9.13)], from which, (15.2.12) and (15.3.11), when $\mathbf{y} = \mathbf{0}$, it follows that

$$B((2,2,2);Q;A_2) = \frac{187}{2755620}.$$

Therefore we obtain from Theorem 15.7 that

$$\zeta_2((2,2,2);PU(3)) = \sum_{\substack{m,n=1\\ m\equiv n \pmod 3}}^{\infty} \frac{1}{m^2n^2(m+n)^2} = \frac{187}{688905}\pi^6. \tag{15.4.6}$$

Similarly, we can compute

$$\zeta_2((4,4,4);PU(3)) = \frac{3279473}{48475988686125}\pi^{12}, \tag{15.4.7}$$

$$\zeta_2((6,6,6);PU(3)) = \frac{53109402098}{3020275543157103456225}\pi^{18}, \tag{15.4.8}$$

$$\zeta_2((8,8,8);PU(3)) = \frac{178778564412743}{3909780002479478774489029687 5}\pi^{24}. \tag{15.4.9}$$

Also, in the case $\mathbf{y} = \lambda_1^\vee = \frac{2}{3}\alpha_1^\vee + \frac{1}{3}\alpha_2^\vee$ (see (15.4.4)), that is, $(y_1,y_2) = \left(\frac{2}{3},\frac{1}{3}\right)$, we can similarly obtain

(15.4.10)

$$\zeta_2((2,2,2),\lambda_1^\vee;SU(3)) = \frac{53}{229635}\pi^6,$$

(15.4.11)

$$\zeta_2((4,4,4),\lambda_1^\vee;SU(3)) = \frac{1078771}{16158662895375}\pi^{12},$$

(15.4.12)
$$\zeta_2((6,6,6),\lambda_1^\vee;SU(3)) = \frac{88392335894}{5033792571928505760375}\pi^{18},$$

(15.4.13)
$$\zeta_2((8,8,8),\lambda_1^\vee;SU(3)) = \frac{1012923518531597}{2215542001405037972210450156 25}\pi^{24}.$$

Note that from the definition, we can confirm

$$\begin{aligned}\zeta_2((2p,2p,2p),\lambda_1^\vee;SU(3)) &= \zeta_2((2p,2p,2p),\lambda_2^\vee;SU(3)) \qquad (p\in\mathbb{N}),\\ \zeta_2((2p,2p,2p),\lambda_1^\vee;PU(3)) &= \zeta_2((2p,2p,2p),\lambda_2^\vee;PU(3))\\ &= \zeta_2((2p,2p,2p);PU(3)) \qquad (p\in\mathbb{N}).\end{aligned}$$

In the next section, we will prove certain functional relations for $\zeta_2(\mathbf{s},\mathbf{0};PU(3))$ including (15.4.6)–(15.4.9).

Remark 15.10. In the above example, $\lambda_1^\vee$ and $\lambda_2^\vee$ are minuscule (see Remark 15.1). The example shows that the case of minuscule coweights is essential in practical computations.

Remark 15.11. In [**189**, Section 5], Subbarao and Sitaramachandrarao proposed a problem of evaluating the double series

$$\sum_{m,n=1}^{\infty}\frac{(-1)^{m-1}}{m^k n^k (m+n)^k},\quad \sum_{m,n=1}^{\infty}\frac{(-1)^{m+n}}{m^k n^k (m+n)^k} \qquad (k\in\mathbb{N}).$$

As for the case of odd k, the third-named author evaluated each series in terms of values of $\zeta(s)$ (see [**196**][**197**]). The case of even k is still open. It follows from (15.4.4) that the above formulas (15.4.10)–(15.4.13) imply certain answers to a problem analogous to that of Subbarao and Sitaramachandrarao.

Example 15.12 ([**118**, Example 4.2]). We consider the type A_3. Let $\Delta=\Delta(A_3)$ with $\Psi=\{\alpha_1,\,\alpha_2,\,\alpha_3\}$, $\Delta_+=\{\alpha_1,\,\alpha_2,\,\alpha_3,\,\alpha_1+\alpha_2,\,\alpha_2+\alpha_3,\,\alpha_1+\alpha_2+\alpha_3\}$, $P=\sum_{j=1}^3\mathbb{Z}\lambda_j$ and $Q=\sum_{j=1}^3\mathbb{Z}\alpha_j$. Analogously to (15.4.2), we have

(15.4.14)
$$\lambda_1=\frac{3}{4}\alpha_1+\frac{1}{2}\alpha_2+\frac{1}{4}\alpha_3,\ \ \lambda_2=\frac{1}{2}\alpha_1+\alpha_2+\frac{1}{2}\alpha_3,\ \ \lambda_3=\frac{1}{4}\alpha_1+\frac{1}{2}\alpha_2+\frac{3}{4}\alpha_3.$$

It is known that $P/Q\simeq\mathbb{Z}/4\mathbb{Z}$ (see [**20**]). Therefore there is a unique intermediate lattice L_1 with $P\supsetneq L_1\supsetneq Q$, satisfying $(L_1:Q)=2$. The group corresponding to P (resp. Q) is $SU(4)$ (resp. $PU(4)$). The group $G=G(L_1)$ is $SU(4)/\{\pm1\}$, which is known to be isomorphic to $SO(6)$.

From (15.3.1), we see that

(15.4.15)

$$\zeta_3(\mathbf{s},\mathbf{y};SU(4)) = \zeta_3(\mathbf{s},\mathbf{y};P;A_3)$$
$$= \sum_{m_1,m_2,m_3=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1+m_2\lambda_2+m_3\lambda_3\rangle}}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(m_1+m_2+m_3)^{s_6}}.$$

Note that $\zeta_3(\mathbf{s},\mathbf{0};SU(4)) = \zeta_3(\mathbf{s};A_3)$ (see (3.3.20)). Similar to the A_2 case (see Example 15.9), from the generating function which was already given in (4.3.9), we can compute $P((2,2,2,2,2,2),\mathbf{y};A_3)$, though it is too long to write it here. Hence we can obtain by (15.2.12) that

$$P((2,2,2,2,2,2),\lambda_1^\vee;P;A_3) = -\frac{19329337}{14283291230208000},$$

where $\lambda_1^\vee = \frac{3}{4}\alpha_1^\vee + \frac{1}{2}\alpha_2^\vee + \frac{1}{4}\alpha_3^\vee$. Therefore we obtain from Theorem 15.7 that

(15.4.16)

$$\zeta_3((2,2,2,2,2,2),\lambda_1^\vee;SU(4)) = \zeta_3((2,2,2,2,2,2),\lambda_1^\vee;P;A_3)$$
$$= \sum_{m_1,m_2,m_3=1}^{\infty} \frac{i^{3l+2m+n}}{m_1^2m_2^2m_3^2(m_1+m_2)^2(m_2+m_3)^2(m_1+m_2+m_3)^2}$$
$$= -\frac{19329337}{2678117105664000}\pi^{12}.$$

Concerning L_1 and Q, similar to (15.4.1), we can show

(15.4.17)

$$(L_1)_+ + \rho = \left\{ \sum_{j=1}^{3} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^3,\ m_1\equiv m_3 \pmod 2 \right\},$$

(15.4.18)

$$Q_+ + \rho = \left\{ \sum_{j=1}^{3} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^3,\ m_1+2m_2+3m_3\equiv 2 \pmod 4 \right\}.$$

In fact, letting $\lambda = \sum_{j=1}^3 m_j\lambda_j \in Q_+ + \rho$ $(m_j\in\mathbb{N})$, we have $\lambda-\rho = \sum_{j=1}^3 n_j\lambda_j \in Q_+$, where $n_j = m_j - 1$ $(1\le j\le 3)$. From (15.4.14) we have

$$\sum_{j=1}^{3} n_j\lambda_j = \frac{3n_1+2n_2+n_3}{4}\alpha_1 + \frac{n_1+2n_2+n_3}{2}\alpha_2 + \frac{n_1+2n_2+3n_3}{4}\alpha_3,$$

which belongs to Q. Therefore

$$\text{(i) } 3n_1+2n_2+n_3\equiv 0 \pmod 4;$$
$$\text{(ii) } n_1+2n_2+n_3\equiv 0 \pmod 2;$$

$$\text{(iii) } n_1 + 2n_2 + 3n_3 \equiv 0 \pmod 4.$$

We see that (iii) implies

$$\text{(iv) } n_1 \equiv n_3 \pmod 2,$$

which automatically implies (ii). Moreover, we find that (iii) and (iv) imply (i). Therefore the only essential condition is (iii), which is equivalent to the congruence condition in (15.4.18).

Next, define the homomorphism $\varsigma : P \cong \mathbb{Z}^3 \to \mathbb{Z}/4\mathbb{Z}$ by

$$\varsigma(n_1, n_2, n_3) = n_1 + 2n_2 + 3n_3 \pmod 4.$$

Then from the above argument we find that $Q = \operatorname{Ker} \varsigma$. Let L_1^* be the set of all (n_1, n_2, n_3) satisfying $n_1 \equiv n_3 \pmod 2$. Then $\{0\} \subsetneq \varsigma(L_1^*) \subsetneq \mathbb{Z}/4\mathbb{Z}$, hence $Q \subsetneq L_1^* \subsetneq P$. Therefore L_1^* should be equal to L_1, which implies (15.4.17).

Let $\mathbf{y} = y_1\alpha_1^\vee + y_2\alpha_2^\vee + y_3\alpha_3^\vee$. From (15.4.17) and (15.4.18) we obtain

(15.4.19)

$$\begin{aligned}\zeta_3(\mathbf{s}, \mathbf{y}; SO(6)) &= \zeta_3(\mathbf{s}, \mathbf{y}; L_1; A_3)\\ &= \sum_{\substack{m_1,m_2,m_3=1\\ m_1\equiv m_3 \pmod 2}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2 + m_3\lambda_3\rangle}}{m_1^{s_1} m_2^{s_2} m_3^{s_3} (m_1+m_2)^{s_4} (m_2+m_3)^{s_5} (m_1+m_2+m_3)^{s_6}},\end{aligned}$$

(15.4.20)

$$\begin{aligned}\zeta_3(\mathbf{s}, \mathbf{y}; PU(4)) &= \zeta_3(\mathbf{s}, \mathbf{y}; Q; A_3)\\ &= \sum_{\substack{m_1,m_2,m_3=1\\ m_1+2m_2+3m_3\equiv 2 \pmod 4}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2 + m_3\lambda_3\rangle}}{m_1^{s_1} m_2^{s_2} m_3^{s_3} (m_1+m_2)^{s_4} (m_2+m_3)^{s_5} (m_1+m_2+m_3)^{s_6}}.\end{aligned}$$

EXAMPLE 15.13 ([**118**, Example 4.3]). We consider the cases of types B_r and C_r. The simply-connected group $\widetilde{G}$ in the B_r case is the spinor group $Spin(2r+1)$, which can be defined over the Clifford algebra, and $\widetilde{G}/\widetilde{Z} = SO(2r+1)$, where $\widetilde{Z}$ is the center of $\widetilde{G}$ (see [**181**, Theorem 1.40]).

In the C_r case, $\widetilde{G} = Sp(r)$, the unitary group over quaternions, which is also isomorphic to the intersection of complex symplectic group and complex unitary group ([**95**, Proposition 1.139]). Then $\widetilde{G}/\widetilde{Z}$ is the projective symplectic group $PSp(r)$. The explicit forms of zeta-functions for $r = 2, 3$ are

$$\zeta_2(\mathbf{s}, \mathbf{y}; Spin(5)) = \zeta_2(\mathbf{s}, \mathbf{y}; P; B_2)$$

$$= \sum_{m_1,m_2=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1+m_2\lambda_2\rangle}}{m_1^{s_1}m_2^{s_2}(m_1+m_2)^{s_3}(2m_1+m_2)^{s_4}},$$

$$\zeta_2(\mathbf{s},\mathbf{y};Sp(2)) = \zeta_2(\mathbf{s},\mathbf{y};P;C_2)$$

$$= \sum_{m_1,m_2=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m\lambda_1+n\lambda_2\rangle}}{m_1^{s_1}m_2^{s_2}(m_1+m_2)^{s_3}(m_1+2m_2)^{s_4}},$$

$$\zeta_3(\mathbf{s},\mathbf{y};Spin(7)) = \zeta_3(\mathbf{s},\mathbf{y};P;B_3)$$

$$= \sum_{m_1,m_2,m_3=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1+m_2\lambda_2+m_3\lambda_3\rangle}}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(2m_2+m_3)^{s_6}}$$

$$\times \frac{1}{(m_1+m_2+m_3)^{s_7}(m_1+2m_2+m_3)^{s_8}(2m_1+2m_2+m_3)^{s_9}},$$

$$\zeta_3(\mathbf{s},\mathbf{y};Sp(3)) = \zeta_3(\mathbf{s},\mathbf{y};P;C_3)$$

$$= \sum_{m_1,m_2,m_3=1}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1+m_2\lambda_2+m_3\lambda_3\rangle}}{m_1^{s_1}m_2^{s_2}m_3^{s_3}(m_1+m_2)^{s_4}(m_2+m_3)^{s_5}(m_2+2m_3)^{s_6}}$$

$$\times \frac{1}{(m_1+m_2+m_3)^{s_7}(m_1+m_2+2m_3)^{s_8}(m_1+2m_2+2m_3)^{s_9}}.$$

We know that $(P:Q)=2$ for the case of types B_r and C_r (see [**20**]). Therefore the lattice L with $P \supset L \supset Q$ coincides with P or Q. We show

$$Q_+(B_2)+\rho = \left\{ \sum_{j=1}^{2} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^2,\ m_2 \equiv 1 \pmod 2 \right\},$$

$$Q_+(C_2)+\rho = \left\{ \sum_{j=1}^{2} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^2,\ m_1 \equiv 1 \pmod 2 \right\},$$

$$Q_+(B_3)+\rho = \left\{ \sum_{j=1}^{3} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^3,\ m_3 \equiv 1 \pmod 2 \right\},$$

$$Q_+(C_3)+\rho = \left\{ \sum_{j=1}^{3} m_j\lambda_j \,\middle|\, (m_j)\in\mathbb{N}^3,\ m_1 \equiv m_3 \pmod 2 \right\}.$$

In fact, we consider, for example, the case of B_3. Analogously to Examples 15.9 and 15.12, we have

(15.4.21)

$$\lambda_1 = \alpha_1+\alpha_2+\alpha_3, \quad \lambda_2 = \alpha_1+2\alpha_2+2\alpha_3, \quad \lambda_3 = \frac{1}{2}\alpha_1+\alpha_2+\frac{3}{2}\alpha_3.$$

Let $\lambda = \sum_{j=1}^{3} m_j\lambda_j \in Q_+ + \rho$ ($m_j \in \mathbb{N}$). Then $\lambda - \rho = \sum_{j=1}^{3} n_j\lambda_j \in Q_+$, where $n_j = m_j - 1$ ($1 \le j \le 3$). it follows from (15.4.21) that

$$\lambda - \rho = \frac{2n_1 + 2n_2 + n_3}{2}\alpha_1 + (n_1 + 2n_2 + n_3)\alpha_2 + \frac{2n_1 + 4n_2 + 3n_3}{2}\alpha_3,$$

which belongs to Q. Therefore $n_3 \equiv 0 \pmod 2$, that is, $m_3 \equiv 1 \pmod 2$. The cases of B_2, C_2 and C_3 can be similarly treated.

Therefore we obtain

(15.4.22)

$$\begin{aligned}\zeta_2(\mathbf{s}, \mathbf{y}; SO(5)) &= \zeta_2(\mathbf{s}, \mathbf{y}; Q; B_2)\\ &= \sum_{\substack{m_1, m_2 = 1\\ m_2 \equiv 1 \pmod 2}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2\rangle}}{m_1^{s_1} m_2^{s_2} (m_1 + m_2)^{s_3} (2m_1 + m_2)^{s_4}},\end{aligned}$$

(15.4.23)

$$\begin{aligned}\zeta_2(\mathbf{s}, \mathbf{y}; PSp(2)) &= \zeta_2(\mathbf{s}, \mathbf{y}; Q; C_2)\\ &= \sum_{\substack{m_1, m_2 = 1\\ m_1 \equiv 1 \pmod 2}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2\rangle}}{m_1^{s_1} m_2^{s_2} (m_1 + m_2)^{s_3} (m_1 + 2m_2)^{s_4}},\end{aligned}$$

(15.4.24)

$$\begin{aligned}\zeta_3(\mathbf{s}, \mathbf{y}; SO(7)) &= \zeta_3(\mathbf{s}, \mathbf{y}; Q; B_3)\\ &= \sum_{\substack{m_1, m_2, m_3 = 1\\ m_3 \equiv 1 \pmod 2}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2 + m_3\lambda_3\rangle}}{m_1^{s_1} m_2^{s_2} m_3^{s_3} (m_1 + m_2)^{s_4} (m_2 + m_3)^{s_5} (2m_2 + m_3)^{s_6}}\\ &\quad \times \frac{1}{(m_1 + m_2 + m_3)^{s_7} (m_1 + 2m_2 + m_3)^{s_8} (2m_1 + 2m_2 + m_3)^{s_9}},\end{aligned}$$

(15.4.25)

$$\begin{aligned}\zeta_3(\mathbf{s}, \mathbf{y}; PSp(3)) &= \zeta_3(\mathbf{s}, \mathbf{y}; Q; C_3)\\ &= \sum_{\substack{m_1, m_2, m_3 = 1\\ m_1 \equiv m_3 \pmod 2}}^{\infty} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}, m_1\lambda_1 + m_2\lambda_2 + m_3\lambda_3\rangle}}{m_1^{s_1} m_2^{s_2} m_3^{s_3} (m_1 + m_2)^{s_4} (m_2 + m_3)^{s_5} (m_2 + 2m_3)^{s_6}}\\ &\quad \times \frac{1}{(m_1 + m_2 + m_3)^{s_7} (m_1 + m_2 + 2m_3)^{s_8} (m_1 + 2m_2 + 2m_3)^{s_9}}.\end{aligned}$$

Now we evaluate special values. Consider the case $G = PSp(2)$. Theorem 15.7 of type C_2 with $\nu = \mathbf{y} = \mathbf{0}$ gives that

$$\zeta_2((2k, 2l, 2l, 2k); PSp(2)) \in \mathbb{Q} \cdot \pi^{4(k+l)} \tag{15.4.26}$$

for $k, l \in \mathbb{N}$. Actually, we have already given the generating function of type C_2 (see (4.3.6)) and also the generalized Bernoulli function $P((2, 2, 2, 2), \mathbf{y}; C_2)$ (see [**105**, Examples 3]). Similarly, we can give explicit forms of $P((2k, 2l, 2l, 2k), \mathbf{y}; C_2)$ ($k, l \in \mathbb{N}$), though it is too

complicated to describe them here. Using these results, we can obtain the following explicit formulas:

$$\zeta_2((2,2,2,2);PSp(2)) = \frac{\pi^8}{322560},$$
$$\zeta_2((2,4,4,2);PSp(2)) = \frac{29}{3832012800}\pi^{12},$$
$$\zeta_2((4,2,2,4);PSp(2)) = \frac{13}{3832012800}\pi^{12},$$
$$\zeta_2((4,4,4,4);PSp(2)) = \frac{479}{55794106368000}\pi^{16}.$$

We will give another type of evaluation formulas in the following sections (see Examples 15.24 and 15.32).

REMARK 15.14. As stated in Remark 4.8, from (15.1.3) and (3.1.6) we see that the original volume formula of Witten is restricted to the case $\mathbf{s} = 2(k_\alpha)_{\alpha\in\Delta_+}$, where all the k_αs are the same. Our Theorem 15.7 covers a wider class of special values, such as the $(2,4,4,2)$ and $(4,2,2,4)$ cases in the above.

15.5. An application to partial multiple zeta values

In Chapter 12, we developed a method of studying MZVs from the viewpoint of the zeta-function of C_r (Theorems 12.5 and 12.22). Further, as a B_r-analogue, we discussed a kind of partial multiple zeta-function defined by (12.3.1). The same idea works in the situation of the present chapter. We first prove the following consequence of Theorem 9.20, which has not been published before.

THEOREM 15.15. *For a compact connected semisimple Lie group G, let $\Delta = \Delta(G)$ be its root system, and $L = L(G)$ be its weight lattice. Let $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta_+} \in \mathbb{N}_0^{\#\Delta_+}$ with $k_\alpha \geq 1$ (for $\alpha \in \Delta^*$) and $k_\alpha = 0$ (for $\alpha \in \Delta_+ \setminus \Delta^*$) satisfying $k_\alpha = k_\beta$ whenever $\|\alpha\| = \|\beta\|$, and let $\mathbf{k}_{\Delta^*} = (k_\alpha)_{\alpha\in\Delta^*}$. Let $\kappa = \sum_{\alpha\in\Delta_+} 2k_\alpha$.*

Then we have for $\nu \in P^\vee/Q^\vee$,

$$\zeta_r(2\mathbf{k},\nu;G) = \zeta_r(2\mathbf{k},\nu;L;\Delta) \in \mathbb{Q}\cdot\pi^\kappa.$$

PROOF. By (9.5.19) (with $\mathbf{k}$ replaced by $2\mathbf{k}$), we have

(15.5.1)

$$\sum_{\mu\in P^\vee/Q^\vee} \widehat{f}(\mu) \sum_{w\in W} \Big(\prod_{\alpha\in\Delta_+\cap w\Delta_-} (-1)^{2k_\alpha} \Big) \zeta_r(w^{-1}(2\mathbf{k}), w^{-1}(\mathbf{y}+\mu+\nu);\Delta)$$

$$= (-1)^{\#(\Delta_+)} \sum_{\mu \in P^\vee/Q^\vee} \widehat{f}(\mu) P_{\Delta^*}((2\mathbf{k})_{\Delta^*}, \mathbf{y} + \mu + \nu; \Delta) \Bigg(\prod_{\alpha \in \Delta^*} \frac{(2\pi i)^{2k_\alpha}}{(2k_\alpha)!} \Bigg).$$

Using the fact $w^{-1}(\mu + \nu) \equiv \mu + \nu \mod Q^\vee$ and (15.2.8), we see that the left-hand side is

$$\sum_{\mu \in P^\vee/Q^\vee} \widehat{f}(\mu) \sum_{w \in W} \zeta_r(w^{-1}(2\mathbf{k}), w^{-1}\mathbf{y} + \mu + \nu; \Delta) \tag{15.5.2}$$
$$= \sum_{w \in W} \zeta_r(w^{-1}(2\mathbf{k}), w^{-1}\mathbf{y} + \nu, f; \Delta).$$

Now put $\mathbf{y} = 0$ and $f = \iota_{L+\rho}$. Since $k_\alpha = k_\beta$ for $\|\alpha\| = \|\beta\|$, the right-hand side of (15.5.2) is $(\#W)\zeta_r(2\mathbf{k}, \nu, \iota_{L+\rho}; \Delta)$, and hence

$$\zeta_r(2\mathbf{k}, \nu; G) = \zeta_r(2\mathbf{k}, \nu, \iota_{L+\rho}; \Delta)$$
$$= \frac{(-1)^{\#(\Delta_+)}}{\#W} \sum_{\mu \in P^\vee/Q^\vee} \widehat{\iota_{L+\rho}}(\mu) P_{\Delta^*}((2\mathbf{k})_{\Delta^*}, \mu + \nu; \Delta) \Bigg(\prod_{\alpha \in \Delta^*} \frac{(2\pi i)^{2k_\alpha}}{(2k_\alpha)!} \Bigg).$$

From Proposition 15.6 we have $\widehat{\iota_{L+\rho}}(\mu) \in \mathbb{Q}$, and from Theorem 5.11 and (9.5.25) we find that $P_{\Delta^*}((2\mathbf{k})_{\Delta^*}, \mu + \nu; \Delta) \in \mathbb{Q}$. Therefore we obtain the assertion. □

In Chapter 12, we applied Theorem 9.20 with $\Delta^* = \Delta_{l+}(C_r)$ or $\Delta^* = \Delta_{s+}(B_r)$ to evaluate the values of $\zeta_{EZ,r}$ or $\zeta_r^\sharp$. Similarly, now, we apply Theorem 15.15 with $\Delta^* = \Delta_{l+}(C_r)$ or $\Delta^* = \Delta_{s+}(B_r)$ $(r = 2, 3)$ to obtain the following evaluation of partial sums (restricted by congruence conditions) of $\zeta_{EZ,r}$ or $\zeta_r^\sharp$. These examples were announced in [108] without the details of the proofs.

EXAMPLE 15.16 ([108, Theorem 5.1]). For $p \in \mathbb{N}$,

$$\sum_{\substack{m,n \geq 1 \\ n \equiv 1 (\mathrm{mod}\ 2)}} \frac{1}{n^{2p}(2m+n)^{2p}} \in \mathbb{Q}\pi^{4p},$$

$$\sum_{\substack{m,n \geq 1 \\ m \equiv 1 (\mathrm{mod}\ 2)}} \frac{1}{n^{2p}(m+n)^{2p}} \in \mathbb{Q}\pi^{4p},$$

$$\sum_{\substack{l,m,n \geq 1 \\ n \equiv 1 (\mathrm{mod}\ 2)}} \frac{1}{n^{2p}(2m+n)^{2p}(2l+2m+n)^{2p}} \in \mathbb{Q}\pi^{6p},$$

$$\sum_{\substack{l,m,n \geq 1 \\ l \equiv n (\mathrm{mod}\ 2)}} \frac{1}{n^{2p}(m+n)^{2p}(l+m+n)^{2p}} \in \mathbb{Q}\pi^{6p}.$$

More explicitly, for example,

EXAMPLE 15.17 ([108, Example 5.2]).

$$\sum_{\substack{m,n\geq 1\\ m\equiv 1(\mathrm{mod}\ 2)}} \frac{1}{n^6(m+n)^6} = \frac{1}{58060800}\pi^{12},$$
$$\sum_{\substack{m,n\geq 1\\ m\equiv 1(\mathrm{mod}\ 2)}} \frac{1}{n^8(m+n)^8} = \frac{17}{390168576000}\pi^{16}.$$

15.6. Functional relations and various evaluation formulas

In Section 15.4, we gave explicit forms of several zeta-functions of Lie groups, and especially gave some evaluation formulas for the cases of types A_2, C_2 and A_3 at even integer points, by computing generating functions of their values. However, it seems a difficult problem to evaluate zeta-functions of Lie groups at arbitrary positive integer points by that method. In this section, we give various evaluation formulas for zeta values for the cases of types A_2 and $C_2(\simeq B_2)$, by proving certain functional relations among them which are analogues to our previous results given in Chapters 8, 9 and 11. The advantage of the method in this section is that it may treat the special values at $\mathbf{s}=\mathbf{l}=(l_\alpha)_{\alpha\in\Delta_+}$, $l_\alpha\in\mathbb{N}$ where some of them are odd.

First, we consider $\zeta_2(\mathbf{s};PU(3))=\zeta_2(\mathbf{s},\mathbf{0};Q;A_2)$ and prove the following theorem.

THEOREM 15.18 ([118, Theorem 5.1]). *For $p,q\in\mathbb{N}$,*

(15.6.1)

$$3\Big\{\zeta_2((p,q,s);PU(3))+(-1)^p\zeta_2((p,s,q);PU(3))+(-1)^q\zeta_2((q,s,p);PU(3))\Big\}$$
$$=-\sum_{\tau=0}^{p}\binom{p+q-\tau-1}{q-1}(-1)^\tau\frac{(2\pi\sqrt{-1})^\tau}{\tau!}\sum_{a=0}^{2}B_\tau\left(\frac{a}{3}\right)\phi(s+p+q-\tau,-a/3)$$
$$-\sum_{\tau=0}^{q}\binom{p+q-\tau-1}{p-1}\frac{(2\pi\sqrt{-1})^\tau}{\tau!}\sum_{a=0}^{2}B_\tau\left(\frac{a}{3}\right)\phi(s+p+q-\tau,a/3)$$

holds for $s\in\mathbb{C}$ except for singularities of functions on both sides.

EXAMPLE 15.19 ([118, Example 5.2]). It should be emphasized that Theorem 15.18 gives evaluation formulas for $\zeta_2((a,b,c);PU(3))$ when $a+b+c$ is odd. For example, putting $(p,q,s)=(1,1,1)$ in

(15.6.1), we have

(15.6.2)

$$3\zeta_2((1,1,1);PU(3)) = \sum_{\tau=0}^{1}(-1)^\tau \frac{(2\pi\sqrt{-1})^\tau}{\tau!}\sum_{a=0}^{2} B_\tau\left(\frac{a}{3}\right)\sum_{m=1}^{\infty}\frac{\varrho^{-ma}}{m^{3-\tau}} + \sum_{\tau=0}^{1}\frac{(2\pi\sqrt{-1})^\tau}{\tau!}\sum_{a=0}^{2} B_\tau\left(\frac{a}{3}\right)\sum_{m=1}^{\infty}\frac{\varrho^{ma}}{m^{3-\tau}}.$$

We can easily check that

$$\sum_{a=0}^{2}\varrho^{la}B_1\left(\frac{a}{3}\right) = \begin{cases} -\frac{1}{2} & (l\equiv 0 \pmod 3)\\ -\frac{1}{2}-\frac{\sqrt{-1}}{2\sqrt{3}} & (l\equiv 1 \pmod 3)\\ -\frac{1}{2}+\frac{\sqrt{-1}}{2\sqrt{3}} & (l\equiv 2 \pmod 3).\end{cases} \tag{15.6.3}$$

Then (15.6.2) can be rewritten to

$$\zeta_2((1,1,1);PU(3)) = \frac{2}{27}\zeta(3) + \frac{2\pi}{3\sqrt{3}}L(2,\chi_3), \tag{15.6.4}$$

where we denote by χ_3 the primitive Dirichlet character of conductor 3. This is an analogue of $\zeta_2((1,1,1);SU(3)) = 2\zeta(3)$ (Example 8.22). Similarly, setting $(p,q,s) = (1,2,2)$ in (15.6.1), and using the relations (15.6.3) and

$$\sum_{a=0}^{2}\varrho^{la}B_2\left(\frac{a}{3}\right) = \begin{cases}\frac{1}{18} & (l\equiv 0 \pmod 3)\\ \frac{2}{9} & (l\equiv 1,2 \pmod 3),\end{cases} \tag{15.6.5}$$

we can obtain

$$\zeta_2((2,2,1);PU(3)) = -\frac{1}{81}\zeta(5) + \frac{35\pi^2}{243}\zeta(3) - \frac{2\pi}{3\sqrt{3}}L(4,\chi_3). \tag{15.6.6}$$

The above formulas (15.6.4) and (15.6.6) can also be deduced by using

(15.6.7)

$$\phi(s,1/3) - \phi(s,2/3) = 2\sqrt{-1}\sum_{m=1}^{\infty}\frac{\sin(2\pi m/3)}{m^s} = \sqrt{-3}L(s,\chi_3)$$

instead of (15.6.3), (15.6.5). A more general result will be given in the next section (see Theorem 15.27).

By the partial fraction decomposition, we have

$$\zeta_2((1,1,1);PU(3)) = \sum_{\substack{m,n=1\\ m\equiv n \pmod 3}}^{\infty}\frac{1}{mn(m+n)} = 2\sum_{\substack{m,n=1\\ m\equiv n \pmod 3}}^{\infty}\frac{1}{m(m+n)^2}.$$

Hence combining with (15.6.4) we obtain

$$\sum_{\substack{m,n=1\\ m\equiv n \pmod 3}}^{\infty}\frac{1}{m(m+n)^2} = \frac{1}{27}\zeta(3) + \frac{\pi}{3\sqrt{3}}L(2,\chi_3). \tag{15.6.8}$$

This can be regarded as a formula for a partial sum of the double zeta value, analogous to the following well-known result given by Euler (see (12.1.1)):

$$\sum_{m,n=1}^{\infty} \frac{1}{m(m+n)^2} = \zeta(3).$$

REMARK 15.20. Setting $(p,q,s) = (2k,2k,2k)$ in Theorem 15.18 and using (4.2.4), we obtain

(15.6.9)

$$\zeta_2((2k,2k,2k); PU(3)) = \frac{(2\pi\sqrt{-1})^{6k}}{9} \sum_{\tau=0}^{2k} \binom{4k-\tau-1}{2k-1} \sum_{a=0}^{2} \frac{B_\tau\left(\frac{a}{3}\right)}{\tau!} \frac{B_{6k-\tau}\left(\frac{a}{3}\right)}{(6k-\tau)!} \quad (k \in \mathbb{N}),$$

which is an explicit form of Theorem 15.7 for $PU(3)$ and includes (15.4.6)-(15.4.9).

Now we give the proof of Theorem 15.18. We first prove the following lemma. This lemma first appeared in [**105**, Lemma 9.1] in the case when p is even, and the full form of the lemma was then stated in [**118**, Lemma 5.4].

LEMMA 15.21. *Let $t \in [0,2\pi) \subset \mathbb{R}$, and $h : \mathbb{N}_0 \to \mathbb{C}$ be a function (which may depend on t). Then, for $p \in \mathbb{N}$,*

$$\sum_{j=0}^{p} \phi(p-j)\varepsilon_{p-j} \sum_{\xi=0}^{j} h(j-\xi) \frac{(\sqrt{-1}(t-\pi))^\xi}{\xi!} = -\frac{1}{2} \sum_{\xi=0}^{p} h(p-\xi) \frac{(2\pi\sqrt{-1})^\xi}{\xi!} B_\xi\left(\frac{t}{2\pi}\right). \tag{15.6.10}$$

PROOF. We have from (12.2.4) that

$$\lim_{N\to\infty} \sum_{\substack{n=-N \\ n\neq 0}}^{N} \frac{(-1)^n e^{\sqrt{-1}n\theta}}{n^p} = 2\sum_{j=0}^{p} \phi(p-j)\epsilon_{p-j} \frac{(\sqrt{-1}\theta)^j}{j!} \tag{15.6.11}$$

for $p \in \mathbb{N}$, $\theta \in (-\pi,\pi)$.

First, assume $t \in (0,2\pi)$. On the left-hand side of (15.6.10), we change variables (j,ξ) to (τ,ν) with $\nu = \xi$ and $\tau = p+\xi-j$, where $0 \le \tau \le p$ and $0 \le \nu \le \tau$. Then the left-hand side is equal to

$$\sum_{\tau=0}^{p} h(p-\tau) \sum_{\nu=0}^{\tau} \phi(\tau-\nu)\epsilon_{\tau-\nu} \frac{(\sqrt{-1}(t-\pi))^\nu}{\nu!}.$$

Applying (15.6.11) to the part $\tau \geq 1$ and noting $\phi(0) = \zeta(0) = -1/2$, we find that the above is equal to

$$\frac{1}{2}\sum_{\tau=1}^{p} h(p-\tau) \lim_{N\to\infty} \sum_{\substack{n=-N \\ n\neq 0}}^{N} \frac{(-1)^n e^{\sqrt{-1}n(t-\pi)}}{n^\tau} - \frac{1}{2}h(p)$$

$$= \frac{1}{2}\sum_{\tau=1}^{p} h(p-\tau) \lim_{N\to\infty} \sum_{\substack{n=-N \\ n\neq 0}}^{N} \frac{e^{\sqrt{-1}nt}}{n^\tau} - \frac{1}{2}h(p),$$

which is, by (4.2.4), further equal to

$$-\frac{1}{2}\sum_{\tau=0}^{p} h(p-\tau)\frac{(2\pi\sqrt{-1})^\tau}{\tau!} B_\tau\left(\frac{t}{2\pi}\right).$$

Therefore we obtain (15.6.10) for $t \in (0, 2\pi)$. Furthermore, each side of (15.6.10) is continuous for $t \in [0, 2\pi)$. Thus we obtain the proof. □

Put $\theta = t - \pi$ $(0 \leq t < 2\pi)$ in (11.3.15) (Lemma 11.6) and multiply by $(-1)^p$ both sides. Then, on the left-hand side we divide the sum into three parts according to the conditions $l > 0$, $-m < l < 0$ and $l < -m$, while on the right-hand side we use Lemma 15.21. Consequently, we obtain the following formula. Note that this can also be derived by a certain transformation of a result of Nakamura [**163**, Theorem 3.1] when $|x| = 1$.

LEMMA 15.22 ([**118**, Lemma 5.5]). *For $p, q \in \mathbb{N}$, $s, t \in \mathbb{R}$ with $s > 1$ and $t \in [0, 2\pi)$, and $x \in \mathbb{C}$ with $|x| \leq 1$,*

(15.6.12)

$$\sum_{l,m=1}^{\infty} \frac{x^{l+m}e^{\sqrt{-1}mt}}{l^p m^q (l+m)^s} + (-1)^p \sum_{l,m=1}^{\infty} \frac{x^m e^{\sqrt{-1}(l+m)t}}{l^p m^s (l+m)^q} + (-1)^q \sum_{l,m=1}^{\infty} \frac{x^m e^{-\sqrt{-1}lt}}{l^q m^s (l+m)^p}$$

$$= -\sum_{\tau=0}^{p} \binom{p+q-\tau-1}{q-1} (-1)^\tau \sum_{m=1}^{\infty} \frac{x^m e^{\sqrt{-1}mt}}{m^{s+p+q-\tau}} \frac{(2\pi\sqrt{-1})^\tau}{\tau!} B_\tau\left(\frac{t}{2\pi}\right)$$

$$-\sum_{\tau=0}^{q} \binom{p+q-\tau-1}{p-1} \sum_{m=1}^{\infty} \frac{x^m}{m^{s+p+q-\tau}} \frac{(2\pi\sqrt{-1})^\tau}{\tau!} B_\tau\left(\frac{t}{2\pi}\right).$$

Using these results, we give the proof of Theorem 15.18 as follows.

PROOF OF THEOREM 15.18. Let $x = e^{-2\sqrt{-1}t}$ and further let $t = 2\pi a/3$ $(a = 0, 1, 2)$ on both sides of (15.6.12). Then, summing up with $a = 0, 1, 2$ and using the fact for $\varrho = e^{2\pi\sqrt{-1}/3}$ that

$$\sum_{a=0}^{2} \varrho^{Na} = \begin{cases} 3 & N \equiv 0 \pmod 3 \\ 0 & N \not\equiv 0 \pmod 3, \end{cases}$$

we have

$$
\begin{aligned}
3\Bigg\{ & \sum_{\substack{l,m\geq 1\\ l\equiv m \pmod 3}} \frac{1}{l^p m^q (l+m)^s} + (-1)^p \sum_{\substack{l,m\geq 1\\ l\equiv m \pmod 3}} \frac{1}{l^p m^s (l+m)^q} \\
& + (-1)^q \sum_{\substack{l,m\geq 1\\ l\equiv m \pmod 3}} \frac{1}{l^q m^s (l+m)^p} \Bigg\} \\
= & -\sum_{\tau=0}^{p} \binom{p+q-\tau-1}{q-1} (-1)^\tau \sum_{a=0}^{2} \sum_{m=1}^{\infty} \frac{\varrho^{-ma}}{m^{s+p+q-\tau}} \frac{(2\pi\sqrt{-1})^\tau}{\tau!} B_\tau\left(\frac{a}{3}\right) \\
& - \sum_{\tau=0}^{q} \binom{p+q-\tau-1}{p-1} \sum_{a=0}^{2} \sum_{m=1}^{\infty} \frac{\varrho^{ma}}{m^{s+p+q-\tau}} \frac{(2\pi\sqrt{-1})^\tau}{\tau!} B_\tau\left(\frac{a}{3}\right)
\end{aligned}
$$

(where in the last sum we use $\varrho^{-2ma} = \varrho^{ma}$). Noting (15.4.5) and using Proposition 15.4, we complete the proof of Theorem 15.18. □

Secondly, we consider the case of type C_2, namely the zeta-function $\zeta_2(\mathbf{s}; PSp(2)) = \zeta_2(\mathbf{s}, \mathbf{0}; Q; C_2)$ defined by (15.4.23) with $\mathbf{y} = \mathbf{0}$. Applying Lemma 15.21 to (8.3.15) with $\theta = t - \pi$ and $x = e^{-2\sqrt{-1}t}$, we obtain

(15.6.13)

$$
\begin{aligned}
& \sum_{\substack{l\geq 1\\ m\geq 1}} \frac{e^{\sqrt{-1}lt}}{l^p m^s (l+m)^q (l+2m)^r} + \sum_{\substack{l\geq 1\\ m\geq 1\\ l\neq m\\ l\neq 2m}} \frac{e^{-\sqrt{-1}lt}}{(-l)^p m^s (-l+m)^q (-l+2m)^r} \\
& + \sum_{\xi=0}^{p} \sum_{\omega=0}^{p-\xi} \binom{\omega+r-1}{\omega} \binom{p+q-1-\xi-\omega}{q-1} \frac{(-1)^{p-\xi}}{2^{r+\omega}} \frac{(2\pi\sqrt{-1})^\xi}{\xi!} \\
& \qquad \times \zeta(s+p+q+r-\xi) B_\xi(t/2\pi) \\
& + \sum_{\xi=0}^{q} \sum_{\omega=0}^{q-\xi} \binom{\omega+r-1}{\omega} \binom{p+q-1-\xi-\omega}{p-1} (-1)^{p-\omega} \frac{(2\pi\sqrt{-1})^\xi}{\xi!} \\
& \qquad \times \phi(s+p+q+r-\xi, -t/2\pi) B_\xi(t/2\pi) \\
& + \sum_{\xi=0}^{r} \sum_{\omega=0}^{p-1} \binom{\omega+r-\xi}{\omega} \binom{p+q-2-\omega}{q-1} \frac{(-1)^p}{2^{r-\xi+\omega+1}} \frac{(2\pi\sqrt{-1})^\xi}{\xi!} \\
& \qquad \times \phi(s+p+q+r-\xi, -t/\pi) B_\xi(t/2\pi) \\
& + \sum_{\xi=0}^{r} \sum_{\omega=0}^{q-1} \binom{\omega+r-\xi}{\omega} \binom{p+q-2-\omega}{p-1} (-1)^{p-\omega+1} \frac{(2\pi\sqrt{-1})^\xi}{\xi!} \\
& \qquad \times \phi(s+p+q+r-\xi, -t/\pi) B_\xi(t/2\pi) = 0
\end{aligned}
$$

for $p,q,r\in\mathbb{N}$ and $s,t\in\mathbb{R}$ with $s>1$ and $t\in[0,2\pi)$. Actually, this equation with (p,q,r) replaced by $(2p,2q,2p)$ coincides with [**105**, (9.8)] in the case $(\eta,\rho,\delta,\tau)=(t,0,0,0)$. Denote the second sum on the left-hand side of (15.6.13) by Σ_2. We split Σ_2 into two parts according to the conditions $l<m$ or $l>m$, and transform variables as $j=m-l(\geq 1)$ when $l<m$, and $j=l-m(\geq 1)$ when $l>m$. In the latter case we further split the sum according to $j<m$ or $j>m$ (that is, $l<2m$ or $l>2m$). Then we obtain

(15.6.14)

$$\begin{aligned}\Sigma_2=&(-1)^p\sum_{\substack{l\geq 1\\ m\geq 1}}\frac{e^{-\sqrt{-1}lt}}{l^pm^q(l+m)^s(l+2m)^r}+(-1)^{p+q}\sum_{\substack{l\geq 1\\ m\geq 1}}\frac{e^{-\sqrt{-1}(l+2m)t}}{l^rm^q(l+m)^s(l+2m)^p}\\&+(-1)^{p+q+r}\sum_{\substack{l\geq 1\\ m\geq 1}}\frac{e^{-\sqrt{-1}(l+2m)t}}{l^rm^s(l+m)^q(l+2m)^p}.\end{aligned}$$

Replacing Σ_2 by (15.6.14) on the left-hand side of (15.6.13), and denote the resulting left-hand side by $H(t)$. Then (15.6.13) implies $H(t)=0$ for any $t\in[0,2\pi)$. Let $t=0,\pi$. We note that $e^{\pm\sqrt{-1}l\pi}=e^{-\sqrt{-1}(l+2m)\pi}=(-1)^l$ for $l,m\in\mathbb{N}$. Also we have $\phi(s,0)=\phi(s,-1)=\zeta(s)$, $\phi(s,-1/2)=\phi(s)=(2^{1-s}-1)\zeta(s)$. Therefore, considering $(H(0)-H(\pi))/2=0$ and noting (15.4.23), similar to Theorem 15.18, we have the following result.

THEOREM 15.23 ([**118**, Theorem 5.6]). *For $p,q,r\in\mathbb{N}$,*

(15.6.15)

$$\begin{aligned}&\zeta_2((p,s,q,r);PSp(2))+(-1)^p\zeta_2((p,q,s,r);PSp(2))\\&\quad+(-1)^{p+q}\zeta_2((r,q,s,p);PSp(2))+(-1)^{p+q+r}\zeta_2((r,s,q,p);PSp(2))\\&+\sum_{\xi=0}^{p}\sum_{\omega=0}^{p-\xi}\binom{\omega+r-1}{\omega}\binom{p+q-1-\xi-\omega}{q-1}\frac{(-1)^{p-\xi}}{2^{r+\omega}}\frac{(2\pi\sqrt{-1})^\xi}{\xi!}\\&\qquad\times\zeta(s+p+q+r-\xi)\frac{B_\xi(0)-B_\xi(1/2)}{2}\\&+\sum_{\xi=0}^{q}\sum_{\omega=0}^{q-\xi}\binom{\omega+r-1}{\omega}\binom{p+q-1-\xi-\omega}{p-1}(-1)^{p-\omega}\frac{(2\pi\sqrt{-1})^\xi}{\xi!}\\&\qquad\times\zeta(s+p+q+r-\xi)\frac{B_\xi(0)-\left(2^{1-s-p-q-r+\xi}-1\right)B_\xi(1/2)}{2}\\&+\sum_{\xi=0}^{r}\sum_{\omega=0}^{p-1}\binom{\omega+r-\xi}{\omega}\binom{p+q-2-\omega}{q-1}\frac{(-1)^p}{2^{r-\xi+\omega+1}}\frac{(2\pi\sqrt{-1})^\xi}{\xi!}\end{aligned}$$

$$\times \zeta(s+p+q+r-\xi)\frac{B_\xi(0)-B_\xi(1/2)}{2}$$
$$+\sum_{\xi=0}^{r}\sum_{\omega=0}^{q-1}\binom{\omega+r-\xi}{\omega}\binom{p+q-2-\omega}{p-1}(-1)^{p-\omega+1}\frac{(2\pi\sqrt{-1})^\xi}{\xi!}$$
$$\times \zeta(s+p+q+r-\xi)\frac{B_\xi(0)-B_\xi(1/2)}{2}=0$$

holds for $s\in\mathbb{C}$ except for singularities.

EXAMPLE 15.24 ([**118**, Example 5.7]). By Theorem 15.23, we can evaluate $\zeta_2((a,b,c,d);PSp(2))$ in some case when $a+b+c+d$ is odd. For example, setting $(p,s,q,r)=(2,1,1,1)$ and $(2,3,3,5)$ in (15.6.15), we have

$$\zeta_2((2,1,1,1);PSp(2))=\frac{3}{8}\zeta(2)\zeta(3)-\frac{31}{64}\zeta(5),$$
$$\zeta_2((2,3,3,5);PSp(2))=-\frac{15}{16}\zeta(4)\zeta(9)-\frac{17379}{4096}\zeta(2)\zeta(11)+\frac{8191}{1024}\zeta(13).$$

In general, it seems to be difficult to evaluate $\zeta_2((a,b,c,d),\mathbf{0};PSp(2))$ for arbitrary $a,b,c,d\in\mathbb{N}$. We will further consider this problem in the next section.

REMARK 15.25. Putting $(p,s,q,r)=(2k,2l,2l,2k)$ $(k,l\in\mathbb{N})$ in (15.6.15), we can see that

$$\zeta_2((2k,2l,2l,2k);PSp(2))$$

can be expressed as a polynomial in $\zeta(4k+4l-\xi)(\sqrt{-1}\pi)^\xi$ with $\mathbb{Q}$-coefficients. Since $\zeta_2((2k,2l,2l,2k);PSp(2))\in\mathbb{R}$, we see that the part consisting of the terms of $\zeta(4k+4l-\xi)(\sqrt{-1}\pi)^\xi$ for odd ξ vanish. On the other hand, for even ξ, each term belongs to $\mathbb{Q}\cdot\pi^{4(k+l)}$. Thus we recover (15.4.26).

REMARK 15.26. In [**114**], we studied the case of A_3 by the same type of argument as in the present section, and proved functional relations for zeta-functions of $SU(4)$, of $SO(6)$, and of $PU(4)$. In particular, a functional relation for $\zeta_3(\mathbf{s};A_3)$ is shown there ([**114**, Theorem 9]), which is not included in the present book. From the viewpoint of the present chapter, the case of type A_r is the most interesting, because $|P(\Delta(A_r))/Q(\Delta(A_r))|\to\infty$ as $r\to\infty$ (and hence, there are many intermediate lattices between P and Q), while $|P/Q|$ remains small for any Lie algebras of other types (see Bourbaki [**20**]).

15.7. Parity results

As stated in Section 12.5, a *parity result* means a property where some multiple zeta value whose weight and depth are of different parity can be written in terms of multiple zeta values of lower depth.

It is an interesting problem to ask what kinds of multiple series have this type of property. We again state some known results by using the notation defined in this chapter. Tornheim (see Corollary 8.21) proved that $\zeta_2((a,b,c);SU(3)) = \zeta_2((a,b,c);P,A_2)$ can be expressed as a polynomial in $\{\zeta(j+1)\,|\,j\in\mathbb{N}\}$ with $\mathbb{Q}$-coefficients if its weight $a+b+c$ is odd. Also the third-named author (see Proposition 8.31) proved that $\zeta_2((a,b,c,d);Sp(2)) = \zeta_2((b,a,c,d);Spin(5))$ has this property.

In this section we first prove the following fact, which is a $PU(3)$-type analogue of Tornheim's result stated above.

THEOREM 15.27 ([118, Theorem 6.1]). *Let $a,b,c\in\mathbb{N}$. If $a+b+c$ is odd then the value $\zeta_2((a,b,c);PU(3))$ can be expressed as a polynomial in $\{\phi(j,a/3)\,|\,a\in\{0,1,2\},\ j\in\mathbb{N}\}$ with $\mathbb{Q}[\pi,\sqrt{-1}]$-coefficients.*

PROOF. Denote by $\mathfrak{X}$ the set of polynomials in $\{\phi(j,a/3)\,|\,a\in\{0,1,2\},\ j\in\mathbb{N}\}$ with $\mathbb{Q}[\pi,\sqrt{-1}]$-coefficients. Then we see that the right-hand side of (15.6.1) with $(p,q,s)=(c,a,b)$ is in $\mathfrak{X}$. First, we consider the case a is odd and b,c is even, namely $a+b+c$ is odd. Then, by (15.6.1), we have

$$\zeta_2((c,a,b);PU(3)) + \zeta_2((c,b,a);PU(3)) - \zeta_2((a,b,c);PU(3)) \in \mathfrak{X}.$$

Also, setting $(p,q,s)=(c,b,a)$ in (15.6.1), we have

$$\zeta_2((c,a,b);PU(3)) + \zeta_2((c,b,a);PU(3)) + \zeta_2((b,a,c);PU(3)) \in \mathfrak{X}.$$

Note that $\zeta_2((p,q,r);PU(3)) = \zeta_2((q,p,r);PU(3))$. Hence, these imply the assertion $\zeta_2((a,b,c);PU(3))\in\mathfrak{X}$. As for other cases, we can similarly prove their assertions. □

EXAMPLE 15.28 ([118, Example 6.2]). Setting $(p,q,s)=(1,3,5)$ and $(1,5,3)$ in (15.6.1), we have

$$\begin{aligned}
&\zeta_2((1,3,5);PU(3)) - \zeta_2((1,5,3);PU(3)) - \zeta_2((3,5,1);PU(3))\\
&\quad = -\frac{4}{3}\zeta(9) + \frac{\pi^2}{9}\zeta(7) - \frac{4}{3}(\phi(9,1/3)+\phi(9,2/3))\\
&\qquad + \frac{2\pi\sqrt{-1}}{9}(\phi(8,1/3)-\phi(8,2/3))\\
&\qquad - \frac{\pi^2}{27}(\phi(7,1/3)+\phi(7,2/3)) + \frac{4\pi^3\sqrt{-1}}{243}(\phi(6,1/3)-\phi(6,2/3)),
\end{aligned}$$

$$\begin{aligned}
&\zeta_2((1,5,3);PU(3)) - \zeta_2((1,3,5);PU(3)) - \zeta_2((5,3,1);PU(3)) \\
&\quad = -2\zeta(9) + \frac{\pi^2}{9}\zeta(7) + \frac{\pi^4}{135}\zeta(5) - 2(\phi(9,1/3) + \phi(9,2/3)) \\
&\qquad + \frac{2\pi\sqrt{-1}}{9}(\phi(8,1/3) - \phi(8,2/3)) - \frac{\pi^2}{27}(\phi(7,1/3) + \phi(7,2/3)) \\
&\qquad + \frac{4\pi^3\sqrt{-1}}{243}(\phi(6,1/3) - \phi(6,2/3)) - \frac{13\pi^4}{3645}(\phi(5,1/3) + \phi(5,2/3)) \\
&\qquad + \frac{4\pi^5\sqrt{-1}}{2187}(\phi(4,1/3) - \phi(4,2/3)).
\end{aligned}$$

Combining these results and noting

$$\zeta_2((3,5,1);PU(3)) = \zeta_2((5,3,1);PU(3)),$$

we have

$$\begin{aligned}
&\zeta_2((3,5,1);PU(3)) \\
&= \frac{5}{3}\zeta(9) - \frac{\pi^2}{9}\zeta(7) - \frac{\pi^4}{270}\zeta(5) \\
&+ \frac{5}{3}(\phi(9,1/3) + \phi(9,2/3)) - \frac{2\pi\sqrt{-1}}{9}(\phi(8,1/3) - \phi(8,2/3)) \\
&+ \frac{\pi^2}{27}(\phi(7,1/3) + \phi(7,2/3)) - \frac{4\pi^3\sqrt{-1}}{243}(\phi(6,1/3) - \phi(6,2/3)) \\
&+ \frac{13\pi^4}{7290}(\phi(5,1/3) + \phi(5,2/3)) - \frac{2\pi^5\sqrt{-1}}{2187}(\phi(4,1/3) - \phi(4,2/3)).
\end{aligned}$$

REMARK 15.29. Applying (15.6.7) and

$$\phi(s,1/3) + \phi(s,2/3) = (3^{1-s} - 1)\zeta(s) \tag{15.7.1}$$

to the above expression, we find that $\zeta_2((3,5,1);PU(3))$ can actually be written in terms of Riemann-zeta values and values of the Dirichlet L-function attached to χ_3. In general, since from (15.6.7) and (15.7.1) we see that $\phi(s,1/3)$ and $\phi(s,2/3)$ can be written in terms of $L(s,\chi_3)$ and $\zeta(s)$, Theorem 15.27 can be reinterpreted that *if $a+b+c$ is odd then $\zeta_2((a,b,c);PU(3))$ can be expressed as a polynomial in $\{\zeta(j), L(j,\chi_3) \mid j \in \mathbb{N}\}$ with $\mathbb{Q}[\pi,\sqrt{-1}]$-coefficients.* This agrees with the results stated in Example 15.19.

Next, consider the C_2 case. Since we have already known the parity result for $\zeta_2((a,b,c,d);Sp(2))$ (Proposition 8.31), it should be not surprising to know that the following parity result for $\zeta_2((a,b,c,d),\mathbf{0};PSp(2))$ holds.

THEOREM 15.30 ([118, Theorem 6.3]). *Let $a, b, c, d \in \mathbb{N}$. If $a + b + c + d$ is odd then $\zeta_2((a, b, c, d), \mathbf{0}; PSp(2))$ can be expressed as a polynomial in $\{\zeta(j + 1) \mid j \in \mathbb{N}\}$ with $\mathbb{Q}$-coefficients.*

In this case, we cannot directly obtain the assertion from Theorem 15.23 unlike the case of $PU(3)$. In fact, even if we use (15.6.15), it seems unable to obtain an expression of $\zeta_2((1, 2, 2, 2); PSp(2))$ because this value vanishes if we set $(p, s, q, r) = (1, 2, 2, 2)$ or $(2, 2, 2, 1)$ in (15.6.15). Hence we use another method as follows. First, we quote the following.

LEMMA 15.31 ([204, Theorem 4.1]). *Let*

$$\mathfrak{T}_{\tau,\mu}(k, l, d) = \sum_{\substack{l \geq 0 \\ m \geq 0}} \frac{1}{(2l + \tau)^a (2m + \mu)^b (2l + 2m + \tau + \mu)^c} \tag{15.7.2}$$

for $k, l, d \in \mathbb{N}$ and $\tau, \mu \in \{1, 2\}$. Suppose $k + l + d$ is odd, then $\mathfrak{T}_{\tau,\mu}(k, l, d)$ can be expressed as a polynomial in $\{\zeta(j + 1) \mid j \in \mathbb{N}\}$ with $\mathbb{Q}$-coefficients.

It should be noted that the assertion in [204, Theorem 4.1] includes the condition $d \geq 2$. However, by examining its proof, we can remove this condition. More precisely, we know that [204, Theorem 4.1] can be derived from [204, Theorem 3.4] which includes a condition $d \geq 2$. We can easily check that [204, Theorem 3.4] holds for $d = 1$ if we interpret the empty sum as 0 in its statement. Thus [204, Theorem 4.1] holds for $d = 1$ which implies the above lemma. By this lemma we can prove Theorem 15.30 as follows.

PROOF OF THEOREM 15.30. First, we use the relation

$$\begin{aligned} \frac{(-1)^c}{X^c (X + Y)^d} = &\sum_{j=1}^{c} \binom{c + d - j - 1}{c - j} (-1)^j \frac{1}{Y^{c+d-j} X^j} \\ &+ \sum_{j=1}^{d} \binom{c + d - j - 1}{d - j} \frac{1}{Y^{c+d-j} (X + Y)^j} \end{aligned} \tag{15.7.3}$$

for $c, d \in \mathbb{N}$, given by replacing (X, Y) with $(-X, X + Y)$ in (12.1.2). Setting $(X, Y) = (2l + 1 + m, m)$ in (15.7.3), we see that

(15.7.4)

$$(-1)^c \zeta_2((a, b, c, d), \mathbf{0}; PSp(2)) = \sum_{\substack{l \geq 0 \\ m \geq 1}} \frac{1}{(2l + 1)^a m^b (2l + 1 + m)^c (2l + 1 + 2m)^d}$$

$$
\begin{aligned}
&= \sum_{j=1}^{c} \binom{c+d-j-1}{c-j} (-1)^j \sum_{\substack{l\geq 0\\ m\geq 1}} \frac{1}{(2l+1)^a m^{b+c+d-j}(2l+1+m)^j} \\
&\quad + \sum_{j=1}^{d} \binom{c+d-j-1}{d-j} \sum_{\substack{l\geq 0\\ m\geq 1}} \frac{1}{(2l+1)^a m^{b+c+d-j}(2l+1+2m)^j} \\
&= \sum_{j=1}^{c} \binom{c+d-j-1}{c-j} \\
&\qquad \times (-1)^j \left\{ \mathfrak{T}_{1,1}(a, b+c+d-j, j) + \mathfrak{T}_{1,2}(a, b+c+d-j, j) \right\} \\
&\quad + \sum_{j=1}^{d} \binom{c+d-j-1}{d-j} 2^{b+c+d-j} \mathfrak{T}_{1,2}(a, b+c+d-j, j).
\end{aligned}
$$

Hence, by Lemma 15.31, we obtain the assertion. □

EXAMPLE 15.32 ([**118**, Example 6.5]). As we noted above, it seems impossible to obtain an expression of $\zeta_2((1,2,2,2); PSp(2))$ in terms of $\zeta(s)$, from (15.6.15). Hence we use (15.7.4). Then we have

$$
\begin{aligned}
(15.7.5) \qquad \zeta_2((1,2,2,2); PSp(2)) &= -2\mathfrak{T}_{1,1}(1,5,1) + 62\mathfrak{T}_{1,2}(1,5,1) \\
&\quad + \mathfrak{T}_{1,1}(1,4,2) + 17\mathfrak{T}_{1,2}(1,4,2).
\end{aligned}
$$

By the method used in [**204**, Section 4], we can obtain

$$
\begin{aligned}
\mathfrak{T}_{1,1}(1,5,1) &= -\frac{105}{128}\zeta(3)\zeta(4) - \frac{93}{128}\zeta(5)\zeta(2) + \frac{381}{128}\zeta(7), \\
\mathfrak{T}_{1,2}(1,5,1) &= -\frac{7}{128}\zeta(3)\zeta(4) - \frac{31}{128}\zeta(5)\zeta(2) + \frac{127}{256}\zeta(7), \\
\mathfrak{T}_{1,1}(1,4,2) &= \frac{105}{128}\zeta(3)\zeta(4) + \frac{279}{128}\zeta(5)\zeta(2) - \frac{1143}{256}\zeta(7), \\
\mathfrak{T}_{1,2}(1,4,2) &= \frac{7}{128}\zeta(3)\zeta(4) + \frac{183}{128}\zeta(5)\zeta(2) - \frac{635}{256}\zeta(7).
\end{aligned}
$$

Substituting these results into (15.7.5), we obtain

$$
\zeta_2((1,2,2,2); PSp(2)) = \frac{827}{64}\zeta(5)\zeta(2) - \frac{1397}{64}\zeta(7).
$$

CHAPTER 16

Lattice sums of hyperplane arrangements

The aim of the present chapter is to prove Theorems 9.10, 9.11, 9.12 and 9.14, stated in Section 9.4. For this purpose, we will introduce certain lattice sums of hyperplane arrangements, and the generating functions of special values of those lattice sums. These are generalizations of $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ and $F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta)$ defined by (9.4.1) and (9.4.10), respectively (see also (9.4.11)). The details of our theory of lattice sums of hyperplane arrangements were originally published in [**115**].

As an alternative application, we calculate special values of affine analogue of zeta-functions of root systems. Although in the cases of affine root systems it is natural to work with the character formulas instead of the dimension formulas, a straightforward generalization is also interesting. We will present some examples in Section 16.2.

Our consideration is not restricted to the case in the domain of absolute convergence; we will study the values of lattice sums outside the domain of absolute convergence. Here we explain this point by simple examples.

Recall (4.2.4), which is valid for $k \in \mathbb{N}$. In the case $k = 0$, then (4.2.4) does not hold straightforward. However, in this case, the following equation holds:

$$
\begin{aligned}
-\frac{(2\pi\sqrt{-1})^k}{k!} B_k(\{y\}) &= -1 \\
&= -e^{2\pi\sqrt{-1}my}|_{m=0} \\
&= -\lim_{N\to\infty} \sum_{\substack{|m|\le N \\ m=0}} e^{2\pi\sqrt{-1}my},
\end{aligned}
\tag{16.0.1}
$$

where $B_0(\{y\}) = 1$. From this observation, we may formulate the following formal procedure: *In the formula valid for $k \geq 1$, replacing the sum $\sum_{m\neq 0}$ by $-\sum_{m=0}$ and putting $k = 0$ with $0^0 = 1$, we arrive at the formula valid for $k = 0$.*[1]

[1] In the present case, this procedure can be explained in view of the meromorphic continuation of the Lerch zeta-functions. In fact, the series expression (4.1.8) at $s = 0$ can not be computed, while it can be evaluated as -1 by (4.1.10).

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Springer Monographs in Mathematics, https://doi.org/10.1007/978-981-99-0910-0_16

This interpretation works well in the multi-dimensional cases. For example, let $\alpha, \beta, \gamma \in \mathbb{C}$ and $k_1, k_2, k_3 \in \mathbb{N}_0$, and consider the sum

(16.0.2) $S((k_1,k_2,k_3),(y_1,y_2))$

$$= \lim_{N\to\infty} \sum_{\substack{m,n\in\mathbb{Z} \\ m+\alpha,n+\beta,m+n+\gamma\neq 0 \\ |m|,|n|\leq N}} \frac{e^{2\pi\sqrt{-1}(my_1+ny_2)}}{(m+\alpha)^{k_1}(n+\beta)^{k_2}(m+n+\gamma)^{k_3}}.$$

This is convergent if $k_1, k_2, k_3 \geq 1$. If some of k_i's are 0, then we modify the series. In the case when only $k_1 = 0$, we replace the condition $m+\alpha \neq 0$ by $m+\alpha = 0$ in the sum with the minus sign and $0^0 = 1$, that is,

(16.0.3)

$$S((0,k_2,k_3),(y_1,y_2)) = -\lim_{N\to\infty} \sum_{\substack{m,n\in\mathbb{Z} \\ |m|,|n|\leq N \\ n+\beta,m+n+\gamma\neq 0 \\ m+\alpha=0}} \frac{e^{2\pi\sqrt{-1}(my_1+ny_2)}}{(n+\beta)^{k_2}(m+n+\gamma)^{k_3}}.$$

By the restriction $m+\alpha = 0$, this sum is 0 if $\alpha \notin \mathbb{Z}$. If $\alpha \in \mathbb{Z}$, then the sum reduces to the one-dimensional sum

(16.0.4)

$$S((0,k_2,k_3),(y_1,y_2)) = -\lim_{N\to\infty} \sum_{\substack{n\in\mathbb{Z} \\ |n|\leq N \\ n+\beta,n+\gamma-\alpha\neq 0}} \frac{e^{2\pi\sqrt{-1}(-\alpha y_1+ny_2)}}{(n+\beta)^{k_2}(n+\gamma-\alpha)^{k_3}}.$$

In the other cases, the sum is similarly modified. Then the special values $S((k_1,k_2,k_3),(y_1,y_2))$ for all $k_1, k_2, k_3 \in \mathbb{N}_0$ are explicitly given by coefficients of a generating function, which will be given in Example 16.6.

In the above arguments the sums are taken over all integers. However, in some special cases, it is possible to treat sums running over only positive integers (Examples 16.7, 16.8), which may be regarded as zeta-functions associated with hyperplane arrangements.

In the next section we will introduce more general lattice sums, and their generating functions.

16.1. Notations and the statement of results on lattice sums

We fix a positive integer d. Let $V' = \mathbb{R}^d$ be a real vector space equipped with the standard inner product[2] $\langle\cdot,\cdot\rangle'$. We regard $f =$

[2]We use the notation V' and $\langle\ ,\ \rangle'$ in order to distinguish them from the r-dimensional V and $\langle\ ,\ \rangle$ which will appear in Section 16.9.

$(\vec{f}, \dot{f}) \in V' \times \mathbb{C}$ with $\vec{f} \in V'$ and $\dot{f} \in \mathbb{C}$ as an affine linear functional on V' by $f(\mathbf{v}) = \langle \vec{f}, \mathbf{v} \rangle' + \dot{f}$ for $\mathbf{v} \in V'$.

We use the following notation: For $X \subset V'$, put $\mathrm{Lh}_{\mathbb{Z}}[X] = \sum_{\mathbf{v} \in X} \mathbb{Z}\mathbf{v}$. For $Y \subset V' \times \mathbb{C}$, put $\vec{Y} = \{\vec{f} \mid f = (\vec{f}, \dot{f}) \in Y\}$.

Let $\Lambda \subset (\mathbb{Z}^d \setminus \{0\}) \times \mathbb{C}$ with $\#\Lambda < \infty$ such that $\operatorname{rank} \mathrm{Lh}_{\mathbb{Z}}[\vec{\Lambda}] = d$. Put $\widetilde{\Lambda} = \{f \in \Lambda \mid \operatorname{rank} \mathrm{Lh}_{\mathbb{Z}}[\vec{\Lambda} \setminus \{\vec{f}\}] \neq d\}$. Let $\mathbf{k} = (k_f)_{f \in \Lambda} \in \mathbb{N}_0^{\#\Lambda}$. For any $k \in \mathbb{N}_0$, define

$$\Lambda_k = \Lambda_k(\mathbf{k}) = \{f \in \Lambda \mid k_f = k\},$$
$$\Lambda_+ = \Lambda_+(\mathbf{k}) = \{f \in \Lambda \mid k_f > 0\}.$$

Obviously

$$\Lambda_+ = \bigcup_{k \geq 1} \Lambda_k \quad \text{and} \quad \Lambda = \Lambda_+ \cup \Lambda_0.$$

For $H \subset \Lambda$ such that $\operatorname{rank} \mathrm{Lh}_{\mathbb{Z}}[\vec{H}] = d - 1$, let $\mathfrak{H}_H = \sum_{g \in H} \mathbb{R}\, \vec{g}$ be the hyperplane passing through $\vec{H} \cup \{0\}$.

The following is the main object in the present chapter, a lattice sum over the hyperplane arrangement given by linear functionals belonging to Λ.

Definition 16.1. For $\mathbf{k} = (k_f)_{f \in \Lambda} \in \mathbb{N}_0^{\#\Lambda}$ and

$$\mathbf{y} \in V' \setminus \bigcup_{f \in \widetilde{\Lambda} \cap \Lambda_1} (\mathfrak{H}_{\Lambda \setminus \{f\}} + \mathbb{Z}^d),$$

we define

$$S(\mathbf{k}, \mathbf{y}; \Lambda) = \lim_{N \to \infty} Z(N; \mathbf{k}, \mathbf{y}; \Lambda), \tag{16.1.1}$$

where

$$Z(N; \mathbf{k}, \mathbf{y}; \Lambda) = (-1)^{\#\Lambda_0} \sum_{\substack{\mathbf{v} = (v_1, \dots, v_d) \in \mathbb{Z}^d \\ |v_j| \leq N \quad (1 \leq j \leq d) \\ f(\mathbf{v}) \neq 0 \quad (f \in \Lambda_+) \\ f(\mathbf{v}) = 0 \quad (f \in \Lambda_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \mathbf{v} \rangle'} \prod_{f \in \Lambda_+} \frac{1}{f(\mathbf{v})^{k_f}} \tag{16.1.2}$$

for $N > 0$.

This $S(\mathbf{k}, \mathbf{y}; \Lambda)$ is a generalization of the notion of "Weyl group symmetric" linear combinations of zeta-functions of root systems $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ (9.4.1) (see also (9.4.2)) in the case $\mathbf{s} = \mathbf{k}$. The first main result in this section is as follows.

Theorem 16.2 ([115, Theorem 2.2]). *The series $S(\mathbf{k}, \mathbf{y}; \Lambda)$ converges and is continuous in $\mathbf{y}$ on $V' \setminus \bigcup_{f \in \widetilde{\Lambda} \cap \Lambda_1} (\mathfrak{H}_{\Lambda \setminus \{f\}} + \mathbb{Z}^d)$.*

In order to define the generating function of $S(\mathbf{k},\mathbf{y};\Lambda)$, we need some more notations. Let $\mathscr{B}=\mathscr{B}(\Lambda)$ be the set of all subsets $B=\{f_1,\dots,f_d\}\subset\Lambda$ such that $\vec{B}$ forms a basis of V'. For $B\in\mathscr{B}$, let $\vec{B}^*=\{\vec{f}_1^{\,B},\dots,\vec{f}_d^{\,B}\}$ be the dual basis of $\vec{B}=\{\vec{f}_1,\dots,\vec{f}_d\}$ in V'. It should be noted that for each $B\in\mathscr{B}$, we have

$$\widetilde{\Lambda}\subset B, \tag{16.1.3}$$

because all elements of $\widetilde{\Lambda}$ are indispensable for constructing a basis.

Next, we define a multi-dimensional generalization of the fractional part $\{\cdot\}$ for real numbers, which was introduced in (4.3.2) in the case of zeta-functions of root systems. Let $\mathscr{R}=\mathscr{R}(\Lambda)$ be the set of all subsets $R=\{g_1,\dots,g_{d-1}\}\subset\Lambda$ such that $\vec{R}=\{\vec{g}_1,\dots,\vec{g}_{d-1}\}$ is linearly independent set. We need to fix a vector

$$\phi_0\in V'\setminus\bigcup_{R\in\mathscr{R}}\mathfrak{H}_R \tag{16.1.4}$$

so that $\langle\phi_0,\vec{f}^{\,B}\rangle'\neq 0$ for all $B\in\mathscr{B}$ and $f\in B$ (because if $\langle\phi_0,\vec{f}^{\,B}\rangle'=0$ for some $B\in\mathscr{B}$ and $f\in B$, then $\phi_0\in\mathfrak{H}_R$ with $R=B\setminus\{f\}\in\mathscr{R}$).

For $\mathbf{y}\in V'$, $B\in\mathscr{B}$ and $f\in B$, we define the multi-dimensional fractional part by

$$\{\mathbf{y}\}'_{B,f}=\begin{cases}\{\langle\mathbf{y},\vec{f}^{\,B}\rangle'\} & (\langle\phi_0,\vec{f}^{\,B}\rangle'>0),\\ 1-\{-\langle\mathbf{y},\vec{f}^{\,B}\rangle'\} & (\langle\phi_0,\vec{f}^{\,B}\rangle'<0)\end{cases} \tag{16.1.5}$$

(recall (4.3.2) and (4.3.4)).

Now we define the generating function of $S(\mathbf{k},\mathbf{y};\Lambda)$ and state its properties.

Definition 16.3. For $\mathbf{y}\in V'$ and $\mathbf{t}=(t_f)_{f\in\Lambda}\in\mathbb{C}^{\#\Lambda}$, we define

(16.1.6)

$$F(\mathbf{t},\mathbf{y};\Lambda)=\sum_{B\in\mathscr{B}(\Lambda)}\Bigg(\prod_{g\in\Lambda\setminus B}\frac{t_g}{t_g-2\pi\sqrt{-1}\dot{g}-\sum_{f\in B}(t_f-2\pi\sqrt{-1}\dot{f})\langle\vec{g},\vec{f}^{\,B}\rangle'}\Bigg)$$
$$\times\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]}\Bigg(\prod_{f\in B}\frac{t_f\exp((t_f-2\pi\sqrt{-1}\dot{f})\{\mathbf{y}+\mathbf{w}\}'_{B,f})}{\exp(t_f-2\pi\sqrt{-1}\dot{f})-1}\Bigg).$$

Theorem 16.4 ([115, Theorem 2.4]). (i) *The function $F(\mathbf{t},\mathbf{y};\Lambda)$ has one-sided continuity in $\mathbf{y}\in V'$ in the direction ϕ_0, that is*

$$\lim_{c\to 0+}F(\mathbf{t},\mathbf{y}+c\phi_0;\Lambda)=F(\mathbf{t},\mathbf{y};\Lambda). \tag{16.1.7}$$

(ii) $F(\mathbf{t},\mathbf{y};\Lambda)$ *is continuous in* $\mathbf{y}$ *on* $V'\setminus\bigcup_{f\in\widetilde{\Lambda}}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$. *In particular, if* $\widetilde{\Lambda}$ *is empty, then* $F(\mathbf{t},\mathbf{y};\Lambda)$ *is continuous on the whole* V' *and is independent of the choice of* ϕ_0.

(iii) $F(\mathbf{t},\mathbf{y};\Lambda)$ *is holomorphic in the neighborhood of the origin in* $\mathbf{t}$.

Write the Taylor expansion of $F(\mathbf{t},\mathbf{y};\Lambda)$ around the origin in $\mathbf{t}$ as

$$F(\mathbf{t},\mathbf{y};\Lambda)=\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Lambda}}C(\mathbf{k},\mathbf{y};\Lambda)\prod_{f\in\Lambda}\frac{t_f^{k_f}}{k_f!}. \tag{16.1.8}$$

THEOREM 16.5 ([115, Theorem 2.5]). *We have*

$$S(\mathbf{k},\mathbf{y};\Lambda)=\Big(\prod_{f\in\Lambda}-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!}\Big)C(\mathbf{k},\mathbf{y};\Lambda) \tag{16.1.9}$$

for $\mathbf{k}=(k_f)_{f\in\Lambda}\in\mathbb{N}_0^{\#\Lambda}$ *and* $\mathbf{y}\in V'\setminus\bigcup_{f\in\widetilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$.

The above results are again generalizations of those proved in Theorems 4.3, 9.10 and 9.11.

Before going into the proofs of the above theorems, in the next section we will give several examples. Then we will start the proofs of theorems from Section 16.3. Section 16.3 is devoted to the proof of Theorem 16.2. Then from Section 16.4 to Section 16.7 we will describe the proof of Theorem 16.4 and Theorem 16.5. As an application, in Section 16.8, we will mention that there is some hierarchy among generating functions. Finally, in Sections 16.9 and 16.10, we will give the postponed proofs of Theorems 9.10, 9.11, 9.12 and 9.14.

16.2. Examples

In this section we apply our theorems to some special cases, and to state explicit expressions of $F(\mathbf{t},\mathbf{y};\Lambda)$, $C(\mathbf{k},\mathbf{y};\Lambda)$ and $S(\mathbf{k},\mathbf{y};\Lambda)$ for those examples.

EXAMPLE 16.6. Let $V'=\mathbb{R}^2$. Let $\alpha,\beta,\gamma\in\mathbb{C}$,

$$\Lambda=\{f_1=((1,0),\alpha),f_2=((0,1),\beta),f_3=((1,1),\gamma)\}, \tag{16.2.1}$$

$$\mathscr{B}=\{\{f_1,f_2\},\{f_1,f_3\},\{f_2,f_3\}\}, \tag{16.2.2}$$

which corresponds to the series in (16.0.2), (16.0.3) and so on. Then the generating function is given by

$$F((t_1,t_2,t_3),(y_1,y_2);\Lambda)= \tag{16.2.3}$$
$$\frac{t_3}{t_3-2\pi\sqrt{-1}\gamma-(t_1-2\pi\sqrt{-1}\alpha)-(t_2-2\pi\sqrt{-1}\beta)}$$

$$
\begin{aligned}
&\times \frac{t_1 e^{(t_1-2\pi\sqrt{-1}\alpha)\{y_1\}}}{e^{(t_1-2\pi\sqrt{-1}\alpha)}-1}\frac{t_2 e^{(t_2-2\pi\sqrt{-1}\beta)\{y_2\}}}{e^{(t_2-2\pi\sqrt{-1}\beta)}-1}\\
&+\frac{t_2}{t_2-2\pi\sqrt{-1}\beta+(t_1-2\pi\sqrt{-1}\alpha)-(t_3-2\pi\sqrt{-1}\gamma)}\\
&\times \frac{t_1 e^{(t_1-2\pi\sqrt{-1}\alpha)\{y_1-y_2\}}}{e^{(t_1-2\pi\sqrt{-1}\alpha)}-1}\frac{t_3 e^{(t_3-2\pi\sqrt{-1}\gamma)\{y_2\}}}{e^{(t_3-2\pi\sqrt{-1}\gamma)}-1}\\
&+\frac{t_1}{t_1-2\pi\sqrt{-1}\alpha+(t_2-2\pi\sqrt{-1}\beta)-(t_3-2\pi\sqrt{-1}\gamma)}\\
&\times \frac{t_2 e^{(t_2-2\pi\sqrt{-1}\beta)(1-\{y_1-y_2\})}}{e^{(t_2-2\pi\sqrt{-1}\beta)}-1}\frac{t_3 e^{(t_3-2\pi\sqrt{-1}\gamma)\{y_1\}}}{e^{(t_3-2\pi\sqrt{-1}\gamma)}-1}.
\end{aligned}
$$

In particular, if $\alpha,\beta,\gamma\notin\mathbb{Z}$ with $\alpha+\beta\neq\gamma$, we have[3]

(16.2.4)
$$
\begin{aligned}
&C((2,1,1),(y_1,y_2);\Lambda)\\
&=\frac{\sqrt{-1}\{y_1-y_2\}e^{2\sqrt{-1}\pi(\alpha-\alpha\{y_1-y_2\}-\gamma\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)(\alpha+\beta-\gamma)}\\
&-\frac{\sqrt{-1}\{y_1-y_2\}e^{2\sqrt{-1}\pi(2\alpha-\alpha\{y_1-y_2\}-\gamma\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)(\alpha+\beta-\gamma)}\\
&-\frac{\sqrt{-1}e^{2\sqrt{-1}\pi(\alpha-\alpha\{y_1-y_2\}-\gamma\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)(\alpha+\beta-\gamma)}\\
&+\frac{e^{2\sqrt{-1}\pi(-\beta+\beta\{y_1-y_2\}-\gamma\{y_1\})}}{2\pi^2\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)(\alpha+\beta-\gamma)^2}\\
&-\frac{\sqrt{-1}\{y_1\}e^{2\sqrt{-1}\pi(\alpha-\alpha\{y_1\}-\beta\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)(\alpha+\beta-\gamma)}\\
&+\frac{\sqrt{-1}\{y_1\}e^{2\sqrt{-1}\pi(2\alpha-\alpha\{y_1\}-\beta\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)(\alpha+\beta-\gamma)}\\
&+\frac{\sqrt{-1}e^{2\sqrt{-1}\pi(\alpha-\alpha\{y_1\}-\beta\{y_2\})}}{\pi\left(-1+e^{2\sqrt{-1}\pi\alpha}\right)^2\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)(\alpha+\beta-\gamma)}\\
&-\frac{e^{-2\sqrt{-1}\pi(\alpha\{y_1\}+\beta\{y_2\})}}{2\pi^2\left(-1+e^{-2\sqrt{-1}\pi\alpha}\right)\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)(\alpha+\beta-\gamma)^2}\\
&+\frac{e^{-2\sqrt{-1}\pi(\alpha\{y_1-y_2\}+\gamma\{y_2\})}}{2\pi^2\left(-1+e^{-2\sqrt{-1}\pi\alpha}\right)\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)(\alpha+\beta-\gamma)^2}
\end{aligned}
$$

[3]In the formulas (16.2.4) and (16.2.6), values of some coefficients were incorrectly stated in the original paper [115]. This inaccuracy was first pointed out by Professor Shin-ya Kadota.

and

(16.2.5)

$$\begin{aligned}
&S((2,1,1),(y_1,y_2);\Lambda)\\
&\quad=\lim_{N\to\infty}\sum_{\substack{m,n\in\mathbb{Z}\\ |m|,|n|\le N}}\frac{e^{2\pi\sqrt{-1}(my_1+ny_2)}}{(m+\alpha)^2(n+\beta)^1(m+n+\gamma)^1}\\
&\quad=\frac{-(2\pi\sqrt{-1})^2}{2!}\frac{-(2\pi\sqrt{-1})^1}{1!}\frac{-(2\pi\sqrt{-1})^1}{1!}C((2,1,1),(y_1,y_2);\Lambda).
\end{aligned}$$

If $\alpha=0$ and $\beta,\gamma\notin\mathbb{Z}$ with $\beta\neq\gamma$, we have

(16.2.6)

$$\begin{aligned}
&C((0,1,2),(y_1,y_2);\Lambda)\\
&=\frac{\sqrt{-1}\{y_2\}e^{-2\sqrt{-1}\pi\gamma\{y_2\}}}{\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)\pi(\beta-\gamma)}-\frac{\sqrt{-1}e^{-2\sqrt{-1}\pi\gamma(1+\{y_2\})}}{\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)^2\pi(\beta-\gamma)}\\
&\quad+\frac{e^{-2\sqrt{-1}\pi\beta\{y_2\}}}{2\left(-1+e^{-2\sqrt{-1}\pi\beta}\right)\pi^2(\beta-\gamma)^2}-\frac{e^{-2\sqrt{-1}\pi\gamma\{y_2\}}}{2\left(-1+e^{-2\sqrt{-1}\pi\gamma}\right)\pi^2(\beta-\gamma)^2}
\end{aligned}$$

and

(16.2.7)

$$\begin{aligned}
&S((0,1,2),(y_1,y_2);\Lambda)\\
&=-\lim_{N\to\infty}\sum_{\substack{m,n\in\mathbb{Z}\\ |m|,|n|\le N\\ m=0}}\frac{e^{2\pi\sqrt{-1}(my_1+ny_2)}}{(n+\beta)^1(m+n+\gamma)^2}\\
&=-\lim_{N\to\infty}\sum_{\substack{n\in\mathbb{Z}\\ |n|\le N}}\frac{e^{2\pi\sqrt{-1}ny_2}}{(n+\beta)^1(n+\gamma)^2}\\
&=\frac{-(2\pi\sqrt{-1})^0}{0!}\frac{-(2\pi\sqrt{-1})^1}{1!}\frac{-(2\pi\sqrt{-1})^2}{2!}C((0,1,2),(y_1,y_2);\Lambda).
\end{aligned}$$

Example 16.7. Let $V'=\mathbb{R}$. Let

$$\Lambda=\Lambda_\alpha=\{f_{-1}=(-1,\alpha),f_0=(1,0),f_1=(1,\alpha)\},\tag{16.2.8}$$

$$\mathscr{B}=\{\{f_{-1}\},\{f_0\},\{f_1\}\},\tag{16.2.9}$$

where $\alpha\in\mathbb{C}\setminus\{0\}$, which corresponds to the series

$$S(\mathbf{k},y;\Lambda_\alpha)=\lim_{N\to\infty}\sum_{\substack{m\in\mathbb{Z}\setminus\{0,\pm\alpha\}\\ |m|\le N}}\frac{e^{2\pi\sqrt{-1}my}}{(-m+\alpha)^{k_{-1}}m^{k_0}(m+\alpha)^{k_1}},\tag{16.2.10}$$

where $k_{-1}, k_0, k_1 \in \mathbb{N}$. Then the generating function is given by

$$
\begin{aligned}
(16.2.11)\quad & F((t_{-1},t_0,t_1),y;\Lambda_\alpha) \\
&= \frac{t_0}{t_0+(t_{-1}-2\pi\sqrt{-1}\alpha)}\frac{t_1}{t_1-2\pi\sqrt{-1}\alpha+(t_{-1}-2\pi\sqrt{-1}\alpha)} \\
&\quad\times\frac{t_{-1}e^{(t_{-1}-2\pi\sqrt{-1}\alpha)(1-\{y\})}}{e^{t_{-1}-2\pi\sqrt{-1}\alpha}-1} \\
&\quad+\frac{t_{-1}}{t_{-1}-2\pi\sqrt{-1}\alpha+t_0}\frac{t_1}{t_1-2\pi\sqrt{-1}\alpha-t_0}\frac{t_0e^{t_0\{y\}}}{e^{t_0}-1} \\
&\quad+\frac{t_{-1}}{t_{-1}-2\pi\sqrt{-1}\alpha+(t_1-2\pi\sqrt{-1}\alpha)}\frac{t_0}{t_0-(t_1-2\pi\sqrt{-1}\alpha)} \\
&\quad\times\frac{t_1e^{(t_1-2\pi\sqrt{-1}\alpha)\{y\}}}{e^{t_1-2\pi\sqrt{-1}\alpha}-1}.
\end{aligned}
$$

We have for $\alpha \notin \mathbb{Z}$,

$$
\begin{aligned}
(16.2.12)\quad & C((2,2,2),y;\Lambda_\alpha) \\
&= -\frac{1}{4\pi^6\alpha^6}+\frac{1}{24\pi^4\alpha^4}-\frac{\{y\}}{4\pi^4\alpha^4}+\frac{\{y\}^2}{4\pi^4\alpha^4} \\
&\quad-\frac{3\sqrt{-1}e^{2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^5\alpha^5\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}-\frac{3\sqrt{-1}e^{2\pi\sqrt{-1}\alpha(1-\{y\})}}{16\pi^5\alpha^5\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2} \\
&\quad+\frac{3\sqrt{-1}e^{2\pi\sqrt{-1}\alpha(2-\{y\})}}{16\pi^5\alpha^5\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}+\frac{3\sqrt{-1}e^{2\pi\sqrt{-1}\alpha(\{y\}+1)}}{16\pi^5\alpha^5\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2} \\
&\quad-\frac{\{y\}e^{2\pi\sqrt{-1}\alpha\{y\}}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}+\frac{\{y\}e^{2\pi\sqrt{-1}\alpha(1-\{y\})}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2} \\
&\quad-\frac{\{y\}e^{2\pi\sqrt{-1}\alpha(2-\{y\})}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}+\frac{\{y\}e^{2\pi\sqrt{-1}\alpha(\{y\}+1)}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2} \\
&\quad-\frac{e^{2\pi\sqrt{-1}\alpha(1-\{y\})}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}-\frac{e^{2\pi\sqrt{-1}\alpha(\{y\}+1)}}{8\pi^4\alpha^4\left(-1+e^{2\pi\sqrt{-1}\alpha}\right)^2}
\end{aligned}
$$

and for $\alpha \in \mathbb{Z}$,

(16.2.13)

$$
\begin{aligned}
& C((2,2,2),y;\Lambda_\alpha) \\
&= -\frac{1}{4\pi^6\alpha^6}+\frac{1}{24\pi^4\alpha^4}-\frac{\{y\}}{4\pi^4\alpha^4}+\frac{\{y\}^2}{4\pi^4\alpha^4}-\frac{3\sqrt{-1}\{y\}e^{-2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^5\alpha^5} \\
&\quad+\frac{3\sqrt{-1}\{y\}e^{2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^5\alpha^5}+\frac{3\sqrt{-1}e^{-2\pi\sqrt{-1}\alpha\{y\}}}{32\pi^5\alpha^5}-\frac{3\sqrt{-1}e^{2\pi\sqrt{-1}\alpha\{y\}}}{32\pi^5\alpha^5} \\
&\quad+\frac{\{y\}^2e^{-2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^4\alpha^4}+\frac{\{y\}^2e^{2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^4\alpha^4}-\frac{\{y\}e^{-2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^4\alpha^4}-\frac{\{y\}e^{2\pi\sqrt{-1}\alpha\{y\}}}{16\pi^4\alpha^4}
\end{aligned}
$$

$$-\frac{23e^{-2\pi\sqrt{-1}\alpha\{y\}}}{128\pi^6\alpha^6}-\frac{23e^{2\pi\sqrt{-1}\alpha\{y\}}}{128\pi^6\alpha^6}+\frac{e^{-2\pi\sqrt{-1}\alpha\{y\}}}{96\pi^4\alpha^4}+\frac{e^{2\pi\sqrt{-1}\alpha\{y\}}}{96\pi^4\alpha^4}.$$

For example, setting $y=0$ and $\alpha=1,2,3$, we obtain

$$S((2,2,2),0;\Lambda_1)=\sum_{m\in\mathbb{Z}\setminus\{0,\pm1\}}\frac{1}{(-m+1)^2m^2(m+1)^2}=\frac{1}{2}\pi^2-\frac{39}{8},$$

$$S((2,2,2),0;\Lambda_2)=\sum_{m\in\mathbb{Z}\setminus\{0,\pm2\}}\frac{1}{(-m+2)^2m^2(m+2)^2}=\frac{1}{32}\pi^2-\frac{39}{512},$$

$$S((2,2,2),0;\Lambda_3)=\sum_{m\in\mathbb{Z}\setminus\{0,\pm3\}}\frac{1}{(-m+3)^2m^2(m+3)^2}=\frac{1}{162}\pi^2-\frac{13}{1944}.$$

Similarly, computing $C((2k,2k,2k),0;\Lambda_\alpha)$, we can obtain

$$S((4,4,4),0;\Lambda_1)=\frac{1}{40}\pi^4+\frac{35}{16}\pi^2-\frac{3075}{128},$$

$$S((6,6,6),0;\Lambda_2)=\frac{11}{20643840}\pi^6+\frac{21}{2097152}\pi^4+\frac{3003}{16777216}\pi^2-\frac{137067}{268435456},$$

$$\begin{aligned}S((8,8,8),0;\Lambda_3)=&\frac{43}{8678218953600}\pi^8+\frac{367}{7810397058240}\pi^6\\&+\frac{581}{1983592903680}\pi^4+\frac{46189}{21422803359744}\pi^2\\&-\frac{2864587}{1028294561267712}.\end{aligned}$$

Here we define the zeta-function associated with Λ_α by

$$\zeta((s_1,s_2,s_3);\Lambda_\alpha)=\sum_{\substack{m=1\\m\neq\pm\alpha}}^{\infty}\frac{1}{(-m+\alpha)^{s_1}m^{s_2}(m+\alpha)^{s_3}},\tag{16.2.14}$$

which can be regarded as a Hurwitz-type analogue of the Riemann zeta-function, that is, with a shifting parameter α. We can easily check that $S((2k,2k,2k),0;\Lambda_\alpha)=2\zeta((2k,2k,2k);\Lambda_\alpha)$ for $k\in\mathbb{N}$. Therefore we obtain from the above results that, for example,

$$\zeta((2,2,2);\Lambda_1)=\frac{1}{4}\pi^2-\frac{39}{16},$$

$$\zeta((4,4,4);\Lambda_1)=\frac{1}{80}\pi^4+\frac{35}{32}\pi^2-\frac{3075}{256},$$

$$\zeta((6,6,6);\Lambda_2)=\frac{11}{41287680}\pi^6+\frac{21}{4194304}\pi^4+\frac{3003}{33554432}\pi^2-\frac{137067}{536870912}.$$

Example 16.8. Let $V'=\mathbb{R}^2$, and $\alpha\in\mathbb{C}\setminus\{0\}$. Let

$$\begin{aligned}\Lambda=\Lambda_\alpha&=\{\{f_{1j}\}_j,\{f_{2j}\}_j,\{f_{3j}\}_j\}\\&=\{\{(-1,0,\alpha),(1,0,0),(1,0,\alpha)\},\ \{(0,-1,\alpha),(0,1,0),(0,1,\alpha)\},\\&\qquad\{(-1,-1,\alpha),(1,1,0),(1,1,\alpha)\}\},\end{aligned}\tag{16.2.15}$$

(16.2.16)
$$\mathscr{B} = \{\{f_{1j}, f_{2l}\}_{j,l}, \{f_{1j}, f_{3l}\}_{j,l}, \{f_{2j}, f_{3l}\}_{j,l}\}.$$

Set $\mathbf{y} = 0$ and

$$\begin{aligned} &S(\{k_j\}_{1\le j\le 9}, 0; \Lambda_\alpha) \\ &= \lim_{N\to\infty} \sum_{\substack{m,n\in\mathbb{Z}\setminus\{0,\pm\alpha\} \\ m+n\neq 0,\pm\alpha \\ |m|,|n|\le N}} \frac{1}{(-m+\alpha)^{k_1} m^{k_2} (m+\alpha)^{k_3} (-n+\alpha)^{k_4} n^{k_5} (n+\alpha)^{k_6}} \\ &\qquad \times \frac{1}{(-(m+n)+\alpha)^{k_7} (m+n)^{k_8} (m+n+\alpha)^{k_9}}. \end{aligned}$$

Then, computing $C(\{k_j\}, 0; \Lambda_\alpha)$, we obtain, for example,

$$S((1,2,2,2,1,1,1,2,2), 0; \Lambda_1) = \frac{1}{1890}\pi^6 + \frac{701}{2160}\pi^4 - \frac{1841}{108}\pi^2 + \frac{2822557}{20736},$$

$$\begin{aligned} S((2,2,2,2,2,2,2,2,2), 0; \Lambda_2) &= \frac{11}{15482880}\pi^6 + \frac{4901}{70778880}\pi^4 \\ &\quad - \frac{26747}{28311552}\pi^2 + \frac{20643217}{10871635968}, \end{aligned}$$

$$S((1,1,1,2,2,2,1,1,1), 0; \Lambda_3) = \frac{2}{295245}\pi^4 - \frac{227}{6377292}\pi^2 + \frac{14183}{459165024}.$$

Similar to Example 16.7, we define the zeta-function associated with Λ by

(16.2.17)
$$\begin{aligned} &\zeta_2(\{s_j\}_{1\le j\le 9}; \Lambda_\alpha) \\ &= \sum_{\substack{m,n=1 \\ m\neq\pm\alpha \\ n\neq\pm\alpha \\ m+n\neq\pm\alpha}}^{\infty} \frac{1}{(-m+\alpha)^{s_1} m^{s_2} (m+\alpha)^{s_3} (-n+\alpha)^{s_4} n^{s_5} (n+\alpha)^{s_6}} \\ &\qquad \times \frac{1}{(-(m+n)+\alpha)^{s_7} (m+n)^{s_8} (m+n+\alpha)^{s_9}}, \end{aligned}$$

which can be regarded as a Hurwitz-type analogue of the zeta-function of the root system of type A_2 defined by

$$\zeta_2((s_1, s_2, s_3); A_2) = \sum_{m,n=1}^{\infty} \frac{1}{m^{s_1} n^{s_2} (m+n)^{s_3}} \tag{16.2.18}$$

(see (3.3.19)). Certain Hurwitz-type analogues of zeta-functions of root systems appeared in the proof of Theorem 14.1.

From the viewpoint of root systems, we can regard $S(\{2k\}_{1\le j\le 9}, 0; \Lambda_\alpha)$ as the sum of zeta values $\zeta_2(\{2k\}_{1\le j\le 9}; \Lambda_\alpha)$ under the action of the

Weyl group of type A_2 ($\simeq S_3$). This implies that

$$S(\{2k\}_{1\le j\le 9}, 0; \Lambda_\alpha) = 6\zeta_2(\{2k\}_{1\le j\le 9}; \Lambda_\alpha) \quad (k \in \mathbb{N}).$$

Therefore, as an analogue of $\zeta_2((2,2,2); A_2) = \pi^6/2835$, we obtain from the above result that

$$\begin{aligned}&\zeta_2((2,2,2,2,2,2,2,2,2); \Lambda_2)\\ &= \frac{11}{92897280}\pi^6 + \frac{4901}{424673280}\pi^4 - \frac{26747}{169869312}\pi^2 + \frac{20643217}{65229815808}.\end{aligned}$$

REMARK 16.9. We give another interpretation of the series (16.2.14) and (16.2.17), that is, we regard each term of these series as a product of positive roots of affine root system $A_1^{(1)}$ and $A_2^{(1)}$ respectively (for the theory of affine root systems, see [78]). Since there are infinitely many positive roots in affine root systems, the product consists of infinitely many factors. In order for the infinite product to make sense, we understand that infinitely many variables are set to be zero and hence the product is truncated.

16.3. Proof of Theorem 16.2

Now we start the proofs of the theorems stated in Section 16.1. First of all, in this section, we prove Theorem 16.2. The main body of the argument is the proof of an evaluation formula (Proposition 16.10) for $S(\mathbf{k}, \mathbf{y}; \Lambda)$.

In this section, the letters K, K', K'' denote positive constants, not necessarily the same at each occurrence.

For $t, b \in \mathbb{C}$ and $y \in \mathbb{R}$, let

$$F(t, y; b) = \frac{te^{(t-2\pi\sqrt{-1}b)y}}{e^{t-2\pi\sqrt{-1}b} - 1} = \sum_{k=0}^{\infty} C(k, y; b)\frac{t^k}{k!}, \tag{16.3.1}$$

where the right-hand side converges when $|t|$ is sufficiently small. In particular, comparing with (4.2.1) we find

$$C(k, y; 0) = B_k(y). \tag{16.3.2}$$

It should be noted that $F(t, \{y\}; b)$ (resp. $C(k, \{y\}; b)$) is just the special case $d = 1$, $\Lambda = \{(1, b)\} = B$ of $F(\mathbf{t}, \mathbf{y}; \Lambda)$ defined by (16.1.6) (resp. $C(\mathbf{k}, \mathbf{y}; \Lambda)$ defined by (16.1.8)).

PROPOSITION 16.10 ([115, Proposition 4.1]). *For* $\mathbf{k} = (k_f)_{f\in\Lambda} \in \mathbb{N}_0^{\#\Lambda}$, $\mathbf{y} \in V' \setminus \bigcup_{f\in\tilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$, *the series* (16.1.1) *converges. For*

a fixed decomposition $\Lambda = B_0 \cup L_0$ *with* $B_0 = \{f_1, \dots, f_d\} \in \mathscr{B}$, *we have*

$$S(\mathbf{k}, \mathbf{y}; \Lambda) = \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])} \prod_{f\in\Lambda}\Big(-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!}\Big) \times \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \int_0^1 \cdots \int_0^1 \Big(\prod_{g\in L_0} C(k_g, x_g; \dot{g}) dx_g\Big) \times \prod_{f\in B_0}\Big(C(k_f, \{\mathbf{y}+\mathbf{w}-\sum_{g\in L_0} x_g\vec{g}\}'_{B_0,f}; \dot{f})\Big). \tag{16.3.3}$$

This is a generalization of Theorem 4.1 (for integral values of $\mathbf{k}$). Only the case in the domain of absolute convergence was considered in Theorem 4.1, so there was no problem of convergence. In our present situation, if $k_f \geq 2$ for all $f \in B$ with some fixed $B \in \mathscr{B}$, then the matter of convergence is again obvious, so it is easy to prove our claims. However, if $k_f = 1$ for sufficiently many $f \in \Lambda$, then there are subtle problems on convergence, and the proof becomes much more complicated. It should be remarked that the key of the convergence of $S(\mathbf{k}, \mathbf{y}; \Lambda)$ is the condition $\operatorname{rank} \mathrm{Lh}_{\mathbb{Z}}[\vec{\Lambda}] = d$.

Since it is difficult to treat (16.1.1) directly, in the following we consider a slightly modified sum

$$S_1(\mathbf{k}, \mathbf{y}; \Lambda; B_0) = \lim_{N\to\infty} Z_1(N; \mathbf{k}, \mathbf{y}; \Lambda; B_0), \tag{16.3.4}$$

where

$$Z_1(N; \mathbf{k}, \mathbf{y}; \Lambda; B_0) = (-1)^{\#\Lambda_0} \sum_{\substack{\mathbf{v}\in\mathbb{Z}^d \\ |\Re f(\mathbf{v})|\leq N \quad (f\in B_0) \\ f(\mathbf{v})\neq 0 \quad (f\in\Lambda_+) \\ f(\mathbf{v})=0 \quad (f\in\Lambda_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \mathbf{v}\rangle'} \prod_{f\in\Lambda_+} \frac{1}{f(\mathbf{v})^{k_f}}. \tag{16.3.5}$$

That is, the condition $|v_j| \leq N$ for $1 \leq j \leq r$ in the definition of $S(\mathbf{k}, \mathbf{y}; \Lambda)$ is replaced by $|\Re f(\mathbf{v})| \leq N$ for $f \in B_0$. At the last stage of the proof we will show that $S(\mathbf{k}, \mathbf{y}; \Lambda) = S_1(\mathbf{k}, \mathbf{y}; \Lambda; B_0)$. In particular, we will find that $S_1(\mathbf{k}, \mathbf{y}; \Lambda; B_0)$ actually does not depend on the choice of B_0.

The proof of Proposition 16.10 consists of three steps.

The first step. We first consider the simplest case of (16.3.4), which corresponds to $d = 1$ and $\Lambda = \{(1, b)\} = B$ with $\mathscr{B} = \{B\}$, in Lemmas 16.11 and 16.13.

LEMMA 16.11 ([115, Lemma 4.2]). *For $b \in \mathbb{C}$, $y \in \mathbb{R}$ and $k \in \mathbb{N}$ with $k \geq 2$, we have*

$$\lim_{N\to\infty}\left(\sum_{\substack{n\in\mathbb{Z}\\ |n+\Re b|\leq N\\ n+b\neq 0}}\frac{e^{2\pi\sqrt{-1}ny}}{(n+b)^k}\right) = -\frac{(2\pi\sqrt{-1})^k}{k!}C(k,\{y\};b), \tag{16.3.6}$$

$$\lim_{N\to\infty}\left(-\sum_{\substack{n\in\mathbb{Z}\\ |n+\Re b|\leq N\\ n+b=0}}e^{2\pi\sqrt{-1}ny}\right) = -C(0,\{y\};b). \tag{16.3.7}$$

(Actually the sum on the left-hand side of (16.3.7) consists of at most one term.) The series above converge absolutely uniformly in y and hence $C(k,\{y\};b)$ and $C(0,\{y\};b)$ are continuous in y.

PROOF. Let $\gamma_{X,Y}$ be the counterclockwise rectangle contour with vertices at $\pm X \pm 2\pi\sqrt{-1}Y$. Applying the Cauchy theorem to the integral

$$\lim_{N\to\infty}\lim_{M\to\infty}\int_{\gamma_{M,N+\epsilon}}F(t,\{y\};b)t^{-k-1}dt \qquad (k\geq 2) \tag{16.3.8}$$

with a sufficiently small $\epsilon > 0$, we see that the sum of all the residues vanishes, namely,

$$\frac{C(k,\{y\};b)}{k!} + \frac{1}{(2\pi\sqrt{-1})^k}\lim_{N\to\infty}\sum_{\substack{n\in\mathbb{Z}\\ |n+\Re b|\leq N\\ n+b\neq 0}}\frac{e^{2\pi\sqrt{-1}ny}}{(n+b)^k} = 0, \tag{16.3.9}$$

and hence (16.3.6).

The left-hand side of (16.3.7) consists of only one term $e^{-2\pi\sqrt{-1}by}$ if $b \in \mathbb{Z}$, and vanishes if $b \notin \mathbb{Z}$, while

$$C(0,\{y\};b) = F(0,\{y\};b) = \begin{cases} e^{-2\pi\sqrt{-1}by} & (b\in\mathbb{Z}),\\ 0 & (b\notin\mathbb{Z}) \end{cases} \tag{16.3.10}$$

and hence (16.3.7).

Both in (16.3.6) and (16.3.7), the absolute uniform convergence of the series in y is clear. □

The case $k = 1$ is more subtle. We first prepare the following:

LEMMA 16.12 ([115, Lemma 4.3]). *For $\mu > 0$ there exists $K > 0$ such that for $a, z > 0$*

$$\int_0^\infty \frac{e^{-xz}}{\sqrt{x^2+a^2}}dx \leq K(az)^{-\frac{1}{\mu+1}}. \tag{16.3.11}$$

PROOF. Rewrite

$$\int_0^\infty \frac{e^{-xz}}{\sqrt{x^2+a^2}}dx = \int_0^\infty \frac{e^{-x}}{\sqrt{x^2+(az)^2}}dx. \tag{16.3.12}$$

By the inequality of weighted arithmetic and geometric means

$$\mu A + B \geq (\mu+1)(A^\mu B)^{\frac{1}{\mu+1}} \qquad (\mu, A, B > 0), \tag{16.3.13}$$

we have

$$\sqrt{x^2+(az)^2} \geq \sqrt{\mu+1}\mu^{-\frac{\mu}{2(\mu+1)}} x^{\frac{\mu}{\mu+1}} (az)^{\frac{1}{\mu+1}} \tag{16.3.14}$$

and

$$\begin{aligned}\int_0^\infty \frac{e^{-x}}{\sqrt{x^2+(az)^2}}dx &\leq \frac{\mu^{\frac{\mu}{2(\mu+1)}}}{\sqrt{\mu+1}}(az)^{-\frac{1}{\mu+1}} \int_0^\infty e^{-x}x^{-\frac{\mu}{\mu+1}}dx \\ &\leq \Big(\frac{\mu^{\frac{\mu}{2(\mu+1)}}}{\sqrt{\mu+1}}\Gamma\Big(\frac{1}{\mu+1}\Big)\Big)(az)^{-\frac{1}{\mu+1}}.\end{aligned} \tag{16.3.15}$$

□

Let

$$\delta_{N<0<M} = \begin{cases} 1 & (N<0<M), \\ 0 & \text{otherwise.}\end{cases} \tag{16.3.16}$$

LEMMA 16.13 ([115, Lemma 4.4]). *For $b \in \mathbb{C}$ and $y \in \mathbb{R}\setminus\mathbb{Z}$ we have*

$$\lim_{N\to\infty}\Bigg(\sum_{\substack{n\in\mathbb{Z}\\ |n+\Re b|\leq N\\ n+b\neq 0}} \frac{e^{2\pi\sqrt{-1}ny}}{n+b}\Bigg) = -2\pi\sqrt{-1}C(1,\{y\};b) \tag{16.3.17}$$

and $C(1,\{y\};b)$ is continuous in y. Moreover, for any $\mu>0$ there exists $K>0$ such that for all $y\in\mathbb{R}\setminus\mathbb{Z}$ and all $M,N\in\mathbb{R}$ with sufficiently large $|M|,|N|$ and $M\geq|N|$,

(16.3.18)

$$\Big|\sum_{\substack{n\in\mathbb{Z}\\ N\leq n+\Re b\leq M\\ n+b\neq 0}} \frac{e^{2\pi\sqrt{-1}ny}}{n+b}\Big| \leq K|N|^{-\frac{1}{\mu+1}}((1-\{y\})^{-\frac{1}{\mu+1}}+\{y\}^{-\frac{1}{\mu+1}})+\delta_{N<0<M}K.$$

PROOF. From (16.3.1) we can easily see that

$$C(1,\{y\};b) = \begin{cases} \Big(\{y\}-\dfrac{1}{2}\Big)e^{-2\pi\sqrt{-1}b\{y\}} & (b\in\mathbb{Z}), \\ \dfrac{e^{-2\pi\sqrt{-1}b\{y\}}}{e^{-2\pi\sqrt{-1}b}-1} & (b\notin\mathbb{Z}),\end{cases} \tag{16.3.19}$$

from which the continuity of $C(1,\{y\};b)$ follows.

Let γ_Y be the horizontal path from $-\infty+2\pi\sqrt{-1}Y$ to $\infty+2\pi\sqrt{-1}Y$. Then for all $y \in \mathbb{R} \setminus \mathbb{Z}$, we have

(16.3.20)
$$\frac{1}{2\pi\sqrt{-1}}\Big(-\int_{\gamma_{M+\epsilon}} F(t,\{y\};b)t^{-2}dt+\int_{\gamma_{N-\epsilon}} F(t,\{y\};b)t^{-2}dt\Big)-\delta_{N<0<M}C(1,\{y\};b)$$
$$=\frac{1}{2\pi\sqrt{-1}}\sum_{\substack{n\in\mathbb{Z}\\ N\le n+\Re b\le M\\ n+b\neq 0}}\frac{e^{2\pi\sqrt{-1}ny}}{n+b}.$$

On the other hand, for $L \in \mathbb{Z}$

(16.3.21)
$$\begin{aligned}\Big|\int_{\gamma_{L\pm\epsilon}} F(t,\{y\};b)t^{-2}dt\Big| &\le \int_{-\infty}^{\infty}\Big|\frac{e^{(x+2\pi\sqrt{-1}(L\pm\epsilon)-2\pi\sqrt{-1}b)\{y\}}}{e^{x+2\pi\sqrt{-1}(L\pm\epsilon)-2\pi\sqrt{-1}b}-1}\frac{1}{x+2\pi\sqrt{-1}(L\pm\epsilon)}\Big|dx\\ &\le \int_{-\infty}^{\infty}\frac{e^{x\{y\}}e^{2\pi\Im b\{y\}}}{|e^{x-2\pi\sqrt{-1}(b\mp\epsilon)}-1|}\frac{dx}{|x+2\pi\sqrt{-1}(L\pm\epsilon)|}\\ &= \int_{-\infty}^{\infty}\frac{e^{x\{y\}}e^{2\pi|\Im b|}}{|e^{x-2\pi\sqrt{-1}(b\mp\epsilon)}-1|}\frac{dx}{\sqrt{x^2+4\pi^2(L\pm\epsilon)^2}}.\end{aligned}$$

It is easy to see that there exists $K' > 0$ independent of x and y such that

$$\frac{e^{x\{y\}}e^{2\pi|\Im b|}}{|e^{x-2\pi\sqrt{-1}(b\mp\epsilon)}-1|} \le g(x,y)=\begin{cases}K'e^{x\{y\}} & (x<0),\\ K'e^{x(\{y\}-1)} & (x\ge 0).\end{cases} \tag{16.3.22}$$

Applying (16.3.22) and Lemma 16.12 to (16.3.21), we have

(16.3.23)
$$\begin{aligned}\Big|\int_{\gamma_{L\pm\epsilon}} F(t,\{y\};b)t^{-2}dt\Big| &\le \int_{-\infty}^{\infty}\frac{g(x,y)}{\sqrt{x^2+4\pi^2(L\pm\epsilon)^2}}dx\\ &\le K''|L|^{-\frac{1}{\mu+1}}((1-\{y\})^{-\frac{1}{\mu+1}}+\{y\}^{-\frac{1}{\mu+1}})\end{aligned}$$

for some $K'' > 0$. Therefore, choosing $N = -L$ and $M = L$ in (16.3.20) and taking the limit $L \to \infty$, we obtain (16.3.17). Moreover, since $C(1,\{y\};b)$ is bounded in y, we obtain (16.3.18). □

The second step. We next consider the higher rank case of (16.3.4) under the special condition $\Lambda = B$ with $\mathscr{B} = \{B\}$ in Lemmas 16.16, 16.17 and 16.19. We first prepare the following algebraic lemma. This statement is included in [**20**, Chapitre 6, Section 1.9], but here we supply a proof.

LEMMA 16.14 ([115, Lemma 4.5]). *Let Q, P be free $\mathbb{Z}$-modules of rank r with $Q \subset P$ so that P/Q is a finite abelian group. Then*

$$\operatorname{Hom}(P/Q, \mathbb{Q}/\mathbb{Z}) \simeq \operatorname{Hom}(Q, \mathbb{Z})/\operatorname{Hom}(P, \mathbb{Z}) \tag{16.3.24}$$

and for $\bar{\lambda} \in P/Q$, we have

$$\frac{1}{\#(P/Q)} \sum_{\bar{f} \in \operatorname{Hom}(Q,\mathbb{Z})/\operatorname{Hom}(P,\mathbb{Z})} e^{2\pi\sqrt{-1}\bar{f}(\bar{\lambda})} = \delta_{\bar{\lambda},0}, \tag{16.3.25}$$

where the right-hand side denotes Kronecker's delta.

PROOF. First, we note that an element of $\operatorname{Hom}(P, \mathbb{Z})$ can be naturally regarded as an element of $\operatorname{Hom}(Q, \mathbb{Z})$. Denote this injection by ι. Next, let $f \in \operatorname{Hom}(Q, \mathbb{Z})$. It is well-known that there exist a basis $\{\lambda_i\}_{i=1}^r$ of P and a basis $\{\lambda_i'\}_{i=1}^r$ of Q such that $\lambda_i' = k_i\lambda_i$ with $k_i \in \mathbb{N}$ and hence

$$P/Q = \operatorname{Lh}_{\mathbb{Z}}[\bar{\lambda}_1] \oplus \cdots \oplus \operatorname{Lh}_{\mathbb{Z}}[\bar{\lambda}_r], \tag{16.3.26}$$

where each $\operatorname{Lh}_{\mathbb{Z}}[\bar{\lambda}_i]$ is a cyclic group of order k_i with $\bar{\lambda}_i \in P/Q$. Define $\varphi(f)$ by the linear extension of

$$\varphi(f)(\bar{\lambda}_i) = p(f(\lambda_i')/k_i), \tag{16.3.27}$$

where p denotes the natural projection $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$. Then $\varphi(f) \in \operatorname{Hom}(P/Q, \mathbb{Q}/\mathbb{Z})$ is well-defined.

We show that the sequence

$$0 \to \operatorname{Hom}(P, \mathbb{Z}) \xrightarrow{\iota} \operatorname{Hom}(Q, \mathbb{Z}) \xrightarrow{\varphi} \operatorname{Hom}(P/Q, \mathbb{Q}/\mathbb{Z}) \to 0 \tag{16.3.28}$$

is exact. First, show the surjectivity of φ. Let $g \in \operatorname{Hom}(P/Q, \mathbb{Q}/\mathbb{Z})$. Then choose a representative $a_i \in \mathbb{Q}$ of $g(\bar{\lambda}_i)$. Since $k_i g(\bar{\lambda}_i) = g(\bar{\lambda}_i') = g(0) = 0$, we have $k_i a_i \in \mathbb{Z}$. Define $f \in \operatorname{Hom}(Q, \mathbb{Z})$ by the linear extension of $f(\lambda_i') = k_i a_i$. Then $\varphi(f)(\bar{\lambda}_i) = p(a_i) = g(\bar{\lambda}_i)$, so $g = \varphi(f)$. Therefore φ is surjective. Next, let $f \in \operatorname{Ker}\varphi$. Then $\varphi(f)(\bar{\lambda}_i) = p(f(\lambda_i')/k_i) = 0$ and so $f(k_i\lambda_i) \in k_i\mathbb{Z}$, that is, $f(\lambda_i) \in \mathbb{Z}$. This implies $f \in \operatorname{Hom}(P, \mathbb{Z})$, and hence the exactness at $\operatorname{Hom}(Q, \mathbb{Z})$ is proved. The assertion (16.3.24) immediately follows from (16.3.28), and (16.3.25) follows from the orthogonality relations of group characters (cf. Apostol [4, Theorem 6.13]). □

LEMMA 16.15 ([115, Lemma 4.6]). *Let $B \in \mathscr{B}$ and $f \in B$. Then*

$$\mathfrak{H}_{B\setminus\{f\}} + \mathbb{Z}^d = \{\mathbf{y} \in V' \mid \langle \mathbf{y}+\mathbf{w}, \vec{f}^B \rangle' \in \mathbb{Z} \text{ for some } \mathbf{w} \in \mathbb{Z}^d\}. \tag{16.3.29}$$

PROOF. We denote the left-hand side by P and the right-hand side by Q respectively. If $\mathbf{y} \in P$, then $\mathbf{y} = \mathbf{y}_0 + \mathbf{w}_0$ with $\langle \mathbf{y}_0, \vec{f}^B \rangle' = 0$ and $\mathbf{w}_0 \in \mathbb{Z}^d$. By setting $\mathbf{w} = -\mathbf{w}_0$ we have

$$\langle \mathbf{y} + \mathbf{w}, \vec{f}^B \rangle' = \langle \mathbf{y}_0, \vec{f}^B \rangle' = 0 \in \mathbb{Z}, \tag{16.3.30}$$

and $\mathbf{y} \in Q$. Conversely, if $\mathbf{y} \in Q$, then

$$\mathbf{y} + \mathbf{w} \in \mathfrak{H}_{B\setminus\{f\}} + \mathbb{Z}\vec{f} \subset \mathfrak{H}_{B\setminus\{f\}} + \mathbb{Z}^d \tag{16.3.31}$$

because $\mathfrak{H}_{B\setminus\{f\}}$ is orthogonal to $\vec{f}^B$. Hence we have $\mathbf{y} \in P$. □

LEMMA 16.16 ([115, Lemma 4.7]). *Assume $\Lambda = B$ with $\mathscr{B} = \{B\}$. For $\mathbf{k} = (k_f)_{f\in\Lambda} \in \mathbb{N}_0^d$ and $\mathbf{y} \in V' \setminus \bigcup_{f\in\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$, the limit (16.3.4) (with $B_0 = B$) converges, and we have*

(16.3.32)

$$S_1(\mathbf{k}, \mathbf{y}; \Lambda; B) = \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \times \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \prod_{f\in\Lambda} \Big(-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!} C(k_f, \{\mathbf{y}+\mathbf{w}\}'_{B,f}; \dot{f}) \Big).$$

PROOF. Let $A = {}^t(\vec{f})_{f\in B}$ be a regular matrix, where $\vec{f}$ are regarded as column vectors. Then $A^{-1} = (\vec{f}^B)_{f\in B}$. For $\mathbf{v} \in \mathbb{Z}^d$, write $\mathbf{u} = (u_f)_{f\in B} = A\mathbf{v}$ so that $u_f = \langle \vec{f}, \mathbf{v} \rangle'$ and $\mathbf{v} = A^{-1}\mathbf{u} = \sum_{f\in B} \vec{f}^B u_f$. We have

(16.3.33)

$$\begin{aligned} &Z_1(N; \mathbf{k}, \mathbf{y}; \Lambda; B) \\ &\quad = (-1)^{\#\Lambda_0} \sum_{\substack{\mathbf{v}\in\mathbb{Z}^d \\ |\Re f(\mathbf{v})|\le N \quad (f\in B) \\ f(\mathbf{v})\ne 0 \quad (f\in\Lambda_+) \\ f(\mathbf{v})=0 \quad (f\in\Lambda_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}, \mathbf{v}\rangle'} \prod_{f\in\Lambda_+} \frac{1}{(\langle \vec{f}, \mathbf{v}\rangle' + \dot{f})^{k_f}} \\ &\quad = (-1)^{\#\Lambda_0} \sum_{\substack{u_f\in\mathbb{Z} \\ |u_f + \Re\dot{f}|\le N \quad (f\in B) \\ u_f + \dot{f}\ne 0 \quad (f\in\Lambda_+) \\ u_f + \dot{f} = 0 \quad (f\in\Lambda_0)}} \iota(\mathbf{u}) e^{2\pi\sqrt{-1}\langle \mathbf{y}, A^{-1}\mathbf{u}\rangle'} \prod_{f\in\Lambda_+} \frac{1}{(u_f + \dot{f})^{k_f}}, \end{aligned}$$

where

$$\iota(\mathbf{u}) = \begin{cases} 1 & (A^{-1}\mathbf{u} \in \mathbb{Z}^d), \\ 0 & (A^{-1}\mathbf{u} \notin \mathbb{Z}^d). \end{cases} \tag{16.3.34}$$

We prove

$$\iota(\mathbf{u}) = \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} e^{2\pi\sqrt{-1}\langle \mathbf{w}, A^{-1}\mathbf{u}\rangle'}. \tag{16.3.35}$$

In fact, using Lemma 16.14 with $Q = \mathbb{Z}^d$, $P = \mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*]$ and noting $\mathrm{Hom}(\mathbb{Z}^d, \mathbb{Z}) \simeq \mathbb{Z}^d$, $\mathrm{Hom}(\mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*], \mathbb{Z}) \simeq \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$ and $\#(\mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*]/\mathbb{Z}^d) = \#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])$, we have

$$\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} e^{2\pi\sqrt{-1}\langle \mathbf{w}, \bar{\lambda}\rangle'} = \delta_{\bar{\lambda},0}$$

for $\bar{\lambda} \in \mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*]/\mathbb{Z}^d$. Since $A^{-1}\mathbf{u} = \sum_{f\in B} \vec{f}^B u_f \in \mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*]$, choosing $\bar{\lambda} = \overline{A^{-1}\mathbf{u}} \in \mathrm{Lh}_{\mathbb{Z}}[\vec{B}^*]/\mathbb{Z}^d$ in the above, we obtain (16.3.35).

Using (16.3.35) we find that (16.3.33) is equal to

(16.3.36)

$$\begin{aligned}
&\frac{(-1)^{\#\Lambda_0}}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \sum_{\substack{u_f\in\mathbb{Z} \\ |u_f+\Re\dot{f}|\le N \quad (f\in B) \\ u_f+\dot{f}\ne 0 \quad (f\in\Lambda_+) \\ u_f+\dot{f}=0 \quad (f\in\Lambda_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, A^{-1}\mathbf{u}\rangle'} \\
&\qquad \times \prod_{f\in\Lambda_+} \frac{1}{(u_f+\dot{f})^{k_f}} \\
&= \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \prod_{f\in\Lambda_0} \Big(- \sum_{\substack{u_f\in\mathbb{Z} \\ |u_f+\Re\dot{f}|\le N \\ u_f+\dot{f}=0}} e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}^B\rangle' u_f} \Big) \\
&\qquad \times \prod_{f\in\Lambda_+} \Big(\sum_{\substack{u_f\in\mathbb{Z} \\ |u_f+\Re\dot{f}|\le N \\ u_f+\dot{f}\ne 0}} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}^B\rangle' u_f}}{(u_f+\dot{f})^{k_f}} \Big),
\end{aligned}$$

where we have used $A^{-1}\mathbf{u} = \sum_{f\in B} \vec{f}^B u_f$. (Note that at present $\Lambda = \Lambda_0 \cup \Lambda_+ = B$.)

By Lemma 16.15 we see that the assumption $\mathbf{y} \in V' \backslash \bigcup_{f\in\Lambda_1} (\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$ implies that $\langle \mathbf{y}+\mathbf{w}, \vec{f}^B\rangle' \notin \mathbb{Z}$ for all $f \in \Lambda_1$ and $\mathbf{w} \in \mathbb{Z}^d$. Therefore the condition of Lemma 16.13 is satisfied (for $y = \langle \mathbf{y}+\mathbf{w}, \vec{f}^B\rangle'$). Therefore letting $N \to \infty$ on the right-hand side of (16.3.36) and applying

Lemmas 16.11 and 16.13, we obtain

(16.3.37)

$$S_1(\mathbf{k},\mathbf{y};\Lambda;B)=\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}\times\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]}\prod_{f\in\Lambda}\Big(-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!}C(k_f,\{\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle'\};\dot{f})\Big).$$

Lastly, we note that the factor $\{\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle'\}$ on the right-hand side of the above can be replaced by $\{\mathbf{y}+\mathbf{w}\}'_{B,f}$. This is because $C(k,\{y\};b)=C(k,1-\{-y\};b)$ for $k\geq 2$ or $k=0$, while $y\notin\mathbb{Z}$ when $k=1$ (cf. (4.3.4)). This completes the proof of the lemma. □

Lemma 16.16 gives the right-hand side of (16.3.3) in the special case $\Lambda=B$ and $\mathscr{B}=\{B\}$, but for $S_1(\mathbf{k},\mathbf{y};\Lambda;B)$ instead of $S(\mathbf{k},\mathbf{y};\Lambda)$. In order to use Lemma 16.16 in the proof of the general case, we need the following inequality.

LEMMA 16.17 ([115, Lemma 4.8]). *Assume $\Lambda=B$ with $\mathscr{B}=\{B\}$. For $\mu>0$ and $\mathbf{k}=(k_f)_{f\in\Lambda}\in\mathbb{N}_0^d$ there exists $K>0$ such that for all $\mathbf{y}\in V'\setminus\bigcup_{f\in\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$ and all sufficiently large $N>0$,*

(16.3.38)

$$|Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B)|\leq K\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]}\prod_{f\in\Lambda_1}\Big(1+(1-\{\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle'\})^{-\frac{1}{\mu+1}}+\{\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle'\}^{-\frac{1}{\mu+1}}\Big).$$

PROOF. From the proof of Lemma 16.16, we have

(16.3.39)

$$\begin{aligned}&|Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B)|\\&\leq\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]}\prod_{f\in\Lambda_0}\Big|-\sum_{\substack{u_f\in\mathbb{Z}\\|u_f+\Re\dot{f}|\leq N\\u_f+\dot{f}=0}}e^{2\pi\sqrt{-1}\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle' u_f}\Big|\\&\quad\times\prod_{f\in\Lambda_+}\Big|\sum_{\substack{u_f\in\mathbb{Z}\\|u_f+\Re\dot{f}|\leq N\\u_f+\dot{f}\neq 0}}\frac{e^{2\pi\sqrt{-1}\langle\mathbf{y}+\mathbf{w},\vec{f}^{B}\rangle' u_f}}{(u_f+\dot{f})^{k_f}}\Big|.\end{aligned}$$

On the right-hand side of (16.3.39), each sum corresponding to $f\in\Lambda_0$ consists of just one term, and each sum corresponding to $k_f\geq 2$ is convergent absolutely as $N\to\infty$, so all of them are bounded. For

$f \in \Lambda_1$, we apply Lemma 16.13 to obtain that the right-hand side of (16.3.39) is

(16.3.40)

$$\le K \sum_{\mathbf{w} \in \mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \prod_{f \in \Lambda_1} (1+(1-\{\langle \mathbf{y}+\mathbf{w}, \vec{f^B}\rangle'\})^{-\frac{1}{\mu+1}} + \{\langle \mathbf{y}+\mathbf{w}, \vec{f^B}\rangle'\}^{-\frac{1}{\mu+1}}).$$

□

To evaluate the difference between $Z(N;\mathbf{k},\mathbf{y};\Lambda;B)$ and $Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B)$ in the final step of the proof, the following two lemmas are necessary. For $B \in \mathscr{B}$ and $R > 0$, let

$$U_R = U_R(B) = \{\mathbf{x} \in \mathbb{R}^d \mid |\Re f(\mathbf{x})| \le R \quad (f \in B)\}, \tag{16.3.41}$$

$$W_R = \{\mathbf{x} \in \mathbb{R}^d \mid |x_j| \le R \quad (1 \le j \le d)\}. \tag{16.3.42}$$

LEMMA 16.18 ([115, Lemma 4.9]). *There exist positive numbers c_1, c_2 with $c_1 > c_2 > 0$ such that*

$$U_{c_2 R}(B) \subset W_R \subset U_{c_1 R}(B) \tag{16.3.43}$$

for all sufficiently large $R > 0$.

PROOF. We show the first inclusion. Each vertex $\mathbf{x}$ of the parallelotope $U_R(B)$ satisfies one of the following equations:

$$A\mathbf{x} + (\Re \dot{f})_{f \in B} = (R_f)_{f \in B} \qquad (R_f \in \{R, -R\}), \tag{16.3.44}$$

where we regard $(\Re \dot{f})_{f \in B}$, $(R_f)_{f \in B}$ and $\vec{f}$ as column vectors respectively and $A = {}^t(\vec{f})_{f \in B}$. Hence we see that the Euclid norm $\|\mathbf{x}\|$ of each vertex $\mathbf{x} = A^{-1}(R_f - \Re \dot{f})_{f \in B}$ satisfies

$$\|\mathbf{x}\| \le \|A^{-1}\| \|(R_f - \Re \dot{f})_{f \in B}\| \le \|A^{-1}\| \max_{f \in B} \sqrt{d} |R \pm \Re \dot{f}| \tag{16.3.45}$$

$$\le R\sqrt{d} \|A^{-1}\| \max_{f \in B} \left|1 \pm \frac{\Re \dot{f}}{R}\right| \le 2\sqrt{d} \|A^{-1}\| R$$

for all sufficiently large $R > 0$, where $\|A\|$ denotes the operator norm of A. Thus choosing

$$c_2 = (2\sqrt{d}\|A^{-1}\|)^{-1} \tag{16.3.46}$$

we find that the vertex $\mathbf{x}$ of $U_{c_2 R}(B)$ satisfies $\|\mathbf{x}\| \le R$, so $U_{c_2 R}(B) \subset W_R$.

We show the second inclusion. Denote by $K(R)$ the ball of radius R whose center is the origin. The distance between the origin and the hyperplane $\{\mathbf{x} \in V' \mid \langle \vec{f}, \mathbf{x}\rangle' + \Re \dot{f} = \pm R\}$ is given by $\frac{|R \pm \Re \dot{f}|}{\|\vec{f}\|}$. Since

$$R \min_{f \in B} \frac{1}{2\|\vec{f}\|} \le \min_{f \in B} \frac{|R \pm \Re \dot{f}|}{\|\vec{f}\|} \tag{16.3.47}$$

holds for all sufficiently large $R > 0$, we find that $K(R\min(2\|\vec{f}\|)^{-1}) \subset U_R(B)$. Therefore, choosing

$$c_1^{-1} = \min_{f\in B} \frac{1}{2\sqrt{d}\|\vec{f}\|}, \tag{16.3.48}$$

we obtain

$$W_R \subset K(\sqrt{d}R) = K(c_1 R\min(2\|\vec{f}\|)^{-1}) \subset U_{c_1R}(B). \tag{16.3.49}$$

□

LEMMA 16.19 ([115, Lemma 4.10]). *Assume* $\Lambda = B = \{f_1, \dots, f_d\}$ *with* $\mathscr{B} = \{B\}$. *Let* c_1, c_2 *be as in Lemma 16.18. For* $\mu > 0$ *and* $\mathbf{k} = (k_f)_{f\in\Lambda} \in \mathbb{N}_0^d$ *there exists* $K > 0$ *such that for all* $\mathbf{y} \in V' \setminus \bigcup_{f\in\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$ *and all sufficiently large* $N \in \mathbb{N}$,

(16.3.50)
$$\Big|\sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap(W_N\setminus U_{c_2N}(B))\\ f(\mathbf{v})\neq 0 \quad (f\in\Lambda_+)\\ f(\mathbf{v})=0 \quad (f\in\Lambda_0)}} e^{2\pi\sqrt{-1}\langle\mathbf{y},\mathbf{v}\rangle'} \prod_{f\in\Lambda_+}\frac{1}{f(\mathbf{v})^{k_f}}\Big|$$
$$\le KN^{-\frac{1}{\mu+1}}(\log N)^d$$
$$\times\Big(1+\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]}\sum_{f\in\Lambda_1}\Big((1-\{\langle\mathbf{y}+\mathbf{w},\vec{f^B}\rangle'\})^{-\frac{1}{\mu+1}}+\{\langle\mathbf{y}+\mathbf{w},\vec{f^B}\rangle'\}^{-\frac{1}{\mu+1}}\Big)\Big).$$

PROOF. For brevity, we put

$$G(\mathbf{y},\mathbf{v}) = e^{2\pi\sqrt{-1}\langle\mathbf{y},\mathbf{v}\rangle'}\prod_{f\in\Lambda_+}\frac{1}{(\langle\vec{f},\mathbf{v}\rangle'+\dot{f})^{k_f}}. \tag{16.3.51}$$

We rearrange $\{f_1,\dots,f_l\} = \Lambda_+$ and $\{f_{l+1},\dots,f_d\} = \Lambda_0$ and decompose

(16.3.52)
$$\left\{\mathbf{v}\in\mathbb{Z}^d\cap(W_N\setminus U_{c_2N}(B)) \;\middle|\; \begin{array}{ll} f(\mathbf{v})\neq 0 & (f\in\Lambda_+),\\ f(\mathbf{v})=0 & (f\in\Lambda_0)\end{array}\right\} = \bigcup_{j=1}^{l} X_j(N)$$

with

(16.3.53)
$$X_j(N) = \left\{\mathbf{v}\in\mathbb{Z}^d \;\middle|\; \begin{array}{ll} |\Re f_1(\mathbf{v})|,\dots,|\Re f_{j-1}(\mathbf{v})|\le c_2N, |\Re f_j(\mathbf{v})|>c_2N,\\ f_i(\mathbf{v})\neq 0 \quad (1\le i\le l),\ f_i(\mathbf{v})=0 \quad (l+1\le i\le d),\\ |v_k|\le N \qquad (1\le k\le d)\end{array}\right\}$$

for $1 \le j \le l$. Let $A = {}^t(\vec{f})_{f\in B}$ with $\vec{f}$ regarded as column vectors. We rewrite the series in terms of $\mathbf{u} = A\mathbf{v}$. Let

(16.3.54)
$$Y_j(N) = \left\{ \mathbf{u}\in\mathbb{Z}^d \;\middle|\; \begin{array}{l} |u_1+\Re\dot{f}_1|,\dots,|u_{j-1}+\Re\dot{f}_{j-1}| \le c_2N, |u_j+\Re\dot{f}_j| > c_2N, \\ u_i+\dot{f}_i \ne 0 \quad (1\le i\le l),\ u_i+\dot{f}_i = 0 \quad (l+1\le i\le d), \\ |\text{each entry of } A^{-1}\mathbf{u}| \le N \end{array} \right\}$$

for $1 \le j \le l$. In $Y_j(N)$ for a fixed $(u_1,\dots,u_{j-1},u_{j+1},\dots,u_d)$ with sufficiently large $N \in \mathbb{N}$, we see that u_j runs over all integers such that $c_2N - \Re\dot{f}_j < u_j < H_j^+$ and $-H_j^- < u_j < -c_2N - \Re\dot{f}_j$ for some $H_j^\pm = H_j^\pm(u_1,\dots,u_{j-1},u_{j+1},\dots,u_d) \ge 0$, where $H_j^\pm$ are determined by the intersection point of the half line $\{(u_1,\dots,u_{j-1},x,u_{j+1},\dots,u_d) \mid \pm x > 0\}$ and the boundaries $\partial W_N = \bigcup_{1\le l\le d}\{A\mathbf{v} \mid |v_k| \le N \quad (k \ne l), v_l = \pm N\}$ of W_N. If there is no intersection point, then we put $H_j^\pm = 0$ accordingly. See Figure 16.3.1 for these sets and parameters.

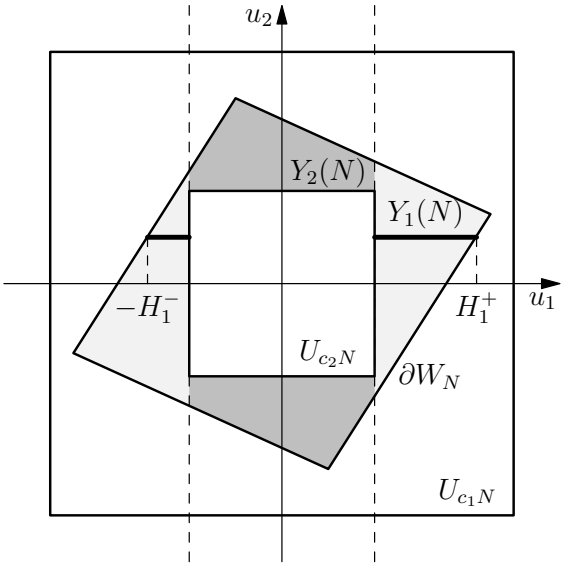


FIGURE 16.3.1.

From the proof of Lemma 16.16 we evaluate

(16.3.55)
$$\Big|\sum_{\mathbf{v}\in X_j(N)} G(\mathbf{y},\mathbf{v})\Big| \le \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \Big|\sum_{\mathbf{u}\in Y_j(N)} \Big(\prod_{i=1}^{l} \frac{e^{2\pi\sqrt{-1}\langle\mathbf{y}+\mathbf{w},\vec{f}_i^{\,B}\rangle' u_i}}{(u_i+\dot{f}_i)^{k_i}}\Big)\Big|.$$

Further, for each j and $\mathbf{w}$, we have

$$
\begin{aligned}
(16.3.56)\qquad & \Bigg| \sum_{\mathbf{u}\in Y_j(N)} \left(\prod_{i=1}^{l} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_i^{\,B}\rangle' u_i}}{(u_i+\dot{f}_i)^{k_i}} \right) \Bigg| \\
&= \Bigg| \sum_{\substack{u_i\in\mathbb{Z},\ u_i+\dot{f}_i\neq 0 \quad (1\le i\le l)\\ |u_i+\Re\dot{f}_i|\le c_2N \quad (1\le i\le j-1)\\ c_2N-\Re\dot{f}_j<u_j<H_j^+ \text{ or}\\ -H_j^-<u_j<-c_2N-\Re\dot{f}_j}} \left(\prod_{i=1}^{l} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_i^{\,B}\rangle' u_i}}{(u_i+\dot{f}_i)^{k_i}} \right) \Bigg| \\
&= \Bigg| \sum_{\substack{u_i\in\mathbb{Z},\ u_i+\dot{f}_i\neq 0 \quad (1\le i\le l, i\neq j)\\ |u_i+\Re\dot{f}_i|\le c_2N \quad (1\le i\le j-1)}} \left(\prod_{\substack{i=1\\ i\neq j}}^{l} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_i^{\,B}\rangle' u_i}}{(u_i+\dot{f}_i)^{k_i}} \right) \\
&\qquad \times \sum_{\substack{c_2N-\Re\dot{f}_j<u_j<H_j^+ \text{ or}\\ -H_j^-<u_j<-c_2N-\Re\dot{f}_j}} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_j^{\,B}\rangle' u_j}}{(u_j+\dot{f}_j)^{k_j}} \Bigg| \\
&\le \sum_{\substack{u_i\in\mathbb{Z},\ u_i+\dot{f}_i\neq 0 \quad (1\le i\le l, i\neq j)\\ |u_i+\Re\dot{f}_i|\le c_2N \quad (1\le i\le j-1)\\ |u_i+\Re\dot{f}_i|\le c_1N \quad (j+1\le i\le l)}} \left(\prod_{\substack{i=1\\ i\neq j}}^{l} \frac{1}{|u_i+\dot{f}_i|^{k_i}} \right) \\
&\qquad \times \Bigg| \sum_{\substack{c_2N-\Re\dot{f}_j<u_j<H_j^+ \text{ or}\\ -H_j^-<u_j<-c_2N-\Re\dot{f}_j}} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_j^{\,B}\rangle' u_j}}{(u_j+\dot{f}_j)^{k_j}} \Bigg|,
\end{aligned}
$$

where in the last member, we added the extra conditions $|u_i+\Re\dot{f}_i|\le c_1N$ for $j+1\le i\le l$, which comes from Lemma 16.18. If $k_j=1$, then by Lemma 16.13, and if $k_j\ge 2$ directly, we obtain

$$
\begin{aligned}
(16.3.57)\qquad & \Bigg| \sum_{\substack{c_2N-\Re\dot{f}_j<u_j<H_j^+\\ -H_j^-<u_j<-c_2N-\Re\dot{f}_j}} \frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w}, \vec{f}_j^{\,B}\rangle' u_j}}{(u_j+\dot{f}_j)^{k_j}} \Bigg| \\
&\le \begin{cases} KN^{-\frac{1}{\mu+1}}\left((1-\{\langle \mathbf{y}+\mathbf{w}, \vec{f}_j^{\,B}\rangle'\})^{-\frac{1}{\mu+1}} + \{\langle \mathbf{y}+\mathbf{w}, \vec{f}_j^{\,B}\rangle'\}^{-\frac{1}{\mu+1}}\right) & (k_j=1)\\ KN^{-1}\le KN^{-\frac{1}{\mu+1}} & (k_j\ge 2) \end{cases} \\
&=: Q_j(N,\mathbf{y},\mathbf{w})
\end{aligned}
$$

for some $K > 0$. Therefore

$$\begin{aligned}(16.3.58)\qquad &\Big|\sum_{\mathbf{u}\in Y_j(N)}\Big(\prod_{i=1}^{l}\frac{e^{2\pi\sqrt{-1}\langle \mathbf{y}+\mathbf{w},\vec{f}_i^{B}\rangle' u_i}}{(u_i+\dot{f}_i)^{k_i}}\Big)\Big|\\ &\le Q_j(N,\mathbf{y},\mathbf{w})\sum_{\substack{u_i\in\mathbb{Z},\ u_i+\dot{f}_i\neq 0 \quad (1\le i\le l, i\neq j)\\ |u_i+\Re\dot{f}_i|\le c_2 N \quad (1\le i\le j-1)\\ |u_i+\Re\dot{f}_i|\le c_1 N \quad (j+1\le i\le l)}}\Big(\prod_{\substack{i=1\\ i\neq j}}^{l}\frac{1}{|u_i+\dot{f}_i|^{k_i}}\Big)\\ &\le K' Q_j(N,\mathbf{y},\mathbf{w})(\log c_2 N)^{j-1}(\log c_1 N)^{l-j}\\ &\le K'' Q_j(N,\mathbf{y},\mathbf{w})(\log N)^{d}\end{aligned}$$

for some $K', K'' > 0$. Substituting (16.3.58) into (16.3.55), we complete the proof. □

The third step. Lastly, we consider the general case. First, we prove several preparatory lemmas.

LEMMA 16.20 ([115, Lemma 4.11]). *Fix a decomposition $\Lambda = B_0 \cup L_0$ with $B_0 \in \mathscr{B}$. Let $\mathbf{y} \in V'$ and $f \in B_0$. If $f \notin \widetilde{\Lambda}$, then there exists $g \in L_0$ such that $\langle \vec{g}, \vec{f}^{B_0}\rangle' \neq 0$. If $f \in \widetilde{\Lambda}$ and $\mathbf{y} \notin \mathfrak{H}_{B_0\setminus\{f\}} + \mathbb{Z}^d$, then there exists $c \in \mathbb{R}\setminus\mathbb{Z}$ such that*

$$(16.3.59)\qquad \langle \mathbf{y} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0}\rangle' = c$$

for all $\mathbf{x} = (x_g)_{g\in L_0} \in \mathbb{R}^{\#L_0}$.

PROOF. The first assertion directly follows from the definition. Assume that $f \in \widetilde{\Lambda}$. Then $\langle \vec{g}, \vec{f}^{B_0}\rangle' = 0$ for all $g \in L_0$ and we have

$$(16.3.60)\qquad \langle \mathbf{y} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0}\rangle' = \langle \mathbf{y}, \vec{f}^{B_0}\rangle',$$

which is a constant function in $\mathbf{x}$. By Lemma 16.15, we find $\langle \mathbf{y}, \vec{f}^{B_0}\rangle' \notin \mathbb{Z}$. This implies the second assertion. □

For $\mathbf{y} \in V'$, a decomposition $\Lambda = B_0 \cup L_0$ with $B_0 \in \mathscr{B}$ and $f \in B_0$, let

$$(16.3.61)\qquad H(f,\mathbf{y}) = \Big\{\mathbf{x} = (x_g)_{g\in L_0} \in \mathbb{R}^{\#L_0} \ \Big|\ \langle \mathbf{y} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0}\rangle' \in \mathbb{Z}\Big\}.$$

LEMMA 16.21 ([115, Lemma 4.12]). *Let $\mathbf{y} \in V'$, $f \in B_0$, and assume that $\mathbf{y} \notin \mathfrak{H}_{B_0\setminus\{f\}} + \mathbb{Z}^d$ if $f \in \widetilde{\Lambda}$. Then the set $H(f,\mathbf{y})$ is empty, or a collection of equally spaced parallel hyperplanes.*

PROOF. Let $U = (\vec{g})_{g\in L_0}$ be an $d \times \#L_0$ matrix and $\mathbf{x} = (x_g)_{g\in L_0}$ be a column vector. Consider the equation

$$\langle \mathbf{y} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0}\rangle' = n \tag{16.3.62}$$

for $n \in \mathbb{Z}$, which is equivalent to

$$\langle U\mathbf{x}, \vec{f}^{B_0}\rangle' = \langle \mathbf{y}, \vec{f}^{B_0}\rangle' - n. \tag{16.3.63}$$

Assume that $f \notin \widetilde{\Lambda}$. Then there exists $g \in L_0$ such that $\langle \vec{g}, \vec{f}^{B_0}\rangle' \neq 0$. We see that (16.3.63) has a solution $\mathbf{x} = \mathbf{x}_0 - n\mathbf{a}$ with

$$(\mathbf{x}_0)_h = \begin{cases} \langle \mathbf{y}, \vec{f}^{B_0}\rangle'/\langle \vec{g}, \vec{f}^{B_0}\rangle' & (h = g) \\ 0 & (h \neq g), \end{cases} \tag{16.3.64}$$

$$(\mathbf{a})_h = \begin{cases} 1/\langle \vec{g}, \vec{f}^{B_0}\rangle' & (h = g) \\ 0 & (h \neq g), \end{cases}$$

and so the equation (16.3.63) is rewritten as

$$\langle U(\mathbf{x} - \mathbf{x}_0 + n\mathbf{a}), \vec{f}^{B_0}\rangle' = 0. \tag{16.3.65}$$

The condition $\langle \vec{g}, \vec{f}^{B_0}\rangle' \neq 0$ also implies that $\dim \operatorname{Ker} \mathscr{U} = \#L_0 - 1$ for the linear functional $\mathscr{U}(\mathbf{v}) = \langle U\mathbf{v}, \vec{f}^{B_0}\rangle'$, and

$$H(f, \mathbf{y}) = (\operatorname{Ker} \mathscr{U} + \mathbf{x}_0) + \mathbb{Z}\mathbf{a} \tag{16.3.66}$$

is a collection of equally spaced parallel hyperplanes.

Assume that $f \in \widetilde{\Lambda}$. Then by Lemma 16.20,

$$\langle \mathbf{y} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0}\rangle' \in \mathbb{R} \setminus \mathbb{Z}, \tag{16.3.67}$$

and hence $H(f, \mathbf{y}) = \emptyset$. □

LEMMA 16.22 ([115, Lemma 4.13]). *Fix a decomposition $\Lambda = B_0 \cup L_0$ with $B_0 \in \mathscr{B}$. Assume $D \subset B_0$. For a fixed $\mathbf{y} \in V' \setminus \bigcup_{f\in\widetilde{\Lambda}\cap D}(\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$, the Lebesgue measure of*

$$M(\mathbf{y}) = \Big\{ (x_g)_{g\in L_0} \in \mathbb{R}^{\#L_0} \;\Big|\; \mathbf{y} - \sum_{g\in L_0} x_g \vec{g} \in \bigcup_{f\in D} (\mathfrak{H}_{B_0\setminus\{f\}} + \mathbb{Z}^d) \Big\} \tag{16.3.68}$$

is zero.

PROOF. By Lemma 16.15, we have $M(\mathbf{y}) = \bigcup_{f\in D} \bigcup_{\mathbf{w}\in\mathbb{Z}^d} H(f, \mathbf{y} + \mathbf{w})$. Further, by (16.1.3) we have

$$\bigcup_{f\in\widetilde{\Lambda}\cap D} (\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d) = \bigcup_{f\in\widetilde{\Lambda}\cap D} (\mathfrak{H}_{B_0\setminus\{f\}} + \mathbb{Z}^d), \tag{16.3.69}$$

so from the assumption $\mathbf{y} \in V' \setminus \bigcup_{f\in\tilde{\Lambda}\cap D}(\mathfrak{H}_{\Lambda\setminus\{f\}} + \mathbb{Z}^d)$ we see that $\mathbf{y}+\mathbf{w} \notin \bigcup_{f\in\tilde{\Lambda}\cap D}(\mathfrak{H}_{B_0\setminus\{f\}} + \mathbb{Z}^d)$ for any $\mathbf{w} \in \mathbb{Z}^d$. Therefore we can apply Lemma 16.21 to find that for each $f \in D$ and $\mathbf{w} \in \mathbb{Z}^d$ the measure of $H(f, \mathbf{y}+\mathbf{w})$ is zero. □

LEMMA 16.23 ([115, Lemma 4.14]). *Let $n \in \mathbb{N}$ and $P, Q \in \mathbb{N}_0$ with $P \geq Q$. Let $a_{ki} \in \mathbb{R}$ for $1 \leq k \leq P$ and $0 \leq i \leq n$ such that for each $k = 1, \dots, P$ there exists $i \geq 1$ such that $a_{ki} \neq 0$. If $\mu \geq P$, then*

$$\int_0^1 dx_1 \cdots \int_0^1 dx_n \Big(\prod_{k=1}^{Q} (1 - \{a_{k0} + \sum_{i=1}^{n} a_{ki}x_i\})^{-\frac{1}{\mu+1}} \Big) \times \Big(\prod_{k=Q+1}^{P} \{a_{k0} + \sum_{i=1}^{n} a_{ki}x_i\}^{-\frac{1}{\mu+1}} \Big) < \infty. \tag{16.3.70}$$

PROOF. For $1 \leq k \leq P$ put

$$L_k(\mathbf{x}) = \sum_{i=1}^{n} a_{ki}x_i. \tag{16.3.71}$$

Since $[0,1]^n$ is compact, by considering a neighborhood of each point $\mathbf{x}_0$ in $[0,1]^n$ and shifting the point $\mathbf{x}_0$ to the origin, we see that it is sufficient to show that

(16.3.72)
$$I = \int_{[-\epsilon,\epsilon]^n} dx_1 \cdots dx_n \Big(\prod_{k=1}^{q} (1 - \{L_k(\mathbf{x})\})^{-\frac{1}{\mu+1}} \Big) \Big(\prod_{k=q+1}^{p} \{L_k(\mathbf{x})\}^{-\frac{1}{\mu+1}} \Big)$$

is finite for a sufficiently small $\epsilon > 0$, where $0 \leq q \leq Q$ and $q \leq p \leq P$. This is estimated as

$$I \leq \int_{[-\epsilon,\epsilon]^n} dx_1 \cdots dx_n \Big(\prod_{k=1}^{p} |L_k(\mathbf{x})|^{-\frac{1}{\mu+1}} \Big). \tag{16.3.73}$$

For this integral, we decompose the region

$$[-\epsilon, \epsilon]^n = \bigcup_{k=1}^{p} U_k, \tag{16.3.74}$$

where

$$U_k = \{\mathbf{x} \in [-\epsilon,\epsilon]^n \mid |L_k(\mathbf{x})| \leq |L_m(\mathbf{x})| \text{ for any } m \neq k\}. \tag{16.3.75}$$

We show that the integral on each U_k is finite. Since on U_k

$$|L_m(\mathbf{x})|^{-\frac{1}{\mu+1}} \leq |L_k(\mathbf{x})|^{-\frac{1}{\mu+1}} \tag{16.3.76}$$

for any $m \neq k$, we have

$$\Big(\prod_{m=1}^{p} |L_m(\mathbf{x})|^{-\frac{1}{\mu+1}} \Big) \leq |L_k(\mathbf{x})|^{-\frac{p}{\mu+1}}. \tag{16.3.77}$$

Fix i such that $a_{ki} \neq 0$. Then by changing variables as $y_i = L_k(\mathbf{x})$ and $y_j = x_j$ for $j \neq i$, we obtain

(16.3.78)
$$\int_{U_k} dx_1 \cdots dx_n \Big(\prod_{m=1}^{p} |L_m(\mathbf{x})|^{-\frac{1}{\mu+1}} \Big) \leq \int_{U_k} dx_1 \cdots dx_n |L_k(\mathbf{x})|^{-\frac{p}{\mu+1}} \\ \leq |a_{ki}| \int_{-r}^{r} |y_i|^{-\frac{p}{\mu+1}} dy_i \int_{[-\epsilon,\epsilon]^{n-1}} \prod_{j \neq i} dy_j$$

for some $r > 0$. By the assumption $\mu \geq P$, the right-hand side is finite because

(16.3.79)
$$-\frac{p}{\mu+1} > -\frac{P}{P+1} > -1.$$

□

Lemma 16.24 ([115, Lemma 4.15]). *For $k \in \mathbb{N}_0$, $m \in \mathbb{Z}$ and $b \in \mathbb{C}$,*

(16.3.80)
$$-\frac{(2\pi\sqrt{-1})^k}{k!} \int_0^1 C(k,x;b) e^{-2\pi\sqrt{-1}mx} dx \\ = \begin{cases} -1 & (m+b=0, k=0), \\ 0 & (m+b \neq 0, k=0), \\ 0 & (m+b=0, k \neq 0), \\ \dfrac{1}{(m+b)^k} & (m+b \neq 0, k \neq 0). \end{cases}$$

Proof. By definition (16.3.1), for $0 \leq x < 1$, we have

(16.3.81)
$$\frac{1}{k!} C(k,x;b) e^{-2\pi\sqrt{-1}mx} = \frac{1}{2\pi\sqrt{-1}} \int_{|t|=\epsilon} \frac{t e^{(t-2\pi\sqrt{-1}(m+b))x}}{e^{t-2\pi\sqrt{-1}b} - 1} t^{-k-1} dt$$

for sufficiently small $\epsilon > 0$. By integrating both sides in the region $0 \leq x < 1$, we obtain

(16.3.82)
$$\frac{1}{k!} \int_0^1 C(k,x;b) e^{-2\pi\sqrt{-1}mx} dx = \frac{1}{2\pi\sqrt{-1}} \int_{|t|=\epsilon} \frac{t}{t - 2\pi\sqrt{-1}(m+b)} t^{-k-1} dt.$$

Since

(16.3.83)
$$\frac{t}{t - 2\pi\sqrt{-1}(m+b)} = \begin{cases} 1 & (m+b=0), \\ -\displaystyle\sum_{l=1}^{\infty} \frac{1}{(2\pi\sqrt{-1}(m+b))^l} t^l & (m+b \neq 0), \end{cases}$$

we obtain the assertion. □

PROOF OF PROPOSITION 16.10. Applying Lemma 16.24 with $m = \langle \vec{g}, \mathbf{v}\rangle'$ and $b = \dot{g}$ (for $g \in L_0$) to (16.3.5), for $N > 0$ we have[4]

(16.3.84)

$$
\begin{aligned}
&Z_1(N; \mathbf{k}, \mathbf{y}; \Lambda; B_0)\\
&= (-1)^{\#\Lambda_0} \sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap U_N(B_0)\\ f(\mathbf{v})\neq 0 \quad (f\in\Lambda_+\cap B_0)\\ f(\mathbf{v})=0 \quad (f\in\Lambda_0\cap B_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y},\mathbf{v}\rangle'} \prod_{f\in\Lambda_+\cap B_0} \frac{1}{f(\mathbf{v})^{k_f}}\\
&\quad \times (-1)^{\#(\Lambda_0\cap L_0)} \prod_{g\in L_0} \Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!} \int_0^1 C(k_g, x_g; \dot{g}) e^{-2\pi\sqrt{-1}\langle \vec{g},\mathbf{v}\rangle' x_g} dx_g \Big)\\
&= \prod_{g\in L_0} \Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!} \Big) \int_0^1 \cdots \int_0^1 \Big(\prod_{g\in L_0} C(k_g, x_g; \dot{g}) dx_g \Big) (-1)^{\#(\Lambda_0\cap B_0)}\\
&\quad \times \sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap U_N(B_0)\\ f(\mathbf{v})\neq 0 \quad (f\in\Lambda_+\cap B_0)\\ f(\mathbf{v})=0 \quad (f\in\Lambda_0\cap B_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}-\sum_{g\in L_0} x_g\vec{g},\mathbf{v}\rangle'} \Big(\prod_{f\in\Lambda_+\cap B_0} \frac{1}{f(\mathbf{v})^{k_f}} \Big)\\
&= \prod_{g\in L_0} \Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!} \Big) \int_0^1 \cdots \int_0^1 \Big(\prod_{g\in L_0} C(k_g, x_g; \dot{g}) dx_g \Big)\\
&\quad \times Z_1(N; \mathbf{k}(B_0), \mathbf{y} - \sum_{g\in L_0} x_g\vec{g}; B_0; B_0),
\end{aligned}
$$

where $\mathbf{k}(B_0) = (k_f)_{f\in B_0}$.

We want to take the limit $N \to \infty$. First, we claim that it is possible to exchange the limit and the integrals. By Lemma 16.17 with $\mu = \#B_0 = d$, we have

(16.3.85)

$$
\begin{aligned}
&\Big| \Big(\prod_{g\in L_0} C(k_g, x_g; \dot{g}) \Big) \sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap U_N(B_0)\\ f(\mathbf{v})\neq 0 \quad (f\in\Lambda_+\cap B_0)\\ f(\mathbf{v})=0 \quad (f\in\Lambda_0\cap B_0)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}-\sum_{g\in L_0} x_g\vec{g},\mathbf{v}\rangle'} \Big(\prod_{f\in\Lambda_+\cap B_0} \frac{1}{f(\mathbf{v})^{k_f}} \Big) \Big|\\
&\le K \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \prod_{f\in\Lambda_1\cap B_0} (1 + (1 - \{\langle \mathbf{y} + \mathbf{w} - \sum_{g\in L_0} x_g\vec{g}, \vec{f}^{B_0}\rangle'\})^{-\frac{1}{d+1}}\\
&\qquad + \{\langle \mathbf{y} + \mathbf{w} - \sum_{g\in L_0} x_g\vec{g}, \vec{f}^{B_0}\rangle'\}^{-\frac{1}{d+1}})\\
&= K \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \prod_{f\in\Lambda_1\cap B_0} X_f,
\end{aligned}
$$

[4]In the second member of (16.3.84), the location of the factor $(-1)^{\#(\Lambda_0\cap L_0)}$ was wrong in [115] and is corrected here.

say. When $f \in \widetilde{\Lambda}$, then under the condition $\mathbf{y} \notin \bigcup_{f \in \widetilde{\Lambda} \cap \Lambda_1 \cap B_0} (\mathfrak{H}_{B_0 \setminus \{f\}} + \mathbb{Z}^d)$, we see that X_f is just a constant because of the second assertion of Lemma 16.20. When $f \notin \widetilde{\Lambda}$, by the first assertion of Lemma 16.20 we see that X_f fulfills the assumption of Lemma 16.23, and hence by the lemma it is integrable since $d \geq \#(\Lambda_1 \cap B_0)$. Therefore our claim follows from Lebesgue's dominated convergence theorem. Note that $\widetilde{\Lambda} \cap \Lambda_1 \cap B_0 = \widetilde{\Lambda} \cap \Lambda_1$ because of (16.1.3).

Therefore from (16.3.84) we now obtain

$$\begin{aligned}(16.3.86)\qquad &\lim_{N\to\infty} Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B_0)\\ &= \prod_{g\in L_0}\Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!}\Big)\int_0^1\cdots\int_0^1\Big(\prod_{g\in L_0} C(k_g,x_g;\dot{g})dx_g\Big)\\ &\qquad\times \lim_{N\to\infty} Z_1(N;\mathbf{k}(B_0),\mathbf{y}-\sum_{g\in L_0} x_g\vec{g};B_0;B_0).\end{aligned}$$

By Lemma 16.22 with $D = \Lambda_1 \cap B_0$ the measure of $M(\mathbf{y})$ is 0, and if $(x_g)_{g\in L_0} \notin M(\mathbf{y})$, then by Lemma 16.16 with $B = B_0$ we see that $Z_1(N;\mathbf{k}(B_0),\mathbf{y}-\sum_{g\in L_0} x_g\vec{g};B_0;B_0)$ converges as $N\to\infty$. That is, the integrand on the right-hand side of (16.3.86) converges almost everywhere, and (16.3.32) of Lemma 16.16 implies

(16.3.87)

$$\begin{aligned}&S_1(\mathbf{k},\mathbf{y};\Lambda;B_0)\\ &= \lim_{N\to\infty} Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B_0)\\ &= \prod_{g\in L_0}\Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!}\Big)\int_0^1\cdots\int_0^1\Big(\prod_{g\in L_0} C(k_g,x_g;\dot{g})dx_g\Big)\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}\\ &\quad\times\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]}\prod_{f\in B_0}\Big(-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!}C(k_f,\{\mathbf{y}+\mathbf{w}-\sum_{g\in L_0}x_g\vec{g}\}'_{B_0,f};\dot{f})\Big)\\ &= \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}\prod_{f\in\Lambda}\Big(-\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!}\Big)\\ &\qquad\times\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]}\int_0^1\cdots\int_0^1\Big(\prod_{g\in L_0}C(k_g,x_g;\dot{g})dx_g\Big)\\ &\qquad\times\prod_{f\in B_0}\Big(C(k_f,\{\mathbf{y}+\mathbf{w}-\sum_{g\in L_0}x_g\vec{g}\}'_{B_0,f};\dot{f})\Big).\end{aligned}$$

The right-hand side of this equation coincides with that of (16.3.3). Therefore, to complete the proof of the proposition, the only remaining task is to show that $S_1(\mathbf{k},\mathbf{y};\Lambda;B_0) = S(\mathbf{k},\mathbf{y};\Lambda)$.

The sum $Z(N;\mathbf{k},\mathbf{y};\Lambda)$ has an expression which is almost the same as (16.3.84): the only difference is that the condition $\mathbf{v}\in\mathbb{Z}^d\cap U_N(B_0)$ is replaced by $\mathbf{v}\in\mathbb{Z}^d\cap W_N$. Therefore, by using Lemmas 16.18 and 16.19, we see that the difference $Z(N;\mathbf{k},\mathbf{y};\Lambda)-Z_1(c_2N;\mathbf{k},\mathbf{y};\Lambda;B_0)$ is evaluated as

(16.3.88)

$$
\begin{aligned}
&|Z(N;\mathbf{k},\mathbf{y};\Lambda)-Z_1(c_2N;\mathbf{k},\mathbf{y};\Lambda;B_0)|\\
&=\Big|(-1)^{\#(\Lambda_0\cap B_0)}\prod_{g\in L_0}\Big(-\frac{(2\pi\sqrt{-1})^{k_g}}{k_g!}\Big)\int_0^1\cdots\int_0^1\Big(\prod_{g\in L_0}C(k_g,x_g;\dot{g})dx_g\Big)\\
&\qquad\times\sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap(W_N\setminus U_{c_2N}(B_0))\\ f(\mathbf{v})\neq 0\quad(f\in\Lambda_+\cap B_0)\\ f(\mathbf{v})=0\quad(f\in\Lambda_0\cap B_0)}}e^{2\pi\sqrt{-1}\langle\mathbf{y}-\sum_{g\in L_0}x_g\vec{g},\mathbf{v}\rangle'}\Big(\prod_{f\in\Lambda_+\cap B_0}\frac{1}{f(\mathbf{v})^{k_f}}\Big)\Big|\\
&\le K\int_0^1\cdots\int_0^1\prod_{g\in L_0}dx_g\\
&\qquad\times\Big|\sum_{\substack{\mathbf{v}\in\mathbb{Z}^d\cap(W_N\setminus U_{c_2N}(B_0))\\ f(\mathbf{v})\neq 0\quad(f\in\Lambda_+\cap B_0)\\ f(\mathbf{v})=0\quad(f\in\Lambda_0\cap B_0)}}e^{2\pi\sqrt{-1}\langle\mathbf{y}-\sum_{g\in L_0}x_g\vec{g},\mathbf{v}\rangle'}\Big(\prod_{f\in\Lambda_+\cap B_0}\frac{1}{f(\mathbf{v})^{k_f}}\Big)\Big|\\
&\le K'N^{-\frac{1}{d+1}}(\log N)^d\int_0^1\cdots\int_0^1\prod_{g\in L_0}dx_g\\
&\qquad\Big(1+\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]}\sum_{f\in\Lambda_1\cap B_0}\Big((1-\{\langle\mathbf{y}+\mathbf{w}-\sum_{g\in L_0}x_g\vec{g},\vec{f}^{B_0}\rangle'\})^{-\frac{1}{d+1}}\\
&\qquad\quad+\{\langle\mathbf{y}+\mathbf{w}-\sum_{g\in L_0}x_g\vec{g},\vec{f}^{B_0}\rangle'\}^{-\frac{1}{d+1}}\Big)\Big)
\end{aligned}
$$

for some $K,K'>0$. Again by Lemmas 16.20 and 16.23, we obtain

$$
|Z(N;\mathbf{k},\mathbf{y};\Lambda)-Z_1(c_2N;\mathbf{k},\mathbf{y};\Lambda;B_0)|\le K''N^{-\frac{1}{d+1}}(\log N)^d \tag{16.3.89}
$$

for some $K''>0$. Hence we have

$$
\begin{aligned}
S(\mathbf{k},\mathbf{y};\Lambda)&=\lim_{N\to\infty}Z(N;\mathbf{k},\mathbf{y};\Lambda)\\
&=\lim_{N\to\infty}Z_1(N;\mathbf{k},\mathbf{y};\Lambda;B_0)=S_1(\mathbf{k},\mathbf{y};\Lambda;B_0).
\end{aligned}
\tag{16.3.90}
$$

□

We have shown the convergence of $S(\mathbf{k},\mathbf{y};\Lambda)$ in Proposition 16.10. Therefore to complete the proof of Theorem 16.2, we have only to show the continuity of $S(\mathbf{k},\mathbf{y};\Lambda)$ in $\mathbf{y}$ on $V'\setminus\bigcup_{f\in\tilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$.

Let $\mathbf{y}_0 \in V' \setminus \bigcup_{f\in\widetilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$, and let $G(\mathbf{y},(x_g))$ be the integrand of (16.3.3). Since $G(\mathbf{y},(x_g))$ is bounded, we have

$$\lim_{\mathbf{y}\to\mathbf{y}_0}\int G(\mathbf{y},(x_g))\prod_{g\in L_0}dx_g=\int\lim_{\mathbf{y}\to\mathbf{y}_0}G(\mathbf{y},(x_g))\prod_{g\in L_0}dx_g. \tag{16.3.91}$$

Thus it is sufficient to show that

$$\lim_{\mathbf{y}\to\mathbf{y}_0}G(\mathbf{y},(x_g))=G(\mathbf{y}_0,(x_g)) \tag{16.3.92}$$

almost everywhere in (x_g). By Lemmas 16.11 and 16.13, we see that $C(k,\{y\};b)$ is continuous in y on $\mathbb{R}$ if $k\neq 1$, and on $\mathbb{R}\setminus\mathbb{Z}$ if $k=1$. Hence if (x_g) satisfies

$$\langle\mathbf{y}_0+\mathbf{w}-\sum_{g\in L_0}x_g\vec{g},\vec{f}^{B_0}\rangle'\notin\mathbb{Z} \tag{16.3.93}$$

for all $f\in\Lambda_1\cap B_0$, then (16.3.92) holds. Therefore it is sufficient to show that (x_g) satisfies (16.3.93) almost everywhere. Since $\mathbf{y}_0+\mathbf{w}\in V'\setminus\bigcup_{f\in\widetilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$, we see that the measure of $M(\mathbf{y}_0+\mathbf{w})$ is zero by Lemma 16.22 with $D=\Lambda_1\cap B_0$.

The proof of Theorem 16.2 is thus complete.

16.4. The structure of the proof of Theorems 16.4 and 16.5

Now we start the proof of Theorem 16.4 and Theorem 16.5. We first prove the assertion (i) of Theorem 16.4. Let

$$\mathfrak{H}_{\mathscr{R}}=\bigcup_{R\in\mathscr{R}}(\mathfrak{H}_R+\mathbb{Z}^d). \tag{16.4.1}$$

LEMMA 16.25 ([115, Lemma 5.1]). *The set $\mathfrak{H}_{\mathscr{R}}$ is a locally finite collection of hyperplanes, that is, for any $\mathbf{y}\in V'$ there exists a neighborhood U of $\mathbf{y}$ such that U intersects only finitely many hyperplanes.*

PROOF. Let $\mathbf{n}_R$ be a normal vector of $\mathfrak{H}_R$. We may assume that $\mathbf{n}_R\in\mathbb{Z}^d$, because $\vec{g}_1,\dots,\vec{g}_{d-1}\in\mathbb{Z}^d$. Then the hyperplane

$$\mathfrak{H}_R+\mathbf{v}=\{\mathbf{y}+\mathbf{v}\mid\langle\mathbf{y},\mathbf{n}_R\rangle'=0\} \tag{16.4.2}$$

with $\mathbf{v}\in\mathbb{Z}^d$ can be rewritten as

$$\{\mathbf{y}\mid\langle\mathbf{y},\mathbf{n}_R\rangle'-\langle\mathbf{v},\mathbf{n}_R\rangle'=0\}=\{\mathbf{y}\mid\langle\mathbf{y}-m\mathbf{e}_R,\mathbf{n}_R\rangle'=0\}, \tag{16.4.3}$$

where $m=\langle\mathbf{v},\mathbf{n}_R\rangle'\in\mathbb{Z}$ and $\mathbf{e}_R=\mathbf{n}_R/\langle\mathbf{n}_R,\mathbf{n}_R\rangle'$. Therefore

$$\mathfrak{H}_R+\mathbf{v}=\mathfrak{H}_R+m\mathbf{e}_R, \tag{16.4.4}$$

and so

$$\mathfrak{H}_{\mathscr{R}}\subset\bigcup_{R\in\mathscr{R}}(\mathfrak{H}_R+\mathbb{Z}\mathbf{e}_R). \tag{16.4.5}$$

Hence the assertion follows from this expression and $\#\mathscr{R} < \infty$. □

LEMMA 16.26 ([115, Lemma 5.2]).

$$\lim_{c\to 0+} \{\mathbf{y} + c\phi_0\}'_{B,f} = \{\mathbf{y}\}'_{B,f} \tag{16.4.6}$$

for $\mathbf{y} \in V'$.

PROOF. By Lemma 16.25, for any $\mathbf{y} \in V'$, we see that $\mathbf{y} + c\phi_0 \notin \mathfrak{H}_{\mathscr{R}}$ and so $\langle \mathbf{y} + c\phi_0, \vec{f}^B\rangle' \notin \mathbb{Z}$ for all sufficiently small $c > 0$. Therefore, if $\langle \mathbf{y}, \vec{f}^B\rangle' \notin \mathbb{Z}$, then

$$\lim_{c\to 0+} \{\mathbf{y} + c\phi_0\}'_{B,f} = \lim_{c\to 0+} \{\langle \mathbf{y} + c\phi_0, \vec{f}^B\rangle'\} = \{\langle \mathbf{y}, \vec{f}^B\rangle'\} = \{\mathbf{y}\}'_{B,f} \tag{16.4.7}$$

by (4.3.4). If $\langle \mathbf{y}, \vec{f}^B\rangle' \in \mathbb{Z}$, then for $\langle \phi_0, \vec{f}^B\rangle' > 0$, we have

$$\begin{aligned} \lim_{c\to 0+} \{\mathbf{y} + c\phi_0\}'_{B,f} &= \lim_{c\to 0+} \{\langle \mathbf{y} + c\phi_0, \vec{f}^B\rangle'\} \\ &= \lim_{c\to 0+} \{c\langle \phi_0, \vec{f}^B\rangle'\} = 0 = \{\langle \mathbf{y}, \vec{f}^B\rangle'\} \end{aligned} \tag{16.4.8}$$

and for $\langle \phi_0, \vec{f}^B\rangle' < 0$, we have

$$\begin{aligned} \lim_{c\to 0+} \{\mathbf{y} + c\phi_0\}'_{B,f} &= \lim_{c\to 0+} 1 - \{-\langle \mathbf{y} + c\phi_0, \vec{f}^B\rangle'\} \\ &= \lim_{c\to 0+} 1 - \{-c\langle \phi_0, \vec{f}^B\rangle'\} = 1 = 1 - \{-\langle \mathbf{y}, \vec{f}^B\rangle'\}. \end{aligned} \tag{16.4.9}$$

Hence we have the assertion. □

By this lemma we immediately obtain

$$\lim_{c\to 0+} F(\mathbf{t}, \mathbf{y} + c\phi_0; \Lambda) = F(\mathbf{t}, \mathbf{y}; \Lambda). \tag{16.4.10}$$

This shows the assertion (i) of Theorem 16.4.

Next, observe that the right-hand side of (16.3.3) can be defined for any $\mathbf{y} \in V'$ (though (16.3.3) itself is valid only under the assumption of Proposition 16.10). Therefore, we can define $\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda)$ for any $\mathbf{y} \in V'$ as the $\left(\prod_{f\in\Lambda} -\frac{k_f!}{(2\pi\sqrt{-1})^{k_f}}\right)$ multiple of the right-hand side of (16.3.3), and we introduce the generating function of $\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda)$ of the form

$$\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda) = \sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Lambda}} \widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda) \prod_{f\in\Lambda} \frac{t_f^{k_f}}{k_f!}, \tag{16.4.11}$$

where $\mathbf{t} = (t_f)_{f\in\Lambda} \in \mathbb{C}^{\#\Lambda}$. A more explicit form of the generating function can be deduced by substituting the formula of Proposition 16.10 into (16.4.11). In fact:

LEMMA 16.27 ([**115**, Lemma 5.3]). *For any* $\mathbf{y} \in V'$, *the series on the right-hand side of* (16.4.11) *is absolutely and uniformly convergent in the neighborhood of the origin with respect to* $\mathbf{t} \in \mathbb{C}^{\#\Lambda}$. *Furthermore we have*

(16.4.12)

$$\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda) = \Big(\prod_{f\in\Lambda} \frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1}\Big) \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}$$
$$\times \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \int_0^1 \cdots \int_0^1 \prod_{g\in L_0} dx_g$$
$$\times \exp\Big(\sum_{g\in L_0} (t_g - 2\pi\sqrt{-1}\dot{g}) x_g + \sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f}) \{\mathbf{y} + \mathbf{w} - \sum_{g\in L_0} x_g \vec{g}\}'_{B_0,f}\Big).$$

PROOF. The following proof is similar to that of Lemma 4.9. By (16.3.81) we see that, for $b \in \mathbb{C}$, there exists a sufficiently small $R_b > 0$ such that

$$\frac{C(k,y;b)}{k!} = \frac{1}{2\pi\sqrt{-1}} \int_{|z|=R_b} \frac{z e^{(z-2\pi\sqrt{-1}b)y}}{e^{z-2\pi\sqrt{-1}b} - 1} \frac{dz}{z^{k+1}} \tag{16.4.13}$$

holds for $y \in \mathbb{R}$. Thus we have for $0 \le y \le 1$

$$\Big|\frac{C(k,y;b)}{k!}\Big| \le \frac{1}{2\pi} \int_{|z|=R_b} \Big|\frac{z e^{(z-2\pi\sqrt{-1}b)y}}{e^{z-2\pi\sqrt{-1}b} - 1}\Big| \frac{|dz|}{R_b^{k+1}} \le \frac{C_b}{R_b^k}, \tag{16.4.14}$$

where

$$C_b = \max\Big\{\Big|\frac{z e^{(z-2\pi\sqrt{-1}b)y}}{e^{z-2\pi\sqrt{-1}b} - 1}\Big| \;\Big|\; |z| = R_b,\ 0 \le y \le 1\Big\}. \tag{16.4.15}$$

Fix r such that $0 < r < \min_{f\in\Lambda} R_{\dot{f}}$. Then for $|t_f| < r$ $(f \in \Lambda)$,

(16.4.16)

$$\Big|\widetilde{C}(\mathbf{k},\mathbf{y};\Lambda) \prod_{f\in\Lambda} \frac{t_f^{k_f}}{k_f!}\Big|$$
$$\le \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \Big(\prod_{f\in\Lambda} |t_f|^{k_f}\Big) \int_0^1 \cdots \int_0^1 \Big|\Big(\prod_{g\in L_0} \frac{C(k_g,x_g;\dot{g})}{k_g!} dx_g\Big)$$
$$\times \prod_{f\in B_0} \Big(\frac{C(k_f, \{\mathbf{y} + \mathbf{w} - \sum_{g\in L_0} x_g \vec{g}\}'_{B_0,f}; \dot{f})}{k_f!}\Big)\Big|$$
$$\le \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]} \Big(\prod_{f\in\Lambda} r^{k_f}\Big) \int_0^1 \cdots \int_0^1 \prod_{f\in\Lambda} \frac{C_{\dot{f}}}{R_{\dot{f}}^{k_f}} \prod_{g\in L_0} dx_g$$
$$= \prod_{f\in\Lambda} C_{\dot{f}} \Big(\frac{r}{R_{\dot{f}}}\Big)^{k_f}.$$

Since

$$\sum_{\mathbf{k}\in\mathbb{N}_0^{\#\Lambda}}\prod_{f\in\Lambda}C_{\dot f}\Big(\frac{r}{R_{\dot f}}\Big)^{k_f}=\prod_{f\in\Lambda}\Big(\frac{C_{\dot f}}{1-r/R_{\dot f}}\Big)<\infty, \tag{16.4.17}$$

we have the uniform and absolute convergence of $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$, which implies the holomorphy of $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$ in the neighborhood of the origin with respect to $\mathbf{t}\in\mathbb{C}^{\#\Lambda}$.

Furthermore, by exchanging the sum and the integral and using (16.3.1) we obtain

(16.4.18)

$$\begin{aligned}\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)&=\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec B_0])}\\&\times\sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec B_0]}\int_0^1\cdots\int_0^1\Big(\prod_{g\in L_0}dx_g\Big)\Big(\prod_{g\in L_0}\frac{t_g\exp((t_g-2\pi\sqrt{-1}\dot g)x_g)}{\exp(t_g-2\pi\sqrt{-1}\dot g)-1}\Big)\\&\times\Big(\prod_{f\in B_0}\frac{t_f\exp((t_f-2\pi\sqrt{-1}\dot f)\{\mathbf{y}+\mathbf{w}-\sum_{g\in L_0}x_g\vec g\}'_{B_0,f})}{\exp(t_f-2\pi\sqrt{-1}\dot f)-1}\Big),\end{aligned}$$

which yields (16.4.12). □

Lemma 16.28 ([115, Lemma 5.4]). *$\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$ is continuous in $\mathbf{y}$ on $V'\setminus\bigcup_{f\in\widetilde\Lambda}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$ and has one-sided continuity in $\mathbf{y}\in V'$ in the direction ϕ_0.*

Proof. The proof is almost the same as that of the continuity of $S(\mathbf{k},\mathbf{y};\Lambda)$ in (16.3.91). We apply the argument to (16.4.12). In this case, the continuity comes from (16.3.93) for all $f\in B_0$. Hence the first assertion follows from Lemma 16.22 with $D=B_0$. The second assertion immediately follows from Lemma 16.26.

□

We have obtained the assertions, corresponding to (i), (ii) and (iii) of Theorem 16.4, for $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$. In the following sections, we will prove

$$\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)=F(\mathbf{t},\mathbf{y};\Lambda) \tag{16.4.19}$$

for $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$. Then $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)=F(\mathbf{t},\mathbf{y};\Lambda)$ on the whole of V' by the one-sided continuity of $F(\mathbf{t},\mathbf{y};\Lambda)$ and $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$, which we have already shown. Thus automatically the assertions (ii) and (iii) of Theorem 16.4 will follow.

Also, comparing (16.1.8) with (16.4.11), we find that

$$\widetilde{C}(\mathbf{k},\mathbf{y};\Lambda)=C(\mathbf{k},\mathbf{y};\Lambda). \tag{16.4.20}$$

By the definition of $\widetilde{C}(\mathbf{k},\mathbf{y};\Lambda)$ and Proposition 16.10, we have

$$\widetilde{C}(\mathbf{k},\mathbf{y};\Lambda)=\Big(\prod_{f\in\Lambda}-\frac{k_f!}{(2\pi\sqrt{-1})^{k_f}}\Big)S(\mathbf{k},\mathbf{y};\Lambda) \tag{16.4.21}$$

for $\mathbf{y}\in V'\setminus\bigcup_{f\in\widetilde{\Lambda}\cap\Lambda_1}(\mathfrak{H}_{\Lambda\setminus\{f\}}+\mathbb{Z}^d)$. Combining this with (16.4.20), we obtain the assertion of Theorem 16.5. Therefore the only remaining task is to show (16.4.19) for $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$.

16.5. The generating function and convex polytopes

The aim of the following three sections is to prove (16.4.19), which will be shown in Section 16.7. The present and the next sections are devoted to the preparations for the proof of (16.4.19), which are connected with the theory of convex polytopes.

First, in this section, we present the following fundamental proposition, which gives an expression of $\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)$ involving integrals over certain convex polytopes. This proposition is a generalization of Theorem 5.10.

Proposition 16.29 ([115, Proposition 6.4]).

(16.5.1)

$$\begin{aligned}\widetilde{F}(\mathbf{t},\mathbf{y};\Lambda)=&\Big(\prod_{f\in\Lambda}\frac{t_f}{\exp(t_f-2\pi\sqrt{-1}\dot{f})-1}\Big)\frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}\\&\times\sum_{\mathbf{m}\in\mathbb{Z}^d}\exp\Big(\sum_{f\in B_0}(t_f-2\pi\sqrt{-1}\dot{f})\langle\mathbf{y}+\mathbf{m},\vec{f}^{B_0}\rangle'\Big)\\&\times\int_{\mathcal{P}(\mathbf{m};\mathbf{y})}\exp\Big(\sum_{g\in L_0}t_g^*x_g\Big)\prod_{g\in L_0}dx_g,\end{aligned}$$

where

$$t_g^*=(t_g-2\pi\sqrt{-1}\dot{g})-\sum_{f\in B_0}(t_f-2\pi\sqrt{-1}\dot{f})\langle\vec{g},\vec{f}^{B_0}\rangle' \tag{16.5.2}$$

and

(16.5.3)

$$\begin{aligned}&\mathcal{P}(\mathbf{m};\mathbf{y})\\&=\left\{\mathbf{x}=(x_g)_{g\in L_0}\;\middle|\;\begin{array}{l}0\le x_g\le 1\quad(g\in L_0)\\ \langle\mathbf{y}+\mathbf{m}-\vec{f},\vec{f}^{B_0}\rangle'\le\displaystyle\sum_{g\in L_0}x_g\langle\vec{g},\vec{f}^{B_0}\rangle'\\ \qquad\qquad\le\langle\mathbf{y}+\mathbf{m},\vec{f}^{B_0}\rangle'\quad(f\in B_0)\end{array}\right\}\end{aligned}$$

is a convex polytope or an empty set.

PROOF. We fix a representative of each $\mathbf{w} \in \mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]$ in $\mathbb{Z}^d$. Let $\mathbf{m} = (m_f)_{f\in B_0} \in \mathbb{Z}^d$, and denote by $\mathcal{Q}(\mathbf{w},\mathbf{m})$ the set of all $\mathbf{x} = (x_g)_{g\in L_0}$ satisfying the conditions $0 \le x_g \le 1$ $(g \in L_0)$ and

$$-m_f \le \Big\langle \mathbf{y} + \mathbf{w} - \sum_{g\in L_0} x_g \vec{g}, \vec{f}^{B_0} \Big\rangle' < -m_f + 1. \tag{16.5.4}$$

This condition is equivalent to

$$\Big\langle \mathbf{y}+\mathbf{w}+\sum_{h\in B_0} m_h\vec{h} - \vec{f}, \vec{f}^{B_0}\Big\rangle' < \sum_{g\in L_0} x_g \langle \vec{g}, \vec{f}^{B_0}\rangle' \le \Big\langle \mathbf{y}+\mathbf{w}+\sum_{h\in B_0} m_h\vec{h}, \vec{f}^{B_0}\Big\rangle', \tag{16.5.5}$$

because $\Big\langle \sum_{h\in B_0} m_h\vec{h}, \vec{f}^{B_0}\Big\rangle' = m_f$. Also we have

$$\begin{aligned}\Big\{\Big\langle \mathbf{y}+\mathbf{w}-\sum_{g\in L_0} x_g\vec{g}, \vec{f}^{B_0}\Big\rangle'\Big\} &= \Big\langle \mathbf{y}+\mathbf{w}-\sum_{g\in L_0} x_g\vec{g}, \vec{f}^{B_0}\Big\rangle' + m_f \\ &= \Big\langle \mathbf{y}+\mathbf{w}+\sum_{h\in B_0} m_h\vec{h}, \vec{f}^{B_0}\Big\rangle' - \sum_{g\in L_0} x_g\langle \vec{g}, \vec{f}^{B_0}\rangle'.\end{aligned} \tag{16.5.6}$$

Denote the multiple integral on the right-hand side of (16.4.12) by $I(\mathbf{w})$, and divide it as

$$I(\mathbf{w}) = \sum_{\mathbf{m}\in\mathbb{Z}^d} \int_{\mathcal{Q}(\mathbf{w},\mathbf{m})} .$$

Applying (16.5.6), we obtain

$$\begin{aligned} I(\mathbf{w}) = \sum_{\mathbf{m}\in\mathbb{Z}^d} &\exp\Big(\sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f})\langle \mathbf{y}+\mathbf{w}+\sum_{h\in B_0} m_h\vec{h}, \vec{f}^{B_0}\rangle'\Big) \\ &\times \int_{\mathcal{Q}(\mathbf{w},\mathbf{m})} \exp\Big(\sum_{g\in L_0} t_g^* x_g\Big) \prod_{g\in L_0} dx_g. \end{aligned} \tag{16.5.7}$$

Note that $\mathbf{w} + \sum_{h\in B_0} m_h\vec{h}$ runs over $\mathbb{Z}^d$, when $\mathbf{w} \in \mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0]$ and $\mathbf{m} \in \mathbb{Z}^d$ run. Therefore, rewriting $\mathbf{w} + \sum_{h\in B_0} m_h\vec{h}$ as $\mathbf{m}$, we obtain the assertion of the proposition. □

REMARK 16.30. For readers' convenience, we give typical pictures of $\mathcal{P}(\mathbf{m};\mathbf{y})$ in the cases $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$ and $\mathbf{y} \notin \mathfrak{H}_{\mathscr{R}}$, which will be treated below in Lemmas 16.31, 16.32, 16.33 and 16.34.

Let $V' = \mathbb{R}^2$ $(r = 2)$. Let $\mathbf{e}_1 = \begin{pmatrix}1\\0\end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix}0\\1\end{pmatrix}$ and

$$\begin{aligned}\Lambda &= \{f_1 = (\mathbf{e}_1, \alpha_1), f_2 = (\mathbf{e}_2, \alpha_2), g = (a\mathbf{e}_1 + b\mathbf{e}_2, \alpha_g), h = (c\mathbf{e}_1 + d\mathbf{e}_2, \alpha_h)\} \\ &= B_0 \cup L_0,\end{aligned}$$

$$B_0 = \{(\mathbf{e}_1, \alpha_1), (\mathbf{e}_2, \alpha_2)\},$$
$$L_0 = \{(a\mathbf{e}_1 + b\mathbf{e}_2, \alpha_g), (c\mathbf{e}_1 + d\mathbf{e}_2, \alpha_h)\},$$
$$\mathfrak{H}_{\mathscr{R}} = (\mathbb{R}\mathbf{e}_1 + \mathbb{Z}^2) \cup (\mathbb{R}\mathbf{e}_2 + \mathbb{Z}^2) \cup (\mathbb{R}(a\mathbf{e}_1 + b\mathbf{e}_2) + \mathbb{Z}^2) \cup (\mathbb{R}(c\mathbf{e}_1 + d\mathbf{e}_2) + \mathbb{Z}^2),$$

where a, b, c, d are positive integers, which correspond to the series (16.5.8)

$$S(\mathbf{k}, \mathbf{y}; \Lambda) = \sum_{n_1, n_2} \frac{e^{2\pi\sqrt{-1}(n_1 y_1 + n_2 y_2)}}{(n_1+\alpha_1)^{k_1}(n_2+\alpha_2)^{k_2}(an_1+bn_2+\alpha_g)^{k_g}(cn_1+dn_2+\alpha_h)^{k_h}},$$

where $\alpha_1, \alpha_2, \alpha_g, \alpha_h \in \mathbb{C}$, $k_1, k_2, k_g, k_h \geq 2$ and n_1, n_2 run over all integers such that the denominator does not vanish. Then we have (16.5.9)

$$\mathcal{P}(\mathbf{m};\mathbf{y}) = \left\{ \mathbf{x} = (x_g, x_h) \;\middle|\; \begin{array}{l} 0 \leq x_g \leq 1, \\ 0 \leq x_h \leq 1, \\ y_1 + m_1 - 1 \leq ax_g + cx_h \leq y_1 + m_1, \\ y_2 + m_2 - 1 \leq bx_g + dx_h \leq y_2 + m_2 \end{array} \right\}.$$

In the case $a = b = c = 1$, $d = 2$, the polytope $\mathcal{P}(\mathbf{m};\mathbf{y})$ is drawn as in Figure 16.5.1 if $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$ and in Figure 16.5.2 if $\mathbf{y} \notin \mathfrak{H}_{\mathscr{R}}$. In the former case, there are more than 2 ($= \#\Lambda - d$) hyperplanes at some vertices, while in the latter case, there are only 2 hyperplanes at each vertex, which ensures that $\mathcal{P}(\mathbf{m};\mathbf{y})$ is a simple polytope in higher dimensions.

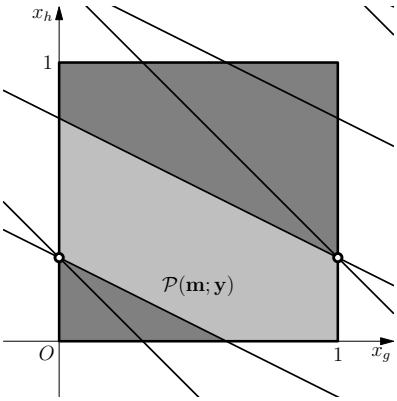


FIGURE 16.5.1. $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$

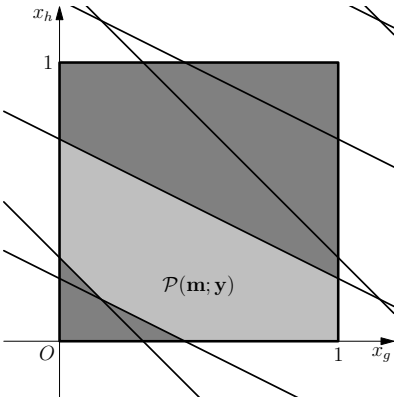


FIGURE 16.5.2. $\mathbf{y} \notin \mathfrak{H}_{\mathscr{R}}$

16.6. Properties of the polytopes $\mathcal{P}(\mathbf{m};\mathbf{y})$

The argument developed in this and the next sections is a generalization of that in [**105**, Section 6].

Let $\mathscr{A} = \{0,1\}^{\#\Lambda - d}$. Let $\mathscr{B}'$ be the set of all subsets of Λ which have r elements, and define $\mathscr{W}' = \mathscr{B}' \times \mathscr{A}$ and $\mathscr{W} = \mathscr{B} \times \mathscr{A}$. Obviously, $\mathscr{W} \subset \mathscr{W}'$. For an element $W = (B, A) \in \mathscr{W}'$, we number $A = (a_g)_{g \in \Lambda \setminus B}$. We fix a decomposition $\Lambda = B_0 \cup L_0$ with $B_0 \in \mathscr{B}$, and for $f \in \Lambda$, $a \in \{0,1\}$, $\mathbf{m} \in \mathbb{Z}^d$ and $\mathbf{y} \in V'$, we define $\mathbf{u}(f,a) \in \mathbb{R}^{\#L_0}$ by

$$\mathbf{u}(f,a)_g = \begin{cases} (-1)^{1-a}\langle \vec{g}, \vec{f}^{B_0} \rangle' & \text{if } f \in B_0, \\ (-1)^a \delta_{g,f} & \text{if } f \notin B_0, \end{cases} \tag{16.6.1}$$

where g runs over L_0 and $\delta_{g,f}$ denotes Kronecker's delta, and define $v(f,a;\mathbf{m};\mathbf{y}) \in \mathbb{R}$ by

$$v(f,a;\mathbf{m};\mathbf{y}) = \begin{cases} (-1)^{1-a}\langle \mathbf{y} + \mathbf{m} - a\vec{f}, \vec{f}^{B_0} \rangle' & \text{if } f \in B_0, \\ (-1)^a a = -a & \text{if } f \notin B_0. \end{cases} \tag{16.6.2}$$

Further, we define the hyperplanes

$$\mathcal{H}(f,a;\mathbf{m};\mathbf{y}) = \{\mathbf{x} = (x_g)_{g \in L_0} \in \mathbb{R}^{\#L_0} \mid \mathbf{u}(f,a) \cdot \mathbf{x} = v(f,a;\mathbf{m};\mathbf{y})\}, \tag{16.6.3}$$

and the half-spaces

$$\mathcal{H}^+(f,a;\mathbf{m};\mathbf{y}) = \{\mathbf{x} = (x_g)_{g \in L_0} \in \mathbb{R}^{\#L_0} \mid \mathbf{u}(f,a) \cdot \mathbf{x} \geq v(f,a;\mathbf{m};\mathbf{y})\}, \tag{16.6.4}$$

where for $\mathbf{w} = (w_g), \mathbf{x} = (x_g) \in \mathbb{R}^{\#L_0}$, we have set

$$\mathbf{w} \cdot \mathbf{x} = \sum_{g \in L_0} w_g x_g. \tag{16.6.5}$$

Then we have

$$\mathcal{P}(\mathbf{m};\mathbf{y}) = \bigcap_{\substack{f \in \Lambda \\ a \in \{0,1\}}} \mathcal{H}^+(f,a;\mathbf{m};\mathbf{y}). \tag{16.6.6}$$

LEMMA 16.31 ([115, Lemma 7.1]). *All vertices of* $\mathcal{P}(\mathbf{m};\mathbf{y})$ *are of the form*

$$\bigcap_{f \in L'} \mathcal{H}(f, a_f; \mathbf{m}; \mathbf{y}), \tag{16.6.7}$$

where $L' \subset \Lambda$ *with* $\#L' = \#L_0 = \#\Lambda - d$ *and* $(a_f)_{f \in L'} \in \mathscr{A}$.

PROOF. By Proposition 5.5, we see that any vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ is obtained as the intersection of $(\#L_0)$ hyperplanes. Since for $f \in \Lambda$, two hyperplanes $\mathcal{H}(f,a;\mathbf{m};\mathbf{y})$ $(a = 0,1)$ are parallel and hence their intersection is empty, we see that a vertex must be of the form (16.6.7). □

Since $\#L' = \#\Lambda - d$, we have $B = \Lambda \setminus L' \in \mathscr{B}'$. Therefore Lemma 16.31 implies that any vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ determines an element $(B, A) \in \mathscr{W}'$. The next lemma is a kind of converse assertion.

LEMMA 16.32 ([115, Lemma 7.2]). *Let $(B, A) \in \mathscr{W}'$ and $L' = \Lambda \setminus B$. The set*

$$\bigcap_{g \in L'} \mathcal{H}(g, a_g; \mathbf{m}; \mathbf{y}) \tag{16.6.8}$$

consists of only one point, which we denote by $\mathbf{p}(\mathbf{m};\mathbf{y};W)$, if and only if $W = (B, A) \in \mathscr{W}$.

PROOF. Let $B = \{f_1, \dots, f_d\} \in \mathscr{B}'$ and $a_f \in \{0, 1\}$ for $f \in L' = \Lambda \setminus B$. Consider the intersection of $(\#\Lambda - d)$ hyperplanes (16.6.8). Then this set consists of the solutions of the system of the $(\#L_0)$ linear equations

(16.6.9)
$$\begin{cases} \sum_{g \in L_0} x_g \langle \vec{g}, \vec{f}^{B_0} \rangle' = \langle \mathbf{y} + \mathbf{m} - a_f \vec{f}, \vec{f}^{B_0} \rangle' & \text{for } f \in B_0 \setminus B, \\ x_f = a_f & \text{for } f \in L_0 \setminus B. \end{cases}$$

The system of the linear equations (16.6.9) has a unique solution if and only if

$$\det(\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \setminus B}^{g \in B \setminus B_0} \neq 0, \tag{16.6.10}$$

and hence if and only if $B \in \mathscr{B}$, since

$$\begin{aligned} &\left| \det \begin{pmatrix} (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \setminus B}^{g \in B \setminus B_0} & (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \cap B}^{g \in B \setminus B_0} \\ (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \setminus B}^{g \in B \cap B_0} & (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \cap B}^{g \in B \cap B_0} \end{pmatrix} \right| \\ &= \left| \det \begin{pmatrix} (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \setminus B}^{g \in B \setminus B_0} & * \\ 0 & E_{\#(B \cap B_0)} \end{pmatrix} \right| \\ &= \left| \det(\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in B_0 \setminus B}^{g \in B \setminus B_0} \right|, \end{aligned} \tag{16.6.11}$$

where E_p is the $p \times p$ identity matrix. □

LEMMA 16.33 ([115, Lemma 7.3]). *Let $W = (B, A) \in \mathscr{W}$. The point $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ is a vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ if and only if*

$$\mathbf{p}(\mathbf{m};\mathbf{y};W) \in \bigcap_{\substack{f \in B \\ a \in \{0,1\}}} \mathcal{H}^+(f, a; \mathbf{m}; \mathbf{y}). \tag{16.6.12}$$

PROOF. By (16.6.6), we see that $\mathcal{P}(\mathbf{m};\mathbf{y})$ is defined by $(\#\Lambda)$ pairs of inequalities. By Proposition 5.4, the point $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ is a vertex of

$\mathcal{P}(\mathbf{m};\mathbf{y})$ if and only if all of these inequalities hold. We see that $(\#L_0)$ pairs among them are automatically satisfied, because

(16.6.13)
$$\{\mathbf{p}(\mathbf{m};\mathbf{y};W)\} = \bigcap_{g\in\Lambda\setminus B} \mathcal{H}(g,a_g;\mathbf{m};\mathbf{y}) \subset \bigcap_{\substack{g\in\Lambda\setminus B\\ a\in\{0,1\}}} \mathcal{H}^+(g,a;\mathbf{m};\mathbf{y}).$$

Therefore $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ is a vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ if and only if the remaining r pairs of inequalities are satisfied, which implies (16.6.12). □

LEMMA 16.34 ([**115**, Lemma 7.4]). *If $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$ and $\mathcal{P}(\mathbf{m};\mathbf{y})$ is not empty, then $\mathcal{P}(\mathbf{m};\mathbf{y})$ is a simple polytope.*

PROOF. By Lemmas 16.31 and 16.32, it is sufficient to check the following claim: *If for $W=(B,A)\in\mathscr{W}$, the point $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ lies on some other hyperplanes of the form (16.6.3) than the defining hyperplanes $\{\mathcal{H}(g,a_g;\mathbf{m};\mathbf{y})\}_{g\in\Lambda\setminus B}$, then $\mathbf{y}\in\mathfrak{H}_{\mathscr{R}}$.* Because this claim implies that if $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$, we can uniquely determine the $(\#L_0)$ hyperplanes on which the point $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ lies, and hence it implies the simplicity of the polytope $\mathcal{P}(\mathbf{m};\mathbf{y})$.

Since

(16.6.14)
$$\mathbf{p}(\mathbf{m};\mathbf{y};W) \notin \bigcup_{g\in\Lambda\setminus B} \mathcal{H}(g,1-a_g;\mathbf{m};\mathbf{y})$$

always holds, it is sufficient to check that

(16.6.15)
$$\mathbf{p}(\mathbf{m};\mathbf{y};W) \in \bigcup_{\substack{f\in B\\ a\in\{0,1\}}} \mathcal{H}(f,a;\mathbf{m};\mathbf{y})$$

implies $\mathbf{y}\in\mathfrak{H}_{\mathscr{R}}$.

First, we show the claim when

(16.6.16)
$$\mathbf{p}(\mathbf{m};\mathbf{y};W) \in \mathcal{H}(h,a_h;\mathbf{m};\mathbf{y})$$

holds for some $h\in B\cap B_0$ and $a_h\in\{0,1\}$. For $\mathbf{x}=\mathbf{p}(\mathbf{m};\mathbf{y};W)$, condition (16.6.16) is equivalent to

(16.6.17)
$$\sum_{g\in L_0} x_g\langle\vec{g},\vec{h}^{B_0}\rangle' = \langle\mathbf{y}+\mathbf{m}-a_h\vec{h},\vec{h}^{B_0}\rangle'.$$

Let $p=\#(B_0\setminus B)=\#(B\setminus B_0)$. Divide the left-hand side of (the first formula of) (16.6.9) and (16.6.17) into two parts according to the conditions $g\in B\setminus B_0$ and $g\in L_0\setminus B$ (noting that $L_0=(B\setminus B_0)\cup(L_0\setminus B)$). Then we obtain an overdetermined system with the p variables x_g for $g\in B\setminus B_0$ and the $(p+1)$ equations

(16.6.18)
$$\sum_{g\in B\setminus B_0} x_g\langle\vec{g},\vec{f}^{B_0}\rangle' = \langle\mathbf{y}+\mathbf{m}-a_f\vec{f},\vec{f}^{B_0}\rangle' + c_f$$

for $f \in (B_0 \setminus B) \cup \{h\}$, where

$$c_f = -\sum_{g \in L_0 \setminus B} a_g \langle \vec{g}, \vec{f}^{B_0} \rangle'. \tag{16.6.19}$$

Hence we have

$$\left((x_g)_{g \in B \setminus B_0} \quad -1\right) M(\mathbf{y}) = \begin{pmatrix} 0 & \cdots & 0 \end{pmatrix}, \tag{16.6.20}$$

where $(x_g)_{g \in B \setminus B_0}$ is a row vector and $M(\mathbf{y})$ is a $(p+1) \times (p+1)$ matrix defined by

$$M(\mathbf{y}) = \begin{pmatrix} (\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in (B_0 \setminus B) \cup \{h\}}^{g \in B \setminus B_0} \\ (\langle \mathbf{y} + \mathbf{m} - a_f \vec{f}, \vec{f}^{B_0} \rangle' + c_f)_{f \in (B_0 \setminus B) \cup \{h\}} \end{pmatrix}. \tag{16.6.21}$$

As the consistency for these equations, we get $\det M(\mathbf{y}) = 0$. We may rewrite

$$\langle \mathbf{y} + \mathbf{m} - a_f \vec{f}, \vec{f}^{B_0} \rangle' + c_f = \Big\langle \mathbf{y} + \mathbf{m} - \sum_{g \in (\Lambda \setminus B) \cup \{h\}} a_g \vec{g}, \vec{f}^{B_0} \Big\rangle', \tag{16.6.22}$$

because

$$\sum_{g \in (\Lambda \setminus B) \cup \{h\}} a_g \vec{g} - \left(\sum_{g \in L_0 \setminus B} a_g \vec{g} + a_f \vec{f} \right) = \sum_{\substack{g \in (B_0 \setminus B) \cup \{h\} \\ g \neq f}} a_g \vec{g}$$

and $\langle \vec{g}, \vec{f}^{B_0} \rangle' = 0$ for all $\vec{g}$ on the right-hand side.

Since the row vectors $(\langle \vec{g}, \vec{f}^{B_0} \rangle')_{f \in (B_0 \setminus B) \cup \{h\}}$ for $g \in B \setminus B_0$ are linearly independent, $\det M(\mathbf{y}) = 0$ implies that the last row vector is written as a linear combination of the other row vectors. That is, for $f \in (B_0 \setminus B) \cup \{h\}$ we have

$$\Big\langle \mathbf{y} + \mathbf{m} - \sum_{g \in (\Lambda \setminus B) \cup \{h\}} a_g \vec{g}, \vec{f}^{B_0} \Big\rangle' = \sum_{g \in B \setminus B_0} q_g \langle \vec{g}, \vec{f}^{B_0} \rangle', \tag{16.6.23}$$

with some $q_g \in \mathbb{R}$, and so

$$\Big\langle \mathbf{y} + \mathbf{m} - \sum_{g \in (\Lambda \setminus B) \cup \{h\}} a_g \vec{g} - \sum_{g \in B \setminus B_0} q_g \vec{g}, \vec{f}^{B_0} \Big\rangle' = 0. \tag{16.6.24}$$

Vectors which are orthogonal to all $\vec{f}^{B_0}$ ($f \in (B_0 \setminus B) \cup \{h\}$) are spanned by $\vec{g}$ ($g \in (B \cap B_0) \setminus \{h\}$). Therefore, since $\mathbf{m} - \sum_{g \in (\Lambda \setminus B) \cup \{h\}} a_g \vec{g} \in \mathbb{Z}^d$, we have

$$\mathbf{y} \in \sum_{g \in B \setminus B_0} \mathbb{R}\vec{g} + \mathbb{Z}^d + \sum_{g \in (B \cap B_0) \setminus \{h\}} \mathbb{R}\vec{g} = \sum_{g \in B \setminus \{h\}} \mathbb{R}\vec{g} + \mathbb{Z}^d \subset \mathfrak{H}_{\mathscr{R}}, \tag{16.6.25}$$

which implies the desired claim.

Next, we consider the condition $\mathbf{p}(\mathbf{m};\mathbf{y};W) \in \mathcal{H}(h,a_h;\mathbf{m};\mathbf{y})$ for some $h \in B \setminus B_0$ and $a_h \in \{0,1\}$. Then similar to the above, we see that $\mathbf{y}$ lies in $\mathfrak{H}_{\mathscr{R}}$ because

$$\det\begin{pmatrix} (\langle \vec{g}, \vec{f}^{B_0}\rangle')_{f\in B_0\setminus B}^{g\in B\setminus(B_0\cup\{h\})} \\ (\langle \mathbf{y}+\mathbf{m}-a_f\vec{f}, \vec{f}^{B_0}\rangle' + d_f)_{f\in B_0\setminus B} \end{pmatrix} = 0, \tag{16.6.26}$$

where

$$d_f = -\sum_{g\in(L_0\setminus B)\cup\{h\}} a_g\langle \vec{g}, \vec{f}^{B_0}\rangle'. \tag{16.6.27}$$

This completes the proof of the lemma. □

REMARK 16.35. We draw the picture of a vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ in the same setting as in Figure 16.5.2. For example, for $W = (B,A) \in \mathscr{B}$ with $B = \{f_2,h\} \in \mathscr{B}$ and $A = (a_{f_1},a_g) = (1,0)$ $(\Lambda\setminus B = \{f_1,g\})$, the associated vertex is $\mathbf{p}(\mathbf{m};\mathbf{y};W) = \mathbf{p}(\mathbf{m};\mathbf{y};(\{f_2,h\},(1,0)))$, which is uniquely determined by the hyperplanes $\mathcal{H}(f_1,1;\mathbf{m};\mathbf{y})$ and $\mathcal{H}(g,0;\mathbf{m};\mathbf{y})$. See Figure 16.6.1.

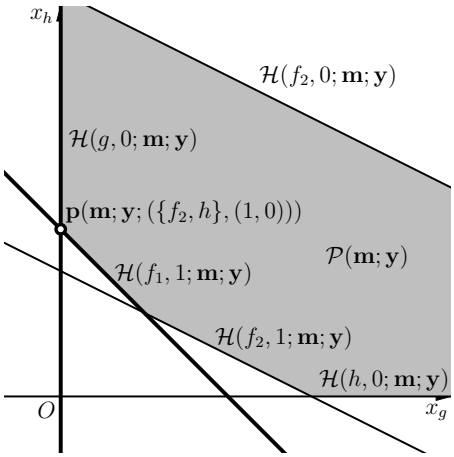


FIGURE 16.6.1. Vertices and hyperplanes

LEMMA 16.36 ([115, Lemma 7.6]). *Let $\mathbf{y} \in V' \setminus \mathfrak{H}_{\mathscr{R}}$ and $W \in \mathscr{W}$. Then we have*

$$\sum_{f\in B_0}(t_f - 2\pi\sqrt{-1}\dot{f})\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' + \mathbf{t}^*\cdot\mathbf{p}(\mathbf{m};\mathbf{y};W) \tag{16.6.28}$$

$$= \sum_{g\in\Lambda\setminus B}(t_g - 2\pi\sqrt{-1}\dot{g})a_g + \sum_{f\in B}(t_f - 2\pi\sqrt{-1}\dot{f})\Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B}\Big\rangle',$$

where $\mathbf{t}^* = (t_g^*)_{g\in L_0}$ with t_g^* defined by (16.5.2).

PROOF. By (16.6.8), the point $\mathbf{p}(\mathbf{m};\mathbf{y};W) = (x_g)_{g\in L_0}$ satisfies

(16.6.29)
$$\begin{cases} \sum_{g\in L_0} x_g\langle \vec{g}, \vec{f}^{B_0}\rangle' = \langle \mathbf{y}+\mathbf{m}-a_f\vec{f}, \vec{f}^{B_0}\rangle' & \text{for } f\in B_0\setminus B,\\ x_f = a_f & \text{for } f\in L_0\setminus B. \end{cases}$$

For $f\in B_0\setminus B$, the equations are rewritten as

(16.6.30)
$$\sum_{h\in B\setminus B_0} x_h\langle \vec{h}, \vec{f}^{B_0}\rangle' = \langle \mathbf{y}+\mathbf{m}-a_f\vec{f}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0\setminus B} a_g\langle \vec{g}, \vec{f}^{B_0}\rangle' = \Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle'.$$

By Lemma 16.32, the system of these equations has a unique solution. We show the explicit solution is given by

(16.6.31)
$$x_h = \begin{cases} \Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{h}^B\Big\rangle' & \text{for } h\in B\setminus B_0,\\ x_h = a_h & \text{for } h\in L_0\setminus B. \end{cases}$$

First, note that for $f\in B_0\cap B$ we have

(16.6.32)
$$\vec{f}^B = \vec{f}^{B_0} - \sum_{h\in B\setminus B_0} \vec{h}^B\langle \vec{h}, \vec{f}^{B_0}\rangle'.$$

In fact, since

(16.6.33)
$$\mathbf{z} = \sum_{h\in B} \vec{h}^B\langle \vec{h}, \mathbf{z}\rangle'$$

holds for any $\mathbf{z}\in V'$, we have

(16.6.34)
$$\vec{f}^{B_0} = \sum_{h\in B\cap B_0} \vec{h}^B\langle \vec{h}, \vec{f}^{B_0}\rangle' + \sum_{h\in B\setminus B_0} \vec{h}^B\langle \vec{h}, \vec{f}^{B_0}\rangle' = \vec{f}^B + \sum_{h\in B\setminus B_0} \vec{h}^B\langle \vec{h}, \vec{f}^{B_0}\rangle',$$

which is (16.6.32).

Using (16.6.33), for $f\in B_0\setminus B$, we have the left-hand side of (16.6.30) as

(16.6.35)
$$\begin{aligned} &\sum_{h\in B\setminus B_0} \Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{h}^B\Big\rangle'\langle \vec{h}, \vec{f}^{B_0}\rangle' \\ &\quad= \sum_{h\in B} \Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{h}^B\Big\rangle'\langle \vec{h}, \vec{f}^{B_0}\rangle' \\ &\quad= \Big\langle \mathbf{y}+\mathbf{m}-\sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle', \end{aligned}$$

which yields the right-hand side of (16.6.30). Thus we see that (16.6.31) is indeed the solution.

Noting $L_0 = (L_0 \setminus B) \cup (B \setminus B_0)$, we have

$$
\begin{aligned}
(16.6.36)\quad & \sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f})\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' + \mathbf{t}^* \cdot \mathbf{p}(\mathbf{m};\mathbf{y};W) \\
&= \sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f})\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' \\
&\quad + \sum_{g\in L_0} \Big((t_g - 2\pi\sqrt{-1}\dot{g}) - \sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f})\langle \vec{g}, \vec{f}^{B_0}\rangle' \Big) x_g \\
&= \sum_{g\in L_0} (t_g - 2\pi\sqrt{-1}\dot{g}) x_g \\
&\quad + \sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f}) \Big(\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0} x_g \langle \vec{g}, \vec{f}^{B_0}\rangle' \Big) \\
&= \sum_{g\in L_0\setminus B} (t_g - 2\pi\sqrt{-1}\dot{g}) a_g + \sum_{h\in B\setminus B_0} (t_h - 2\pi\sqrt{-1}\dot{h}) x_h \\
&\quad + \sum_{f\in B_0\cap B} (t_f - 2\pi\sqrt{-1}\dot{f}) \\
&\qquad \times \Big(\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0\setminus B} a_g \langle \vec{g}, \vec{f}^{B_0}\rangle' - \sum_{h\in B\setminus B_0} x_h \langle \vec{h}, \vec{f}^{B_0}\rangle' \Big) \\
&\quad + \sum_{f\in B_0\setminus B} (t_f - 2\pi\sqrt{-1}\dot{f}) \\
&\qquad \times \Big(\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0\setminus B} a_g \langle \vec{g}, \vec{f}^{B_0}\rangle' - \sum_{h\in B\setminus B_0} x_h \langle \vec{h}, \vec{f}^{B_0}\rangle' \Big).
\end{aligned}
$$

By the first equality of (16.6.30), the last term on the last member of (16.6.36) is equal to

$$
\begin{aligned}
(16.6.37)\quad & \sum_{f\in B_0\setminus B} (t_f - 2\pi\sqrt{-1}\dot{f}) \Big(\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0\setminus B} a_g \langle \vec{g}, \vec{f}^{B_0}\rangle' - \sum_{h\in B\setminus B_0} x_h \langle \vec{h}, \vec{f}^{B_0}\rangle' \Big) \\
&= \sum_{f\in B_0\setminus B} (t_f - 2\pi\sqrt{-1}\dot{f}) \langle a_f \vec{f}, \vec{f}^{B_0}\rangle' = \sum_{f\in B_0\setminus B} (t_f - 2\pi\sqrt{-1}\dot{f}) a_f.
\end{aligned}
$$

On the other hand, for $f \in B_0 \cap B$, we have

$$
(16.6.38)\quad \langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' - \sum_{g\in L_0\setminus B} a_g \langle \vec{g}, \vec{f}^{B_0}\rangle' - \sum_{h\in B\setminus B_0} x_h \langle \vec{h}, \vec{f}^{B_0}\rangle'
$$

$$
\begin{aligned}
&= \Big\langle \mathbf{y}+\mathbf{m}- \sum_{g\in L_0\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle' - \sum_{h\in B\setminus B_0} \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{h}^{B}\Big\rangle' \langle \vec{h}, \vec{f}^{B_0}\rangle' \\
&= \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0} - \sum_{h\in B\setminus B_0} \vec{h}^{B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\Big\rangle' \\
&= \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B}\Big\rangle'
\end{aligned}
$$

by (16.6.31) and (16.6.32). Therefore we finally obtain (16.6.28). □

LEMMA 16.37 ([**115**, Lemma 7.7]). *Let $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$ and $W\in\mathscr{W}$. Then the point $\mathbf{p}(\mathbf{m};\mathbf{y};W)$ is a vertex of $\mathcal{P}(\mathbf{m};\mathbf{y})$ if and only if*

$$
0 \le \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B}\Big\rangle' \le 1 \tag{16.6.39}
$$

for all $f\in B$.

PROOF. By Lemma 16.33, the point $\mathbf{p}(\mathbf{m};\mathbf{y};W)=(x_g)_{g\in L_0}$ is indeed a vertex if and only if

(16.6.40)
$$
\begin{cases}
\langle \mathbf{y}+\mathbf{m}-\vec{f}, \vec{f}^{B_0}\rangle' \le \sum_{g\in L_0} x_g\langle \vec{g}, \vec{f}^{B_0}\rangle' \le \langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle', & (f\in B\cap B_0),\\
0\le x_f\le 1, & (f\in B\setminus B_0).
\end{cases}
$$

For $f\in B\cap B_0$, applying (16.6.29) (the second equality) and (16.6.31), we have

(16.6.41)
$$
\begin{aligned}
\sum_{g\in L_0} x_g\langle \vec{g}, \vec{f}^{B_0}\rangle' &= \sum_{g\in L_0\setminus B} x_g\langle \vec{g}, \vec{f}^{B_0}\rangle' + \sum_{h\in B\setminus B_0} x_h\langle \vec{h}, \vec{f}^{B_0}\rangle' \\
&= \Big\langle \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle' + \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \sum_{h\in B\setminus B_0} \vec{h}^{B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\Big\rangle'.
\end{aligned}
$$

Therefore, noting (16.6.32), we see that the first pair of inequalities of (16.6.40) is

$$
\begin{aligned}
\Big\langle \mathbf{y}+\mathbf{m}-\vec{f} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle' &\le \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0} - \vec{f}^{B}\Big\rangle' \\
&\le \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B_0}\Big\rangle',
\end{aligned}
$$

which is equivalent to

$$
0 \le \Big\langle \mathbf{y}+\mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g\vec{g}, \vec{f}^{B}\Big\rangle' \le 1. \tag{16.6.42}
$$

For $f \in B \setminus B_0$, noting (16.6.31) we see that the second pair of inequalities of (16.6.40) is again of the same form as (16.6.42). Therefore the desired assertion follows. □

Fix $W = (B, A) \in \mathscr{W}$. Let U be the $(\#L_0) \times (\#L_0)$ matrix whose f-th column consists of $\mathbf{u}(f, a_f)$ for $f \in \Lambda \setminus B$ and $U(h, \mathbf{v})$ be the same as the matrix U but with the h-th column replaced by $\mathbf{v}$. Then we have the following two lemmas.

LEMMA 16.38 ([115, Lemma 7.8]).

$$|\det U| = \frac{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}. \tag{16.6.43}$$

PROOF. By rearranging rows and columns we see that

$$\begin{aligned} |\det U| &= \left|\det(\mathbf{u}(f,a_f)_g)_{f\in\Lambda\setminus B}^{g\in L_0}\right| \\ &= \left|\det\begin{pmatrix} (\mathbf{u}(f,a_f)_g)_{f\in B_0\setminus B}^{g\in B\setminus B_0} & (\mathbf{u}(f,a_f)_g)_{f\in L_0\setminus B}^{g\in B\setminus B_0} \\ (\mathbf{u}(f,a_f)_g)_{f\in B_0\setminus B}^{g\in L_0\setminus B} & (\mathbf{u}(f,a_f)_g)_{f\in L_0\setminus B}^{g\in L_0\setminus B} \end{pmatrix}\right| \\ &= \left|\det\begin{pmatrix} (\langle \vec{g}, \vec{f}^{B_0}\rangle')_{f\in B_0\setminus B}^{g\in B\setminus B_0} & 0 \\ * & E_{\#(L_0\setminus B)} \end{pmatrix}\right| \\ &= \left|\det(\langle \vec{g}, \vec{f}^{B_0}\rangle')_{f\in B_0\setminus B}^{g\in B\setminus B_0}\right| \\ &= \left|\det(\langle \vec{g}, \vec{f}^{B_0}\rangle')_{f\in B_0}^{g\in B}\right| \end{aligned} \tag{16.6.44}$$

by (16.6.11). Further

$$\begin{aligned} \left|\det(\langle \vec{g}, \vec{f}^{B_0}\rangle')_{f\in B_0}^{g\in B}\right| &= \left|\det(\vec{g})_{g\in B}\det(\vec{f}^{B_0})_{f\in B_0}\right| \\ &= \frac{\left|\det(\vec{g})_{g\in B}\right|}{\left|\det(\vec{f})_{f\in B_0}\right|} = \frac{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])}. \end{aligned} \tag{16.6.45}$$

□

LEMMA 16.39 ([115, Lemma 7.9]). *For $f \in \Lambda \setminus B$, we have*

$$\frac{\det U(f, \mathbf{t}^*)}{\det U} = (-1)^{a_f}\Big((t_f - 2\pi\sqrt{-1}\dot{f}) - \sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{f}, \vec{g}^B\rangle'\Big). \tag{16.6.46}$$

PROOF. We show that $\mathbf{x} = (x_f)_{f\in\Lambda\setminus B}$ defined by

$$x_f = (-1)^{a_f}\Big((t_f - 2\pi\sqrt{-1}\dot{f}) - \sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{f}, \vec{g}^B\rangle'\Big) \tag{16.6.47}$$

is a unique solution of the linear equation

$$U\mathbf{x} = \mathbf{t}^*. \tag{16.6.48}$$

Then the statement follows from Cramer's rule.

Now we show that (16.6.47) satisfies (16.6.48). First, observe

(16.6.49)
$$\begin{aligned}(U\mathbf{x})_h = &\sum_{f\in B_0\setminus B}(-1)^{1-a_f}\langle \vec{h}, \vec{f}^{B_0}\rangle'(-1)^{a_f}\\ &\qquad\times\Big((t_f-2\pi\sqrt{-1}\dot{f})-\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{f},\vec{g}^{B}\rangle'\Big)\\ &+\sum_{f\in L_0\setminus B}(-1)^{a_f}\delta_{h,f}(-1)^{a_f}\\ &\qquad\times\Big((t_f-2\pi\sqrt{-1}\dot{f})-\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{f},\vec{g}^{B}\rangle'\Big)\\ =&-\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\Big((t_f-2\pi\sqrt{-1}\dot{f})-\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{f},\vec{g}^{B}\rangle'\Big)\\ &+\sum_{f\in L_0\setminus B}\delta_{h,f}\Big((t_h-2\pi\sqrt{-1}\dot{h})-\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{h},\vec{g}^{B}\rangle'\Big).\end{aligned}$$

Assume $h\in B\setminus B_0$. Then $\delta_{h,f}=0$ for all $f\in L_0\setminus B$, and hence

(16.6.50)
$$\begin{aligned}(U\mathbf{x})_h =&-\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\Big((t_f-2\pi\sqrt{-1}\dot{f})-\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{f},\vec{g}^{B}\rangle'\Big)\\ =&-\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'(t_f-2\pi\sqrt{-1}\dot{f})\\ &+\sum_{f\in B_0\setminus B}\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f},\vec{g}^{B}\rangle'.\end{aligned}$$

From (16.6.33) we have

$$\mathbf{z}=\sum_{f\in B_0\setminus B}\vec{f}^{B_0}\langle \vec{f},\mathbf{z}\rangle'+\sum_{f\in B_0\cap B}\vec{f}^{B_0}\langle \vec{f},\mathbf{z}\rangle',$$

and hence

$$\langle \vec{h},\mathbf{z}\rangle'=\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f},\mathbf{z}\rangle'+\sum_{f\in B_0\cap B}\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f},\mathbf{z}\rangle'. \tag{16.6.51}$$

Putting $\mathbf{z}=\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\vec{g}^{B}$ in (16.6.51), we obtain

(16.6.52)
$$\begin{aligned}&\sum_{f\in B_0\setminus B}\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f},\vec{g}^{B}\rangle'\\ &=\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\delta_{h,g}-\sum_{f\in B_0\cap B}\sum_{g\in B}(t_g-2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f},\vec{g}^{B}\rangle'.\end{aligned}$$

Substituting this into the right-hand side of (16.6.50), we obtain

(16.6.53)
$$\begin{aligned}(U\mathbf{x})_h &= -\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'(t_f - 2\pi\sqrt{-1}\dot{f}) + (t_h - 2\pi\sqrt{-1}\dot{h})\\ &\quad - \sum_{f\in B_0\cap B}\sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f}, \vec{g}^{B}\rangle'\\ &= (t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'(t_f - 2\pi\sqrt{-1}\dot{f})\\ &\quad - \sum_{f\in B_0\cap B}(t_f - 2\pi\sqrt{-1}\dot{f})\langle \vec{h}, \vec{f}^{B_0}\rangle' = t_h^*.\end{aligned}$$

Assume $h \in L_0 \setminus B$. Then again using (16.6.51), we have

(16.6.54)
$$\begin{aligned}(U\mathbf{x})_h &= -\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'(t_f - 2\pi\sqrt{-1}\dot{f})\\ &\quad + \sum_{f\in B_0\setminus B}\sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f}, \vec{g}^{B}\rangle'\\ &\quad + (t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{g}^{B}\rangle'\\ &= -\sum_{f\in B_0\setminus B}\langle \vec{h}, \vec{f}^{B_0}\rangle'(t_f - 2\pi\sqrt{-1}\dot{f})\\ &\quad - \sum_{f\in B_0\cap B}\sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{f}^{B_0}\rangle'\langle \vec{f}, \vec{g}^{B}\rangle'\\ &\quad + (t_h - 2\pi\sqrt{-1}\dot{h})\\ &= t_h^*.\end{aligned}$$

□

16.7. Completion of the proof of Theorems 16.4 and 16.5

In this section we prove (16.4.19) to complete the proof of Theorems 16.4 and 16.5. First, we show an elementary lemma.

Let $n \in \mathbb{N}$ and let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Let $M = (m_{ij})_{1\le j\le n}^{1\le i\le n}$ be an $n \times n$ matrix. For $1 \le j \le n$, and $\mathbf{v} \in \mathbb{R}^n$, let $M(j, \mathbf{v})$ be the same as the matrix M but with the j-th column replaced by $\mathbf{v}$. Let

(16.7.1)
$$\begin{aligned}\mathbf{u}(j) &= \det M(j, (\mathbf{e}_i)_{1\le i\le n})\\ &= \det\begin{pmatrix} m_{11} & \dots & m_{1\,j-1} & \mathbf{e}_1 & m_{1\,j+1} & \dots & m_{1n}\\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots\\ m_{n1} & \dots & m_{n\,j-1} & \mathbf{e}_n & m_{n\,j+1} & \dots & m_{nn}\end{pmatrix}\end{aligned}$$

$$= \sum_{i=1}^{n} b_{ij}\mathbf{e}_i \in \mathbb{R}^n,$$

where b_{ij} is the cofactor of (i,j)-entry of M. The following lemma (properties of cofactor matrices) is shown by elementary linear algebra, and we omit the proof.

LEMMA 16.40 ([115, Lemma 8.1]). (1) *For $\mathbf{v} \in \mathbb{R}^n$, we have*

$$\mathbf{u}(j) \cdot \mathbf{v} = \det M(j, \mathbf{v}). \tag{16.7.2}$$

(2) *Let $U = (\mathbf{u}(j))_{1\le j\le n}$ be the $n \times n$ matrix whose j-th column consists of $\mathbf{u}(j)$. Then*

$$\det U = (\det M)^{n-1}. \tag{16.7.3}$$

Now we start the proof of (16.4.19), that is, $F(\mathbf{t}, \mathbf{y}; \Lambda) = \widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda)$ for $\mathbf{y} \in V' \setminus \mathfrak{H}_{\mathscr{R}}$.

Since all the polytopes $\mathcal{P}(\mathbf{m}; \mathbf{y})$ for $\mathbf{y} \in V' \setminus \mathfrak{H}_{\mathscr{R}}$ are empty or simple by Lemma 16.34, we may apply Lemma 5.7 to the right-hand side of Proposition 16.29. In the present case the vertices are of the form $\mathbf{p}(\mathbf{m}; \mathbf{y}; W)$, satisfying (16.6.39) by Lemma 16.37. Therefore we have (16.7.4)

$$\begin{aligned}\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda) = {} & \Big(\prod_{f\in\Lambda} \frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1}\Big) \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}_0])} \sum_{\mathbf{m}\in\mathbb{Z}^d} \sum_{W} \\ & \times \exp\Big(\sum_{f\in B_0} (t_f - 2\pi\sqrt{-1}\dot{f})\langle \mathbf{y}+\mathbf{m}, \vec{f}^{B_0}\rangle' + \mathbf{t}^* \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W)\Big) \\ & \times \frac{|\det\big(\mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W')\big)_{W'\in E(\mathbf{m};W)}|}{\prod_{W'\in E(\mathbf{m};W)} \mathbf{t}^* \cdot \big(\mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W')\big)},\end{aligned}$$

where $E(\mathbf{m}; W)$ is the set of all indices W' such that

$$\mathrm{Conv}(\{\mathbf{p}(\mathbf{m}; \mathbf{y}; W), \mathbf{p}(\mathbf{m}; \mathbf{y}; W')\})$$

is an edge of $\mathcal{P}(\mathbf{m}; \mathbf{y})$, and for each $\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{Z}^d$, $W = (B, A) \in \mathscr{W}$ runs over those satisfying

$$0 \le \Big\langle \mathbf{y} + \mathbf{m} - \sum_{g\in\Lambda\setminus B} a_g \vec{g}, \vec{f}^{B} \Big\rangle' \le 1. \tag{16.7.5}$$

Recall that a vertex $\mathbf{p}(\mathbf{m}; \mathbf{y}; W)$ satisfies $(\#\Lambda - d)$ equations of the form

$$\mathbf{u}(f, a_f) \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W) = v(f, a_f; \mathbf{m}; \mathbf{y}) \tag{16.7.6}$$

for $f \in \Lambda \setminus B$ with $W = (B, A)$ (see (16.6.13)). For $W' = (B', A') \in E(\mathbf{m}; W)$, we see that the two distinct vertices $\mathbf{p}(\mathbf{m}; \mathbf{y}; W)$ and $\mathbf{p}(\mathbf{m}; \mathbf{y};$

W') share common $(\#\Lambda - d - 1)$ hyperplanes, that is, there exists $h \in \Lambda \setminus B$ such that $\Lambda \setminus (B \cup \{h\}) \subset \Lambda \setminus B'$ and $a_f = a'_f$ for $f \in \Lambda \setminus (B \cup \{h\})$, which implies

$$\begin{aligned} \mathbf{u}(f, a_f) \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W) &= v(f, a_f; \mathbf{m}; \mathbf{y}), \\ \mathbf{u}(f, a_f) \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W') &= v(f, a_f; \mathbf{m}; \mathbf{y}) \end{aligned} \tag{16.7.7}$$

for $f \in \Lambda \setminus (B \cup \{h\})$ and

$$\begin{aligned} \mathbf{u}(h, a_h) \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W) &= v(h, a_h; \mathbf{m}; \mathbf{y}), \\ \mathbf{u}(h, a_h) \cdot \mathbf{p}(\mathbf{m}; \mathbf{y}; W') &> v(h, a_h; \mathbf{m}; \mathbf{y}). \end{aligned} \tag{16.7.8}$$

This h is unique because otherwise we have $\mathbf{p}(\mathbf{m}; \mathbf{y}; W) = \mathbf{p}(\mathbf{m}; \mathbf{y}; W')$. Since $\#E(\mathbf{m}; W) = \#(\Lambda \setminus B) = \#\Lambda - d$ (because $\mathcal{P}(\mathbf{m}; \mathbf{y})$ is simple), we find that there is a one-to-one correspondence between $E(\mathbf{m}; W)$ and $\Lambda \setminus B$. By (16.7.7), we see that the set of the equations and the inequality with respect to $\mathbf{v}$

$$\begin{cases} \mathbf{u}(f, a_f) \cdot \mathbf{v} = 0 & (f \in \Lambda \setminus (B \cup \{h\})), \\ \mathbf{u}(h, a_h) \cdot \mathbf{v} < 0 \end{cases} \tag{16.7.9}$$

has a solution $\mathbf{v} = \mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W')$.

We construct another vector $\mathbf{e}(\mathbf{m}; W, W')$ satisfying (16.7.9) so that

$$\mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W') = c(\mathbf{m}; \mathbf{y}; W, W')\mathbf{e}(\mathbf{m}; W, W'), \tag{16.7.10}$$

where $c(\mathbf{m}; \mathbf{y}; W, W') > 0$. Then it follows that

$$\begin{aligned} &\frac{|\det(\mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W'))_{W' \in E(\mathbf{m}; W)}|}{\prod_{W' \in E(\mathbf{m}; W)} \mathbf{t}^* \cdot \big(\mathbf{p}(\mathbf{m}; \mathbf{y}; W) - \mathbf{p}(\mathbf{m}; \mathbf{y}; W')\big)} \\ &\quad = \frac{\big|\prod_{W' \in E(\mathbf{m}; W)} c(\mathbf{m}; \mathbf{y}; W, W')\big| |\det(\mathbf{e}(\mathbf{m}; W, W'))_{W' \in E(\mathbf{m}; W)}|}{\prod_{W' \in E(\mathbf{m}; W)} c(\mathbf{m}; \mathbf{y}; W, W')\mathbf{t}^* \cdot \mathbf{e}(\mathbf{m}; W, W')} \\ &\quad = \frac{|\det(\mathbf{e}(\mathbf{m}; W, W'))_{W' \in E(\mathbf{m}; W)}|}{\prod_{W' \in E(\mathbf{m}; W)} \mathbf{t}^* \cdot \mathbf{e}(\mathbf{m}; W, W')}. \end{aligned} \tag{16.7.11}$$

The construction of $\mathbf{e}(\mathbf{m}; W, W')$ is as follows. Let $\mathbf{e}_g$ for $g \in L_0$ be the standard orthonormal basis of $\mathbb{R}^{\#L_0}$. Let U be the $(\#\Lambda - d) \times (\#\Lambda - d)$ matrix whose f-th column consists of $\mathbf{u}(f, a_f)$ for $f \in \Lambda \setminus B$ with $W = (B, A)$. For $h \in \Lambda \setminus B$, let $U(h, \mathbf{v})$ be the matrix U with only the h-th column replaced by $\mathbf{v}$. Note that $\det U \neq 0$ by Lemma 16.38. Define

(16.7.12)

$$\mathbf{e}(\mathbf{m}; W, W') = -(\operatorname{sgn} \det U) \det U(h, (\mathbf{e}_g)_{g \in L_0}) = -(\operatorname{sgn} \det U) \sum_{g \in L_0} b_{gh} \mathbf{e}_g$$

(the second equality is due to (16.7.1)), where $W' = (B', A')$ such that $\Lambda \setminus (B \cup \{h\}) \subset \Lambda \setminus B'$ and b_{gh} is the cofactor of (g, h)-entry of U. Then by Lemma 16.40 (1), we have

(16.7.13)

$$\begin{aligned}\mathbf{e}(\mathbf{m}; W, W') \cdot \mathbf{u}(f, a_f) &= -(\operatorname{sgn}\det U)\det U\big(h, \mathbf{u}(f, a_f)\big) = 0,\\ \mathbf{e}(\mathbf{m}; W, W') \cdot \mathbf{u}(h, a_h) &= -(\operatorname{sgn}\det U)\det U\big(h, \mathbf{u}(h, a_h)\big) = -(\operatorname{sgn}\det U)\det U\\ &= -|\det U| < 0\end{aligned}$$

for $f \in \Lambda \setminus (B \cup \{h\})$ as required.

We observed that h runs over $\Lambda \setminus B$ when W' runs over $E(\mathbf{m}; W)$. Therefore by Lemma 16.40 (2) we see that (16.7.12) implies

$$|\det(\mathbf{e}(\mathbf{m}; W, W'))_{W' \in E(\mathbf{m};W)}| = |\det U|^{\#\Lambda - d - 1}. \tag{16.7.14}$$

Also, from (16.7.12) we have

$$\mathbf{t}^* \cdot \mathbf{e}(\mathbf{m}; W, W') = -(\operatorname{sgn}\det U)\sum_{g \in L_0} b_{gh} t_g^* = -(\operatorname{sgn}\det U)\det U(h, \mathbf{t}^*).$$

Therefore

(16.7.15)

$$\begin{aligned}\frac{|\det(\mathbf{e}(\mathbf{m}; W, W'))_{W' \in E(\mathbf{m};W)}|}{\prod_{W' \in E(\mathbf{m};W)} \mathbf{t}^* \cdot \mathbf{e}(\mathbf{m}; W, W')} &= (-1)^{\#\Lambda - d}(\operatorname{sgn}\det U)^{\#\Lambda - d}\frac{|\det U|^{\#\Lambda - d - 1}}{\prod_{h \in \Lambda \setminus B} \det U(h, \mathbf{t}^*)}\\ &= (-1)^{\#\Lambda - d}\frac{1}{|\det U|}\frac{(\det U)^{\#\Lambda - d}}{\prod_{h \in \Lambda \setminus B} \det U(h, \mathbf{t}^*)}.\end{aligned}$$

Substituting this into the right-hand side of (16.7.4) and using Lemmas 16.36, 16.38 and 16.39, we have

(16.7.16)

$$\begin{aligned}\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda) &= (-1)^{\#\Lambda - d}\Big(\prod_{f \in \Lambda} \frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1}\Big)\sum_{\mathbf{m} \in \mathbb{Z}^d}\sum_W \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])}\\ &\times \exp\Big(\sum_{g \in \Lambda \setminus B}(t_g - 2\pi\sqrt{-1}\dot{g})a_g + \sum_{f \in B}(t_f - 2\pi\sqrt{-1}\dot{f})\Big\langle \mathbf{y} + \mathbf{m} - \sum_{g \in \Lambda \setminus B} a_g \vec{g}, \vec{f}^B \Big\rangle'\Big)\\ &\times \prod_{h \in \Lambda \setminus B} \frac{(-1)^{a_h}}{(t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{g \in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{g}^B \rangle'}.\end{aligned}$$

We rewrite the double sum on the first line of the above so as to exchange the order of the sums with respect to $W \in \mathscr{W}$ and $\mathbf{m} \in \mathbb{Z}^d$. For each $W \in \mathscr{W}$, we see that

$$\mathbf{m} - \sum_{g \in \Lambda \setminus B} a_g \vec{g} \tag{16.7.17}$$

runs over $\mathbb{Z}^d$ when $\mathbf{m}$ runs over $\mathbb{Z}^d$. Thus (16.7.5) can be rewritten in terms of $\mathbf{v} \in \mathbb{Z}^d$, that is, $\mathbf{v}$ runs over those satisfying

$$0 \le \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' \le 1 \tag{16.7.18}$$

for all $f \in B$. If there exist $f \in B$ and $\mathbf{v} \in \mathbb{Z}^d$ such that $c = \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' \in \mathbb{Z}$, then we can write $\mathbf{y} + \mathbf{v} = \sum_{g \in R} c_g \vec{g} + c\vec{f}$, where $R = B \setminus \{f\} \in \mathscr{R}$ and $c_g \in \mathbb{R}$. Therefore we have $\mathbf{y} + \mathbf{v} - c\vec{f} \in \mathfrak{H}_R$, hence $\mathbf{y} \in \mathfrak{H}_{\mathscr{R}}$, which contradicts with the assumption. Thus condition (16.7.18) can be replaced by

$$0 < \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' < 1. \tag{16.7.19}$$

Let $G = \{\mathbf{v} \in \mathbb{Z}^d \mid 0 < \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' < 1 \text{ for all } f \in B\}$. We show that the natural projection $g : G \to \mathbb{Z}^d / \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$ is bijective. If $g(\mathbf{v}) = g(\mathbf{v}')$ for $\mathbf{v}, \mathbf{v}' \in G$, then $\mathbf{v} = \mathbf{v}' + \mathbf{x}$ for some $\mathbf{x} \in \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$. Since $\langle \mathbf{x}, \vec{f}^B \rangle' = \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' - \langle \mathbf{y} + \mathbf{v}', \vec{f}^B \rangle'$, we have $-1 < \langle \mathbf{x}, \vec{f}^B \rangle' < 1$ for all $f \in B$, which implies $\mathbf{x} = 0$. Conversely, for any $\mathbf{v} \in \mathbb{Z}^d$, putting $\mathbf{x} = \sum_{g \in B} c_g \vec{g} \in \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$ with $c_g = -[\langle \mathbf{y} + \mathbf{v}, \vec{g}^B \rangle'] \in \mathbb{Z}$, we have

$$\langle \mathbf{y} + \mathbf{v} + \mathbf{x}, \vec{f}^B \rangle' = \langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle' + c_f = \{\langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle'\}$$

and so $0 < \langle \mathbf{y} + \mathbf{v} + \mathbf{x}, \vec{f}^B \rangle' < 1$ because $\mathbf{y} \notin \mathfrak{H}_{\mathscr{R}}$. This implies the assertion. Hence replacing $\langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle'$ by $\{\langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle'\}$, we see that $\mathbf{v}$ runs over all representatives of $\mathbb{Z}^d / \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$.

Therefore by exchanging the order of the sums with respect to $W = (B, A) \in \mathscr{W} = \mathscr{B} \times \mathscr{A}$ and $\mathbf{m} \in \mathbb{Z}^d$, and summing with respect to $\mathbf{v}$, we have

(16.7.20)

$$\begin{aligned}
\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda) &= (-1)^{\#\Lambda - d} \Big(\prod_{f \in \Lambda} \frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1} \Big) \sum_{W \in \mathscr{W}} \frac{1}{\#(\mathbb{Z}^d / \mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \\
&\quad \times \sum_{\mathbf{v} \in \mathbb{Z}^d / \mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \exp\Big(\sum_{g \in \Lambda \setminus B} (t_g - 2\pi\sqrt{-1}\dot{g}) a_g \\
&\qquad\qquad + \sum_{f \in B} (t_f - 2\pi\sqrt{-1}\dot{f}) \{\langle \mathbf{y} + \mathbf{v}, \vec{f}^B \rangle'\} \Big) \\
&\quad \times \prod_{h \in \Lambda \setminus B} \frac{(-1)^{a_h}}{(t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{g \in B} (t_g - 2\pi\sqrt{-1}\dot{g}) \langle \vec{h}, \vec{g}^B \rangle'} \\
&= (-1)^{\#\Lambda - d} \Big(\prod_{f \in \Lambda} \frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1} \Big) \\
&\quad \times \sum_{B \in \mathscr{B}} \sum_{A \in \mathscr{A}} \Big(\prod_{h \in \Lambda \setminus B} (-1)^{a_h} \exp((t_h - 2\pi\sqrt{-1}\dot{h}) a_h) \Big)
\end{aligned}$$

$$\times \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{v}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \exp\Big(\sum_{f\in B}(t_f - 2\pi\sqrt{-1}\dot{f})\{\langle \mathbf{y}+\mathbf{v}, \vec{f}^B\rangle'\}\Big)$$
$$\times \prod_{h\in\Lambda\setminus B} \frac{1}{(t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{g}^B\rangle'}.$$

Since

(16.7.21)
$$(-1)^{a_h}\exp((t_h - 2\pi\sqrt{-1}\dot{h})a_h) = \begin{cases} -\exp(t_h - 2\pi\sqrt{-1}\dot{h}) & \text{if } a_h = 1,\\ 1 & \text{if } a_h = 0,\end{cases}$$

we have

(16.7.22)
$$\begin{aligned}\sum_{A\in\mathscr{A}}\Big(\prod_{h\in\Lambda\setminus B}(-1)^{a_h}\exp((t_h - 2\pi\sqrt{-1}\dot{h})a_h)\Big)&\\ = \prod_{h\in\Lambda\setminus B}(1 - \exp(t_h - 2\pi\sqrt{-1}\dot{h}))&\\ = (-1)^{\#\Lambda - d}\prod_{h\in\Lambda\setminus B}(\exp(t_h - 2\pi\sqrt{-1}\dot{h}) - 1).&\end{aligned}$$

Therefore the rightmost side of (16.7.20) is finally equal to

(16.7.23)
$$\Big(\prod_{f\in\Lambda}\frac{t_f}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1}\Big)\sum_{B\in\mathscr{B}}\Big(\prod_{h\in\Lambda\setminus B}(\exp(t_h - 2\pi\sqrt{-1}\dot{h}) - 1)\Big)$$
$$\times \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{v}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} \exp\Big(\sum_{f\in B}(t_f - 2\pi\sqrt{-1}\dot{f})\{\langle \mathbf{y}+\mathbf{v}, \vec{f}^B\rangle'\}\Big)$$
$$\times \prod_{h\in\Lambda\setminus B} \frac{1}{(t_h - 2\pi\sqrt{-1}\dot{h}) - \sum_{g\in B}(t_g - 2\pi\sqrt{-1}\dot{g})\langle \vec{h}, \vec{g}^B\rangle'},$$

and coincides with (16.1.6) for $\mathbf{y}\in V'\setminus\mathfrak{H}_{\mathscr{R}}$. This completes the proof of (16.4.19), and hence the proof of Theorems 16.4 and 16.5.

16.8. A hierarchy and differential equations

This section is a little digression. We give a theorem which asserts that the family of our generating functions has a hierarchy. Let

(16.8.1) $$\mathbf{\Lambda}_d = \{\Lambda\subset(\mathbb{Z}^d\setminus\{0\})\times\mathbb{C} \mid \#\Lambda<\infty, \operatorname{rank}\mathrm{Lh}_{\mathbb{Z}}[\vec{\Lambda}] = d\}.$$

For $\mathbf{v} = (v_1,\dots,v_d)\in V'$, let[5]

(16.8.2) $$\partial'_{\mathbf{v}} = v_1\partial_{y_1} + \cdots + v_d\partial_{y_d},$$

[5]This $\partial'_{\mathbf{v}}$ is similar to $\partial_{\mathbf{v}}$ defined by (9.5.1). In fact, if we identify V to $\mathbb{R}^r$ by $\mathbf{v} = \sum_{j=1}^r v_j\alpha_j^\vee$ and $\mathbf{y} = \sum_{j=1}^r y_j\alpha_j^\vee$, then $\partial_{\mathbf{v}} = v_1\partial_{y_1} + \cdots + v_r\partial_{y_r}$.

where ∂_{y_j} is the j-th partial differential operator acting on $\mathbf{y} = (y_j)_{1\le j\le d} \in V'$. For $g = (\vec{g}, \dot{g}) \in (\mathbb{Z}^d \setminus \{0\}) \times \mathbb{C}$, define

$$D_g = \frac{t_g - 2\pi\sqrt{-1}\dot{g}}{t_g} - \frac{1}{t_g}\partial'_{\vec{g}}. \tag{16.8.3}$$

THEOREM 16.41 ([115, Theorem 9.1]). *Let $\Lambda, \Lambda' \in \mathbf{\Lambda}_d$ with $\Lambda' \subset \Lambda$, and $\mathbf{t} = (t_g)_{g\in\Lambda}$, $\mathbf{t}' = (t_g)_{g\in\Lambda'}$. We have*

$$\Big(\prod_{g\in\Lambda\setminus\Lambda'} D_g\Big) F(\mathbf{t}, \mathbf{y}; \Lambda) = F(\mathbf{t}', \mathbf{y}; \Lambda'), \tag{16.8.4}$$

where on the left-hand side D_g is understood to act at $\mathbf{y} \in V' \setminus \mathfrak{H}_{\mathscr{R}(\Lambda)}$ and the resulting function, to be continuously extended by the one-sided limit along ϕ_0.

PROOF. It is sufficient to show the assertion in the case $\Lambda\setminus\Lambda' = \{g\}$ and $\mathbf{y} \in V' \setminus \mathfrak{H}_{\mathscr{R}(\Lambda)}$. For $B \in \mathscr{B}(\Lambda)$ and $\mathbf{w} \in \mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]$, we put

$$F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda) = \Big(\prod_{h\in\Lambda\setminus B} K(\mathbf{t}, h)^{-1}\Big)\Big(\prod_{f\in B} \frac{t_f \exp((t_f - 2\pi\sqrt{-1}\dot{f})\{\mathbf{y}+\mathbf{w}\}'_{B,f})}{\exp(t_f - 2\pi\sqrt{-1}\dot{f}) - 1}\Big), \tag{16.8.5}$$

where for $h \in \Lambda$,

$$K(\mathbf{t}, h) = \frac{t_h - 2\pi\sqrt{-1}\dot{h} - \sum_{f\in B}(t_f - 2\pi\sqrt{-1}\dot{f})\langle\vec{h}, \vec{f}^B\rangle'}{t_h}, \tag{16.8.6}$$

so that

$$F(\mathbf{t}, \mathbf{y}; \Lambda) = \sum_{B\in\mathscr{B}(\Lambda)} \frac{1}{\#(\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}])} \sum_{\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]} F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda). \tag{16.8.7}$$

By simple computations we obtain

$$D_g F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda) = K(\mathbf{t}, g) F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda). \tag{16.8.8}$$

If $g \in \Lambda \setminus B$, then the factor $K(\mathbf{t}, g)$ cancels with the factor $K(\mathbf{t}, g)^{-1}$ appearing on the right-hand side of (16.8.5), and so the variable t_g disappears. Thus we have

$$D_g F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda) = F_{B,\mathbf{w}}(\mathbf{t}', \mathbf{y}; \Lambda \setminus \{g\}). \tag{16.8.9}$$

In this case $B \in \mathscr{B}(\Lambda')$.

If $g \in B$, then

$$\sum_{f\in B}(t_f - 2\pi\sqrt{-1}\dot{f})\langle\vec{g}, \vec{f}^B\rangle' = t_g - 2\pi\sqrt{-1}\dot{g} \tag{16.8.10}$$

and hence $K(\mathbf{t}, g) = 0$ and

$$D_g F_{B,\mathbf{w}}(\mathbf{t}, \mathbf{y}; \Lambda) = 0. \tag{16.8.11}$$

Thus the sum runs over all $\mathscr{B}(\Lambda')$ and

$$D_g F(\mathbf{t}, \mathbf{y}; \Lambda) = F(\mathbf{t}', \mathbf{y}; \Lambda \setminus \{g\}). \tag{16.8.12}$$

□

16.9. Proof of Theorems 9.10, 9.11 and 9.12

Now we proceed to prove fundamental results stated in Chapter 9. In this section we prove Theorems 9.10, 9.11 and 9.12.

Let Λ be as defined in Section 16.1, and $\mathbf{k} = (k_f)_{f\in\Lambda} \in \mathbb{N}^{\#\Lambda}$. (Note that $k_f \in \mathbb{N}_0$ in Section 16.1, but here we assume $k_f \geq 1$.) Then $\Lambda = \bigcup_{k\geq 1} \Lambda_k(\mathbf{k})$. Assume

$$\widetilde{\Lambda} \cap \Lambda_1(\mathbf{k}) = \emptyset. \tag{16.9.1}$$

For the above $\mathbf{k}$ and $\mathbf{y}' \in V' = \mathbb{R}^d$, (16.1.1) and (16.1.2) can be read as

$$S(\mathbf{k}, \mathbf{y}'; \Lambda) = \lim_{N\to\infty} \sum_{\substack{\mathbf{u}=(u_1,\dots,u_d)\in\mathbb{Z}^d \\ |u_j|\leq N\ (1\leq j\leq d) \\ f(\mathbf{u})\neq 0\ (f\in\Lambda)}} e^{2\pi\sqrt{-1}\langle \mathbf{y}', \mathbf{u}\rangle'} \prod_{f\in\Lambda} \frac{1}{f(\mathbf{u})^{k_f}}. \tag{16.9.2}$$

Under the assumption (16.9.1), by Theorem 16.2, the right-hand side of (16.9.2) converges for any $\mathbf{y}' \in V'$.

Theorem 16.5 asserts

$$S(\mathbf{k}, \mathbf{y}'; \Lambda) = \left(\prod_{f\in\Lambda} -\frac{(2\pi\sqrt{-1})^{k_f}}{k_f!} \right) C(\mathbf{k}, \mathbf{y}'; \Lambda), \tag{16.9.3}$$

where $C(\mathbf{k}, \mathbf{y}'; \Lambda)$ is defined by (16.1.8). This is the key formula for the proof of Theorem 9.11.

Proof of Theorem 9.11. For $\lambda \in P_{I+}$ and $\mu \in P_{I^c}$, the relation $\lambda + \mu \notin H_{\Delta_+^\vee} = H_{\Delta_{I+}^\vee} \cup H_{\Delta^{(I)\vee}}$ is equivalent to

$$\lambda \notin H_{\Delta_{I+}^\vee}, \qquad \text{and} \qquad \lambda + \mu \notin H_{\Delta^{(I)\vee}}. \tag{16.9.4}$$

In fact, if $\lambda + \mu \notin H_{\Delta_+^\vee}$, then $\langle \alpha^\vee, \lambda + \mu\rangle \neq 0$ for $\alpha \in \Delta_{I+}$. This implies $\langle \alpha^\vee, \lambda\rangle \neq 0$ since $\langle \alpha^\vee, \mu\rangle = 0$, and so $\lambda \notin H_{\Delta_{I+}^\vee}$. The converse implication is similar.

Now, let $\mathbf{s} = (s_\alpha)$ be as in the statement of Theorem 9.11; that is, $\mathbf{s} \in \mathcal{S}$ and $s_\alpha = k_\alpha \in \mathbb{N}$ for $\alpha \in \Delta^{(I)}$. Then for $\mathbf{y} \in V$, $S(\mathbf{s}, \mathbf{y}; I; \Delta)$ defined by (9.4.1) is absolutely convergent (by Theorem 9.7), and by (16.9.4) it can be written as

$$S(\mathbf{s}, \mathbf{y}; I; \Delta) \tag{16.9.5}$$

$$= \sum_{\lambda\in P_{I+}} \sum_{\substack{\mu\in P_{I^c}\\ \lambda+\mu\notin H_{\Delta_+^\vee}}} \left(\prod_{\alpha\in\Delta_{I+}} \frac{1}{\langle\alpha^\vee,\lambda+\mu\rangle^{s_\alpha}}\right) e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle}$$
$$\times \left(e^{2\pi\sqrt{-1}\langle\mathbf{y},\mu\rangle} \prod_{\alpha\in\Delta^{(I)}} \frac{1}{\langle\alpha^\vee,\lambda+\mu\rangle^{k_\alpha}}\right)$$
$$= \sum_{\lambda\in P_{I++}} \left(\prod_{\alpha\in\Delta_{I+}} \frac{1}{\langle\alpha^\vee,\lambda\rangle^{s_\alpha}}\right) e^{2\pi\sqrt{-1}\langle\mathbf{y},\lambda\rangle} S_\lambda(\mathbf{k},\mathbf{y};I;\Delta),$$

where

(16.9.6)
$$S_\lambda(\mathbf{k},\mathbf{y};I;\Delta) = \sum_{\substack{\mu\in P_{I^c}\\ \lambda+\mu\notin H_{\Delta^{(I)\vee}}}} \left(e^{2\pi\sqrt{-1}\langle\mathbf{y},\mu\rangle} \prod_{\alpha\in\Delta^{(I)}} \frac{1}{\langle\alpha^\vee,\lambda+\mu\rangle^{k_\alpha}}\right).$$

We apply the results in Section 16.1 to find the generating function of $S_\lambda(\mathbf{k},\mathbf{y};I;\Delta)$. For this purpose, we need to identify each symbol appearing here and in Section 16.1. We identify $V' = \mathbb{R}^d \simeq \bigoplus_{i\in I^c} \mathbb{R}\alpha_i^\vee$ ($d = \#I^c$) and through this identification, we define the projection $\pi_{I^c} : V \to V'$ by

$$\pi_{I^c}(v) = (\langle v,\lambda_i\rangle)_{i\in I^c} \quad (v\in V). \tag{16.9.7}$$

For $\alpha\in\Delta^{(I)}$, we set the affine linear form (depending on λ)

$$f_\alpha(\mu) = \langle\alpha^\vee,\mu\rangle + \langle\alpha^\vee,\lambda\rangle \tag{16.9.8}$$

for $\mu\in\bigoplus_{i\in I^c}\mathbb{R}\lambda_i$.

We prove several necessary claims. First, we note:

Claim 16.42. *f_α is naturally regarded as an affine linear form on V' via $V'\simeq\bigoplus_{i\in I^c}\mathbb{R}\lambda_i$.*

In fact, for $\alpha^\vee = \sum_{i\in I^c} a_i\alpha_i^\vee$ and $\mu = \sum_{i\in I^c} b_i\lambda_i$, we have

(16.9.9)
$$\langle\alpha^\vee,\mu\rangle = \sum_{i\in I^c} a_ib_i = \langle(a_i)_{i\in I^c},(b_i)_{i\in I^c}\rangle' = \langle(\langle\alpha^\vee,\lambda_i\rangle)_{i\in I^c},(b_i)_{i\in I^c}\rangle'$$
$$= \langle\pi_{I^c}(\alpha^\vee),(b_i)_{i\in I^c}\rangle',$$

hence the claim.

Thus we may choose

$$\Lambda = \Lambda(\lambda) = \{f_\alpha = (\vec{f}_\alpha,\dot{f}_\alpha) \mid \alpha\in\Delta^{(I)}\} \subset (\mathbb{Z}^d\setminus\{0\})\times\mathbb{Z}, \tag{16.9.10}$$

where $\vec{f}_\alpha = \pi_{I^c}(\alpha^\vee)$ and $\dot{f}_\alpha = \langle\alpha^\vee,\lambda\rangle$. Here we show:

CLAIM 16.43. *The pair* $(\Lambda, \mathbf{k})$, *where* Λ *is given by* (16.9.10) *and* $\mathbf{k} = (k_\alpha)_{\alpha\in\Delta^{(I)}}$ *is as in the statement of Theorem 9.11, satisfies* (16.9.1). (*Therefore the convergence of* (16.9.2) *is established for this pair* $(\Lambda, \mathbf{k})$.)

In fact, if $I = \{1, 2, \dots, r\}$, then $\Delta^{(I)} = \emptyset$, so $\Lambda = \emptyset$ and the claim is clear. Let $I \neq \{1, 2, \dots, r\}$. If Δ is of type A_1, the claim follows from the assumption $\mathbf{s} \in \mathcal{S}$ of Theorem 9.11. If Δ is not of type A_1, we assert that for any $\alpha \in \Delta^{(I)}$, the set $\Delta^{(I)\vee} \setminus \{\alpha^\vee\}$ spans all linear functionals on P_{I^c}. Since $\Psi_{I^c}^\vee$ spans all linear functionals, the assertion is clear if $\alpha \notin \Psi_{I^c}$. When $\alpha = \alpha_j \in \Psi_{I^c}$, then there exists $\beta^\vee (\neq \alpha_j^\vee) \in \Delta^{(I)\vee}$ which includes $\alpha_j^\vee$ in its linear combination expression in terms of fundamental coroots (because, if not, then all coroots in $\Delta^\vee$ except $\pm\alpha_j^\vee$ do not include $\alpha_j^\vee$ in their linear combination expressions, which contradicts the assumption that Δ is irreducible). Then $(\Delta^{(I)\vee} \setminus \{\alpha^\vee\}) \cup \beta^\vee$ spans all linear functionals. The claim is proved.

Next, let $C \subset \Delta^{(I)}$ with $\#C = d$ and $B = \{f_\alpha \mid \alpha \in C\} \subset \Lambda$. Thirdly, we show:

CLAIM 16.44. $B \in \mathscr{B}(\Lambda)$ *if and only if* $\mathbf{V} = C \cup \Psi_I \in \mathscr{V}_I$.

Since the matrix of the pairings $\langle \beta^\vee, \lambda_i\rangle$ for $\beta \in C \cup \Psi_I$ and $1 \le i \le r$ is written as the form

$$\begin{pmatrix} (\langle \beta^\vee, \lambda_i\rangle)_{\beta\in C, i\in I^c} & (\langle \beta^\vee, \lambda_i\rangle)_{\beta\in C, i\in I} \\ (\langle \beta^\vee, \lambda_i\rangle)_{\beta\in \Psi_I, i\in I^c} & (\langle \beta^\vee, \lambda_i\rangle)_{\beta\in \Psi_I, i\in I} \end{pmatrix} = \begin{pmatrix} A & * \\ 0 & \mathrm{id}_{\#I} \end{pmatrix}, \tag{16.9.11}$$

it is sufficient to show that that A is regular if and only if $B \in \mathscr{B}(\Lambda)$. The definition of $\mathscr{B}(\Lambda)$ implies that $B \in \mathscr{B}(\Lambda)$ if and only if

$$A' = (\langle \vec{f}_\alpha, e_i\rangle')_{\alpha\in C, i\in I^c} \tag{16.9.12}$$

is a regular matrix, where $\{e_i\}_{i\in I^c}$ is the standard basis of $\mathbb{R}^d$. Here by (16.9.9), we have

$$\langle \vec{f}_\alpha, e_i\rangle' = \langle \pi_{I^c}(\alpha^\vee), e_i\rangle' = \langle \alpha^\vee, \lambda_i\rangle \tag{16.9.13}$$

and so A' is rewritten as

$$A' = (\langle \alpha^\vee, \lambda_i\rangle)_{\alpha\in C, i\in I^c} = A. \tag{16.9.14}$$

Thus we showed the desired claim.

When these equivalent conditions hold, we write $C = \mathbf{V}_I$, which agrees with the notation in Section 9.4. In this case we have

$$Q^\vee/L(\mathbf{V}^\vee) = \Big(\bigoplus_{i=1}^r \mathbb{Z}\alpha_i^\vee\Big)/\Big(\bigoplus_{\beta\in\mathbf{V}_I} \mathbb{Z}\beta^\vee \oplus \bigoplus_{i\in I} \mathbb{Z}\alpha_i^\vee\Big) \tag{16.9.15}$$

$$\simeq \pi_{I^c}\Big(\bigoplus_{i\in I^c}\mathbb{Z}\alpha_i^\vee\Big)/\pi_{I^c}\Big(\bigoplus_{\beta\in\mathbf{V}_I}\mathbb{Z}\beta^\vee\Big).$$

Since $\pi_{I^c}(\beta^\vee)=\vec{f}_\beta$, we have $\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]\simeq Q^\vee/L(\mathbf{V}^\vee)$.

The next claim is:

CLAIM 16.45. *$\langle\vec{f}_\gamma,\vec{f}^{\,B}_\alpha\rangle'=\langle\gamma^\vee,\mu^{\mathbf{V}}_\alpha\rangle$ for $\gamma\in\Delta^{(I)}$, $B=\{f_\alpha\mid\alpha\in\mathbf{V}_I\}\in\mathscr{B}(\Lambda)$ and $\alpha\in\mathbf{V}_I$.*

To prove this claim, we show that for any $v\in V$,

$$\langle\pi_{I^c}(v),\vec{f}^{\,B}_\alpha\rangle'=\langle v,\mu^{\mathbf{V}}_\alpha\rangle \tag{16.9.16}$$

holds. Since $\mathbf{V}=\mathbf{V}_I\cup\Psi_I\in\mathscr{V}_I$ by Claim 16.44, we can write

$$v=\sum_{\beta\in\Psi_I}b_\beta\beta^\vee+\sum_{\beta\in\mathbf{V}_I}c_\beta\beta^\vee. \tag{16.9.17}$$

Then

$$\pi_{I^c}(v)=\sum_{\beta\in\mathbf{V}_I}c_\beta\pi_{I^c}(\beta^\vee). \tag{16.9.18}$$

Since $\langle\pi_{I^c}(\beta^\vee),\vec{f}^{\,B}_\alpha\rangle'=\langle\vec{f}_\beta,\vec{f}^{\,B}_\alpha\rangle'=\delta_{\beta,\alpha}$ for $\beta\in\mathbf{V}_I$, we have

$$\langle\pi_{I^c}(v),\vec{f}^{\,B}_\alpha\rangle'=\sum_{\beta\in\mathbf{V}_I}c_\beta\langle\pi_{I^c}(\beta^\vee),\vec{f}^{\,B}_\alpha\rangle'=c_\alpha=\langle v,\mu^{\mathbf{V}}_\alpha\rangle, \tag{16.9.19}$$

which is (16.9.16). Therefore $\langle\vec{f}_\gamma,\vec{f}^{\,B}_\alpha\rangle'=\langle\pi_{I^c}(\gamma^\vee),\vec{f}^{\,B}_\alpha\rangle'=\langle\gamma^\vee,\mu^{\mathbf{V}}_\alpha\rangle$ as desired.

From the above arguments, we see that (16.1.6) under the choice (16.9.10) now reads as

(16.9.20)

$$\begin{aligned}
&F(\mathbf{t},\mathbf{y}';\Lambda(\lambda))\\
&=\sum_{\mathbf{V}\in\mathscr{V}_I}\left(\prod_{\gamma\in\Delta^{(I)}\setminus\mathbf{V}_I}\frac{t_\gamma}{t_\gamma-2\pi\sqrt{-1}\langle\gamma^\vee,\lambda\rangle-\sum_{\beta\in\mathbf{V}_I}(t_\beta-2\pi\sqrt{-1}\langle\beta^\vee,\lambda\rangle)\langle\gamma^\vee,\mu^{\mathbf{V}}_\beta\rangle}\right)\\
&\times\frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))}\sum_{q\in Q^\vee/L(\mathbf{V}^\vee)}\prod_{\beta\in\mathbf{V}_I}\frac{t_\beta\exp((t_\beta-2\pi\sqrt{-1}\langle\beta^\vee,\lambda\rangle)\{\mathbf{y}'+q\}'_{B,f})}{\exp(t_\beta-2\pi\sqrt{-1}\langle\beta^\vee,\lambda\rangle)-1}
\end{aligned}$$

for $\mathbf{y}'\in V'$. In fact, there are the following one-to-one correspondences:

$$\begin{aligned}
B\in\mathscr{B}(\Lambda)&\longleftrightarrow\mathbf{V}\in\mathscr{V}_I,\\
g\in\Lambda\setminus B&\longleftrightarrow\gamma\in\Delta^{(I)}\setminus\mathbf{V}_I,\\
f\in B&\longleftrightarrow\beta\in\mathbf{V}_I,\\
\mathbf{w}\in\mathbb{Z}^d/\mathrm{Lh}_{\mathbb{Z}}[\vec{B}]&\longleftrightarrow q\in Q^\vee/L(\mathbf{V}^\vee),
\end{aligned}$$

hence $\langle \vec{g}, \vec{f}^B \rangle'$ corresponds to $\langle \vec{f}_\gamma, \vec{f}^B_\beta \rangle'$, the latter being equal to $\langle \gamma^\vee, \mu^{\mathbf{V}}_\beta \rangle$ as we have already seen.

Now choose $\mathbf{y}' = \pi_{I^c}(\mathbf{y})$. Then from (16.9.9) we see that

$$\langle \mathbf{y}, \mu \rangle = \langle \pi_{I^c}(\mathbf{y}), (b_i)_{i \in I^c} \rangle' = \langle \mathbf{y}', (b_i)_{i \in I^c} \rangle'. \tag{16.9.21}$$

Therefore comparing (16.9.2) and (16.9.6) we find that

$$S_\lambda(\mathbf{k}, \mathbf{y}; I; \Delta) = S(\mathbf{k}, \pi_{I^c}(\mathbf{y}); \Lambda(\lambda)). \tag{16.9.22}$$

Next, consider some factors on the right-hand side of (16.9.20). First,

$$\begin{aligned} &t_\gamma - 2\pi\sqrt{-1}\langle \gamma^\vee, \lambda \rangle - \sum_{\beta \in \mathbf{V}_I} (t_\beta - 2\pi\sqrt{-1}\langle \beta^\vee, \lambda \rangle)\langle \gamma^\vee, \mu^{\mathbf{V}}_\beta \rangle \\ &= t_\gamma - \sum_{\beta \in \mathbf{V}_I} t_\beta \langle \gamma^\vee, \mu^{\mathbf{V}}_\beta \rangle - 2\pi\sqrt{-1}\Big(\langle \gamma^\vee, \lambda \rangle - \sum_{\beta \in \mathbf{V}_I} \langle \beta^\vee, \lambda \rangle \langle \gamma^\vee, \mu^{\mathbf{V}}_\beta \rangle \Big) \\ &= t_\gamma - \sum_{\beta \in \mathbf{V}_I} t_\beta \langle \gamma^\vee, \mu^{\mathbf{V}}_\beta \rangle - 2\pi\sqrt{-1}\langle \gamma^\vee, p_{\mathbf{V}_I^\perp}(\lambda) \rangle \end{aligned} \tag{16.9.23}$$

by (9.4.8). Secondly, by (16.9.16) we have $\{\pi_{I^c}(\mathbf{y}) + q\}'_{B,f} = \{\mathbf{y} + q\}_{\mathbf{V},\beta}$, and

$$\begin{aligned} &\exp\Big(-2\pi\sqrt{-1} \sum_{\beta \in \mathbf{V}_I} \langle \beta^\vee, \lambda \rangle \{\mathbf{y} + q\}_{\mathbf{V},\beta} \Big) \\ &\quad = \exp\Big(-2\pi\sqrt{-1} \sum_{\beta \in \mathbf{V}_I} \langle \beta^\vee, \lambda \rangle \langle \mathbf{y} + q, \mu^{\mathbf{V}}_\beta \rangle \Big) \\ &\quad = \exp(-2\pi\sqrt{-1}\langle \mathbf{y} + q, \lambda \rangle) \exp\Big(2\pi\sqrt{-1}\langle \mathbf{y} + q, p_{\mathbf{V}_I^\perp}(\lambda) \rangle \Big) \\ &\quad = \exp(-2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle) \exp\Big(2\pi\sqrt{-1}\langle \mathbf{y} + q, p_{\mathbf{V}_I^\perp}(\lambda) \rangle \Big) \end{aligned} \tag{16.9.24}$$

by (9.4.8) again and the facts $\langle \beta^\vee, \lambda \rangle, \langle q, \lambda \rangle \in \mathbb{Z}$. (To show the first equality we notice (4.3.4).) Collecting these facts and comparing with (9.4.10), we obtain[6]

$$F(\mathbf{t}, \pi_{I^c}(\mathbf{y}); \Lambda(\lambda)) = \exp(-2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle) F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta) \tag{16.9.25}$$

and hence

$$C(\mathbf{k}, \pi_{I^c}(\mathbf{y}); \Lambda(\lambda)) = \exp(-2\pi\sqrt{-1}\langle \mathbf{y}, \lambda \rangle) P(\mathbf{k}, \mathbf{y}, \lambda; I; \Delta). \tag{16.9.26}$$

Therefore by (16.9.3) and (16.9.22) we have

$$S_\lambda(\mathbf{k}, \mathbf{y}; I; \Delta) = \left(\prod_{\alpha \in \Delta^{(I)}} -\frac{(2\pi\sqrt{-1})^{k_\alpha}}{k_\alpha!} \right)$$

[6]Here, Λ on the left-hand side of (16.9.25) and (16.9.26) is misprinted as λ in [**119**].

$$\times \exp(-2\pi\sqrt{-1}\langle \mathbf{y}, \lambda\rangle) P(\mathbf{k}, \mathbf{y}, \lambda; I; \Delta).$$

Substituting this into (16.9.5), we arrive at the assertion of the theorem. □

Now we can also prove Theorem 9.10. In fact, if Δ is not of type A_1, then $\widetilde{\Lambda}$ is empty (see the proof of Claim 16.43). Therefore Theorem 9.10 immediately follows from Theorem 16.4 and the above (16.9.25).

Proof of Theorem 9.12. Consider the case $I = \emptyset$, and hence $\lambda = 0$. Then $\mathscr{V}_I = \mathscr{V}$, $\Delta^{(I)} = \Delta_+$, $\mathbf{V}_I = \mathbf{V}$, $\mathbf{y}' = \mathbf{y}$, so from (9.4.13) and (16.9.20) we find that $F(\mathbf{t}, \mathbf{y}, 0; \emptyset; \Delta) = F(\mathbf{t}, \mathbf{y}; \Lambda(0))$. Combining this with (16.4.19), we obtain

$$F(\mathbf{t}, \mathbf{y}, 0; \emptyset; \Delta) = \widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda(0)). \tag{16.9.27}$$

We see from the definition (16.4.11) that $\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda(0))$ is the generating function of $\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda(0))$. On the other hand, (4.2.11) shows that $F(\mathbf{t}, \mathbf{y}; \Delta)$ is the generating function of $P(\mathbf{k}, \mathbf{y}; \Delta)$. Therefore, if we can show

$$\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda(0)) = P(\mathbf{k}, \mathbf{y}; \Delta), \tag{16.9.28}$$

then it follows that $\widetilde{F}(\mathbf{t}, \mathbf{y}; \Lambda(0)) = F(\mathbf{t}, \mathbf{y}; \Delta)$. This and (16.9.27) imply the assertion of the theorem.

In view of (16.4.21), we find that the explicit form of $\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda(0))$ is given by the right-hand side of (16.3.3) (with removal of the factor $\prod_{f\in\Lambda}(-(2\pi\sqrt{-1})^{k_f}/k_f!)$).

We choose $B_0 = \{f_\alpha \mid \alpha \in \Psi\}$ in Proposition 16.10. Then by (16.9.7) we see that $\vec{B}_0$ is the standard basis of $V' = \mathbb{R}^r$, so $B_0 \in \mathscr{B}$. Under this choice of B_0, we have $\mathbb{Z}^r/\mathrm{Lh}_{\mathbb{Z}}[B_0] = \{0\}$, and hence from (16.3.3) we obtain

$$\widetilde{C}(\mathbf{k}, \mathbf{y}; \Lambda(0)) = \int_0^1 \cdots \int_0^1 \Big(\prod_{g\in L_0} C(k_g, x_g; 0) dx_g \Big) \times \prod_{f\in B_0} \Big(C(k_f, \{\mathbf{y} - \sum_{g\in L_0} x_g \vec{g}\}'_{B_0,f}; 0) \Big). \tag{16.9.29}$$

Noting (16.3.2), we see that the right-hand side of (16.9.29) coincides with the right-hand side of (4.2.2). Therefore we obtain (16.9.28), and hence the theorem. □

16.10. Proof of Theorem 9.14

In this final section we prove Theorem 9.14. In what follows, let $\mathrm{Lh}_{\mathbb{R}}[A]$ denote the $\mathbb{R}$-span of A. Let $d(I) = \#\Delta_{I+}$ for brevity. Fix an

order $\Delta_{I+} = \{\beta_1, \beta_2, \dots, \beta_{d(I)}\}$ and put $A_j = \{\beta_1, \dots, \beta_j\}$ for $0 \le j \le d(I)$. Note that

$$\emptyset = A_0 \subset A_1 \subset \cdots \subset A_{d(I)} = \Delta_{I+}. \tag{16.10.1}$$

Fix $\mathbf{V} \in \mathscr{V}$ and $q \in Q^\vee/L(\mathbf{V}^\vee)$. Consider the following condition for j:

$$\beta_k \in \mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{k-1}] \text{ for all } k \le j. \tag{16.10.2}$$

Under this condition it follows that, in the sequence of nondecreasing linear spaces

$$\{0\} = \mathrm{Lh}_{\mathbb{R}}[A_0] \subset \mathrm{Lh}_{\mathbb{R}}[A_1] \subset \cdots \subset \mathrm{Lh}_{\mathbb{R}}[A_{d(I)}] = \mathrm{Lh}_{\mathbb{R}}[\Delta_{I+}], \tag{16.10.3}$$

if $\mathrm{Lh}_{\mathbb{R}}[A_{k-1}] \subsetneq \mathrm{Lh}_{\mathbb{R}}[A_k]$, then $\beta_k \notin \mathrm{Lh}_{\mathbb{R}}[A_{k-1}]$ and hence $\beta_k \in \mathbf{V}$.

Let $\mathbf{t}_j = (t_\gamma)_{\gamma \in \Delta_+ \setminus A_j}$ (hence $\mathbf{t}_0 = \mathbf{t}$), $\mathbf{t}_{j,\mathbf{V}} = (t_\beta)_{\beta \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_j]}$, and

$$S_{j,\gamma}(t_\gamma, \mathbf{t}_{j,\mathbf{V}}) = t_\gamma - \sum_{\beta \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_j]} t_\beta \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle - 2\pi\sqrt{-1} \sum_{\beta \in \mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_j]} \langle \beta^\vee, \lambda \rangle \langle \gamma^\vee, \mu_\beta^{\mathbf{V}} \rangle.$$

For j satisfying (16.10.2), define

$$\begin{aligned} f_j(\mathbf{t}_j; \mathbf{V}, q) &= \left(\prod_{\gamma \in \Delta_+ \setminus (\mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_j])} \frac{t_\gamma}{S_{j,\gamma}(t_\gamma, \mathbf{t}_{j,\mathbf{V}})} \right) \\ &\quad \times \left(\prod_{\gamma \in (\Delta_+ \cap \mathrm{Lh}_{\mathbb{R}}[A_j]) \setminus (\mathbf{V} \cup A_j)} \frac{t_\gamma}{t_\gamma - 2\pi\sqrt{-1}\langle \gamma^\vee, \lambda \rangle} \right) \\ &\quad \times \exp\left(2\pi\sqrt{-1} \sum_{\beta \in \mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_j]} \langle \beta^\vee, \lambda \rangle \langle \mathbf{y} + q, \mu_\beta^{\mathbf{V}} \rangle \right) \\ &\quad \times \left(\prod_{\gamma \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_j]} \frac{t_\gamma \exp(t_\gamma \{\mathbf{y} + q\}_{\mathbf{V},\gamma})}{e^{t_\gamma} - 1} \right) \\ &= f_{j1} \times f_{j2} \times f_{j3} \times f_{j4}, \end{aligned} \tag{16.10.4}$$

say. If j does not satisfy (16.10.2), we let $f_j(\mathbf{t}_j; \mathbf{V}, q) = 0$. Then

$$F(\mathbf{t}, \mathbf{y}; \Delta) = \sum_{\mathbf{V} \in \mathscr{V}} \frac{1}{\#(Q^\vee/L(\mathbf{V}^\vee))} \sum_{q \in Q^\vee/L(\mathbf{V}^\vee)} f_0(\mathbf{t}; \mathbf{V}, q). \tag{16.10.5}$$

Lemma 16.46 ([119, Lemma 9.1]). *We have*

$$f_j(\mathbf{t}_j; \mathbf{V}, q) = \operatorname*{Res}_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} \frac{f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q)}{t_{\beta_j}}, \tag{16.10.6}$$

where the limit procedure of calculating the residue is performed in the sense of Remark 9.15.

PROOF. We begin with two preparatory statements in relation to (16.10.2). First, we show that under the condition (16.10.2),

$$\text{either } \beta_j \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \text{ or } \beta_j \in \mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \setminus \mathbf{V} \tag{16.10.7}$$

holds. Under the condition (16.10.2) we have $\beta_j \in \mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$. Assume $\beta_j \in \mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$. Then there exists $k \le j-1$ such that $\beta_j \in \mathrm{Lh}_{\mathbb{R}}[A_k] \setminus \mathrm{Lh}_{\mathbb{R}}[A_{k-1}]$. Since $\mathrm{Lh}_{\mathbb{R}}[A_k] \setminus \mathrm{Lh}_{\mathbb{R}}[A_{k-1}] \neq \emptyset$, we have $\beta_k \in \mathbf{V}$ by the argument just below (16.10.3). Then $\beta_j, \beta_k \in (\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_k]) \setminus (\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_{k-1}])$ and $\#(\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_k]) - \#(\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_{k-1}]) \ge 2$. Since the elements of $\mathbf{V}$ are linearly independent, this contradicts $\dim \mathrm{Lh}_{\mathbb{R}}[A_k] - \dim \mathrm{Lh}_{\mathbb{R}}[A_{k-1}] \le 1$. Therefore (16.10.7) holds.

Next, we show that under the condition (16.10.2),

$$\mathrm{Lh}_{\mathbb{R}}[A_j] = \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_j]] \tag{16.10.8}$$

holds. Since $\mathrm{Lh}_{\mathbb{R}}[A_j] \supset \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_j]]$ is trivial, we show the opposite inclusion. If $\mathrm{Lh}_{\mathbb{R}}[A_{k-1}] \subsetneq \mathrm{Lh}_{\mathbb{R}}[A_k]$, then $\beta_k \in \mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_k]$ (again by the argument just below (16.10.3)). Hence $\dim \mathrm{Lh}_{\mathbb{R}}[A_j] = \#(\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_j])$, which implies (16.10.8).

Now we start to prove (16.10.6).

Case 1 (The case when j satisfies (16.10.2)). By (16.10.7) either $\beta_j \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$ or $\beta_j \in \mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \setminus \mathbf{V}$ holds.

Case 1-1. Assume $\beta_j \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$. Then $\mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \subsetneq \mathrm{Lh}_{\mathbb{R}}[A_j]$ and we see that, in the expression (16.10.4) of $f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q)$, β_j appears only in $f_{j-1,1}$ and $f_{j-1,4}$. When $t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle$, the singularity appears only on the factor corresponding to $\gamma = \beta_j$ in $f_{j-1,4}$, whose contribution to the residue (16.10.6) is

$$\begin{aligned} \operatorname*{Res}_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} \frac{1}{t_{\beta_j}} \frac{t_{\beta_j} \exp(t_{\beta_j} \{\mathbf{y}+q\}_{\mathbf{V},\beta_j})}{e^{t_{\beta_j}} - 1} \\ = \exp(2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle \{\mathbf{y}+q\}_{\mathbf{V},\beta_j}) \\ = \exp(2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle \langle \mathbf{y}+q, \mu_{\beta_j}^{\mathbf{V}} \rangle) \end{aligned} \tag{16.10.9}$$

because $\langle \beta_j^\vee, \lambda \rangle \in \mathbb{Z}$, which becomes a member of f_{j3}. Therefore the contribution of $f_{j-1,3} f_{j-1,4}$ to the residue (16.10.6) is

$$\begin{aligned} &= \exp\Big(2\pi\sqrt{-1} \sum_{\beta \in \mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]} \langle \beta^\vee, \lambda \rangle \langle \mathbf{y}+q, \mu_\beta^{\mathbf{V}} \rangle\Big) \\ &\times \exp(2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle \langle \mathbf{y}+q, \mu_{\beta_j}^{\mathbf{V}} \rangle) \prod_{\gamma \in \mathbf{V} \setminus (\mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \cup \{\beta_j\})} \frac{t_\gamma \exp(t_\gamma \{\mathbf{y}+q\}_{\mathbf{V},\gamma})}{e^{t_\gamma} - 1} \end{aligned} \tag{16.10.10}$$

$$= \exp\Big(2\pi\sqrt{-1}\sum_{\beta\in\mathbf{V}\cap\mathrm{Lh}_{\mathbb{R}}[A_j]}\langle\beta^\vee,\lambda\rangle\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\Big)\prod_{\gamma\in\mathbf{V}\setminus\mathrm{Lh}_{\mathbb{R}}[A_j]}\frac{t_\gamma\exp(t_\gamma\{\mathbf{y}+q\}_{\mathbf{V},\gamma})}{e^{t_\gamma}-1}$$
$$= f_{j3}f_{j4}.$$

Next, consider the denominator of $f_{j-1,1}$ for

$$\gamma\in\Delta_+\setminus(\mathbf{V}\cup\mathrm{Lh}_{\mathbb{R}}[A_{j-1}])$$

at $t_{\beta_j}=2\pi\sqrt{-1}\langle\beta_j^\vee,\lambda\rangle$. It is

(16.10.11)

$$\begin{aligned}
&S_{j-1,\gamma}(t_\gamma,\mathbf{t}_{j-1,\mathbf{V}})\Big|_{t_{\beta_j}=2\pi\sqrt{-1}\langle\beta_j^\vee,\lambda\rangle}\\
&= t_\gamma-\sum_{\beta\in\mathbf{V}\setminus\mathrm{Lh}_{\mathbb{R}}[A_{j-1}]}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle\Big|_{t_{\beta_j}=2\pi\sqrt{-1}\langle\beta_j^\vee,\lambda\rangle}\\
&\qquad-2\pi\sqrt{-1}\sum_{\beta\in\mathbf{V}\cap\mathrm{Lh}_{\mathbb{R}}[A_{j-1}]}\langle\beta^\vee,\lambda\rangle\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle\\
&= t_\gamma-\sum_{\beta\in\mathbf{V}\setminus\mathrm{Lh}_{\mathbb{R}}[A_j]}t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle-2\pi\sqrt{-1}\sum_{\beta\in\mathbf{V}\cap\mathrm{Lh}_{\mathbb{R}}[A_j]}\langle\beta^\vee,\lambda\rangle\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle\\
&= S_{j,\gamma}(t_\gamma,\mathbf{t}_{j,\mathbf{V}}).
\end{aligned}$$

For $\gamma\in\mathrm{Lh}_{\mathbb{R}}[A_j]$, this is further

$$= t_\gamma-2\pi\sqrt{-1}\langle\gamma^\vee,\lambda\rangle, \tag{16.10.12}$$

because by (16.10.8), $\gamma^\vee$ is a linear combination of $\mathbf{V}^\vee\cap\mathrm{Lh}_{\mathbb{R}}[A_j]$, so $\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle=0$ for $\beta\in\mathbf{V}\setminus\mathrm{Lh}_{\mathbb{R}}[A_j]$ and by writing $\gamma^\vee=\sum_{\beta\in\mathbf{V}\cap\mathrm{Lh}_{\mathbb{R}}[A_j]}a_\beta\beta^\vee$, we have

$$\sum_{\beta\in\mathbf{V}\cap\mathrm{Lh}_{\mathbb{R}}[A_j]}\langle\beta^\vee,\lambda\rangle\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle=\langle\gamma^\vee,\lambda\rangle.$$

Thus each corresponding member of $f_{j-1,1}$ becomes a member of f_{j2}. Therefore

(16.10.13)

$$\begin{aligned}
&f_{j-1,1}f_{j-1,2}\Big|_{t_{\beta_j}=2\pi\sqrt{-1}\langle\beta_j^\vee,\lambda\rangle}\\
&=\prod_{\gamma\in(\Delta_+\setminus\mathrm{Lh}_{\mathbb{R}}[A_j])\setminus(\mathbf{V}\cup\mathrm{Lh}_{\mathbb{R}}[A_{j-1}])}\frac{t_\gamma}{S_{j,\gamma}(t_\gamma,\mathbf{t}_{j,\mathbf{V}})}\times\prod_{\gamma\in\mathcal{A}_{j-1}}\frac{t_\gamma}{t_\gamma-2\pi\sqrt{-1}\langle\gamma^\vee,\lambda\rangle},
\end{aligned}$$

where

$$\mathcal{A}_{j-1}=\big((\Delta_+\cap\mathrm{Lh}_{\mathbb{R}}[A_j])\setminus(\mathbf{V}\cup\mathrm{Lh}_{\mathbb{R}}[A_{j-1}])\big)\cup\big((\Delta_+\cap\mathrm{Lh}_{\mathbb{R}}[A_{j-1}])\setminus(\mathbf{V}\cup A_{j-1})\big).$$

We note that

$$\mathcal{A}_{j-1} = (\Delta_+ \cap \mathrm{Lh}_{\mathbb{R}}[A_j]) \setminus (\mathbf{V} \cup A_{j-1}) \tag{16.10.14}$$

holds. In fact, it is obvious that $\mathcal{A}_{j-1}$ is included in the right-hand side, while an element of the right-hand side which does not belong to $\Delta_+ \cap \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$ does not belong to $\mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$ either. Therefore (16.10.14) holds, and noting $\mathbf{V} \cup A_{j-1} = \mathbf{V} \cup A_j$, we further find that $\mathcal{A}_{j-1} = (\Delta_+ \cap \mathrm{Lh}_{\mathbb{R}}[A_j]) \setminus (\mathbf{V} \cup A_j)$. As for the first product on the right-hand side of (16.10.13), we see that $(\Delta_+ \setminus \mathrm{Lh}_{\mathbb{R}}[A_j]) \setminus (\mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]) = \Delta_+ \setminus (\mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_j])$. Therefore (16.10.13) can be read as

$$f_{j-1,1} f_{j-1,2}\Big|_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} = f_{j1} f_{j2}. \tag{16.10.15}$$

Combining (16.10.10) and (16.10.15) we arrive at (16.10.6) in the present case.

Case 1-2. Assume $\beta_j \in \mathrm{Lh}_{\mathbb{R}}[A_{j-1}] \setminus \mathbf{V}$. Then $\mathrm{Lh}_{\mathbb{R}}[A_{j-1}] = \mathrm{Lh}_{\mathbb{R}}[A_j]$ and we see that β_j appears only in $f_{j-1,2}$ in the expression (16.10.4) of $f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q)$. Furthermore

$$\operatorname*{Res}_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} \frac{1}{t_{\beta_j}} \frac{t_{\beta_j}}{t_{\beta_j} - 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} = 1 \tag{16.10.16}$$

and hence

$$\operatorname*{Res}_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} \frac{f_{j-1,2}}{t_{\beta_j}} = f_{j2}, \tag{16.10.17}$$

$$f_{j-1,i}\Big|_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} = f_{ji} \qquad (i = 1, 3, 4). \tag{16.10.18}$$

Therefore (16.10.6) holds.

Case 2 (The case when j does not satisfy (16.10.2)). There exists $k \leq j$ such that $\beta_k \notin \mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{k-1}]$. Take the minimum k among them. If $k \leq j-1$, then $f_j(\mathbf{t}_j; \mathbf{V}, q) = f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q) = 0$ and (16.10.6) holds. Assume $k = j$. Then $f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q)$ is given by (16.10.4) and $f_j(\mathbf{t}_j; \mathbf{V}, q) = 0$. In this case β_j appears only in $f_{j-1,1}$ in the expression (16.10.4) of $f_{j-1}(\mathbf{t}_{j-1}; \mathbf{V}, q)$. If $\langle \beta_j^\vee, \mu_\beta^{\mathbf{V}} \rangle = 0$ for all $\beta \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$, then $\beta_j \in \mathrm{Lh}_{\mathbb{R}}[\mathbf{V} \cap \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]] \subset \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$, which contradicts $\beta_j \notin \mathbf{V} \cup \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$. Hence at least one of the coefficients of t_β for $\beta \in \mathbf{V} \setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]$ does not vanish in

$$S_{j-1,\gamma}(t_\gamma, \mathbf{t}_{j-1,\mathbf{V}})\Big|_{t_{\beta_j} = 2\pi\sqrt{-1}\langle \beta_j^\vee, \lambda \rangle} \tag{16.10.19}$$

$$= t_{\beta_j} - \sum_{\beta\in\mathbf{V}\setminus \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]} t_\beta \langle \beta_j^\vee, \mu_\beta^{\mathbf{V}}\rangle$$
$$- 2\pi\sqrt{-1} \sum_{\beta\in\mathbf{V}\cap \mathrm{Lh}_{\mathbb{R}}[A_{j-1}]} \langle \beta^\vee, \lambda\rangle \langle \beta_j^\vee, \mu_\beta^{\mathbf{V}}\rangle \Big|_{t_{\beta_j}=2\pi\sqrt{-1}\langle \beta_j^\vee,\lambda\rangle}.$$

Therefore the denominator of $f_{j-1,1}$ is not identically 0, which implies that the residue of $f_{j-1,1}$ and hence of $f_{j-1}(\mathbf{t}_{j-1};\mathbf{V},q)$ is 0 (by the calculation in the sense of Remark 9.15) and so (16.10.6) holds.

Thus we showed that (16.10.6) holds in any case. □

By the repeated use of Lemma 16.46, we obtain

(16.10.20)
$$f_{d(I)}(\mathbf{t}_{d(I)};\mathbf{V},q) = \operatorname*{Res}_{t_{\beta_{d(I)}}=2\pi\sqrt{-1}\langle \beta_{d(I)}^\vee,\lambda\rangle} \cdots \operatorname*{Res}_{t_{\beta_1}=2\pi\sqrt{-1}\langle \beta_1^\vee,\lambda\rangle} \Big(\prod_{\alpha\in\Delta_{I+}} \frac{1}{t_\alpha}\Big) f_0(\mathbf{t};\mathbf{V},q).$$

Proof of Theorem 9.14. Let $\Phi_I = \{\beta_k \mid \beta_k \notin \mathrm{Lh}_{\mathbb{R}}[A_{k-1}]\} \subset \Delta_{I+}$. Then Φ_I is a linearly independent set with $\#\Phi_I = \#I$.

Assume $\Phi_I \subset \mathbf{V}$. Let $\mathbf{V}_I = \mathbf{V}\setminus\Phi_I$. Since $\mathrm{Lh}_{\mathbb{R}}[A_{d(I)}] = \mathrm{Lh}_{\mathbb{R}}[\Delta_{I+}]$, we see that $\mathbf{V}\setminus \mathrm{Lh}_{\mathbb{R}}[A_{d(I)}] = \mathbf{V}_I$,

$$\Delta_+ \setminus (\mathbf{V}\cup \mathrm{Lh}_{\mathbb{R}}[A_{d(I)}]) = \Delta_+ \setminus (\mathbf{V}_I \cup \mathrm{Lh}_{\mathbb{R}}[\Delta_{I+}]) = \Delta^{(I)} \setminus \mathbf{V}_I,$$

$$(\Delta_+ \cap \mathrm{Lh}_{\mathbb{R}}[A_{d(I)}]) \setminus (\mathbf{V}\cup A_{d(I)}) = \Delta_{I+} \setminus (\mathbf{V}\cup\Delta_{I+}) = \emptyset.$$

Further, we have

$$\mathbf{V}\cap \mathrm{Lh}_{\mathbb{R}}[A_{d(I)}] = \mathbf{V}\setminus\mathbf{V}_I = \Phi_I.$$

Therefore (16.10.4) for $j = d(I)$ implies

(16.10.21)
$$f_{d(I)}(\mathbf{t}_{d(I)};\mathbf{V},q)$$
$$= \prod_{\gamma\in\Delta^{(I)}\setminus\mathbf{V}_I} \frac{t_\gamma}{t_\gamma - \sum_{\beta\in\mathbf{V}_I} t_\beta\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle - 2\pi\sqrt{-1}\sum_{\beta\in\Phi_I}\langle\beta^\vee,\lambda\rangle\langle\gamma^\vee,\mu_\beta^{\mathbf{V}}\rangle}$$
$$\times \exp\Big(2\pi\sqrt{-1}\sum_{\beta\in\Phi_I}\langle\beta^\vee,\lambda\rangle\langle\mathbf{y}+q,\mu_\beta^{\mathbf{V}}\rangle\Big) \prod_{\gamma\in\mathbf{V}_I} \frac{t_\gamma \exp(t_\gamma\{\mathbf{y}+q\}_{\mathbf{V},\gamma})}{e^{t_\gamma}-1}.$$

If $\Phi_I \not\subset \mathbf{V}$, then for $\beta_k \in \Phi_I\setminus\mathbf{V}$, we have $\beta_k \notin \mathbf{V}\cup\mathrm{Lh}_{\mathbb{R}}[A_{k-1}]$ and hence $f_{d(I)}(\mathbf{t}_{d(I)};\mathbf{V},q) = 0$.

Applying Lemma 9.8 to the right-hand side of (9.4.10), and noting that there is a one-to-one correspondence between

(16.10.22) $\{\mathbf{V}\in\mathscr{V} \mid \Phi_I\subset\mathbf{V}\}$ and $\{\mathbf{V}_I \mid \mathbf{V}_I\cup\Psi_I\in\mathscr{V}_I\}$,

we obtain

(16.10.23)

$$F(\mathbf{t}_I, \mathbf{y}, \lambda; I; \Delta) = \sum_{\mathbf{V} \in \mathscr{V}} \frac{1}{\#(Q^\vee / L(\mathbf{V}^\vee))} \sum_{q \in Q^\vee / L(\mathbf{V}^\vee)} f_{d(I)}(\mathbf{t}_{d(I)}; \mathbf{V}, q)$$

by using $\mathbf{t}_{d(I)} = \mathbf{t}_I$. Combining this with (16.10.20), and using (16.10.5), we finally obtain the formula (9.4.17) stated in Theorem 9.14.

So far our discussion has been under the fixed choice of the order $\beta_1, \dots, \beta_{d(I)}$ of the elements of Δ_{I+}. The right-hand side of (9.4.17) seems to depend on this choice. However, the left-hand side of (9.4.17), defined by (9.4.10), does not depend on the choice of the order. Neither does the right-hand side of (9.4.17). This implies the independence of the choice of the order of iterated residues, and hence the proof of Theorem 9.14 is thus complete.

□

REMARK 16.47. A key formula in the proof of Theorem 9.11 is (16.9.5), whose right-hand side includes the quantity $S_\lambda(\mathbf{k}, \mathbf{y}; I; \Delta)$. This is equal to $S(\mathbf{k}, \pi_{I^c}(\mathbf{y}); \Lambda)$ by (16.9.22), where Λ is given by (16.9.10). If $I = \emptyset$, then $P_{\emptyset++} = \{0\}$, so the only λ appearing on the right-hand side of (16.9.5) is $\lambda = 0$. Then $\dot{f}_\alpha = \langle \alpha^\vee, \lambda \rangle = 0$, so the elements of Λ have no affine-shift term. However, when we consider general I, we have to treat non-zero λ, and hence the elements of Λ may have an affine-shift term $\dot{f}_\alpha$. This is the reason why in this chapter we develop a rather long general theory on lattice sums of hyperplane arrangements, and consider the lattice sum (16.1.2) with affine-shift terms in the denominators.

CHAPTER 17

Miscellaneous results

In the previous chapters we discussed various aspects of zeta-functions of root systems, but still, there remain several topics which we have not yet discussed. In this last chapter we mention those topics briefly, just stating definitions and results without explaining the details of their proofs.

17.1. Functional equations

The Riemann zeta-function $\zeta(s)$ satisfies the beautiful symmetric functional equation

$$\pi^{s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{(1-s)/2}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s). \tag{17.1.1}$$

In the case of Hurwitz zeta-functions $\zeta(s,\alpha)$, we cannot expect such symmetry. The functional equation for $\zeta(s,\alpha)$ is of the form

$$\zeta(s,\alpha) = \frac{\Gamma(1-s)}{\sqrt{-1}(2\pi)^{1-s}}\left\{e^{\pi\sqrt{-1}s/2}\phi(1-s,\alpha) - e^{-\pi\sqrt{-1}s/2}\phi(1-s,-\alpha)\right\}, \tag{17.1.2}$$

where $\phi(s,\alpha)$ is the Lerch zeta-function defined by (4.1.9).

How about the case of multiple zeta-functions? The first discovery in this direction is the second-named author's article [**135**], in which a certain *functional equation* for $\zeta_{EZ,2}(s_1,s_2)$ was obtained. Let

$$F_{\pm}(s_1,s_2) = \sum_{k=1}^{\infty}\sigma_{s_1+s_2-1}(k)\Psi(s_2,s_1+s_2;\pm 2\pi\sqrt{-1}k), \tag{17.1.3}$$

where $\sigma_a(k) = \sum_{d|k} d^a$ and Ψ is the confluent hypergeometric function defined by

$$\Psi(b,c;x) = \frac{1}{\Gamma(b)}\int_0^{e^{\sqrt{-1}\phi}\infty} e^{-xy}y^{b-1}(1+y)^{c-b-1}dy \tag{17.1.4}$$

for $\Re b > 0$, $-\pi < \phi < \pi$ and $|\phi + \arg x| < \pi/2$. Define

$$g_2(s_1,s_2) = \zeta_{EZ,2}(s_1,s_2) - \frac{\Gamma(1-s_1)}{\Gamma(s_2)}\Gamma(s_1+s_2-1)\zeta(s_1+s_2-1). \tag{17.1.5}$$

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In [135], the formula

(17.1.6)

$$g_2(s_1,s_2) = (2\pi)^{s_1+s_2-1}\Gamma(1-s_1) \times \left\{ e^{\pi\sqrt{-1}(s_1+s_2-1)/2}F_+(s_1,s_2) + e^{\pi\sqrt{-1}(1-s_1-s_2)/2}F_-(s_1,s_2) \right\}$$

and the symmetric property

$$F_\pm(1-s_2,1-s_1) = (\pm 2\pi\sqrt{-1})^{s_1+s_2-1}F_\pm(s_1,s_2) \tag{17.1.7}$$

for $F_\pm$ were shown (actually in a slightly generalized form). Combining (17.1.6) and (17.1.7), we readily obtain:

THEOREM 17.1. *It holds that*

(17.1.8)

$$g_2(s_1,s_2) = \Gamma(1-s_1)\left\{F_+(1-s_2,1-s_1) + F_-(1-s_2,1-s_1)\right\}.$$

This form is similar to (17.1.2). Moreover, since $\Psi(s_2, s_1+s_2; \pm 2\pi\sqrt{-1}k)$ has the asymptotic expansion whose first term is $(\pm 2\pi\sqrt{-1}k)^{-s_2}$, we may regard that $F_\pm(s_1,s_2)$ is a kind of generalized Dirichlet series. From this viewpoint, we may understand that (17.1.8) is a functional equation for g_2, hence for $\zeta_{EZ,2}$.

Theorem 17.1 is not explicitly stated in [135]; it first appeared in [138]. On the other hand, in [135], the formula

(17.1.9)

$$\frac{g_2(s_1,s_2)}{(2\pi)^{s_1+s_2-1}\Gamma(1-s_1)} = \frac{g_2(1-s_2,1-s_1)}{(\sqrt{-1})^{s_1+s_2-1}\Gamma(s_2)} + 2\sqrt{-1}\sin\left(\frac{\pi}{2}(s_1+s_2-1)\right)F_+(s_1,s_2)$$

was shown. Later in [107] it was pointed out that the following theorem is immediate from (17.1.9):

THEOREM 17.2. *The symmetric form of the functional equation*

$$\frac{g_2(s_1,s_2)}{(2\pi)^{s_1+s_2-1}\Gamma(1-s_1)} = \frac{g_2(1-s_2,1-s_1)}{(\sqrt{-1})^{s_1+s_2-1}\Gamma(s_2)} \tag{17.1.10}$$

is valid on the hyperplane $\{(s_1,s_2)\in\mathbb{C}^2 \mid s_1+s_2 = 2k+1\}$, *where* k *is any non-zero integer.*

Therefore we may interpret (17.1.9) as a "meta" functional equation which produces (17.1.10) for all $k \neq 0$.

A generalization of Theorem 17.2 to the case of double L-functions twisted by Dirichlet characters was given in [111].

As we discussed in Chapter 3, the Mordell–Tornheim multiple zeta-function $\zeta_{MT,r}$ is a specialization of the zeta-function of the root system

of type A_r (see (3.2.6)). In this case, a functional equation can be shown not only for $r = 2$, but also for all $r \geq 2$. This was done by Okamoto and Onozuka [**169**], who proved an analogue of (17.1.9) for $\zeta_{MT,r}$, $r \geq 2$.

Another type of functional equation in the general r-fold case, due to Hirose and Sato [**62**], is also to be mentioned; this is valid for normalized Shintani multiple L-functions.

17.2. The desingularization

In the case of the Riemann zeta-function $\zeta(s)$, the explicit values at negative integer points are important from various viewpoints, such as the p-adic theory. Therefore it is natural to study the values of multiple zeta-functions at negative integer points. However, as we remarked in Chapter 7 (see Remark 7.2), multi-variable multiple zeta-functions in general have many singular analytic sets. For example, $\zeta_{EZ,r}(s_1, \ldots, s_r)$ has infinitely many singular hyperplanes (1.3.6). Almost all negative integer points are located on those singularities, and hence those points are the points of indeterminacy of $\zeta_{EZ,r}(s_1, \ldots, s_r)$.

This fact makes the matter of values of multiple zeta-functions at negative integer points very complicated. In fact, let $\mathbf{k} = (k_1, \ldots, k_r)$ be one of such points. Then the "value" $\zeta_{EZ,r}(\mathbf{k})$ may be only defined as the limit

$$\lim_{\mathbf{s} \to \mathbf{k}} \zeta_{EZ,r}(\mathbf{s}) \qquad (\mathbf{s} = (s_1, \ldots, s_r)), \tag{17.2.1}$$

but this value may depend on the choice of the way how $\mathbf{s}$ approaches $\mathbf{k}$. The explicit evaluation of this value under various choices of the approaching way was first studied by Akiyama, Egami and Tanigawa [**1**], Akiyama and Tanigawa [**3**], and then by subsequent papers ([**179**] [**180**] [**97**] [**98**] [**176**] [**40**] [**41**] [**144**]). Among them, the first-named author's article [**98**] treats a rather general class of multiple zeta-functions, involving zeta-functions of root systems. Therefore [**98**] includes the study of the values of zeta-functions of root systems at negative integer points in the above "limit" sense.

On the other hand, there are also attempts to find some suitable way of regularization, which will give a more definite meaning of the "value" at $\mathbf{k}$. Some people developed the method of *renormalization* in the frame of the Hopf algebra structure ([**54**] [**129**] [**34**] [**35**]). Another method is *desingularization*, due to Furusho and the authors ([**44**] [**46**]),

which is of a more analytic flavor.[1] In [44] we only studied the case of (a slight generalization of) the Euler–Zagier multiple zeta-functions, but in [46], more generally, we treated multiple zeta-functions of the form $\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b})$ defined by (7.2.25) (actually, moreover, with twisted factors on the numerators).

The *desingularized zeta-function* associated with $\zeta_r(s_1,\dots,s_n;\mathscr{A},\mathbf{b})$ is defined as follows:

(17.2.2)

$$\zeta_r^{\mathrm{des}}(s_1,\dots,s_n;\mathscr{A},\mathbf{b}) = \lim_{c\to 1}\prod_{k=1}^{r}(1-c)^{-1}$$
$$\times\frac{1}{\prod_{j=1}^{n}(e^{2\pi\sqrt{-1}s_j}-1)\Gamma(s_j)}\int_{\mathcal{C}^n}\prod_{j=1}^{n}x_j^{s_j-1}\exp\left(\left(\sum_{k=1}^{r}a_{jk}-b_j\right)x_j\right)dx_j$$
$$\times\prod_{k=1}^{r}\left(\frac{1}{\exp\left(\sum_{j=1}^{n}a_{jk}x_j\right)-1}-\frac{c}{\exp\left(c\sum_{j=1}^{n}a_{jk}x_j\right)-1}\right),$$

where the limit is taken for $c\in\mathbb{R}$, and $\mathcal{C}$ is the Hankel contour, the same as defined in (8.1.4).

This definition[2] seems complicated, but we can show that the desingularized zeta-function ζ_r^{des} is actually a finite linear combination of ζ_r. Assume that there exists a set of constants c_{mj} $(1\le m\le r, 1\le j\le n)$ such that

$$\sum_{j=1}^{n}c_{mj}a_{jk}=\delta_{m,k} \tag{17.2.3}$$

for all k,m, where $\delta_{m,k}$ is Kronecker's delta. Let

$$G(\mathbf{u},\mathbf{v})=\prod_{k=1}^{r}\left\{1-\left(1+\sum_{j=1}^{n}c_{kj}(v_j^{-1}-b_j)\right)\left(\sum_{j=1}^{n}a_{jk}u_jv_j\right)\right\},$$

where $\mathbf{u}=(u_j),\mathbf{v}=(v_j)$ $(j=1,\dots,n)$ are indeterminates. Expanding the right-hand side, we may write

$$G(\mathbf{u},\mathbf{v})=\sum_{\mathbf{l},\mathbf{m}}\alpha_{\mathbf{l},\mathbf{m}}\prod_{j=1}^{n}u_j^{l_j}v_j^{m_j},$$

[1] The principles of these two methods were originally quite different from each other, but actually these are closely connected; see Komiyama [96].

[2] The original motivation for introducing this definition is connected with the theory of *p-adic multiple L-functions*; see [111] [45] and [46].

where $\mathbf{l} = (l_1, \dots, l_n) \in \mathbb{N}_0^n$, $\mathbf{m} = (m_1, \dots, m_n) \in \mathbb{Z}^n$, and $\alpha_{\mathbf{l},\mathbf{m}}$ are coefficients. The sum on the right-hand side is obviously a finite sum.

THEOREM 17.3. ([46, Theorem 2.7]) *Under the assumption* (17.2.3)*, we have*

$$\zeta_r^{\mathrm{des}}(s_1, \dots, s_n; \mathscr{A}, \mathbf{b}) = \sum_{\mathbf{l},\mathbf{m}} \alpha_{\mathbf{l},\mathbf{m}} \left(\prod_{j=1}^{n} (s_j)_{l_j} \right) \zeta_r(s_1 + m_1, \dots, s_n + m_n; \mathscr{A}, \mathbf{b}), \tag{17.2.4}$$

where $(s)_k = s(s+1)\cdots(s+k-1)$ *stands for the Pochhammer symbol.*

THEOREM 17.4. ([46, Theorem 2.2]) $\zeta_r^{\mathrm{des}}(s_1, \dots, s_n; \mathscr{A}, \mathbf{b})$ *can be continued to the whole space* $\mathbb{C}^n$ *as an entire function.*

Theorem 17.4 justifies the name of the desingularized zeta-function. Theorem 17.3 implies that just a finite linear combination eliminates all of the infinitely many singularities. In particular, the values of $\zeta_r^{\mathrm{des}}(s_1, \dots, s_n; \mathscr{A}, \mathbf{b})$ at negative integer points are definitely determined, and we can write down those values explicitly:

THEOREM 17.5. ([46, Theorem 2.6]) *Let* $\lambda_1, \dots, \lambda_n \in \mathbb{N}_0$*. Then we have*

$$\begin{aligned} \zeta_r^{\mathrm{des}}(-\lambda_1, \dots, -\lambda_n; \mathscr{A}, \mathbf{b}) &= \prod_{j=1}^{n} (-1)^{\lambda_j} \lambda_j! \sum_{\substack{m_j + \nu_{j1} + \cdots + \nu_{jr} = \lambda_j \\ (1 \le j \le n)}} \left(\prod_{k=1}^{r} B_{\nu_{1k} + \cdots + \nu_{nk} + 1} \right) \\ &\times \left(\prod_{j=1}^{n} \frac{(\sum_{k=1}^{r} a_{jk} - b_j)^{m_j}}{m_j!} \right) \left(\prod_{j=1}^{n} \prod_{k=1}^{r} \frac{a_{jk}^{\nu_{jk}}}{\nu_{jk}!} \right). \end{aligned} \tag{17.2.5}$$

Zeta-functions of root systems are special cases of $\zeta_r(s_1, \dots, s_n; \mathscr{A}, \mathbf{b})$, with $n = \#(\Delta_+)$, $\mathscr{A} = (\langle \alpha^\vee, \lambda_k \rangle)_{\alpha \in \Delta_+, 1 \le k \le r}$, and $\mathbf{b} = (\langle \alpha^\vee, \rho \rangle)_{\alpha \in \Delta_+}$. The assumption (17.2.3) is satisfied with $(c_{m\alpha}) = (I_r \ \mathbf{0})$, where I_r is the $r \times r$ identity matrix, $\mathbf{0}$ is the $(n-r) \times r$ zero-matrix (provided that we fix the order in the set Δ_+ so that the first r elements are fundamental roots). Then

$$G(\mathbf{u}, \mathbf{v}) = \prod_{k=1}^{r} \left(1 - \sum_{\alpha \in \Delta_+} \langle \alpha^\vee, \lambda_k \rangle u_\alpha v_\alpha v_{\alpha_k}^{-1} \right)$$

(see [46, (3.12)]). By Theorems 17.4 and 17.5, we find that $\zeta_r^{\mathrm{des}}(\mathbf{s}; \Delta)$ is entire, and its values at negative integer points can be explicitly computed.

REMARK 17.6. Another variant of zeta-functions of root systems is their q-analogue, which was introduced and studied by Kato [88].

17.3. The number of representations of $SU(3)$

Let $p_1(n)$ be the number (up to equivalence) of n-dimensional representations of $\mathfrak{sl}(2)$, or equivalently, of $SU(2)$. Since $SU(2)$ has just one irreducible representation of dimension k for each $k \in \mathbb{N}$, we see that $p_1(n)$ is equal to the number of partitions of n. Therefore the asymptotic behavior of $p_1(n)$ can be described by the classical Hardy–Ramanujan formula [55]:

$$p_1(n) = \frac{1}{4\sqrt{3}n} e^{\pi\sqrt{2n/3}}(1 + o(1)) \qquad (n \to \infty). \tag{17.3.1}$$

Romik [177] considered the analogue of this matter in the case of $\mathfrak{sl}(3)$, or $SU(3)$. Let $p_2(n)$ be the number of n-dimensional representations of $\mathfrak{sl}(3)$. From (1.5.2) we find that

$$\zeta_W(s; \mathfrak{sl}(3)) = \sum_{\varphi} (\dim \varphi)^{-s} = \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \left(\frac{m_1 m_2 (m_1 + m_2)}{2} \right)^{-s}. \tag{17.3.2}$$

This is because the irreducible representations of $\mathfrak{sl}(3)$ are a family of representations U_{m_1,m_2} parametrized by $m_1, m_2 \in \mathbb{N}$, with

$$\dim U_{m_1,m_2} = \frac{1}{2} m_1 m_2 (m_1 + m_2).$$

Therefore from (17.3.2) we see that $p_2(n)$ satisfies the generating function identity

$$\sum_{n=0}^{\infty} p_2(n) X^n = \prod_{m_1,m_2=1}^{\infty} \frac{1}{1 - X^{m_1 m_2 (m_1+m_2)/2}}.$$

From this observation we may expect that $\zeta_W(s; \mathfrak{sl}(3))$ will be useful to study $p_2(n)$. In fact, inspired by the method developed in [131], Romik studied the analytic properties of $\zeta_2((s,s,s); \mathfrak{sl}(3))$ in detail, and (combining other ideas from asymptotic analysis, the theory of theta functions, probability theory etc.) he proved the following asymptotic formula:

THEOREM 17.7. (Romik [177]) *As $n \to \infty$, we have*

$$p_2(n) = \frac{K}{n^{3/5}} \exp\left(A_1 n^{2/5} - A_2 n^{3/10} - A_3 n^{1/5} - A_4 n^{1/10}\right)(1 + o(1)), \tag{17.3.3}$$

where $A_1 = 5\xi^2$, $A_2 = \xi^{-1}\eta$, $A_3 = (3/80)\xi^{-4}\eta^2$, $A_4 = (11/3200)\xi^{-7}\eta^3$ *and*

$$K = \frac{2\sqrt{3\pi}}{\sqrt{5}}\xi^{1/3}\exp\left(-\frac{1}{2560}\xi^{-10}\eta^4\right)$$

with

$$\xi = \left(\frac{1}{9}\Gamma(1/3)^2\zeta(5/3)\right)^{3/10}, \qquad \eta = -\sqrt{\pi}\zeta(1/2)\zeta(3/2).$$

Further researches on the topic discussed in this section have been done recently by Bringmann et al. [22][23][24].

17.4. A connection with Schur multiple zeta-functions

Let $\lambda = (\lambda_1, \dots, \lambda_m)$ be a partition of $n \in \mathbb{N}$, that is, $\{\lambda_i\}$ is a non-increasing sequence of positive integers whose sum is equal to n. The *Young diagram* of λ is a collection of n square boxes, with λ_i boxes in the ith row $(1 \le i \le m)$. This can be identified with the set

$$D(\lambda) = \{(i,j) \in \mathbb{Z}^2 \mid 1 \le i \le m, 1 \le j \le \lambda_i\}.$$

Let X be a set, and make a *Young tableau* of the diagram by filling any box (i,j) with some element $x_{ij} \in X$. If $X = \mathbb{N}$ and the entries in each row are weakly increasing from left to right and those in each column are strictly increasing from top to bottom, we call it a *semi-standard Young tableau* of shape λ. Denote the set of all semi-standard Young tableaux of shape λ by $\mathrm{SSTY}(\lambda)$.

The *Schur multiple zeta-function* of λ is defined by

$$\zeta_\lambda(\mathbf{s}) = \sum_{M \in \mathrm{SSYT}(\lambda)} \frac{1}{M^{\mathbf{s}}}, \tag{17.4.1}$$

where

$$M^{\mathbf{s}} = \prod_{(i,j)\in D(\lambda)} m_{ij}^{s_{ij}} \quad \text{for} \quad M = (m_{ij}) \in \mathrm{SSYT}(\lambda).$$

This notion was introduced by Nakasuji, Phuksuwan and Yamasaki [164].

When $D(\lambda)$ is just one column, then $\zeta_\lambda(\mathbf{s})$ is nothing but the Euler–Zagier multiple zeta-function $\zeta_{EZ,n}(s_{11}, \dots, s_{n1})$. On the other hand, when $D(\lambda)$ is just one row, then $\zeta_\lambda(\mathbf{s})$ becomes

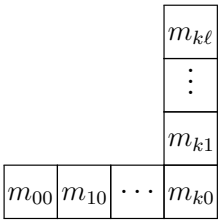
$$\zeta^\star_{EZ,n}(s_{11}, \dots, s_{1n}) = \sum_{1 \le m_1 \le \cdots \le m_n} \frac{1}{m_1^{s_{11}} m_2^{s_{12}} \cdots m_n^{s_{1n}}} \tag{17.4.2}$$

which is usually called the *multiple zeta-star function*. Special values of (17.4.2), called *multiple zeta-star values* (MZSV), are important objects

in the theory of multiple zeta values. Therefore Schur multiple zeta-functions, which interpolate these two notions $\zeta_{EZ,n}$ and $\zeta^{\star}_{EZ,n}$, should also be interesting.

A skew Young diagram is a diagram obtained as a set difference of two Young diagrams λ and μ, with $\mu_i \leq \lambda_i$ for all i. In particular, when $\mu = (k, \ldots, k)$ (ℓ-times) and $\lambda = (k+1, \ldots, k+1)$ ($(\ell+1)$-times), we call the corresponding diagram a Young diagram of anti-hook type, and denote it by $\mathrm{rib}(k|\ell)$. The associated Young tableau can be illustrated as follows:

(17.4.3)



where we renumbered the indices in boxes as above.

The corresponding Schur multiple zeta-function $\zeta_{\mathrm{rib}(k|\ell)}(\mathbf{s})$ can be defined similarly as above. The second-named author and Nakasuji [**142**] discovered that $\zeta_{\mathrm{rib}(k|\ell)}(\mathbf{s})$ can be written in terms of a certain generalized form of zeta-functions of root systems of type A.

For $r > 0$ and $0 \leq d \leq r$, let

(17.4.4)

$$\zeta^{\bullet}_{r,d}(\underline{\mathbf{s}}, A_r) = \underbrace{\left(\sum_{m_1=0}^{\infty} \cdots \sum_{m_d=0}^{\infty}\right)'}_{d \text{ times}} \underbrace{\sum_{m_{d+1}=1}^{\infty} \cdots \sum_{m_r=1}^{\infty}}_{r-d \text{ times}} \prod_{1 \leq i < j \leq r+1} (m_i + \cdots + m_{j-1})^{-s(i,j)},$$

where $\underline{\mathbf{s}} = (s(i,j))$ and the prime means that the terms $(m_i + \cdots + m_{j-1})^{-s(i,j)}$, with $1 \leq i < j \leq d+1$ and $m_i = \cdots = m_{j-1} = 0$, are omitted. Obviously, $\zeta^{\bullet}_{r,0}(\underline{\mathbf{s}}, A_r) = \zeta_r(\underline{\mathbf{s}}, A_r)$. We also introduce, for $x > 0$,

(17.4.5)

$$\zeta^{H}_{r}(\underline{\mathbf{s}}, x, A_r) = \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{1 \leq i < j \leq r+1} (x + m_i + \cdots + m_{j-1})^{-s(i,j)},$$

and

(17.4.6)

$$\zeta^{\bullet,H}_{r,d}(\underline{\mathbf{s}}, x, A_r) = \underbrace{\left(\sum_{m_1=0}^{\infty} \cdots \sum_{m_d=0}^{\infty}\right)}_{d \text{ times}} \underbrace{\sum_{m_{d+1}=1}^{\infty} \cdots \sum_{m_r=1}^{\infty}}_{r-d \text{ times}}$$

$$\prod_{1\le i<j\le r+1}(x+m_i+\cdots+m_{j-1})^{-s(i,j)},$$

so $\zeta^{\bullet,H}_{r,0}(\underline{\mathbf{s}},x,A_r)=\zeta^H_r(\underline{\mathbf{s}},x,A_r)$. When $r=0$, we set $\zeta_0=\zeta^H_0=\zeta^\bullet_{0,0}=\zeta^{\bullet,H}_{0,0}=1$. We can say that $\zeta^\bullet_{r,d}$ may be regarded as a combination of zeta-functions and zeta-"star"-functions of root systems, and ζ^H_r and $\zeta^{\bullet,H}_{r,d}$ may be regarded as examples of zeta-functions of root systems of "Hurwitz type". Note that these are special cases of multiple zeta-functions (7.2.25) studied in Chapter 7. Then:

THEOREM 17.8. ([**142**, Theorem 4.1]) *We have*

$$\zeta_{\mathrm{rib}(k|\ell)}(\mathbf{s})=\zeta^\bullet_{k+\ell+1,k+1}(\mathfrak{u},A_{k+\ell+1})+\sum_{\mu=0}^{k-1}\sum_{\nu=0}^{\ell-1}(-1)^{\mu+\nu}Z_{\mu,\nu}(\mathbf{s};(k,\ell)),$$

where

$$\begin{aligned}Z_{\mu,\nu}(\mathbf{s};(k,\ell))=&\sum_{m_{00},m_{k\ell}\ge1}m_{00}^{-s_{00}}m_{k\ell}^{-s_{k\ell}}\\&\times\zeta^{\bullet,H}_{k-(\mu+1),k-(\mu+1)}(\mathfrak{s}_1,m_{00},A_{k-(\mu+1)})\\&\times\zeta^{\bullet,H}_{\mu+\nu+1,\nu}(\mathfrak{v},m_{00}+m_{k\ell},A_{\mu+\nu+1})\zeta^H_{\ell-(\nu+1)}(\mathfrak{s}_2,m_{k\ell},A_{\ell-(\nu+1)}).\end{aligned}$$

Here, $\mathfrak{u}$, $\mathfrak{v}$, $\mathfrak{s}_1$, $\mathfrak{s}_2$ *are certain vectors consisting of* s_{ij} *and* 0, *whose definitions we omit.*

REMARK 17.9. A further study in this direction has been done in [**143**].

REMARK 17.10. It should be noted that the double series $Z_{\mu,\nu}(\mathbf{s};(k,\ell))$ has some similarity with "Weyl group multiple Dirichlet series" in the sense of Bump [**27**]. Therefore the above theorem may shed light on Bump's question: *whether there are some undiscovered connections between the theory of Weyl group multiple Dirichlet series and the theory of zeta-functions of root systems.*

17.5. Recent results on functional relations

Functional relations among zeta-functions of root systems are one of the main topics of the present text. In this final section we briefly mention recent developments on this theme.

Hirose, Murahara and Onozuka [**61**] gave a new class of functional relations for zeta-functions of root systems of type A_r which interpolate Ohno's relations for MZVs (see (17.5.1) below).

Let $\mathbf{k}=(k_1,\dots,k_r)\in\mathbb{N}^r$ with $k_r>1$ and $\mathbf{k}^\dagger$ its dual index, namely if

$$\mathbf{k}=(\underbrace{1,\dots,1}_{a_1-1},b_1+1,\dots,\underbrace{1,\dots,1}_{a_\ell-1},b_\ell+1)$$

then $\mathbf{k}^\dagger$ is defined by

$$\mathbf{k}^\dagger=(\underbrace{1,\dots,1}_{b_\ell-1},a_\ell+1,\dots,\underbrace{1,\dots,1}_{b_1-1},a_1+1)$$

for each a_i, $b_i\in\mathbb{N}$. Ohno proved the following formula which is now called *Ohno's relation* ([**165**, Theorem 1]):

$$\sum_{\substack{e_1+\cdots+e_r=m\\ e_1,\dots,e_r\ge 0}}\zeta_{EZ,r}(k_1+e_1,\dots,k_r+e_r)=\sum_{\substack{e'_1+\cdots+e'_{r'}=m\\ e'_1,\dots,e'_{r'}\ge 0}}\zeta_{EZ,r'}(k'_1+e'_1,\dots,k'_{r'}+e'_{r'}) \tag{17.5.1}$$

for $\mathbf{k}^\dagger=(k'_1,\dots,k'_{r'})$ and $m\in\mathbb{N}_0$. It is noted that (17.5.1) includes two well-known formulas for MZVs. One is the sum formula given by Granville [**52**] and Zagier (unpublished) (see (1.3.5)):

$$\sum_{\substack{k_1,\dots,k_r\in\mathbb{N},k_r\ge 2\\ k_1+\cdots+k_r=k}}\zeta_{EZ,r}(k_1,\dots,k_r)=\zeta(k),$$

which coincides with the case $\mathbf{k}=(\underbrace{1,\dots,1}_{r-1},2)$ and $\mathbf{k}^\dagger=(r+1)$ of (17.5.1). Another is the *duality relation*

$$\zeta_{EZ,r}(\mathbf{k})=\zeta_{EZ,r'}(\mathbf{k}^\dagger), \tag{17.5.2}$$

which coincides with the case $m=0$ of (17.5.1).

The proof of (17.5.1) is based on Zagier's proof of sum formulas, the idea of which is to construct the generating function of sum formulas. For the details, see [**165**].

The duality (17.5.2) was originally shown from the Drinfel'd integral (13.1.2) whose original form is

$$\mathcal{I}(\varepsilon_1,\dots,\varepsilon_\ell)=\int\cdots\int_{0<t_1<\cdots<t_\ell<1}\frac{dt_1}{A_{\varepsilon_1}(t_1)}\cdots\frac{dt_\ell}{A_{\varepsilon_\ell}(t_\ell)} \tag{17.5.3}$$

for $\varepsilon_i\in\{0,1\}$ with $\varepsilon_1=1$, $\varepsilon_\ell=0$ and $A_0(t)=t$, $A_1(t)=1-t$. Then (as M. Kontsevich observed) we have

$$\zeta(k_1,\dots,k_r)=\mathcal{I}(\underbrace{1,0,\dots,0}_{k_1},\underbrace{1,0,\dots,0}_{k_2},\dots,\underbrace{1,0,\dots,0}_{k_r}),$$

by using $(1-t)^{-1} = 1 + t + t^2 + \cdots$ and integrating the power series iteratively. By the change of variables $(t'_1, \dots, t'_\ell) = (1-t_\ell, \cdots, 1-t_1)$, we obtain

$$\mathcal{I}(\varepsilon_1, \dots, \varepsilon_\ell) = \mathcal{I}(1-\varepsilon_\ell, \dots, 1-\varepsilon_1),$$

which implies (17.5.2) (see Zagier [**215**]).

As a functional interpolation of the left-hand side of (17.5.1), Hirose, Murahara and Onozuka defined

$$I_{\mathbf{k}}(s) = \sum_{\nu=1}^{r} \sum_{0<n_1<\cdots<n_r} \frac{1}{n_1^{k_1}\cdots n_r^{k_r}} \frac{1}{n_\nu^s} \prod_{\substack{\mu=1 \\ \mu\neq\nu}}^{r} \frac{n_\mu}{n_\mu - n_\nu}, \tag{17.5.4}$$

and called it Ohno's function. They showed that $I_{\mathbf{k}}(m)$ coincides with the left-hand side of (17.5.1), and proved the following.

THEOREM 17.11. ([**61**, Theorem 1.3])

$$I_{\mathbf{k}}(s) = I_{\mathbf{k}^\dagger}(s). \tag{17.5.5}$$

We can easily see that

$$I_{\mathbf{k}}(s) = \sum_{\nu=1}^{r} (-1)^{\nu-1} \zeta_r(\mathbf{s}_\nu; A_r), \tag{17.5.6}$$

where $\mathbf{s}_\nu = (s_{ij}^{(\nu)})$ $(1 \le \nu \le r)$ with

$$s_{ij}^{(\nu)} = \begin{cases} k_{j-1} - 1 & (i = 1,\ 2 \le j \le r+1,\ j \neq \nu+1) \\ k_\nu + s & (i = 1,\ j = \nu+1) \\ 1 & (2 \le i \le \nu,\ j = \nu+1) \\ 1 & (i = \nu+1,\ \nu+2 \le j \le r+1) \\ 0 & (\text{otherwise}) \end{cases}$$

(for the order of indices, see (3.2.2)). Therefore (17.5.5) gives an answer to Problem 1.1. Corollary 7.5 gives that $I_{\mathbf{k}}(s)$ can be continued meromorphically to the whole space. For further analytic properties of $I_{\mathbf{k}}(s)$, see Kamano and Onozuka [**81**].

As for the proof of Theorem 17.11, Hirose, Murahara and Onozuka first confirmed that the validity of (17.5.5) for any $s \in \mathbb{N}_0$ is reduced to (17.5.1). Then, since both sides of (17.5.5) are linear forms of zeta-functions of type A_r, namely are finite sums of Dirichlet series, using [4, Theorem 11.3] they showed that (17.5.5) holds for $s \in \mathbb{C}$.

It seems that the functional relation (17.5.5) does not belong to the class of functional relations given by the general theory in Chapter 9, but we are not sure whether (17.5.5) can be captured in the frame of the theory of the present text, or not.

Here we consider functional relations given from (17.5.5) with $r, r' \leq 2$, namely the case $\mathbf{k} = (1,2)$ and $\mathbf{k}' = (3)$. We can easily obtain from (17.5.6) that

$$\begin{aligned} I_{(1,2)}(s) &= \sum_{n_1<n_2} \frac{1}{n_1^{s+1}(n_2-n_1)n_2} - \sum_{n_1<n_2} \frac{1}{(n_2-n_1)n_2^{s+2}} \\ &= \zeta_2(1, s+1, 1; A_2) - \zeta_{EZ,2}(1, s+2), \\ I_{(3)}(s) &= \zeta(s+3). \end{aligned}$$

Hence $I_{(1,2)}(s) = I_{(3)}(s)$ implies

$$\zeta_2(1, s+1, 1; A_2) - \zeta_{EZ,2}(1, s+2) = \zeta(s+3). \tag{17.5.7}$$

On the other hand, (17.5.7) can be directly obtained from Theorem 8.3 with $(k,l) = (1,1)$, which is (12.1.3), by applying $1/mn = (1/m + 1/n)/(m+n)$ to the first sum on the left-hand side. Setting $s = m \in \mathbb{N}_0$ in (17.5.7), we obtain the sum formula for double zeta values. This discussion is just what we stated in the proof of Proposition 12.1. Therefore (17.5.5) can be regarded as a certain extension of the interpolation relation (17.5.7) (for the zeta-functions of type A_2) to the general case of type A_r ($r \geq 3$).

There are some relevant results. Hirose, Murahara and Onozuka [**60**] gave another type of functional relations (involving certain infinite sums of zeta-functions) which interpolate sum formulas. We also mention a result of Murahara and Onozuka [**160**] on the interpolation of cyclic relations. Inspired by an integral expression of Schur multiple zeta values obtained by Nakasuji, Phuksuwan and Yamasaki [**164**] (which itself is a generalization of the integral-series identity due to Kaneko and Yamamoto [**86**]), Hirose, Murahara and Murakami [**59**] defined *cyclic multiple zeta values*, proved their integral-series identity, and obtained some relations valid among them. Those relations imply the sum formula, the cyclic sum formula ([**68**] [**166**]) and the derivation relation [**72**]. Murahara and Onozuka [**160**] discovered a formula among certain multiple zeta-functions, which interpolates those cyclic relations.

Now a lot of functional relations are known. However, it is still a difficult open problem to determine all functional relations among zeta-functions of root systems, and to discover some underlying structure of them.

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